```
-- src/games/MM2/Blatant/KnifeAura.lua
local Module={Name='MM2 Knife Aura',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Knife Aura requires the MM2 core module')Module.Runtime=runtime local getPlayerRole:any=core.getPlayerRole local
getPlayerWeapon:any=core.getPlayerWeapon local isPlayerAlive:any=core.isPlayerAlive local isProtectedTarget:any=core.isProtectedTarget local trajectoryCalibration:any=core.trajectoryCalibration type
KnifeRuntimeState={masterEnabled:boolean,auraWanted:boolean,throwWanted:boolean,lastAuraSwing:number}local knifeSettings:any={aura=false,auraRange=14,holdMouse=false}state.mm2KnifeRuntime={
masterEnabled=false,auraWanted=false,throwWanted=false,lastAuraSwing=-math.huge}::KnifeRuntimeState local function getEffectiveRange():number return knifeSettings.auraRange or 14 end local function
getEquippedWeapon(name:string,_tag:string):Tool?return getPlayerWeapon(LocalPlayer,name,true)end local function canUseEquippedKnife(requireReady:boolean?):(Tool?,BasePart?)local character:Model?=
LocalPlayer.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')::Humanoid?local root:BasePart?=character and character:FindFirstChild('HumanoidRootPart')::
BasePart?local knife:Tool?=getEquippedWeapon('Knife','Weapon_Knife')if not character or not humanoid or humanoid.Health<=0 or not root or not knife or(requireReady==true and knife.Enabled==false)or
knife:GetAttribute('Disabled')==true or getPlayerRole(LocalPlayer)~='Murderer'then return nil,nil end return knife,root end local function isValidKnifeTarget(player:Player):boolean if player==
LocalPlayer or isProtectedTarget(player)or not isPlayerAlive(player)or getPlayerRole(player)=='Murderer'then return false end return player.Character~=nil and player.Character:FindFirstChild(
'HumanoidRootPart')~=nil end local function findKnifeAuraTarget(localRoot:BasePart,maxDistance:number):Player?local closest:Player?=nil local closestDistance:number=maxDistance for _,player:Player in
ipairs(Players:GetPlayers())do if isValidKnifeTarget(player)then local targetRoot:BasePart?=player.Character and player.Character:FindFirstChild('HumanoidRootPart')::BasePart?if targetRoot then local
distance:number=(targetRoot.Position-localRoot.Position).Magnitude if distance<=closestDistance then closest=player closestDistance=distance end end end end return closest end local
knifeControllerGeneration:number=0 local function processKnifeAura(manual:boolean?):()if knifeSettings.holdMouse and manual~=true then return end local swingInterval:number=if manual==true then 0.3
else 0.86 if os.clock()-state.mm2KnifeRuntime.lastAuraSwing<swingInterval then return end local knife:Tool?,localRoot:BasePart?=canUseEquippedKnife(true)if not knife or not localRoot then return end
local currentRange:number=getEffectiveRange()local target:Player?=findKnifeAuraTarget(localRoot,currentRange)local targetCharacter:Model?=target and target.Character local targetRoot:BasePart?=
targetCharacter and targetCharacter:FindFirstChild('HumanoidRootPart')::BasePart?if not target or not targetCharacter or not targetRoot or(targetRoot.Position-localRoot.Position).Magnitude>
currentRange or os.clock()-state.mm2KnifeRuntime.lastAuraSwing<swingInterval then return end local handle:BasePart?=knife:FindFirstChild('Handle')::BasePart?if not handle or not handle:IsA('BasePart')
then return end local events:Instance?=knife:FindFirstChild('Events')local handleTouched:Instance?=events and events:FindFirstChild('HandleTouched')local activated:boolean=pcall(function():()knife:
Activate()end)if not activated then return end state.mm2KnifeRuntime.lastAuraSwing=os.clock()local swingGeneration:number=knifeControllerGeneration task.delay(0.025,function():()local currentKnife:
Tool?,currentLocalRoot:BasePart?=canUseEquippedKnife(false)local currentCharacter:Model?=target.Character local currentRoot:BasePart?=currentCharacter and currentCharacter:FindFirstChild(
'HumanoidRootPart')::BasePart?local currentHandle:BasePart?=knife:FindFirstChild('Handle')::BasePart?if not knifeSettings.aura or not state.mm2KnifeRuntime.masterEnabled or swingGeneration~=
knifeControllerGeneration or currentKnife~=knife or not currentLocalRoot or currentCharacter~=targetCharacter or not isValidKnifeTarget(target)or not currentRoot or not currentHandle or not
currentHandle:IsA('BasePart')or(currentRoot.Position-currentLocalRoot.Position).Magnitude>getEffectiveRange()then return end local sent:boolean=false if handleTouched and handleTouched:IsA(
'RemoteEvent')then sent=pcall(function():()(handleTouched::RemoteEvent):FireServer(currentRoot)end)end if not sent and type(state.fireTouchInterest)=='function'then pcall(state.fireTouchInterest,
currentHandle,currentRoot,0)pcall(state.fireTouchInterest,currentHandle,currentRoot,1)end pcall(trajectoryCalibration.noteAuthoredShot,trajectoryCalibration,{kind='stab',toolName='Knife',originPos=
currentLocalRoot.Position,aimPos=currentRoot.Position,target=target})end)end local holdConnection:RBXScriptConnection?=nil local function stopHoldMouse():()if holdConnection then pcall(function():()(
holdConnection::RBXScriptConnection):Disconnect()end)holdConnection=nil end end local function refreshHoldMouse():()stopHoldMouse()if not knifeSettings.holdMouse or not knifeSettings.aura or not state
.mm2KnifeRuntime.masterEnabled then return end local inputService:any=UserInputService if not inputService or not inputService.InputBegan then return end holdConnection=inputService.InputBegan:
Connect(function(input:any,gameProcessed:boolean):()if gameProcessed then return end local kind:any=input.UserInputType local clicked:boolean=kind==Enum.UserInputType.MouseButton1 local touched:
boolean=kind==Enum.UserInputType.Touch if not clicked and not touched then return end if inputService:GetFocusedTextBox()~=nil then return end local menu:any=services and services.menu or nil if menu
and type(menu.isCapturingInput)=='function'then local ok:boolean,capturing:any=pcall(menu.isCapturingInput)if ok and capturing==true then return end end processKnifeAura(true)end)end local function
refreshKnifeController():()knifeControllerGeneration+=1 disconnectFeatureConnection('MM2KnifeVirtualHitbox')disconnectFeatureConnection('MM2KnifeAura')disconnectFeatureConnection('MM2KnifeController')
refreshHoldMouse()if not knifeSettings.aura or not state.mm2KnifeRuntime.masterEnabled then return end local elapsed:number=0 featureConnections.MM2KnifeController=TaskManager:Connect(function(
deltaTime:number):()elapsed+=math.max(0,deltaTime)local interval:number=1/20 if elapsed+1e-9<interval then return end elapsed=elapsed%interval processKnifeAura()end,nil,'MM2 Knife Aura')end
local function toggleKnifeAura(enabled:boolean):()knifeSettings.aura=enabled state.mm2KnifeRuntime.auraWanted=enabled state.mm2KnifeRuntime.lastAuraSwing=-math.huge refreshKnifeController()end local
killedCharacterByPlayer:{[Player]:Model}={}local killPlayerOption:any=nil local function isLivingInnocentCandidate(player:Player):boolean if player==LocalPlayer or isProtectedTarget(player)or not
isPlayerAlive(player)then return false end local character:Model?=player.Character if not character or killedCharacterByPlayer[player]==character then return false end local humanoid:Humanoid?=
character:FindFirstChildOfClass('Humanoid')::Humanoid?local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?if not humanoid or humanoid.Health<=0 or not root then return false
end local role:any=getPlayerRole(player)if role=='Murderer'or role=='Dead'or role=='None'then return false end return true end local function getLivingInnocentNames():{string}local names:{string}={}
for _,player:Player in ipairs(Players:GetPlayers())do if isLivingInnocentCandidate(player)then table.insert(names,player.Name)end end table.sort(names)return names end local function equipKnifeForKill
():(Model?,Humanoid?,BasePart?,Tool?)if getPlayerRole(LocalPlayer)~='Murderer'then if type(notify)=='function'then notify([[Kill Aura: Only the Murderer can use Kill All / Kill Player.]])end return
nil,nil,nil,nil end local character:Model?=LocalPlayer.Character local humanoid:Humanoid?=character and(character:FindFirstChildOfClass('Humanoid')::Humanoid?)local myRoot:BasePart?=character and(
character:FindFirstChild('HumanoidRootPart')::BasePart?)if not character or not humanoid or humanoid.Health<=0 or not myRoot then return nil,nil,nil,nil end local knife:Tool?=getPlayerWeapon(
LocalPlayer,'Knife',false)if not knife then if type(notify)=='function'then notify('Kill Aura: No Knife found.')end return nil,nil,nil,nil end if knife.Parent~=character then pcall(function():()
humanoid:EquipTool(knife::Tool)end)local deadline:number=os.clock()+0.3 while knife.Parent~=character and os.clock()<deadline do task.wait()end end if knife.Parent~=character then return nil,nil,nil,
nil end return character,humanoid,myRoot,knife end local function executeKnifeKillOnTargets(targets:{Player}):()if#targets==0 then return end local _,_,myRoot:BasePart?,knife:Tool?=equipKnifeForKill()
if not myRoot or not knife then return end local handle:BasePart?=knife:FindFirstChild('Handle')::BasePart?local victimRoots:{BasePart}={}for _,target:Player in ipairs(targets)do if
isLivingInnocentCandidate(target)then local targetChar:Model?=target.Character local victimRoot:BasePart?=targetChar and(targetChar:FindFirstChild('HumanoidRootPart')::BasePart?)if targetChar and
victimRoot then killedCharacterByPlayer[target]=targetChar table.insert(victimRoots,victimRoot)pcall(function():()victimRoot.Anchored=true victimRoot.CFrame=myRoot.CFrame+myRoot.CFrame.LookVector*1
end)end end end if killPlayerOption and type(killPlayerOption.Refresh)=='function'then pcall(function():()killPlayerOption:Refresh()end)end if#victimRoots==0 then return end task.wait(0.1)local stab:
Instance?=knife:FindFirstChild('Stab')if stab and stab:IsA('RemoteEvent')then pcall(function():()(stab::RemoteEvent):FireServer('Slash')end)end local events:Instance?=knife:FindFirstChild('Events')
local handleTouched:Instance?=events and events:FindFirstChild('HandleTouched')pcall(function():()knife:Activate()end)local fireTouch:any=state.fireTouchInterest for _,victimRoot:BasePart in ipairs(
victimRoots)do if handleTouched and handleTouched:IsA('RemoteEvent')then pcall(function():()(handleTouched::RemoteEvent):FireServer(victimRoot)end)end if handle and handle:IsA('BasePart')and type(
fireTouch)=='function'then pcall(fireTouch,handle,victimRoot,0)pcall(fireTouch,handle,victimRoot,1)end task.wait(0.04)end task.delay(0.45,function():()for _,victimRoot:BasePart in ipairs(victimRoots)
do pcall(function():()victimRoot.Anchored=false end)end end)end local function killAllPlayersAction():()task.spawn(function():()local targets:{Player}={}for _,player:Player in ipairs(Players:
GetPlayers())do if isLivingInnocentCandidate(player)then table.insert(targets,player)end end executeKnifeKillOnTargets(targets)end)end local aura:any=state.framework and state.framework:GetModule(
'KillAura')assert(aura,'MM2 Knife requires the existing KillAura module')local wasEnabled:boolean=aura.Enabled local migratedShortcut:any=nil if type(configData)=='table'and not configData.values[
'MM2.Knife.MigratedToKillAura']then for _,label:string in ipairs({'KnifeAura'})do local old:any=configData.states['MM2.Knife.'..label]local key:string='Universal.KillAura.'..label if configData.states
[key]==nil and old~=nil then configData.states[key]=old end end for _,pair in ipairs({{'MM2.Knife.AuraRange','Universal.KillAura.AuraRange'},{'Shortcut.MM2Knife','Shortcut.UniversalKillAura'}})do if
configData.values[pair[2] ]==nil then configData.values[pair[2] ]=configData.values[pair[1] ]if pair[2]=='Shortcut.UniversalKillAura'then migratedShortcut=configData.values[pair[2] ]end end end if
configData.states['Universal.KillAura']==nil then local previous=configData.states['MM2.Knife']configData.states['Universal.KillAura']=previous wasEnabled=wasEnabled or previous==true end configData.
values['MM2.Knife.MigratedToKillAura']=true if type(queueConfigSave)=='function'then queueConfigSave()end end local shortcut:any=state.shortcutBindings and state.shortcutBindings['Universal.KillAura']
if shortcut and type(migratedShortcut)=='string'then local ok:boolean,key:any=pcall(function():any return(Enum.KeyCode::any)[migratedShortcut]end)if ok and key then shortcut.key=key if type(shortcut.
refresh)=='function'then shortcut.refresh()end end end if wasEnabled then aura:Toggle(false)end local KnifeFeature:any=aura.Feature local originalDescription:string=KnifeFeature.description
KnifeFeature.description=[[MM2 melee combat: Knife Aura stabs with your held knife (or click with Hold Mouse). Includes Kill All and Kill Player.]]KnifeFeature.tooltipText=KnifeFeature.description if
type(KnifeFeature.SetName)=='function'then KnifeFeature:SetName('Knife Stab')end aura.CombatAdapter=function(enabled:boolean):()state.mm2KnifeRuntime.masterEnabled=enabled refreshKnifeController()aura
:SetStatus(enabled and'Knife'or nil)end aura.OptionFilter=function(name:string):boolean return name=='Knife Aura'or name=='Aura Range'or name=='Hold Mouse'or name=='Kill All'or name=='Kill Player'end
aura:RefreshVisibility()state.mm2KnifeAuraModule=aura state.mm2KnifeFeature=KnifeFeature aura:CreateToggle({Name='Knife Aura',Default=true,Function=toggleKnifeAura,Tooltip=
'MM2 stabs with your equipped knife while Knife Stab is ON.'})aura:CreateSlider({Name='Aura Range',Min=7,Max=30,Default=14,Show={Option='Knife Aura',Values={true}},Function=function(value:number):()
knifeSettings.auraRange=value end})aura:CreateToggle({Name='Hold Mouse',Default=false,Show={Option='Knife Aura',Values={true}},Tooltip=
[[No automatic stabs: the knife strikes where you click (or tap), ]]..'with the same reach as Aura Range.',Function=function(value:boolean):()knifeSettings.holdMouse=value state.mm2KnifeRuntime.
lastAuraSwing=-math.huge refreshKnifeController()end})aura:CreateButton({Name='Kill All',Function=killAllPlayersAction,Tooltip=
[[As Murderer, equip your Knife and eliminate all living Innocent/Sheriff players.]]})local killPlayerReady:boolean=false local suppressKillPlayerCallback:boolean=false killPlayerOption=aura:
CreateList({Name='Kill Player',Items=getLivingInnocentNames,Single=true,EmptyText='No living innocents',Tooltip=
[[Click to pick a living Innocent player to kill. Dead or eliminated players are removed from the list.]],Function=function(_selected:any,names:{string}):()if not killPlayerReady or
suppressKillPlayerCallback then return end local chosenName:string?=names and names[1]if not chosenName or chosenName==''then return end local targetPlayer:Player?=nil for _,player:Player in ipairs(
Players:GetPlayers())do if player.Name==chosenName and isLivingInnocentCandidate(player)then targetPlayer=player break end end suppressKillPlayerCallback=true if killPlayerOption and type(
killPlayerOption.Set)=='function'then pcall(function():()killPlayerOption:Set(chosenName,false)end)end suppressKillPlayerCallback=false if targetPlayer then local victim:Player=targetPlayer::Player
task.spawn(function():()executeKnifeKillOnTargets({victim})end)end end})if killPlayerOption and type(killPlayerOption.Selected)=='table'and#killPlayerOption.Selected>0 then suppressKillPlayerCallback=
true for _,staleName:string in ipairs(table.clone(killPlayerOption.Selected))do pcall(function():()killPlayerOption:Set(staleName,false)end)end suppressKillPlayerCallback=false end killPlayerReady=
true if wasEnabled then if not KnifeFeature.enabled and type(KnifeFeature.activate)=='function'then KnifeFeature.activate()else aura:Toggle(true)end end activeCleanup=function():()stopHoldMouse()if
aura.Enabled then aura:Toggle(false)end if KnifeFeature.enabled then KnifeFeature.enabled=false refreshFeatureToggle(KnifeFeature)end toggleKnifeAura(false)table.clear(killedCharacterByPlayer)state.
mm2KnifeRuntime.masterEnabled=false state.mm2KnifeFeature=nil state.mm2KnifeAuraModule=nil aura.CombatAdapter=nil KnifeFeature.description=originalDescription KnifeFeature.tooltipText=
originalDescription if type(KnifeFeature.SetName)=='function'then KnifeFeature:SetName('KillAura')end aura.OptionFilter=function(name:string):boolean return name~='Knife Aura'and name~='Aura Range'and
name~='Hold Mouse'and name~='Kill All'and name~='Kill Player'end aura:RefreshVisibility()end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if
not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Blatant/Shoot.lua
local Module={Name='MM2 Shoot',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized then
return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Shoot requires the MM2 core module')local prediction:any=state.mm2Prediction assert(type(prediction)=='table',
'MM2 Shoot requires MM2 Prediction')local trajectoryCalibration:any=core.trajectoryCalibration Module.Runtime=runtime local connectGunFiredSignal:any=core.connectGunFiredSignal local
disconnectGunFiredObserver:any=core.disconnectGunFiredObserver local findMurderer:any=core.findMurderer local getGunOriginCFrame:any=core.getGunOriginCFrame local getPlayerRole:any=core.getPlayerRole
local getPlayerWeapon:any=core.getPlayerWeapon local getWeaponServiceModule:any=core.getWeaponServiceModule local isPlayerAlive:any=core.isPlayerAlive local isProtectedTarget:any=core.
isProtectedTarget local isWeaponPathClear:any=core.isWeaponPathClear local describeRayHit:any=core.describeRayHit local mm2Settings:any=core.mm2Settings local validateGunShot:any=core.validateGunShot
local shootMacro:boolean=false local lastGunOccluder:any=nil local function macroUnequip(gun:Tool?):()if not shootMacro or not gun then return end task.delay(0.01,function():()pcall(function():()local
backpack:Instance?=LocalPlayer:FindFirstChildOfClass('Backpack')local character:Model?=LocalPlayer.Character if backpack and character and gun.Parent==character then gun.Parent=backpack end end)end)
end local function getSelectedShootTarget():Player?local target if mm2Settings.shootMode=='Custom'and mm2Settings.shootTarget~=''then target=findPlayerByText(mm2Settings.shootTarget)else target=
findMurderer()end if not target or target==LocalPlayer or not isPlayerAlive(target)or isProtectedTarget(target)then return nil end if getPlayerRole(target)~='Murderer'then return nil end return target
end local SILENT_ORIGIN_STUDS:number=2.5 local SILENT_PIERCE_STUDS:number=1 local SILENT_LIFT_STUDS:number=1.6 local MAX_SILENT_SWEEP:number=1.5 local POINT_BLANK_STUDS:number=6 local function
silentLift(info:any,predicted:Vector3):number if type(info)=='table'and typeof(info.rootPosition)=='Vector3'then local aimOffsetY:number=predicted.Y-info.rootPosition.Y if aimOffsetY>0.5 then return
math.clamp(aimOffsetY+0.35,1.2,2.4)end end return SILENT_LIFT_STUDS end local function getSilentSweepDistance(speed:number?):number local configured:number=tonumber(mm2Settings.silentSweep)or 0 local
sweep:number=math.clamp(configured,0,MAX_SILENT_SWEEP)local uncertainty:number=0 if type(prediction)=='table'and type(prediction.getLeadUncertainty)=='function'then local ok:boolean,seconds:any=pcall(
prediction.getLeadUncertainty,'CFrame')if ok and type(seconds)=='number'and seconds==seconds then uncertainty=math.clamp(seconds,0,0.15)end end if uncertainty<=0 then return sweep end local travel:
number=math.max(tonumber(speed)or 0,0)local extra:number=uncertainty*travel return math.clamp(sweep+extra,0,MAX_SILENT_SWEEP)end local function buildSilentShot(_target:Player,info:any,muzzleOrigin:
CFrame?,variant:string):(CFrame?,CFrame?,number,number,string)local predicted:Vector3=info.endpoint local liftStuds:number=silentLift(info,predicted)local velocity:Vector3=info.velocity local
horizontal:Vector3=Vector3.new(velocity.X,0,velocity.Z)local direction:Vector3?=nil local sweep:number=0 if horizontal.Magnitude>1.5 then direction=horizontal.Unit sweep=getSilentSweepDistance(
horizontal.Magnitude)end local sweepStart:Vector3=direction and-direction*sweep or Vector3.zero local sweepEnd:Vector3=direction and direction*sweep or Vector3.zero local approach:Vector3?=nil local
approachOffset:number=SILENT_ORIGIN_STUDS if muzzleOrigin then local toTarget:Vector3=predicted-muzzleOrigin.Position local flat:Vector3=Vector3.new(toTarget.X,0,toTarget.Z)if flat.Magnitude>0.05 then
approach=flat.Unit approachOffset=math.min(SILENT_ORIGIN_STUDS,math.max(1,flat.Magnitude*0.6))end end local startPoint:Vector3 local endPoint:Vector3 local resolvedVariant:string=variant local
pierceAxis:Vector3?=direction or approach if(variant=='Through'or variant==nil or variant=='')and pierceAxis then resolvedVariant='Through'local captured:Vector3=info.capturedPosition or predicted
startPoint=captured-pierceAxis*1+sweepStart endPoint=predicted+pierceAxis*SILENT_PIERCE_STUDS+sweepEnd elseif variant=='Top'then resolvedVariant='Top'startPoint=predicted+Vector3.new(0,liftStuds,0)+
sweepStart endPoint=predicted+sweepEnd elseif variant=='Behind'and approach then resolvedVariant='Behind'startPoint=predicted+approach*approachOffset+sweepStart endPoint=predicted+sweepEnd elseif
approach then resolvedVariant='Front'startPoint=predicted-approach*approachOffset+sweepStart endPoint=predicted+sweepEnd elseif direction then resolvedVariant='Motion'startPoint=predicted+sweepStart
endPoint=predicted+sweepEnd else resolvedVariant='Top'startPoint=predicted+Vector3.new(0,liftStuds,0)endPoint=predicted end if(endPoint-startPoint).Magnitude<=0.05 then startPoint=predicted+Vector3.
new(0,liftStuds,0)endPoint=predicted resolvedVariant='Top'sweep=0 end return CFrame.lookAt(startPoint,endPoint),CFrame.new(endPoint),approachOffset,sweep,resolvedVariant end type FireOptions={silent:
boolean?,quiet:boolean?,whenReady:boolean?,continuous:boolean?,wallCheck:boolean?,redirect:((CFrame,CFrame)->())?,nativeRedirect:boolean?,nativeOrigin:CFrame?}local function chooseExperimentalGunMode(
target:Player?,origin:Vector3?):string local character:Model?=target and target.Character local root:BasePart?=character and character:FindFirstChild('HumanoidRootPart')::BasePart?if not root or not
origin then return'Bullet'end local speed:number=0 pcall(function():()local velocity:Vector3=root.AssemblyLinearVelocity speed=Vector3.new(velocity.X,0,velocity.Z).Magnitude end)if(root.Position-
origin).Magnitude>=30 and speed>=8 then return'CFrame'end return'Bullet'end local SHOT_CONFIRM_WINDOW:number=1.5 local ShootFeature:any=nil local lastRejectReason:string=''local AIM_BODY_PARTS:{string
}={'UpperTorso','Torso','HumanoidRootPart','Head','RightUpperArm','LeftUpperArm','RightUpperLeg','LeftUpperLeg','Right Arm','Left Arm','Right Leg','Left Leg'}local function ghostOnLine(muzzle:Vector3,
endpoint:Vector3,targetCharacter:Model?):any if type(describeRayHit)~='function'then return nil end local ok:boolean,hit:any=pcall(describeRayHit,muzzle,endpoint,targetCharacter)if not ok or type(hit)
~='table'or hit.onTarget then return nil end if hit.transparency~=1 or hit.canCollide~=false then return nil end local reach:number=(endpoint-muzzle).Magnitude if type(hit.distance)~='number'or hit.
distance>=reach-0.5 then return nil end return hit end local GHOST_PROBE_INTERVAL:number=0.15 local GHOST_PROBE_CANDIDATES:number=4 local ghostProbe:any={target=nil,at=-math.huge,partName=nil}
local function avoidGhostAim(target:Player,info:any,muzzle:Vector3):(Vector3?,string?)local character:Model?=target.Character if not character or not info or typeof(info.endpoint)~='Vector3'then
return nil,nil end local now:number=os.clock()if target~=ghostProbe.target or now-ghostProbe.at>=GHOST_PROBE_INTERVAL then ghostProbe.target=target ghostProbe.at=now ghostProbe.partName=nil if
ghostOnLine(muzzle,info.endpoint,character)then local captured:Vector3?=info.capturedPosition local probeLead:Vector3=captured and(info.endpoint-captured)or Vector3.zero local tried:number=0 for _,
partName:string in ipairs(AIM_BODY_PARTS)do if tried>=GHOST_PROBE_CANDIDATES then break end local part:Instance?=character:FindFirstChild(partName)if part and part:IsA('BasePart')then tried+=1 local
candidate:Vector3=(part::BasePart).Position+probeLead if not ghostOnLine(muzzle,candidate,character)then ghostProbe.partName=partName break end end end end end local partName:string?=ghostProbe.
partName if not partName then return nil,nil end local part:Instance?=character:FindFirstChild(partName)if not part or not part:IsA('BasePart')then ghostProbe.partName=nil return nil,nil end local
captured:Vector3?=info.capturedPosition local lead:Vector3=captured and(info.endpoint-captured)or Vector3.zero return(part::BasePart).Position+lead,partName end local function fireGunAtTarget(target:
Player?,options:FireOptions?):boolean local opts:FireOptions=options or{}local silent:boolean=mm2Settings.silentAim local experimental:boolean=opts.silent==nil and mm2Settings.experimentalAim==true if
experimental then local localCharacter:Model?=LocalPlayer and LocalPlayer.Character local ownRoot:BasePart?=localCharacter and localCharacter:FindFirstChild('HumanoidRootPart')::BasePart?silent=
chooseExperimentalGunMode(target,ownRoot and ownRoot.Position)=='CFrame'elseif opts.silent~=nil then silent=opts.silent end local quiet:boolean=opts.quiet==true local continuous:boolean=opts.
continuous==true local nativeRedirect:boolean=opts.nativeRedirect==true and type(opts.redirect)=='function'local useRaw:boolean=mm2Settings.shootRaw or(continuous and silent)local useWallCheck:boolean
=if opts.wallCheck~=nil then opts.wallCheck==true else mm2Settings.shootWallCheck local function reject(message:string):boolean lastRejectReason=message if not quiet and ShootFeature then ShootFeature
:SetStatus(message)end return false end local character,humanoid=getCharacterParts()if not target or not target.Character or target==LocalPlayer or not isPlayerAlive(target)or isProtectedTarget(target
)or getPlayerRole(target)~='Murderer'then return reject('No selected shoot target was found.')end if not character or not humanoid then return reject('Your character is not available.')end local
targetCharacter=target.Character local gun:Tool?=getPlayerWeapon(LocalPlayer,'Gun',true)if not gun or gun.Parent~=character then return reject('Equip the sheriff gun yourself to shoot.')end if not
nativeRedirect and(gun.Enabled==false or gun:GetAttribute('Disabled')==true or gun:GetAttribute('Reloading')==true or gun:GetAttribute('IsReloading')==true)then return reject(
'The gun is disabled or reloading.')end local remote:Instance?=gun and gun:FindFirstChild('Shoot')local okOrigin:boolean,foundOrigin:any=pcall(getGunOriginCFrame,character,gun)local origin:CFrame?=
okOrigin and foundOrigin or nil if not origin and nativeRedirect and typeof(opts.nativeOrigin)=='CFrame'then origin=opts.nativeOrigin end local pendingShot:any=state.mm2ShotFeedback.pending if not
continuous and not nativeRedirect and type(pendingShot)=='table'and type(pendingShot.queuedAt)=='number'and os.clock()-pendingShot.queuedAt<SHOT_CONFIRM_WINDOW then local remaining:number=
SHOT_CONFIRM_WINDOW-(os.clock()-pendingShot.queuedAt)return reject(string.format('Shot gate: waiting for the previous shot (%.1f s left).',remaining))end local trackFeedback:boolean=not continuous and
not(nativeRedirect and type(pendingShot)=='table')if not remote or not remote:IsA('RemoteEvent')or not origin then return reject(not origin and'The gun has no usable origin.'or
'The murderer is unavailable or obstructed.')end local info:any=nil local reason:string?=nil if origin then local aimMode:string=silent and'CFrame'or'Bullet'local ok:boolean,endpoint:any,aimInfo:any,
aimReason:any=pcall(prediction.gunAim,target,origin.Position,aimMode)if not ok then return reject('Aim or visibility check failed.')end if endpoint==nil then reason=aimReason or'no samples'else info=
aimInfo end end if(useRaw or continuous or nativeRedirect or opts.whenReady==true)and not info and target.Character then local targetRoot:BasePart?=target.Character:FindFirstChild('HumanoidRootPart')
if targetRoot then local vel:Vector3=Vector3.zero pcall(function():()vel=targetRoot.AssemblyLinearVelocity or Vector3.zero end)local fallbackEndpoint:Vector3=targetRoot.Position+vel*0.08 info={
endpoint=fallbackEndpoint,velocity=vel,leadSeconds=0.08,motionSource='raw',model='raw',sampleAge=0,samples=0,capturedPosition=targetRoot.Position,rtt=0,echoDelay=0,rttEngine='raw',echoSamples=0,
turnRate=0,brakeSeconds=0,leadCorrection=0,rootPosition=targetRoot.Position}end end if info and origin and targetCharacter then local cleaner:Vector3?,cleanerPart:string?=avoidGhostAim(target,info,
origin.Position)if cleaner then info.endpoint=cleaner info.targetPosition=cleaner info.ghostAvoided=cleanerPart end end if origin and info then local root:BasePart?=nil pcall(function():()root=
targetCharacter:FindFirstChild('HumanoidRootPart')::BasePart?end)local aimVector:Vector3=info.endpoint-origin.Position local toRoot:Vector3=root and(root.Position-origin.Position)or aimVector local
inverted:boolean=aimVector.Magnitude<0.05 or aimVector:Dot(toRoot)<=0 if root and(inverted or aimVector.Magnitude<POINT_BLANK_STUDS)then info.capturedPosition=root.Position info.endpoint=root.Position
info.aimPartChosen='HumanoidRootPart'info.pointBlank=true end end local rejectMessages:{[string]:string}={['no body']='The murderer has no usable body to aim at.',['no samples']=
'The murderer has no fresh motion samples yet.',['stale sample']=[[The murderer's position is stale; waiting for a fresh sample.]],['unreachable']='The murderer is moving too fast to intercept.'}if
not remote or not remote:IsA('RemoteEvent')or not origin or not info then local message:string=rejectMessages[reason or'']or(not origin and'The gun has no usable origin.'or
'The murderer is unavailable or obstructed.')return reject(message)end local function safeToSend(shotOrigin:CFrame,shotEnd:CFrame,muzzle:CFrame?):boolean if gun.Parent~=character or(not nativeRedirect
and(gun.Enabled==false or gun:GetAttribute('Disabled')==true or gun:GetAttribute('Reloading')==true or gun:GetAttribute('IsReloading')==true))then return reject('The gun is disabled or reloading.')end
if useRaw then if not useWallCheck or not isWeaponPathClear then return true end local rawMuzzle:Vector3=(muzzle or shotOrigin).Position local rawClear,rawBlocker=isWeaponPathClear(rawMuzzle,shotEnd.
Position,targetCharacter,nil,{ignoreTarget=true})if not rawClear then if type(rawBlocker)=='table'and trajectoryCalibration and type(trajectoryCalibration.noteBlockedShot)=='function'then pcall(
trajectoryCalibration.noteBlockedShot,trajectoryCalibration,rawBlocker)end return reject('A wall or another body blocks the shot.')end return true end local muzzlePosition:Vector3=(muzzle or
shotOrigin).Position local ok,allowed,why,blocker,occluder=pcall(validateGunShot,target,character,targetCharacter,muzzlePosition,shotOrigin.Position,shotEnd.Position,useWallCheck)if not ok or not
allowed then if type(blocker)=='table'and trajectoryCalibration and type(trajectoryCalibration.noteBlockedShot)=='function'then pcall(trajectoryCalibration.noteBlockedShot,trajectoryCalibration,
blocker)end return reject(why or'Shot safety check failed.')end if type(occluder)=='table'then lastGunOccluder=occluder else lastGunOccluder=nil end return true end if silent then local variant:string
='Through'local configuredSpawn:any=mm2Settings.silentSpawn if configuredSpawn=='Top'or configuredSpawn=='Behind'or configuredSpawn=='Front'then variant=configuredSpawn end local shotOrigin:CFrame?,
shotEnd:CFrame?,offset:number,sweep:number,resolvedVariant:string if info.pointBlank and origin then shotOrigin=origin shotEnd=CFrame.new(info.endpoint)offset=0 sweep=0 resolvedVariant='PointBlank'
else shotOrigin,shotEnd,offset,sweep,resolvedVariant=buildSilentShot(target,info,origin,variant)end if not shotOrigin or not shotEnd then if continuous and target.Character then local tr:BasePart?=
target.Character:FindFirstChild('HumanoidRootPart')if tr then local ep:Vector3=info.endpoint or tr.Position local capPt:Vector3=info.capturedPosition or tr.Position local startPt:Vector3=capPt+Vector3
.new(0,1.2,-1.5)shotOrigin=CFrame.lookAt(startPt,ep)shotEnd=CFrame.new(ep)offset=1.5 sweep=0 resolvedVariant='Through'end end if not shotOrigin or not shotEnd then return reject(
'Could not construct the silent shot.')end end variant=resolvedVariant if not useRaw and not isWeaponPathClear(shotOrigin.Position,shotEnd.Position,targetCharacter,nil,{ignoreTarget=true})then
shotOrigin,shotEnd=origin,CFrame.new(info.endpoint)variant='Muzzle'offset=0 sweep=0 end if not safeToSend(shotOrigin,shotEnd,origin)then return false end if trackFeedback then pcall(state.
mm2ShotFeedback.queue,target,gun,shotOrigin,shotEnd)end state.lastSilentShotInfo={variant=variant,at=os.clock(),offset=math.floor(offset*10)/10}if opts.redirect then opts.redirect(shotOrigin,shotEnd)
else remote:FireServer(shotOrigin,shotEnd)end local function recordSilent():()pcall(function():()if not trajectoryCalibration or type(trajectoryCalibration.noteAuthoredShot)~='function'then return end
trajectoryCalibration:noteAuthoredShot({kind='gun',source=nativeRedirect and'native-redirect'or'menu',tool=gun,toolName=gun and gun.Name or'Gun',originPos=shotOrigin.Position,aimPos=shotEnd.Position,
target=target,silent=true,variant=variant,offset=offset,settings={mode='Silent',spawn=variant,sweep=sweep,wallCheck=useWallCheck,leadSeconds=info.leadSeconds,motionSource=info.motionSource,model=info.
model,sampleAge=info.sampleAge,samples=info.samples,referencePoint=info.capturedPosition and{info.capturedPosition.X,info.capturedPosition.Y,info.capturedPosition.Z},predictedPoint={info.endpoint.X,
info.endpoint.Y,info.endpoint.Z},sampledVelocity={info.velocity.X,info.velocity.Y,info.velocity.Z},rtt=info.rtt,echoDelay=info.echoDelay,rttEngine=info.rttEngine,rttData=info.rttData,occluder=
lastGunOccluder,echoSamples=info.echoSamples,turnRate=info.turnRate,brakeSeconds=info.brakeSeconds,leadCorrection=info.leadCorrection,leadCorrectionTarget=info.leadCorrectionTarget,leadSamples=info.
leadSamples,aimMode=info.aimMode,ghostAvoided=info.ghostAvoided}})end)end if nativeRedirect and task and type(task.defer)=='function'then task.defer(recordSilent)else recordSilent()end if not quiet
and ShootFeature then ShootFeature:SetStatus(experimental and'Experimental \183 CFrame \183 request sent'or nil)end if not continuous then macroUnequip(gun)end return true end local shotEndCFrame:
CFrame=CFrame.new(info.endpoint)if not safeToSend(origin,shotEndCFrame)then return false end if trackFeedback then pcall(state.mm2ShotFeedback.queue,target,gun,origin,shotEndCFrame)end local
shotOriginCFrame:CFrame=CFrame.lookAt(origin.Position,info.endpoint)if opts.redirect then opts.redirect(shotOriginCFrame,shotEndCFrame)else remote:FireServer(shotOriginCFrame,shotEndCFrame)end
local function recordBullet():()pcall(function():()if not trajectoryCalibration or type(trajectoryCalibration.noteAuthoredShot)~='function'then return end trajectoryCalibration:noteAuthoredShot({kind=
'gun',source=nativeRedirect and'native-redirect'or'menu',tool=gun,toolName=gun and gun.Name or'Gun',originPos=origin.Position,aimPos=info.endpoint,target=target,silent=false,settings={mode='Normal',
wallCheck=useWallCheck,leadSeconds=info.leadSeconds,motionSource=info.motionSource,model=info.model,sampleAge=info.sampleAge,samples=info.samples,referencePoint=info.capturedPosition and{info.
capturedPosition.X,info.capturedPosition.Y,info.capturedPosition.Z},predictedPoint={info.endpoint.X,info.endpoint.Y,info.endpoint.Z},sampledVelocity={info.velocity.X,info.velocity.Y,info.velocity.Z},
rtt=info.rtt,echoDelay=info.echoDelay,rttEngine=info.rttEngine,rttData=info.rttData,occluder=lastGunOccluder,echoSamples=info.echoSamples,turnRate=info.turnRate,brakeSeconds=info.brakeSeconds,
leadCorrection=info.leadCorrection,leadCorrectionTarget=info.leadCorrectionTarget,leadSamples=info.leadSamples,aimMode=info.aimMode,ghostAvoided=info.ghostAvoided}})end)end if nativeRedirect and task
and type(task.defer)=='function'then task.defer(recordBullet)else recordBullet()end if not quiet and ShootFeature then ShootFeature:SetStatus(experimental and
'Experimental \183 Bullet \183 request sent'or nil)end if not continuous then macroUnequip(gun)end return true end local shootOptionsReady:boolean=false local shootFeatureActive:boolean=false local
autoShootActive:boolean=false local autoShootWanted:boolean=false local transientShotActive:boolean=false local transientShotGeneration:number=0 local updateAutoShootConnection:(()->())?=nil local
observerGeneration:number=0 local cachedShootRemote:Instance?=nil local lastCacheProbeAt:number=-math.huge local function refreshShootRemoteCache():()local remote:Instance?=nil local okGun:boolean,
foundGun:any=pcall(getPlayerWeapon,LocalPlayer,'Gun',true)local gun:Tool?=if okGun then foundGun else nil if gun then pcall(function():()local found:Instance?=gun:FindFirstChild('Shoot')if found and
found:IsA('RemoteEvent')then remote=found end end)end cachedShootRemote=remote end local function refreshShootInfrastructure():()observerGeneration+=1 local generation:number=observerGeneration local
wantActive:boolean=shootFeatureActive or autoShootActive or transientShotActive state.mm2ShotFeedback.shootActive=wantActive if wantActive then local function observe():()local weaponService:any=
getWeaponServiceModule()if generation~=observerGeneration or not state.mm2ShotFeedback.shootActive then return end if type(weaponService)=='table'then connectGunFiredSignal(weaponService)end end if
state.isMobile then observe()else task.spawn(observe)end refreshShootRemoteCache()return end disconnectGunFiredObserver()state.mm2ShotFeedback.pending=nil state.mm2ShotFeedback.lastAccepted=nil state.
mm2ShotFeedback.hide()refreshShootRemoteCache()end local shotHookInstalled:boolean=false local shotHookWrapper:any=nil local originalNamecall:any=nil local function installNativeGunHook():boolean if
state.isMobile then return true end if shotHookInstalled then return true end local hook:any=state.hookMetamethod local getMethod:any=state.getNamecallMethod local checkCaller:any=state.checkCaller if
type(hook)~='function'or type(getMethod)~='function'or type(checkCaller)~='function'then return false end if type(task)~='table'or type(task.defer)~='function'then return false end local predecessor:
any=nil local pendingRedirects:number=0 local MAX_PENDING_REDIRECTS:number=2 local sendingRedirect:boolean=false local function sendAsExecutor(remote:any,arguments:any):boolean local previous:boolean=
sendingRedirect sendingRedirect=true local ok:boolean,failure:any=pcall(function():()remote:FireServer(table.unpack(arguments,1,arguments.n))end)sendingRedirect=previous if not ok then warn(
'[Wurst] native gun redirect failed: '..tostring(failure))end return ok end local function redirectDeferred(remote:any,arguments:any):()if not shootFeatureActive or not shootOptionsReady then
sendAsExecutor(remote,arguments)return end local shotOrigin:CFrame?=nil local shotEnd:CFrame?=nil local ok:boolean=pcall(function():any lastRejectReason=''return fireGunAtTarget(
getSelectedShootTarget(),{quiet=true,nativeRedirect=true,nativeOrigin=arguments[1],redirect=function(origin:CFrame,endpoint:CFrame):()shotOrigin,shotEnd=origin,endpoint end})end)if shotOrigin and
shotEnd then if ShootFeature then pcall(function():()ShootFeature:SetStatus('Native click redirected')end)end local redirected:any=table.pack(shotOrigin,shotEnd)local copied:number=2 for index:number=
3,arguments.n do copied+=1 redirected[copied]=arguments[index]end redirected.n=copied sendAsExecutor(remote,redirected)return end if ShootFeature then pcall(function():()ShootFeature:SetStatus(if not
ok then'Native redirect failed; original click kept'else(lastRejectReason~=''and lastRejectReason or'Native click unchanged'))end)end sendAsExecutor(remote,arguments)end local function
redirectNamecall(remote:any,...:any):any if sendingRedirect then return predecessor(remote,...)end if not shootFeatureActive or not shootOptionsReady then return predecessor(remote,...)end if remote~=
cachedShootRemote then local now:number=os.clock()if now-lastCacheProbeAt>=0.5 then lastCacheProbeAt=now task.defer(refreshShootRemoteCache)end return predecessor(remote,...)end local okMethod:boolean
,method:any=pcall(getMethod)if not okMethod or method~='FireServer'then return predecessor(remote,...)end local okCaller:boolean,fromExecutor:any=pcall(checkCaller)if not okCaller or fromExecutor~=
false then return predecessor(remote,...)end local arguments:any=table.pack(...)if typeof(arguments[1])~='CFrame'or typeof(arguments[2])~='CFrame'then return predecessor(remote,...)end if
pendingRedirects>=MAX_PENDING_REDIRECTS then return predecessor(remote,...)end pendingRedirects+=1 task.defer(function():()local handled:boolean,failure:any=pcall(redirectDeferred,remote,arguments)
pendingRedirects-=1 if not handled then warn('[Wurst] native gun redirect error: '..tostring(failure))end end)return nil end local callback:any=redirectNamecall if type(state.newCClosure)=='function'
then local ok:boolean,wrapped:any=pcall(state.newCClosure,callback)if ok and type(wrapped)=='function'then callback=wrapped end end local ok:boolean,old:any=pcall(hook,game,'__namecall',callback)if
not ok or type(old)~='function'then return false end predecessor=old originalNamecall,shotHookWrapper,shotHookInstalled=old,callback,true return true end local function uninstallNativeGunHook():()if
not shotHookInstalled or type(state.hookMetamethod)~='function'then return end local ok:boolean,displaced:any=pcall(state.hookMetamethod,game,'__namecall',originalNamecall)if not ok then return end if
displaced==shotHookWrapper then shotHookInstalled,shotHookWrapper,originalNamecall=false,nil,nil elseif type(displaced)=='function'then pcall(state.hookMetamethod,game,'__namecall',displaced)end end
local manualShotPending:boolean=false local function triggerManualShot():()if state.isMobile and not shootFeatureActive then notify('Enable Gun Aimbot first.')return end if manualShotPending then
return end local character,humanoid=getCharacterParts()local gun:Tool?=getPlayerWeapon(LocalPlayer,'Gun')if not character or not humanoid or not gun then notify('No sheriff gun is available.')return
end manualShotPending=true local ok:boolean,message:any=pcall(function():()if gun.Parent~=character then humanoid:EquipTool(gun)RunService.Heartbeat:Wait()end if(not state.isMobile or
shootFeatureActive)and LocalPlayer.Character==character and gun.Parent==character then if not shootFeatureActive and not state.isMobile then transientShotGeneration+=1 local generation:number=
transientShotGeneration transientShotActive=true refreshShootInfrastructure()task.delay(2,function():()if generation~=transientShotGeneration then return end transientShotActive=false
refreshShootInfrastructure()end)end fireGunAtTarget(getSelectedShootTarget(),{whenReady=true})end end)manualShotPending=false if not ok then notify('Shoot failed: '..tostring(message))end end
local function toggleShootMurderer(enabled:boolean):()shootFeatureActive=enabled state.mm2SilentAimEnabled=enabled if state.mm2KnifeThrowRefresh then state.mm2KnifeThrowRefresh()end autoShootActive=
shootFeatureActive and autoShootWanted if enabled and not state.isMobile then if not installNativeGunHook()then notify(
[[Native click redirect needs executor hookmetamethod, getnamecallmethod and checkcaller; the keybind can still fire.]])end elseif not enabled then uninstallNativeGunHook()end
refreshShootInfrastructure()if updateAutoShootConnection then updateAutoShootConnection()end end local lastAutoStatus:string?=nil local function autoStatus(message:string):()if ShootFeature and
lastAutoStatus~=message then lastAutoStatus=message ShootFeature:SetStatus(message)end end local function autoShootTick():()if not shootFeatureActive or not autoShootActive then return end local gun:
Tool?=getPlayerWeapon(LocalPlayer,'Gun',true)if not gun or gun.Parent~=LocalPlayer.Character then autoStatus('Equip gun')return end if state and state.mm2ShotFeedback and state.mm2ShotFeedback.pending
then autoStatus('Esperando confirmaci\243n')return end local target:Player?=getSelectedShootTarget()if not target then autoStatus('No target')return end lastRejectReason=''local sent:boolean=
fireGunAtTarget(target,{whenReady=true,quiet=true})autoStatus(sent and'Auto: request sent'or lastRejectReason)end updateAutoShootConnection=function():()disconnectFeatureConnection('MM2AutoShoot')
autoShootActive=shootOptionsReady and shootFeatureActive and autoShootWanted if not autoShootActive then lastAutoStatus=nil if ShootFeature then ShootFeature:SetStatus(nil)end return end
featureConnections.MM2AutoShoot=TaskManager:Connect(autoShootTick,nil,'MM2 Auto Shoot')autoShootTick()end local function toggleAutoShoot(enabled:boolean):()autoShootWanted=enabled autoShootActive=
shootFeatureActive and autoShootWanted refreshShootInfrastructure()if updateAutoShootConnection then updateAutoShootConnection()end end if type(configData)=='table'then local changed:boolean=false if
configData.states['MM2.Shoot.AutoShoot']==nil and configData.states['MM2.AutoShoot']~=nil then local enabled:boolean=configData.states['MM2.AutoShoot']==true configData.states['MM2.Shoot.AutoShoot']=
enabled if enabled then configData.states['MM2.Shoot']=true end changed=true end local oldKey:any=configData.values['Shortcut.MM2AutoShoot']if configData.values['Shortcut.MM2ShootAutoShoot']==nil and
type(oldKey)=='string'then configData.values['Shortcut.MM2ShootAutoShoot']=oldKey changed=true end if changed and type(queueConfigSave)=='function'then queueConfigSave()end end if type(configData)==
'table'and configData.values['MM2.Shoot.Aimmethod']==nil then configData.values['MM2.Shoot.Aimmethod']=configData.values['MM2.Shoot.Mode']end ShootFeature=createUniversalFeature('Gun Aimbot',state.
isMobile and[[Gun Aimbot with Normal(Bullet), Silent(Cframe), and Experimental modes. The assigned action fires/equips Gun; WallCheck is configurable per mode.]]or
[[Redirect native clicks with Bullet or Silent(CFrame). One optional keybind fires even while the module is off.]],4,toggleShootMurderer,{categoryName='Exploits',configKey='MM2.Shoot',parent=MM2Scroll
,registry=mm2Features})state.mm2SilentAimFeature=ShootFeature state.mm2ChooseExperimentalAim=chooseExperimentalGunMode state.mm2FireSilentShot=function(target:Player?,shotOpts:any):boolean local opts:
any=type(shotOpts)=='table'and shotOpts or{}return fireGunAtTarget(target,{silent=true,quiet=opts.quiet~=false,continuous=opts.continuous~=false,wallCheck=opts.wallCheck})end state.mm2FireGunShot=
function(target:Player?,shotOpts:any):boolean local opts:any=type(shotOpts)=='table'and shotOpts or{}return fireGunAtTarget(target,{silent=opts.silent==true,wallCheck=opts.wallCheck,quiet=opts.quiet~=
false,continuous=opts.continuous~=false})end if addSectionOption then addSectionOption(ShootFeature,'Gun')end local states:any=type(configData)=='table'and configData.states or{}local values:any=type(
configData)=='table'and configData.values or{}local function modeKey(value:any):string if value=='Experimental'then return'Experimental'end if value=='CFrame'or value=='Silent(Cframe)'or value==
'Silent(CFrame)'then return'CFrame'end return'Bullet'end local savedMode:any=values['MM2.Shoot.Aimmethod']local initialMode:string=modeKey(savedMode or(mm2Settings.experimentalAim and'Experimental'or(
mm2Settings.silentAim and'CFrame'or'Bullet')))if state.isMobile then if savedMode=='Bullet'or savedMode=='CFrame'or savedMode=='Silent(CFrame)'then values['MM2.Shoot.Aimmethod']=(savedMode=='CFrame'or
savedMode=='Silent(CFrame)')and'Silent(Cframe)'or'Normal(Bullet)'if type(queueConfigSave)=='function'then queueConfigSave()end end else if initialMode=='Experimental'then initialMode='Bullet'end local
normalized:string=initialMode=='CFrame'and'Silent(CFrame)'or'Bullet'if values['MM2.Shoot.Aimmethod']~=normalized then values['MM2.Shoot.Aimmethod']=normalized if type(queueConfigSave)=='function'then
queueConfigSave()end end end local function modeConfig(mode:string):any local prefix:string='MM2.Shoot.'..mode..'.'local legacy:boolean=mode==initialMode local wall:any=states[prefix..'WallCheck']
local macro:any=states[prefix..'Macro']local targetMode:any=values[prefix..'Target']local targetName:any=values[prefix..'TargetPlayer']if legacy then if wall==nil then wall=states[
'MM2.Shoot.WallCheck']end if macro==nil then macro=states['MM2.Shoot.Macro']end if targetMode==nil then targetMode=values['MM2.Shoot.Target']end if targetName==nil then targetName=values[
'MM2.Shoot.Targetplayer']end end return{wallCheck=wall==nil or wall==true,macro=macro==true,targetMode=targetMode=='Custom'and'Custom'or'Murder',targetName=type(targetName)=='string'and targetName or
''}end local modeConfigs:any={Bullet=modeConfig('Bullet'),CFrame=modeConfig('CFrame'),Experimental=modeConfig('Experimental')}local activeModeKey:string=initialMode local switching:boolean=false local
optionsReady:boolean=false local ShootTargetBox local function refreshShootOptions():()if ShootTargetBox then setOptionVisible(ShootFeature,ShootTargetBox.Parent::GuiObject,mm2Settings.shootMode==
'Custom')end end local function syncMode():()local cfg:any=modeConfigs[activeModeKey]or modeConfigs.Bullet switching=true mm2Settings.shootWallCheck=cfg.wallCheck shootMacro,mm2Settings.shootMode,
mm2Settings.shootTarget=cfg.macro,cfg.targetMode,cfg.targetName if state and type(state.toggleControls)=='table'then for _,item:any in ipairs({{'WallCheck',cfg.wallCheck},{'Macro',cfg.macro}})do local
control:any=state.toggleControls['MM2.Shoot.'..item[1] ]if control and control.setChecked then pcall(control.setChecked,item[2])end end end if state and type(state.valueControls)=='table'then local
target:any=state.valueControls['MM2.Shoot.Target']local name:any=state.valueControls['MM2.Shoot.Targetplayer']if target and target.apply then pcall(target.apply,cfg.targetMode)end if name and name.
apply then pcall(name.apply,cfg.targetName)end end switching=false refreshShootOptions()end local aimMethods:{string}=state.isMobile and{'Normal(Bullet)','Silent(Cframe)','Experimental'}or{'Bullet',
'Silent(CFrame)'}addCycleOption(ShootFeature,'Aim method',aimMethods,initialMode=='Experimental'and 3 or(initialMode=='CFrame'and 2 or 1),function(value:string):()activeModeKey=modeKey(value)
mm2Settings.experimentalAim=activeModeKey=='Experimental'mm2Settings.silentAim=activeModeKey=='CFrame'syncMode()end,state.isMobile and
[[Experimental selects one route based on movement and range; hit-rate improvement is unverified. Options are saved per mode.]]or
[[One keybind is shared; WallCheck and target settings are saved per mode.]])addCycleOption(ShootFeature,'Target',{'Murder','Custom'},1,function(value:string):()mm2Settings.shootMode=value if
optionsReady and not switching then modeConfigs[activeModeKey].targetMode=value values['MM2.Shoot.'..activeModeKey..'.Target']=value end refreshShootOptions()end)if state.bindFeatureActivationKey then
state.bindFeatureActivationKey(ShootFeature,mm2Settings.shootKey,triggerManualShot)end if not state.isMobile then local binding:any=state.shortcutBindings and state.shortcutBindings[ShootFeature.
configKey]if binding then binding.feedback=nil end if type(state.addFeatureShortcutOption)=='function'then state.addFeatureShortcutOption(ShootFeature,'Keybind',
[[Optional: click the box and press a key. It fires in either mode, even when Gun Aimbot is off.]])end end addToggleOption(ShootFeature,'WallCheck',true,function(value:boolean):()mm2Settings.
shootWallCheck=value==true if optionsReady and not switching then modeConfigs[activeModeKey].wallCheck=value==true states['MM2.Shoot.'..activeModeKey..'.WallCheck']=value==true end end,
'Check for walls; saved independently per mode.')addToggleOption(ShootFeature,'Macro',false,function(value:boolean):()shootMacro=value==true if optionsReady and not switching then modeConfigs[
activeModeKey].macro=value==true states['MM2.Shoot.'..activeModeKey..'.Macro']=value==true end end,'Pistol macro: hides the gun after the shot; saved per mode.')ShootTargetBox=addTextOption(
ShootFeature,'Target player','',function(value:string):()mm2Settings.shootTarget=value if optionsReady and not switching then modeConfigs[activeModeKey].targetName=value values['MM2.Shoot.'..
activeModeKey..'.TargetPlayer']=value if type(queueConfigSave)=='function'then queueConfigSave()end end end,false)optionsReady=true syncMode()shootOptionsReady=true if shootFeatureActive and not state
.isMobile and not shotHookInstalled then if not installNativeGunHook()then notify(
[[Native click redirect needs executor hookmetamethod, getnamecallmethod and checkcaller; the keybind can still fire.]])end end if updateAutoShootConnection then updateAutoShootConnection()end
activeCleanup=function():()transientShotGeneration+=1 transientShotActive=false toggleShootMurderer(false)state.mm2SilentAimFeature=nil state.mm2FireSilentShot=nil state.mm2FireGunShot=nil state.
mm2ChooseExperimentalAim=nil toggleAutoShoot(false)state.mm2ShotFeedback.hide()end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.
Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Combat/AutoShoot.lua
local Module={Name='MM2 Auto Shoot',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Auto Shoot requires the MM2 core module')local prediction:any=state.mm2Prediction assert(type(prediction)=='table',
'MM2 Auto Shoot requires MM2 Prediction')Module.Runtime=runtime local findMurderer:any=core.findMurderer local getPlayerRole:any=core.getPlayerRole local getPlayerWeapon:any=core.getPlayerWeapon local
isPlayerAlive:any=core.isPlayerAlive local isProtectedTarget:any=core.isProtectedTarget local isWeaponPathClear:any=core.isWeaponPathClear local getGunOriginCFrame:any=core.getGunOriginCFrame local
settings={mode='Bullet',wallCheck=true,proximity=false,aura=25}local autoShootActive:boolean=false local AutoShootFeature:any=nil local toolConnections:{RBXScriptConnection}={}local lastShotSentAt:
number=0 local AUTO_SHOT_PACE_SECONDS:number=0.22 local function clearToolConnections():()for _,conn:RBXScriptConnection in ipairs(toolConnections)do pcall(function():()conn:Disconnect()end)end table.
clear(toolConnections)end local function findLocalGun():(Tool?,Model?,Humanoid?,BasePart?)local character:Model?=LocalPlayer and LocalPlayer.Character if not character then return nil,nil,nil,nil end
local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?if not humanoid or humanoid.Health<=0 or not root then
return nil,character,humanoid,root end local gun:Tool?=nil if type(getPlayerWeapon)=='function'then gun=getPlayerWeapon(LocalPlayer,'Gun',false)end if not gun then local charGun:Instance?=character:
FindFirstChild('Gun')if charGun and charGun:IsA('Tool')then gun=charGun end end if not gun then local backpack:Instance?=LocalPlayer:FindFirstChildOfClass('Backpack')local bpGun:Instance?=backpack and
backpack:FindFirstChild('Gun')if bpGun and bpGun:IsA('Tool')then gun=bpGun end end return gun,character,humanoid,root end local function ensureGunEquipped(character:Model,humanoid:Humanoid,gun:Tool):
boolean if gun.Parent==character then return true end pcall(function():()humanoid:EquipTool(gun)end)if gun.Parent~=character then pcall(function():()gun.Parent=character end)end return gun.Parent==
character end local function resolveMurdererTarget(localRoot:BasePart,enforceProximity:boolean):(Player?,BasePart?)local murderer:Player?=nil if type(findMurderer)=='function'then murderer=
findMurderer()end if not murderer then local playersService:Players=game:GetService('Players')for _,player:Player in ipairs(playersService:GetPlayers())do if player~=LocalPlayer then local role:any=
type(getPlayerRole)=='function'and getPlayerRole(player)or nil local hasKnife:boolean=type(getPlayerWeapon)=='function'and getPlayerWeapon(player,'Knife',false)~=nil if role=='Murderer'or hasKnife
then murderer=player break end end end end if not murderer or murderer==LocalPlayer then return nil,nil end if type(isPlayerAlive)=='function'and not isPlayerAlive(murderer)then return nil,nil end if
type(isProtectedTarget)=='function'and isProtectedTarget(murderer)then return nil,nil end local targetChar:Model?=murderer.Character local targetRoot:BasePart?=targetChar and targetChar:
FindFirstChild('HumanoidRootPart')::BasePart?if not targetRoot then return nil,nil end if enforceProximity then local okDist:boolean,distance:any=pcall(function():any return(targetRoot.Position-
localRoot.Position).Magnitude end)if okDist and type(distance)=='number'and distance>settings.aura then return nil,nil end end return murderer,targetRoot end local function computeTargetEndpoint(
target:Player,targetRoot:BasePart,originPos:Vector3,aimMode:string?):Vector3 if type(prediction)=='table'and type(prediction.gunAim)=='function'then local ok:boolean,endpoint:any=pcall(prediction.
gunAim,target,originPos,aimMode)if ok and typeof(endpoint)=='Vector3'then return endpoint end end local vel:Vector3=Vector3.zero pcall(function():()vel=targetRoot.AssemblyLinearVelocity end)return
targetRoot.Position+vel*0.08 end local function fireCFrameShotImmediate(target:Player,targetRoot:BasePart,character:Model,localRoot:BasePart,gun:Tool):boolean local shootRemote:Instance?=gun:
FindFirstChild('Shoot')if not(shootRemote and shootRemote:IsA('RemoteEvent'))then return false end if gun.Enabled==false or gun:GetAttribute('Disabled')==true or gun:GetAttribute('Reloading')==true or
gun:GetAttribute('IsReloading')==true then return false end local originCF:CFrame=(type(getGunOriginCFrame)=='function'and getGunOriginCFrame(character,gun))or localRoot.CFrame local endpoint:Vector3=
computeTargetEndpoint(target,targetRoot,originCF.Position,'CFrame')local capturedPos:Vector3=targetRoot.Position local chordFlat:Vector3=Vector3.new(endpoint.X-capturedPos.X,0,endpoint.Z-capturedPos.Z
)local pierceAxis:Vector3 if chordFlat.Magnitude>0.15 then pierceAxis=chordFlat.Unit else local toTargetFlat:Vector3=Vector3.new(endpoint.X-originCF.Position.X,0,endpoint.Z-originCF.Position.Z)if
toTargetFlat.Magnitude>0.05 then pierceAxis=toTargetFlat.Unit else pierceAxis=Vector3.new(0,0,-1)end end local startPoint:Vector3=capturedPos-pierceAxis*1.5 local endPoint:Vector3=endpoint+pierceAxis*
1.5 if(endPoint-startPoint).Magnitude<0.2 then startPoint=endpoint+Vector3.new(0,2,-1.5)endPoint=endpoint end local shotOrigin:CFrame=CFrame.lookAt(startPoint,endPoint)local shotEnd:CFrame=CFrame.new(
endPoint)local firedSilent:boolean=false if type(state.mm2FireSilentShot)=='function'then local okSilent:boolean,res:any=pcall(state.mm2FireSilentShot,target,{continuous=true,quiet=true,wallCheck=
settings.wallCheck})firedSilent=okSilent and res==true if not okSilent then lastShotSentAt=os.clock()end if not firedSilent then return false end end if not firedSilent then local clear:boolean=false
if settings.wallCheck and type(isWeaponPathClear)=='function'then local okClear:boolean,allowed:any=pcall(isWeaponPathClear,originCF.Position,endpoint,target.Character,nil,{ignoreTarget=true})clear=
okClear and allowed==true else clear=true end if clear then local okFire:boolean=pcall(function():()(shootRemote::RemoteEvent):FireServer(shotOrigin,shotEnd)end)if not okFire then lastShotSentAt=os.
clock()return false end else lastShotSentAt=0 return false end end lastShotSentAt=os.clock()return true end local function fireBulletShotImmediate(target:Player,targetRoot:BasePart,character:Model,
localRoot:BasePart,gun:Tool):boolean local shootRemote:Instance?=gun:FindFirstChild('Shoot')if not(shootRemote and shootRemote:IsA('RemoteEvent'))then return false end if gun.Enabled==false or gun:
GetAttribute('Disabled')==true or gun:GetAttribute('Reloading')==true or gun:GetAttribute('IsReloading')==true then return false end local originCF:CFrame=(type(getGunOriginCFrame)=='function'and
getGunOriginCFrame(character,gun))or localRoot.CFrame local endpoint:Vector3=computeTargetEndpoint(target,targetRoot,originCF.Position,'Bullet')if settings.wallCheck and type(isWeaponPathClear)==
'function'then local okClear:boolean,clear:any=pcall(isWeaponPathClear,originCF.Position,endpoint,target.Character)if not okClear or clear~=true then return false end end local firedCore:boolean=false
if type(state.mm2FireGunShot)=='function'then local okShot:boolean,res:any=pcall(state.mm2FireGunShot,target,{silent=false,wallCheck=settings.wallCheck,continuous=true,quiet=true})firedCore=okShot and
res==true if not okShot then lastShotSentAt=os.clock()end if not firedCore then return false end end if not firedCore then local shotOrigin:CFrame=CFrame.lookAt(originCF.Position,endpoint)local
shotEnd:CFrame=CFrame.new(endpoint)local okFire:boolean=pcall(function():()(shootRemote::RemoteEvent):FireServer(shotOrigin,shotEnd)end)if not okFire then lastShotSentAt=os.clock()return false end end
lastShotSentAt=os.clock()return true end local function executeAutoShootCFrameOnce(enforceProximity:boolean):boolean local gun:Tool?,character:Model?,humanoid:Humanoid?,localRoot:BasePart?=
findLocalGun()if not gun or not character or not humanoid or not localRoot then lastShotSentAt=0 return false end local target:Player?,targetRoot:BasePart?=resolveMurdererTarget(localRoot,
enforceProximity)if not target or not targetRoot then return false end if not ensureGunEquipped(character,humanoid,gun)then return false end if os.clock()-lastShotSentAt<AUTO_SHOT_PACE_SECONDS then
return true end return fireCFrameShotImmediate(target,targetRoot,character,localRoot,gun)end state.mm2AutoShootCFrameOnce=executeAutoShootCFrameOnce local function autoShootTick():()if not
autoShootActive then return end local gun:Tool?,character:Model?,humanoid:Humanoid?,localRoot:BasePart?=findLocalGun()if not gun or not character or not humanoid or not localRoot then lastShotSentAt=0
return end local target:Player?,targetRoot:BasePart?=resolveMurdererTarget(localRoot,settings.proximity==true)if not target or not targetRoot then return end if not ensureGunEquipped(character,
humanoid,gun)then return end if os.clock()-lastShotSentAt<AUTO_SHOT_PACE_SECONDS then return end local isCFrame:boolean=settings.mode=='CFrame'or settings.mode=='Cframe'or settings.mode=='Silent AIM'
if settings.mode=='Experimental'then local mode:string='Bullet'if type(state.mm2ChooseExperimentalAim)=='function'then local origin:CFrame=(type(getGunOriginCFrame)=='function'and getGunOriginCFrame(
character,gun))or localRoot.CFrame local okMode:boolean,chosen:any=pcall(state.mm2ChooseExperimentalAim,target,origin.Position)if okMode and chosen=='CFrame'then mode='CFrame'end end local sent:
boolean=false local uncertain:boolean=false if type(state.mm2FireGunShot)=='function'then local ok:boolean,result:any=pcall(state.mm2FireGunShot,target,{silent=mode=='CFrame',wallCheck=settings.
wallCheck,continuous=true,quiet=true})sent=ok and result==true uncertain=not ok if uncertain then lastShotSentAt=os.clock()end end if sent then lastShotSentAt=os.clock()end if AutoShootFeature then
AutoShootFeature:SetStatus('Experimental \183 '..mode..(sent and' \183 request sent'or(uncertain and' \183 send uncertain'or' \183 not sent')))end elseif isCFrame then fireCFrameShotImmediate(target,
targetRoot,character,localRoot,gun)else fireBulletShotImmediate(target,targetRoot,character,localRoot,gun)end end local function watchToolContainer(container:Instance?):()if not container or type((
container::any).ChildAdded)~='table'and typeof((container::any).ChildAdded)~='RBXScriptSignal'then return end pcall(function():()local conn:RBXScriptConnection=(container::any).ChildAdded:Connect(
function(child:Instance):()if autoShootActive and child and child.Name=='Gun'then autoShootTick()end end)table.insert(toolConnections,conn)end)end local function setupInstantGunWatchers():()
clearToolConnections()if not LocalPlayer then return end watchToolContainer(LocalPlayer.Character)watchToolContainer(LocalPlayer:FindFirstChildOfClass('Backpack'))pcall(function():()if LocalPlayer.
CharacterAdded then table.insert(toolConnections,LocalPlayer.CharacterAdded:Connect(function(newChar:Model):()watchToolContainer(newChar)watchToolContainer(LocalPlayer:FindFirstChildOfClass('Backpack'
))end))end if LocalPlayer.ChildAdded then table.insert(toolConnections,LocalPlayer.ChildAdded:Connect(function(child:Instance):()if child and child:IsA('Backpack')then watchToolContainer(child)end end
))end end)end local function updateStatus():()if not AutoShootFeature or not autoShootActive then return end if settings.proximity then AutoShootFeature:SetStatus(settings.mode..' \183 '..tostring(
settings.aura)..' studs')else AutoShootFeature:SetStatus(settings.mode)end end local CONNECTION_KEY:string='MM2AutoShootCard'local function toggleAutoShoot(enabled:boolean):()autoShootActive=enabled==
true if enabled then updateStatus()setupInstantGunWatchers()disconnectFeatureConnection(CONNECTION_KEY)featureConnections[CONNECTION_KEY]=TaskManager:Connect(autoShootTick,nil,'Auto Shoot')
autoShootTick()else clearToolConnections()disconnectFeatureConnection(CONNECTION_KEY)if AutoShootFeature then AutoShootFeature:SetStatus(nil)end end end AutoShootFeature=createUniversalFeature(
'Auto Shoot',
[[Automatically equips the Gun the instant you receive it (as Sheriff or Hero) and shoots the Murderer continuously using Bullet (respects WallCheck) or CFrame (Silent AIM). Optional Proximity radius filter.]]
,4,toggleAutoShoot,{categoryName='Utilities',parent=MM2Scroll,registry=mm2Features})state.mm2AutoShootFeature=AutoShootFeature local auraSlider:GuiObject?=nil local states:any=type(configData)==
'table'and configData.states or{}local values:any=type(configData)=='table'and configData.values or{}local initialMode:string=values['MM2.AutoShoot.Mode']or'Bullet'if initialMode~='CFrame'and
initialMode~='Experimental'then initialMode='Bullet'end local function modeConfig(mode:string):any local prefix:string='MM2.AutoShoot.'..mode..'.'local legacy:boolean=mode==initialMode local wall:any=
states[prefix..'WallCheck']local proximity:any=states[prefix..'Proximity']local aura:any=values[prefix..'Aura']if wall==nil and legacy then wall=states['MM2.AutoShoot.WallCheck']end if proximity==nil
and legacy then proximity=states['MM2.AutoShoot.Proximity']end if aura==nil and legacy then aura=values['MM2.AutoShoot.Aura']end return{wallCheck=wall==nil or wall==true,proximity=proximity==true,aura
=type(aura)=='number'and aura or 25}end local modeConfigs:any={Bullet=modeConfig('Bullet'),CFrame=modeConfig('CFrame'),Experimental=modeConfig('Experimental')}local switching:boolean=false local
optionsReady:boolean=false local function refreshProximityOption():()if auraSlider and type(setOptionVisible)=='function'then setOptionVisible(AutoShootFeature,auraSlider::GuiObject,settings.proximity
==true)end updateStatus()end local function syncMode():()local cfg:any=modeConfigs[settings.mode]or modeConfigs.Bullet switching=true settings.wallCheck,settings.proximity,settings.aura=cfg.wallCheck,
cfg.proximity,cfg.aura if state and type(state.toggleControls)=='table'then local wc:any=state.toggleControls['MM2.AutoShoot.WallCheck']local px:any=state.toggleControls['MM2.AutoShoot.Proximity']if
wc and wc.setChecked then pcall(wc.setChecked,cfg.wallCheck)end if px and px.setChecked then pcall(px.setChecked,cfg.proximity)end end if state and type(state.valueControls)=='table'then local au:any=
state.valueControls['MM2.AutoShoot.Aura']if au and au.apply then pcall(au.apply,cfg.aura)end end switching=false refreshProximityOption()end addCycleOption(AutoShootFeature,'Mode',{'Bullet','CFrame',
'Experimental'},1,function(value:string):()settings.mode=value syncMode()end,
[[Experimental selects one path based on range and movement; the heuristic does not guarantee hits. Options are independent per mode.]])addToggleOption(AutoShootFeature,'WallCheck',true,function(value
:boolean):()settings.wallCheck=value==true if optionsReady and not switching then modeConfigs[settings.mode].wallCheck=settings.wallCheck states['MM2.AutoShoot.'..settings.mode..'.WallCheck']=settings
.wallCheck end end,'Check for walls; saved independently per mode.')addToggleOption(AutoShootFeature,'Proximity',false,function(value:boolean):()settings.proximity=value==true if optionsReady and not
switching then modeConfigs[settings.mode].proximity=settings.proximity states['MM2.AutoShoot.'..settings.mode..'.Proximity']=settings.proximity end refreshProximityOption()end,
[[Only shoot inside Aura radius; saved independently per mode.]])auraSlider=addNumberOption(AutoShootFeature,'Aura',25,5,150,function(value:number):()settings.aura=value if optionsReady and not
switching then modeConfigs[settings.mode].aura=value values['MM2.AutoShoot.'..settings.mode..'.Aura']=value end updateStatus()end,'Maximum distance with Proximity enabled, saved per mode.',1)
optionsReady=true syncMode()activeCleanup=function():()toggleAutoShoot(false)state.mm2AutoShootFeature=nil state.mm2AutoShootCFrameOnce=nil end Module.Events=featureConnections Module.Initialized=true
return Module end function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil
end return Module
-- src/games/MM2/Combat/KnifeThrow.lua
local Module={Name='MM2 Knife Throw',PlaceId=142823291,Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if
Module.Initialized then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Knife Throw requires the MM2 core module')Module.Runtime=runtime local findMurderer:any=core.
findMurderer local getPlayerRole:any=core.getPlayerRole local getPlayerWeapon:any=core.getPlayerWeapon local isPlayerAlive:any=core.isPlayerAlive local isProtectedTarget:any=core.isProtectedTarget
local getEstimatedLatency:any=core.getEstimatedLatency local isWeaponPathClear:any=core.isWeaponPathClear local prediction:any=state.mm2Prediction local settings={enabled=false,autoThrow=false,
wallCheck=true,knifeTracer=false,delay=0.6}local lastThrowAttempt:number=0 local KnifeThrowFeature:any=nil local lastKnifeStatus:string?=nil local function setKnifeStatus(message:string?):()if
KnifeThrowFeature and lastKnifeStatus~=message then lastKnifeStatus=message KnifeThrowFeature:SetStatus(message)end end local tracerPart:BasePart?=nil local targetDotPart:BasePart?=nil local
hookedWeaponService:any=nil local origGetMouseTargetCFrame:any=nil local origGetTargetPosition:any=nil local lastRedirectInstallAttempt:number=-math.huge local function destroyTracerVisuals():()if
tracerPart then pcall(function():()tracerPart:Destroy()end)tracerPart=nil end if targetDotPart then pcall(function():()targetDotPart:Destroy()end)targetDotPart=nil end end local function isValidTarget
(player:Player):boolean if not player or player==LocalPlayer then return false end if type(isProtectedTarget)=='function'and isProtectedTarget(player)then return false end if type(isPlayerAlive)==
'function'and not isPlayerAlive(player)then return false end if type(getPlayerRole)=='function'and getPlayerRole(player)=='Murderer'then return false end local character:Model?=player.Character if not
character then return false end local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')if not root or not
humanoid or humanoid.Health<=0 then return false end return true end local function getKnife():Tool?local character:Model?=LocalPlayer and LocalPlayer.Character if not character then return nil end if
type(getPlayerWeapon)=='function'then local equippedKnife:Tool?=getPlayerWeapon(LocalPlayer,'Knife',true)if equippedKnife then return equippedKnife end local anyKnife:Tool?=getPlayerWeapon(LocalPlayer
,'Knife',false)if anyKnife then return anyKnife end end for _,item:Instance in ipairs(character:GetChildren())do if item:IsA('Tool')then local name:string=item.Name:lower()if name:find('knife')or item
:FindFirstChild('Events')then local events:Instance?=item:FindFirstChild('Events')if name:find('knife')or(events and events:FindFirstChild('KnifeThrown'))then return item::Tool end end end end local
backpack:Instance?=LocalPlayer:FindFirstChildOfClass('Backpack')if backpack then for _,item:Instance in ipairs(backpack:GetChildren())do if item:IsA('Tool')then local name:string=item.Name:lower()
local events:Instance?=item:FindFirstChild('Events')if name:find('knife')or(events and events:FindFirstChild('KnifeThrown'))then return item::Tool end end end end return nil end local originHandleTool
:Tool?=nil local originHandle:BasePart?=nil local function getThrowOriginCFrame(knife:Tool?):CFrame?local character:Model?=LocalPlayer and LocalPlayer.Character if not character then return nil end if
knife and knife.Parent==character then if originHandleTool~=knife or not originHandle or originHandle.Parent~=knife then local handle:Instance?=knife:FindFirstChild('Handle')originHandleTool=knife
originHandle=handle and handle:IsA('BasePart')and(handle::BasePart)or nil end if originHandle then return originHandle.CFrame end end local hand:Instance?=character:FindFirstChild('RightHand')or
character:FindFirstChild('Right Arm')or character:FindFirstChild('HumanoidRootPart')if hand and hand:IsA('BasePart')then return(hand::BasePart).CFrame end return nil end local CORRIDOR_STUDS:number=
0.45 local function raycastClear(from:Vector3,to:Vector3,ignoreList:{Instance},requireBody:Model?):boolean local ok:boolean,clear:any=pcall(function():boolean local params:RaycastParams=RaycastParams.
new()params.FilterType=Enum.RaycastFilterType.Exclude params.FilterDescendantsInstances=ignoreList params.RespectCanCollide=true params.IgnoreWater=true local hit:RaycastResult?=workspace:Raycast(from
,to-from,params)if requireBody then return hit~=nil and hit.Instance:IsDescendantOf(requireBody)end return hit==nil end)return ok and clear==true end local function sharedPathClear(from:Vector3,to:
Vector3,targetChar:Model?,requireBody:boolean):boolean?if type(isWeaponPathClear)~='function'or not targetChar then return nil end local options:any=requireBody and{requireTarget=true}or{ignoreTarget=
true}local ok:boolean,clear:any=pcall(isWeaponPathClear,from,to,targetChar,true,options)if not ok or type(clear)~='boolean'then return nil end return clear end local function ignoreListFor(targetChar:
Model?):{Instance}local list:{Instance}={}local localChar:Model?=LocalPlayer and LocalPlayer.Character if localChar then table.insert(list,localChar)end if targetChar then table.insert(list,targetChar
)end return list end local function bodyVisible(originPos:Vector3,bodyPos:Vector3,targetChar:Model?):boolean if not settings.wallCheck then return true end local from:Vector3=originPos local to:
Vector3=bodyPos if(to-from).Magnitude<0.05 then return true end local shared:boolean?=sharedPathClear(from,to,targetChar,true)if shared~=nil then return shared end return raycastClear(from,to,
ignoreListFor(nil),targetChar)end local function checkLineOfSight(originPos:Vector3,targetPos:Vector3,targetChar:Model?):boolean if not settings.wallCheck then return true end local direction:Vector3=
targetPos-originPos local length:number=direction.Magnitude if length<0.05 then return true end local unit:Vector3=direction/length local side:Vector3=unit:Cross(Vector3.new(0,1,0))if side.Magnitude<
0.05 then side=Vector3.new(1,0,0)end side=side.Unit local up:Vector3=side:Cross(unit).Unit local offsets:{Vector3}={Vector3.zero,side*CORRIDOR_STUDS,side*-CORRIDOR_STUDS,up*CORRIDOR_STUDS,up*-
CORRIDOR_STUDS}for _index:number,offset:Vector3 in ipairs(offsets)do local from:Vector3=originPos+offset local to:Vector3=targetPos+offset local shared:boolean?=sharedPathClear(from,to,targetChar,
false)if shared~=nil then if not shared then return false end elseif not raycastClear(from,to,ignoreListFor(targetChar),nil)then return false end end return true end local function resolveKnifeAim(
target:Player,originPos:Vector3,knife:Tool?,cheap:boolean?):(Vector3?,number?)local character:Model?=target.Character if not character then return nil,nil end local root:BasePart?=character:
FindFirstChild('HumanoidRootPart')::BasePart?if not root then return nil,nil end if not bodyVisible(originPos,root.Position,character)then return nil,nil end local speed:number=96 if knife then pcall(
function():()local attrSpeed:any=knife:GetAttribute('ThrowSpeed')if type(attrSpeed)=='number'and attrSpeed>=24 and attrSpeed<=300 then speed=attrSpeed end end)end local predicted:Vector3?=nil local
aimInfo:any=nil if type(prediction)=='table'and type(prediction.knifeAim)=='function'then local okPred:boolean,aimPoint:any,info:any=pcall(prediction.knifeAim,target,originPos,speed)if okPred and
typeof(aimPoint)=='Vector3'then predicted=aimPoint::Vector3 aimInfo=info end end if not predicted then local velocity:Vector3=root.AssemblyLinearVelocity if velocity.X~=velocity.X or velocity.Y~=
velocity.Y or velocity.Z~=velocity.Z or math.abs(velocity.X)==math.huge or math.abs(velocity.Y)==math.huge or math.abs(velocity.Z)==math.huge then velocity=Vector3.zero end local latency:number=0.06
if type(getEstimatedLatency)=='function'then local okLat:boolean,l:any=pcall(getEstimatedLatency)if okLat and type(l)=='number'and l==l then latency=math.clamp(l,0,0.5)end end local horizVel:Vector3=
Vector3.new(velocity.X,0,velocity.Z)local observed:Vector3=root.Position+horizVel*latency local rel:Vector3=observed-originPos local a:number=horizVel:Dot(horizVel)-speed*speed local b:number=2*rel:
Dot(horizVel)local c:number=rel:Dot(rel)local flight:number?=nil if c<0.0025 then flight=0 elseif math.abs(a)<1e-6 then if b<-1e-6 then flight=-c/b end else local disc:number=b*b-4*a*c if disc>=0 then
local sqrtDisc:number=math.sqrt(disc)local t1:number=(-b-sqrtDisc)/(2*a)local t2:number=(-b+sqrtDisc)/(2*a)if t1>0 then flight=t1 end if t2>0 and(not flight or t2<flight)then flight=t2 end end end if
flight and flight==flight then flight=math.clamp(flight,0,1.5)local yLead:number=math.clamp(velocity.Y*math.min(latency+flight,0.28),-2,4)predicted=observed+horizVel*flight+Vector3.new(0,yLead,0)else
local yLead:number=math.clamp(velocity.Y*math.min(latency,0.28),-2,4)predicted=observed+Vector3.new(0,yLead,0)end end if not predicted then return nil,nil end local flight:number?=nil if aimInfo and
type(aimInfo.flightSeconds)=='number'then flight=math.max(0,aimInfo.flightSeconds)else flight=(predicted-originPos).Magnitude/speed end if cheap or checkLineOfSight(originPos,predicted,character)then
return predicted,flight end local leadOffset:Vector3=predicted-root.Position if aimInfo and typeof(aimInfo.capturedPosition)=='Vector3'then leadOffset=predicted-(aimInfo.capturedPosition::Vector3)end
local candidateNames:{string}={'UpperTorso','Torso','HumanoidRootPart','Head','RightUpperArm','LeftUpperArm','Right Arm','Left Arm'}for _,partName:string in ipairs(candidateNames)do local part:
Instance?=character:FindFirstChild(partName)if part and part:IsA('BasePart')then local partPosition:Vector3=(part::BasePart).Position local candidatePoint:Vector3=partPosition+leadOffset if
bodyVisible(originPos,partPosition,character)and(cheap or checkLineOfSight(originPos,candidatePoint,character))then return candidatePoint,flight end end end return nil,nil end local function
findTarget():Player?local localChar:Model?=LocalPlayer and LocalPlayer.Character if not localChar then return nil end local localRoot:BasePart?=localChar:FindFirstChild('HumanoidRootPart')::BasePart?
if not localRoot then return nil end local knife:Tool?=getKnife()local originCF:CFrame?=getThrowOriginCFrame(knife)local originPos:Vector3=originCF and originCF.Position or localRoot.Position local
bestTarget:Player?=nil local bestScore:number=math.huge for _,player:Player in ipairs(Players:GetPlayers())do if isValidTarget(player)then local targetChar:Model?=player.Character local targetRoot:
BasePart?=targetChar and targetChar:FindFirstChild('HumanoidRootPart')::BasePart?if targetRoot then local distance:number=(targetRoot.Position-originPos).Magnitude if distance<=250 then local aimPoint
:Vector3?,flight:number?=resolveKnifeAim(player,originPos,knife,true)if aimPoint then local score:number=flight or(distance/96)if type(getPlayerRole)=='function'then local role:string?=getPlayerRole(
player)if role=='Sheriff'or role=='Hero'then score-=0.08 end end if score<bestScore then bestScore=score bestTarget=player end end end end end end return bestTarget end local TRACER_SMOOTH_RATE:number
=26 local TRACER_SNAP_DISTANCE:number=20 local tracerPoint:Vector3?=nil local tracerTarget:Player?=nil local tracerShown:boolean=false local tracerOpacity:number=0 local drawnOpacity:number=-1 local
drawnOrigin:Vector3?=nil local drawnPoint:Vector3?=nil local function hideTracerParts():()if tracerPart and tracerPart.Parent then tracerPart.Transparency=1 end if targetDotPart and targetDotPart.
Parent then targetDotPart.Transparency=1 end end local function hideTracerVisuals():()if tracerShown then pcall(hideTracerParts)end tracerShown=false tracerOpacity=0 drawnOpacity=-1 tracerPoint=nil
tracerTarget=nil drawnOrigin=nil drawnPoint=nil end local function drawTracerParts(originPos:Vector3,point:Vector3,delta:Vector3,dist:number,updateGeometry:boolean,opacity:number,updateOpacity:boolean
):()if not tracerPart or not tracerPart.Parent then local beamPart:Part=Instance.new('Part')beamPart.Name='WurstKnifeTracer'beamPart.Anchored=true beamPart.CanCollide=false beamPart.CanQuery=false
beamPart.CanTouch=false beamPart.Material=Enum.Material.Neon beamPart.Color=Color3.fromRGB(255,65,65)beamPart.Transparency=1 beamPart.Parent=workspace tracerPart=beamPart tracerShown=false end if not
targetDotPart or not targetDotPart.Parent then local dot:Part=Instance.new('Part')dot.Name='WurstKnifeAimDot'dot.Shape=Enum.PartType.Ball dot.Size=Vector3.new(1.1,1.1,1.1)dot.Anchored=true dot.
CanCollide=false dot.CanQuery=false dot.CanTouch=false dot.Material=Enum.Material.Neon dot.Color=Color3.fromRGB(255,220,60)dot.Transparency=1 dot.Parent=workspace targetDotPart=dot tracerShown=false
end if updateGeometry then tracerPart.Size=Vector3.new(0.14,0.14,dist)tracerPart.CFrame=CFrame.lookAt(originPos+delta*0.5,point)targetDotPart.CFrame=CFrame.new(point)end if updateOpacity or not
tracerShown then tracerPart.Transparency=1-0.75*opacity targetDotPart.Transparency=1-0.85*opacity tracerShown=true end end local function updateTracerVisual(originPos:Vector3?,aimPos:Vector3?,target:
Player?,deltaTime:number?):()if not settings.enabled or not settings.knifeTracer then hideTracerVisuals()destroyTracerVisuals()return end if not originPos or not aimPos or not target then
hideTracerVisuals()return end local dt:number=tonumber(deltaTime)or(1/60)if dt~=dt then dt=1/60 end local alpha:number=1-math.exp(-TRACER_SMOOTH_RATE*math.clamp(dt,0,0.25))local previous:Vector3?=
tracerPoint if not previous or tracerTarget~=target or(aimPos-previous).Magnitude>TRACER_SNAP_DISTANCE then tracerPoint=aimPos tracerOpacity=0 else local nextPoint:Vector3=previous+(aimPos-previous)*
alpha tracerPoint=(aimPos-nextPoint).Magnitude<0.002 and aimPos or nextPoint end tracerOpacity+=(1-tracerOpacity)*alpha if tracerOpacity>0.998 then tracerOpacity=1 end tracerTarget=target local point:
Vector3=tracerPoint::Vector3 local delta:Vector3=point-originPos local dist:number=delta.Magnitude if dist<0.25 or dist>400 then hideTracerVisuals()return end local missing:boolean=not tracerPart or
not tracerPart.Parent or not targetDotPart or not targetDotPart.Parent local geometryChanged:boolean=missing or not drawnOrigin or not drawnPoint or(originPos-(drawnOrigin::Vector3)).Magnitude>=0.002
or(point-(drawnPoint::Vector3)).Magnitude>=0.002 local opacityChanged:boolean=missing or not tracerShown or math.abs(tracerOpacity-drawnOpacity)>=0.002 if not geometryChanged and not opacityChanged
then return end local ok:boolean=pcall(drawTracerParts,originPos,point,delta,dist,geometryChanged,tracerOpacity,opacityChanged)if ok then if geometryChanged then drawnOrigin=originPos drawnPoint=point
end if opacityChanged then drawnOpacity=tracerOpacity end else pcall(hideTracerParts)tracerShown=false drawnOpacity=-1 drawnOrigin=nil drawnPoint=nil end end local equipRequestedAt:number=-math.huge
local equipRequestedTool:Tool?=nil local function ensureEquipped(knife:Tool,humanoid:Humanoid):boolean local character:Model?=LocalPlayer and LocalPlayer.Character if not character then return false
end if knife.Parent==character then equipRequestedAt=-math.huge equipRequestedTool=nil return true end local now:number=os.clock()if equipRequestedTool~=knife or now-equipRequestedAt>=0.25 then
equipRequestedAt=now equipRequestedTool=knife pcall(function():()humanoid:EquipTool(knife)end)end return false end local function throwKnife(target:Player?,knife:Tool):(boolean,string?)local character
:Model?=LocalPlayer and LocalPlayer.Character if not character then return false,'Character unavailable'end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')if not humanoid or
humanoid.Health<=0 then return false,'Character unavailable'end if not ensureEquipped(knife,humanoid)then return false,'Equipping knife\u{2026}'end local originCF:CFrame?=getThrowOriginCFrame(knife)if
not originCF then return false,'No throw origin'end if not target or not isValidTarget(target)then return false,'No visible target'end local aimPos:Vector3?=resolveKnifeAim(target,originCF.Position,
knife)if not aimPos then return false,'Target blocked \183 WallCheck'end local fired:boolean=false local noted:boolean=false local function noteThrowAttempt():()if noted then return end noted=true
local recorder:any=core and core.trajectoryCalibration if recorder and type(recorder.noteAuthoredShot)=='function'then local root:BasePart?=target.Character and target.Character:FindFirstChild(
'HumanoidRootPart')::BasePart?pcall(recorder.noteAuthoredShot,recorder,{kind='knife',tool=knife,toolName=knife.Name,originPos=originCF.Position,aimPos=aimPos,target=target,aimVelocity=root and root.
AssemblyLinearVelocity or nil,source='menu-auto-throw',settings={mode='Auto Knife Throw',wallCheck=settings.wallCheck}})end end local eventsFolder:Instance?=knife:FindFirstChild('Events')local
knifeThrownRemote:Instance?=eventsFolder and eventsFolder:FindFirstChild('KnifeThrown')if knifeThrownRemote and knifeThrownRemote:IsA('RemoteEvent')then noteThrowAttempt()local okFire:boolean=pcall(
function():()(knifeThrownRemote::RemoteEvent):FireServer(originCF,CFrame.new(aimPos))end)if okFire then fired=true end end if not fired then pcall(function():()local rs:Instance?=game:GetService(
'ReplicatedStorage')local remotes:Instance?=rs and rs:FindFirstChild('Remotes')local throwRemote:Instance?=remotes and remotes:FindFirstChild('Event')if throwRemote and throwRemote:IsA('RemoteEvent')
then noteThrowAttempt()local dir:Vector3=(aimPos-originCF.Position).Unit local rem:RemoteEvent=throwRemote::RemoteEvent rem:FireServer('Throw',aimPos,dir)fired=true end end)end if not fired then
return false,'KnifeThrown remote unavailable'end return true,nil end local function installWeaponServiceRedirect():()if hookedWeaponService then return end pcall(function():()local rs:Instance?=game:
GetService('ReplicatedStorage')local clientServices:Instance?=rs and rs:FindFirstChild('ClientServices')local wsModule:Instance?=clientServices and clientServices:FindFirstChild('WeaponService')if not
wsModule then return end local okReq:boolean,wsTable:any=pcall(require,wsModule)if not okReq or type(wsTable)~='table'then return end origGetMouseTargetCFrame=wsTable.GetMouseTargetCFrame
origGetTargetPosition=wsTable.GetTargetPosition hookedWeaponService=wsTable if type(origGetMouseTargetCFrame)=='function'then wsTable.GetMouseTargetCFrame=function(self:any,...:any):any if settings.
enabled then local knife:Tool?=getKnife()local character:Model?=LocalPlayer and LocalPlayer.Character if knife and character and knife.Parent==character then local originCF:CFrame?=
getThrowOriginCFrame(knife)local target:Player?=findTarget()if originCF and target then local aim:Vector3?=resolveKnifeAim(target,originCF.Position,knife)if aim then return CFrame.new(aim)end end end
end return origGetMouseTargetCFrame(self,...)end end if type(origGetTargetPosition)=='function'then wsTable.GetTargetPosition=function(self:any,...:any):any if settings.enabled then local knife:Tool?=
getKnife()local character:Model?=LocalPlayer and LocalPlayer.Character if knife and character and knife.Parent==character then local originCF:CFrame?=getThrowOriginCFrame(knife)local target:Player?=
findTarget()if originCF and target then local aim:Vector3?=resolveKnifeAim(target,originCF.Position,knife)if aim then return CFrame.new(aim)end end end end return origGetTargetPosition(self,...)end
end end)end local function uninstallWeaponServiceRedirect():()if not hookedWeaponService then return end pcall(function():()if origGetMouseTargetCFrame then hookedWeaponService.GetMouseTargetCFrame=
origGetMouseTargetCFrame end if origGetTargetPosition then hookedWeaponService.GetTargetPosition=origGetTargetPosition end end)hookedWeaponService=nil origGetMouseTargetCFrame=nil
origGetTargetPosition=nil end local AIM_CACHE_SECONDS:number=1/15 local TRACER_CACHE_SECONDS:number=1/10 local cachedTarget:Player?=nil local cachedAim:Vector3?=nil local cachedOrigin:CFrame?=nil
local cachedVisualRoot:BasePart?=nil local cachedVisualPosition:Vector3?=nil local cachedAt:number=-math.huge local function solveAim(knife:Tool,throwDue:boolean):(Player?,Vector3?,CFrame?)local now:
number=os.clock()local interval:number=throwDue and AIM_CACHE_SECONDS or TRACER_CACHE_SECONDS if now-cachedAt>=interval then cachedAt=now cachedTarget=findTarget()cachedAim=nil cachedVisualRoot=nil
cachedVisualPosition=nil cachedOrigin=getThrowOriginCFrame(knife)if cachedOrigin and cachedTarget then cachedAim=resolveKnifeAim(cachedTarget,cachedOrigin.Position,knife)if cachedAim and settings.
knifeTracer then local character:Model?=cachedTarget.Character local root:BasePart?=character and character:FindFirstChild('HumanoidRootPart')::BasePart?if root then cachedVisualRoot=root
cachedVisualPosition=root.Position end end end elseif cachedOrigin then cachedOrigin=getThrowOriginCFrame(knife)or cachedOrigin end return cachedTarget,cachedAim,cachedOrigin end local function
invalidateAim():()cachedAt=-math.huge cachedTarget=nil cachedAim=nil cachedOrigin=nil cachedVisualRoot=nil cachedVisualPosition=nil end local loopKnife:Tool?=nil local loopKnifeParent:Instance?=nil
local loopKnifeCharacter:Model?=nil local loopKnifeAt:number=-math.huge local function resetLoopKnife():()loopKnife=nil loopKnifeParent=nil loopKnifeCharacter=nil loopKnifeAt=-math.huge
originHandleTool=nil originHandle=nil end local function getLoopKnife(now:number,throwDue:boolean):Tool?local character:Model?=LocalPlayer and LocalPlayer.Character if character~=loopKnifeCharacter
then resetLoopKnife()end loopKnifeCharacter=character if not throwDue then if loopKnife and loopKnife.Parent==character and character then return loopKnife end local age:number=now-loopKnifeAt if age
>=0 and age<TRACER_CACHE_SECONDS then if not loopKnife then return nil end if loopKnifeParent and loopKnife.Parent==loopKnifeParent then return loopKnife end end end loopKnife=getKnife()
loopKnifeParent=loopKnife and loopKnife.Parent or nil loopKnifeAt=now return loopKnife end local function tracerGoal(target:Player?,aimPos:Vector3?):Vector3?if not target or not aimPos then return nil
end local root:BasePart?=cachedVisualRoot local captured:Vector3?=cachedVisualPosition if not root or not captured then return aimPos end if target~=cachedTarget or root.Parent~=target.Character then
invalidateAim()return nil end local moved:Vector3=root.Position-captured if moved.Magnitude>8 then invalidateAim()return nil end return aimPos+moved end local function knifeThrowLoop(deltaTime:number?
):()if not settings.enabled then setKnifeStatus(nil)hideTracerVisuals()destroyTracerVisuals()invalidateAim()return end local now:number=os.clock()if not hookedWeaponService and now-
lastRedirectInstallAttempt>=1 then lastRedirectInstallAttempt=now installWeaponServiceRedirect()end local cooldown:number=math.clamp(tonumber(settings.delay)or 0,0,10)local wantThrow:boolean=settings.
autoThrow and now-lastThrowAttempt>=cooldown if not settings.knifeTracer and not wantThrow then hideTracerVisuals()destroyTracerVisuals()if not settings.autoThrow then setKnifeStatus(nil)
invalidateAim()end return end local knife:Tool?=getLoopKnife(now,wantThrow)if not knife then setKnifeStatus('Waiting for knife')hideTracerVisuals()invalidateAim()return end local target,aimPos,
originCF=solveAim(knife,wantThrow)if not aimPos then setKnifeStatus(target and'Target blocked \183 WallCheck'or'No visible target')hideTracerVisuals()return end setKnifeStatus('Predicted target')if
settings.knifeTracer and originCF then updateTracerVisual(originCF.Position,tracerGoal(target,aimPos),target,deltaTime)else hideTracerVisuals()destroyTracerVisuals()end if wantThrow then local thrown:
boolean,why:string?=throwKnife(target,knife)if thrown then setKnifeStatus('Throw sent \183 hit unconfirmed')lastThrowAttempt=now invalidateAim()elseif why then setKnifeStatus(why)end end end
local function toggleKnifeThrow(enabled:boolean):()settings.enabled=enabled disconnectFeatureConnection('MM2KnifeThrow')if not enabled then uninstallWeaponServiceRedirect()setKnifeStatus(nil)
hideTracerVisuals()destroyTracerVisuals()invalidateAim()resetLoopKnife()return end installWeaponServiceRedirect()featureConnections.MM2KnifeThrow=TaskManager:Connect(knifeThrowLoop,nil,
'MM2 Knife Throw')end KnifeThrowFeature=createUniversalFeature('Knife Throw',[[Redirects PC & Mobile knife throws to the Knife Predictor's aim point, with optional Auto Knife Throw and Knife Tracer.]]
,15,toggleKnifeThrow,{categoryName='Exploits',parent=MM2Scroll,registry=mm2Features})addToggleOption(KnifeThrowFeature,'WallCheck',true,function(value:boolean):()settings.wallCheck=value end,
[[Checks the blade's path from the hand to the target, including a thin corridor and invisible solid barriers. With nobody visible, nothing is thrown.]])addToggleOption(KnifeThrowFeature,
'Auto Knife Throw',false,function(value:boolean):()settings.autoThrow=value==true end,
[[Throws at the predicted intercept of a visible target. With nobody visible, or with the path blocked, no knife is spent. A sent throw is not a confirmed hit.]])addNumberOption(KnifeThrowFeature,
'Delay',0.6,0,10,function(value:number):()settings.delay=math.clamp(tonumber(value)or 0,0,10)end,[[Delay in seconds between automatic knife throws (0 to 10 seconds).]],0.1)addInformationOption(
KnifeThrowFeature,'Seconds')addToggleOption(KnifeThrowFeature,'Knife Tracer',false,function(value:boolean):()settings.knifeTracer=value==true if not settings.knifeTracer then hideTracerVisuals()
destroyTracerVisuals()end end,[[Smooth 3D preview from your hand to the predicted point. The actual throw is re-solved separately; the preview hides when no target passes WallCheck.]])activeCleanup=
function():()toggleKnifeThrow(false)end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then return end Module.
Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Fun/BlurtRoles.lua
local Module={Name='MM2 Blurt Roles',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.
Initialized then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Blurt Roles requires the MM2 core module')Module.Runtime=runtime local findMurderer:any=core.
findMurderer local findSheriff:any=core.findSheriff local hasActiveRoundRoles:any=core.hasActiveRoundRoles local mm2Settings:any=core.mm2Settings local blurtEnabled:boolean=false local blurtSignature:
string=''local blurtPending:boolean=false local blurtGeneration:number=0 local lastBlurtAt:number=-math.huge local function cancelPendingBlurt():()blurtGeneration+=1 blurtPending=false end
local function sendPublicChat(message:string):boolean local sent:boolean=false local textChatOk:boolean=pcall(function():()local textChatService:TextChatService=game:GetService('TextChatService')if
textChatService.ChatVersion~=Enum.ChatVersion.TextChatService then return end local channels:Instance?=textChatService:FindFirstChild('TextChannels')local channel:Instance?=channels and(channels:
FindFirstChild('RBXGeneral')or channels:FindFirstChildWhichIsA('TextChannel'))if channel and channel:IsA('TextChannel')then(channel::TextChannel):SendAsync(message)sent=true end end)if sent then
return true end local legacyOk:boolean=pcall(function():()local events:Instance?=game:GetService('ReplicatedStorage'):FindFirstChild('DefaultChatSystemChatEvents')local say:Instance?=events and events
:FindFirstChild('SayMessageRequest')if say and say:IsA('RemoteEvent')then(say::RemoteEvent):FireServer(message,'All')sent=true end end)return sent and(textChatOk or legacyOk)end local function
buildBlurtMessage():(string?,string?)if not hasActiveRoundRoles()then return nil,nil end local murderer:Player?=findMurderer()local sheriff:Player?=findSheriff()if not murderer and not sheriff then
return nil,nil end local murdererName:string=murderer and murderer.Name or'?'local sheriffName:string=sheriff and sheriff.Name or'?'local message:string local signature:string if mm2Settings.blurtFake
then message=string.format('Murder; "%s" Sheriff; "%s"',sheriffName,murdererName)signature='fake/'..sheriffName..'/'..murdererName else message=string.format('Murder; "%s" Sheriff; "%s" | Wurst',
murdererName,sheriffName)signature=murdererName..'/'..sheriffName end return message,signature end local function blurtRoles(manual:boolean):()local message:string?,signature:string?=
buildBlurtMessage()if not message then if manual then notify('MM2 \183 no roles to blurt yet.')end return end if not manual then if signature==blurtSignature and not mm2Settings.blurtRepeat then
return end if os.clock()-lastBlurtAt<2.5 then return end end blurtSignature=signature::string lastBlurtAt=os.clock()task.spawn(function():()if sendPublicChat(message::string)then notify(
'MM2 \183 blurted: '..(message::string))else notify('MM2 \183 the chat refused the message (filtered or disabled).')end end)end local function scheduleBlurt():()if not blurtEnabled or blurtPending
then return end local message:string?,signature:string?=buildBlurtMessage()if not message then return end if signature==blurtSignature and not mm2Settings.blurtRepeat then return end blurtPending=true
local generation:number=blurtGeneration task.delay(math.max(mm2Settings.blurtDelay,0),function():()if generation~=blurtGeneration then return end blurtPending=false if blurtEnabled then blurtRoles(
false)end end)end local unsubscribe:()->()=core.onRoundRoles(function(active:boolean):()if not active then cancelPendingBlurt()blurtSignature=''return end scheduleBlurt()end)local function
toggleBlurtRoles(enabled:boolean):()blurtEnabled=enabled cancelPendingBlurt()disconnectFeatureConnection('MM2BlurtRoles')if not enabled then return end blurtSignature=''local elapsed:number=0
featureConnections.MM2BlurtRoles=TaskManager:Connect(function(deltaTime:number):()elapsed+=deltaTime if elapsed<1 then return end elapsed=0 if not hasActiveRoundRoles()then blurtSignature=''return end
scheduleBlurt()end,nil,'MM2 Blurt Roles')end local BlurtFeature=createUniversalFeature('Blurt Roles',[[Types the murderer and sheriff in the public chat. Click the card to blurt right now.]],3,
function():()blurtRoles(true)end,{categoryName='World',parent=MM2Scroll,registry=mm2Features,kind='action'})addToggleOption(BlurtFeature,'Auto blurt',false,function(value:boolean):()toggleBlurtRoles(
value)end,[[Blurt automatically once per round, after the Delay below. Off by default: the card alone is an instant, one-shot blurt.]])addNumberOption(BlurtFeature,'Delay',mm2Settings.blurtDelay,0,15,
function(value:number):()mm2Settings.blurtDelay=value end,'Seconds to wait after the roles are known before typing.',0.5)addToggleOption(BlurtFeature,'Repeat on change',mm2Settings.blurtRepeat,
function(value:boolean):()mm2Settings.blurtRepeat=value end,[[Blurt again when the sheriff dies and the gun changes hands.]])addToggleOption(BlurtFeature,'Fake Blurt Roles',mm2Settings.blurtFake,
function(value:boolean):()mm2Settings.blurtFake=value end,[[Announces the roles SWAPPED (and without the Wurst tag) so the chat ]]..'lies about who the murderer and sheriff are.')addInformationOption(
BlurtFeature,[[This types in the real public chat: everyone reads it and you will be]]..' reported. Roblox also rate-limits and filters messages.')activeCleanup=function():()pcall(unsubscribe)
toggleBlurtRoles(false)end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false
activeCleanup()activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Movement/Sprint.lua
local Module={Name='MM2 Sprint',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Sprint requires the MM2 core module')Module.Runtime=runtime local sprintTrailObjects:{Instance}={}local function
clearSprintTrail():()for _,object:Instance in ipairs(sprintTrailObjects)do if object.Parent then object:Destroy()end end table.clear(sprintTrailObjects)end local function ensureSprintTrail(root:
BasePart):Trail local existing:Instance?=root:FindFirstChild('Wurst_SprintTail')if existing and existing:IsA('Trail')then return existing end clearSprintTrail()local left:Attachment=Instance.new(
'Attachment')left.Name='Wurst_SprintLeft'left.Position=Vector3.new(-0.8,-0.5,0.75)left.Parent=root local right:Attachment=Instance.new('Attachment')right.Name='Wurst_SprintRight'right.Position=Vector3
.new(0.8,-0.5,0.75)right.Parent=root local trail:Trail=Instance.new('Trail')trail.Name='Wurst_SprintTail'trail.Attachment0=left trail.Attachment1=right trail.Color=ColorSequence.new(Color3.fromRGB(255
,220,55),Color3.fromRGB(175,115,20))trail.Lifetime=0.38 trail.LightEmission=0.25 trail.FaceCamera=true trail.Transparency=NumberSequence.new({NumberSequenceKeypoint.new(0,0.2),NumberSequenceKeypoint.
new(1,1)})trail.Enabled=false trail.Parent=root sprintTrailObjects={trail,left,right}return trail end local speed:any=state.framework and(state.framework:GetModule('Speed Settings')or state.framework:
GetModule('Speed Hack'))assert(speed,'Sprint option requires the Speed module')speed.ClearSprint=clearSprintTrail speed.SprintFrame=function(active:boolean,root:BasePart):()if not active then
clearSprintTrail()return end local _,humanoid:Humanoid?=getCharacterParts()ensureSprintTrail(root).Enabled=humanoid~=nil and humanoid.MoveDirection.Magnitude>0.05 end local function releaseSprint():()
disconnectFeatureConnection('MM2SprintHold')speed.SetSprintHeld(false)clearSprintTrail()end speed:CreateButton({Name='Sprint',Hold=true,Function=function():()speed.SetSprintHeld(true)if not
featureConnections.MM2SprintHold then featureConnections.MM2SprintHold=TaskManager:Connect(speed.StepSprint,nil,'MM2 Sprint')end end,OnRelease=releaseSprint,Tooltip=
[[Hold this button, its assigned key or the touch action to sprint. Works with Speed OFF; release restores normal Speed or your original speed. No toggle state is saved.]]})activeCleanup=function():()
releaseSprint()speed.ClearSprint=nil speed.SprintFrame=nil end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then
return end Module.Initialized=false activeCleanup()activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Movement/Teleport.lua
local Module={Name='MM2 Teleport',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Teleport requires the MM2 core module')Module.Runtime=runtime local findDroppedGun:any=core.findDroppedGun local
findMM2Map:any=core.findMM2Map local getPlayerRole:any=core.getPlayerRole local isRoundLifecycleActive:any=core.isRoundLifecycleActive local mm2Settings:any=core.mm2Settings local function
canAutoGetGun():boolean return getPlayerRole(LocalPlayer)=='Innocent'and isRoundLifecycleActive()==true and findMM2Map()~=nil end local function touchDroppedGun(gunDrop:Instance):boolean if type(state
.fireTouchInterest)~='function'then return false end local character:Model?=LocalPlayer.Character local root:Instance?=character and character:FindFirstChild('HumanoidRootPart')if not root then return
false end local touchPart:Instance?=nil if gunDrop:IsA('BasePart')then touchPart=gunDrop elseif gunDrop:IsA('Model')then touchPart=gunDrop:FindFirstChild('Handle',true)or gunDrop:
FindFirstChildWhichIsA('BasePart',true)end if not touchPart then return false end pcall(state.fireTouchInterest,root,touchPart,0)pcall(state.fireTouchInterest,root,touchPart,1)local backpack:Instance?
=LocalPlayer:FindFirstChildOfClass('Backpack')return(backpack~=nil and backpack:FindFirstChild('Gun')~=nil)or(character~=nil and character:FindFirstChild('Gun')~=nil)end local autoGetGunBusy=false
local function toggleAutoGetGun(enabled)disconnectFeatureConnection('MM2AutoGetGun')autoGetGunBusy=false state.mm2AutoGetGunEnabled=enabled==true if not enabled then return end local function
autoGetGunTick():()if autoGetGunBusy then return end local character=LocalPlayer.Character local backpack=LocalPlayer:FindFirstChildOfClass('Backpack')local gunDrop=findDroppedGun()if not character or
not gunDrop or not canAutoGetGun()or(backpack and backpack:FindFirstChild('Gun'))or character:FindFirstChild('Gun')then return end autoGetGunBusy=true task.spawn(function()pcall(touchDroppedGun,
gunDrop)autoGetGunBusy=false end)end featureConnections.MM2AutoGetGun=TaskManager:Connect(autoGetGunTick,nil,'MM2 Get Gun')autoGetGunTick()end local function moveCharacterTo(character:any,destination:
CFrame):()local root:BasePart?=character:FindFirstChild('HumanoidRootPart')if root and root:IsA('BasePart')then root.Anchored=false root.AssemblyLinearVelocity=Vector3.zero root.
AssemblyAngularVelocity=Vector3.zero else root=nil end character:PivotTo(destination)if root then root.CFrame=destination end end local function teleportToMap()local map=findMM2Map()local spawns=map
and(map:FindFirstChild('Spawns')or map:FindFirstChild('Spawns',true))local character=LocalPlayer.Character if not map or not character then notify('No active MM2 map was found.')return end local
target=nil if spawns then local spawnList=spawns:GetChildren()target=spawnList[math.random(1,math.max(1,#spawnList))]if target and not target:IsA('BasePart')then target=target:FindFirstChildWhichIsA(
'BasePart',true)end end if(not target or not target:IsA('BasePart'))and spawns then target=spawns:FindFirstChildWhichIsA('BasePart',true)end if not target or not target:IsA('BasePart')then target=map:
FindFirstChildWhichIsA('BasePart',true)end if target and target:IsA('BasePart')then moveCharacterTo(character,target.CFrame+Vector3.new(0,3,0))else notify('No active MM2 map spawn was found.')end end
local function findLobbySpawn()local lobby=workspace:FindFirstChild('Lobby')or workspace:FindFirstChild('MainLobby')or workspace:FindFirstChild('RegularLobby')if not lobby then for _,child in ipairs(
workspace:GetChildren())do local lowerName=string.lower(tostring(child.Name))if string.find(lowerName,'lobby',1,true)~=nil or child:FindFirstChild('VotePads',true)~=nil or child:FindFirstChild(
'VotePad1',true)~=nil or child:FindFirstChild('VoteIcons',true)~=nil then lobby=child break end end end if lobby then local spawns=lobby:FindFirstChild('Spawns')or lobby:FindFirstChild('Spawns',true)
if spawns then local spawn=spawns:FindFirstChildWhichIsA('SpawnLocation')or spawns:FindFirstChildWhichIsA('BasePart')or spawns:FindFirstChildWhichIsA('BasePart',true)if spawn then return spawn end end
local directSpawn=lobby:FindFirstChildWhichIsA('SpawnLocation',true)if directSpawn then return directSpawn end local votePads=lobby:FindFirstChild('VotePads',true)or lobby:FindFirstChild('VotePad1',
true)if votePads then local padPart=votePads:FindFirstChildWhichIsA('BasePart',true)if padPart then return padPart end end local anyPart=lobby:FindFirstChildWhichIsA('BasePart',true)if anyPart then
return anyPart end end local worldSpawns=workspace:FindFirstChild('Spawns')if worldSpawns then local spawn=worldSpawns:FindFirstChildWhichIsA('SpawnLocation')or worldSpawns:FindFirstChildWhichIsA(
'BasePart')if spawn then return spawn end end return workspace:FindFirstChildWhichIsA('SpawnLocation',true)end local function teleportToLobby()local spawn=findLobbySpawn()local character=LocalPlayer.
Character if spawn and character then moveCharacterTo(character,spawn.CFrame+Vector3.new(0,3,0))else notify('The MM2 lobby spawn was not found.')end end local teleportTargetName=''local function
teleportToPlayer()local character=LocalPlayer.Character if not character then notify('Your character is not available.')return end local needle=string.lower(string.gsub(teleportTargetName or'','%s+',
''))if needle==''then notify('Type a player name in the Player name box first.')return end local exact,partial=nil,nil for _,player in ipairs(Players:GetPlayers())do if player~=LocalPlayer then local
name=string.lower(player.Name)if string.gsub(name,'%s+','')==needle then exact=player break end if not partial and string.find(name,needle,1,true)then partial=player end end end local target=exact or
partial local targetRoot=target and target.Character and target.Character:FindFirstChild('HumanoidRootPart')if targetRoot then character:PivotTo(targetRoot.CFrame+Vector3.new(0,3,0))elseif target then
notify(target.Name..' has no character to teleport to.')else notify('No player named "'..teleportTargetName..'" was found.')end end local function grabDroppedGun()local gunDrop=findDroppedGun()if
gunDrop and LocalPlayer.Character then touchDroppedGun(gunDrop)else notify('No dropped gun was found.')end end local teleport=state.framework:GetModule('Teleport')assert(teleport,
'MM2 extends the universal Teleport module')for _,id in ipairs({'GetGunkey','Playername'})do local key='Universal.ClickTeleport.'..id if configData.values[key]==nil then configData.values[key]=
configData.values['MM2.Teleport.'..id]end end local autoKey='Universal.ClickTeleport.AutoGetGun'if configData.states[autoKey]==nil then configData.states[autoKey]=configData.states[
'MM2.Teleport.AutoGetGun']end local show={Option='Destination',Values={'MM2'}}teleport:CreateButton({Name='Teleport to Map',Show=show,Function=teleportToMap})teleport:CreateButton({Name=
'Teleport to Lobby',Show=show,Function=teleportToLobby})teleport:CreateTextBox({Name='Player name',Default='',Show=show,Function=function(value)teleportTargetName=value end})teleport:CreateButton({
Name='Teleport to Player',Show=show,Function=teleportToPlayer})teleport:CreateButton({Name='Get Gun',Show=show,Function=grabDroppedGun})local key=teleport:CreateBind({Name='Get Gun key',Default=
mm2Settings.getGunKey,Show=show,Function=function(value)mm2Settings.getGunKey=value end})if state.bindMobileActionPlacement and key.Object then state.bindMobileActionPlacement(key.Object,'MM2GetGun',
'GUN',grabDroppedGun)end featureConnections.MM2GetGunKey=UserInputService.InputBegan:Connect(function(input,gameProcessed)if gameProcessed or state.keyCaptureCallback or state.waitingForKey or
UserInputService:GetFocusedTextBox()then return end if input.UserInputType==Enum.UserInputType.Keyboard and input.KeyCode==mm2Settings.getGunKey then task.spawn(grabDroppedGun)end end)teleport:
CreateToggle({Name='Auto Get Gun',Default=false,Show=show,Function=toggleAutoGetGun})activeCleanup=function():()toggleAutoGetGun(false)disconnectFeatureConnection('MM2GetGunKey')end Module.Events=
featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function()
:()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Other/AutoFarm.lua
local Module={Name='MM2 Auto Farm',Events={},Initialized=false}local activeCleanup:()->()=function():()end function Module.init(_runtime:any):any assert(state.mm2Core,
'MM2 Auto Farm requires the MM2 core')local core:any=state.mm2Core local getPlayerRole:any=core.getPlayerRole local getPlayerWeapon:any=core.getPlayerWeapon local isPlayerAlive:any=core.isPlayerAlive
local findMurderer:any=core.findMurderer local findSheriff:any=core.findSheriff local findDroppedGun:any=core.findDroppedGun local isRoundLifecycleActive:any=core.isRoundLifecycleActive local
findMM2Map:any=core.findMM2Map local getEstimatedLatency:any=core.getEstimatedLatency if type(configData)=='table'and type(configData.states)=='table'then local rename={[
'MM2.AutoFarm.FlingMurderafterreset']='MM2.AutoFarm.FlingMurderafterfullbag',['MM2.AutoFarm.FlingSheriffafterreset']='MM2.AutoFarm.FlingSheriffafterfullbag'}for oldKey:string,newKey:string in pairs(
rename)do if configData.states[newKey]==nil and configData.states[oldKey]~=nil then configData.states[newKey]=configData.states[oldKey]end configData.states[oldKey]=nil end end local settings={mode=
'Walk',speed=16,stayUnderOnStart=true,safeZone=true,resetOnFull=true,flingMurder=false,flingSheriff=false,grabGunShoot=false,killAllOnFull=false}local totalCoinsCollected:number=0 local
farmElapsedSeconds:number=0 local lastFormattedSecond:number=-1 local coinsCollectedStat:any=nil local farmTimeStat:any=nil local safeZonePlatform:Part?=nil local leaveSafeZone:((boolean?,boolean?)->
boolean)?=nil local function formatFarmTime(totalSeconds:number):string local whole:number=math.max(0,math.floor(totalSeconds))local hours:number=math.floor(whole/3600)local mins:number=math.floor((
whole%3600)/60)local secs:number=whole%60 if hours>0 then return string.format('%02d:%02d:%02d',hours,mins,secs)end return string.format('%02d:%02d',mins,secs)end local function refreshFarmOverlay():(
)if coinsCollectedStat and type(coinsCollectedStat.set)=='function'then coinsCollectedStat.set(tostring(totalCoinsCollected))end local wholeSec:number=math.max(0,math.floor(farmElapsedSeconds))if
wholeSec~=lastFormattedSecond then lastFormattedSecond=wholeSec if farmTimeStat and type(farmTimeStat.set)=='function'then farmTimeStat.set(formatFarmTime(wholeSec))end end end local function
destroySafeZonePlatform():()if safeZonePlatform then pcall(function():()safeZonePlatform:Destroy()end)safeZonePlatform=nil end end local function ensureSafeZonePlatform(originPos:Vector3):Vector3?
local voidCenter:Vector3=Vector3.new(originPos.X+650,math.max(originPos.Y+280,380),originPos.Z+650)if not safeZonePlatform or not safeZonePlatform.Parent then pcall(function():()local part:Part=
Instance.new('Part')part.Name='Wurst_AutoFarmSafeZone'part.Anchored=true part.CanCollide=true part.CanQuery=false part.CanTouch=false part.CastShadow=false part.Size=Vector3.new(64,4,64)part.Material=
Enum.Material.SmoothPlastic part.Color=Color3.fromRGB(35,42,55)part.Transparency=0.15 part.CFrame=CFrame.new(voidCenter)part.Parent=workspace safeZonePlatform=part end)end if safeZonePlatform and
safeZonePlatform.Parent then return safeZonePlatform.Position+Vector3.new(0,5,0)end return nil end local BELOW_OFFSET:number=3 local HITBOX_SIZE:Vector3=Vector3.new(4,8,4)local BLACKLIST_SECONDS:
number=6 local function characterParts():(Model?,Humanoid?,BasePart?)if type(getCharacterParts)=='function'then local ok:boolean,character:any,humanoid:any,root:any=pcall(getCharacterParts)if ok and
character then return character,humanoid,root end end local character:Model?=LocalPlayer and LocalPlayer.Character if not character then return nil,nil,nil end local humanoid:Humanoid?=character:
FindFirstChildOfClass('Humanoid')::Humanoid?local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?return character,humanoid,root end local function rootOf(player:Player?):
BasePart?local character:Model?=player and player.Character if not character then return nil end return character:FindFirstChild('HumanoidRootPart')::BasePart?end local fallbackScanAt:number=-math.
huge local fallbackScanMap:Instance?=nil local fallbackPickups:{Instance}={}local function coinPickups():{Instance}local map:Instance?=type(findMM2Map)=='function'and findMM2Map()local out:{Instance}=
{}local function consider(object:Instance):()if not object.Parent or not object:IsA('BasePart')then return end if map and type(object.IsDescendantOf)=='function'and not object:IsDescendantOf(map)then
return end local ok:boolean,usable:any=pcall(function():boolean local parent:Instance?=object.Parent local server:Instance=if parent and parent.Name=='Coin_Server'then parent else object return object
:GetAttribute('Collected')~=true and object:GetAttribute('Delete')~=true and server:GetAttribute('Collected')~=true and server:GetAttribute('Delete')~=true end)if ok and usable then table.insert(out,
object)end end if type(state.mm2CoinPickups)=='function'then local ok:boolean,list:any=pcall(state.mm2CoinPickups)if ok and type(list)=='table'then for _,object:Instance in ipairs(list)do consider(
object)end if#out>0 then return out end end end local now:number=os.clock()if map==fallbackScanMap and now-fallbackScanAt<0.5 then return fallbackPickups end fallbackScanAt=now fallbackScanMap=map
local okTagged:boolean,tagged:any=pcall(function():any return game:GetService('CollectionService'):GetTagged('CoinVisual')end)if okTagged and type(tagged)=='table'then for _,object:Instance in ipairs(
tagged)do consider(object)end end if#out==0 then local ok:boolean,descendants:any=pcall(function():any if map and type(map.GetDescendants)=='function'then return map:GetDescendants()end return
workspace:GetDescendants()end)if ok and type(descendants)=='table'then for _,object:Instance in ipairs(descendants)do if object.Name=='Coin_Server'then consider(object)end end end end fallbackPickups=
out return out end local function pingMillis():number if type(getEstimatedLatency)=='function'then local ok:boolean,value:any=pcall(getEstimatedLatency)if ok and type(value)=='number'and value==value
then return math.round(math.clamp(value,0.02,0.6)*1000)end end return 0 end local function setStatus(text:string?):()local card:any=state.mm2AutoFarmFeature if card then card:SetStatus(text)end end
local function killAll(isCurrent:()->boolean):()if not isCurrent()then return end if type(getPlayerRole)~='function'or getPlayerRole(LocalPlayer)~='Murderer'then notify(
'Auto Farm: Kill All only works as the Murderer.')return end local character:Model?,humanoid:Humanoid?,myRoot:BasePart?=characterParts()if not character or not humanoid or not myRoot then return end
local knife:Tool?=nil if type(getPlayerWeapon)=='function'then knife=getPlayerWeapon(LocalPlayer,'Knife')end if not knife then notify('Auto Farm: no Knife found for Kill All.')return end if knife.
Parent~=character then pcall(function():()humanoid:EquipTool(knife::Tool)end)local equipDeadline:number=os.clock()+0.3 while knife.Parent~=character and os.clock()<equipDeadline and isCurrent()do task
.wait()end end if not isCurrent()or knife.Parent~=character then return end local victims:{BasePart}={}for _,player:Player in ipairs(Players:GetPlayers())do if player~=LocalPlayer and(type(
isPlayerAlive)~='function'or isPlayerAlive(player))then local victimRoot:BasePart?=rootOf(player)if victimRoot then table.insert(victims,victimRoot)pcall(function():()victimRoot.Anchored=true
victimRoot.CFrame=myRoot.CFrame+myRoot.CFrame.LookVector*1 end)end end end if#victims==0 then return end local function releaseVictims():()for _,victimRoot:BasePart in ipairs(victims)do pcall(function
():()victimRoot.Anchored=false end)end end task.wait(0.15)if not isCurrent()then releaseVictims()return end local stab:Instance?=knife:FindFirstChild('Stab')if stab and stab:IsA('RemoteEvent')then
pcall(function():()(stab::RemoteEvent):FireServer('Slash')end)end local events:Instance?=knife:FindFirstChild('Events')local handleTouched:Instance?=events and events:FindFirstChild('HandleTouched')
pcall(function():()(knife::Tool):Activate()end)for _,victimRoot:BasePart in ipairs(victims)do if not isCurrent()then break end if handleTouched and handleTouched:IsA('RemoteEvent')then pcall(function(
):()(handleTouched::RemoteEvent):FireServer(victimRoot)end)end task.wait(0.05)end if isCurrent()then task.delay(0.5,releaseVictims)notify('Auto Farm: Kill All fired at '..#victims..' players.')else
releaseVictims()end end local function flingPlayer(target:Player?,isCurrent:()->boolean):()if not target or not isCurrent()then return end if type(state.mm2PerformFling)=='function'then pcall(state.
mm2PerformFling,target,{silent=true,duration=3.5,blocking=true,shouldStop=function():boolean return not isCurrent()end,skipReturnOnStop=true})return end local _,_,myRoot:BasePart?=characterParts()if
not myRoot then return end local saved:CFrame=myRoot.CFrame local startedAt:number=os.clock()while os.clock()-startedAt<2.2 and isCurrent()do local targetRoot:BasePart?=rootOf(target)if not targetRoot
then break end pcall(function():()myRoot.CFrame=targetRoot.CFrame myRoot.AssemblyAngularVelocity=Vector3.new(0,400,0)myRoot.AssemblyLinearVelocity=Vector3.zero end)task.wait()end pcall(function():()
myRoot.AssemblyAngularVelocity=Vector3.zero if isCurrent()then myRoot.CFrame=saved end end)end local lastShotAt:number=-math.huge local function shootMurderer(murderer:Player):boolean if type(state.
mm2FireSilentShot)~='function'then setStatus('Silent shot unavailable')return false end local now:number=os.clock()if now-lastShotAt<0.22 then return false end lastShotAt=now local ok:boolean,sent:any
=pcall(state.mm2FireSilentShot,murderer,{continuous=true,quiet=true,wallCheck=false})return ok and sent==true end local farmGeneration:number=0 local function toggleAutoFarm(enabled:boolean):()
farmGeneration+=1 local enableGeneration:number=farmGeneration disconnectFeatureConnection('MM2AutoFarmSafeZoneRetry')disconnectFeatureConnection('MM2AutoFarmTick')disconnectFeatureConnection(
'MM2AutoFarmStepped')disconnectFeatureConnection('MM2AutoFarmCoinCollected')disconnectFeatureConnection('MM2AutoFarmCoinAdded')disconnectFeatureConnection('MM2AutoFarmRoundEnd')
disconnectFeatureConnection('MM2AutoFarmRoundStart')for _,name:string in ipairs({'RoleSelect','ShowRoleSelect','ShowRoleSelectNew','CoinsStarted'})do disconnectFeatureConnection('MM2AutoFarmStart_'..
name)end disconnectFeatureConnection('MM2AutoFarmRestore')if not enabled then setStatus(nil)return end if leaveSafeZone and not leaveSafeZone(false)then setStatus(
'Auto Farm: waiting in Safe Zone for the original map')local retryElapsed:number=0 featureConnections.MM2AutoFarmSafeZoneRetry=TaskManager:Connect(function(delta:number):()if farmGeneration~=
enableGeneration then return end retryElapsed+=math.max(0,tonumber(delta)or 0)if retryElapsed<1 then return end retryElapsed=0 if leaveSafeZone and leaveSafeZone(false,true)then toggleAutoFarm(true)
end end,nil,'MM2 Auto Farm Safe Zone Retry')return end leaveSafeZone=nil fallbackScanAt=-math.huge fallbackScanMap=nil fallbackPickups={}local bagAmount:number=0 local bagLimit:number=50 local busy:
boolean=false local fullHandled:boolean=false local roundFullReached:boolean=false local teleportedHomeThisRound:boolean=false local wasInActiveRound:boolean=false local currentRoundMap:Instance?=nil
local busyLobbyPart:BasePart?=nil local busyLobbyScanAt:number=0 local endedMap:Instance?=nil local waitingForNextRound:boolean=false local roundGeneration:number=0 local scanCache:{Instance}={}local
scanAt:number=0 local DWELL_SECONDS:number=0.3 local RETRY_SECONDS:number=1.5 local RETRY_LIMIT:number=3 local retryAt:{[Instance]:number}={}local attempts:{[Instance]:number}={}local blacklist:{[
Instance]:number}={}local stuckPart:Instance?=nil local stuckSince:number=0 local lastTouchPulse:number=0 local travelPart:Instance?=nil local travelBestDistance:number=math.huge local
travelProgressAt:number=0 local TRAVEL_STALL_SECONDS:number=3.5 local engaged:boolean=false local engagedCharacter:Model?=nil local homeCFrame:CFrame?=nil local homeCharacter:Model?=nil local
parkedCharacter:Model?=nil local parkedMap:Instance?=nil local parkedReturn:CFrame?=nil local parkedPosition:Vector3?=nil local hiddenUnderMap:boolean=false local savedCollide:{[Instance]:boolean}={}
local engagedMode:string?=nil local savedWalkSpeed:number?=nil local savedRootSize:Vector3?=nil local walkLastDistance:number=math.huge local walkStuckSince:number=0 local walkBodyReady:boolean=false
local walkCarryDirection:Vector3?=nil local walkCarryUntil:number=0 local hoverTarget:Vector3?=nil local hoverActive:boolean=false local lastSteppedPosition:Vector3?=nil local function yawOf(frame:
CFrame):number local look:Vector3=frame.LookVector return math.atan2(-look.X,-look.Z)end local function findLobbyReferencePart():BasePart?local okChildren:boolean,children:any=pcall(function():any
return workspace:GetChildren()end)if not okChildren or type(children)~='table'then return nil end for _,child:Instance in ipairs(children)do local lowerName:string=string.lower(tostring(child.Name))
local isLobby:boolean=string.find(lowerName,'lobby',1,true)~=nil or child:FindFirstChild('VotePads',true)~=nil or child:FindFirstChild('VotePad1',true)~=nil or child:FindFirstChild('VoteIcons',true)~=
nil if isLobby then local votePads:Instance?=child:FindFirstChild('VotePads',true)if votePads then local padPart:BasePart?=votePads:FindFirstChildWhichIsA('BasePart',true)if padPart then return
padPart end end local spawns:Instance?=child:FindFirstChild('Spawns',true)if spawns then local spawnPart:BasePart?=spawns:FindFirstChildWhichIsA('BasePart',true)if spawnPart then return spawnPart end
end local anyPart:BasePart?=child:FindFirstChildWhichIsA('BasePart',true)if anyPart then return anyPart end end end return nil end local function isNearLobby(root:BasePart?,lobby:BasePart?):boolean if
not root or not lobby then return false end local ok:boolean,near:any=pcall(function():boolean return(root.Position-lobby.Position).Magnitude<=350 end)return ok and near==true end local function
resolveSafeAboveGroundPosition(candidatePos:Vector3,character:Model?):Vector3 if homeCFrame then local dx:number=candidatePos.X-homeCFrame.Position.X local dz:number=candidatePos.Z-homeCFrame.Position
.Z if(dx*dx+dz*dz)<=36 then return Vector3.new(candidatePos.X,homeCFrame.Position.Y,candidatePos.Z)end end local resolvedY:number=candidatePos.Y local wasUnderground:boolean=hiddenUnderMap or
engagedMode=='Below'local probeRaise:number=if wasUnderground then(BELOW_OFFSET+4.5)else 2.5 local foundFloor:boolean=false pcall(function():()if type(workspace.Raycast)=='function'and type(
RaycastParams)=='table'then local params:RaycastParams=RaycastParams.new()params.FilterType=Enum.RaycastFilterType.Exclude local excluded:{Instance}={}if character then table.insert(excluded,character
)end if safeZonePlatform then table.insert(excluded,safeZonePlatform)end params.FilterDescendantsInstances=excluded params.RespectCanCollide=true params.IgnoreWater=true local rayOrigin:Vector3=
Vector3.new(candidatePos.X,candidatePos.Y+probeRaise,candidatePos.Z)local hit:any=workspace:Raycast(rayOrigin,Vector3.new(0,-(probeRaise+8),0),params)if hit and typeof(hit.Position)=='Vector3'then
resolvedY=hit.Position.Y+3.2 foundFloor=true end end end)if not foundFloor and wasUnderground then if homeCFrame then resolvedY=homeCFrame.Position.Y else resolvedY=candidatePos.Y+BELOW_OFFSET+4 end
end return Vector3.new(candidatePos.X,resolvedY,candidatePos.Z)end local function forceNoClipCharacter(character:Model):()for _,object:Instance in ipairs(character:GetDescendants())do if object:IsA(
'BasePart')then if savedCollide[object]==nil then savedCollide[object]=(object::BasePart).CanCollide end pcall(function():()(object::BasePart).CanCollide=false end)end end end local MAX_BELOW_DEPTH:
number=90 local VOID_RECOVER_DEPTH:number=130 local function destroyHeightY():number local destroyHeight:any=nil pcall(function():()destroyHeight=workspace.FallenPartsDestroyHeight end)if type(
destroyHeight)~='number'or destroyHeight~=destroyHeight or math.abs(destroyHeight)>1e6 then return-500 end return destroyHeight end local function currentHomeY(root:BasePart):number if homeCFrame then
return homeCFrame.Position.Y end return root.Position.Y end local function floorBudgetY(root:BasePart):number return math.max(destroyHeightY()+40,currentHomeY(root)-MAX_BELOW_DEPTH)end local function
clampBelowTarget(position:Vector3,root:BasePart):Vector3 local lowest:number=floorBudgetY(root)local highest:number=currentHomeY(root)+80 return Vector3.new(position.X,math.clamp(position.Y,lowest,
highest),position.Z)end local function applyBelowControl(root:BasePart,humanoid:Humanoid?):()if not hoverActive then return end local speed:number=math.clamp(settings.speed,5,60)local target:Vector3?=
hoverTarget if target then target=clampBelowTarget(target::Vector3,root)end local velocityX:number=0 local velocityY:number=0 local velocityZ:number=0 if target then local delta:Vector3=(target::
Vector3)-root.Position local flat:Vector3=Vector3.new(delta.X,0,delta.Z)local flatDistance:number=flat.Magnitude if flatDistance>0.35 then local direction:Vector3=flat/flatDistance velocityX=direction
.X*speed velocityZ=direction.Z*speed end velocityY=math.clamp(delta.Y*10,-speed,speed)end pcall(function():()root.AssemblyLinearVelocity=Vector3.new(velocityX,velocityY,velocityZ)root.
AssemblyAngularVelocity=Vector3.zero end)if humanoid then pcall(function():()if not humanoid.PlatformStand then humanoid.PlatformStand=true end if humanoid.AutoRotate then humanoid.AutoRotate=false
end end)end end local function holdAltitude(root:BasePart):()local targetY:number=homeCFrame and homeCFrame.Position.Y or root.Position.Y if hoverTarget then targetY=hoverTarget.Y elseif
hiddenUnderMap then targetY=currentHomeY(root)-BELOW_OFFSET end hoverTarget=clampBelowTarget(Vector3.new(root.Position.X,targetY,root.Position.Z),root)hoverActive=true end local function holdUnderMap(
character:Model,humanoid:Humanoid,root:BasePart):()hiddenUnderMap=true forceNoClipCharacter(character)local basePosition:Vector3=if homeCFrame then homeCFrame.Position else root.Position local hidePos
:Vector3=clampBelowTarget(basePosition-Vector3.new(0,BELOW_OFFSET+4,0),root)hoverTarget=hidePos hoverActive=settings.mode~='Walk'pcall(function():()humanoid.PlatformStand=true humanoid.AutoRotate=
false humanoid:Move(Vector3.zero,false)if settings.mode~='Walk'then if savedRootSize==nil then savedRootSize=root.Size end root.Size=HITBOX_SIZE end if(root.Position-hidePos).Magnitude>6 then root.
CFrame=CFrame.new(hidePos)*CFrame.Angles(0,yawOf(root.CFrame),0)root.AssemblyLinearVelocity=Vector3.zero root.AssemblyAngularVelocity=Vector3.zero end end)if not lastSteppedPosition then
lastSteppedPosition=root.Position end end local function recoverFromVoid(character:Model,humanoid:Humanoid,root:BasePart):()local homeX:number=homeCFrame and homeCFrame.Position.X or root.Position.X
local homeZ:number=homeCFrame and homeCFrame.Position.Z or root.Position.Z local homeY:number=currentHomeY(root)if settings.mode=='Walk'then local safePos:Vector3=if homeCFrame then homeCFrame.
Position else Vector3.new(homeX,destroyHeightY()+120,homeZ)hoverActive=false hoverTarget=nil walkCarryDirection=nil walkCarryUntil=0 pcall(function():()if savedRootSize then root.Size=savedRootSize
savedRootSize=nil end character:PivotTo(CFrame.new(safePos))root.CFrame=CFrame.new(safePos)root.AssemblyLinearVelocity=Vector3.zero root.AssemblyAngularVelocity=Vector3.zero humanoid.PlatformStand=
false humanoid.AutoRotate=true humanoid:ChangeState(Enum.HumanoidStateType.Running)end)walkBodyReady=false return end local recoverY:number=math.min(homeY-BELOW_OFFSET,homeY+8)recoverY=math.max(
recoverY,destroyHeightY()+60)local target:Vector3=Vector3.new(homeX,recoverY,homeZ)hiddenUnderMap=true forceNoClipCharacter(character)hoverTarget=target hoverActive=true pcall(function():()if
savedRootSize==nil then savedRootSize=root.Size end root.Size=HITBOX_SIZE humanoid.PlatformStand=true humanoid.AutoRotate=false root.CFrame=CFrame.new(target)*CFrame.Angles(0,yawOf(root.CFrame),0)root
.AssemblyLinearVelocity=Vector3.zero root.AssemblyAngularVelocity=Vector3.zero end)end local function isCharacterInActiveRound(root:BasePart,knownMap:Instance?):boolean if type(findMM2Map)~='function'
then return false end local map:Instance?=knownMap or findMM2Map()if not map then return false end if type(getPlayerRole)=='function'then local role:any=getPlayerRole(LocalPlayer)if role=='Dead'then
return false end if(role=='None'or role==nil)and type(isRoundLifecycleActive)=='function'and isRoundLifecycleActive()==false then local lobbyPartCheck:BasePart?=findLobbyReferencePart()if
lobbyPartCheck then local okLD:boolean,ld:any=pcall(function():any return(root.Position-lobbyPartCheck.Position).Magnitude end)if okLD and type(ld)=='number'and ld<=350 then return false end end end
end local lobbyPart:BasePart?=findLobbyReferencePart()if lobbyPart then local okLobbyDist:boolean,lobbyDist:any=pcall(function():any return(root.Position-lobbyPart.Position).Magnitude end)if
okLobbyDist and type(lobbyDist)=='number'and lobbyDist<=350 then return false end end if safeZonePlatform and safeZonePlatform.Parent then local okZoneDist:boolean,zoneDist:any=pcall(function():any
return(root.Position-safeZonePlatform.Position).Magnitude end)if okZoneDist and type(zoneDist)=='number'and zoneDist<=120 then return true end end local refPart:BasePart?=nil local spawns:Instance?=(
map::Instance):FindFirstChild('Spawns')if spawns then refPart=spawns:FindFirstChildWhichIsA('BasePart',true)end if not refPart then local coinContainer:Instance?=(map::Instance):FindFirstChild(
'CoinContainer')if coinContainer then refPart=coinContainer:FindFirstChildWhichIsA('BasePart',true)end end if not refPart then refPart=(map::Instance):FindFirstChildWhichIsA('BasePart',true)end if
refPart then local okDist:boolean,dist:any=pcall(function():any return(root.Position-refPart.Position).Magnitude end)if okDist and type(dist)=='number'and dist>1500 then return false end return true
end if type(isRoundLifecycleActive)=='function'and isRoundLifecycleActive()~=true then return false end return true end local function restoreCharacter(returnHome:boolean,noReposition:boolean?):()
local wasMovedByFarm:boolean=engaged or hiddenUnderMap local character:Model?,humanoid:Humanoid?,root:BasePart?=characterParts()local movedCharacter:Model?=engagedCharacter or(hiddenUnderMap and
homeCharacter)if wasMovedByFarm and movedCharacter and movedCharacter~=character then character=movedCharacter humanoid=movedCharacter:FindFirstChildOfClass('Humanoid')::Humanoid?root=movedCharacter:
FindFirstChild('HumanoidRootPart')::BasePart?end for part:Instance,was:boolean in pairs(savedCollide)do pcall(function():()(part::BasePart).CanCollide=was end)end table.clear(savedCollide)if humanoid
and wasMovedByFarm then pcall(function():()humanoid.PlatformStand=false humanoid.AutoRotate=true if humanoid.Sit then humanoid.Sit=false end if savedWalkSpeed then humanoid.WalkSpeed=savedWalkSpeed
end humanoid:Move(Vector3.zero,false)humanoid:ChangeState(Enum.HumanoidStateType.Running)end)end savedWalkSpeed=nil if root and wasMovedByFarm then pcall(function():()if savedRootSize then root.Size=
savedRootSize end root.AssemblyLinearVelocity=Vector3.zero root.AssemblyAngularVelocity=Vector3.zero if noReposition then return end local allowReturnHome:boolean=returnHome and not
teleportedHomeThisRound and(bagAmount>=50 or not enabled)and homeCFrame~=nil and(not homeCharacter or homeCharacter==character)if allowReturnHome and homeCFrame then local destPosition:Vector3=
resolveSafeAboveGroundPosition(homeCFrame.Position,character)local mapReturn:CFrame=CFrame.new(destPosition)*CFrame.Angles(0,yawOf(homeCFrame),0)local zoneDestination:Vector3?=nil if enabled and
settings.safeZone and bagAmount>=50 then zoneDestination=ensureSafeZonePlatform(homeCFrame.Position)if zoneDestination then destPosition=zoneDestination end end local targetCF:CFrame=CFrame.new(
destPosition)*CFrame.Angles(0,yawOf(homeCFrame),0)if character then pcall(function():()character:PivotTo(targetCF)end)end root.CFrame=targetCF if zoneDestination then parkedCharacter,parkedMap=
character,currentRoundMap parkedReturn,parkedPosition=mapReturn,zoneDestination end if bagAmount>=50 then teleportedHomeThisRound=true homeCFrame=nil end elseif not teleportedHomeThisRound then local
safePos:Vector3=if hiddenUnderMap and homeCFrame then homeCFrame.Position else resolveSafeAboveGroundPosition(root.Position,character)local targetCF:CFrame=CFrame.new(safePos)*CFrame.Angles(0,yawOf(
root.CFrame),0)if character then pcall(function():()character:PivotTo(targetCF)end)end root.CFrame=targetCF end end)end hiddenUnderMap=false savedRootSize=nil engaged=false engagedMode=nil
engagedCharacter=nil hoverActive=false hoverTarget=nil lastSteppedPosition=nil walkCarryDirection=nil walkCarryUntil=0 walkBodyReady=false end local function stillParked():boolean local character:
Model?,humanoid:Humanoid?,root:BasePart?=characterParts()if not parkedReturn or not parkedPosition or character~=parkedCharacter or not root or not humanoid or humanoid.Health<=0 then return false end
return(root.Position-(parkedPosition::Vector3)).Magnitude<=120 end leaveSafeZone=function(cancelBusy:boolean?,quiet:boolean?):boolean if not stillParked()then parkedCharacter,parkedMap,parkedReturn,
parkedPosition=nil,nil,nil,nil destroySafeZonePlatform()return true end if cancelBusy then roundGeneration+=1 busy=false end local character:Model?,_,root:BasePart?=characterParts()local okMap:boolean
,map:any=pcall(function():any return type(findMM2Map)=='function'and findMM2Map()or nil end)if not okMap or not parkedMap or map~=parkedMap or not parkedMap.Parent or not root then if not quiet then
notify([[Auto Farm: waiting in Safe Zone; the original map is unavailable.]])end return false end local destination:CFrame=parkedReturn::CFrame local moved:boolean=pcall(function():()if character then
pcall(function():()character:PivotTo(destination)end)end root.CFrame=destination root.AssemblyLinearVelocity=Vector3.zero root.AssemblyAngularVelocity=Vector3.zero end)if not moved then return false
end parkedCharacter,parkedMap,parkedReturn,parkedPosition=nil,nil,nil,nil destroySafeZonePlatform()return true end local function clearRound():()roundGeneration+=1 busy=false if engaged or
hiddenUnderMap then local character:Model?,humanoid:Humanoid?,root:BasePart?=characterParts()local movedCharacter:Model?=engagedCharacter or(hiddenUnderMap and homeCharacter)local sameMap:boolean=
currentRoundMap~=nil and type(findMM2Map)=='function'and findMM2Map()==currentRoundMap local mayReposition:boolean=sameMap and character~=nil and character==movedCharacter and humanoid~=nil and
humanoid.Health>0 and root~=nil and isCharacterInActiveRound(root::BasePart,currentRoundMap)restoreCharacter(false,not mayReposition)end wasInActiveRound=false currentRoundMap=nil busyLobbyPart=nil
busyLobbyScanAt=0 bagAmount=0 bagLimit=50 fullHandled=false roundFullReached=false teleportedHomeThisRound=false homeCFrame=nil homeCharacter=nil scanCache={}scanAt=0 table.clear(retryAt)table.clear(
attempts)table.clear(blacklist)stuckPart=nil stuckSince=0 lastTouchPulse=0 travelPart=nil travelBestDistance=math.huge travelProgressAt=0 if not stillParked()then parkedCharacter,parkedMap,
parkedReturn,parkedPosition=nil,nil,nil,nil destroySafeZonePlatform()end end local function enterRound(character:Model,root:BasePart,map:Instance):()if wasInActiveRound and currentRoundMap==map then
return end clearRound()wasInActiveRound=true currentRoundMap=map homeCFrame=CFrame.new(root.Position)*CFrame.Angles(0,yawOf(root.CFrame),0)homeCharacter=character end local function engage(character:
Model,humanoid:Humanoid,root:BasePart):()if homeCharacter~=character or not homeCFrame then local basePos:Vector3=if hiddenUnderMap then root.Position+Vector3.new(0,BELOW_OFFSET+4,0)else root.Position
homeCFrame=CFrame.new(basePos)*CFrame.Angles(0,yawOf(root.CFrame),0)homeCharacter=character end if engaged and engagedMode~=nil and engagedMode~=settings.mode then restoreCharacter(false)end
engagedMode=settings.mode if engagedCharacter~=nil and engagedCharacter~=character then restoreCharacter(false)engagedMode=settings.mode if homeCharacter~=character or not homeCFrame then homeCFrame=
root.CFrame homeCharacter=character end end engagedCharacter=character if not engaged then engaged=true walkLastDistance=math.huge walkStuckSince=os.clock()if settings.mode~='Walk'then hiddenUnderMap=
false forceNoClipCharacter(character)pcall(function():()local startPos:Vector3=root.Position if homeCFrame and startPos.Y>=homeCFrame.Position.Y-1 then startPos=Vector3.new(startPos.X,homeCFrame.
Position.Y-BELOW_OFFSET,startPos.Z)end root.CFrame=CFrame.new(startPos)*CFrame.Angles(0,yawOf(root.CFrame),0)humanoid.PlatformStand=true humanoid.AutoRotate=false if savedRootSize==nil then
savedRootSize=root.Size end root.Size=HITBOX_SIZE end)end lastSteppedPosition=root.Position end if settings.mode=='Walk'then hoverActive=false hoverTarget=nil if not walkBodyReady then walkBodyReady=
true hiddenUnderMap=false walkCarryDirection=nil walkCarryUntil=0 for part:Instance,was:boolean in pairs(savedCollide)do pcall(function():()((part::any)::BasePart).CanCollide=was end)end table.clear(
savedCollide)pcall(function():()if savedRootSize then root.Size=savedRootSize savedRootSize=nil end humanoid.PlatformStand=false humanoid.AutoRotate=true humanoid:Move(Vector3.zero,false)if humanoid.
Sit then humanoid.Sit=false end end)end if savedWalkSpeed==nil then savedWalkSpeed=humanoid.WalkSpeed end pcall(function():()humanoid.WalkSpeed=math.clamp(settings.speed,5,60)end)return end
walkBodyReady=false forceNoClipCharacter(character)hoverActive=true end local function moveTowards(root:BasePart,destination:Vector3,_deltaTime:number?):number local target:Vector3=clampBelowTarget(
destination,root)hoverTarget=target hoverActive=true applyBelowControl(root,nil)return(target-root.Position).Magnitude end local function walkTowards(humanoid:Humanoid,root:BasePart,destination:
Vector3):number hoverActive=false hoverTarget=nil local delta:Vector3=destination-root.Position local flat:Vector3=Vector3.new(delta.X,0,delta.Z)local distance:number=flat.Magnitude local now:number=
os.clock()if distance<1.4 then walkCarryDirection=nil walkCarryUntil=0 pcall(function():()humanoid:Move(Vector3.zero,false)end)return distance end if distance<walkLastDistance-0.3 then
walkLastDistance=distance walkStuckSince=now walkCarryUntil=0 elseif now-walkStuckSince>2 then walkStuckSince=now walkCarryUntil=now+1 walkCarryDirection=flat/distance pcall(function():()humanoid.Jump
=true end)end if walkCarryUntil>now and walkCarryDirection then local speed:number=math.clamp(settings.speed,5,60)local direction:Vector3=walkCarryDirection::Vector3 pcall(function():()local current:
Vector3=root.AssemblyLinearVelocity root.AssemblyLinearVelocity=Vector3.new(direction.X*speed,math.min(current.Y,0),direction.Z*speed)humanoid:Move(direction,false)end)else walkCarryDirection=nil
pcall(function():()humanoid:Move(flat.Unit,false)end)end return distance end local function serverPartOf(part:Instance):Instance?if part.Name=='Coin_Server'then return part end local parent:Instance?=
part.Parent if parent and parent.Name=='Coin_Server'then return parent end return part end local function stillCollectable(part:Instance):boolean if not part.Parent then return false end local server:
Instance?=serverPartOf(part)local ok:boolean,usable:any=pcall(function():boolean return part:GetAttribute('Collected')~=true and part:GetAttribute('Delete')~=true and server~=nil and server:
GetAttribute('Collected')~=true and server:GetAttribute('Delete')~=true end)return ok and usable==true end local function canGrabGun():boolean if type(getPlayerRole)~='function'then return false end
local role:any=getPlayerRole(LocalPlayer)if role~='Innocent'and role~='Sheriff'and role~='Hero'then return false end if type(isRoundLifecycleActive)~='function'or isRoundLifecycleActive()~=true then
return false end if type(findMM2Map)~='function'or not findMM2Map()then return false end return true end local function touchDroppedGun(drop:Instance,root:BasePart):boolean if type(state.
fireTouchInterest)~='function'then return false end local touchPart:Instance?=nil if drop:IsA('BasePart')then touchPart=drop elseif drop:IsA('Model')then touchPart=drop:FindFirstChild('Handle',true)or
drop:FindFirstChildWhichIsA('BasePart',true)end if not touchPart then return false end pcall(state.fireTouchInterest,root,touchPart,0)pcall(state.fireTouchInterest,root,touchPart,1)return type(
getPlayerWeapon)=='function'and getPlayerWeapon(LocalPlayer,'Gun')~=nil end local function grabGunAndShoot(isCurrent:()->boolean):()if not isCurrent()or not canGrabGun()then return end local
shootLoopDeadline:number=os.clock()+30 while os.clock()<shootLoopDeadline and isCurrent()and settings.grabGunShoot do if not canGrabGun()then return end local murderer:Player?=nil if type(findMurderer
)=='function'then murderer=findMurderer()end if not murderer or(type(isPlayerAlive)=='function'and not isPlayerAlive(murderer))then return end local character:Model?,humanoid:Humanoid?,startRoot:
BasePart?=characterParts()if not character or not humanoid or not startRoot or humanoid.Health<=0 then return end local gun:Tool?=nil if type(getPlayerWeapon)=='function'then gun=getPlayerWeapon(
LocalPlayer,'Gun')end if not gun and type(findDroppedGun)=='function'then local firstDrop:Instance?=findDroppedGun()if firstDrop and touchDroppedGun(firstDrop,startRoot)then gun=getPlayerWeapon(
LocalPlayer,'Gun')end local pickupDeadline:number=os.clock()+5 while os.clock()<pickupDeadline and not gun do if not isCurrent()or not canGrabGun()or not settings.grabGunShoot then return end local
curChar:Model?,curHum:Humanoid?,root:BasePart?=characterParts()local drop:Instance?=findDroppedGun()if not curChar or not curHum or not root or not drop then break end if touchDroppedGun(drop,root)
then gun=getPlayerWeapon(LocalPlayer,'Gun')if gun then break end end if teleportedHomeThisRound or roundFullReached then break end local dropPosition:Vector3?=nil pcall(function():()if drop:IsA(
'BasePart')then dropPosition=(drop::BasePart).Position elseif drop:IsA('Model')then dropPosition=(drop::Model):GetPivot().Position end end)if not dropPosition then break end engage(curChar,curHum,root
)local remaining:number if settings.mode=='Walk'then remaining=walkTowards(curHum,root,dropPosition::Vector3)else remaining=moveTowards(root,(dropPosition::Vector3)+Vector3.new(0,1,0),0.05)end if
remaining<2.5 then task.wait(0.05)gun=getPlayerWeapon(LocalPlayer,'Gun')if gun then break end end task.wait()end end if not gun then task.wait(0.15)continue end character,humanoid=characterParts()if
not character or not humanoid or humanoid.Health<=0 then return end if gun.Parent~=character then pcall(function():()humanoid:EquipTool(gun::Tool)end)local equipWait:number=os.clock()+0.15 while gun.
Parent~=character and os.clock()<equipWait and isCurrent()do task.wait()end end if isCurrent()and gun.Parent==character and canGrabGun()then shootMurderer(murderer)end task.wait(0.12)end end
local function runFullSequence(isCurrent:()->boolean):()local ok:boolean,err:any=pcall(function():()if not isCurrent()then return end if settings.killAllOnFull then killAll(isCurrent)task.wait()end if
not isCurrent()then return end restoreCharacter(true)if settings.grabGunShoot then grabGunAndShoot(isCurrent)task.wait(0.15)end if not isCurrent()then return end if settings.resetOnFull then local
oldCharacter:Model?=LocalPlayer.Character local _,humanoid:Humanoid?=characterParts()if humanoid then pcall(function():()humanoid.Health=0 end)end local deadline:number=os.clock()+10 while os.clock()<
deadline and isCurrent()do local character:Model?,newHumanoid:Humanoid?,newRoot:BasePart?=characterParts()if character and character~=oldCharacter and newHumanoid and newHumanoid.Health>0 and newRoot
then break end task.wait(0.1)end if not isCurrent()then return end end if settings.flingMurder and type(findMurderer)=='function'then flingPlayer(findMurderer(),isCurrent)end if isCurrent()
and settings.flingSheriff and type(findSheriff)=='function'then flingPlayer(findSheriff(),isCurrent)end end)if not ok then warn('Auto Farm full-bag sequence failed: '..tostring(err))end end
local function startBusySequence(action:(()->boolean)->()):()if busy then return end busy=true busyLobbyPart=findLobbyReferencePart()busyLobbyScanAt=os.clock()+0.5 local sequenceRound:number=
roundGeneration local sequenceMap:Instance?=currentRoundMap local observedCharacter:Model?=LocalPlayer and LocalPlayer.Character local mapChanged:boolean=false local function isCurrent():boolean if
farmGeneration~=enableGeneration or roundGeneration~=sequenceRound then return false end local character:Model?,_,root:BasePart?=characterParts()if sequenceMap and not sequenceMap.Parent then return
false end if character~=observedCharacter then observedCharacter=character local map:Instance?=type(findMM2Map)=='function'and findMM2Map()if map and map~=sequenceMap then mapChanged=true end end
return not mapChanged and not isNearLobby(root,busyLobbyPart)end task.spawn(function():()local ok:boolean,err:any=pcall(action,isCurrent)if not ok then warn('Auto Farm sequence failed: '..tostring(err
))end if farmGeneration==enableGeneration and roundGeneration==sequenceRound then if isCurrent()then busy=false else local _,_,root:BasePart?=characterParts()local inLobby:boolean=isNearLobby(root,
busyLobbyPart)clearRound()setStatus(if inLobby then'Auto Farm: waiting in lobby'else'Auto Farm: waiting for round')end end end)end if workspace.DescendantAdded then featureConnections.
MM2AutoFarmCoinAdded=workspace.DescendantAdded:Connect(function(object:Instance):()if object:IsA('BasePart')and(object.Name=='Coin_Server'or object.Name=='CoinVisual')then fallbackScanAt=-math.huge
scanAt=-math.huge end end)end local remotes:any=core.mm2GameplayRemotes local roundEnd:Instance?=remotes and remotes:FindFirstChild('RoundEndFade')if roundEnd and roundEnd:IsA('RemoteEvent')then
featureConnections.MM2AutoFarmRoundEnd=(roundEnd::any).OnClientEvent:Connect(function():()endedMap=currentRoundMap or(type(findMM2Map)=='function'and findMM2Map())waitingForNextRound=true clearRound()
end)end local roundStart:Instance?=remotes and remotes:FindFirstChild('RoundStart')if roundStart and roundStart:IsA('RemoteEvent')then featureConnections.MM2AutoFarmRoundStart=(roundStart::any).
OnClientEvent:Connect(function(duration:any):()if waitingForNextRound and tonumber(duration)and tonumber(duration)>0 then waitingForNextRound=false endedMap=nil end end)end for _,name:string in
ipairs({'RoleSelect','ShowRoleSelect','ShowRoleSelectNew','CoinsStarted'})do local remote:Instance?=remotes and remotes:FindFirstChild(name)if remote and remote:IsA('RemoteEvent')then
featureConnections['MM2AutoFarmStart_'..name]=(remote::any).OnClientEvent:Connect(function():()if waitingForNextRound then waitingForNextRound=false endedMap=nil end end)end end local collected:
Instance?=remotes and remotes:FindFirstChild('CoinCollected')if collected and collected:IsA('RemoteEvent')then featureConnections.MM2AutoFarmCoinCollected=(collected::any).OnClientEvent:Connect(
function(_bag:any,amount:any,limit:any):()local character:Model?,humanoid:Humanoid?,root:BasePart?=characterParts()local map:Instance?=type(findMM2Map)=='function'and findMM2Map()if not character or
not humanoid or humanoid.Health<=0 or not root or not map or not isCharacterInActiveRound(root,map)or(waitingForNextRound and map==endedMap)then return end if waitingForNextRound then
waitingForNextRound=false endedMap=nil end enterRound(character,root,map)if type(limit)=='number'and limit>=50 then bagLimit=limit end if type(amount)=='number'then if amount>bagAmount then
totalCoinsCollected+=(amount-bagAmount)refreshFarmOverlay()end bagAmount=amount end local targetLimit:number=math.max(50,bagLimit)if bagAmount<targetLimit then fullHandled=false roundFullReached=false
end end)end if RunService and RunService.Stepped then featureConnections.MM2AutoFarmStepped=RunService.Stepped:Connect(function():()if not hiddenUnderMap and not engaged then return end local
character:Model?=LocalPlayer and LocalPlayer.Character local movedCharacter:Model?=engagedCharacter or(hiddenUnderMap and homeCharacter)if not character or(movedCharacter and movedCharacter~=character
)then return end local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?if not root or not root.Parent then return end local position:Vector3=root.Position if lastSteppedPosition
and(position-lastSteppedPosition).Magnitude>60 then restoreCharacter(false,true)return end lastSteppedPosition=position if hiddenUnderMap or settings.mode~='Walk'then forceNoClipCharacter(character)
end if not hoverActive or settings.mode=='Walk'then return end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')if humanoid and humanoid.Health>0 then applyBelowControl(root,
humanoid)end end)end local function farmTick(deltaTime:number):()if type(deltaTime)=='number'and deltaTime>0 and deltaTime<5 then farmElapsedSeconds+=deltaTime refreshFarmOverlay()end local character:
Model?,humanoid:Humanoid?,root:BasePart?=characterParts()if not character or not humanoid or not root or humanoid.Health<=0 then return end local map:Instance?=type(findMM2Map)=='function'and
findMM2Map()if busy and(not map or map==currentRoundMap)then if not busyLobbyPart and os.clock()>=busyLobbyScanAt then busyLobbyPart=findLobbyReferencePart()busyLobbyScanAt=os.clock()+0.5 end if
isNearLobby(root,busyLobbyPart)then clearRound()setStatus('Auto Farm: waiting in lobby')end return end if stillParked()and not settings.safeZone and leaveSafeZone then if not leaveSafeZone(false,true)
then setStatus('Auto Farm: waiting in Safe Zone for the original map')return end end if stillParked()and parkedMap~=map then setStatus('Auto Farm: waiting for the round teleport')return end local
inActiveRound:boolean=map~=nil and isCharacterInActiveRound(root,map)if not inActiveRound then if wasInActiveRound or engaged or hiddenUnderMap then clearRound()end if waitingForNextRound then local
lobby:BasePart?=findLobbyReferencePart()local reachedLobby:boolean=lobby~=nil and(root.Position-(lobby::BasePart).Position).Magnitude<=350 if not map or map~=endedMap or reachedLobby then
waitingForNextRound=false endedMap=nil end end setStatus('Auto Farm: waiting in lobby')return end if waitingForNextRound and map==endedMap then setStatus('Auto Farm: waiting for next round')return end
if waitingForNextRound then waitingForNextRound=false endedMap=nil end enterRound(character,root,map::Instance)if busy then return end local targetLimit:number=math.max(50,bagLimit)if bagAmount>=
targetLimit then if not fullHandled then fullHandled=true roundFullReached=true if settings.killAllOnFull or settings.resetOnFull or settings.flingMurder or settings.flingSheriff or settings.
grabGunShoot then startBusySequence(runFullSequence)else restoreCharacter(true)end elseif settings.grabGunShoot and canGrabGun()then startBusySequence(function(isCurrent:()->boolean):()
grabGunAndShoot(isCurrent)end)else setStatus(string.format('Auto Farm: full bag (%d/%d) - stopped at start position',bagAmount,targetLimit))end return end if roundFullReached then return end local
voidThreshold:number=math.max(destroyHeightY()+40,currentHomeY(root)-VOID_RECOVER_DEPTH)if root.Position.Y<=voidThreshold then recoverFromVoid(character,humanoid,root)setStatus(
'Auto Farm: recovered from the void - back inside the map')return end local now:number=os.clock()local scanInterval:number=if#scanCache==0 then 0.12 else 0.35 if now-scanAt>scanInterval then scanAt=
now scanCache=coinPickups()end local nearest:Instance?=nil local nearestPosition:Vector3?=nil local nearestDistance:number=math.huge for _,part:Instance in ipairs(scanCache)do if part.Parent and
stillCollectable(part)and(not blacklist[part]or blacklist[part]<now)and(not retryAt[part]or retryAt[part]<now)then if part:IsA('BasePart')then local position:Vector3=(part::BasePart).Position local
distance:number=(position-root.Position).Magnitude if distance<nearestDistance then nearest=part nearestPosition=position nearestDistance=distance end end end end if not nearest or not nearestPosition
then local bodyOnMap:boolean=homeCFrame~=nil and root.Position.Y>=homeCFrame.Position.Y-1 if bagAmount==0 and homeCFrame and not hiddenUnderMap and bodyOnMap then local dx:number=root.Position.X-
homeCFrame.Position.X local dz:number=root.Position.Z-homeCFrame.Position.Z if math.sqrt(dx*dx+dz*dz)>14 and root.Position.Y>destroyHeightY()+60 then local safeSpawnPos:Vector3=Vector3.new(root.
Position.X,root.Position.Y,root.Position.Z)homeCFrame=CFrame.new(safeSpawnPos)*CFrame.Angles(0,yawOf(root.CFrame),0)homeCharacter=character end end travelPart=nil if settings.stayUnderOnStart and not
engaged then holdUnderMap(character,humanoid,root)elseif hiddenUnderMap then restoreCharacter(false)pcall(function():()humanoid:Move(Vector3.zero,false)end)elseif settings.mode=='Walk'then pcall(
function():()humanoid:Move(Vector3.zero,false)end)else holdAltitude(root)end setStatus(string.format([[Auto Farm (%s): waiting for coin - bag %d/%d - speed %d u/s - ping %dms]],settings.mode,bagAmount
,targetLimit,math.round(math.clamp(settings.speed,5,60)),pingMillis()))return end if hiddenUnderMap then restoreCharacter(false)end engage(character,humanoid,root)local coinPosition:Vector3=
nearestPosition::Vector3 if travelPart~=nearest then travelPart=nearest travelBestDistance=math.huge travelProgressAt=now walkLastDistance=math.huge walkStuckSince=now walkCarryUntil=0
walkCarryDirection=nil end local remaining:number if settings.mode=='Walk'then remaining=walkTowards(humanoid,root,coinPosition)else remaining=moveTowards(root,coinPosition+Vector3.new(0,-BELOW_OFFSET
,0),deltaTime)end local reach:number=if settings.mode=='Walk'then 2 else 3.5 local arrived:boolean=remaining<=reach if not arrived then if remaining<travelBestDistance-0.35 then travelBestDistance=
remaining travelProgressAt=now elseif now-travelProgressAt>=TRAVEL_STALL_SECONDS then blacklist[nearest]=now+BLACKLIST_SECONDS travelPart=nil stuckPart=nil return end end if arrived then if stuckPart
~=nearest then stuckPart=nearest stuckSince=now lastTouchPulse=0 if not attempts[nearest]then attempts[nearest]=0 end attempts[nearest]=attempts[nearest]+1 end if now-lastTouchPulse>0.12 then
lastTouchPulse=now local touchPart:Instance?=serverPartOf(nearest)if type(state.fireTouchInterest)=='function'and touchPart then pcall(state.fireTouchInterest,root,touchPart,0)pcall(state.
fireTouchInterest,root,touchPart,1)end end if now-stuckSince>DWELL_SECONDS and stillCollectable(nearest)then retryAt[nearest]=now+RETRY_SECONDS stuckPart=nil if(attempts[nearest]or 0)>=RETRY_LIMIT
then blacklist[nearest]=now+BLACKLIST_SECONDS attempts[nearest]=nil end end else stuckPart=nil end setStatus(string.format([[Auto Farm (%s): %d coins - bag %d/%d - speed %d u/s - ping %dms]],settings.
mode,#scanCache,bagAmount,targetLimit,math.round(math.clamp(settings.speed,5,60)),pingMillis()))end local retryTickAt:number=0 local warnTickAt:number=-math.huge featureConnections.MM2AutoFarmTick=
TaskManager:Connect(function(deltaTime:number):()local now:number=os.clock()if now<retryTickAt then return end local ok:boolean,err:any=pcall(farmTick,deltaTime)if not ok then retryTickAt=now+0.25 if
now>=warnTickAt then warnTickAt=now+5 warn('Auto Farm tick failed; retrying: '..tostring(err))setStatus('Auto Farm: retrying after a temporary error')end end end,nil,'Auto Farm tick')
featureConnections.MM2AutoFarmRestore={Disconnect=function():()if leaveSafeZone then leaveSafeZone(false)end restoreCharacter(true)end}end local AutoFarmFeature:any=createUniversalFeature('Auto Farm',
[[Coin farming in Walk or Below (Hitbox) mode with full-bag actions.]],6,toggleAutoFarm,{categoryName='Utilities',parent=MM2Scroll,registry=mm2Features})state.mm2AutoFarmFeature=AutoFarmFeature if
type(state.addSectionOption)=='function'then state.addSectionOption(AutoFarmFeature,'Overlay Info')end if type(state.addStatOption)=='function'then coinsCollectedStat=state.addStatOption(
AutoFarmFeature,'Coins Collected','0')farmTimeStat=state.addStatOption(AutoFarmFeature,'Time','00:00')end addCycleOption(AutoFarmFeature,'Farm Mode',{'Walk','Below (Hitbox)'},1,function(value:string):
()settings.mode=if value=='Walk'then'Walk'else'Below'end,
[[Walk: your character walks on the ground to each coin (nothing about the body is changed; the slider is WalkSpeed). Below (Hitbox): upright, NoClip, flying under the floor with the root hitbox grown so it still touches the coin from below.]]
)addNumberOption(AutoFarmFeature,'Farm Speed',16,5,60,function(value:number):()settings.speed=value end,
[[Walk mode: WalkSpeed. Below mode: flight speed in studs/second -- if the server kicks you (Invalid position), lower it; raise it slowly to find your fastest safe speed.]],1)addToggleOption(
AutoFarmFeature,'Stay Under When Round starts',true,function(value:boolean):()settings.stayUnderOnStart=value==true end,
[[Hides your character under the map while the round is starting (both farm modes) and starts farming normally as soon as the first coins spawn. Turn it off to wait on the surface at your spawn point.]]
)addToggleOption(AutoFarmFeature,'Safe Zone',true,function(value:boolean):()settings.safeZone=value==true if not settings.safeZone and leaveSafeZone then leaveSafeZone(true)end end,
[[After 50 coins, use a safe platform. Turning this option or Auto Farm off returns the same character to the current map.]])addToggleOption(AutoFarmFeature,'Reset When Full Coin Bag',true,function(
value:boolean):()settings.resetOnFull=value end,'Reset the character as soon as the coin bag fills.')addToggleOption(AutoFarmFeature,'Fling Murder after full bag',false,function(value:boolean):()
settings.flingMurder=value end,[[When the coin bag fills (50): fling the Murderer. Works with or without Reset When Full Coin Bag.]])addToggleOption(AutoFarmFeature,'Fling Sheriff after full bag',
false,function(value:boolean):()settings.flingSheriff=value end,[[When the coin bag fills (50): fling the Sheriff. Works with or without Reset When Full Coin Bag.]])addToggleOption(AutoFarmFeature,
'Auto Grab gun and Shoot',false,function(value:boolean):()settings.grabGunShoot=value end,
[[Only when the coin bag fills (50), only as Innocent and only during an active round (never as Murderer, never in the lobby): try an instant touch for the dropped sheriff gun, otherwise walk/fly to it, equip it, and fire Silent shots at the Murderer without WallCheck. Never during ordinary farming.]]
)addToggleOption(AutoFarmFeature,'Auto Kill All When full coin bag',false,function(value:boolean):()settings.killAllOnFull=value end,
[[As Murderer, when the bag fills: pull everyone in front of you and slash (YARHM Kill All).]])addInformationOption(AutoFarmFeature,
[[The body moves at full Farm Speed to the nearest coin, touches it and moves on to the next one the same frame it is collected (a coin that does not pay is retried later, never waited on). Below (Hitbox) skims under the floor with the root hitbox grown so coins pay from underneath; Walk does it on the ground. If Below ever kicks you for Invalid position, lower Farm Speed. Turning it off restores collisions, hitbox and WalkSpeed; from Safe Zone it returns you to the current map.]]
)activeCleanup=function():()state.mm2AutoFarmFeature=nil toggleAutoFarm(false)end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.
Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Other/AutoUnbox.lua
local Module={Name='MM2 Auto Unbox',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end type BoxEntry={id:string,name:string,category:string,prices
:{[string]:number}}local AUTO_CURRENCY:string='Auto'local REVEAL_ON:string='On'local REVEAL_OFF:string='Off'local DEFAULT_BOX_LABEL:string='Mystery Box #1'local FALLBACK_BOXES:{BoxEntry}={{id=
'MysteryBox1',name='Mystery Box #1',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='MysteryBox2',name='Mystery Box #2',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id=
'KnifeBox1',name='Knife Box #1',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='KnifeBox2',name='Knife Box #2',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='KnifeBox3',
name='Knife Box #3',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='KnifeBox4',name='Knife Box #4',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='KnifeBox5',name=
'Knife Box #5',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='MLG Box',name='Rainbow Box',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='GunBox1',name='Gun Box #1',
category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='GunBox2',name='Gun Box #2',category='MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='GunBox3',name='Gun Box #3',category=
'MysteryBox',prices={Coins=1000,Gems=100,Key=1}},{id='Egg',name='Common Egg',category='Eggs',prices={Coins=1000,Gems=100}},{id='Christmas2024Box',name="Christmas Box '24",category='MysteryBox',prices=
{SnowTokens2024=800,SnowKey2024=1}},{id='Summer2025Box',name="Summer Box '25",category='MysteryBox',prices={BeachBalls2025=800,SummerKey2025=1}},{id='Halloween2025Box',name="Halloween Box '25",
category='MysteryBox',prices={Candies2025=800,HalloweenKey2025=1}}}local FAMILY_CURRENCIES:{[string]:{string}}={Coin={'Coins'},Candy={'Candies2025','Candies2024','Candies2022'},SnowToken={
'SnowTokens2024','SnowTokens2023','SnowTokens2022','SnowTokens2020'},BeachBall={'BeachBalls2025'},Egg={'Eggs2025','RareEggs2025'},Heart={'Hearts2023'}}local BAG_TO_FAMILY:{[string]:string}={Coin=
'Coin',Candy='Candy',SnowToken='SnowToken',BeachBall='BeachBall',Egg='Egg'}local function lower(value:string):string return string.lower(value)end local function contains(value:string,needle:string):
boolean return string.find(value,needle,1,true)~=nil end local function sortedKeys(map:{[string]:any}):{string}local keys:{string}={}for key:string in pairs(map)do table.insert(keys,key)end table.
sort(keys)return keys end function Module.init(runtime:any):any if Module.Initialized then return Module end local core:any=state.mm2Core assert(type(core)=='table',
'MM2 Auto Unbox requires the MM2 core module')Module.Runtime=runtime local settings={boxLabel=DEFAULT_BOX_LABEL,currency=AUTO_CURRENCY,delay=1.5,maxBoxes=10,reveal=REVEAL_ON}local session={opened=0,
failures=0,consecutiveFailures=0,nextAt=0,finished=false}local serverFamily:string?=nil local function safeRequire(instance:Instance?):any?if not instance then return nil end local ok:boolean,result:
any=pcall(function():any return(state::any).requireModule(instance)end)if ok then return result end return nil end local function priceOf(boxId:string,sync:any):{[string]:number}?if type(sync)==
'table'and type(sync.NewShop)=='table'then local newShop:any=sync.NewShop local holders:{any}={}if type(newShop[boxId])=='table'then table.insert(holders,newShop[boxId])end if type(newShop.Weapons)==
'table'and type(newShop.Weapons[boxId])=='table'then table.insert(holders,newShop.Weapons[boxId])end if type(newShop.MysteryBox)=='table'and type(newShop.MysteryBox[boxId])=='table'then table.insert(
holders,newShop.MysteryBox[boxId])end for _,holder:any in ipairs(holders)do if type(holder.Price)=='table'then return holder.Price end end end for _,entry:BoxEntry in ipairs(FALLBACK_BOXES)do if entry
.id==boxId then return entry.prices end end return nil end local function eventBoxModules():{Instance}local found:{Instance}={}local replicated:Instance=game:GetService('ReplicatedStorage')local
shared:Instance?=replicated:FindFirstChild('SharedServices')local service:Instance?=shared and shared:FindFirstChild('EventInfoService1')if not service then return found end local direct:Instance?=
service:FindFirstChild('MysteryBoxes')if direct then for _,child:Instance in ipairs(direct:GetChildren())do if child:IsA('ModuleScript')then table.insert(found,child)end end end local eventList:
Instance?=service:FindFirstChild('EventList')if eventList then for _,event:Instance in ipairs(eventList:GetChildren())do local boxModule:Instance?=event:FindFirstChild('MysteryBox')if boxModule and
boxModule:IsA('ModuleScript')then table.insert(found,boxModule)end end end return found end local function buildCatalog():({BoxEntry},{string})local catalog:{BoxEntry}={}local seen:{[string]:boolean}=
{}local function add(entry:BoxEntry):()if seen[entry.id]then return end seen[entry.id]=true table.insert(catalog,entry)end local replicated:Instance=game:GetService('ReplicatedStorage')local database:
Instance?=replicated:FindFirstChild('Database')local sync:any=safeRequire(database and database:FindFirstChild('Sync'))if type(sync)=='table'then for _,bankName:string in ipairs({'MysteryBox','Eggs'})
do local bank:any=(sync::any)[bankName]if type(bank)=='table'then for id:string,data:any in pairs(bank)do add({id=id,name=(type(data)=='table'and type(data.Name)=='string'and data.Name)or id,category=
bankName=='Eggs'and'Eggs'or'MysteryBox',prices=priceOf(id,sync)or{}})end end end end for _,boxModule:Instance in ipairs(eventBoxModules())do local data:any=safeRequire(boxModule)local display:string?=
nil if type(data)=='table'then if type(data.ItemData)=='table'and type(data.ItemData.Name)=='string'then display=data.ItemData.Name elseif type(data.Name)=='string'and data.Name~=boxModule.Name then
display=data.Name end end add({id=boxModule.Name,name=display or boxModule.Name,category='MysteryBox',prices=priceOf(boxModule.Name,sync)or{}})end for _,entry:BoxEntry in ipairs(FALLBACK_BOXES)do add(
entry)end table.sort(catalog,function(a:BoxEntry,b:BoxEntry):boolean return lower(a.name)<lower(b.name)end)local labelCounts:{[string]:number}={}for _,entry:BoxEntry in ipairs(catalog)do labelCounts[
entry.name]=(labelCounts[entry.name]or 0)+1 end local labels:{string}={}for _,entry:BoxEntry in ipairs(catalog)do if labelCounts[entry.name]>1 then table.insert(labels,entry.name..' ('..entry.id..')')
else table.insert(labels,entry.name)end end return catalog,labels end local catalog:{BoxEntry},boxLabels:{string}=buildCatalog()local labelToEntry:{[string]:BoxEntry}={}for index:number,entry:BoxEntry
in ipairs(catalog)do labelToEntry[boxLabels[index] ]=entry end if not labelToEntry[settings.boxLabel]then settings.boxLabel=boxLabels[1]or''end local function currentEntry():BoxEntry?return
labelToEntry[settings.boxLabel]or catalog[1]end local currencyChoices:{string}={AUTO_CURRENCY}do local seen:{[string]:boolean}={[AUTO_CURRENCY]=true}for _,entry:BoxEntry in ipairs(catalog)do for
currency:string in pairs(entry.prices)do if not seen[currency]then seen[currency]=true table.insert(currencyChoices,currency)end end end end local function familyFromWorkspace():string?local eventName
:string=lower(tostring(workspace:GetAttribute('CurrentEvent')or''))if contains(eventName,'halloween')then return'Candy'elseif contains(eventName,'christmas')then return'SnowToken'elseif contains(
eventName,'summer')then return'BeachBall'elseif contains(eventName,'easter')then return'Egg'elseif contains(eventName,'valentine')then return'Heart'end for attribute:string,family:string in pairs({
isValentines='Heart',isChristmas='SnowToken',isChristmasEnding='SnowToken',IsSummer='BeachBall',IsEaster='Egg'})do if workspace:GetAttribute(attribute)==true then return family end end return nil end
local function activeFamily():string?return serverFamily or familyFromWorkspace()end local function readSyncDatabase():any local replicated:Instance=game:GetService('ReplicatedStorage')local database:
Instance?=replicated:FindFirstChild('Database')return safeRequire(database and database:FindFirstChild('Sync'))end local function readProfileData():any local replicated:Instance=game:GetService(
'ReplicatedStorage')local modules:Instance?=replicated:FindFirstChild('Modules')return safeRequire(modules and modules:FindFirstChild('ProfileData'))end local function getPlayerCurrencyBalance(
currency:string):number?if currency=='Robux'then return nil end local profile:any=readProfileData()if type(profile)~='table'then return nil end local direct:any=profile[currency]if type(direct)==
'number'then return direct end local materials:any=profile.Materials if type(materials)=='table'and type(materials.Owned)=='table'then local matOwned:any=materials.Owned[currency]if type(matOwned)==
'number'then return matOwned end end local weapons:any=profile.Weapons if type(weapons)=='table'and type(weapons.Owned)=='table'then local wepOwned:any=weapons.Owned[currency]if type(wepOwned)==
'number'then return wepOwned end end if type(profile.Coins)=='number'or type(profile.Gems)=='number'or type(materials)=='table'or type(weapons)=='table'then return 0 end return nil end local function
canAffordCurrency(entry:BoxEntry,currency:string):boolean local price:number?=entry.prices and entry.prices[currency]or nil if type(price)~='number'or price<=0 then return true end local balance:
number?=getPlayerCurrencyBalance(currency)if balance==nil then return true end return balance>=price end local function autoCurrency(entry:BoxEntry):string?local prices:{[string]:number}=entry.prices
or{}local orderedCandidates:{string}={}local seenCandidate:{[string]:boolean}={}local function pushCandidate(c:string):()if prices[c]and not seenCandidate[c]then seenCandidate[c]=true table.insert(
orderedCandidates,c)end end local family:string?=activeFamily()if family then for _,currency:string in ipairs(FAMILY_CURRENCIES[family]or{})do pushCandidate(currency)end end for _,currency:string in
ipairs({'Coins','Gems'})do pushCandidate(currency)end for _,currency:string in ipairs(sortedKeys(prices))do if not contains(lower(currency),'key')then pushCandidate(currency)end end for _,currency:
string in ipairs(sortedKeys(prices))do pushCandidate(currency)end for _,currency:string in ipairs(orderedCandidates)do if canAffordCurrency(entry,currency)then return currency end end return
orderedCandidates[1]end local function resolveCurrency(entry:BoxEntry):(string?,boolean)if settings.currency~=AUTO_CURRENCY then if entry.prices and entry.prices[settings.currency]then return settings
.currency,false end return autoCurrency(entry),true end return autoCurrency(entry),false end local function shopRemotes():Instance?local replicated:Instance=game:GetService('ReplicatedStorage')local
remotes:Instance?=replicated:FindFirstChild('Remotes')return remotes and remotes:FindFirstChild('Shop')or nil end local function report(text:string):()local card:any=state.mm2AutoUnboxFeature if card
then card:SetStatus(text)end end local function summary():string local entry:BoxEntry?=currentEntry()local currency:string?=entry and resolveCurrency(entry)or nil local text:string=(entry and entry.
name or'no box')..' \183 pay '..(currency or'nothing')local family:string?=activeFamily()if family then text..=' \183 server coin: '..family end if session.opened>0 or session.failures>0 then text..=
' \183 '..tostring(session.opened)..' opened'if session.failures>0 then text..=' \183 '..tostring(session.failures)..' failed'end end if session.finished then text..=' \183 done'end return text end
local function turnCardOff():()local card:any=state.mm2AutoUnboxFeature if not card then return end card.enabled=false pcall(function():()card.onToggle(false)end)if type(refreshFeatureToggle)==
'function'then pcall(refreshFeatureToggle,card)end end local function stop(reason:string?):()disconnectFeatureConnection('MM2AutoUnbox')if reason then notify(reason)end turnCardOff()end local
openingBusy:boolean=false local revealUntil:number=0 local REVEAL_SECONDS:number=5 local function clearStuckUnboxOverlays():()pcall(function():()local playerGui:Instance?=LocalPlayer and LocalPlayer:
FindFirstChildOfClass('PlayerGui')if not playerGui then return end for _,gui:Instance in ipairs(playerGui:GetChildren())do if gui.Name=='MysteryBoxOpen'or gui.Name=='Unboxing2'then gui:Destroy()elseif
gui.Name=='MainGUI'then for _,subName:string in ipairs({'Lobby','Game'})do local sub:Instance?=gui:FindFirstChild(subName)local proc:Instance?=sub and sub:FindFirstChild('Processing')if proc and proc:
IsA('GuiObject')then(proc::GuiObject).Visible=false end end end end end)end local function isValidReward(reward:any,category:string):boolean if type(reward)~='string'or reward==''or reward=='false'or
reward=='nil'then return false end local sync:any=readSyncDatabase()if type(sync)=='table'then if category=='Eggs'then local pets:any=sync.Pets if type(pets)=='table'and next(pets)~=nil then return
pets[reward]~=nil end else local weapons:any=sync.Weapons local items:any=sync.Item if type(weapons)=='table'and next(weapons)~=nil then if weapons[reward]~=nil then return true end if type(items)==
'table'and items[reward]~=nil then return true end return false end end end return true end local function canPlayBoxAnimationSafely(boxId:string,reward:string):boolean local sync:any=
readSyncDatabase()if type(sync)~='table'then return true end local weapons:any=sync.Weapons if type(weapons)~='table'or next(weapons)==nil then return true end if type(weapons[reward])~='table'then
return false end local mysteryBoxes:any=sync.MysteryBox if type(mysteryBoxes)=='table'and next(mysteryBoxes)~=nil then local boxDef:any=mysteryBoxes[boxId]if type(boxDef)~='table'or type(boxDef.
Contents)~='table'then return false end end return true end local function reveal(boxId:string,category:string,reward:any):()local remotes:Instance?=shopRemotes()if not remotes then return end if
settings.reveal==REVEAL_ON then if category=='Eggs'then revealUntil=os.clock()+REVEAL_SECONDS local eggController:Instance?=remotes:FindFirstChild('EggController')if eggController then pcall(function(
):()(eggController::any):Fire(boxId,reward)end)end return elseif canPlayBoxAnimationSafely(boxId,tostring(reward))then revealUntil=os.clock()+REVEAL_SECONDS local boxController:Instance?=remotes:
FindFirstChild('BoxController')if boxController then pcall(function():()(boxController::any):Fire({{MysteryBoxId=boxId,RewardedItemId=reward}})end)end return end end local crateComplete:Instance?=
remotes:FindFirstChild('CrateComplete')if crateComplete then pcall(function():()(crateComplete::any):FireServer(reward)end)end end local function openOnce():boolean if openingBusy then return false
end openingBusy=true local entry:BoxEntry?=currentEntry()if not entry then session.finished=true openingBusy=false stop('Auto Unbox: no box catalogue found.')return false end local currency:string?,
forced:boolean=resolveCurrency(entry)if not currency then session.finished=true openingBusy=false stop('Auto Unbox: '..entry.name..' has no known price currency.')return false end local requiredPrice:
number?=entry.prices and entry.prices[currency]or nil local currentBalance:number?=getPlayerCurrencyBalance(currency)if type(requiredPrice)=='number'and requiredPrice>0 and type(currentBalance)==
'number'and currentBalance<requiredPrice then clearStuckUnboxOverlays()openingBusy=false report(summary()..' \183 waiting for '..tostring(currency)..' ('..tostring(currentBalance)..'/'..tostring(
requiredPrice)..')')return false end local remotes:Instance?=shopRemotes()local openCrate:Instance?=remotes and remotes:FindFirstChild('OpenCrate')if not(openCrate and openCrate:IsA('RemoteFunction'))
then openingBusy=false report(summary()..' \183 waiting for shop')return false end local note:string=forced and(' \183 '..settings.currency..' rejected, using '..currency)or''local ok:boolean,reward:
any=pcall(function():any return(openCrate::any):InvokeServer(entry.id,entry.category,currency)end)if ok and isValidReward(reward,entry.category)then reveal(entry.id,entry.category,reward)session.
opened+=1 session.consecutiveFailures=0 report(summary()..note)openingBusy=false return true end clearStuckUnboxOverlays()session.failures+=1 session.consecutiveFailures+=1 report(summary()..note..
' \183 waiting for '..tostring(currency))openingBusy=false return false end local function tick(_deltaTime:number):()local now:number=os.clock()if now<session.nextAt or now<revealUntil or openingBusy
then return end local opened:boolean=openOnce()local settle:number=(opened and settings.reveal==REVEAL_ON)and REVEAL_SECONDS or 0 session.nextAt=os.clock()+settings.delay+settle end local
AutoUnboxFeature:any=createUniversalFeature('Auto Unbox',
[[Opens any mystery box or egg of the game through its own OpenCrate remote, paying with the currency you choose or the coin the server is giving.]],4,function(enabled:boolean):()
disconnectFeatureConnection('MM2AutoUnbox')if os.clock()>=revealUntil then clearStuckUnboxOverlays()end if not enabled then return end session.opened=0 session.failures=0 session.consecutiveFailures=0
session.nextAt=math.max(os.clock(),revealUntil)session.finished=false report(summary())featureConnections.MM2AutoUnbox=TaskManager:Connect(tick,nil,'MM2 Auto Unbox')end,{categoryName='Utilities',
parent=MM2Scroll,registry=mm2Features})state.mm2AutoUnboxFeature=AutoUnboxFeature local startedRemote:Instance?=core.mm2GameplayRemotes and core.mm2GameplayRemotes:FindFirstChild('CoinsStarted')if
startedRemote and startedRemote:IsA('RemoteEvent')then featureConnections.MM2AutoUnboxCoinsStarted=(startedRemote::any).OnClientEvent:Connect(function(bagTable:any):()if type(bagTable)=='table'then
for bag:any in pairs(bagTable)do local family:string?=BAG_TO_FAMILY[tostring(bag)]if family then serverFamily=family end end end if state.mm2AutoUnboxFeature then report(summary())end end)end local
defaultBoxIndex:number=table.find(boxLabels,DEFAULT_BOX_LABEL)or 1 addCycleOption(AutoUnboxFeature,'Box to open',boxLabels,defaultBoxIndex,function(value:string):()settings.boxLabel=value end,
'Every box and egg of the game, including the event boxes.')addCycleOption(AutoUnboxFeature,'Pay with',currencyChoices,1,function(value:string):()settings.currency=value end,
[[Auto pays with the currency the server's current coin fills.]])addCycleOption(AutoUnboxFeature,'Reveal animation',{REVEAL_ON,REVEAL_OFF},1,function(value:string):()settings.reveal=value end,
[[On replays the game's own opening animation and crate report.]])addNumberOption(AutoUnboxFeature,'Delay between boxes',settings.delay,0.8,5,function(value:number):()settings.delay=value end)
addNumberOption(AutoUnboxFeature,'Max boxes per run',settings.maxBoxes,1,50,function(value:number):()settings.maxBoxes=value end)addActionOption(AutoUnboxFeature,'Report boxes and coin',function():()
local entry:BoxEntry?=currentEntry()local prices:{string}={}if entry then for _,currency:string in ipairs(sortedKeys(entry.prices))do table.insert(prices,currency..'='..tostring(entry.prices[currency]
))end end notify('Boxes found: '..tostring(#catalog)..' \183 chosen: '..(entry and entry.name or'none')..' ('..(entry and entry.id or'?')..')'..' \183 price: '..(#prices>0 and table.concat(prices,', '
)or'unknown')..' \183 server coin: '..tostring(activeFamily()or'unknown'))end)addInformationOption(AutoUnboxFeature,
[[Uses the shop's own OpenCrate remote with the box id and currency the game expects, then the game's own reveal. Robux prompts are never pressed.]])activeCleanup=function():()state.
mm2AutoUnboxFeature=nil disconnectFeatureConnection('MM2AutoUnbox')disconnectFeatureConnection('MM2AutoUnboxCoinsStarted')end Module.Events=featureConnections Module.Initialized=true return Module end
function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Other/AutoVote.lua
local Module={Name='MM2 Auto Vote',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Auto Vote requires the MM2 core module')Module.Runtime=runtime local TICK_INTERVAL:number=0.1 local
MAP_REFRESH_INTERVAL:number=0.5 local VOTE_HOLD_BEFORE_RESET:number=0.22 local RESET_RETRY_INTERVAL:number=0.25 local MAX_LOBBY_PAD_DISTANCE:number=250 local DEFAULT_MAP_CHOICES:{string}={'House 2',
'Hospital 3','Police Station'}local mapChoices:{string}={DEFAULT_MAP_CHOICES[1],DEFAULT_MAP_CHOICES[2],DEFAULT_MAP_CHOICES[3]}local selectedSlot:number=1 local selectedMapName:string=mapChoices[1]
local selectMapButton:TextButton?=nil local selectMapOptionKey:string='MM2.AutoVote.SelectMap'local AutoVoteFeature:any=nil local cachedLobby:Instance?=nil local cachedPadModels:{[number]:Instance?}={
}local cachedVotePadsFolder:Instance?=nil local cachedVoteIconsFolder:Instance?=nil local lastLobbyCheckAt:number=-math.huge local lastMapRefreshAt:number=-math.huge local savedDetectorCFrames:{[
BasePart]:CFrame}={}local roundBlockingVote:boolean=false local lastBillboardVotingState:boolean=false local function trim(s:string):string return(string.gsub(s,'^%s*(.-)%s*$','%1'))end local function
isPlaceholderMapLabel(title:string?):boolean if type(title)~='string'then return true end local clean:string=string.upper(trim(title))return clean==''or clean=='MAP NAME'or clean=='LOADING'or clean==
'NONE'or clean=='MAP 1'or clean=='MAP 2'or clean=='MAP 3'end local function isValidMapTitle(title:string?):boolean return not isPlaceholderMapLabel(title)end local function isLobbyContainer(candidate:
Instance?):boolean if not candidate or not candidate.Parent then return false end return candidate:FindFirstChild('VotePad1')~=nil or candidate:FindFirstChild('VotePads')~=nil or candidate:
FindFirstChild('VoteIcons')~=nil end local function findLobbyModel():Instance?local now:number=os.clock()if cachedLobby and cachedLobby:IsDescendantOf(workspace)and now-lastLobbyCheckAt<1 then return
cachedLobby end lastLobbyCheckAt=now for _,name:string in ipairs({'RegularLobby','Lobby','MainLobby','HalloweenLobby'})do local candidate:Instance?=workspace:FindFirstChild(name)if isLobbyContainer(
candidate)then cachedLobby=candidate return candidate end end local okChildren:boolean,children:any=pcall(function():any return workspace:GetChildren()end)if okChildren and type(children)=='table'then
for _,child:Instance in ipairs(children)do if(child:IsA('Model')or child:IsA('Folder'))and isLobbyContainer(child)then cachedLobby=child return child end end end local okTagged:boolean,tagged:any=
pcall(function():any return game:GetService('CollectionService'):GetTagged('VotePad')end)if okTagged and type(tagged)=='table'then for _,inst:Instance in ipairs(tagged)do if inst and inst:
IsDescendantOf(workspace)then local slotNum:number?=tonumber(string.match(inst.Name,'^VotePad(%d)$'))if slotNum then cachedPadModels[slotNum]=inst end if inst.Parent and inst.Parent~=workspace then
cachedLobby=inst.Parent end end end end if cachedLobby and cachedLobby:IsDescendantOf(workspace)then return cachedLobby end end local okDeep:boolean,deepPad:any=pcall(function():any return workspace:
FindFirstChild('VotePads',true)or workspace:FindFirstChild('VotePad1',true)or workspace:FindFirstChild('VoteIcons',true)end)if okDeep and deepPad and(deepPad::Instance):IsDescendantOf(workspace)then
local parent:Instance?=(deepPad::Instance).Parent if parent and parent.Name=='VoteIcons'and parent.Parent then parent=parent.Parent end cachedLobby=parent return cachedLobby end cachedLobby=nil return
nil end local function getPadModel(lobby:Instance?,slot:number):Instance?local cached:Instance?=cachedPadModels[slot]if cached and cached:IsDescendantOf(workspace)and not(cached.Parent and cached.
Parent.Name=='VoteIcons')then return cached end local padName:string='VotePad'..tostring(slot)if lobby then local direct:Instance?=lobby:FindFirstChild(padName)if direct and not(direct.Parent and
direct.Parent.Name=='VoteIcons')then cachedPadModels[slot]=direct return direct end local okDeep:boolean,deep:any=pcall(function():any return lobby:FindFirstChild(padName,true)end)if okDeep and deep
and not((deep::Instance).Parent and(deep::Instance).Parent.Name=='VoteIcons')then cachedPadModels[slot]=deep::Instance return deep::Instance end end return nil end local function getVoteIconsFolder(
lobby:Instance?):Instance?if cachedVoteIconsFolder and cachedVoteIconsFolder:IsDescendantOf(workspace)then return cachedVoteIconsFolder end if lobby then local direct:Instance?=lobby:FindFirstChild(
'VoteIcons')or lobby:FindFirstChild('VoteIcons',true)if direct then cachedVoteIconsFolder=direct return direct end end return nil end local function getVotePadsFolder(lobby:Instance?):Instance?if
cachedVotePadsFolder and cachedVotePadsFolder:IsDescendantOf(workspace)then return cachedVotePadsFolder end if lobby then local direct:Instance?=lobby:FindFirstChild('VotePads')or lobby:
FindFirstChild('VotePads',true)if direct then cachedVotePadsFolder=direct return direct end end return nil end local function readSlotMapName(lobby:Instance?,slot:number):string?if not lobby then
return nil end local padModel:Instance?=getPadModel(lobby,slot)if padModel then local voteInfoGui:Instance?=padModel:FindFirstChild('VoteInfoGui')local container:Instance?=voteInfoGui and voteInfoGui:
FindFirstChild('Container')local mapLabel:Instance?=(container and container:FindFirstChild('MapName'))or padModel:FindFirstChild('MapName',true)if mapLabel and mapLabel:IsA('TextLabel')then local
text:string=(mapLabel::TextLabel).Text if isValidMapTitle(text)then return trim(text)end end end local voteIcons:Instance?=getVoteIconsFolder(lobby)local iconPart:Instance?=voteIcons and voteIcons:
FindFirstChild('VotePad'..tostring(slot))if iconPart then local mapLabel:Instance?=iconPart:FindFirstChild('MapName',true)if mapLabel and mapLabel:IsA('TextLabel')then local text:string=(mapLabel::
TextLabel).Text if isValidMapTitle(text)then return trim(text)end end local mapObj:Instance?=iconPart:FindFirstChild('Map')if mapObj and mapObj:IsA('ObjectValue')then local val:Instance?=(mapObj::
ObjectValue).Value if val and isValidMapTitle(val.Name)then return trim(val.Name)end end end return nil end local function readSlotVoteCount(lobby:Instance?,slot:number):number?if not lobby then
return nil end local voteIcons:Instance?=getVoteIconsFolder(lobby)local iconPart:Instance?=voteIcons and voteIcons:FindFirstChild('VotePad'..tostring(slot))if iconPart then local votesVal:Instance?=
iconPart:FindFirstChild('Votes')if votesVal and votesVal:IsA('IntValue')then return(votesVal::IntValue).Value end end local padModel:Instance?=getPadModel(lobby,slot)if padModel then local voteInfoGui
:Instance?=padModel:FindFirstChild('VoteInfoGui')local container:Instance?=voteInfoGui and voteInfoGui:FindFirstChild('Container')local votesLabel:Instance?=(container and container:FindFirstChild(
'Votes'))or padModel:FindFirstChild('Votes',true)if votesLabel and votesLabel:IsA('TextLabel')then local numStr:string?=string.match((votesLabel::TextLabel).Text,'(%d+)')if numStr then return
tonumber(numStr)end end end return nil end local function syncSelectMapControl():()local control:any=state.valueControls and state.valueControls[selectMapOptionKey]if type(control)=='table'and type(
control.setIndex)=='function'then pcall(control.setIndex,selectedSlot)elseif selectMapButton then pcall(function():()selectMapButton.Text=mapChoices[selectedSlot]end)end end local function
refreshMapChoices(force:boolean?):()local now:number=os.clock()if not force and now-lastMapRefreshAt<MAP_REFRESH_INTERVAL then return end lastMapRefreshAt=now local lobby:Instance?=findLobbyModel()
local changed:boolean=false for slot:number=1,3 do local title:string?=readSlotMapName(lobby,slot)local current:string?=mapChoices[slot]local nextLabel:string=title or(isValidMapTitle(current)and(
current::string)or DEFAULT_MAP_CHOICES[slot])if mapChoices[slot]~=nextLabel then mapChoices[slot]=nextLabel changed=true end end if isValidMapTitle(selectedMapName)then local matchedSlot:number?=nil
local targetLower:string=string.lower(selectedMapName)for slot:number=1,3 do if string.lower(mapChoices[slot])==targetLower then matchedSlot=slot break end end if matchedSlot and selectedSlot~=
matchedSlot then selectedSlot=matchedSlot changed=true end end selectedSlot=math.clamp(selectedSlot,1,3)selectedMapName=mapChoices[selectedSlot]if changed or force then syncSelectMapControl()end end
local function getSlotDetectorParts(lobby:Instance?,slot:number):{BasePart}local parts:{BasePart}={}if not lobby then return parts end local padModel:Instance?=getPadModel(lobby,slot)if padModel then
local padPart:Instance?=padModel:FindFirstChild('Pad')or padModel:FindFirstChild('Pad',true)if padPart and padPart:IsA('BasePart')then parts[#parts+1]=padPart::BasePart end end local votePads:Instance
?=getVotePadsFolder(lobby)if votePads then local detector:Instance?=votePads:FindFirstChild('Detector'..tostring(slot))or votePads:FindFirstChild('Detector'..tostring(slot),true)if detector and
detector:IsA('BasePart')then parts[#parts+1]=detector::BasePart end end return parts end local function rememberDetectorPart(part:BasePart):()if not savedDetectorCFrames[part]then savedDetectorCFrames
[part]=part.CFrame end end local function restoreAllDetectorParts():()for part,cf in pairs(savedDetectorCFrames)do if part and part.Parent then pcall(function():()part.CFrame=cf end)end end table.
clear(savedDetectorCFrames)end local function areVotingBillboardsActive(lobby:Instance?):boolean if not lobby then return false end for slot:number=1,3 do local padModel:Instance?=getPadModel(lobby,
slot)if padModel then local voteGui:Instance?=padModel:FindFirstChild('VoteInfoGui')if voteGui and voteGui:IsA('SurfaceGui')and(voteGui::SurfaceGui).Enabled==true then return true end local mapGui:
Instance?=padModel:FindFirstChild('MapInfoGui')if mapGui and mapGui:IsA('SurfaceGui')and(mapGui::SurfaceGui).Enabled==true then return true end end end local voteIcons:Instance?=getVoteIconsFolder(lobby)
local iconPart:Instance?=voteIcons and voteIcons:FindFirstChild('VotePad'..tostring(slot))if iconPart then local votingFlag:Instance?=iconPart:FindFirstChild('Voting')if votingFlag and votingFlag:IsA(
'BoolValue')and(votingFlag::BoolValue).Value==true then return true end local surface:Instance?=iconPart:FindFirstChild('Surface')if surface and surface:IsA('SurfaceGui')and(surface::SurfaceGui).
Enabled==true then return true end local mapInfo:Instance?=iconPart:FindFirstChild('MapInfo')local infoSurface:Instance?=mapInfo and mapInfo:FindFirstChild('Surface')if infoSurface and infoSurface:
```

```
IsA('SurfaceGui')and(infoSurface::SurfaceGui).Enabled==true then return true end end end return false end local function isVotingIntermissionActive(root:BasePart?,detector:BasePart?):boolean local
lobby:Instance?=findLobbyModel()if not lobby then return false end local billboardsOn:boolean=areVotingBillboardsActive(lobby)if billboardsOn and not lastBillboardVotingState then roundBlockingVote=
false elseif not billboardsOn and lastBillboardVotingState then roundBlockingVote=true end lastBillboardVotingState=billboardsOn if roundBlockingVote then return false end local timerPart:Instance?=
workspace:FindFirstChild('RoundTimerPart')local timerIndicatesLobby:boolean=false if timerPart then local okTime:boolean,roundTime:any=pcall(function():any return timerPart:GetAttribute('Time')end)if
okTime and type(roundTime)=='number'then if roundTime>0 then return false end timerIndicatesLobby=true end end local isNearPad:boolean=false if root and detector then local okDist:boolean,dist:any=
pcall(function():any return(root.Position-detector.Position).Magnitude end)if okDist and type(dist)=='number'then if dist>MAX_LOBBY_PAD_DISTANCE then return false end isNearPad=true end end if not
isNearPad and type(core.isRoundLifecycleActive)=='function'and core.isRoundLifecycleActive()==true then return false end if billboardsOn then return true end if isNearPad or timerIndicatesLobby then
for slot:number=1,3 do if readSlotMapName(lobby,slot)~=nil then return true end end end return false end local function resolveTargetSlot(lobby:Instance?):number refreshMapChoices(false)if lobby and
isValidMapTitle(selectedMapName)then local targetLower:string=string.lower(selectedMapName)for slot:number=1,3 do local nameOnBoard:string?=readSlotMapName(lobby,slot)if nameOnBoard and string.lower(
nameOnBoard)==targetLower then selectedSlot=slot return slot end end end return math.clamp(selectedSlot,1,3)end local function pulseVotePad(root:BasePart,detectors:{BasePart}):()local primaryPad:
BasePart?=detectors[1]if primaryPad then local saved:CFrame?=savedDetectorCFrames[primaryPad]local padPos:Vector3=if saved then saved.Position else primaryPad.Position pcall(function():()local
currentCF:CFrame=root.CFrame local rotOnly:CFrame=currentCF-currentCF.Position root.CFrame=CFrame.new(padPos)*rotOnly root.AssemblyLinearVelocity=Vector3.zero end)end local fireTouch:any=state.
fireTouchInterest local rootCF:CFrame=root.CFrame for _,part:BasePart in ipairs(detectors)do rememberDetectorPart(part)pcall(function():()part.CanCollide=false part.CFrame=rootCF end)if type(fireTouch
)=='function'then pcall(fireTouch,root,part,0)pcall(fireTouch,root,part,1)end end end local function resetLocalCharacter(character:Model,humanoid:Humanoid):()pcall(function():()humanoid:ChangeState(
Enum.HumanoidStateType.Dead)end)pcall(function():()humanoid.Health=0 end)pcall(function():()if humanoid.Health>0 then character:BreakJoints()end end)end local lastStatus:string?=nil local function
setStatus(text:string?):()if text==lastStatus then return end lastStatus=text if AutoVoteFeature then AutoVoteFeature:SetStatus(text)end end local function connectGameplayRemotes():()local okRs:
boolean,rs:any=pcall(function():any return game:GetService('ReplicatedStorage')end)if not okRs or not rs then return end local remotes:Instance?=rs:FindFirstChild('Remotes')local gameplay:Instance?=
remotes and remotes:FindFirstChild('Gameplay')if not gameplay then return end for _,stopName:string in ipairs({'LoadingMap','Fade','RoleSelect','ShowRoleSelect','ShowRoleSelectNew','RoundStart',
'CoinsStarted'})do local rem:Instance?=gameplay:FindFirstChild(stopName)if rem and rem:IsA('RemoteEvent')then local key:string='MM2AutoVoteStop_'..stopName disconnectFeatureConnection(key)
featureConnections[key]=(rem::RemoteEvent).OnClientEvent:Connect(function():()roundBlockingVote=true restoreAllDetectorParts()end)end end for _,endName:string in ipairs({'RoundEndFade','VictoryScreen'
,'GameOver'})do local rem:Instance?=gameplay:FindFirstChild(endName)if rem and rem:IsA('RemoteEvent')then local key:string='MM2AutoVoteEnd_'..endName disconnectFeatureConnection(key)featureConnections
[key]=(rem::RemoteEvent).OnClientEvent:Connect(function():()roundBlockingVote=false refreshMapChoices(true)end)end end end local function disconnectAllAutoVoteConnections():()
disconnectFeatureConnection('MM2AutoVote')for key in pairs(featureConnections)do if string.sub(key,1,11)=='MM2AutoVote'and key~='MM2AutoVoteMapSync'then disconnectFeatureConnection(key)end end end
local function toggleAutoVote(enabled:boolean):()disconnectAllAutoVoteConnections()restoreAllDetectorParts()if not enabled then setStatus(nil)return end roundBlockingVote=false
lastBillboardVotingState=false refreshMapChoices(true)connectGameplayRemotes()local lastTickAt:number=-math.huge local currentChar:Model?=nil local firstTouchAt:number=0 local lastResetAttemptAt:
number=0 local voteCountBeforeTouch:number?=nil local function autoVoteTick():()local now:number=os.clock()if now-lastTickAt<TICK_INTERVAL then return end lastTickAt=now refreshMapChoices(false)local
lobby:Instance?=findLobbyModel()local slot:number=resolveTargetSlot(lobby)local targetMapLabel:string=mapChoices[slot]or DEFAULT_MAP_CHOICES[slot]local detectors:{BasePart}=getSlotDetectorParts(lobby,
slot)local primaryDetector:BasePart?=detectors[1]local character:Model?=LocalPlayer and LocalPlayer.Character local root:BasePart?=character and(character:FindFirstChild('HumanoidRootPart')::BasePart?
)local humanoid:Humanoid?=character and(character:FindFirstChildOfClass('Humanoid')::Humanoid?)if not isVotingIntermissionActive(root,primaryDetector)then restoreAllDetectorParts()firstTouchAt=0
lastResetAttemptAt=0 setStatus('Waiting for vote \183 '..tostring(targetMapLabel))return end if#detectors==0 then setStatus('No vote pad')return end if not character or not root or not humanoid or
humanoid.Health<=0 then restoreAllDetectorParts()return end if character~=currentChar then currentChar=character firstTouchAt=0 lastResetAttemptAt=0 voteCountBeforeTouch=readSlotVoteCount(lobby,slot)
end pulseVotePad(root,detectors)if firstTouchAt==0 then firstTouchAt=now end local currentVotes:number?=readSlotVoteCount(lobby,slot)local voteIncremented:boolean=(currentVotes~=nil and
voteCountBeforeTouch~=nil and currentVotes>voteCountBeforeTouch)setStatus('Voting '..targetMapLabel..(currentVotes~=nil and(' ('..tostring(currentVotes)..')')or''))if(voteIncremented or(now-
firstTouchAt>=VOTE_HOLD_BEFORE_RESET))and(now-lastResetAttemptAt>=RESET_RETRY_INTERVAL)then restoreAllDetectorParts()if isVotingIntermissionActive(root,primaryDetector)then lastResetAttemptAt=now
resetLocalCharacter(character,humanoid)end end end featureConnections.MM2AutoVote=TaskManager:Connect(autoVoteTick,nil,'MM2 Auto Vote')autoVoteTick()end pcall(function():()if type(configData)=='table'
and type(configData.values)=='table'then local saved:any=configData.values[selectMapOptionKey]if isPlaceholderMapLabel(saved)then configData.values[selectMapOptionKey]=nil end end end)pcall(
refreshMapChoices,true)AutoVoteFeature=createUniversalFeature('Auto Vote',
[[Detects the 3 maps on the lobby voting billboard, votes on the wooden floor pad for your selected map, and resets repeatedly to stack votes until voting intermission ends.]],7,toggleAutoVote,{
categoryName='Utilities',parent=MM2Scroll,registry=mm2Features})state.mm2AutoVoteFeature=AutoVoteFeature selectMapButton=addCycleOption(AutoVoteFeature,'Select Map',mapChoices,1,function(value:string)
:()refreshMapChoices(false)selectedMapName=tostring(value)for slot:number=1,3 do if mapChoices[slot]==selectedMapName then selectedSlot=slot break end end if AutoVoteFeature and AutoVoteFeature.
enabled then setStatus('Voting '..tostring(mapChoices[selectedSlot]or selectedMapName))end end,'Select which of the 3 lobby billboard maps to vote for.')pcall(function():()local control:any=state.
valueControls and state.valueControls[selectMapOptionKey]if type(control)=='table'then control.onBeforeOpen=function():()refreshMapChoices(true)end end end)local lastBgSyncAt:number=-math.huge
disconnectFeatureConnection('MM2AutoVoteMapSync')featureConnections.MM2AutoVoteMapSync=TaskManager:Connect(function():()local now:number=os.clock()if now-lastBgSyncAt<1 then return end lastBgSyncAt=
now refreshMapChoices(false)end,nil,'Auto Vote sync')activeCleanup=function():()disconnectFeatureConnection('MM2AutoVoteMapSync')toggleAutoVote(false)end Module.Events=featureConnections Module.
Initialized=true return Module end function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}
Module.Runtime=nil end return Module
-- src/games/MM2/Other/AutoWin.lua
local Module={Name='MM2 Auto Win',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Auto Win requires the MM2 core module')local prediction:any=state.mm2Prediction Module.Runtime=runtime local
findDroppedGun:any=core.findDroppedGun local findMM2Map:any=core.findMM2Map local findMurderer:any=core.findMurderer local findSheriff:any=core.findSheriff local getGunOriginCFrame:any=core.
getGunOriginCFrame local getPlayerRole:any=core.getPlayerRole local getPlayerWeapon:any=core.getPlayerWeapon local isPlayerAlive:any=core.isPlayerAlive local isProtectedTarget:any=core.
isProtectedTarget local isRoundLifecycleActive:any=core.isRoundLifecycleActive local ABOVE_MAP_OFFSET:number=120 local MAP_FLING_WARMUP_SECONDS:number=5 local SHERIFF_READY_DELAY_SECONDS:number=2
local SHERIFF_FLING_COOLDOWN_SECONDS:number=1.2 local settings={instaKillMurder=true,instaKillSheriff=true,flingSheriffGetGun=false,flingMurderIfDead=false}local autoWinActive:boolean=false local
AutoWinFeature:any=nil local lastStatus:string?=nil local sheriffWins:number=0 local murderWins:number=0 local innocentWins:number=0 local sheriffWinsStat:any=nil local murderWinsStat:any=nil local
innocentWinsStat:any=nil local wasOnMap:boolean=false local mapEnteredAt:number=0 local sheriffFirstSeenAt:number=0 local lastSheriffFlingAt:number=0 local participatedInRound:boolean=false local
roundMyRole:string?=nil local roundSawMurderer:boolean=false local roundCountedWin:boolean=false local flungMurderThisDeath:boolean=false local flingBusy:boolean=false local safeAbovePosition:Vector3?
=nil local savedCollide:{[Instance]:boolean}={}local savedCustomProps:{[BasePart]:PhysicalProperties?}={}local victimSnapshots:{[BasePart]:{cframe:CFrame,anchored:boolean,canCollide:boolean}}={}local
toolConnections:{RBXScriptConnection}={}local flingAngle:number=0 local function updateWinOverlay():()if sheriffWinsStat and type(sheriffWinsStat.set)=='function'then sheriffWinsStat.set(tostring(
sheriffWins))end if murderWinsStat and type(murderWinsStat.set)=='function'then murderWinsStat.set(tostring(murderWins))end if innocentWinsStat and type(innocentWinsStat.set)=='function'then
innocentWinsStat.set(tostring(innocentWins))end end local function awardRoundWin(roleCategory:string?):()if roundCountedWin then return end local cat:string=roleCategory or roundMyRole or'Innocent'
roundCountedWin=true if cat=='Murderer'then murderWins+=1 elseif cat=='Sheriff'or cat=='Hero'then sheriffWins+=1 else innocentWins+=1 end updateWinOverlay()end local function setStatus(text:string?):(
)if text==lastStatus then return end lastStatus=text if AutoWinFeature then AutoWinFeature:SetStatus(text)end end local function yawOf(frame:CFrame):number local look:Vector3=frame.LookVector return
math.atan2(-look.X,-look.Z)end local function characterParts():(Model?,Humanoid?,BasePart?)local character:Model?=LocalPlayer and LocalPlayer.Character if not character then return nil,nil,nil end
local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?if not humanoid or not root then return nil,nil,nil end
return character,humanoid,root end local function findLobbyReferencePart():BasePart?for _,child:Instance in ipairs(workspace:GetChildren())do local lowerName:string=string.lower(tostring(child.Name))
local isLobby:boolean=string.find(lowerName,'lobby',1,true)~=nil or child:FindFirstChild('VotePads',true)~=nil or child:FindFirstChild('VotePad1',true)~=nil or child:FindFirstChild('VoteIcons',true)~=
nil if isLobby then local votePads:Instance?=child:FindFirstChild('VotePads',true)if votePads then local padPart:BasePart?=votePads:FindFirstChildWhichIsA('BasePart',true)if padPart then return
padPart end end local spawns:Instance?=child:FindFirstChild('Spawns',true)if spawns then local spawnPart:BasePart?=spawns:FindFirstChildWhichIsA('BasePart',true)if spawnPart then return spawnPart end
end local anyPart:BasePart?=child:FindFirstChildWhichIsA('BasePart',true)if anyPart then return anyPart end end end return nil end local function isCharacterOnMap(root:BasePart):(boolean,Vector3?)if
type(findMM2Map)~='function'then return false,nil end local map:Instance?=findMM2Map()if not map then return false,nil end local rootPos:Vector3=root.Position local lobbyPart:BasePart?=
findLobbyReferencePart()if lobbyPart and(rootPos-lobbyPart.Position).Magnitude<=350 then return false,nil end local refPart:BasePart?=nil local spawns:Instance?=(map::Instance):FindFirstChild('Spawns'
)if spawns then refPart=spawns:FindFirstChildWhichIsA('BasePart',true)end if not refPart then local coinContainer:Instance?=(map::Instance):FindFirstChild('CoinContainer')if coinContainer then refPart
=coinContainer:FindFirstChildWhichIsA('BasePart',true)end end if not refPart then refPart=(map::Instance):FindFirstChildWhichIsA('BasePart',true)end if refPart then if(rootPos-refPart.Position).
Magnitude>1500 then return false,nil end return true,refPart.Position end return true,rootPos end local function restoreVictims():()for part:BasePart,snap in pairs(victimSnapshots)do if part and part.
Parent then pcall(function():()part.Anchored=snap.anchored part.CanCollide=snap.canCollide part.CFrame=snap.cframe end)end end table.clear(victimSnapshots)end local function restoreLocalCharacter():()
restoreVictims()if services and services.activity then pcall(function():()services.activity.set('fling',false)end)end for part:Instance,wasCollide:boolean in pairs(savedCollide)do if part and part.
Parent then pcall(function():()(part::BasePart).CanCollide=wasCollide end)end end table.clear(savedCollide)for part:BasePart,oldProps:PhysicalProperties?in pairs(savedCustomProps)do if part and part.
Parent then pcall(function():()part.CustomPhysicalProperties=oldProps end)end end table.clear(savedCustomProps)local character:Model?,humanoid:Humanoid?,root:BasePart?=characterParts()if humanoid then
pcall(function():()humanoid.PlatformStand=false humanoid.AutoRotate=true humanoid:SetStateEnabled(Enum.HumanoidStateType.FallingDown,true)humanoid:SetStateEnabled(Enum.HumanoidStateType.Seated,true)
humanoid:ChangeState(Enum.HumanoidStateType.Running)end)end if root then pcall(function():()root.AssemblyLinearVelocity=Vector3.zero root.AssemblyAngularVelocity=Vector3.zero if safeAbovePosition then
local groundPos:Vector3=safeAbovePosition-Vector3.new(0,ABOVE_MAP_OFFSET-4,0)local targetCF:CFrame=CFrame.new(groundPos)*CFrame.Angles(0,yawOf(root.CFrame),0)if character then pcall(function():()
character:PivotTo(targetCF)end)end root.CFrame=targetCF end end)end end local function disableCharacterCollisions(character:Model):()for _,object:Instance in ipairs(character:GetDescendants())do if
object:IsA('BasePart')then if savedCollide[object]==nil then savedCollide[object]=(object::BasePart).CanCollide end pcall(function():()(object::BasePart).CanCollide=false end)end end end
local function enableCharacterFlingCollisions(character:Model):()local heavyProps:PhysicalProperties=PhysicalProperties.new(100,0.3,0.5,1,1)for _,object:Instance in ipairs(character:GetChildren())do
if object:IsA('BasePart')then local part:BasePart=object::BasePart if savedCollide[part]==nil then savedCollide[part]=part.CanCollide end if savedCustomProps[part]==nil then pcall(function():()
savedCustomProps[part]=part.CustomPhysicalProperties end)end pcall(function():()part.CanCollide=true part.CustomPhysicalProperties=heavyProps end)end end end local function holdAboveMap(character:
Model,humanoid:Humanoid,root:BasePart):()if services and services.activity then pcall(function():()services.activity.set('fling',false)end)end disableCharacterCollisions(character)local holdPos:
Vector3=safeAbovePosition or(root.Position+Vector3.new(0,ABOVE_MAP_OFFSET,0))pcall(function():()humanoid.PlatformStand=true humanoid.AutoRotate=false root.AssemblyLinearVelocity=Vector3.zero root.
AssemblyAngularVelocity=Vector3.zero root.CFrame=CFrame.new(holdPos)*CFrame.Angles(0,yawOf(root.CFrame),0)end)end local function findLocalTool(toolName:string):Tool?local character:Model?=LocalPlayer
and LocalPlayer.Character if type(getPlayerWeapon)=='function'then local found:Tool?=getPlayerWeapon(LocalPlayer,toolName,false)if found then return found end end if character then local inChar:
Instance?=character:FindFirstChild(toolName)if inChar and inChar:IsA('Tool')then return inChar end end local backpack:Instance?=LocalPlayer and LocalPlayer:FindFirstChildOfClass('Backpack')if backpack
then local inBp:Instance?=backpack:FindFirstChild(toolName)if inBp and inBp:IsA('Tool')then return inBp end end return nil end local function ensureToolEquipped(character:Model,humanoid:Humanoid,tool:
Tool):boolean if tool.Parent==character then return true end pcall(function():()humanoid:EquipTool(tool)end)if tool.Parent~=character then pcall(function():()tool.Parent=character end)end return tool.
Parent==character end local function findActiveMurderer():Player?local playersService:Players=game:GetService('Players')if type(getPlayerWeapon)=='function'then for _,player:Player in ipairs(
playersService:GetPlayers())do if player~=LocalPlayer and(type(isPlayerAlive)~='function'or isPlayerAlive(player))and getPlayerWeapon(player,'Gun',false)==nil and getPlayerWeapon(player,'Knife',false)
~=nil then return player end end end if type(findMurderer)=='function'then local target:Player?=findMurderer()if target and target~=LocalPlayer and(type(isPlayerAlive)~='function'or isPlayerAlive(
target))and(type(getPlayerWeapon)~='function'or getPlayerWeapon(target,'Gun',false)==nil)then return target end end for _,player:Player in ipairs(playersService:GetPlayers())do if player~=LocalPlayer
and(type(isPlayerAlive)~='function'or isPlayerAlive(player))and(type(getPlayerWeapon)~='function'or getPlayerWeapon(player,'Gun',false)==nil)then local role:any=type(getPlayerRole)=='function'and
getPlayerRole(player)or nil if role=='Murderer'then return player end end end return nil end local function findActiveSheriffOrHero():Player?local playersService:Players=game:GetService('Players')if
type(getPlayerWeapon)=='function'then for _,player:Player in ipairs(playersService:GetPlayers())do if player~=LocalPlayer and(type(isPlayerAlive)~='function'or isPlayerAlive(player))and
getPlayerWeapon(player,'Knife',false)==nil and getPlayerWeapon(player,'Gun',false)~=nil then return player end end end if type(findSheriff)=='function'then local target:Player?=findSheriff()if target
and target~=LocalPlayer and(type(isPlayerAlive)~='function'or isPlayerAlive(target))and(type(getPlayerWeapon)~='function'or getPlayerWeapon(target,'Knife',false)==nil)then return target end end return
nil end local function runInstaKillAsMurder(character:Model,humanoid:Humanoid,root:BasePart,knife:Tool):boolean holdAboveMap(character,humanoid,root)if not ensureToolEquipped(character,humanoid,knife)
then return false end local handle:BasePart?=(knife:FindFirstChild('Handle')::BasePart?)or knife:FindFirstChildWhichIsA('BasePart')local slashRemote:Instance?=knife:FindFirstChild('Slash')local
stabRemote:Instance?=knife:FindFirstChild('Stab')local eventsFolder:Instance?=knife:FindFirstChild('Events')local touchedRemote:Instance?=eventsFolder and eventsFolder:FindFirstChild('HandleTouched')
local attackCF:CFrame=(handle and handle.CFrame)or(root.CFrame*CFrame.new(0,0,-2.5))local playersService:Players=game:GetService('Players')local hitAny:boolean=false for _,player:Player in ipairs(
playersService:GetPlayers())do if player~=LocalPlayer and(type(isPlayerAlive)~='function'or isPlayerAlive(player))and(type(isProtectedTarget)~='function'or not isProtectedTarget(player))then local
victimChar:Model?=player.Character local victimRoot:BasePart?=victimChar and victimChar:FindFirstChild('HumanoidRootPart')::BasePart?if victimChar and victimRoot then local onVictimMap:boolean=
isCharacterOnMap(victimRoot)if onVictimMap then hitAny=true if not victimSnapshots[victimRoot]then pcall(function():()victimSnapshots[victimRoot]={cframe=victimRoot.CFrame,anchored=victimRoot.Anchored
,canCollide=victimRoot.CanCollide}end)end pcall(function():()victimRoot.Anchored=true victimRoot.CanCollide=false victimRoot.AssemblyLinearVelocity=Vector3.zero victimRoot.AssemblyAngularVelocity=
Vector3.zero victimRoot.CFrame=attackCF end)if slashRemote and slashRemote:IsA('RemoteEvent')then pcall(function():()(slashRemote::RemoteEvent):FireServer('Slash')end)end if stabRemote and stabRemote:
IsA('RemoteEvent')then pcall(function():()(stabRemote::RemoteEvent):FireServer('Slash')end)end if touchedRemote and touchedRemote:IsA('RemoteEvent')then pcall(function():()(touchedRemote::RemoteEvent)
:FireServer(victimRoot)end)end if handle and type(state.fireTouchInterest)=='function'then pcall(state.fireTouchInterest,handle,victimRoot,0)pcall(state.fireTouchInterest,handle,victimRoot,1)end end
end end end if not hitAny then restoreVictims()awardRoundWin('Murderer')end return hitAny end local function runInstaKillAsSheriff(character:Model,humanoid:Humanoid,root:BasePart,gun:Tool):boolean
holdAboveMap(character,humanoid,root)if not ensureToolEquipped(character,humanoid,gun)then return false end local murderer:Player?=findActiveMurderer()if not murderer then if roundSawMurderer then
awardRoundWin('Sheriff')end return false end roundSawMurderer=true if type(state.mm2AutoShootCFrameOnce)=='function'then local okHelper:boolean,helperRes:any=pcall(state.mm2AutoShootCFrameOnce,false)
if okHelper and helperRes==true then return true end end if type(isProtectedTarget)=='function'and isProtectedTarget(murderer)then return false end local targetChar:Model?=murderer.Character local
targetRoot:BasePart?=targetChar and targetChar:FindFirstChild('HumanoidRootPart')::BasePart?if not targetRoot then return false end if type(state.mm2FireSilentShot)=='function'then local okSilent:
boolean,res:any=pcall(state.mm2FireSilentShot,murderer,{continuous=true,quiet=true,wallCheck=true})if okSilent and res==true then return true end end local shootRemote:Instance?=gun:FindFirstChild(
'Shoot')if not(shootRemote and shootRemote:IsA('RemoteEvent'))then return false end local originCF:CFrame=(type(getGunOriginCFrame)=='function'and getGunOriginCFrame(character,gun))or root.CFrame
local endpoint:Vector3=targetRoot.Position if type(prediction)=='table'and type(prediction.gunAim)=='function'then local okAim:boolean,aimPt:any=pcall(prediction.gunAim,murderer,originCF.Position,
'Bullet')if okAim and typeof(aimPt)=='Vector3'then endpoint=aimPt else local vel:Vector3=Vector3.zero pcall(function():()vel=targetRoot.AssemblyLinearVelocity end)endpoint=targetRoot.Position+vel*0.08
end end local startPoint:Vector3=endpoint+Vector3.new(0,1.5,-2.5)local shotOrigin:CFrame=CFrame.lookAt(startPoint,endpoint)local shotEnd:CFrame=CFrame.new(endpoint)if type(isWeaponPathClear)==
'function'then local okClear:boolean,allowed:any=pcall(isWeaponPathClear,originCF.Position,endpoint,targetChar,nil,{ignoreTarget=true})if not okClear or allowed~=true then return false end end pcall(
function():()(shootRemote::RemoteEvent):FireServer(shotOrigin,shotEnd)end)return true end local function grabDroppedGunTouchOnly(character:Model,humanoid:Humanoid,root:BasePart,drop:Instance):boolean
holdAboveMap(character,humanoid,root)if type(state.fireTouchInterest)~='function'then return false end local touchPart:BasePart?=nil if drop:IsA('BasePart')then touchPart=drop::BasePart elseif drop:
IsA('Model')then touchPart=(drop:FindFirstChild('Handle',true)::BasePart?)or drop:FindFirstChildWhichIsA('BasePart',true)end if not touchPart then return false end pcall(state.fireTouchInterest,root,
touchPart,0)pcall(state.fireTouchInterest,root,touchPart,1)return findLocalTool('Gun')~=nil end local function flingSheriffStep(character:Model,humanoid:Humanoid,root:BasePart,sheriff:Player):boolean
local sheriffChar:Model?=sheriff.Character local sheriffHum:Humanoid?=sheriffChar and sheriffChar:FindFirstChildOfClass('Humanoid')local sheriffRoot:BasePart?=sheriffChar and sheriffChar:
FindFirstChild('HumanoidRootPart')::BasePart?if not sheriffChar or not sheriffHum or sheriffHum.Health<=0 or sheriffHum.Sit or not sheriffRoot then holdAboveMap(character,humanoid,root)return false
end local destroyHeight:number=workspace.FallenPartsDestroyHeight if destroyHeight~=destroyHeight or math.abs(destroyHeight)>1e6 then destroyHeight=-500 end local sheriffPos:Vector3=sheriffRoot.
Position if sheriffPos.Y<=destroyHeight+60 then holdAboveMap(character,humanoid,root)return false end local sheriffOnMap:boolean=isCharacterOnMap(sheriffRoot)if not sheriffOnMap then holdAboveMap(
character,humanoid,root)return false end if type(state.mm2PerformFling)=='function'then if os.clock()-lastSheriffFlingAt<SHERIFF_FLING_COOLDOWN_SECONDS then holdAboveMap(character,humanoid,root)return
true end flingBusy=true lastSheriffFlingAt=os.clock()local returnPos:Vector3=safeAbovePosition or(root.Position+Vector3.new(0,ABOVE_MAP_OFFSET,0))task.spawn(function():()pcall(state.mm2PerformFling,
sheriff,{silent=true,duration=2.6,blocking=true,returnCFrame=CFrame.new(returnPos),shouldStop=function():boolean if not autoWinActive or not settings.flingSheriffGetGun then return true end if
findLocalTool('Gun')~=nil then return true end if type(findDroppedGun)=='function'and findDroppedGun()~=nil then return true end return false end})lastSheriffFlingAt=os.clock()local cNow:Model?,hNow:
Humanoid?,rNow:BasePart?=characterParts()if cNow and hNow and rNow and autoWinActive then holdAboveMap(cNow,hNow,rNow)end flingBusy=false end)return true end if type(state.mm2FlingStep)=='function'
then local okStep:boolean,res:any=pcall(state.mm2FlingStep,character,humanoid,root,sheriff,savedCollide,savedCustomProps)if okStep and res==true then return true end holdAboveMap(character,humanoid,
root)return false end if services and services.activity then pcall(function():()services.activity.set('fling',true)end)end enableCharacterFlingCollisions(character)flingAngle=(flingAngle+100)%360
local phase:number=math.floor(flingAngle/100)%4 local sweepOffset:CFrame=if phase==0 then CFrame.new(0,1.5,0)elseif phase==1 then CFrame.new(0,-1.5,0)elseif phase==2 then CFrame.new(2,1.2,-2)else
CFrame.new(-2,-1.2,2)local sheriffVel:Vector3=Vector3.zero pcall(function():()sheriffVel=sheriffRoot.AssemblyLinearVelocity end)local targetCF:CFrame=CFrame.new(sheriffPos+sheriffVel*0.08)*sweepOffset
*CFrame.Angles(math.rad(flingAngle),math.rad(flingAngle),0)pcall(function():()if humanoid.Sit then humanoid.Sit=false end humanoid.AutoRotate=false root.CFrame=targetCF character:PivotTo(targetCF)root
.AssemblyLinearVelocity=Vector3.new(10000,12000,10000)root.AssemblyAngularVelocity=Vector3.new(12000,12000,12000)end)return true end local function autoWinTick():()if not autoWinActive or flingBusy
then return end local character:Model?,humanoid:Humanoid?,root:BasePart?=characterParts()if not character or not humanoid or not root or humanoid.Health<=0 then return end local onMap:boolean,
mapRefPos:Vector3?=isCharacterOnMap(root)local activeMurderer:Player?=findActiveMurderer()if activeMurderer then roundSawMurderer=true end if not onMap then if wasOnMap then wasOnMap=false
mapEnteredAt=0 sheriffFirstSeenAt=0 lastSheriffFlingAt=0 safeAbovePosition=nil restoreLocalCharacter()end local roundStillRunning:boolean=type(findMM2Map)=='function'and findMM2Map()~=nil and
activeMurderer~=nil and(type(isRoundLifecycleActive)~='function'or isRoundLifecycleActive()~=false)if not roundStillRunning then if participatedInRound and not roundCountedWin and roundSawMurderer
then local myRoleNow:any=type(getPlayerRole)=='function'and getPlayerRole(LocalPlayer)or nil if myRoleNow~='Dead'then awardRoundWin(roundMyRole)end end participatedInRound=false roundMyRole=nil
roundSawMurderer=false roundCountedWin=false flungMurderThisDeath=false setStatus('Auto Win: waiting for map')return end local myRole:any=type(getPlayerRole)=='function'and getPlayerRole(LocalPlayer)
or nil local diedInRound:boolean=participatedInRound or myRole=='Dead'if settings.flingMurderIfDead and diedInRound and not flungMurderThisDeath and activeMurderer then flungMurderThisDeath=true if
type(state.mm2PerformFling)=='function'then flingBusy=true setStatus('Auto Win: Flinging Murderer (killed us)')task.spawn(function():()pcall(state.mm2PerformFling,activeMurderer,{silent=true,duration=
4,blocking=true})flingBusy=false end)return end end setStatus('Auto Win: waiting in lobby')return end if not wasOnMap or not safeAbovePosition then wasOnMap=true mapEnteredAt=os.clock()
sheriffFirstSeenAt=0 lastSheriffFlingAt=0 participatedInRound=true flungMurderThisDeath=false local anchor:Vector3=mapRefPos or root.Position safeAbovePosition=anchor+Vector3.new(0,ABOVE_MAP_OFFSET,0)
end if roundSawMurderer and not activeMurderer and not roundCountedWin then awardRoundWin(roundMyRole)end local knife:Tool?=findLocalTool('Knife')if knife then roundMyRole='Murderer'if settings.
instaKillMurder then runInstaKillAsMurder(character,humanoid,root,knife)setStatus('Auto Win: Insta Kill as Murder')else holdAboveMap(character,humanoid,root)setStatus(
'Auto Win: safe above map (Murderer)')end return end local gun:Tool?=findLocalTool('Gun')if gun then roundMyRole='Sheriff'if settings.instaKillSheriff then runInstaKillAsSheriff(character,humanoid,
root,gun)setStatus('Auto Win: Insta Kill As Sheriff (CFrame)')else holdAboveMap(character,humanoid,root)setStatus('Auto Win: safe above map (Gun ready)')end return end if not roundMyRole then local
roleNow:any=type(getPlayerRole)=='function'and getPlayerRole(LocalPlayer)or nil if roleNow=='Murderer'then roundMyRole='Murderer'elseif roleNow=='Sheriff'or roleNow=='Hero'then roundMyRole='Sheriff'
else roundMyRole='Innocent'end end local autoGrabExternal:boolean=state.mm2GunAuraEnabled==true or state.mm2AutoGetGunEnabled==true if settings.flingSheriffGetGun or autoGrabExternal then local drop:
Instance?=type(findDroppedGun)=='function'and findDroppedGun()or nil if drop then local grabbed:boolean=grabDroppedGunTouchOnly(character,humanoid,root,drop)setStatus('Auto Win: touching dropped gun')
if grabbed then roundMyRole='Sheriff'local freshGun:Tool?=findLocalTool('Gun')if freshGun and settings.instaKillSheriff then runInstaKillAsSheriff(character,humanoid,root,freshGun)end end return end
end if settings.flingSheriffGetGun then local nowClock:number=os.clock()local mapElapsed:number=nowClock-mapEnteredAt if mapElapsed<MAP_FLING_WARMUP_SECONDS then holdAboveMap(character,humanoid,root)
setStatus('Auto Win: loading map before fling...')return end local sheriff:Player?=findActiveSheriffOrHero()if sheriff and sheriff~=LocalPlayer and(type(isProtectedTarget)~='function'or not
isProtectedTarget(sheriff))then if sheriffFirstSeenAt==0 then sheriffFirstSeenAt=nowClock end if nowClock-sheriffFirstSeenAt<SHERIFF_READY_DELAY_SECONDS then holdAboveMap(character,humanoid,root)
setStatus('Auto Win: waiting for Sheriff to settle...')return end if flingSheriffStep(character,humanoid,root,sheriff)then setStatus('Auto Win: flinging Sheriff for gun')return end else
sheriffFirstSeenAt=0 end end holdAboveMap(character,humanoid,root)setStatus('Auto Win: safe above map')end local function clearToolConnections():()for _,conn:RBXScriptConnection in ipairs(
toolConnections)do pcall(function():()conn:Disconnect()end)end table.clear(toolConnections)end local function watchToolContainer(container:Instance?):()if not container or(type((container::any).
ChildAdded)~='table'and typeof((container::any).ChildAdded)~='RBXScriptSignal')then return end pcall(function():()local conn:RBXScriptConnection=(container::any).ChildAdded:Connect(function(child:
Instance):()if autoWinActive and child and(child.Name=='Gun'or child.Name=='Knife')then autoWinTick()end end)table.insert(toolConnections,conn)end)end local function setupInstantWeaponWatchers():()
clearToolConnections()if not LocalPlayer then return end watchToolContainer(LocalPlayer.Character)watchToolContainer(LocalPlayer:FindFirstChildOfClass('Backpack'))pcall(function():()if LocalPlayer.
CharacterAdded then table.insert(toolConnections,LocalPlayer.CharacterAdded:Connect(function(newChar:Model):()watchToolContainer(newChar)watchToolContainer(LocalPlayer:FindFirstChildOfClass('Backpack'
))end))end if LocalPlayer.ChildAdded then table.insert(toolConnections,LocalPlayer.ChildAdded:Connect(function(child:Instance):()if child and child:IsA('Backpack')then watchToolContainer(child)end end
))end end)end local function toggleAutoWin(enabled:boolean):()autoWinActive=enabled==true disconnectFeatureConnection('MM2AutoWin')clearToolConnections()if not enabled then wasOnMap=false
participatedInRound=false roundMyRole=nil roundSawMurderer=false roundCountedWin=false flungMurderThisDeath=false restoreLocalCharacter()safeAbovePosition=nil setStatus(nil)return end
setupInstantWeaponWatchers()featureConnections.MM2AutoWin=TaskManager:Connect(autoWinTick,nil,'MM2 Auto Win')autoWinTick()end AutoWinFeature=createUniversalFeature('Auto Win',
[[Keeps your character safely above the map as soon as you enter the map (even before the round starts) so nobody can kill you, with Insta Kill as Murder, Insta Kill As Sheriff (AutoShoot CFrame), Fling Sheriff and Get Gun, and Fling Murder if Dead.]]
,8,toggleAutoWin,{categoryName='Utilities',parent=MM2Scroll,registry=mm2Features})state.mm2AutoWinFeature=AutoWinFeature if type(state.addSectionOption)=='function'then state.addSectionOption(
AutoWinFeature,'Overlay Info')end if type(state.addStatOption)=='function'then sheriffWinsStat=state.addStatOption(AutoWinFeature,'Sheriff Wins','0')murderWinsStat=state.addStatOption(AutoWinFeature,
'Murder Wins','0')innocentWinsStat=state.addStatOption(AutoWinFeature,'Innocent Wins','0')end addToggleOption(AutoWinFeature,'Insta Kill as Murder',true,function(value:boolean):()settings.
instaKillMurder=value==true end,[[Automatically kills all living players as soon as you receive the Knife as Murderer.]])addToggleOption(AutoWinFeature,'Insta Kill As Sheriff',true,function(value:
boolean):()settings.instaKillSheriff=value==true end,[[Automatically equips the Gun and shoots the Murderer using AutoShoot CFrame the instant you receive the Gun (as Sheriff or Hero).]])
addToggleOption(AutoWinFeature,'Fling Sheriff and Get Gun',false,function(value:boolean):()settings.flingSheriffGetGun=value==true end,
[[Safely flies to the Sheriff with Fling without killing yourself, then automatically grabs the dropped gun via TouchInterest once the Sheriff dies and shoots the Murderer.]])addToggleOption(
AutoWinFeature,'Fling Murder if Dead',false,function(value:boolean):()settings.flingMurderIfDead=value==true end,
[[If the Murderer kills you and sends you to the lobby while the round is still active, automatically flings the Murderer.]])activeCleanup=function():()toggleAutoWin(false)state.mm2AutoWinFeature=nil
end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)
activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Other/CoinsAura.lua
local Module={Name='MM2 Coins Aura',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Coins Aura requires the MM2 core module')Module.Runtime=runtime local MAX_SERVER_REACH:number=15 local
MAX_SERVER_REACH_SQ:number=MAX_SERVER_REACH*MAX_SERVER_REACH local TICK_INTERVAL:number=0.08 local CACHE_INTERVAL:number=0.5 local savedCFrames:{[BasePart]:CFrame}={}local savedPositions:{[BasePart]:
Vector3}={}local activeCoin:BasePart?=nil local function rememberCoin(serverPart:BasePart):Vector3?local pos:Vector3?=savedPositions[serverPart]if pos then return pos end local cf:CFrame=serverPart.
CFrame savedCFrames[serverPart]=cf savedPositions[serverPart]=cf.Position return cf.Position end local function restoreCoin(serverPart:BasePart?):()if not serverPart then return end local cf:CFrame?=
savedCFrames[serverPart]if cf and serverPart.Parent then pcall(function():()serverPart.CFrame=cf end)end if activeCoin==serverPart then activeCoin=nil end end local function restoreAllCoins():()for
serverPart,cf in pairs(savedCFrames)do if serverPart and serverPart.Parent then pcall(function():()serverPart.CFrame=cf end)end end table.clear(savedCFrames)table.clear(savedPositions)activeCoin=nil
end local function isCoinCollectable(serverPart:BasePart):boolean if not serverPart or not serverPart.Parent then return false end if serverPart:GetAttribute('Collected')==true or serverPart:
GetAttribute('Delete')==true then return false end local visual:Instance?=serverPart:FindFirstChild('CoinVisual')if visual and(visual:GetAttribute('Collected')==true or visual:GetAttribute('Delete')==
true)then return false end return true end local function detachVisualWeld(serverPart:BasePart):()local visual:Instance?=serverPart:FindFirstChild('CoinVisual')if not visual or not visual:IsA(
'BasePart')then return end local vPart:BasePart=visual::BasePart vPart.Anchored=true local weld:Instance?=vPart:FindFirstChildOfClass('WeldConstraint')if weld then(weld::WeldConstraint).Enabled=false
end end local function pulseCoinTouch(root:BasePart,serverPart:BasePart):()rememberCoin(serverPart)detachVisualWeld(serverPart)serverPart.CanCollide=false serverPart.CFrame=root.CFrame local fireTouch
:any=state.fireTouchInterest if type(fireTouch)=='function'then pcall(fireTouch,root,serverPart,0)pcall(fireTouch,root,serverPart,1)end end local CoinsAuraFeature:any=nil local function
toggleCoinsAura(enabled:boolean):()disconnectFeatureConnection('MM2CoinsAura')restoreAllCoins()if not enabled then if CoinsAuraFeature then CoinsAuraFeature:SetStatus(nil)end return end local
cachedCoins:{BasePart}={}local lastCacheAt:number=-math.huge local lastTickAt:number=-math.huge if CoinsAuraFeature then CoinsAuraFeature:SetStatus('Max reach')end local function refreshCoins(now:
number):()if now-lastCacheAt<CACHE_INTERVAL then return end lastCacheAt=now local list:{BasePart}={}local seen:{[BasePart]:boolean}={}local function addCoin(candidate:Instance?):()if not candidate
then return end local serverPart:BasePart?=nil if candidate:IsA('BasePart')then if candidate.Name=='Coin_Server'or candidate.Name=='ServerCoinPart'then serverPart=candidate::BasePart elseif candidate.
Parent and candidate.Parent:IsA('BasePart')and(candidate.Parent.Name=='Coin_Server'or candidate.Parent.Name=='ServerCoinPart')then serverPart=candidate.Parent::BasePart else serverPart=candidate::
BasePart end elseif candidate:IsA('Model')then local child:Instance?=candidate:FindFirstChild('Coin_Server')or candidate:FindFirstChildWhichIsA('BasePart')if child and child:IsA('BasePart')then
serverPart=child::BasePart end end if serverPart and not seen[serverPart]and isCoinCollectable(serverPart)then seen[serverPart]=true rememberCoin(serverPart)list[#list+1]=serverPart end end if type(
core.findMM2Map)=='function'then local map:Instance?=core.findMM2Map()if map then local container:Instance?=map:FindFirstChild('CoinContainer')or map:FindFirstChild('CoinAreas')or map:FindFirstChild(
'Coins')if container then for _,child:Instance in ipairs(container:GetChildren())do addCoin(child)end end end end if#list==0 and type(state.mm2CoinPickups)=='function'then local okPickups:boolean,
pickups:any=pcall(state.mm2CoinPickups)if okPickups and type(pickups)=='table'then for _,pickup:Instance in ipairs(pickups)do addCoin(pickup)end end end cachedCoins=list end local function auraTick():
()local now:number=os.clock()if now-lastTickAt<TICK_INTERVAL then return end lastTickAt=now local character:Model?=LocalPlayer and LocalPlayer.Character local root:BasePart?=character and character:
FindFirstChild('HumanoidRootPart')::BasePart?if not root then if activeCoin then restoreCoin(activeCoin)end return end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')::Humanoid?if
humanoid and humanoid.Health<=0 then if activeCoin then restoreCoin(activeCoin)end return end refreshCoins(now)local rootPos:Vector3=root.Position local bestCoin:BasePart?=nil local bestDistSq:number=
MAX_SERVER_REACH_SQ for i:number=1,#cachedCoins do local coin:BasePart=cachedCoins[i]if isCoinCollectable(coin)then local coinPos:Vector3?=savedPositions[coin]or rememberCoin(coin)if coinPos then
local diff:Vector3=coinPos-rootPos local distSq:number=diff.X*diff.X+diff.Y*diff.Y+diff.Z*diff.Z if distSq<=bestDistSq then bestDistSq=distSq bestCoin=coin end end elseif coin==activeCoin then
restoreCoin(coin)end end if activeCoin and activeCoin~=bestCoin then restoreCoin(activeCoin)end if bestCoin then activeCoin=bestCoin pcall(pulseCoinTouch,root,bestCoin)end end featureConnections.
MM2CoinsAura=TaskManager:Connect(auraTick,nil,'MM2 Coins Aura')auraTick()end CoinsAuraFeature=createUniversalFeature('Coins Aura',
[[Automatically collects nearby coins at the maximum reach allowed by the MM2 server with minimal CPU usage.]],7,toggleCoinsAura,{categoryName='Utilities',parent=MM2Scroll,registry=mm2Features})state.
mm2CoinsAuraFeature=CoinsAuraFeature activeCleanup=function():()toggleCoinsAura(false)end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not
Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Other/Fling.lua
local Module={Name='MM2 Fling',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end local MAX_POWER:number=12 local MAX_RETRIES:number=3 local
FLING_DURATION:number=3 local AUTO_FLING_INTERVAL:number=0.35 local ROLE_ORDER:{[string]:number}={Murderer=1,Sheriff=2,Hero=3,Innocent=4,Dead=5}function Module.init(runtime:any):any if Module.
Initialized then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Fling requires the MM2 core module')Module.Runtime=runtime local localPlayer:any=LocalPlayer local
playerService:any=Players local gameBridge:any=services and services.gameBridge or nil local protectedTargets:any=services and services.protectedTargets or nil local TaskManager:any=TaskManager local
featureConnections:any=featureConnections local disconnectFeatureConnection:any=disconnectFeatureConnection local findMurderer:any=core.findMurderer local findSheriff:any=core.findSheriff local
getPlayerRole:any=core.getPlayerRole local getPlayerWeapon:any=core.getPlayerWeapon local isPlayerAlive:any=core.isPlayerAlive local settings={power=MAX_POWER,retries=MAX_RETRIES,autoFling=false,
autoFlingTarget=nil::any}local FlingFeature:any=nil local listReady:boolean=false local selectedPlayer:any=nil local selectedLabel:string=''local allInnocentsSequence:number=0 local autoElapsed:number
=0 local function antiFlingOn():boolean return state.antiFlingActive==true end local function setStatus(text:string?):()if FlingFeature and type(FlingFeature.SetStatus)=='function'then pcall(function(
):()FlingFeature:SetStatus(text)end)end end local function notifyUser(text:string):()if type(notify)=='function'then pcall(notify,text)end end local function roster():{any}local ok:boolean,players:any
=pcall(function():any return playerService:GetPlayers()end)if not ok or type(players)~='table'then return{}end return players end local function livingCharacter(player:any):(any?,any?)local character:
any=player and player.Character if not character or not character.Parent then return nil,nil end local humanoid:any=character:FindFirstChildOfClass('Humanoid')if not humanoid or humanoid.Health<=0
then return nil,nil end return character,humanoid end local function rootOf(player:any):any?local character:any=player and player.Character if not character then return nil end local root:any=
character:FindFirstChild('HumanoidRootPart')if not root or root.Parent==nil then return nil end return root end local function isLivePlayer(player:any):boolean if not player or player==localPlayer
then return false end local classOk:boolean,className:any=pcall(function():any return player.ClassName end)if not classOk or className~='Player'then return false end local parentOk:boolean,parent:any=
pcall(function():any return player.Parent end)return parentOk and parent~=nil end local function canFling(player:any):boolean if not isLivePlayer(player)then return false end if protectedTargets and
type(protectedTargets.isProtected)=='function'then local ok:boolean,protected:any=pcall(protectedTargets.isProtected,player)if ok and protected==true then return false end end if type(isPlayerAlive)==
'function'then local ok:boolean,alive:any=pcall(isPlayerAlive,player)if ok and alive==false then return false end end local character:any=livingCharacter(player)return character~=nil and rootOf(player
)~=nil end local function hasWeapon(player:any,toolName:string):boolean if type(getPlayerWeapon)~='function'then return false end local ok:boolean,weapon:any=pcall(getPlayerWeapon,player,toolName,
false)return ok and weapon~=nil end local function roleOf(player:any):string?if not player then return nil end local role:any=nil if type(getPlayerRole)=='function'then local ok:boolean,value:any=
pcall(getPlayerRole,player)if ok then role=value end end if not role and gameBridge and type(gameBridge.playerRole)=='function'then local ok:boolean,value:any=pcall(gameBridge.playerRole,player)if ok
then role=value end end if role=='Murderer'or role=='Sheriff'or role=='Hero'or role=='Innocent'or role=='Dead'then return role end if hasWeapon(player,'Knife')then return'Murderer'end if hasWeapon(
player,'Gun')then return'Sheriff'end if role==nil then local character:any=player.Character local humanoid:any=character and character:FindFirstChildOfClass('Humanoid')if humanoid and humanoid.Health
<=0 then return'Dead'end end return role end local function rolesPublished():boolean if type(core.hasActiveRoundRoles)=='function'then local ok:boolean,published:any=pcall(core.hasActiveRoundRoles)if
ok and published==true then return true end end for _,player:any in ipairs(roster())do if player~=localPlayer and roleOf(player)~=nil then return true end end return false end local function
refreshRoles():()if type(state.clearRoleMemo)=='function'then pcall(state.clearRoleMemo)end if type(core.requestRoleRefresh)=='function'then pcall(core.requestRoleRefresh,false)end end local function
rosterEntries(withRefresh:boolean?):{{player:any,label:string}}if withRefresh~=false then refreshRoles()end local entries:{{player:any,label:string}}={}for _,player:any in ipairs(roster())do if player
~=localPlayer and canFling(player)then local role:string?=roleOf(player)table.insert(entries,{player=player,label=tostring(player.Name)..(role and(' ['..role..']')or'')})end end table.sort(entries,
function(left:any,right:any):boolean local leftRole:number=ROLE_ORDER[(string.match(left.label,'%[(%a+)%]')or'')]or 9 local rightRole:number=ROLE_ORDER[(string.match(right.label,'%[(%a+)%]')or'')]or 9
if leftRole~=rightRole then return leftRole<rightRole end return string.lower(left.label)<string.lower(right.label)end)return entries end local function resolveMurdererTarget():any?refreshRoles()for _
,player:any in ipairs(roster())do if player~=localPlayer and canFling(player)and not hasWeapon(player,'Gun')and hasWeapon(player,'Knife')then return player end end if type(findMurderer)=='function'
then local ok:boolean,target:any=pcall(findMurderer)if ok and target and canFling(target)and roleOf(target)~='Dead'then return target end end for _,player:any in ipairs(roster())do if player~=
localPlayer and canFling(player)and roleOf(player)=='Murderer'then return player end end return nil end local function resolveSheriffTarget():any?refreshRoles()for _,player:any in ipairs(roster())do
if player~=localPlayer and canFling(player)and not hasWeapon(player,'Knife')and hasWeapon(player,'Gun')then return player end end if type(findSheriff)=='function'then local ok:boolean,target:any=
pcall(findSheriff)if ok and target and canFling(target)then local role:string?=roleOf(target)if role~='Murderer'and role~='Dead'then return target end end end for _,player:any in ipairs(roster())do if
player~=localPlayer and canFling(player)then local role:string?=roleOf(player)if role=='Sheriff'or role=='Hero'then return player end end end return nil end state.mm2FlingRoster=function():{any}local
snapshot:{any}={}for _,entry:any in ipairs(rosterEntries(false))do table.insert(snapshot,{name=tostring(entry.player.Name),label=entry.label})end return snapshot end state.mm2FlingSelection=function()
:string?return if selectedPlayer then tostring(selectedPlayer.Name)else nil end local function performFling(target:any,options:{[string]:any}?):boolean if antiFlingOn()then if not(type(options)==
'table'and options.silent==true)then notifyUser('Anti Fling is ON: turn it off before flinging someone.')end setStatus('Anti Fling is ON')return false end if not canFling(target)then if not(type(
options)=='table'and options.silent==true)then notifyUser('No living, valid fling target is available.')setStatus('No target')end return false end local perform:any=state.mm2PerformFling if type(
perform)~='function'then setStatus('Fling engine unavailable')return false end local call:{[string]:any}={power=settings.power,retries=settings.retries,duration=FLING_DURATION}if type(options)==
'table'then for key:string,value:any in pairs(options)do call[key]=value end end if call.power==nil then call.power=settings.power end if call.retries==nil then call.retries=settings.retries end
setStatus('Flinging '..tostring(target.Name))local started:boolean=false local ok:boolean,result:any=pcall(perform,target,call)if ok then started=result==true end if not started then setStatus(
'Fling busy or refused')end return started end local function flingSelected(label:string?):boolean local target:any=nil if label and label~=''then for _,entry:any in ipairs(rosterEntries(false))do if
entry.label==label then target=entry.player break end end end if not target then target=if canFling(selectedPlayer)then selectedPlayer else nil end if not target then setStatus('Pick a target')if
label and label~=''then notifyUser('That player is not a living fling target any more.')end return false end return performFling(target)end local function flingAllInnocents():()if antiFlingOn()then
notifyUser('Anti Fling is ON: turn it off before flinging someone.')setStatus('Anti Fling is ON')return end allInnocentsSequence+=1 local sequence:number=allInnocentsSequence refreshRoles()task.spawn(
function():()local flung:number=0 for _,player:any in ipairs(roster())do if sequence~=allInnocentsSequence then break end if canFling(player)and roleOf(player)=='Innocent'then if performFling(player,{
blocking=true,silent=true,duration=FLING_DURATION})then flung+=1 end task.wait(0.05)end end if sequence==allInnocentsSequence then setStatus(flung>0 and('Flung '..tostring(flung))or nil)end end)end
local function stopAutoFling():()settings.autoFling=false settings.autoFlingTarget=nil autoElapsed=0 disconnectFeatureConnection('MM2FlingAuto')end local function startAutoFling():()
disconnectFeatureConnection('MM2FlingAuto')autoElapsed=0 featureConnections.MM2FlingAuto=TaskManager:Connect(function(deltaTime:number):()autoElapsed+=math.max(0,deltaTime)if autoElapsed<
AUTO_FLING_INTERVAL then return end autoElapsed=0 if not settings.autoFling or antiFlingOn()then return end local target:any=settings.autoFlingTarget if not canFling(target)then target=if canFling(
selectedPlayer)then selectedPlayer else nil end if not target then return end performFling(target,{silent=true,duration=FLING_DURATION})end,nil,'MM2 Fling Auto')end FlingFeature=
createUniversalFeature('Fling',[[MM2 only: throws the murderer, the sheriff, every innocent or the player you pick, at maximum power.]],9,function():()end,{categoryName='Exploits',parent=MM2Scroll,
registry=mm2Features,kind='group'})if type(addActionOption)=='function'then addActionOption(FlingFeature,'Fling Murderer',function():()local target:any=resolveMurdererTarget()if target then
performFling(target)else notifyUser(if rolesPublished()then'No living murderer was found in this round.'else[[MM2 has not published the roles yet. Wait for the round to start.]])end end)
addActionOption(FlingFeature,'Fling Sheriff',function():()local target:any=resolveSheriffTarget()if target then performFling(target)else notifyUser(if rolesPublished()then
'No living sheriff or hero was found in this round.'else[[MM2 has not published the roles yet. Wait for the round to start.]])end end)addActionOption(FlingFeature,'Fling All Innocents',function():()
flingAllInnocents()end)end local listHandle:any=nil if type(addListOption)=='function'then listHandle=addListOption(FlingFeature,'Fling target',{items=function():{string}local labels:{string}={}for _,
entry:any in ipairs(rosterEntries())do table.insert(labels,entry.label)end return labels end,single=true,default={},emptyText='No other players',hint=
[[Living players, ordered by role. Anti Fling must be OFF: it cancels the contact a fling needs.]],onChanged=function(_selected:any,names:{string}):()local label:string=names and names[1]or''if label
==''then return end selectedLabel=label selectedPlayer=nil for _,entry:any in ipairs(rosterEntries(false))do if entry.label==label then selectedPlayer=entry.player settings.autoFlingTarget=entry.
player break end end if not listReady then return end if selectedPlayer then local target:any=selectedPlayer local key:string=tostring(FlingFeature.configKey or'MM2.Fling')..'.Flingtarget'local
controls:any=state.listControls and state.listControls[key]if controls then listReady=false pcall(function():()controls.set(label,false)end)listReady=true end performFling(target)end end})listReady=
true end if type(addActionOption)=='function'then addActionOption(FlingFeature,'Fling',function():()flingSelected(selectedLabel)end)end if type(addToggleOption)=='function'then addToggleOption(
FlingFeature,'Auto Fling Selected Target',false,function(value:boolean):()settings.autoFling=value==true if settings.autoFling then if not canFling(selectedPlayer)then settings.autoFling=false
notifyUser('Pick a living player in "Fling target" first.')if state.setToggleChecked then pcall(state.setToggleChecked,tostring(FlingFeature.configKey or'MM2.Fling')..'.AutoFlingSelectedTarget',false)
end return end settings.autoFlingTarget=selectedPlayer startAutoFling()else stopAutoFling()end end)end if type(addNumberOption)=='function'then addNumberOption(FlingFeature,'Power',settings.power,1,
MAX_POWER,function(value:number):()settings.power=math.clamp(math.floor(tonumber(value)or MAX_POWER),1,MAX_POWER)end,
[[Fling strength. 12 is the maximum the engine accepts: four times YARHM's 9e7/9e8 pose velocity, which is what carries a fling through a laggy or fast server.]],1)addNumberOption(FlingFeature,
'Retries',settings.retries,0,MAX_RETRIES,function(value:number):()settings.retries=math.clamp(math.floor(tonumber(value)or MAX_RETRIES),0,MAX_RETRIES)end,
[[Extra passes that swap the contact part when the server misses the first one. 3 is the maximum.]],1)end if type(addInformationOption)=='function'then addInformationOption(FlingFeature,
[[MM2 only. Every button refuses while Anti Fling is ON: its collision ]]..'shield cancels the contact a fling needs.')end activeCleanup=function():()allInnocentsSequence+=1 stopAutoFling()state.
mm2FlingFeature=nil state.mm2FlingSettings=nil end state.mm2FlingFeature=FlingFeature state.mm2FlingSettings=settings Module.Events=featureConnections Module.Initialized=true return Module end
function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Other/FlingEngine.lua
local Module={Name='MM2 Fling Engine',Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.init(runtime:any):any if Module.Initialized then return Module end
local host:any=state.frameworkEnvironment local services:any=runtime and runtime.Services local Players:any=host.Players local LocalPlayer:any=host.LocalPlayer local getCharacterParts:any=host.
getCharacterParts local notify:any=host.notify local activity:any=services and services.activity local protectedTargets:any=services and services.protectedTargets local configData:any=host.configData
local workspace:any=host.workspace local function protectedCheck(player:Player?):boolean if not protectedTargets or type(protectedTargets.isProtected)~='function'then return false end local ok:boolean
,result:any=pcall(protectedTargets.isProtected,player)return ok and result==true end if configData and type(configData.states)=='table'then configData.states['Universal.Fling']=false configData.states
['Universal.Fling.AutoFlingSelectedTarget']=nil configData.states['MM2.Fling.AutoFlingSelectedTarget']=nil end local FLING_VELOCITY_BASE:Vector3=Vector3.new(9e7,9e7*10,9e7)local
FLING_ROT_VELOCITY_BASE:Vector3=Vector3.new(9e8,9e8,9e8)local PIN_VELOCITY_BASE:Vector3=Vector3.new(9e8,9e8,9e8)local PIN_FORCE:Vector3=Vector3.new(9e9,9e9,9e9)local FLING_POWER:number=6 local
FLING_DURATION:number=3 local FLING_PREDICTION:number=0.15 local flingRetries:number=2 local flingBoost:number=1 local function powerScale():number return(FLING_POWER/3)*flingBoost end local function
scaledVelocity(base:Vector3):Vector3 local scale:number=powerScale()return Vector3.new(base.X*scale,base.Y*scale,base.Z*scale)end local flingRunning:boolean=false local engineActive:boolean=true local
fullFlingPin:any=nil local stepPhase:number=0 local stepAngle:number=0 local lastOldPos:CFrame?=nil local stepPin:any=nil local stepPinToken:number=0 local function setFlingRunning(running:boolean):()
flingRunning=running if activity and type(activity.set)=='function'then activity.set('fling',running)end end local function targetParts(targetPlayer:Player?):(Model?,Humanoid?,BasePart?,BasePart?,
BasePart?)local targetCharacter:Model?=targetPlayer and targetPlayer.Character if not targetCharacter then return nil,nil,nil,nil,nil end local targetHumanoid:Humanoid?=targetCharacter:
FindFirstChildOfClass('Humanoid')local targetRoot:BasePart?=targetHumanoid and targetHumanoid.RootPart or nil local targetHead:BasePart?=targetCharacter:FindFirstChild('Head')::BasePart?return
targetCharacter,targetHumanoid,targetRoot,targetHead,nil end local function fpos(character:Model,root:BasePart,basePart:BasePart,offset:CFrame,angles:CFrame,lead:Vector3?):()local origin:Vector3=
basePart.Position if lead then origin=origin+lead end local targetCFrame:CFrame=CFrame.new(origin)*offset*angles root.CFrame=targetCFrame pcall(function():()character:SetPrimaryPartCFrame(targetCFrame
)end)local vel:Vector3=scaledVelocity(FLING_VELOCITY_BASE)local rot:Vector3=scaledVelocity(FLING_ROT_VELOCITY_BASE)local hum:Humanoid?=character:FindFirstChildOfClass('Humanoid')if hum and hum.
ChangeState then pcall(function():()hum:ChangeState(Enum.HumanoidStateType.Physics)end)end root.Velocity=vel root.RotVelocity=rot pcall(function():()for _,part:Instance in ipairs(character:
GetChildren())do if part:IsA('BasePart')then(part::BasePart).CanCollide=false end end end)end local function transportSeconds():number local ping:any=0 pcall(function():()local player:any=LocalPlayer
if player and type(player.GetNetworkPing)=='function'then ping=player:GetNetworkPing()end end)if type(ping)~='number'or ping~=ping or ping<0 then ping=0 end if ping>2 then ping=ping/1000 end return
math.clamp(ping*0.5,0,0.5)end local function leadFor(targetPart:BasePart?):Vector3?if not targetPart then return nil end local velocity:any=Vector3.zero pcall(function():()velocity=targetPart.
AssemblyLinearVelocity end)if type(velocity)~='userdata'and type(velocity)~='table'then return nil end local seconds:number=math.clamp(transportSeconds()+FLING_PREDICTION,0,0.6)if seconds<=0.001 then
return nil end local lead:any=Vector3.new(velocity.X,0,velocity.Z)*seconds return lead::Vector3 end local function liveTargetParts(targetPlayer:Player,targetCharacter:Model):(Model?,Humanoid?,BasePart
?,BasePart?)local character:Model?=targetPlayer.Character or targetCharacter if not character or not character.Parent then return nil,nil,nil,nil end local humanoid:Humanoid?=character:
FindFirstChildOfClass('Humanoid')local root:BasePart?=humanoid and humanoid.RootPart or nil local head:BasePart?=character:FindFirstChild('Head')::BasePart?return character,humanoid,root,head end
local function partSpeed(part:BasePart?):number return if part then part.Velocity.Magnitude else 0 end local function playerStillHere(player:Player):boolean return Players:FindFirstChild(player.Name)
~=nil end local MIN_POSES:number=12 local function sfBasePart(character:Model,humanoid:Humanoid,root:BasePart,basePart:BasePart,targetCharacter:Model,targetPlayer:Player,targetHumanoid:Humanoid?,
targetRoot:BasePart?,timeToWait:number,shouldStop:(()->boolean)?,stats:any):()local startedAt:number=os.clock()local angle:number=0 local liveBase:BasePart=basePart local function currentBasePart():
BasePart?local characterNow:Model?,humanoidNow:Humanoid?,rootNow:BasePart?,headNow:BasePart?=liveTargetParts(targetPlayer,targetCharacter)if not characterNow then return nil end targetCharacter=
characterNow targetHumanoid=humanoidNow or targetHumanoid targetRoot=rootNow or targetRoot local candidate:BasePart?=rootNow if rootNow and headNow then if(rootNow.Position-headNow.Position).Magnitude
>5 then candidate=headNow end elseif headNow then candidate=headNow end if candidate and candidate.Parent then liveBase=candidate end return liveBase end local function pose(offset:CFrame,angles:
CFrame):boolean if shouldStop and shouldStop()then return false end local part:BasePart?=currentBasePart()if not part then return false end fpos(character,root,part,offset,angles,leadFor(part))stats.
poses+=1 task.wait()return true end repeat if not root or not targetHumanoid then break end if shouldStop and shouldStop()then break end local part:BasePart?=currentBasePart()if not part then break
end local speed:number=partSpeed(part)local moveDirection:Vector3=Vector3.zero local walkSpeed:number=16 pcall(function():()moveDirection=targetHumanoid.MoveDirection walkSpeed=targetHumanoid.
WalkSpeed end)if speed<50 then angle=angle+100 local lead:Vector3=moveDirection*(speed/1.25)local spin:CFrame=CFrame.Angles(math.rad(angle),0,0)pose(CFrame.new(0,1.5,0)+lead,spin)pose(CFrame.new(0,-
1.5,0)+lead,spin)pose(CFrame.new(2.25,1.5,-2.25)+lead,spin)pose(CFrame.new(-2.25,-1.5,2.25)+lead,spin)pose(CFrame.new(0,1.5,0)+moveDirection,spin)pose(CFrame.new(0,-1.5,0)+moveDirection,spin)else
local downLead:number=partSpeed(targetRoot)/1.25 pose(CFrame.new(0,1.5,walkSpeed),CFrame.Angles(math.rad(90),0,0))pose(CFrame.new(0,-1.5,-walkSpeed),CFrame.Angles(0,0,0))pose(CFrame.new(0,1.5,
walkSpeed),CFrame.Angles(math.rad(90),0,0))pose(CFrame.new(0,1.5,downLead),CFrame.Angles(math.rad(90),0,0))pose(CFrame.new(0,-1.5,-downLead),CFrame.Angles(0,0,0))pose(CFrame.new(0,1.5,downLead),CFrame
.Angles(math.rad(90),0,0))pose(CFrame.new(0,-1.5,0),CFrame.Angles(math.rad(90),0,0))pose(CFrame.new(0,-1.5,0),CFrame.Angles(0,0,0))pose(CFrame.new(0,-1.5,0),CFrame.Angles(math.rad(-90),0,0))pose(
CFrame.new(0,-1.5,0),CFrame.Angles(0,0,0))end until(stats.poses>=MIN_POSES and partSpeed(liveBase)>500)or not liveBase.Parent or targetPlayer.Character~=targetCharacter or not playerStillHere(
targetPlayer)or(targetHumanoid~=nil and targetHumanoid.Sit)or humanoid.Health<=0 or os.clock()>startedAt+timeToWait or stats.poses>=600 end local function createPin(root:BasePart):any local pin:any=
Instance.new('BodyVelocity')pin.Name='EpixVel'pin.Velocity=scaledVelocity(PIN_VELOCITY_BASE)pin.MaxForce=PIN_FORCE pin.Parent=root return pin end local function destroyPin(pin:any):()if pin then pin:
Destroy()end end local function flingSingleStep(character:Model,humanoid:Humanoid,root:BasePart,targetPlayer:Player,_savedCollide:any,_savedProps:any):boolean local targetCharacter:Model?,
targetHumanoid:Humanoid?,targetRoot:BasePart?,targetHead:BasePart?=targetParts(targetPlayer)if not targetCharacter or not targetHead and not targetRoot then return false end if targetHumanoid and(
targetHumanoid.Health<=0 or targetHumanoid.Sit)then return false end if not targetHumanoid then return false end local basePart:BasePart=targetRoot or targetHead::BasePart if targetRoot and targetHead
then if(targetRoot.Position-targetHead.Position).Magnitude>5 then basePart=targetHead end end if activity and type(activity.set)=='function'then activity.set('fling',true)end if not stepPin then
stepPin=createPin(root)end stepPinToken+=1 local myToken:number=stepPinToken task.delay(0.5,function():()if myToken==stepPinToken then destroyPin(stepPin)stepPin=nil if not flingRunning and activity
and type(activity.set)=='function'then activity.set('fling',false)end end end)stepAngle=(stepAngle+100)%360 local speed:number=0 pcall(function():()speed=basePart.Velocity.Magnitude end)if speed<50
then stepPhase=(stepPhase+1)%6 local offsets:{CFrame}={CFrame.new(0,1.5,0),CFrame.new(0,-1.5,0),CFrame.new(2.25,1.5,-2.25),CFrame.new(-2.25,-1.5,2.25),CFrame.new(0,1.5,0),CFrame.new(0,-1.5,0)}fpos(
character,root,basePart,offsets[stepPhase+1],CFrame.Angles(math.rad(stepAngle),0,0))else fpos(character,root,basePart,CFrame.new(0,1.5,0),CFrame.Angles(math.rad(90),0,0))end return true end
local function performFling(targetPlayer:Player?,options:{[string]:any}?):boolean if not Module.Initialized or not targetPlayer then return false end if protectedCheck(targetPlayer)then notify(
'player protected')return false end local silent:boolean=type(options)=='table'and options.silent==true local maxDuration:number=(type(options)=='table'and tonumber(options.duration))or FLING_DURATION
local requestedStop:(()->boolean)?=type(options)=='table'and type(options.shouldStop)=='function'and options.shouldStop or nil local function shouldStop():boolean return not engineActive or(
requestedStop~=nil and requestedStop())end local skipReturnOnStop:boolean=type(options)=='table'and options.skipReturnOnStop==true local returnCFrame:CFrame?=type(options)=='table'and typeof(options.
returnCFrame)=='CFrame'and options.returnCFrame or nil local requestedPower:number?=nil local requestedRetries:number?=nil if type(options)=='table'then local power:number?=tonumber(options.power)if
power then requestedPower=math.clamp(power,1,12)end local retries:number?=tonumber(options.retries)if retries then requestedRetries=math.clamp(math.floor(retries),0,3)end end if flingRunning then if
not silent then notify('fling already running')end return false end setFlingRunning(true)local stats:any={poses=0,speed=0,pinned=false,retries=0}local function runFling():boolean local character:Model
?,humanoid:Humanoid?,root:BasePart?=getCharacterParts()if not character or not humanoid or not root then setFlingRunning(false)if not silent then notify('character not ready')end return false end
local targetCharacter:Model?,targetHumanoid:Humanoid?,targetRoot:BasePart?,targetHead:BasePart?=targetParts(targetPlayer)if not targetCharacter or not targetCharacter:FindFirstChildWhichIsA('BasePart'
)then setFlingRunning(false)if not silent then notify('No valid character of said target player. May have died.')end return false end if not targetHumanoid then setFlingRunning(false)if not silent
then notify('the target has no humanoid')end return false end if partSpeed(root)<50 then lastOldPos=root.CFrame end local oldPos:CFrame=returnCFrame or lastOldPos or root.CFrame local
savedDestroyHeight:any=nil pcall(function():()savedDestroyHeight=workspace.FallenPartsDestroyHeight end)local camera:Camera?=workspace.CurrentCamera local pin:any=nil local success:boolean,
errorMessage:any=pcall(function():()if camera then pcall(function():()local accessory:any=targetCharacter:FindFirstChildOfClass('Accessory')camera.CameraSubject=targetHead or(accessory and accessory:
FindFirstChild('Handle'))or targetHumanoid end)end pcall(function():()workspace.FallenPartsDestroyHeight=0/0 end)pin=createPin(root)fullFlingPin=pin stats.pinned=true pcall(function():()humanoid:
SetStateEnabled(Enum.HumanoidStateType.Seated,false)end)local basePart:BasePart?=nil if targetRoot and targetHead then if(targetRoot.Position-targetHead.Position).Magnitude>5 then basePart=targetHead
else basePart=targetRoot end elseif targetRoot then basePart=targetRoot elseif targetHead then basePart=targetHead end local maxPasses:number=math.clamp(math.floor(tonumber(flingRetries)or 0),0,3)+1
local used:BasePart?=nil local pass:number=0 while pass<maxPasses do if shouldStop and shouldStop()then break end pass+=1 local partNow:BasePart?=nil if pass==1 then partNow=basePart else local _c:
Model?,_h:Humanoid?,rootNow:BasePart?,headNow:BasePart?=liveTargetParts(targetPlayer,targetCharacter)targetHumanoid=_h or targetHumanoid targetRoot=rootNow or targetRoot local primary:BasePart?=
rootNow if rootNow and headNow then if(rootNow.Position-headNow.Position).Magnitude>5 then primary=headNow end elseif headNow then primary=headNow end if primary==used then partNow=if primary==rootNow
then headNow else rootNow else partNow=primary end if partNow then flingBoost=1.5 stats.retries=(stats.retries or 0)+1 end end if not partNow or not partNow.Parent then flingBoost=1 break end used=
partNow sfBasePart(character,humanoid,root,partNow,targetCharacter,targetPlayer,targetHumanoid,targetRoot,if pass>1 then maxDuration*0.5 else maxDuration,shouldStop,stats)flingBoost=1 stats.speed=math
.max(stats.speed or 0,partSpeed(targetRoot),partSpeed(targetHead))if stats.speed>100 then break end end stats.speed=math.max(stats.speed or 0,partSpeed(targetRoot),partSpeed(targetHead))end)flingBoost
=1 destroyPin(pin)if fullFlingPin==pin then fullFlingPin=nil end pcall(function():()humanoid:SetStateEnabled(Enum.HumanoidStateType.Seated,true)end)pcall(function():()workspace.
FallenPartsDestroyHeight=savedDestroyHeight end)local finalCharacter:Model?,finalHumanoid:Humanoid?,finalRoot:BasePart?=getCharacterParts()local cancelled:boolean=not engineActive or(skipReturnOnStop
and shouldStop())if camera then pcall(function():()camera.CameraSubject=if cancelled then(finalHumanoid or humanoid)else humanoid end)end local function stopBodyMotion(body:Model):()for _,descendant:
Instance in ipairs(body:GetChildren())do if descendant:IsA('BasePart')then local bodyPart:BasePart=descendant::BasePart bodyPart.Velocity=Vector3.zero bodyPart.RotVelocity=Vector3.zero end end end if
cancelled then pcall(stopBodyMotion,character)elseif finalCharacter and finalRoot and finalRoot.Parent then local attempts:number=0 repeat if not engineActive or(skipReturnOnStop and shouldStop())then
pcall(stopBodyMotion,character)break end attempts+=1 pcall(function():()finalRoot.CFrame=oldPos*CFrame.new(0,0.5,0)finalCharacter:SetPrimaryPartCFrame(oldPos*CFrame.new(0,0.5,0))if finalHumanoid then
pcall(function():()finalHumanoid:ChangeState(Enum.HumanoidStateType.GettingUp)end)end stopBodyMotion(finalCharacter)end)task.wait()until(finalRoot.Position-oldPos.Position).Magnitude<25 or attempts>=
60 end setFlingRunning(false)local touchedTarget:boolean=stats.speed>100 pcall(function():()state.mm2LastFling={target=tostring(targetPlayer.Name),poses=stats.poses,speed=stats.speed,pinned=stats.
pinned,retries=stats.retries,power=powerScale(),ok=success}end)if not Module.Initialized then return touchedTarget end if not silent then local speedText:string=tostring(math.floor(stats.speed))if not
success then notify('fling failed: '..tostring(errorMessage))elseif touchedTarget then notify(string.format('fling: %s sent flying (%s studs/s, %d poses%s)',tostring(targetPlayer.Name),speedText,stats
.poses,(stats.retries or 0)>0 and string.format(', %d retries',stats.retries)or''))else notify(string.format('fling: %s did not move (%d poses, %s studs/s)',tostring(targetPlayer.Name),stats.poses,
speedText))end end return touchedTarget end local function guardedRun():boolean local savedPower:number=FLING_POWER local savedRetries:number=flingRetries if requestedPower then FLING_POWER=
requestedPower end if requestedRetries then flingRetries=requestedRetries end local ok:boolean,result:any=pcall(runFling)FLING_POWER,flingRetries=savedPower,savedRetries if not ok then
setFlingRunning(false)if not silent then notify('fling error: '..tostring(result))end return false end return result==true end if type(options)=='table'and options.blocking==true then return
guardedRun()end task.spawn(guardedRun)return true end state.mm2PerformFling=performFling state.mm2FlingStep=flingSingleStep activeCleanup=function():()engineActive=false destroyPin(fullFlingPin)
fullFlingPin=nil flingRunning=false stepPinToken+=1 destroyPin(stepPin)stepPin=nil setFlingRunning(false)state.mm2PerformFling=nil state.mm2FlingStep=nil end Module.Initialized=true return Module end
function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/MM2/Other/GetAllEmotes.lua
local Module={Name='MM2 Get All Emotes',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.
Initialized then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Get All Emotes requires the MM2 core module')Module.Runtime=runtime type EmoteDef={id:string,label:
string,command:string,animationId:string,looped:boolean?}local EMOTES:{EmoteDef}={{id='sit',label='Sit',command='sit2',animationId='rbxassetid://2431845940',looped=true},{id='ninja',label='Ninja Rest'
,command='ninja',animationId='rbxassetid://2431864798',looped=true},{id='zen',label='Zen',command='zen',animationId='rbxassetid://2431812646',looped=true},{id='dab',label='Dab',command='dab',
animationId='rbxassetid://2445521505'},{id='floss',label='Floss',command='floss',animationId='rbxassetid://2452938820',looped=true},{id='zombie',label='Zombie',command='zombie',animationId=
'rbxassetid://2513692312',looped=true},{id='headless',label='Headless',command='headless',animationId='rbxassetid://2513664073'}}local LABELS:{string}={}local ID_BY_LABEL:{[string]:string}={}local
KNOWN_IDS:{[string]:boolean}={}local DEFAULT_IDS:{[string]:boolean}={back=true,wave=true,cheer=true,laugh=true,point=true,dance=true,dance1=true,dance2=true,dance3=true}for _,def:EmoteDef in ipairs(
EMOTES)do table.insert(LABELS,def.label)ID_BY_LABEL[def.label]=def.id KNOWN_IDS[def.id]=true end local selectedLabel:string='Floss'local selectionTouched:boolean=false local activeTrack:any=nil local
feature:any=nil local generation:number=0 local grantInFlight:boolean=false local catalogInFlight:boolean=false local activated:boolean=false local function stopActiveTrack():()if activeTrack then
pcall(function():()activeTrack:Stop(0.1)end)activeTrack=nil end end local function animationUri(value:any):string?if type(value)=='number'and value>0 and value<math.huge then return'rbxassetid://'..
string.format('%.0f',value)end if type(value)=='string'and value~=''then return string.match(value,'^%d+$')and('rbxassetid://'..value)or value end return nil end local function discoverLiveEmotes(
ticket:number):()local rs:Instance=game:GetService('ReplicatedStorage')local db:Instance?=rs:FindFirstChild('Database')local sync:Instance?=db and db:FindFirstChild('Sync')local emotesModule:Instance?
=sync and sync:FindFirstChild('Emotes')if not emotesModule then return end local ok:boolean,catalog:any=pcall(require,emotesModule)if ticket~=generation or not Module.Initialized or not ok or type(
catalog)~='table'then return end local extra:{string}={}for id:any,live:any in pairs(catalog)do if type(id)=='string'and not KNOWN_IDS[id]and not DEFAULT_IDS[string.lower(id)]and type(live)=='table'
and type(live.Command)=='string'and live.Command~=''and not DEFAULT_IDS[string.lower(live.Command)]and animationUri(live.AnimationID)then table.insert(extra,id)end end table.sort(extra)for _,id:string
in ipairs(extra)do local live:any=catalog[id]local label:string=type(live.Name)=='string'and live.Name~=''and live.Name or id if ID_BY_LABEL[label]then label..=' ('..id..')'end local def:EmoteDef={id=
id,label=label,command=live.Command,animationId=animationUri(live.AnimationID)::string,looped=live.Loop==true}table.insert(EMOTES,def)table.insert(LABELS,label)ID_BY_LABEL[label]=id KNOWN_IDS[id]=true
end local values:any=type(configData)=='table'and configData.values or nil local saved:any=type(values)=='table'and values['MM2.GetAllEmotes.SelectEmote']or nil local index:number?=type(saved)==
'string'and table.find(LABELS,saved)or nil if not selectionTouched and index and ID_BY_LABEL[saved]then selectedLabel=saved local control:any=state.valueControls and state.valueControls[
'MM2.GetAllEmotes.SelectEmote']if control and type(control.setIndex)=='function'then control.setIndex(index)end end end local function emoteDefinition(id:string):EmoteDef?local result:EmoteDef?=nil
for _,def:EmoteDef in ipairs(EMOTES)do if def.id==id then result=table.clone(def)break end end if not result then return nil end pcall(function():()local rs:Instance=game:GetService(
'ReplicatedStorage')local db:Instance?=rs:FindFirstChild('Database')local sync:Instance?=db and db:FindFirstChild('Sync')local emotesModule:Instance?=sync and sync:FindFirstChild('Emotes')if not
emotesModule then return end local ok:boolean,catalog:any=pcall(require,emotesModule)local live:any=ok and type(catalog)=='table'and catalog[id]or nil if type(live)~='table'then return end local def:
EmoteDef=result::EmoteDef if type(live.Command)=='string'and live.Command~=''then def.command=live.Command end local asset:string?=animationUri(live.AnimationID)if asset then def.animationId=asset end
def.looped=live.Loop==true end)return result end local function refreshLegacyGui(rs:Instance,owned:{string}):boolean local globals:any=_G::any local gui:any=globals and globals.EmoteFrame if not gui
or type(globals.EmoteController)~='table'then return false end local valid:boolean,pages:any=pcall(function():any return gui:FindFirstChild('EmotePages')end)if not valid or not pages then return false
end local modules:Instance?=rs:FindFirstChild('Modules')local scriptModule:Instance?=modules and modules:FindFirstChild('EmoteModule')if not scriptModule then return false end local ok:boolean,
emoteModule:any=pcall(require,scriptModule)if not ok or type(emoteModule)~='table'or type(emoteModule.GenerateEmotes)~='function'then return false end emoteModule.EmoteGUI=gui return pcall(emoteModule
.GenerateEmotes,owned,gui)end local function grantShopEmotes(ticket:number):(boolean,number,boolean)if ticket~=generation or not Module.Initialized then return false,0,false end local ok:boolean,ready
:any,added:any,refreshed:any=pcall(function():(boolean,number,boolean)local rs:Instance=game:GetService('ReplicatedStorage')local modules:Instance?=rs:FindFirstChild('Modules')local profileModule:
Instance?=modules and modules:FindFirstChild('ProfileData')if not profileModule then return false,0,false end local profileOk:boolean,profile:any=pcall(require,profileModule)if ticket~=generation or
not Module.Initialized then return false,0,false end if not profileOk or type(profile)~='table'or type(profile.Emotes)~='table'or type(profile.Emotes.Owned)~='table'then return false,0,false end local
owned:{string}=profile.Emotes.Owned local known:{[string]:boolean}={}for _,existing:any in ipairs(owned)do if type(existing)=='string'then known[existing]=true end end local newIds:{string}={}for _,
def:EmoteDef in ipairs(EMOTES)do if not known[def.id]then table.insert(owned,def.id)known[def.id]=true table.insert(newIds,def.id)end end local remotes:Instance?=rs:FindFirstChild('Remotes')local
inventory:Instance?=remotes and remotes:FindFirstChild('Inventory')local changed:any=inventory and inventory:FindFirstChild('InventoryDataChanged')local signalled:boolean=false if changed then local
wakeIds:{string}=#newIds>0 and newIds or{EMOTES[1].id}for _,id:string in ipairs(wakeIds)do local fired:boolean=pcall(function():()changed:Fire('Emotes',id,1)end)signalled=signalled or fired end end if
not signalled then signalled=refreshLegacyGui(rs,owned)end return true,#newIds,signalled end)if not ok then return false,0,false end return ready==true,tonumber(added)or 0,refreshed==true end
local function requestGrant(showNotification:boolean):()if grantInFlight then if showNotification and type(notify)=='function'then pcall(notify,'Get All Emotes: waiting for the MM2 profile.')end
return end grantInFlight=true local ticket:number=generation task.spawn(function():()local ready:boolean,added:number,refreshed:boolean=grantShopEmotes(ticket)if ticket~=generation or not Module.
Initialized then return end grantInFlight=false if not ready then if feature then feature:SetStatus('profile not ready')end if showNotification and type(notify)=='function'then pcall(notify,
[[Get All Emotes: MM2 profile unavailable or not ready; try again after the game loads.]])end return end local count:number=#EMOTES if feature then feature:SetStatus(refreshed and(tostring(count)..
' local emotes')or(tostring(count)..' local \183 UI not refreshed'))end if showNotification and type(notify)=='function'then local suffix:string=refreshed and'The emote UI was notified.'or
[[The MM2 emote UI could not be notified; try again after it loads.]]pcall(notify,string.format([[Get All Emotes: %d local entries (%d new). %s Server ownership is not verified.]],count,added,suffix))
end if not catalogInFlight then catalogInFlight=true task.spawn(function():()if ticket~=generation or not Module.Initialized then catalogInFlight=false return end local previous:number=#EMOTES pcall(
discoverLiveEmotes,ticket)catalogInFlight=false if ticket==generation and Module.Initialized and#EMOTES>previous then requestGrant(false)end end)end end)end local function activate():()activated=true
if featureConnections.MM2GetAllEmotesRespawn or not(LocalPlayer and LocalPlayer.CharacterAdded)then return end featureConnections.MM2GetAllEmotesRespawn=LocalPlayer.CharacterAdded:Connect(function():(
)local ticket:number=generation task.delay(0.5,function():()if ticket==generation and Module.Initialized and activated then requestGrant(false)end end)end)end local function requestEmotePlayback(id:
string):boolean stopActiveTrack()local def:EmoteDef?=emoteDefinition(id)if not def then return false end local fired:boolean=false pcall(function():()local rs:Instance=game:GetService(
'ReplicatedStorage')local remotes:Instance?=rs:FindFirstChild('Remotes')local misc:Instance?=remotes and remotes:FindFirstChild('Misc')local event:any=misc and misc:FindFirstChild('PlayEmote')if event
then fired=pcall(function():()event:Fire(id)end)end end)if fired then return true end if def.animationId==''then return false end local played:boolean=false pcall(function():()local character:Model?=
LocalPlayer and LocalPlayer.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')if not humanoid or humanoid.Health<=0 then return end local anim:Animation=
Instance.new('Animation')anim.Name=def.command anim.AnimationId=def.animationId local animator:Animator?=humanoid:FindFirstChildOfClass('Animator')local loaded:boolean,track:any=pcall(function():any
return if animator then animator:LoadAnimation(anim)else(humanoid::any):LoadAnimation(anim)end)if loaded and track then activeTrack=track played=pcall(function():()track.Looped=def.looped==true track:
Play(0.1)end)end anim:Destroy()end)return played end local function playSelected():()activate()requestGrant(false)local id:string=ID_BY_LABEL[selectedLabel]or'floss'if not requestEmotePlayback(id)and
type(notify)=='function'then pcall(notify,[[Could not request that emote; check the character and animation permissions.]])end end feature=createUniversalFeature('Get All Emotes',
[[Adds MM2's playable non-default emotes locally, including Sit, Ninja Rest, Zen, Dab, Floss, Zombie and Headless. ]]..
[[It does not grant server ownership; playback and replication depend on the game.]],11,function():()activate()requestGrant(true)end,{categoryName='Utilities',parent=MM2Scroll,registry=mm2Features,
action=true,noOptions=false,configKey='MM2.GetAllEmotes'})state.mm2GetAllEmotesFeature=feature local cycleRestored:boolean=false addCycleOption(feature,'Select Emote',LABELS,5,function(value:string):(
)selectedLabel=value if not cycleRestored then cycleRestored=true return end selectionTouched=true playSelected()end,[[Select and request an MM2 emote; server ownership is not changed.]])
addActionOption(feature,'Play Emote',playSelected,'Request the selected emote again.')addInformationOption(feature,
[[MM2's playable emotes are added locally. Existing inventory is preserved; avatar defaults ]]..[[are not added, and the server controls ownership and replication.]])activeCleanup=function():()
generation+=1 activated=false disconnectFeatureConnection('MM2GetAllEmotesRespawn')stopActiveTrack()if state.mm2GetAllEmotesFeature==feature then state.mm2GetAllEmotesFeature=nil end end Module.Events
=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function(
):()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Other/GunAura.lua
local Module={Name='MM2 Gun Aura',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Gun Aura requires the MM2 core module')Module.Runtime=runtime local settings={studs=25,showHitbox=true}local
TICK_INTERVAL:number=0.08 local SCAN_INTERVAL:number=0.4 local ESP_INTERVAL:number=0.5 local gunAuraActive=false local gunHitboxes:{[Instance]:BasePart}={}local espParts:{[Player]:BasePart}={}local
savedGunStates:{[BasePart]:{cframe:CFrame,canCollide:boolean}}={}local trackedDrops:{[Instance]:boolean}={}local lastScanAt:number=-math.huge local lastTickAt:number=-math.huge local lastEspAt:number=
-math.huge local function resolveGunTouchPart(gun:Instance?):BasePart?if not gun then return nil end if gun:IsA('BasePart')then return gun::BasePart end local handle:Instance?=gun:FindFirstChild(
'Handle')or gun:FindFirstChild('GunDrop')or gun:FindFirstChildWhichIsA('BasePart')if handle and handle:IsA('BasePart')then return handle::BasePart end return nil end local function
rememberGunPartState(part:BasePart):{cframe:CFrame,canCollide:boolean}local existing=savedGunStates[part]if existing then return existing end local snap={cframe=part.CFrame,canCollide=part.CanCollide}
savedGunStates[part]=snap return snap end local function restoreGunPartState(part:BasePart):()local snap=savedGunStates[part]if not snap then return end if part and part.Parent then pcall(function():(
)part.CFrame=snap.cframe part.CanCollide=snap.canCollide end)end savedGunStates[part]=nil end local function restoreAllGunPartStates():()for part in pairs(savedGunStates)do restoreGunPartState(part)
end table.clear(savedGunStates)end local function trueGunPartPosition(part:BasePart):Vector3 local snap=savedGunStates[part]return if snap then snap.cframe.Position else part.Position end
local function canGrabGun():boolean local character=LocalPlayer and LocalPlayer.Character if not character then return false end local humanoid=character:FindFirstChildOfClass('Humanoid')if not
humanoid or humanoid.Health<=0 then return false end if type(core.getPlayerRole)=='function'then local role=core.getPlayerRole(LocalPlayer)if role=='Murderer'or role=='Sheriff'or role=='Hero'then
return false end end local backpack=LocalPlayer:FindFirstChildOfClass('Backpack')if(backpack and(backpack:FindFirstChild('Gun')or backpack:FindFirstChild('Knife')))or character:FindFirstChild('Gun')or
character:FindFirstChild('Knife')then return false end return true end local function isValidGunDrop(obj:Instance?):boolean if not obj or not obj.Parent then return false end if obj.Name=='GunDrop'and
(obj:IsA('BasePart')or obj:IsA('Model'))then return resolveGunTouchPart(obj)~=nil end return false end local function createGunHitbox(gun:Instance):BasePart?local handle=resolveGunTouchPart(gun)if not
handle then return nil end rememberGunPartState(handle)local pos=trueGunPartPosition(handle)if not pos then return nil end local part=Instance.new('Part')part.Name='WurstGunAuraHitbox'part.Size=
Vector3.new(settings.studs,settings.studs,settings.studs)part.Transparency=0.65 part.BrickColor=BrickColor.new('White')part.Material=Enum.Material.Neon part.CanCollide=false part.CanQuery=false part.
CanTouch=false part.Anchored=true part.CFrame=CFrame.new(pos)part.Parent=workspace return part end local function refreshTrackedDrops(now:number):()if now-lastScanAt<SCAN_INTERVAL then return end
lastScanAt=now local wsDrop:Instance?=workspace:FindFirstChild('GunDrop')if wsDrop and isValidGunDrop(wsDrop)then trackedDrops[wsDrop]=true end local okChildren:boolean,children:any=pcall(function():
any return workspace:GetChildren()end)if okChildren and type(children)=='table'then for _,child:Instance in ipairs(children)do if child.Name=='GunDrop'and isValidGunDrop(child)then trackedDrops[child]
=true elseif child:IsA('Model')or child:IsA('Folder')then local gd:Instance?=child:FindFirstChild('GunDrop')if gd and isValidGunDrop(gd)then trackedDrops[gd]=true end end end end if type(core.
findDroppedGun)=='function'then local okCore:boolean,drop:any=pcall(core.findDroppedGun)if okCore and drop and isValidGunDrop(drop)then trackedDrops[drop]=true end end end local function isWithinAura(
truePos:Vector3,rootPos:Vector3):boolean local diff=rootPos-truePos local radius=settings.studs if diff.X*diff.X+diff.Y*diff.Y+diff.Z*diff.Z<=radius*radius then return true end local half=radius*0.5
return math.abs(diff.X)<=half and math.abs(diff.Y)<=math.max(half,8)and math.abs(diff.Z)<=half end local function tryGrabGun(root:BasePart,handle:BasePart):()rememberGunPartState(handle)handle.
CanCollide=false local rootCF:CFrame?=root.CFrame if rootCF then handle.CFrame=rootCF end local fireTouch:any=state.fireTouchInterest if type(fireTouch)=='function'then pcall(fireTouch,root,handle,0)
pcall(fireTouch,root,handle,1)end end local function clearAllHitboxes():()for gun,hitbox in pairs(gunHitboxes)do if hitbox and hitbox.Parent then pcall(function():()hitbox:Destroy()end)end gunHitboxes
[gun]=nil end end local function clearAllEsp():()for player,esp in pairs(espParts)do if esp and esp.Parent then pcall(function():()esp:Destroy()end)end espParts[player]=nil end end local function
updateSheriffESP(player:Player):()local character=player.Character local root=character and character:FindFirstChild('HumanoidRootPart')local humanoid=character and character:FindFirstChildOfClass(
'Humanoid')if not character or not root or not root:IsA('BasePart')or not humanoid or humanoid.Health>0 then local existing=espParts[player]if existing then if existing.Parent then pcall(function():()
existing:Destroy()end)end espParts[player]=nil end return end local esp=espParts[player]if not esp or not esp.Parent then esp=Instance.new('Part')esp.Name='WurstDeadSheriffESP_'..player.Name esp.Size=
Vector3.new(3,3,3)esp.Transparency=0.3 esp.BrickColor=BrickColor.new('White')esp.Material=Enum.Material.Neon esp.CanCollide=false esp.CanQuery=false esp.CanTouch=false esp.Anchored=true esp.Parent=
workspace espParts[player]=esp end esp.CFrame=(root::BasePart).CFrame end local GunAuraFeature:any=nil local lastStatus:string?=nil local function setStatus(text:string?):()if text==lastStatus then
return end lastStatus=text if GunAuraFeature then GunAuraFeature:SetStatus(text)end end local function gunAuraTick():()if not gunAuraActive then return end local now:number=os.clock()if now-lastTickAt
<TICK_INTERVAL then return end lastTickAt=now refreshTrackedDrops(now)local character=LocalPlayer and LocalPlayer.Character local root:BasePart?=character and character:FindFirstChild(
'HumanoidRootPart')::BasePart?local canGrab=canGrabGun()local activeGuns:{[Instance]:boolean}={}local touchedParts:{[BasePart]:boolean}={}local gunCount:number=0 for gun in pairs(trackedDrops)do if
isValidGunDrop(gun)then local handle:BasePart?=resolveGunTouchPart(gun)if handle then gunCount+=1 activeGuns[gun]=true rememberGunPartState(handle)local truePos:Vector3?=trueGunPartPosition(handle)if
settings.showHitbox then local hitbox=gunHitboxes[gun]if not hitbox or not hitbox.Parent then hitbox=createGunHitbox(gun)if hitbox then gunHitboxes[gun]=hitbox end elseif truePos then local targetSize
=Vector3.new(settings.studs,settings.studs,settings.studs)if hitbox.Size~=targetSize then hitbox.Size=targetSize end hitbox.CFrame=CFrame.new(truePos)end end if canGrab and root and truePos and
isWithinAura(truePos,root.Position)then touchedParts[handle]=true pcall(tryGrabGun,root,handle)end end else trackedDrops[gun]=nil end end for part in pairs(savedGunStates)do if not touchedParts[part]
then restoreGunPartState(part)end end for gun,hitbox in pairs(gunHitboxes)do if not activeGuns[gun]then if hitbox and hitbox.Parent then pcall(function():()hitbox:Destroy()end)end gunHitboxes[gun]=nil
end end if gunCount>0 then setStatus('Gun detected \183 '..tostring(math.round(settings.studs))..' studs')else setStatus(tostring(math.round(settings.studs))..' studs')end if now-lastEspAt>=
ESP_INTERVAL and type(core.getPlayerRole)=='function'then lastEspAt=now local seenSheriffs:{[Player]:boolean}={}for _,player in ipairs(game:GetService('Players'):GetPlayers())do if player~=LocalPlayer
then local role=core.getPlayerRole(player)if role=='Sheriff'then seenSheriffs[player]=true updateSheriffESP(player)end end end for player,esp in pairs(espParts)do if not seenSheriffs[player]then if
esp and esp.Parent then pcall(function():()esp:Destroy()end)end espParts[player]=nil end end end end local function toggleGunAura(enabled:boolean):()gunAuraActive=enabled state.mm2GunAuraEnabled=
enabled==true disconnectFeatureConnection('MM2GunAura')disconnectFeatureConnection('MM2GunAuraDescendant')if not enabled then clearAllHitboxes()clearAllEsp()restoreAllGunPartStates()table.clear(
trackedDrops)setStatus(nil)return end lastScanAt=-math.huge lastTickAt=-math.huge lastEspAt=-math.huge if workspace.DescendantAdded then featureConnections.MM2GunAuraDescendant=workspace.
DescendantAdded:Connect(function(obj:Instance):()if obj.Name=='GunDrop'then trackedDrops[obj]=true end end)end setStatus(tostring(math.round(settings.studs))..' studs')featureConnections.MM2GunAura=
TaskManager:Connect(gunAuraTick,nil,'MM2 Gun Aura')gunAuraTick()end GunAuraFeature=createUniversalFeature('Gun Aura',
[[Detects dropped guns in the map with event-driven tracking, draws a white aura box around them, and automatically grabs the gun within Studs range.]],7,toggleGunAura,{categoryName='Utilities',parent
=MM2Scroll,registry=mm2Features})state.mm2GunAuraFeature=GunAuraFeature addNumberOption(GunAuraFeature,'Studs',25,5,250,function(value:number):()settings.studs=math.clamp(tonumber(value)or 25,5,250)if
GunAuraFeature.enabled then setStatus(tostring(math.round(settings.studs))..' studs')end end,[[Distance in studs around each dropped gun to automatically grab it.]],1)addToggleOption(GunAuraFeature,
'Show Hitbox',true,function(value:boolean):()settings.showHitbox=value==true if not settings.showHitbox then clearAllHitboxes()end end,
[[Shows or hides the white aura hitbox square around dropped guns.]])activeCleanup=function():()toggleGunAura(false)end Module.Events=featureConnections Module.Initialized=true return Module end
function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Other/Notifications.lua
local Module={Name='MM2 Notifications',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.
Initialized then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Notifications requires the MM2 core')Module.Runtime=runtime if type(configData)=='table'and type(
configData.states)=='table'then local states:any=configData.states local oldKey:string='MM2.BlurtRoles.InstantRoles'local newKey:string='MM2.Notifications.InstantRoles'local old:any=states[oldKey]
local changed:boolean=false if states[newKey]==nil and type(old)=='boolean'then states[newKey]=old if old and states['MM2.Notifications']==nil then states['MM2.Notifications']=true end changed=true
end if old~=nil then states[oldKey]=nil changed=true end if changed and type(queueConfigSave)=='function'then queueConfigSave()end end local settings:any={instantRoles=false,gun=true,murderClose=true,
yourRole=true,roundEvents=true,murderDefeated=true,sheriffDown=true,closeRadius=30,closeCooldown=8}local enabled:boolean=false local phase:string=type(core.getRoundPhase)=='function'and core.
getRoundPhase()or'unknown'local elapsed:number=0 local murdererSeen:boolean=false local sheriffSeen:boolean=false local lastLocalRole:string?=nil local lastMurderer:Player?=nil local lastSheriff:
Player?=nil local defeatNotified:boolean=false local sheriffNotified:boolean=false local nearMurderer:boolean=false local lastCloseAt:number=-math.huge local gunWasDropped:boolean=false local
gunLastSeenAt:number=-math.huge local gunDropPreviousHolder:Player?=nil local lastGunHolder:Player?=nil local gunHolderInitialized:boolean=false local function resetRound():()murdererSeen=false
sheriffSeen=false lastLocalRole=nil lastMurderer=nil lastSheriff=nil defeatNotified=false sheriffNotified=false nearMurderer=false lastCloseAt=-math.huge gunWasDropped=false gunLastSeenAt=-math.huge
gunDropPreviousHolder=nil lastGunHolder=nil gunHolderInitialized=false end local function inRound():boolean if type(core.isRoundLifecycleActive)=='function'and core.isRoundLifecycleActive()==false
then return false end if phase=='lobby'then return false end return phase=='map'or(type(core.hasActiveRoundRoles)=='function'and core.hasActiveRoundRoles())end local function rootOf(player:Player?):
BasePart?local character:Model?=player and player.Character return character and character:FindFirstChild('HumanoidRootPart')::BasePart?or nil end local function isAlive(player:Player?):boolean if not
player then return false end if type(core.isPlayerAlive)=='function'then return core.isPlayerAlive(player)==true end return rootOf(player)~=nil end local function showRoles(murderer:Player?,sheriff:
Player?):()if not enabled or not settings.instantRoles or not inRound()then return end if not murdererSeen and not sheriffSeen and murderer and sheriff then murdererSeen=true sheriffSeen=true notify(
'Roles ready - Murderer: '..murderer.Name..' | Sheriff/Hero: '..sheriff.Name)return end if murderer and not murdererSeen then murdererSeen=true notify('Murderer: '..murderer.Name)end if sheriff and
not sheriffSeen then sheriffSeen=true notify('Sheriff/Hero: '..sheriff.Name)end end local function gunDistance(drop:Instance?):number?local root:BasePart?=rootOf(LocalPlayer)if not root or not drop
then return nil end local ok:boolean,distance:any=pcall(function():number local position:Vector3=if drop:IsA('BasePart')then(drop::BasePart).Position else(drop::Model):GetPivot().Position return(
position-root.Position).Magnitude end)return if ok and type(distance)=='number'then distance else nil end local function updateGun(sheriff:Player?):()if not settings.gun or type(core.findDroppedGun)~=
'function'then return end local now:number=os.clock()local drop:Instance?=core.findDroppedGun()local holder:Player?=nil if sheriff and isAlive(sheriff)and type(core.getPlayerWeapon)=='function'and
core.getPlayerWeapon(sheriff,'Gun')~=nil then holder=sheriff end if drop then if not gunWasDropped then gunDropPreviousHolder=lastGunHolder or holder local distance:number?=gunDistance(drop)notify(
'Gun dropped'..(distance and(' \183 '..tostring(math.round(distance))..' studs away')or''),'danger')end gunWasDropped=true gunLastSeenAt=now elseif gunWasDropped and holder and holder~=
gunDropPreviousHolder then notify('Gun picked up by '..holder.Name,'good')gunWasDropped=false gunDropPreviousHolder=nil elseif gunWasDropped and now-gunLastSeenAt>=2 then gunWasDropped=false
gunDropPreviousHolder=nil elseif holder and gunHolderInitialized and lastGunHolder and holder~=lastGunHolder then notify('Gun is now held by '..holder.Name)end if holder then lastGunHolder=holder
gunHolderInitialized=true end end local function confirmedDead(player:Player?):boolean if not player then return false end if type(core.getRoundRole)=='function'and core.getRoundRole(player)=='Dead'
then return true end local character:Model?=player.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')::Humanoid?return humanoid~=nil and humanoid.Health<=0
end local COMPASS:{string}={'N','NE','E','SE','S','SW','W','NW'}local function compassOf(x:number?,z:number?):string local flatX:number=tonumber(x)or 0 local flatZ:number=tonumber(z)or 0 if math.abs(
flatX)+math.abs(flatZ)<1e-3 then return'here'end local angle:number=math.deg(math.atan2(flatX,flatZ))local index:number=math.floor(((angle+360+22.5)%360)/45)+1 return COMPASS[index]end local function
updateProximity(murderer:Player?):()if not settings.murderClose or not murderer or murderer==LocalPlayer or not isAlive(LocalPlayer)or not isAlive(murderer)then nearMurderer=false return end local
localRoot:BasePart?=rootOf(LocalPlayer)local murderRoot:BasePart?=rootOf(murderer)if not localRoot or not murderRoot then nearMurderer=false return end local delta:Vector3=murderRoot.Position-
localRoot.Position local distance:number=delta.Magnitude local radius:number=math.clamp(tonumber(settings.closeRadius)or 30,10,120)local cooldown:number=math.clamp(tonumber(settings.closeCooldown)or 8
,2,30)local now:number=os.clock()if distance<=radius and now-lastCloseAt>=cooldown then nearMurderer=true lastCloseAt=now notify('Murderer close: '..murderer.Name..' \183 '..tostring(math.round(
distance))..' studs'..' '..compassOf(delta.X,delta.Z)..(distance<=radius*0.4 and' \183 VERY close'or''),'danger')elseif distance>radius*1.25 then nearMurderer=false end end local function sample():()
if not enabled or not inRound()then return end local murderer:Player?=type(core.findMurderer)=='function'and core.findMurderer()or nil local sheriff:Player?=type(core.findSheriff)=='function'and core.
findSheriff()or nil if murderer then lastMurderer=murderer end if sheriff then lastSheriff=sheriff end showRoles(murderer,sheriff)if settings.yourRole and type(core.getPlayerRole)=='function'then
local myRole:any=core.getPlayerRole(LocalPlayer)if(myRole=='Innocent'or myRole=='Sheriff'or myRole=='Hero'or myRole=='Murderer')and myRole~=lastLocalRole then lastLocalRole=myRole notify('Your role: '
..myRole)end end updateGun(sheriff)updateProximity(murderer)if settings.murderDefeated and not defeatNotified and confirmedDead(lastMurderer)then defeatNotified=true notify('Murderer defeated: '..(
lastMurderer::Player).Name,'good')end if settings.sheriffDown and not sheriffNotified and lastSheriff~=nil and lastSheriff~=LocalPlayer and confirmedDead(lastSheriff)then sheriffNotified=true notify(
'Sheriff down: '..(lastSheriff::Player).Name..' \183 the gun is loose','danger')end end local nextWarningAt:number=-math.huge local sampling:boolean=false local sampleAgain:boolean=false
local function trySample():()if sampling then sampleAgain=true return end sampling=true for _pass:number=1,2 do sampleAgain=false local ok:boolean,err:any=pcall(sample)if not ok and os.clock()>=
nextWarningAt then nextWarningAt=os.clock()+5 warn('MM2 Notifications: '..tostring(err))end if not sampleAgain or not enabled then break end end sampling=false end local unsubscribeRoles:()->()=core.
onRoundRoles(function(active:boolean):()if active and enabled then trySample()end end)local unsubscribePhase:()->()=core.onRoundPhase(function(nextPhase:string,reason:string):()local previous:string=
phase phase=nextPhase if nextPhase=='lobby'and previous~='lobby'then if enabled and settings.roundEvents and previous=='map'and reason~='subscribe'then notify('Round ended')end resetRound()elseif
nextPhase=='map'and previous~='map'then resetRound()if enabled and settings.roundEvents and reason~='subscribe'then notify('Entered the map - round starting')end if enabled then trySample()end end end
)local function toggleNotifications(value:boolean):()enabled=value==true disconnectFeatureConnection('MM2NotificationsTick')elapsed=0 if not enabled then return end trySample()featureConnections.
MM2NotificationsTick=TaskManager:Connect(function(deltaTime:number):()elapsed+=math.max(0,deltaTime)if elapsed<0.25 then return end elapsed=0 trySample()end,nil,'MM2 Notifications')end local feature:
any=createUniversalFeature('Notifications',[[Private MM2 alerts for roles, the gun, nearby danger and rounds.]],4,toggleNotifications,{categoryName='Utilities',parent=MM2Scroll,registry=mm2Features})
addToggleOption(feature,'Instant Roles',false,function(value:boolean):()settings.instantRoles=value core.mm2Settings.instantRoleNotify=value if enabled and value then trySample()end end,
[[Private role alerts as soon as each role is known; never sends public chat.]])addToggleOption(feature,'Gun Notification',true,function(value:boolean):()settings.gun=value if not value then
gunWasDropped=false gunDropPreviousHolder=nil lastGunHolder=nil gunHolderInitialized=false elseif enabled then trySample()end end,'Announces a dropped gun and a confirmed new holder.')addToggleOption(
feature,'Murder Close',true,function(value:boolean):()settings.murderClose=value if enabled and value then trySample()end end,
[[Warns when the living Murderer gets close, with the distance and the compass side. Off = no proximity alerts.]])addNumberOption(feature,'Close radius',30,10,120,function(value:number):()settings.
closeRadius=math.clamp(tonumber(value)or 30,10,120)nearMurderer=false lastCloseAt=-math.huge if enabled and settings.murderClose then trySample()end end,
[[Studs at which the Murder Close warning fires. The alert repeats when you get closer or after the cooldown.]],10,true)addNumberOption(feature,'Close cooldown',8,2,30,function(value:number):()
settings.closeCooldown=math.clamp(tonumber(value)or 8,2,30)end,'Minimum seconds between two Murder Close warnings.',1,true)addToggleOption(feature,'Sheriff Down',true,function(value:boolean):()
settings.sheriffDown=value if enabled and value then trySample()end end,[[Announces the known Sheriff/Hero death: that is when the gun is loose.]])addToggleOption(feature,'Your Role',true,function(
value:boolean):()settings.yourRole=value if enabled and value then trySample()end end,'Notifies you when your role becomes known or changes.')addToggleOption(feature,'Round Events',true,function(value
:boolean):()settings.roundEvents=value end,'Private alerts for entering the map and the round ending.')addToggleOption(feature,'Murderer Defeated',true,function(value:boolean):()settings.
murderDefeated=value if enabled and value then trySample()end end,'Warns only when the known Murderer is confirmed dead.')addInformationOption(feature,
[[These alerts are private. Blurt Roles is the separate public-chat feature.]])activeCleanup=function():()toggleNotifications(false)pcall(unsubscribeRoles)pcall(unsubscribePhase)core.mm2Settings.
instantRoleNotify=false end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false
activeCleanup()activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Prediction.lua
local Module={Name='MM2 Prediction',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}type Sample={at:number,position:Vector3,velocity:Vector3,state:string,grounded:boolean,aimOffset:
Vector3,aimPart:string,moveDirection:Vector3}type Track={character:Model,humanoid:Humanoid,root:BasePart,samples:{Sample},aimPart:BasePart?,aimName:string?,partsAt:number?,spare:{Sample}?,
knifeFloorCache:{any}?,knifeFloorNext:number?,knifeFloorSerial:number?}type Fit={velocity:Vector3,residual:number,count:number,span:number,center:number}type Motion={velocity:Vector3,horizontal:
Vector3,vertical:number,airborne:boolean,sampleAge:number,samples:number,residual:number,source:string,turnRate:number,longitudinalAccel:number,accelX:number,accelZ:number,brakeSeconds:number?,fitLag:
number,verticalLead:boolean,moveDirection:Vector3?,reportedHorizontal:Vector3?,hasInputHistory:boolean?}type AimInfo={endpoint:Vector3,targetPosition:Vector3,capturedPosition:Vector3,rootPosition:
Vector3,velocity:Vector3,leadSeconds:number,turnRate:number,brakeSeconds:number?,sampleAge:number,samples:number,residual:number,motionSource:string,model:string,rtt:number,rttEngine:number?,rttData:
number?,echoDelay:number?,echoSamples:number,leadCorrection:number,leadCorrectionTarget:number?,leadSamples:number?,aimMode:string?,flightSeconds:number?}local MAX_SAMPLE_AGE:number=0.35 local
FIT_WINDOW:number=0.16 local FIT_TAU:number=0.055 local HISTORY_SECONDS:number=0.6 local SAMPLE_HZ:number=60 local MAX_HORIZONTAL_SPEED:number=60 local DEFAULT_KNIFE_SPEED:number=96 local MAX_ECHOES:
number=8 local MIN_ECHO_SAMPLES:number=3 local MIN_ECHO_DELAY:number=0.05 local MAX_ECHO_DELAY:number=1.6 local ECHO_MAX_AGE:number=30 local ECHO_DIVERGENCE_RATIO:number=1.5 local MAX_HORIZON:number=1
local MAX_AUTO_LEAD:number=0.6 local TURN_SPAN:number=0.1 local TURN_MIN_SPEED:number=2 local TURN_MAX_RATE:number=10 local TURN_MIN_RATE:number=0.4 local TURN_MAX_ANGLE:number=math.rad(45)local
TURN_RESIDUAL:number=0.2 local VERTICAL_MIN_SPEED:number=8 local MAX_FLOOR_LIFT_STUDS:number=6 local RTT_SAMPLES:number=8 local RTT_SAMPLE_INTERVAL:number=0.25 local BRAKE_MIN_ACCEL:number=6 local
BRAKE_MAX_ACCEL:number=120 local LEAD_FEEDBACK_MIN_SAMPLES:number=3 local LEAD_FEEDBACK_MAX_SAMPLES:number=8 local LEAD_FEEDBACK_SAMPLE_LIMIT:number=0.15 local MAX_LEAD_CORRECTION:number=0.06 local
LEAD_SLEW_PER_SAMPLE:number=0.02 local settings={autoLead=true,extraLead=0,interpolation=1/40}local activeCleanup:()->()=function():()end local function finiteNumber(value:any):boolean return type(
value)=='number'and value==value and math.abs(value)<math.huge end local function finiteVector(value:any):boolean return value~=nil and finiteNumber(value.X)and finiteNumber(value.Y)and finiteNumber(
value.Z)end local function humanoidStateName(humanoid:Humanoid):string local state:any=humanoid:GetState()return state.Name or tostring(state)end local function bodyIsAlive(humanoid:Humanoid?):boolean
if not humanoid or humanoid.Health<=0 then return false end local currentState:Enum.HumanoidStateType=humanoid:GetState()if currentState==Enum.HumanoidStateType.Dead or currentState==Enum.
HumanoidStateType.Physics then return false end if humanoid:GetAttribute('Dead')==true then return false end local parent:Instance?=humanoid.Parent return not parent or parent:GetAttribute('Dead')~=
true end local function targetHumanoid(target:Player):any local ok:boolean,humanoid:any=pcall(function():any local character=target.Character return character and character:FindFirstChildOfClass(
'Humanoid')end)if not ok then return nil end return humanoid end local AIM_PARTS:{string}={'UpperTorso','Torso','HumanoidRootPart','Head'}local function findAimPart(character:Model):(BasePart?,string)
for _,partName:string in ipairs(AIM_PARTS)do local part:Instance?=character:FindFirstChild(partName)if part and part:IsA('BasePart')then return part::BasePart,partName end end local root:Instance?=
character:FindFirstChild('HumanoidRootPart')if root and root:IsA('BasePart')then return root::BasePart,'HumanoidRootPart'end return nil,''end function Module.init(runtime:any):any if Module.
Initialized then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Prediction requires the MM2 core module')local predictionMath:any=core.predictionMath assert(type(
predictionMath)=='table','MM2 Prediction requires predictionMath')Module.Runtime=runtime local tracks:{[Player]:Track}=setmetatable({},{__mode='k'})::any local echoes:{number}={}local echoTimes:{
number}={}local elapsed:number=0 local PART_REFRESH_SECONDS:number=0.5 local function refreshTrackParts(track:Track,character:Model,now:number):boolean local refresh:boolean=now-(track.partsAt or-math
.huge)>=PART_REFRESH_SECONDS or track.root==nil or track.humanoid==nil if not refresh then return true end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')::Humanoid?local root:
BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?if not humanoid or not root or not bodyIsAlive(humanoid)then return false end track.humanoid=humanoid track.root=root local aimPart:
BasePart?,aimName:string=findAimPart(character)track.aimPart=aimPart track.aimName=aimName track.partsAt=now return true end local function ensureTrack(player:Player,now:number):Track?local character:
Model?=player.Character if not character then tracks[player]=nil return nil end local track:Track?=tracks[player]if not track or track.character~=character then track={character=character,humanoid=nil
,root=nil,samples={},partsAt=-math.huge}tracks[player]=track end if not refreshTrackParts(track,character,now)then tracks[player]=nil return nil end return track end local function samplePlayer(player
:Player,now:number):()local track:Track?=ensureTrack(player,now)if not track then return end local humanoid:Humanoid?=track.humanoid if not bodyIsAlive(humanoid)then tracks[player]=nil return end
local root:BasePart=track.root local position:Vector3=root.Position local velocity:Vector3=root.AssemblyLinearVelocity if not finiteVector(velocity)then velocity=Vector3.zero end local stateName:
string=humanoidStateName(track.humanoid)local grounded:boolean=stateName~='Jumping'and stateName~='Freefall'local move:any=track.humanoid.MoveDirection local moveDirection:Vector3=if typeof(move)==
'Vector3'and finiteVector(move)then move else Vector3.zero local aimPart:BasePart?=track.aimPart local aimPartName:string=track.aimName or''local aimOffset:Vector3=aimPart and(aimPart.Position-
position)or Vector3.zero local samples:{Sample}=track.samples local last:Sample?=samples[#samples]if last then local delta:number=now-last.at if delta>0 and delta<0.4 and(position-last.position).
Magnitude>8+120*delta then local spare:{Sample}=track.spare or{}track.spare=spare for _,dropped:Sample in ipairs(samples)do if#spare<8 then table.insert(spare,dropped)end end table.clear(samples)end
end local spare:{Sample}=track.spare or{}track.spare=spare local sample:Sample=table.remove(spare)or{}::Sample sample.at=now sample.position=position sample.velocity=velocity sample.state=stateName
sample.grounded=grounded sample.aimOffset=aimOffset sample.aimPart=aimPartName sample.moveDirection=moveDirection table.insert(samples,sample)while#samples>2 and now-samples[1].at>HISTORY_SECONDS do
local retired:Sample=table.remove(samples,1)if#spare<8 then table.insert(spare,retired)end end end local function fitVelocity(track:Track,reference:number):Fit?local samples:{Sample}=track.samples if#
samples<3 then return nil end local collected:{Sample}={}for index:number=#samples,1,-1 do local sample:Sample=samples[index]if sample.at<=reference+1e-4 then if reference-sample.at>FIT_WINDOW then
break end table.insert(collected,1,sample)end end if#collected<3 then return nil end local anchor:number=collected[#collected].at local sw,st,stt:number=0,0,0 local sx,sy,sz:number=0,0,0 local stx,sty
,stz:number=0,0,0 for _,sample:Sample in ipairs(collected)do local dt:number=sample.at-anchor local age:number=math.max(0,reference-sample.at)local weight:number=math.exp(-age/FIT_TAU)sw+=weight st+=
weight*dt stt+=weight*dt*dt sx+=weight*sample.position.X sy+=weight*sample.position.Y sz+=weight*sample.position.Z stx+=weight*dt*sample.position.X sty+=weight*dt*sample.position.Y stz+=weight*dt*
sample.position.Z end if sw<=0 then return nil end local denominator:number=stt-st*st/sw if denominator<=1e-8 then return nil end local meanT:number=st/sw local meanX:number=sx/sw local meanY:number=
sy/sw local meanZ:number=sz/sw local vx:number=(stx-st*sx/sw)/denominator local vy:number=(sty-st*sy/sw)/denominator local vz:number=(stz-st*sz/sw)/denominator local squared:number=0 for _,sample:
Sample in ipairs(collected)do local dt:number=sample.at-anchor local dx:number=sample.position.X-(meanX+vx*(dt-meanT))local dy:number=sample.position.Y-(meanY+vy*(dt-meanT))local dz:number=sample.
position.Z-(meanZ+vz*(dt-meanT))squared+=dx*dx+dy*dy+dz*dz end return{velocity=Vector3.new(vx,vy,vz),residual=math.sqrt(squared/#collected),count=#collected,span=collected[#collected].at-collected[1].
at,center=anchor+meanT}end local function measureTurn(track:Track,now:number,fit:Fit):(number,number,number?,number,number,number)local newest:Sample?=track.samples[#track.samples]local fitLag:number=
newest and math.max(0,newest.at-fit.center)or 0 if fit.count<3 or fit.span<0.06 then return 0,0,nil,fitLag,0,0 end local previous:Fit?=fitVelocity(track,now-TURN_SPAN)if not previous or previous.count
<3 or previous.span<0.04 then return 0,0,nil,fitLag,0,0 end local worstResidual:number=math.max(fit.residual,previous.residual)local confidence:number=1 if worstResidual>TURN_RESIDUAL then if
worstResidual>TURN_RESIDUAL*2.25 then return 0,0,nil,fitLag,0,0 end confidence=0.5 end local v0:Vector3=Vector3.new(previous.velocity.X,0,previous.velocity.Z)local v1:Vector3=Vector3.new(fit.velocity.
X,0,fit.velocity.Z)if v0.Magnitude<TURN_MIN_SPEED or v1.Magnitude<TURN_MIN_SPEED then return 0,0,nil,fitLag,0,0 end local dt:number=math.max(0.04,fit.center-previous.center)local cross:number=v0.Z*v1.
X-v0.X*v1.Z local dot:number=v0.X*v1.X+v0.Z*v1.Z local rate:number=math.clamp(math.atan2(cross,dot)/dt,-TURN_MAX_RATE,TURN_MAX_RATE)if math.abs(rate)<TURN_MIN_RATE then rate=0 end local ax:number=(v1.
X-v0.X)/dt*confidence local az:number=(v1.Z-v0.Z)/dt*confidence rate*=confidence if math.abs(rate)<TURN_MIN_RATE then rate=0 end local speed:number=v1.Magnitude local along:number=math.clamp((ax*v1.X+
az*v1.Z)/speed,-BRAKE_MAX_ACCEL,BRAKE_MAX_ACCEL)local brakeSeconds:number?=nil if along<=-BRAKE_MIN_ACCEL then local currentSpeed:number=math.max(0,speed+along*fitLag)brakeSeconds=currentSpeed/-along
end local accelMagnitude:number=math.sqrt(ax*ax+az*az)if accelMagnitude>BRAKE_MAX_ACCEL then local scale:number=BRAKE_MAX_ACCEL/accelMagnitude ax*=scale az*=scale end return rate,along,brakeSeconds,
fitLag,ax,az end local function estimateMotion(track:Track,now:number):Motion?local samples:{Sample}=track.samples local last:Sample?=samples[#samples]if not last then return nil end local age:number=
math.max(0,now-last.at)if age>MAX_SAMPLE_AGE then return nil end local fit:Fit?=fitVelocity(track,now)local reported:Vector3=last.velocity local horizontal:Vector3 local source:string local turnRate:
number=0 local longitudinalAccel:number=0 local accelX:number=0 local accelZ:number=0 local brakeSeconds:number?=nil local fitLag:number=0 local verticalLead:boolean=false if fit and fit.count>=3 and
fit.span>=0.04 and fit.residual<=1.5 then horizontal=Vector3.new(fit.velocity.X,0,fit.velocity.Z)source='history'else horizontal=Vector3.new(reported.X,0,reported.Z)source='replicated'end if not
finiteVector(horizontal)then horizontal=Vector3.zero elseif horizontal.Magnitude>MAX_HORIZONTAL_SPEED then horizontal=horizontal.Unit*MAX_HORIZONTAL_SPEED end local airborne:boolean=last.state==
'Jumping'or last.state=='Freefall'or math.abs(reported.Y)>=15 local vertical:number=0 if airborne then vertical=reported.Y if not finiteNumber(vertical)then vertical=0 elseif math.abs(vertical)<0.5
and fit then vertical=fit.velocity.Y end vertical=math.clamp(vertical,-120,120)end if fit then turnRate,longitudinalAccel,brakeSeconds,fitLag,accelX,accelZ=measureTurn(track,now,fit)end if airborne
and math.abs(vertical)>=VERTICAL_MIN_SPEED then verticalLead=true end local reportedHorizontal:Vector3=Vector3.new(reported.X,0,reported.Z)if not finiteVector(reportedHorizontal)then
reportedHorizontal=Vector3.zero elseif reportedHorizontal.Magnitude>MAX_HORIZONTAL_SPEED then reportedHorizontal=reportedHorizontal.Unit*MAX_HORIZONTAL_SPEED end local moveDir:Vector3=last.
moveDirection or Vector3.zero local moveHorizontal:Vector3=Vector3.new(moveDir.X,0,moveDir.Z)if not finiteVector(moveHorizontal)then moveHorizontal=Vector3.zero end local hasInputHistory:boolean=
moveHorizontal.Magnitude>0.08 if not hasInputHistory then for idx:number=math.max(1,#samples-15),#samples do local s:Sample=samples[idx]local md:Vector3=s.moveDirection or Vector3.zero if math.sqrt(md
.X*md.X+md.Z*md.Z)>0.08 then hasInputHistory=true break end end end return{velocity=Vector3.new(horizontal.X,vertical,horizontal.Z),horizontal=horizontal,vertical=vertical,airborne=airborne,sampleAge=
age,samples=#samples,residual=fit and fit.residual or math.huge,source=source,turnRate=turnRate,longitudinalAccel=longitudinalAccel,accelX=accelX,accelZ=accelZ,brakeSeconds=brakeSeconds,fitLag=fitLag,
verticalLead=verticalLead,moveDirection=moveHorizontal,reportedHorizontal=reportedHorizontal,hasInputHistory=hasInputHistory}end local function leadHorizontal(motion:Motion,horizon:number):Vector3 if
horizon<=0 then return Vector3.zero end local moveDir:Vector3?=motion.moveDirection local moveMag:number=moveDir and moveDir.Magnitude or 0 if motion.hasInputHistory then local repH:Vector3=motion.
reportedHorizontal or motion.horizontal local fitH:Vector3=motion.horizontal local v0:Vector3=if repH.Magnitude>0.05 then repH*0.65+fitH*0.35 else fitH local v0Mag:number=v0.Magnitude local effHorizon
:number=horizon*0.88 local tau:number=motion.airborne and 0.17 or 0.11 if moveMag>0.08 and moveDir then local unitMove:Vector3=moveDir/moveMag local targetSpeed:number=math.clamp(math.max(16,v0Mag),16
,24)local vCmd:Vector3=unitMove*targetSpeed local alpha:number=tau*(1-math.exp(-effHorizon/tau))return vCmd*effHorizon+(v0-vCmd)*alpha else local stopTau:number=0.1 local alpha:number=stopTau*(1-math.
exp(-effHorizon/stopTau))return v0*alpha end end local speed:number=motion.horizontal.Magnitude if speed<1e-4 then return Vector3.zero end local direction:Vector3=motion.horizontal.Unit local
accelMagnitude:number=math.sqrt(motion.accelX*motion.accelX+motion.accelZ*motion.accelZ)if accelMagnitude>=4 then local applied:Vector3=Vector3.new(motion.accelX,0,motion.accelZ)if accelMagnitude>60
then applied*=60/accelMagnitude end local displacement:Vector3=motion.horizontal*horizon+applied*(motion.fitLag*horizon+0.5*horizon*horizon)if motion.brakeSeconds and horizon>motion.brakeSeconds then
local alongDistance:number=displacement:Dot(direction)local along:number=math.min(0,motion.longitudinalAccel)local speedNow:number=math.max(0,speed+along*motion.fitLag)local stopDistance:number=
speedNow*motion.brakeSeconds*0.5 if alongDistance>stopDistance then displacement-=direction*(alongDistance-stopDistance)end end return displacement end if math.abs(motion.turnRate)>1e-3 then local
angle:number=math.clamp(motion.turnRate*(horizon*0.5+motion.fitLag),-TURN_MAX_ANGLE,TURN_MAX_ANGLE)local cosine:number=math.cos(angle)local sine:number=math.sin(angle)direction=Vector3.new(direction.X
*cosine+direction.Z*sine,0,-direction.X*sine+direction.Z*cosine)end local along:number=math.min(0,motion.longitudinalAccel)local speedNow:number=math.max(0,speed+along*motion.fitLag)local length:
number=speedNow*horizon+0.5*along*horizon*horizon length=math.max(0,length)if motion.brakeSeconds and horizon>motion.brakeSeconds then length=math.min(length,speedNow*motion.brakeSeconds*0.5)end
return direction*length end local function predictedRoot(track:Track,motion:Motion,horizon:number):Vector3 local last:Sample=track.samples[#track.samples]local lead:Vector3=leadHorizontal(motion,
horizon)local x:number=last.position.X+lead.X local z:number=last.position.Z+lead.Z local y:number=last.position.Y if motion.verticalLead then y+=motion.vertical*horizon-0.5*workspace.Gravity*horizon*
horizon if y<last.position.Y then for idx:number=#track.samples,1,-1 do local s:Sample=track.samples[idx]if s.grounded then if s.position.Y-y<=MAX_FLOOR_LIFT_STUDS then y=math.max(y,s.position.Y)end
break end end end end return Vector3.new(x,y,z)end local function currentRtt():number local latency:any=nil if type(core.getLatencyReadings)=='function'then local ok:boolean,engine:any,data:any=pcall(
core.getLatencyReadings)if ok and finiteNumber(data)then latency=data elseif ok and finiteNumber(engine)then latency=engine end end if not finiteNumber(latency)and type(core.getEstimatedLatency)==
'function'then local ok:boolean,value:any=pcall(core.getEstimatedLatency)if ok then latency=value end end if not finiteNumber(latency)then latency=0.08 end return math.clamp(latency,0.02,0.6)end local
rttSamples:{number}={}local lastRttSampleAt:number=-math.huge local function robustRtt():number local now:number=os.clock()if now-lastRttSampleAt>=RTT_SAMPLE_INTERVAL then lastRttSampleAt=now table.
insert(rttSamples,currentRtt())while#rttSamples>RTT_SAMPLES do table.remove(rttSamples,1)end end local sorted:{number}=table.clone(rttSamples)table.insert(sorted,currentRtt())table.sort(sorted)return
sorted[math.floor((#sorted+1)/2)]end local function pruneEchoes(now:number):()if not finiteNumber(now)then return end while#echoes>0 and now-echoTimes[1]>ECHO_MAX_AGE do table.remove(echoes,1)table.
remove(echoTimes,1)end end local function echoMedian():number?pruneEchoes(os.clock())local count:number=#echoes if count<MIN_ECHO_SAMPLES then return nil end local sorted:{number}=table.clone(echoes)
table.sort(sorted)local middle:number=(count+1)/2 local lower:number=math.floor(middle)local upper:number=math.ceil(middle)return(sorted[lower]+sorted[upper])/2 end local function noteEcho(sentAt:
number,receivedAt:number):()if not finiteNumber(sentAt)or not finiteNumber(receivedAt)then return end local delay:number=receivedAt-sentAt if delay<MIN_ECHO_DELAY or delay>MAX_ECHO_DELAY then return
end local plausible:number=math.clamp(currentRtt()*2.5+0.05,0.3,MAX_ECHO_DELAY)if delay>plausible then return end pruneEchoes(receivedAt)table.insert(echoes,delay)table.insert(echoTimes,receivedAt)
while#echoes>MAX_ECHOES do table.remove(echoes,1)table.remove(echoTimes,1)end local median:number?=echoMedian()if median and median>currentRtt()*ECHO_DIVERGENCE_RATIO then table.clear(echoes)table.
clear(echoTimes)end end local function manualLead():number return settings.autoLead and 0 or settings.extraLead end local leadErrors:{[string]:{number}}={}local leadApplied:{[string]:number}={}
local function modeKey(mode:any):string local text:string=type(mode)=='string'and mode or''local lower:string=string.lower(text)if lower=='normal'or lower=='bullet'then return'Bullet'end if lower==
'silent'or lower=='cframe'or lower=='silent aim'then return'CFrame'end if lower=='knife'or lower=='blade'then return'Knife'end return'default'end local function leadBucket(mode:any):{number}local key:
string=modeKey(mode)local bucket:{number}?=leadErrors[key]if not bucket then bucket={}leadErrors[key]=bucket end return bucket::{number}end local function medianOf(values:{number}):number?local count:
number=#values if count==0 then return nil end local sorted:{number}=table.clone(values)table.sort(sorted)if count%2==1 then return sorted[(count+1)//2]end return(sorted[count//2]+sorted[count//2+1])*
0.5 end local function noteLeadError(seconds:any,mode:any):()if not finiteNumber(seconds)then return end if math.abs(seconds)>LEAD_FEEDBACK_SAMPLE_LIMIT then return end local key:string=modeKey(mode)
local bucket:{number}=leadBucket(key)table.insert(bucket,seconds)while#bucket>LEAD_FEEDBACK_MAX_SAMPLES do table.remove(bucket,1)end if#bucket<LEAD_FEEDBACK_MIN_SAMPLES then return end local median:
number?=medianOf(bucket)if not median then return end local target:number=math.clamp(median,-MAX_LEAD_CORRECTION,MAX_LEAD_CORRECTION)local current:number=leadApplied[key]or 0 local step:number=math.
clamp(target-current,-LEAD_SLEW_PER_SAMPLE,LEAD_SLEW_PER_SAMPLE)leadApplied[key]=current+step end local function leadCorrection(mode:any):number local bucket:{number}?=leadErrors[modeKey(mode)]if not
bucket or#bucket<LEAD_FEEDBACK_MIN_SAMPLES then return 0 end return leadApplied[modeKey(mode)]or 0 end local function leadTarget(mode:any):number local bucket:{number}?=leadErrors[modeKey(mode)]if not
bucket or#bucket<LEAD_FEEDBACK_MIN_SAMPLES then return 0 end local median:number?=medianOf(bucket)if not median then return 0 end return math.clamp(median,-MAX_LEAD_CORRECTION,MAX_LEAD_CORRECTION)end
local function exportLeadState():{[string]:any}local out:{[string]:any}={}for key:string,bucket:{number}in pairs(leadErrors)do out[key]={samples=#bucket,median=medianOf(bucket),applied=leadApplied[key
]or 0}end return out end local function seedLeadError(mode:any,seconds:any,samples:any):()if not finiteNumber(seconds)then return end local count:number=finiteNumber(samples)and math.floor(samples)or
0 if count<LEAD_FEEDBACK_MIN_SAMPLES then return end local key:string=modeKey(mode)local clamped:number=math.clamp(seconds,-LEAD_FEEDBACK_SAMPLE_LIMIT,LEAD_FEEDBACK_SAMPLE_LIMIT)local bucket:{number}=
leadBucket(key)table.clear(bucket)for _index:number=1,math.min(count,LEAD_FEEDBACK_MAX_SAMPLES)do table.insert(bucket,clamped)end leadApplied[key]=math.clamp(clamped,-MAX_LEAD_CORRECTION,
MAX_LEAD_CORRECTION)end local function leadSamples(mode:any):number local bucket:{number}?=leadErrors[modeKey(mode)]return bucket and#bucket or 0 end local function leadUncertainty(mode:any):number
```

```
local bucket:{number}?=leadErrors[modeKey(mode)]if not bucket or#bucket<LEAD_FEEDBACK_MIN_SAMPLES then return 0 end local absolute:{number}={}for _,seconds:number in ipairs(bucket)do table.insert(
absolute,math.abs(seconds))end return medianOf(absolute)or 0 end local function gunHorizon(mode:any):(number,number,number?)local rtt:number=robustRtt()local echo:number?=echoMedian()local pipeline:
number=echo and(rtt*0.6+echo*0.4)or rtt local horizon:number=pipeline+settings.interpolation+manualLead()+leadCorrection(mode)if settings.autoLead then horizon=math.min(horizon,MAX_AUTO_LEAD)end
return math.clamp(horizon,0.005,MAX_HORIZON),rtt,echo end local function knifeHorizon():(number,number,number?)local rtt:number=robustRtt()local echo:number?=echoMedian()local pipeline:number=echo and
(rtt*0.6+echo*0.4)or rtt local horizon:number=pipeline+settings.interpolation+manualLead()+leadCorrection('knife')if settings.autoLead then horizon=math.min(horizon,MAX_AUTO_LEAD)end return math.
clamp(horizon,0.005,MAX_HORIZON),rtt,echo end local function gunAim(target:Player,_muzzle:Vector3,mode:any):(Vector3?,AimInfo?,string?)local aimMode:string=modeKey(mode)local humanoid:any=
targetHumanoid(target)if humanoid~=nil and not bodyIsAlive(humanoid)then tracks[target]=nil return nil,nil,'dead target'end local track:Track?=tracks[target]if not track then samplePlayer(target,os.
clock())track=tracks[target]end if not track or#track.samples==0 then return nil,nil,'no samples'end local now:number=os.clock()local motion:Motion?=estimateMotion(track,now)if not motion then return
nil,nil,'stale sample'end local pipeline:number,rtt:number,echo:number?=gunHorizon(aimMode)local rttEngine:number?,rttData:number?=nil,nil if type(core.getLatencyReadings)=='function'then local ok:
boolean,engine:any,data:any=pcall(core.getLatencyReadings)if ok then rttEngine=finiteNumber(engine)and engine or nil rttData=finiteNumber(data)and data or nil end end local horizon:number=math.clamp(
motion.sampleAge+pipeline,0.005,MAX_HORIZON)local rootPosition:Vector3=predictedRoot(track,motion,horizon)local last:Sample=track.samples[#track.samples]local aimVelocity:Vector3=motion.velocity if
motion.hasInputHistory and horizon>1e-4 then local chord:Vector3=rootPosition-last.position aimVelocity=Vector3.new(chord.X/horizon,motion.velocity.Y,chord.Z/horizon)end local aimOffset:Vector3=last.
aimOffset local freshAimPart:BasePart?=findAimPart(track.character)if freshAimPart then aimOffset=freshAimPart.Position-track.root.Position end local endpoint:Vector3=rootPosition+aimOffset return
endpoint,{endpoint=endpoint,targetPosition=endpoint,capturedPosition=last.position+aimOffset,rootPosition=rootPosition,velocity=aimVelocity,leadSeconds=horizon,sampleAge=motion.sampleAge,samples=
motion.samples,residual=motion.residual,motionSource=motion.source,model='kinematic-v51',rtt=rtt,rttEngine=rttEngine,rttData=rttData,echoDelay=echo,echoSamples=#echoes,turnRate=motion.turnRate,
brakeSeconds=motion.brakeSeconds,leadCorrection=leadCorrection(aimMode),leadCorrectionTarget=leadTarget(aimMode),leadSamples=leadSamples(aimMode),aimMode=aimMode},nil end local KNIFE_FLOOR_DEPTH:
number=80 local KNIFE_FLOOR_CACHE_SECONDS:number=0.1 local KNIFE_FLOOR_XZ_TOLERANCE:number=0.35 local knifeFloorParams:RaycastParams?=nil local knifeFloorParamsTried:boolean=false local
knifeFloorFilter:{Instance}={}local function knifeStandingHeight(track:Track):number?local humanoid:Humanoid=track.humanoid local root:BasePart=track.root if not humanoid or not root then return nil
end local hip:number=humanoid.HipHeight local size:Vector3=root.Size if not finiteNumber(hip)or not finiteVector(size)then return nil end local height:number=hip+size.Y*0.5 local rig:any=humanoid.
RigType if rig==Enum.HumanoidRigType.R6 then local leg:Instance?=track.character:FindFirstChild('Left Leg')or track.character:FindFirstChild('Right Leg')if not leg or not leg:IsA('BasePart')then
return nil end height+=(leg::BasePart).Size.Y elseif rig~=Enum.HumanoidRigType.R15 then return nil end if not finiteNumber(height)or height<0.5 or height>10 then return nil end return height end
local function makeKnifeFloorParams():RaycastParams local params:RaycastParams=RaycastParams.new()params.FilterType=Enum.RaycastFilterType.Exclude params.RespectCanCollide=true params.IgnoreWater=true
return params end local function castKnifeFloor(origin:Vector3,params:RaycastParams):RaycastResult?return workspace:Raycast(origin,Vector3.new(0,-KNIFE_FLOOR_DEPTH,0),params)end local function
knifeFloorAt(track:Track,projected:Vector3,now:number):(boolean,number?)local okHeight:boolean,height:any=pcall(knifeStandingHeight,track)if not okHeight or not finiteNumber(height)then return false,
nil end local last:Sample=track.samples[#track.samples]local topY:number=last.position.Y+height local cache:{any}=track.knifeFloorCache or{}track.knifeFloorCache=cache local newest:any=nil for _,probe
:any in ipairs(cache)do local age:number=now-probe.at local dx:number=projected.X-probe.x local dz:number=projected.Z-probe.z if age>=0 and age<=KNIFE_FLOOR_CACHE_SECONDS and dx*dx+dz*dz<=
KNIFE_FLOOR_XZ_TOLERANCE^2 and math.abs(topY-probe.topY)<=0.25 and math.abs(height-probe.height)<=0.05 and(probe.part==nil or(probe.part.Parent~=nil and(probe.partFrame==nil or(probe.part.CFrame==
probe.partFrame and probe.part.CanCollide==probe.partCanCollide))and(probe.partSize==nil or probe.part.Size==probe.partSize)))and(not newest or probe.serial>newest.serial)then newest=probe end end if
newest then return newest.known,newest.supportY end if not knifeFloorParamsTried then knifeFloorParamsTried=true local okParams:boolean,params:any=pcall(makeKnifeFloorParams)if okParams then
knifeFloorParams=params end end local params:RaycastParams?=knifeFloorParams if not params then return false,nil end table.clear(knifeFloorFilter)if LocalPlayer.Character then table.insert(
knifeFloorFilter,LocalPlayer.Character)end table.insert(knifeFloorFilter,track.character)params.FilterDescendantsInstances=knifeFloorFilter local from:Vector3=Vector3.new(projected.X,topY,projected.Z)
local okRay:boolean,hit:any=pcall(castKnifeFloor,from,params)local known:boolean=okRay local supportY:number?=nil local part:Instance?=nil local partFrame:CFrame?=nil local partSize:Vector3?=nil local
partCanCollide:boolean?=nil if okRay and hit then part=hit.Instance local okHit:boolean,valid:boolean=pcall(function():boolean if not part or not finiteVector(hit.Position)or not finiteVector(hit.
Normal)or hit.Normal.Y<0.65 then return false end if part:IsA('BasePart')then partFrame=(part::BasePart).CFrame partSize=(part::BasePart).Size partCanCollide=(part::BasePart).CanCollide if not
partCanCollide then return false end end local model:Model?=part:FindFirstAncestorOfClass('Model')if model and model:FindFirstChildOfClass('Humanoid')then return false end return true end)if not okHit
or not valid then known=false else local candidate:number=hit.Position.Y+height if candidate>last.position.Y+0.75 then known=false else supportY=math.min(candidate,last.position.Y)end end end local
index:number=track.knifeFloorNext or 1 local probe:any=cache[index]or{}track.knifeFloorSerial=(track.knifeFloorSerial or 0)+1 probe.serial=track.knifeFloorSerial probe.at,probe.x,probe.z,probe.topY=
now,projected.X,projected.Z,topY probe.height,probe.known,probe.supportY,probe.part=height,known,supportY,part probe.partFrame,probe.partSize,probe.partCanCollide=partFrame,partSize,partCanCollide
cache[index]=probe track.knifeFloorNext=index==1 and 2 or 1 return known,supportY end local function knifeRootAt(track:Track,motion:Motion,span:number,floorKnown:boolean,floorY:number?):Vector3 if
floorKnown and motion.verticalLead then local last:Sample=track.samples[#track.samples]local lead:Vector3=leadHorizontal(motion,span)local y:number=last.position.Y+motion.vertical*span-0.5*workspace.
Gravity*span*span if floorY then y=math.max(y,floorY)end return Vector3.new(last.position.X+lead.X,y,last.position.Z+lead.Z)end return predictedRoot(track,motion,span)end local function
knifeFloorIntercept(track:Track,motion:Motion,horizon:number,flight:number,origin:Vector3,speed:number,floorY:number):(Vector3,number)local last:Sample=track.samples[#track.samples]local root:Vector3=
knifeRootAt(track,motion,horizon+flight,true,floorY)for _=1,5 do local revised:number=(root+last.aimOffset-origin).Magnitude/speed local delta:number=math.abs(revised-flight)flight=revised root=
knifeRootAt(track,motion,horizon+flight,true,floorY)if delta<=1e-4 then break end end return root,(root+last.aimOffset-origin).Magnitude/speed end local function knifeAirborneIntercept(track:Track,
motion:Motion,horizon:number,origin:Vector3,speed:number,floorKnown:boolean):(Vector3?,number?)local last:Sample=track.samples[#track.samples]local lowTime:number=0 local highTime:number=5 if motion.
verticalLead and workspace.Gravity>0 then local gravity:number=workspace.Gravity local allowance:number=floorKnown and 0 or MAX_FLOOR_LIFT_STUDS local c:number=last.position.Y+last.aimOffset.Y-origin.
Y+motion.vertical*horizon-0.5*gravity*horizon*horizon+allowance local b:number=motion.vertical-gravity*horizon+speed local discriminant:number=b*b+2*gravity*c if not finiteNumber(discriminant)or
discriminant<0 then return nil,nil end local root:number=math.sqrt(discriminant)lowTime=math.max(0,(b-root)/gravity)highTime=math.min(highTime,(b+root)/gravity)end if highTime<lowTime or highTime<=0
then return nil,nil end local segments:number=math.clamp(math.ceil((highTime-lowTime)/0.025),1,64)local previous:number=lowTime local previousRoot:Vector3=knifeRootAt(track,motion,horizon+previous,
floorKnown,nil)local previousGap:number=(previousRoot+last.aimOffset-origin).Magnitude-speed*previous if previousGap<=1e-6 then local actual:number=(previousRoot+last.aimOffset-origin).Magnitude/speed
if math.abs(actual-previous)>0.002 then return nil,nil end return previousRoot,actual end for index:number=1,segments do local time:number=lowTime+(highTime-lowTime)*index/segments local root:Vector3=
knifeRootAt(track,motion,horizon+time,floorKnown,nil)local gap:number=(root+last.aimOffset-origin).Magnitude-speed*time if gap<=0 then local left:number,right:number=previous,time for _=1,12 do local
middle:number=(left+right)*0.5 local midRoot:Vector3=knifeRootAt(track,motion,horizon+middle,floorKnown,nil)local midGap:number=(midRoot+last.aimOffset-origin).Magnitude-speed*middle if midGap<=0 then
right=middle else left=middle end end local solved:Vector3=knifeRootAt(track,motion,horizon+right,floorKnown,nil)local solvedFlight:number=(solved+last.aimOffset-origin).Magnitude/speed if math.abs(
solvedFlight-right)>0.002 then return nil,nil end return solved,solvedFlight end previous=time end return nil,nil end local function knifeAim(target:Player,origin:Vector3,speed:number?):(Vector3?,
AimInfo?,string?)local humanoid:any=targetHumanoid(target)if humanoid~=nil and not bodyIsAlive(humanoid)then tracks[target]=nil return nil,nil,'dead target'end local track:Track?=tracks[target]if
track and track.character~=target.Character then tracks[target]=nil track=nil end if not track then samplePlayer(target,os.clock())track=tracks[target]end if not track or#track.samples==0 then return
nil,nil,'no samples'end local now:number=os.clock()local motion:Motion?=estimateMotion(track,now)if not motion then return nil,nil,'stale sample'end local resolvedSpeed:number=tonumber(speed)or
DEFAULT_KNIFE_SPEED if resolvedSpeed<24 or resolvedSpeed>300 then resolvedSpeed=DEFAULT_KNIFE_SPEED end local pipeline:number,rtt:number,echo:number?=knifeHorizon()local rttEngine:number?,rttData:
number?=nil,nil if type(core.getLatencyReadings)=='function'then local ok:boolean,engine:any,data:any=pcall(core.getLatencyReadings)if ok then rttEngine=finiteNumber(engine)and engine or nil rttData=
finiteNumber(data)and data or nil end end local horizon:number=math.clamp(motion.sampleAge+pipeline,0.005,MAX_HORIZON)local last:Sample=track.samples[#track.samples]local aimStart:Vector3=last.
position+last.aimOffset local initial:Vector3=motion.velocity local flight:number?=predictionMath.interceptTime(aimStart+initial*horizon-origin,initial,resolvedSpeed)local horizontalSeed:boolean=false
if not flight and motion.airborne and motion.vertical<0 then local lateral:Vector3=Vector3.new(initial.X,0,initial.Z)flight=predictionMath.interceptTime(aimStart+lateral*horizon-origin,lateral,
resolvedSpeed)horizontalSeed=flight~=nil end if not flight then return nil,nil,'unreachable'end local span:number=horizon+flight local lead:Vector3=leadHorizontal(motion,span)local vertical:number=
motion.verticalLead and motion.vertical or 0 if motion.verticalLead then vertical-=0.5*workspace.Gravity*span end local effective:Vector3=Vector3.new(lead.X,vertical*span,lead.Z)/span for _iteration:
number=1,4 do local revised:number?=predictionMath.interceptTime(aimStart+effective*horizon-origin,effective,resolvedSpeed)if not revised then break end local delta:number=math.abs(revised-flight)
flight=revised if delta<=1e-4 then break end span=horizon+flight lead=leadHorizontal(motion,span)vertical=motion.verticalLead and motion.vertical or 0 if motion.verticalLead then vertical-=0.5*
workspace.Gravity*span end effective=Vector3.new(lead.X,vertical*span,lead.Z)/span end local endpoint:Vector3=aimStart+effective*(horizon+flight)if motion.airborne then local lateral:Vector3=Vector3.
new(initial.X,0,initial.Z)local lateralFlight:number?=predictionMath.interceptTime(aimStart+lateral*horizon-origin,lateral,resolvedSpeed)local probeFlight:number=lateralFlight or flight local
spanAtProbe:number=horizon+probeFlight local probePoint:Vector3=predictedRoot(track,motion,spanAtProbe)local possibleSpan:number=horizon+math.max(probeFlight,flight)local rawY:number=last.position.Y+
motion.vertical*possibleSpan-0.5*workspace.Gravity*possibleSpan*possibleSpan local known:boolean,floorY:number?=false,nil local probes:number=0 if motion.verticalLead and rawY<last.position.Y-0.25
then known,floorY=knifeFloorAt(track,probePoint,now)probes=1 end local solvedRoot:Vector3?=nil local solvedFlight:number?=nil if known and floorY then solvedRoot,solvedFlight=knifeFloorIntercept(track
,motion,horizon,probeFlight,origin,resolvedSpeed,floorY)local dx:number=(solvedRoot::Vector3).X-probePoint.X local dz:number=(solvedRoot::Vector3).Z-probePoint.Z if dx*dx+dz*dz>
KNIFE_FLOOR_XZ_TOLERANCE^2 then probePoint=solvedRoot::Vector3 known,floorY=knifeFloorAt(track,probePoint,now)probes+=1 if known and floorY then solvedRoot,solvedFlight=knifeFloorIntercept(track,
motion,horizon,solvedFlight::number,origin,resolvedSpeed,floorY)dx=(solvedRoot::Vector3).X-probePoint.X dz=(solvedRoot::Vector3).Z-probePoint.Z if dx*dx+dz*dz>KNIFE_FLOOR_XZ_TOLERANCE^2 then return
nil,nil,'uncertain landing'end else solvedRoot,solvedFlight=nil,nil end end end if not solvedRoot then if horizontalSeed then return nil,nil,'unreachable'end local first:Vector3=knifeRootAt(track,
motion,horizon+flight,known,nil)local actual:number=(first+last.aimOffset-origin).Magnitude/resolvedSpeed if finiteNumber(actual)and actual<=5 and math.abs(actual-flight)<=0.002 then solvedRoot,
solvedFlight=first,actual else solvedRoot,solvedFlight=knifeAirborneIntercept(track,motion,horizon,origin,resolvedSpeed,known)end if not solvedRoot or not solvedFlight then return nil,nil,
'unreachable'end if probes>0 and known then local dx:number=solvedRoot.X-probePoint.X local dz:number=solvedRoot.Z-probePoint.Z if dx*dx+dz*dz>KNIFE_FLOOR_XZ_TOLERANCE^2 then if probes>=2 then return
nil,nil,'uncertain landing'end probePoint=solvedRoot known,floorY=knifeFloorAt(track,probePoint,now)probes+=1 if not known then return nil,nil,'uncertain landing'end if floorY then solvedRoot,
solvedFlight=knifeFloorIntercept(track,motion,horizon,solvedFlight,origin,resolvedSpeed,floorY)dx=solvedRoot.X-probePoint.X dz=solvedRoot.Z-probePoint.Z if dx*dx+dz*dz>KNIFE_FLOOR_XZ_TOLERANCE^2 then
return nil,nil,'uncertain landing'end end end end if not solvedRoot or not solvedFlight or not finiteVector(solvedRoot)or not finiteNumber(solvedFlight)or solvedFlight>5 then return nil,nil,
'unreachable'end local consistentRoot:Vector3=knifeRootAt(track,motion,horizon+solvedFlight,known,floorY)if(consistentRoot-solvedRoot).Magnitude>0.25 then return nil,nil,'uncertain intercept'end if
horizontalSeed then local correctedSpan:number=horizon+solvedFlight local unboundedY:number=last.position.Y+motion.vertical*correctedSpan-0.5*workspace.Gravity*correctedSpan*correctedSpan if not
floorY or unboundedY>floorY then return nil,nil,'unreachable'end end endpoint=solvedRoot+last.aimOffset flight=(endpoint-origin).Magnitude/resolvedSpeed end return endpoint,{endpoint=endpoint,
targetPosition=endpoint,capturedPosition=aimStart,rootPosition=endpoint-last.aimOffset,velocity=motion.velocity,leadSeconds=horizon,flightSeconds=flight,sampleAge=motion.sampleAge,samples=motion.
samples,residual=motion.residual,motionSource=motion.source,model='kinematic-v164-knife',rtt=rtt,rttEngine=rttEngine,rttData=rttData,echoDelay=echo,echoSamples=#echoes,turnRate=motion.turnRate,
brakeSeconds=motion.brakeSeconds,leadCorrection=0},nil end local function playerStatus(target:Player?):string local rtt:number=robustRtt()local echo:number?=echoMedian()local parts:{string}={string.
format('RTT %.0f ms',rtt*1000)}if echo then table.insert(parts,string.format('echo %.0f ms (%d)',echo*1000,#echoes))end table.insert(parts,string.format('interp %.0f ms',settings.interpolation*1000))
if settings.autoLead then table.insert(parts,echo and'auto lead (echo)'or'auto lead (ping)')elseif settings.extraLead~=0 then table.insert(parts,string.format('manual %+.0f ms',settings.extraLead*1000
))end local horizon:number=(echo or rtt)+settings.interpolation+manualLead()if target then local track:Track?=tracks[target]local preview:Motion?=track and estimateMotion(track,os.clock())if preview
then horizon+=preview.sampleAge end end local capped:boolean=false if settings.autoLead and horizon>MAX_AUTO_LEAD then horizon=MAX_AUTO_LEAD capped=true end if horizon>MAX_HORIZON then horizon=
MAX_HORIZON capped=true end table.insert(parts,string.format('lead %.0f ms%s',horizon*1000,capped and' (capped)'or''))if target then local track:Track?=tracks[target]local motion:Motion?=track and
estimateMotion(track,os.clock())if motion then table.insert(parts,string.format('%.1f studs/s',motion.horizontal.Magnitude))table.insert(parts,string.format('%d samples',motion.samples))table.insert(
parts,string.format('age %.0f ms',motion.sampleAge*1000))if motion.airborne then table.insert(parts,'airborne')end if math.abs(motion.turnRate)>0.05 then table.insert(parts,string.format(
'turn %+.0f deg/s',math.deg(motion.turnRate)))end if motion.brakeSeconds then table.insert(parts,string.format('brake %.2f s',motion.brakeSeconds))end else table.insert(parts,'no fresh samples')end
end return table.concat(parts,' \183 ')end local controller:any={settings=settings,noteEcho=noteEcho,getHistory=function(player:Player,seconds:number?):({any}?,number?)local track:Track?=tracks[player
]if not track or#track.samples<2 then return nil,0 end local span:number=math.clamp(seconds or HISTORY_SECONDS,0.05,HISTORY_SECONDS)local newest:Sample=track.samples[#track.samples]local age:number=
math.max(0,os.clock()-newest.at)local points:{any}={}for _,sample:Sample in ipairs(track.samples)do local at:number=sample.at-newest.at if at>=-span-1e-6 then table.insert(points,{at=at,position=
sample.position,velocity=sample.velocity,state=sample.state,grounded=sample.grounded,moveDirection=sample.moveDirection,aimOffset=sample.aimOffset,aimPart=sample.aimPart})end end if#points<2 then
return nil,age end return points,age end,gunAim=gunAim,knifeAim=knifeAim,status=playerStatus,getRtt=currentRtt,getEchoDelay=echoMedian,predict=function(player:Player,horizon:number):Vector3?local
track:Track?=tracks[player]if not track or#track.samples==0 or not finiteNumber(horizon)then return nil end local motion:Motion?=estimateMotion(track,os.clock())if not motion then return nil end
return predictedRoot(track,motion,math.clamp(horizon,0,2))end,getMotion=function(player:Player):Motion?local track:Track?=tracks[player]if not track then return nil end return estimateMotion(track,os.
clock())end,noteLeadError=noteLeadError,exportLeadState=exportLeadState,seedLeadError=seedLeadError,getLeadCorrection=leadCorrection,getLeadCorrectionTarget=leadTarget,getLeadUncertainty=
leadUncertainty,getLeadSamples=leadSamples,reset=function():()table.clear(echoes)table.clear(echoTimes)table.clear(tracks)table.clear(rttSamples)for _,bucket:{number}in pairs(leadErrors)do table.
clear(bucket)end table.clear(leadErrors)table.clear(leadApplied)lastRttSampleAt=-math.huge end}local roster:{Player}={}local rosterAt:number=-math.huge local function tick(deltaTime:number):()elapsed
+=math.max(0,deltaTime or 0)local interval:number=1/SAMPLE_HZ if elapsed+1e-9<interval then return end elapsed=elapsed%interval local now:number=os.clock()if now-rosterAt>=0.25 or#roster==0 then
rosterAt=now roster=Players:GetPlayers()end for _,player:Player in ipairs(roster)do if player~=LocalPlayer then samplePlayer(player,now)end end end state.mm2Prediction=controller featureConnections.
MM2PredictionSampler=TaskManager:Connect(tick,nil,'Prediction sampler')activeCleanup=function():()disconnectFeatureConnection('MM2PredictionSampler')if state.mm2Prediction==controller then state.
mm2Prediction=nil end table.clear(echoes)table.clear(echoTimes)table.clear(tracks)end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not
Module.Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Render/CoinEsp.lua
local Module={Name='MM2 Coin ESP',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end local COIN_TAG:string='CoinVisual'local VISUAL_NAME:string=
'CoinVisual'local SERVER_NAME:string='Coin_Server'local FAMILY_COIN:string='Coin'local FAMILY_CANDY:string='Candy'local FAMILY_SNOW:string='SnowToken'local FAMILY_BEACH:string='BeachBall'local
FAMILY_EGG:string='Egg'local FAMILY_HEART:string='Heart'local FAMILY_ALL:string='All'local FAMILY_AUTO:string='Auto (server)'local FAMILY_ORDER:{string}={FAMILY_EGG,FAMILY_BEACH,FAMILY_SNOW,
FAMILY_CANDY,FAMILY_HEART,FAMILY_COIN}local COUNT_ORDER:{string}={FAMILY_COIN,FAMILY_CANDY,FAMILY_SNOW,FAMILY_BEACH,FAMILY_EGG,FAMILY_HEART}local FAMILY_NOUNS:{[string]:{string}}={[FAMILY_COIN]={
'coin','coins'},[FAMILY_CANDY]={'candy','candies'},[FAMILY_SNOW]={'snowtoken','snowtokens'},[FAMILY_BEACH]={'beachball','beachballs'},[FAMILY_EGG]={'egg','eggs'},[FAMILY_HEART]={'heart','hearts'}}
local FAMILY_CHOICES:{string}={FAMILY_AUTO,FAMILY_COIN,FAMILY_CANDY,FAMILY_SNOW,FAMILY_BEACH,FAMILY_EGG,FAMILY_HEART,FAMILY_ALL}local BAG_TO_FAMILY:{[string]:string}={Coin=FAMILY_COIN,Candy=
FAMILY_CANDY,SnowToken=FAMILY_SNOW,BeachBall=FAMILY_BEACH,Egg=FAMILY_EGG}local FAMILY_PHRASES:{[string]:{string}}={[FAMILY_EGG]={'egg','eggs'},[FAMILY_BEACH]={'beachball','beach ball','beachballs',
'beach balls'},[FAMILY_SNOW]={'snowtoken','snow token','snowtokens','snow tokens','snowball','snow ball'},[FAMILY_CANDY]={'candy','candies','candied','candycorn','candy corn'},[FAMILY_HEART]={'heart',
'hearts'},[FAMILY_COIN]={'maincoin','coinold','decalpart'}}local RESCAN_PERIOD:number=2 local function lower(value:string):string return string.lower(value)end local function contains(value:string,
needle:string):boolean return string.find(value,needle,1,true)~=nil end function Module.init(runtime:any):any if Module.Initialized then return Module end local core:any=state.mm2Core assert(type(core
)=='table','MM2 Coin ESP requires the MM2 core module')Module.Runtime=runtime local CoinEffects:any=core.CoinEffects local clearEffects:any=core.clearEffects local mm2Settings:any=core.mm2Settings
local collectionService:any=game:GetService('CollectionService')local settings={family=FAMILY_AUTO}local serverFamily:string?=nil local lastRaw={tagged=0,coinId=0,named=0}local coinTags:{[Instance]:{
box:Instance}}={}local lastFamilies:{[Instance]:string}={}state.mm2CoinTagParts={}local function clearTags():()for part:Instance,entry in pairs(coinTags)do coinTags[part]=nil pcall(function():()entry.
box:Destroy()end)end table.clear(lastFamilies)state.mm2CoinTagParts={}end local function attribute(object:Instance,name:string):any return object:GetAttribute(name)end local function hasTag(object:
Instance):boolean return collectionService:HasTag(object,COIN_TAG)end local function visualOf(part:Instance):Instance?if part.Name==SERVER_NAME then local child:Instance?=part:FindFirstChild(
VISUAL_NAME)if child and child:IsA('BasePart')then return child end return nil end return part end local function passesFilters(part:Instance):boolean if not part:IsA('BasePart')or not part:
IsDescendantOf(workspace)then return false end local current:Instance?=part.Parent while current and current~=workspace do if current:FindFirstChildOfClass('Humanoid')then return false end current=
current.Parent end return true end local function familyFromWorkspace():string?local eventName:string=lower(tostring(attribute(workspace,'CurrentEvent')or''))if contains(eventName,'halloween')then
return FAMILY_CANDY elseif contains(eventName,'christmas')then return FAMILY_SNOW elseif contains(eventName,'summer')then return FAMILY_BEACH elseif contains(eventName,'easter')then return FAMILY_EGG
elseif contains(eventName,'valentine')then return FAMILY_HEART end for attrName:string,family:string in pairs({isValentines=FAMILY_HEART,isChristmas=FAMILY_SNOW,isChristmasEnding=FAMILY_SNOW,IsSummer=
FAMILY_BEACH,IsEaster=FAMILY_EGG})do if attribute(workspace,attrName)==true then return family end end return nil end local function activeFamily():string?return serverFamily or familyFromWorkspace()
end local familyCache:{[Instance]:string}={}local function familyOf(part:Instance):string local cached:string?=familyCache[part]if cached then return cached end local names:{string}={}for _,object:
Instance in ipairs(part:GetDescendants())do table.insert(names,object.Name)end local haystack:string=lower(table.concat(names,' '))local active:string?=activeFamily()for _,family:string in ipairs(
FAMILY_ORDER)do if family~=FAMILY_COIN or active==nil or active==FAMILY_COIN then for _,phrase:string in ipairs(FAMILY_PHRASES[family])do if contains(haystack,phrase)then if family~=FAMILY_COIN then
familyCache[part]=family end return family end end end end return active or FAMILY_COIN end local filterCache:{[Instance]:boolean}={}local function collectAll():({[Instance]:string},{[string]:number})
local found:{[Instance]:string}={}local counts:{[string]:number}={}lastRaw={tagged=0,coinId=0,named=0}local function passesFiltersCached(part:Instance):boolean if attribute(part,'Collected')==true or
attribute(part,'Delete')==true then return false end local cached:boolean?=filterCache[part]if cached~=nil then return cached end local result:boolean=passesFilters(part)filterCache[part]=result
return result end local function consider(part:Instance?):()if not part then return end local visual:Instance?=visualOf(part)if not visual or found[visual]then return end if not passesFiltersCached(
visual)then return end local family:string=familyOf(visual)found[visual]=family counts[family]=(counts[family]or 0)+1 end local tagged:{Instance}=collectionService:GetTagged(COIN_TAG)lastRaw.tagged=#
tagged for _,object:Instance in ipairs(tagged)do if object:IsA('BasePart')then consider(object)end end for _,object:Instance in ipairs(workspace:GetDescendants())do if object:IsA('BasePart')then if
attribute(object,'CoinID')~=nil then lastRaw.coinId+=1 consider(object)elseif object.Name==VISUAL_NAME or object.Name==SERVER_NAME then lastRaw.named+=1 consider(object)end end end return found,counts
end local function wanted(family:string):boolean local choice:string=settings.family if choice==FAMILY_ALL then return true end if choice==FAMILY_AUTO then local active:string?=activeFamily()return
active==nil or family==active end return family==choice end local function describe(counts:{[string]:number}):string local parts:{string}={}for _,family:string in ipairs(COUNT_ORDER)do local amount:
number=counts[family]or 0 if amount>0 then local nouns:{string}=FAMILY_NOUNS[family]table.insert(parts,tostring(amount)..' '..(amount==1 and nouns[1]or nouns[2]))end end if#parts==0 then return
'no coins yet'end return table.concat(parts,' \183 ')end local pickupCache:{at:number,list:{Instance}}={at=-math.huge,list={}}local function refreshPickupCache(found:{[Instance]:string}):()local list:
{Instance}={}for part:Instance,family:string in pairs(found)do if wanted(family)then table.insert(list,part)end end pickupCache={at=os.clock(),list=list}end state.mm2CoinPickups=function():{Instance}
if os.clock()-pickupCache.at<0.5 then return pickupCache.list end local found:{[Instance]:string}=collectAll()refreshPickupCache(found)return pickupCache.list end local function toggleCoinEsp(enabled:
boolean):()disconnectFeatureConnection('MM2CoinRefresh')disconnectFeatureConnection('MM2CoinCollected')disconnectFeatureConnection('MM2CoinsStarted')disconnectFeatureConnection('MM2CoinTagAdded')
disconnectFeatureConnection('MM2CoinTagRemoved')disconnectFeatureConnection('MM2CoinAdded')clearEffects(CoinEffects)clearTags()if not enabled then local card:any=state.mm2CoinFeature if card then card
:SetStatus(nil)end return end local elapsed:number=0 local lastStatus:string?=nil local dirty:boolean=true local function setStatus(text:string):()if text==lastStatus then return end lastStatus=text
local card:any=state.mm2CoinFeature if card then card:SetStatus(text)end end local function visualOf(part:Instance):BasePart local main:Instance?=part:FindFirstChild('MainCoin',true)if main and main:
IsA('BasePart')then return main::BasePart end local visual:Instance?=part:FindFirstChild('CoinVisual',true)if visual and visual:IsA('BasePart')then return visual::BasePart end return part::BasePart
end local tagsDirty:boolean=true local function markerTransparency():number return math.clamp(mm2Settings.coinTransparency,0,0.6)end local function makeTag(part:Instance):any local visual:BasePart=
visualOf(part)local box:BoxHandleAdornment=Instance.new('BoxHandleAdornment')box.Name='WurstCoinBox'box.AlwaysOnTop=true box.ZIndex=5 box.Adornee=visual box.Color3=mm2Settings.coinColor box.
Transparency=markerTransparency()box.Size=visual.Size*1.3 box.Parent=part return{box=box}end local function syncTags():()for part:Instance in pairs(lastFamilies)do if part.Parent and not coinTags[part
]then coinTags[part]=makeTag(part)tagsDirty=true end end for part:Instance,entry:any in pairs(coinTags)do if not part.Parent or not lastFamilies[part]then coinTags[part]=nil pcall(function():()entry.
box:Destroy()end)tagsDirty=true end end if tagsDirty then tagsDirty=false local published:{[Instance]:boolean}={}for part:Instance in pairs(coinTags)do published[part]=true end state.mm2CoinTagParts=
published end end local lastTagColourAt:number=-math.huge local function drawOverlay():()local now:number=os.clock()if now-lastTagColourAt<0.5 then return end lastTagColourAt=now for part:Instance,
entry:any in pairs(coinTags)do if not part.Parent then coinTags[part]=nil entry.box:Destroy()tagsDirty=true else pcall(function():()entry.box.Color3=mm2Settings.coinColor entry.box.Transparency=
markerTransparency()end)end end if tagsDirty then tagsDirty=false local published:{[Instance]:boolean}={}for part:Instance in pairs(coinTags)do published[part]=true end state.mm2CoinTagParts=published
end end local function refresh():()local found:{[Instance]:string},counts:{[string]:number}=collectAll()local visible:{[Instance]:boolean}={}table.clear(lastFamilies)for object:Instance,family:string
in pairs(found)do if wanted(family)then visible[object]=true lastFamilies[object]=family end end if settings.family==FAMILY_AUTO and activeFamily()~=nil then local anyWanted:boolean=false for object:
Instance in pairs(visible)do anyWanted=true break end if not anyWanted then for object:Instance,family:string in pairs(found)do visible[object]=true lastFamilies[object]=family end end end
refreshPickupCache(found)syncTags()local shown:number=0 for _ in pairs(visible)do shown+=1 end local foundTotal:number=0 for _ in pairs(found)do foundTotal+=1 end if foundTotal==0 then setStatus(
string.format('no coins yet (tag %d \183 id %d \183 named %d)',lastRaw.tagged,lastRaw.coinId,lastRaw.named))else setStatus(describe(counts)..' \183 '..shown..' shown')end dirty=false end
featureConnections.MM2CoinTagAdded=collectionService:GetInstanceAddedSignal(COIN_TAG):Connect(function():()dirty=true end)featureConnections.MM2CoinTagRemoved=collectionService:
GetInstanceRemovedSignal(COIN_TAG):Connect(function():()dirty=true end)featureConnections.MM2CoinAdded=workspace.DescendantAdded:Connect(function(object:Instance):()if object:IsA('BasePart')and(object
.Name==VISUAL_NAME or object.Name==SERVER_NAME or attribute(object,'CoinID')~=nil or hasTag(object))then dirty=true end end)local started:Instance?=core.mm2GameplayRemotes and core.mm2GameplayRemotes:
FindFirstChild('CoinsStarted')if started and started:IsA('RemoteEvent')then featureConnections.MM2CoinsStarted=(started::any).OnClientEvent:Connect(function(bagTable:any):()if type(bagTable)=='table'
then for bag:any in pairs(bagTable)do local family:string?=BAG_TO_FAMILY[tostring(bag)]if family then serverFamily=family end end dirty=true end)end local remotes:any=core.mm2GameplayRemotes local
collected:Instance?=remotes and remotes:FindFirstChild('CoinCollected')if collected and collected:IsA('RemoteEvent')then featureConnections.MM2CoinCollected=(collected::any).OnClientEvent:Connect(
function():()dirty=true end)end refresh()featureConnections.MM2CoinRefresh=TaskManager:Connect(function(deltaTime:number):()drawOverlay()elapsed+=deltaTime if not dirty and elapsed<RESCAN_PERIOD then
return end elapsed=0 refresh()end,nil,'Coin ESP')end local CoinFeature:any=createUniversalFeature('Coin ESP',
[[Draws one small box, exactly the size of the coin, candy, snow token, beach ball or egg mesh, on every real pickup and reports how many of each family exist.]],3,toggleCoinEsp,{categoryName='Render'
,parent=MM2Scroll,registry=mm2Features})state.mm2CoinFeature=CoinFeature addCycleOption(CoinFeature,'Coin in use',FAMILY_CHOICES,1,function(value:string):()settings.family=value if CoinFeature.enabled
then toggleCoinEsp(true)end end,[[Auto follows the coin family the server announces through CoinsStarted.]])addColorOption(CoinFeature,'Coin colour',mm2Settings.coinColor,function(value:Color3):()
mm2Settings.coinColor=value if CoinFeature.enabled then toggleCoinEsp(true)end end)addNumberOption(CoinFeature,'Coin transparency',mm2Settings.coinTransparency,0,1,function(value:number):()mm2Settings
.coinTransparency=value if CoinFeature.enabled then toggleCoinEsp(true)end end)addInformationOption(CoinFeature,
[[Pickups are recognised by the game's own signatures only: the CoinVisual tag, the CoinID attribute and the exact server/visual part pair. Each pickup gets a single BoxHandleAdornment sized exactly to its visible mesh (no labels), so a hundred coins cost no per-frame Lua. Auto follows the family the server announces on CoinsStarted.]]
)activeCleanup=function():()state.mm2CoinFeature=nil toggleCoinEsp(false)end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.
Initialized then return end Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/Render/RoundEsp.lua
local Module={Name='MM2 Round ESP',Events={}::{[string]:any},Initialized=false,Runtime=nil::any}local activeCleanup:()->()=function():()end function Module.init(runtime:any):any if Module.Initialized
then return Module end local core:any=state.mm2Core assert(type(core)=='table','MM2 Round ESP requires the MM2 core module')Module.Runtime=runtime local GunEffects:any=core.GunEffects local
TrapEffects:any=core.TrapEffects local clearEffects:any=core.clearEffects local createMM2Cham:any=core.createMM2Cham local createMM2Marker:any=core.createMM2Marker local findDroppedGun:any=core.
findDroppedGun local findMM2Map:any=core.findMM2Map local mm2Settings:any=core.mm2Settings local function toggleGunEsp(enabled:boolean):()disconnectFeatureConnection('MM2GunESP')clearEffects(
GunEffects)if not enabled then return end local elapsed:number=1 local markedGun:Instance?=nil featureConnections.MM2GunESP=TaskManager:Connect(function(deltaTime:number):()elapsed+=deltaTime if
elapsed<0.4 then return end elapsed=0 local gun:any=findDroppedGun()if gun==markedGun and(gun==nil or gun.Parent~=nil)then return end markedGun=gun clearEffects(GunEffects)if gun then createMM2Marker(
GunEffects,gun,'Gun',mm2Settings.gunColor,mm2Settings.gunTransparency)end end,nil,'MM2 Gun ESP')end local savedTrapTransparency:{[Instance]:number}=setmetatable({},{__mode='k'})::any local function
restoreTrapTransparency():()for part:Instance,orig:number in pairs(savedTrapTransparency)do(part::BasePart).Transparency=orig end table.clear(savedTrapTransparency)end local function revealTrapPart(
part:Instance):()if not part:IsA('BasePart')then return end local current:number=(part::BasePart).Transparency if current>=0.95 then if savedTrapTransparency[part]==nil then savedTrapTransparency[part
]=current end(part::BasePart).Transparency=0.35 end end local function toggleTrapEsp(enabled:boolean):()disconnectFeatureConnection('MM2TrapAdded')restoreTrapTransparency()clearEffects(TrapEffects)if
not enabled then return end local marked=setmetatable({},{__mode='k'})local function addTrap(object:Instance):()if marked[object]then return end if(object.Name=='TrapVisual'or object.Name=='Trap')and(
object:IsA('Model')or object:IsA('BasePart'))then local parent:Instance?=(object::any).Parent if parent and marked[parent]then revealTrapPart(object)return end marked[object]=true revealTrapPart(
object)for _,child:Instance in ipairs(object:GetDescendants())do revealTrapPart(child)end createMM2Cham(TrapEffects,object,mm2Settings.trapColor,mm2Settings.trapTransparency)end end local map=
findMM2Map()for _,object in ipairs((map or workspace):GetDescendants())do addTrap(object)end if map then for _,child:Instance in ipairs(workspace:GetChildren())do addTrap(child)end end
featureConnections.MM2TrapAdded=workspace.DescendantAdded:Connect(addTrap)end core.toggleGunEsp=toggleGunEsp core.toggleTrapEsp=toggleTrapEsp local unregister={}local itemEsp=services and services.
itemEsp assert(itemEsp and type(itemEsp.register)=='function','MM2 item ESP requires the provider service')for _,provider in ipairs({{name='Gun ESP',tooltip=
'Highlights the dropped MM2 gun while ItemESP is ON.',setEnabled=toggleGunEsp},{name='Trap ESP',tooltip=[[Highlights existing and newly placed MM2 traps while ItemESP is ON.]],setEnabled=toggleTrapEsp
}})do local release=itemEsp.register(provider)if release then table.insert(unregister,release)end end activeCleanup=function():()for _,release in ipairs(unregister)do pcall(release)end table.clear(
unregister)toggleGunEsp(false)toggleTrapEsp(false)end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then return end
Module.Initialized=false pcall(activeCleanup)activeCleanup=function():()end Module.Events={}Module.Runtime=nil end return Module
-- src/games/MM2/base.lua
local Module={Name='MM2',Events={}::{[string]:any},Initialized=false,Menu=nil::any,Runtime=nil::any}local moduleCleanup:()->()=function():()end local function buildMM2Features()local host:any=getfenv(
)local trajectoryLogPrefix:string='['..tostring(host.PRODUCT.logPrefix)..':MM2:Trajectory]'local MM2Effects=create('Folder',{Parent=ScreenGui,Name='MM2Effects'})local GunEffects=create('Folder',{
Parent=MM2Effects,Name='GunESP'})local TrapEffects=create('Folder',{Parent=MM2Effects,Name='TrapESP'})local CoinEffects=create('Folder',{Parent=MM2Effects,Name='CoinChams'})local mm2RoundData:{[any]:
any}={}local mm2GameplayRemotes:Instance?=nil local mm2RoleRemote:RemoteFunction?=nil local roleRefreshRunning:boolean=false local lastRoleRefreshAt:number=-math.huge local roundLifecycleActive:
boolean?=nil local roleRefreshGeneration:number=0 pcall(function()local replicatedStorage:ReplicatedStorage=game:GetService('ReplicatedStorage')local remotes:Instance?=replicatedStorage:WaitForChild(
'Remotes',5)local gameplay:Instance?=remotes and remotes:WaitForChild('Gameplay',5)if not gameplay then return end mm2GameplayRemotes=gameplay local getCurrentPlayerData:Instance?=gameplay:
FindFirstChild('GetCurrentPlayerData')if getCurrentPlayerData and getCurrentPlayerData:IsA('RemoteFunction')then mm2RoleRemote=getCurrentPlayerData end end)local mm2Settings={shootMode='Manual',
shootWallCheck=true,shootRaw=false,silentAim=false,shootKey=Enum.KeyCode.Unknown,shootTarget='',showMissCooldown=false,predictionRtt=0.08,silentSweep=0.5,silentSpawn='Through',getGunKey=Enum.KeyCode.
Unknown,instantRoleNotify=false,roleEspAll=false,roleEspRoundOnly=false,roleEspInnocent=false,roleEspMurderer=false,roleEspSheriff=false,innocentColor=Color3.fromRGB(0,255,8),innocentTransparency=0.62
,deadColor=Color3.fromRGB(170,170,180),deadTransparency=0.72,murdererColor=Color3.fromRGB(255,0,4),murdererTransparency=0.5,sheriffColor=Color3.fromRGB(0,153,255),sheriffTransparency=0.5,heroColor=
Color3.fromRGB(255,196,0),heroTransparency=0.5,roleTagsAll=false,blurtDelay=1.5,blurtRepeat=false,blurtFake=false,coinColor=Color3.fromRGB(230,220,65),coinTransparency=0.35,trapColor=Color3.fromRGB(
145,25,25),trapTransparency=0.7,gunColor=Color3.fromRGB(255,215,0),gunTransparency=0.55,autoPlayId=''}local function isProtectedTarget(player:Player?):boolean local runtime:any=Module.Runtime local
targets:any=runtime and runtime.Services and runtime.Services.protectedTargets return targets~=nil and targets.isProtected(player)==true end local function clearEffects(folder)for _,child in ipairs(
folder:GetChildren())do if child:IsA('ObjectValue')and child.Value then child.Value:Destroy()end child:Destroy()end end local function isWeaponPathClear(origin:Vector3,endpoint:Vector3,target:Model?,
strict:boolean?,options:{ignoreTarget:boolean?,requireTarget:boolean?}?):(boolean,any?)local direction:Vector3=endpoint-origin if direction.Magnitude<0.05 then return true,nil end local excluded:{
Instance}={}if LocalPlayer.Character then table.insert(excluded,LocalPlayer.Character)end if options and options.ignoreTarget and target then table.insert(excluded,target)end for _,object:Instance in
ipairs(CollectionService:GetTagged('WeaponPassthrough'))do table.insert(excluded,object)end local params:RaycastParams=RaycastParams.new()params.FilterType=Enum.RaycastFilterType.Exclude params.
IgnoreWater=true for _=1,32 do params.FilterDescendantsInstances=excluded local hit:RaycastResult?=workspace:Raycast(origin,direction,params)if not hit then return not(options and options.
requireTarget==true),nil end if target and hit.Instance:IsDescendantOf(target)then return true,nil end local part:Instance=hit.Instance local info:any={name=nil,className=nil,transparency=nil,
canCollide=nil,distance=nil,skipped=false}pcall(function():()info.name=string.sub(part:GetFullName(),1,160)info.className=part.ClassName local hitPosition:any=(hit::any).Position if hitPosition then
info.distance=(hitPosition-origin).Magnitude end end)if part:IsA('BasePart')then info.transparency=part.Transparency info.canCollide=part.CanCollide end if part:IsA('BasePart')and part.Transparency==1
and(not strict or part.CanCollide==false)then info.skipped=true table.insert(excluded,part)else return false,info end end return false,{name='raycast budget',skipped=false}end local function
describeRayHit(origin:Vector3,endpoint:Vector3,target:Model?):any local direction:Vector3=endpoint-origin if direction.Magnitude<0.05 then return nil end local excluded:{Instance}={}if LocalPlayer.
Character then table.insert(excluded,LocalPlayer.Character)end for _,object:Instance in ipairs(CollectionService:GetTagged('WeaponPassthrough'))do table.insert(excluded,object)end local params:
RaycastParams=RaycastParams.new()params.FilterType=Enum.RaycastFilterType.Exclude params.IgnoreWater=true params.FilterDescendantsInstances=excluded local hit:RaycastResult?=workspace:Raycast(origin,
direction,params)if not hit then return nil end local part:Instance=hit.Instance local info:any={onTarget=target~=nil and part:IsDescendantOf(target)}pcall(function():()info.name=string.sub(part:
GetFullName(),1,160)info.className=part.ClassName local position:any=(hit::any).Position if position then info.distance=(position-origin).Magnitude end end)if part:IsA('BasePart')then info.
transparency=part.Transparency info.canCollide=part.CanCollide end return info end local function getWeaponFromContainer(container:Instance?,weaponName:string,weaponTag:string):Tool?if not container
then return nil end for _,child:Instance in ipairs(container:GetChildren())do if child:IsA('Tool')then if CollectionService:HasTag(child,weaponTag)then return child end if weaponName=='Gun'then if
child.Name=='Gun'then return child end if child.Name~='FakeGun'then local shootRemote:Instance?=child:FindFirstChild('Shoot')if shootRemote and shootRemote:IsA('RemoteEvent')then return child end end
elseif weaponName=='Knife'then if child.Name=='Knife'then return child end local slashRemote:Instance?=child:FindFirstChild('Slash')local stabRemote:Instance?=child:FindFirstChild('Stab')if(
slashRemote and slashRemote:IsA('RemoteEvent'))or(stabRemote and stabRemote:IsA('RemoteEvent'))then return child end local events:Instance?=child:FindFirstChild('Events')local knifeThrown:Instance?=
events and events:FindFirstChild('KnifeThrown')local knifeStabbed:Instance?=events and events:FindFirstChild('KnifeStabbed')if knifeThrown and knifeThrown:IsA('RemoteEvent')and knifeStabbed and
knifeStabbed:IsA('RemoteEvent')then return child end end end end return nil end local function getPlayerWeapon(player:Player,weaponName:string,equippedOnly:boolean?):Tool?local tag:string='Weapon_'..
weaponName local equipped:Tool?=getWeaponFromContainer(player.Character,weaponName,tag)if equipped or equippedOnly then return equipped end return getWeaponFromContainer(player:FindFirstChildOfClass(
'Backpack'),weaponName,tag)end local function isPlayerAlive(player:Player):boolean local character:Model?=player.Character if not character then return false end local humanoid:Humanoid?=character:
FindFirstChildOfClass('Humanoid')::Humanoid?if not humanoid or humanoid.Health<=0 then return false end local okDeadAttribute:boolean,deadAttribute:any=pcall(function():any return character:
GetAttribute('Dead')end)if okDeadAttribute and deadAttribute==true then return false end local okState:boolean,currentState:any=pcall(function():any return humanoid:GetState()end)if okState and
currentState~=nil then local deadState:any=nil local physicsState:any=nil pcall(function():()deadState=Enum.HumanoidStateType.Dead physicsState=Enum.HumanoidStateType.Physics end)if currentState==
deadState or currentState==physicsState then return false end end return true end local function playerFromRoundKey(key:any,data:any):Player?if typeof(key)=='Instance'and key:IsA('Player')then return
key end if type(key)=='number'then local resolved=Players:GetPlayerByUserId(key)if resolved then return resolved end end local player:Player?=nil local namedPlayer:Instance?=Players:FindFirstChild(
tostring(key))if namedPlayer and namedPlayer:IsA('Player')then player=namedPlayer end if type(data)=='table'then if data.Name then local namedFromData:Instance?=Players:FindFirstChild(tostring(data.
Name))if namedFromData and namedFromData:IsA('Player')then player=namedFromData::Player end end if not player and data.UserId then local userId=tonumber(data.UserId)if userId then local success,
resolvedPlayer=pcall(Players.GetPlayerByUserId,Players,userId)player=success and resolvedPlayer or nil end end end return player end local function isRoundEntryDead(data:any):boolean if type(data)~=
'table'then return false end return data.Dead==true or data.Killed==true end local function getRoundRole(player:Player):string?for key:any,data:any in pairs(mm2RoundData)do if playerFromRoundKey(key,
data)==player and type(data)=='table'then if isRoundEntryDead(data)then return'Dead'end local role:any=data.Role if role=='Murderer'or role=='Sheriff'or role=='Hero'or role=='Innocent'then return role
end end end return nil end local roundRolesListeners:{(boolean)->()}={}local publishedRoundRolesActive:boolean=false local function emitRoundRoles(active:boolean):()publishedRoundRolesActive=active
for _,listener:(boolean)->()in ipairs(roundRolesListeners)do pcall(listener,active)end end local function onRoundRoles(listener:(boolean)->()):()->()table.insert(roundRolesListeners,listener)task.
defer(function():()if table.find(roundRolesListeners,listener)then pcall(listener,publishedRoundRolesActive)end end)return function():()local index:number?=table.find(roundRolesListeners,listener)if
index then table.remove(roundRolesListeners,index)end end end local invalidateRoleCaches:()->()local rebuildRoleIndex:()->()local function hasValidRoundRoleData(dataTable:any):boolean if type(
dataTable)~='table'then return false end for _,entry:any in pairs(dataTable)do if type(entry)=='table'and entry.Dead~=true then local role:any=entry.Role if role=='Murderer'or role=='Sheriff'or role==
'Hero'or role=='Innocent'then return true end end end return false end local function requestRoleRefresh(force:boolean?):()if not mm2RoleRemote or(not force and roleRefreshRunning)or(not force and os.
clock()-lastRoleRefreshAt<0.8)then return end roleRefreshRunning=true local generation:number=roleRefreshGeneration task.spawn(function():()local success:boolean,result:any=pcall(mm2RoleRemote.
InvokeServer,mm2RoleRemote)roleRefreshRunning=false lastRoleRefreshAt=os.clock()if success and generation==roleRefreshGeneration and type(result)=='table'and(roundLifecycleActive~=false or
hasValidRoundRoleData(result))then if hasValidRoundRoleData(result)and roundLifecycleActive==nil then roundLifecycleActive=true end mm2RoundData=result invalidateRoleCaches()if rebuildRoleIndex then
rebuildRoleIndex()end end end)end local roleIndex:{roles:{[Player]:string?},murderer:Player?,sheriff:Player?,rolesKnown:boolean,gunHolders:{[Player]:boolean}?}={roles={},murderer=nil,sheriff=nil,
rolesKnown=false,gunHolders={}}local roleIndexDirty:boolean=true local roleIndexAt:number=-math.huge local roleIndexRefreshQueued:boolean=false local roleWatchers:{[Player]:{RBXScriptConnection}}={}
function invalidateRoleCaches():()roleIndexDirty=true if type(state.clearRoleMemo)=='function'then state.clearRoleMemo()end if roleIndexRefreshQueued then return end roleIndexRefreshQueued=true task.
defer(function():()roleIndexRefreshQueued=false if roleIndexDirty and rebuildRoleIndex then rebuildRoleIndex()end end)end local function hasMurdererCollisionGroup(character:Model?):boolean if not
character then return false end for _,child:Instance in ipairs(character:GetChildren())do if child:IsA('BasePart')and child.CollisionGroup=='Murderer'then return true end end return false end
local function buildRoundRoleMap():{[Player]:string}local map:{[Player]:string}={}for key:any,data:any in pairs(mm2RoundData)do if type(data)=='table'then local player:Player?=playerFromRoundKey(key,
data)if player then if isRoundEntryDead(data)then map[player]='Dead'else local role:any=data.Role if role=='Murderer'or role=='Sheriff'or role=='Hero'or role=='Innocent'then map[player]=role end end
end end end return map end rebuildRoleIndex=function():()roleIndexDirty=false roleIndexAt=os.clock()local roles:{[Player]:string?}={}local roundRoleByPlayer:{[Player]:string}=buildRoundRoleMap()local
murderer:Player?=nil local murdererHasKnife:boolean=false local sheriff:Player?=nil local sheriffHasGun:boolean=false local alive:{[Player]:boolean}={}local special:{[Player]:string}={}local
gunHolders:{[Player]:boolean}={}for _,player:Player in ipairs(Players:GetPlayers())do local character:Model?=player.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass(
'Humanoid')local roundRole:string?=roundRoleByPlayer[player]local isAlive:boolean=humanoid~=nil and humanoid.Health>0 and roundRole~='Dead'alive[player]=isAlive local hasGun:boolean=getPlayerWeapon(
player,'Gun')~=nil local hasKnife:boolean=not hasGun and getPlayerWeapon(player,'Knife')~=nil local resolved:string?=nil if hasGun then gunHolders[player]=true if roundRole=='Innocent'or roundRole==
'Hero'or roundRole=='Dead'then resolved='Hero'else resolved='Sheriff'end elseif hasKnife then resolved='Murderer'elseif roundRole=='Dead'then resolved='Dead'elseif roundRole and roundRole~='Innocent'
then resolved=roundRole elseif roundRole==nil and hasMurdererCollisionGroup(character)then resolved='Murderer'end if resolved and resolved~='Dead'then special[player]=resolved if resolved=='Murderer'
then local better:boolean=not murderer or(isAlive and not alive[murderer])or(isAlive==alive[murderer]and hasKnife and not murdererHasKnife)if better then murderer=player murdererHasKnife=hasKnife end
elseif resolved=='Sheriff'or resolved=='Hero'then local better:boolean=not sheriff or(isAlive and not alive[sheriff])or(isAlive==alive[sheriff]and hasGun and not sheriffHasGun)if better then sheriff=
player sheriffHasGun=hasGun end end end end local rolesKnown:boolean=next(special)~=nil if roundLifecycleActive==false then rolesKnown=false end for _,player:Player in ipairs(Players:GetPlayers())do
local resolved:string?=special[player]if not resolved and rolesKnown then resolved=alive[player]and'Innocent'or'Dead'end roles[player]=resolved end local rolesChanged:boolean=roleIndex.rolesKnown~=
rolesKnown or roleIndex.murderer~=murderer or roleIndex.sheriff~=sheriff if not rolesChanged then for player:Player,role:string?in pairs(roles)do if roleIndex.roles[player]~=role then rolesChanged=
true break end end if not rolesChanged then for player:Player in pairs(roleIndex.roles)do if roles[player]==nil and roleIndex.roles[player]~=nil then rolesChanged=true break end end end end roleIndex.
roles=roles roleIndex.murderer=murderer roleIndex.sheriff=sheriff roleIndex.rolesKnown=rolesKnown local gunHoldersChanged:boolean=false do local previous:{[Player]:boolean}=roleIndex.gunHolders or{}
for player in pairs(gunHolders)do if not previous[player]then gunHoldersChanged=true end end for player in pairs(previous)do if not gunHolders[player]then gunHoldersChanged=true end end end roleIndex.
gunHolders=gunHolders if gunHoldersChanged and roundLifecycleActive~=false then requestRoleRefresh()end if rolesChanged then if type(state.invalidateGameRoles)=='function'then state.
invalidateGameRoles()elseif type(state.clearRoleMemo)=='function'then state.clearRoleMemo()end emitRoundRoles(rolesKnown)end if not rolesKnown and roundLifecycleActive~=false then requestRoleRefresh()
end end local function ensureRoleIndex():()if roleIndexDirty or os.clock()-roleIndexAt>1 then rebuildRoleIndex()end end local function unwatchPlayer(player:Player):()local connections:{
RBXScriptConnection}?=roleWatchers[player]if not connections then return end roleWatchers[player]=nil for _,connection:RBXScriptConnection in ipairs(connections)do pcall(function():()connection:
Disconnect()end)end end local function watchPlayer(player:Player):()unwatchPlayer(player)local connections:{RBXScriptConnection}={}local watchedTools:{[Tool]:boolean}=setmetatable({},{__mode='k'})::
any local function watchTool(tool:Tool):()if watchedTools[tool]then return end watchedTools[tool]=true table.insert(connections,tool.DescendantAdded:Connect(invalidateRoleCaches))table.insert(
connections,tool.DescendantRemoving:Connect(invalidateRoleCaches))end local function watchContainer(container:Instance?):()if not container then return end for _,child:Instance in ipairs(container:
GetChildren())do if child:IsA('Tool')then watchTool(child)end end table.insert(connections,container.ChildAdded:Connect(function(child:Instance):()if child:IsA('Tool')then watchTool(child)end
invalidateRoleCaches()end))table.insert(connections,container.ChildRemoved:Connect(invalidateRoleCaches))end local function watchCharacter(character:Model):()invalidateRoleCaches()watchContainer(
character)local function watchCollisionPart(part:BasePart):()table.insert(connections,part:GetPropertyChangedSignal('CollisionGroup'):Connect(invalidateRoleCaches))end for _,child:Instance in ipairs(
character:GetChildren())do if child:IsA('BasePart')then watchCollisionPart(child)end end table.insert(connections,character.ChildAdded:Connect(function(child:Instance):()if child:IsA('BasePart')then
watchCollisionPart(child)end end))local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')if humanoid then table.insert(connections,humanoid.Died:Connect(invalidateRoleCaches))end end
table.insert(connections,player.CharacterAdded:Connect(watchCharacter))table.insert(connections,player.CharacterRemoving:Connect(invalidateRoleCaches))table.insert(connections,player.ChildAdded:
Connect(function(child:Instance):()if child:IsA('Backpack')then watchContainer(child)invalidateRoleCaches()end end))watchContainer(player:FindFirstChildOfClass('Backpack'))if player.Character then
watchCharacter(player.Character)end roleWatchers[player]=connections end for _,player:Player in ipairs(Players:GetPlayers())do watchPlayer(player)end featureConnections.MM2RoleWatchAdded=Players.
PlayerAdded:Connect(function(player:Player):()watchPlayer(player)invalidateRoleCaches()end)featureConnections.MM2RoleWatchRemoving=Players.PlayerRemoving:Connect(function(player:Player):()
unwatchPlayer(player)invalidateRoleCaches()end)for _,weaponTag:string in ipairs({'Weapon_Knife','Weapon_Gun'})do featureConnections['MM2RoleTagAdded'..weaponTag]=CollectionService:
GetInstanceAddedSignal(weaponTag):Connect(invalidateRoleCaches)featureConnections['MM2RoleTagRemoved'..weaponTag]=CollectionService:GetInstanceRemovedSignal(weaponTag):Connect(invalidateRoleCaches)end
local function roundRolesKnown():boolean ensureRoleIndex()return roleIndex.rolesKnown end local function getPlayerRole(player:Player):string?ensureRoleIndex()return roleIndex.roles[player]end
local function findMM2Role(toolName:string,excludedPlayer:Player?):Player?ensureRoleIndex()local wanted:string=toolName=='Knife'and'Murderer'or'Sheriff'for player:Player,role:string?in pairs(roleIndex
.roles)do if player~=excludedPlayer and(role==wanted or(wanted=='Sheriff'and role=='Hero'))then return player end end return nil end local function findPlayerByRoundRole(role:string):Player?for key,
data in pairs(mm2RoundData)do if type(data)=='table'and not isRoundEntryDead(data)and data.Role==role then local player:Player?=playerFromRoundKey(key,data)if player and isPlayerAlive(player)then if
role=='Murderer'and getPlayerWeapon(player,'Gun')~=nil then continue end if(role=='Sheriff'or role=='Hero')and getPlayerWeapon(player,'Knife')~=nil then continue end return player end end end return
nil end local function hasActiveRoundRoles():boolean return roundRolesKnown()end local function findMurderer():Player?ensureRoleIndex()for _,player:Player in ipairs(Players:GetPlayers())do if
isPlayerAlive(player)and getPlayerWeapon(player,'Gun')==nil and getPlayerWeapon(player,'Knife')~=nil then return player end end local candidate:Player?=roleIndex.murderer or findPlayerByRoundRole(
'Murderer')if candidate and getPlayerWeapon(candidate,'Gun')~=nil then return nil end return candidate end local function findSheriff():Player?ensureRoleIndex()for _,player:Player in ipairs(Players:
GetPlayers())do if isPlayerAlive(player)and getPlayerWeapon(player,'Knife')==nil and getPlayerWeapon(player,'Gun')~=nil then return player end end local candidate:Player?=roleIndex.sheriff or
findPlayerByRoundRole('Sheriff')or findPlayerByRoundRole('Hero')if candidate and getPlayerWeapon(candidate,'Knife')~=nil then return nil end return candidate end local MM2_MAP_COIN_CONTAINERS:{string}
={'CoinContainer','CoinAreas','Coins'}local LOBBY_MODEL_NAMES:{string}={'Lobby','MainLobby','RegularLobby'}local function isLobbyName(name:string):boolean for _,candidate:string in ipairs(
LOBBY_MODEL_NAMES)do if name==candidate then return true end end return string.find(string.lower(tostring(name)),'lobby',1,true)~=nil end local function isLobbyModel(object:Instance):boolean if
isLobbyName(object.Name)then return true end if object:FindFirstChild('VotePad1')or object:FindFirstChild('VotePads')or object:FindFirstChild('VoteIcons')then return true end return false end
local function isMM2MapModel(object:Instance):boolean if not object:IsA('Model')or isLobbyModel(object)then return false end for _,containerName:string in ipairs(MM2_MAP_COIN_CONTAINERS)do if object:
FindFirstChild(containerName)then return true end end if object:FindFirstChild('GunDrop')then return true end return false end local mm2MapCache:{map:Instance?,at:number}={map=nil,at=-math.huge}
local function findMM2MapUncached():Instance?for _,object:Instance in ipairs(workspace:GetChildren())do if isMM2MapModel(object)then return object end end for _,object:Instance in ipairs(workspace:
GetChildren())do if object:IsA('Model')and not isLobbyModel(object)and object:FindFirstChild('Spawns')then return object end end return nil end local function findMM2Map():Instance?local cached:
Instance?=mm2MapCache.map if cached and cached.Parent==workspace and isMM2MapModel(cached)then return cached end if os.clock()-mm2MapCache.at<0.25 then return(cached and cached.Parent==workspace)and
cached or nil end mm2MapCache.at=os.clock()mm2MapCache.map=findMM2MapUncached()return mm2MapCache.map end local droppedGunCache:{gun:(Model|BasePart)?,at:number}={gun=nil,at=-math.huge}local
lastGunSweepAt:number=-math.huge local function isLooseGun(object:Instance):boolean if not object:IsA('Tool')then return false end if object.Name~='Gun'and not CollectionService:HasTag(object,
'Weapon_Gun')then return false end return object:IsDescendantOf(workspace)and not object:FindFirstAncestorOfClass('Backpack')and(object.Parent==workspace or object.Parent==findMM2Map()or object.Parent
~=nil and not object.Parent:FindFirstChildOfClass('Humanoid'))end local function resolveGunPart(object:Instance):(Model|BasePart)?if object:IsA('Model')or object:IsA('BasePart')then return object::any
end local handle:Instance?=object:FindFirstChild('Handle')or object:FindFirstChildWhichIsA('BasePart')return(handle and handle:IsA('BasePart'))and handle or nil end local function
findDroppedGunUncached():(Model|BasePart)?local map:Instance?=findMM2Map()local candidate:Instance?=map and map:FindFirstChild('GunDrop')if not candidate then candidate=workspace:FindFirstChild(
'GunDrop')end if candidate and(candidate:IsA('Model')or candidate:IsA('BasePart'))then return candidate::any end for _,taggedDrop:Instance in ipairs(CollectionService:GetTagged('GunDrop'))do if
taggedDrop:IsDescendantOf(workspace)and(taggedDrop:IsA('Model')or taggedDrop:IsA('BasePart'))then return taggedDrop::any end end for _,object:Instance in ipairs(workspace:GetChildren())do if object:
IsA('Model')or object:IsA('Folder')then local childDrop:Instance?=object:FindFirstChild('GunDrop')if childDrop and(childDrop:IsA('Model')or childDrop:IsA('BasePart'))then return childDrop::any end end
end for _,object:Instance in ipairs(CollectionService:GetTagged('Weapon_Gun'))do if isLooseGun(object)then local part:(Model|BasePart)?=resolveGunPart(object)if part then return part end end end for _
,object:Instance in ipairs(workspace:GetChildren())do if isLooseGun(object)then local part:(Model|BasePart)?=resolveGunPart(object)if part then return part end end end if os.clock()-lastGunSweepAt<1.5
then return nil end lastGunSweepAt=os.clock()local deepCandidate:Instance?=(map and map:FindFirstChild('GunDrop',true))or workspace:FindFirstChild('GunDrop',true)if deepCandidate and(deepCandidate:
IsA('Model')or deepCandidate:IsA('BasePart'))then return deepCandidate::any end return nil end local function findDroppedGun():(Model|BasePart)?local cached:(Model|BasePart)?=droppedGunCache.gun if
cached and cached.Parent and cached:IsDescendantOf(workspace)then return cached end if os.clock()-droppedGunCache.at<0.05 then return nil end droppedGunCache.at=os.clock()droppedGunCache.gun=
findDroppedGunUncached()return droppedGunCache.gun end local function createMM2Marker(folder,adornee,labelText,color,transparency)local highlight=Instance.new('Highlight')highlight.Name=labelText..
'Highlight'highlight.Adornee=adornee highlight.DepthMode=Enum.HighlightDepthMode.AlwaysOnTop highlight.FillColor=color highlight.FillTransparency=transparency or 0.68 highlight.OutlineColor=color
highlight.OutlineTransparency=0 highlight.Parent=adornee local highlightReference=Instance.new('ObjectValue')highlightReference.Name=labelText..'HighlightReference'highlightReference.Value=highlight
highlightReference.Parent=folder local head=adornee:IsA('Model')and(adornee:FindFirstChild('Head')or adornee:FindFirstChild('HumanoidRootPart')or adornee:FindFirstChild('Handle',true)or adornee:
FindFirstChildWhichIsA('BasePart',true))or adornee if head and head:IsA('BasePart')then local billboard=Instance.new('BillboardGui')billboard.Name=labelText..'Label'billboard.Adornee=head billboard.
AlwaysOnTop=true billboard.LightInfluence=0 billboard.MaxDistance=1500 billboard.Size=UDim2.fromOffset(150,22)billboard.StudsOffset=Vector3.new(0,3.5,0)billboard.Parent=head local billboardReference=
Instance.new('ObjectValue')billboardReference.Name=labelText..'LabelReference'billboardReference.Value=billboard billboardReference.Parent=folder local text=Instance.new('TextLabel')text.
BackgroundTransparency=1 text.BorderSizePixel=0 text.Font=Enum.Font.GothamBold text.Size=UDim2.fromScale(1,1)text.Text=labelText text.TextColor3=color text.TextSize=14 text.TextStrokeColor3=Color3.
```

```
fromRGB(0,0,0)text.TextStrokeTransparency=0.3 text.Parent=billboard end end local function createMM2Cham(folder,adornee,color,transparency)local highlight=Instance.new('Highlight')highlight.Name=
'WurstCham'highlight.Adornee=adornee highlight.DepthMode=Enum.HighlightDepthMode.AlwaysOnTop highlight.FillColor=color highlight.FillTransparency=transparency highlight.OutlineColor=color highlight.
OutlineTransparency=math.clamp(transparency+0.15,0,1)highlight.Parent=adornee local reference=Instance.new('ObjectValue')reference.Name='ChamReference'reference.Value=highlight reference.Parent=folder
end local function getPlayerPerk(player:Player):string?local char:Model?=player.Character if char then local perkAttr:any=char:GetAttribute('Perk')or player:GetAttribute('Perk')if type(perkAttr)==
'string'and perkAttr~=''then return perkAttr end end for key:any,data:any in pairs(mm2RoundData)do if playerFromRoundKey(key,data)==player and type(data)=='table'then local perk:any=data.Perk or data.
EquippedPerk or data.perk if type(perk)=='string'and perk~=''then return perk end end end if char then if char:FindFirstChild('Trap')or(player:FindFirstChild('Backpack')and player.Backpack:
FindFirstChild('Trap'))then return'Trap'end if char:FindFirstChild('FakeGun')or(player:FindFirstChild('Backpack')and player.Backpack:FindFirstChild('FakeGun'))then return'FakeGun'end end return nil
end local mm2BridgeProvider:any=nil local function unregisterMM2Roles():()if mm2BridgeProvider and type(state.unregisterRoleProvider)=='function'then state.unregisterRoleProvider(mm2BridgeProvider)end
mm2BridgeProvider=nil end local function registerMM2Roles():()mm2BridgeProvider={Name='MM2',ReplacesTeams=true,DeadRoles={'Dead'},Roles={'Murderer','Sheriff','Hero','Innocent','Dead'},Colors={Murderer
=mm2Settings.murdererColor,Sheriff=mm2Settings.sheriffColor,Hero=mm2Settings.heroColor,Innocent=mm2Settings.innocentColor,Dead=mm2Settings.deadColor},Get=function(player:Player):string?return
getPlayerRole(player)end,GetPerk=function(player:Player):string?return getPlayerPerk(player)end,GetColor=function(roleName:string):Color3?if roleName=='Murderer'then return mm2Settings.murdererColor
end if roleName=='Sheriff'then return mm2Settings.sheriffColor end if roleName=='Hero'then return mm2Settings.heroColor end if roleName=='Innocent'then return mm2Settings.innocentColor end if roleName
=='Dead'then return mm2Settings.deadColor end return nil end,SetColor=function(roleName:string,colour:Color3):()if roleName=='Murderer'then mm2Settings.murdererColor=colour elseif roleName=='Sheriff'
then mm2Settings.sheriffColor=colour elseif roleName=='Hero'then mm2Settings.heroColor=colour elseif roleName=='Innocent'then mm2Settings.innocentColor=colour elseif roleName=='Dead'then mm2Settings.
deadColor=colour end end}registerRoleProvider(mm2BridgeProvider)end local primeRoundState:()->()local setRoundPhase:(string,string)->()local lastPhase:string='unknown'if mm2GameplayRemotes then local
playerDataChanged=mm2GameplayRemotes:FindFirstChild('PlayerDataChanged')if playerDataChanged and playerDataChanged:IsA('RemoteEvent')then featureConnections.MM2PlayerDataChanged=playerDataChanged.
OnClientEvent:Connect(function(newData)if type(newData)=='table'and hasValidRoundRoleData(newData)then roundLifecycleActive=true mm2MapCache.at=-math.huge mm2RoundData=newData lastRoleRefreshAt=os.
clock()invalidateRoleCaches()if rebuildRoleIndex then rebuildRoleIndex()end if setRoundPhase and findMM2Map()then lastPhase='map'setRoundPhase('map','PlayerDataChanged')end elseif type(newData)~=
'table'or next(newData)==nil or roundLifecycleActive==false then mm2RoundData={}invalidateRoleCaches()if rebuildRoleIndex then rebuildRoleIndex()end else for key:any,value:any in pairs(newData)do
mm2RoundData[key]=value end lastRoleRefreshAt=os.clock()invalidateRoleCaches()if rebuildRoleIndex then rebuildRoleIndex()end end end)end local roleSelect=mm2GameplayRemotes:FindFirstChild('RoleSelect'
)if roleSelect and roleSelect:IsA('RemoteEvent')then featureConnections.MM2RoleSelect=roleSelect.OnClientEvent:Connect(function(localRole:any)roundLifecycleActive=true mm2MapCache.at=-math.huge
mm2RoundData={}if type(localRole)=='string'and(localRole=='Murderer'or localRole=='Sheriff'or localRole=='Hero'or localRole=='Innocent')then mm2RoundData[LocalPlayer.Name]={Name=LocalPlayer.Name,
UserId=LocalPlayer.UserId,Role=localRole}end invalidateRoleCaches()if rebuildRoleIndex then rebuildRoleIndex()end if primeRoundState then primeRoundState()end if setRoundPhase then lastPhase='map'
setRoundPhase('map','RoleSelect')end end)end for _,remoteName:string in ipairs({'ShowRoleSelect','ShowRoleSelectNew'})do local showRole=mm2GameplayRemotes:FindFirstChild(remoteName)if showRole and
showRole:IsA('RemoteEvent')then featureConnections['MM2'..remoteName]=showRole.OnClientEvent:Connect(function()roundLifecycleActive=true mm2MapCache.at=-math.huge invalidateRoleCaches()if
primeRoundState then primeRoundState()end if setRoundPhase then lastPhase='map'setRoundPhase('map',remoteName)end end)end end end local roundTimer:{endsAt:number?}={endsAt=nil}local timerRemotes=game:
GetService('ReplicatedStorage'):FindFirstChild('Remotes')local timerGameplay=timerRemotes and timerRemotes:FindFirstChild('Gameplay')local timerRoundStart=timerGameplay and timerGameplay:
FindFirstChild('RoundStart')local timerRoundEnd=timerGameplay and timerGameplay:FindFirstChild('RoundEndFade')local sessionRoundActive=false if timerRoundStart and timerRoundStart:IsA('RemoteEvent')
then featureConnections.MM2TimerTrackerStart=timerRoundStart.OnClientEvent:Connect(function(duration,startPlayerData)duration=tonumber(duration)roundTimer.endsAt=duration and os.clock()+duration or
nil if not duration or duration<=0 then mm2RoundData={}sessionRoundActive=false elseif type(startPlayerData)=='table'and hasValidRoundRoleData(startPlayerData)then mm2RoundData=startPlayerData end
invalidateRoleCaches()roundLifecycleActive=duration~=nil and duration>0 roleRefreshGeneration+=1 if duration and duration>0 then mm2MapCache.at=-math.huge primeRoundState()lastPhase='map'
setRoundPhase('map','RoundStart')end if duration and duration>0 and not sessionRoundActive and state.sessionInfo and state.sessionInfo.addRound then sessionRoundActive=true state.sessionInfo.addRound(
)end end)end if timerRoundEnd and timerRoundEnd:IsA('RemoteEvent')then featureConnections.MM2TimerTrackerEnd=timerRoundEnd.OnClientEvent:Connect(function()roundTimer.endsAt=nil sessionRoundActive=
false roundLifecycleActive=false invalidateRoleCaches()roleRefreshGeneration+=1 mm2RoundData={}setRoundPhase('lobby','RoundEndFade')end)end local roundPhase:string='unknown'local roundPhaseListeners:{
(string,string)->()}={}local function emitRoundPhase(phase:string,reason:string):()for _,listener:(string,string)->()in ipairs(roundPhaseListeners)do pcall(listener,phase,reason)end end setRoundPhase=
function(phase:string,reason:string):()if roundPhase==phase then return end roundPhase=phase emitRoundPhase(phase,reason)end local function onRoundPhase(listener:(string,string)->()):()->()table.
insert(roundPhaseListeners,listener)task.defer(function():()if table.find(roundPhaseListeners,listener)then pcall(listener,roundPhase,'subscribe')end end)return function():()local index:number?=table.
find(roundPhaseListeners,listener)if index then table.remove(roundPhaseListeners,index)end end end local lobbyCache:{model:Instance?,at:number}={model=nil,at=-math.huge}local LOBBY_CACHE_SECONDS:
number=0.75 local function findLobbyModelUncached():Instance?for _,object:Instance in ipairs(workspace:GetChildren())do if isLobbyName(object.Name)or isLobbyModel(object)then return object end end
return nil end local function findLobbyModel():Instance?local cached:Instance?=lobbyCache.model if cached and cached.Parent then return cached end local now:number=os.clock()if now-lobbyCache.at<
LOBBY_CACHE_SECONDS then return nil end lobbyCache.at=now lobbyCache.model=findLobbyModelUncached()return lobbyCache.model end local lobbyRefCache:{model:Instance?,part:BasePart?,at:number}={model=nil
,part=nil,at=-math.huge}local function lobbyReferencePart(lobby:Instance):BasePart?local now:number=os.clock()if lobbyRefCache.model==lobby and lobbyRefCache.part and lobbyRefCache.part.Parent then
return lobbyRefCache.part end if lobbyRefCache.model==lobby and now-lobbyRefCache.at<1 then return lobbyRefCache.part end lobbyRefCache.at=now lobbyRefCache.model=lobby local votePads:Instance?=lobby:
FindFirstChild('VotePads',true)local refPart:BasePart?=nil if votePads then refPart=votePads:FindFirstChildWhichIsA('BasePart',true)end if not refPart then local spawns:Instance?=lobby:FindFirstChild(
'Spawns',true)if spawns then refPart=spawns:FindFirstChildWhichIsA('BasePart',true)end end if not refPart then refPart=lobby:FindFirstChildWhichIsA('BasePart',true)end lobbyRefCache.part=refPart
return refPart end local function isRootNearLobby(root:BasePart,lobby:Instance):boolean local refPart:BasePart?=lobbyReferencePart(lobby)return refPart~=nil and(root.Position-refPart.Position).
Magnitude<=350 end local function detectRoundPhase():string local character:Model?=LocalPlayer.Character if not character then return'lobby'end local root:BasePart?=character:FindFirstChild(
'HumanoidRootPart')::BasePart?if not root then return'lobby'end local lobby:Instance?=findLobbyModel()if lobby and(root:IsDescendantOf(lobby)or isRootNearLobby(root,lobby))then return'lobby'end if
findMM2Map()then return'map'end return roundPhase=='map'and'map'or'lobby'end primeRoundState=function():()roleRefreshGeneration+=1 local generation:number=roleRefreshGeneration requestRoleRefresh(true
)task.spawn(function():()for _,delay:number in ipairs({0.05,0.15,0.3,0.5,0.8,1.2,2,3})do task.wait(delay)if generation~=roleRefreshGeneration or roundRolesKnown()then break end requestRoleRefresh(true
)end end)end local phaseElapsed:number=0 lastPhase=detectRoundPhase()if lastPhase=='map'then roundLifecycleActive=true primeRoundState()end featureConnections.MM2RoundPhase=TaskManager:Connect(
function(deltaTime:number):()phaseElapsed+=deltaTime if phaseElapsed<0.1 then return end phaseElapsed=0 local phase:string=detectRoundPhase()if phase==lastPhase then return end lastPhase=phase if
phase=='map'then if not hasValidRoundRoleData(mm2RoundData)then mm2RoundData={}end roundLifecycleActive=true invalidateRoleCaches()primeRoundState('map')setRoundPhase('map','teleport')else
mm2RoundData={}roundLifecycleActive=false roundTimer.endsAt=nil invalidateRoleCaches()roleRefreshGeneration+=1 setRoundPhase('lobby','teleport')end end,nil,'MM2 round phase')setRoundPhase(lastPhase,
'boot')local roundTimerPartCache:{part:Instance?,at:number}={part=nil,at=-math.huge}local function findRoundTimerPart():Instance?local cached:Instance?=roundTimerPartCache.part if cached and cached.
Parent then return cached end if os.clock()-roundTimerPartCache.at<1 then return cached end roundTimerPartCache.at=os.clock()local found:Instance?=workspace:FindFirstChild('RoundTimerPart')
roundTimerPartCache.part=found return found end local function readRoundTimerPartSeconds():number?local part:Instance?=findRoundTimerPart()if not part then return nil end local remaining:number?=
tonumber(part:GetAttribute('Time')::any)if not remaining or remaining<=0 or remaining>600 then return nil end return remaining end local hudTimerCache:{frame:Instance?,label:Instance?,at:number}={
frame=nil,label=nil,at=-math.huge}local function findHudTimerLabel():Instance?local cached:Instance?=hudTimerCache.label if cached and cached.Parent then return cached end if os.clock()-hudTimerCache.
at<1 then return cached end hudTimerCache.at=os.clock()local playerGui:PlayerGui?=LocalPlayer:FindFirstChildOfClass('PlayerGui')local mainGui:Instance?=playerGui and playerGui:FindFirstChild('MainGUI'
)local gameFrame:Instance?=mainGui and mainGui:FindFirstChild('Game')local frame:Instance?=gameFrame and gameFrame:FindFirstChild('Timer')if not frame then frame=mainGui and mainGui:FindFirstChild(
'Timer',true)end hudTimerCache.frame=(frame and frame:IsA('GuiObject'))and frame or nil local label:Instance?=frame and(frame:FindFirstChild('XPText')or frame:FindFirstChild('Timer')or frame:
FindFirstChildWhichIsA('TextLabel',true))hudTimerCache.label=(label and label:IsA('TextLabel'))and label or nil return hudTimerCache.label end local function parseHudTimerSeconds(text:string?):number?
if type(text)~='string'or text==''then return nil end local minutes:string?,seconds:string?=text:match('^%s*(%d+):(%d%d)s?%s*$')if minutes and seconds then return tonumber(minutes)*60+tonumber(seconds
)end local whole:string?=text:match('^%s*(%d%d?)s?%s*$')if whole then return tonumber(whole)end return nil end local function isHudTimerVisible(label:Instance):boolean local ancestor:Instance?=label
while ancestor do if ancestor:IsA('GuiObject')and ancestor.Visible==false then return false end ancestor=ancestor.Parent end return true end local function readHudTimerSeconds():number?local label:
Instance?=findHudTimerLabel()if not label then return nil end if not isHudTimerVisible(label)then return nil end local remaining:number?=parseHudTimerSeconds((label::any).Text)if not remaining or
remaining<=0 or remaining>600 then return nil end return remaining end local lastAttributeRemaining:number?=nil local lastAttributeChangeAt:number=-math.huge local clockElapsed:number=0
featureConnections.MM2RoundClock=TaskManager:Connect(function(deltaTime:number):()clockElapsed+=deltaTime if clockElapsed<0.25 then return end clockElapsed=0 local remaining:number?=
readRoundTimerPartSeconds()or readHudTimerSeconds()if not remaining or roundPhase=='lobby'then lastAttributeRemaining=nil return end if lastAttributeRemaining==nil or math.abs(remaining-
lastAttributeRemaining)>0.01 then lastAttributeChangeAt=os.clock()end lastAttributeRemaining=remaining if os.clock()-lastAttributeChangeAt>3 then return end roundTimer.endsAt=os.clock()+remaining if
roundLifecycleActive==nil then roundLifecycleActive=true end end,nil,'MM2 round clock')requestRoleRefresh(true)local predictionMath={}function predictionMath.finite(value:any):boolean return type(
value)=='number'and value==value and math.abs(value)<math.huge end function predictionMath.interceptTime(offset:Vector3,velocity:Vector3,speed:number):number?if not predictionMath.finite(speed)or
speed<=1 then return nil end local a:number=velocity:Dot(velocity)-speed*speed local b:number=2*offset:Dot(velocity)local c:number=offset:Dot(offset)if not predictionMath.finite(a)or not
predictionMath.finite(b)or not predictionMath.finite(c)then return nil end if c<1e-8 then return 0 end local time:number?=nil if math.abs(a)<1e-6 then if b<-1e-6 then time=-c/b end else local
discriminant:number=b*b-4*a*c if discriminant>=0 then local root:number=math.sqrt(discriminant)local first:number=(-b-root)/(2*a)local second:number=(-b+root)/(2*a)if first>=0 then time=first end if
second>=0 and(not time or second<time)then time=second end end end return time and predictionMath.finite(time)and time<=5 and time or nil end local lastLatencyReadAt:number=-math.huge local function
readLatencyReadings():(number?,number?)local engine:number?=nil pcall(function():()local direct:any=(LocalPlayer::any).GetNetworkPing if type(direct)~='function'then return end local okDirect:boolean,
seconds:any=pcall(direct,LocalPlayer)if okDirect and predictionMath.finite(seconds)and(seconds::number)>0.001 and(seconds::number)<=2 then engine=seconds::number end end)return engine,nil end
local function getEstimatedLatency():number local now:number=os.clock()if now-lastLatencyReadAt<0.1 then return math.clamp(mm2Settings.predictionRtt,0.02,0.6)end local elapsed:number=now-
lastLatencyReadAt lastLatencyReadAt=now local measuredRtt:number=mm2Settings.predictionRtt pcall(function():()local engine:number?,data:number?=readLatencyReadings()local lower:number?=nil if engine
and data then lower=math.min(engine,data)else lower=engine or data end if lower then measuredRtt=lower end end)local alpha:number=1-math.exp(-math.min(elapsed,1)/0.5)mm2Settings.predictionRtt+=(
measuredRtt-mm2Settings.predictionRtt)*alpha return math.clamp(mm2Settings.predictionRtt,0.02,0.6)end local trajectoryCalibration:any=nil local function getGunOriginCFrame(character:Model,gun:Tool?):
CFrame?local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?local attachment:Attachment?=root and root:FindFirstChild('GunRaycastAttachment')::Attachment?if attachment and
attachment:IsA('Attachment')then return attachment.WorldCFrame end if gun then for _,partName:string in ipairs({'Muzzle','Handle'})do local part:Instance?=gun:FindFirstChild(partName)if part and part:
IsA('BasePart')then return part.CFrame end end end for _,partName:string in ipairs({'RightHand','Right Arm','Head'})do local part:Instance?=character:FindFirstChild(partName)if part and part:IsA(
'BasePart')then return part.CFrame end end local camera:Camera?=workspace.CurrentCamera if camera and root then return CFrame.new(root.Position+Vector3.new(0,1.5,0))*(camera.CFrame-camera.CFrame.
Position)end return root and root.CFrame or nil end local function createTrajectoryCalibration():any type TrajectoryPoint={dt:number,serverTime:number,pingMs:number,position:{number},velocity:{number}
,moveDirection:{number},humanoidState:string,grounded:boolean,jumping:boolean}type HistoryPoint={dt:number,position:{number},velocity:{number},humanoidState:string,grounded:boolean,moveDirection:{
number}?,aimOffset:{number}?}type RunningStats={count:number,mean:number,m2:number,minimum:number,maximum:number}type PendingGun={shotId:string?,source:string?,attemptCounted:boolean?,tool:Tool,
activatedAt:number,origin:Vector3?,requestedAim:Vector3?,pingMs:number,targetUserId:number?,targetName:string?,targetTrajectory:{TrajectoryPoint},targetHistory:{HistoryPoint},lastTrajectorySampleAt:
number,confirmedAt:number?,latencyMs:number?,confirmedEndpoint:Vector3?,confirmedHitTarget:boolean?,missReason:string?,serverActivatedAt:number?,serverConfirmedAt:number?,shotMode:string?,silentSpawn:
string?,confirmedServerOrigin:Vector3?,confirmedHitPart:string?,confirmedHitPartPosition:Vector3?,confirmedHitPartSize:Vector3?,rootAtConfirmation:{[string]:any}?,lastSeenPosition:Vector3?,
targetHealthAtAim:number?,targetHealthAtConfirmation:number?,targetHealthAfter:number?,targetHealthLatest:number?,targetHealthDropAt:number?,targetHealthDropServerTime:number?,targetDiedAt:number?,
targetDiedServerTime:number?,healthConnection:RBXScriptConnection?,samplingStoppedAt:number?,samplingStoppedReason:string?,trajectoryCapped:boolean?,rttEngineAtConfirmation:number?,
rttDataAtConfirmation:number?}type PendingKnife={shotId:string?,attemptCounted:boolean?,tool:Tool,activatedAt:number,origin:Vector3?,requestedAim:Vector3?,requestedRoot:Vector3?,aimVelocity:Vector3?,
aimLeadSeconds:number?,pingMs:number,targetUserId:number?,targetName:string?,targetTrajectory:{TrajectoryPoint},targetHistory:{HistoryPoint},lastTrajectorySampleAt:number,serverActivatedAt:number?,
samplingStoppedAt:number?,samplingStoppedReason:string?,trajectoryCapped:boolean?,lastSeenPosition:Vector3?,targetHealthAtAim:number?,targetHealthAfter:number?,targetHealthLatest:number?,
healthConnection:RBXScriptConnection?}type KnifeTrack={shotId:string?,attemptCounted:boolean?,startedAt:number,previousAt:number?,previousPosition:Vector3?,speeds:{number},rawSpeed:number?,
spawnDelayMs:number,pingMs:number,targetUserId:number?,targetName:string?,requestedAim:Vector3?,requestedRoot:Vector3?,aimVelocity:Vector3?,aimLeadSeconds:number?,activatedAt:number?,approachRoot:
Vector3?,approachVelocity:Vector3?,approachAt:number?,targetTrajectory:{TrajectoryPoint},targetHistory:{HistoryPoint},lastTrajectorySampleAt:number,destroyConnection:RBXScriptConnection?,
closestApproach:number?,closestApproachAt:number?,bodyRadius:number?}type CalibrationEvent={shotId:string?,settings:any?,phase:string?,sessionId:string?,buildStamp:string?,kind:string,serverTime:
number,pingMs:number,latencyMs:number?,distance:number?,speed:number?,rawSpeed:number?,samples:number?,reason:string?,targetUserId:number?,targetName:string?,targetState:string?,targetJumped:boolean?,
targetTrajectory:{TrajectoryPoint}?,targetHistory:{HistoryPoint}?,observedDisplacement:{number}?,constantVelocityResidual:{number}?,horizontalLeadSeconds:number?,verticalLeadSeconds:number?,endpoint:{
number}?,origin:{number}?,observedTargetPosition:{number}?,endpointToObservedTarget:number?,shotRange:number?,hitPartPath:string?,evidence:string?,targetPositionScope:string?,hitTarget:boolean?,
pingBucket:string?,bestMotionModel:string?,motionModelErrors:{[string]:number}?,evaluationSeconds:number?,blockedName:string?,blockedTransparency:number?,blockedCanCollide:boolean?,blockedSkipped:
boolean?,blockedDistance:number?,rttEngine:number?,rttData:number?,occluder:any?}type Estimates={gunAcceptanceMs:number?,knifeSpeed:number?,confirmedShots:number,confirmedThrows:number,
liveGunAcceptanceMs:number?,liveGunSamples:number,maxStabDistance:number?,knifeSpawnDelayMs:number?,confirmedSpawnDelays:number,predictionErrorSamples:number,predictionErrorMedianStuds:number?,
predictionErrorMeanStuds:number?}type Controller={prepare:(self:Controller)->(),start:(self:Controller)->(),save:(self:Controller,reason:string?)->boolean,reset:(self:Controller)->(),destroy:(self:
Controller)->(),status:(self:Controller)->string,getEstimates:(self:Controller)->Estimates,noteAuthoredShot:(self:Controller,info:any)->(),noteTracerData:(self:Controller,info:any)->(),getConsent:(
self:Controller)->boolean?,setConsent:(self:Controller,value:boolean?)->boolean}local OUTPUT_ROOT:string=host.PRODUCT.storageFolder local OUTPUT_FOLDER:string=OUTPUT_ROOT..'/Telemetry'local
OUTPUT_PATH:string=OUTPUT_FOLDER..'/MM2_Trajectory_Calibration.json'local MAX_EVENTS:number=2400 local MAX_PENDING_AGE:number=2.5 local TRAJECTORY_SAMPLE_HZ:number=60 local MAX_TRAJECTORY_POINTS:
number=200 local environment:{[string]:any}=getfenv()::any local playerMouse:Mouse=LocalPlayer:GetMouse()local runtime:any={active=false,observationGeneration=0,snapshotInProgress=false,
resaveRequested=false,lastSnapshotBytes=nil::string?,recoverySequence=0,historyLoaded=false,dirty=false,revision=0,saveScheduled=false,startedAt=os.clock(),startedAtUnix=os.time(),sessionId=
HttpService:GenerateGUID(false),buildStamp=tostring(host.SOURCE_STAMP or'unknown'),shotSequence=0,historyBackup=nil::string?,pendingHistoryBackup=nil::string?,historyExpectedAbsent=false,
persistenceBlocked=false,rawKnifeActivations=0,unmatchedGunSignals=0,sessionStabAttempts=0,sessionGunAttempts=0,sessionGunConfirmed=0,sessionGunHitTarget=0,sessionGunKills=0,sessionGunMisses=0,
missReasons={}::{[string]:number},sessionKnifeAttempts=0,sessionKnifeConfirmed=0,sessionKnifeObserved=0,pendingGuns={}::{PendingGun},lastBlockedAt=0,leadSeeds=nil::any,leadSeedsApplied=false,
evictedEvents=0,archivedEvents=0,archiveFiles={},archiveSequence=0,archiveInProgress=false,pendingKnives={}::{PendingKnife},tracks={}::{[Instance]:KnifeTrack},observedTools=setmetatable({},{__mode='k'
})::{[Tool]:boolean},connections={}::{RBXScriptConnection},events={}::{CalibrationEvent},freshGunLatency={}::{{at:number,ms:number}},gunAcceptance=nil::RunningStats?,knifeSpeed=nil::RunningStats?,
knifeSpawnDelay=nil::RunningStats?,motionBuckets={}::{[string]:any},trajectorySampler=nil::any}local controller:Controller local saveSnapshot:(reason:string)->boolean local archiveEvents:()->boolean
local compactPending:(queue:{any},now:number)->()local finalizeGunCapture:(pending:PendingGun)->()local function nextShotId():string runtime.shotSequence+=1 return runtime.sessionId..':'..tostring(
runtime.shotSequence)end local function emitShot(eventType:string,info:any):()info.sessionId=runtime.sessionId info.buildStamp=runtime.buildStamp if type(state.emitGameBridgeEvent)=='function'then
state.emitGameBridgeEvent(eventType,info)end end local function newStats():RunningStats return{count=0,mean=0,m2=0,minimum=math.huge,maximum=-math.huge}end runtime.gunAcceptance=newStats()runtime.
knifeSpeed=newStats()runtime.knifeSpawnDelay=newStats()local function finite(value:number):boolean return value==value and value>-math.huge and value<math.huge end local function applyStatsPayload(
stats:RunningStats,payload:{[string]:number}):()stats.count=tonumber(payload.count)or 0 stats.mean=tonumber(payload.mean)or 0 stats.m2=tonumber(payload.m2)or 0 stats.minimum=stats.count>0 and(
tonumber(payload.minimum)or 0)or math.huge stats.maximum=stats.count>0 and(tonumber(payload.maximum)or 0)or-math.huge end local function restoreMotionBucket(bucket:any):any local count:number=
tonumber(bucket.count)or 0 local residual:any=bucket.meanConstantVelocityResidual or{}local errors:any=bucket.meanModelError or{}return{count=count,durationSum=bucket.durationSum or(bucket.
meanDurationSeconds or 0)*count,horizontalLeadCount=bucket.horizontalLeadCount or(bucket.meanHorizontalLeadSeconds and count or 0),horizontalLeadSum=bucket.horizontalLeadSum or(bucket.
meanHorizontalLeadSeconds or 0)*count,verticalLeadCount=bucket.verticalLeadCount or(bucket.meanVerticalLeadSeconds and count or 0),verticalLeadSum=bucket.verticalLeadSum or(bucket.
meanVerticalLeadSeconds or 0)*count,residualXSum=bucket.residualXSum or(residual[1]or 0)*count,residualYSum=bucket.residualYSum or(residual[2]or 0)*count,residualZSum=bucket.residualZSum or(residual[3
]or 0)*count,constantVelocityErrorSum=bucket.constantVelocityErrorSum or(errors.constantVelocity or 0)*count,observedAccelerationErrorSum=bucket.observedAccelerationErrorSum or(errors.
observedAcceleration or 0)*count,airborneBallisticErrorSum=bucket.airborneBallisticErrorSum or(errors.airborneBallistic or 0)*count}end local function applyLearnedSeeds():boolean if runtime.
leadSeedsApplied then return true end local seeds:any=runtime.leadSeeds if type(seeds)~='table'then return false end local prediction:any=state.mm2Prediction if not prediction or type(prediction.
seedLeadError)~='function'then return false end for mode:string,entry:any in pairs(seeds)do if type(entry)=='table'and predictionMath.finite(entry.median)then pcall(prediction.seedLeadError,mode,entry
.median,entry.samples)end end runtime.leadSeedsApplied=true return true end local function loadSavedHistory():()if type(environment.readfile)~='function'then runtime.persistenceBlocked=true return end
local okRead:boolean,raw:any=pcall(environment.readfile,OUTPUT_PATH)if not okRead or type(raw)~='string'then local okExists,exists=pcall(function()return environment.isfile(OUTPUT_PATH)end)runtime.
persistenceBlocked=not(okExists and exists==false)runtime.historyExpectedAbsent=not runtime.persistenceBlocked return end runtime.pendingHistoryBackup=raw if#raw==0 then return end local ok,decoded=
pcall(HttpService.JSONDecode,HttpService,raw)if not ok or type(decoded)~='table'then return end local aggregate:any=type(decoded.aggregate)=='table'and decoded.aggregate or{}local function importStats
(stats:RunningStats,payload:any,low:number,high:number):()if type(payload)~='table'then return end for _,key in ipairs({'count','mean','m2','minimum','maximum'})do if not predictionMath.finite(payload
[key])then return end end if payload.count<1 or payload.count>1000000 or payload.count%1~=0 or payload.m2<0 or payload.m2>payload.count*(high-low)^2 or payload.minimum<low or payload.maximum>high or
payload.mean<payload.minimum or payload.mean>payload.maximum then return end applyStatsPayload(stats,payload)end importStats(runtime.gunAcceptance,aggregate.gunAcceptanceMs,0,2500)importStats(runtime.
knifeSpeed,aggregate.knifeSpeedStudsPerSecond,24,300)importStats(runtime.knifeSpawnDelay,aggregate.knifeSpawnDelayMs,0,5000)local learned:any=type(decoded.predictor)=='table'and decoded.predictor or{}
local learnedBuckets:any=type(learned.leadErrors)=='table'and learned.leadErrors or nil if learnedBuckets then runtime.leadSeeds=learnedBuckets runtime.leadSeedsApplied=false applyLearnedSeeds()end
end local function vectorArray(value:Vector3):{number}return{value.X,value.Y,value.Z}end local function pingBucketKey(pingMs:number):string if pingMs>250 then return'250+'end local lower:number=math.
clamp(math.floor(pingMs/5)*5,0,245)return string.format('%03d-%03d',lower,lower+5)end local function getTargetRoot(player:Player?):BasePart?return player and player.Character and player.Character:
FindFirstChild('HumanoidRootPart')::BasePart?end local function getTargetHumanoid(player:Player?):Humanoid?return player and player.Character and player.Character:FindFirstChildOfClass('Humanoid')::
Humanoid?end local function captureTrajectoryPoint(player:Player,activatedAt:number,pingMs:number):TrajectoryPoint?local root:BasePart?=getTargetRoot(player)local humanoid:Humanoid?=getTargetHumanoid(
player)if not root or not humanoid or humanoid.Health<=0 then return nil end local stateName:string=humanoid:GetState().Name local grounded:boolean=humanoid.FloorMaterial~=Enum.Material.Air local
jumping:boolean=stateName=='Jumping'or stateName=='Freefall'or not grounded return{dt=os.clock()-activatedAt,serverTime=workspace:GetServerTimeNow(),pingMs=pingMs,position=vectorArray(root.Position),
velocity=vectorArray(root.AssemblyLinearVelocity),moveDirection=vectorArray(humanoid.MoveDirection),humanoidState=stateName,grounded=grounded,jumping=jumping}end local function watchPendingHealth(
pending:any,target:Player?):()pending.targetHealthAtAim=nil pending.targetHealthAfter=nil pending.targetHealthLatest=nil pending.targetHealthDropAt=nil pending.targetDiedAt=nil local character:Model?=
target and target.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')if not humanoid then return end pending.targetHealthAtAim=humanoid.Health pending.
targetHealthLatest=humanoid.Health if pending.healthConnection then pending.healthConnection:Disconnect()pending.healthConnection=nil end pending.healthConnection=humanoid.HealthChanged:Connect(
function(health:number):()if not predictionMath.finite(health)then return end local previous:number?=pending.targetHealthLatest pending.targetHealthLatest=health if previous~=nil and health<previous
and(pending.targetHealthAfter==nil or health<pending.targetHealthAfter)then pending.targetHealthAfter=health if pending.targetHealthDropAt==nil then pending.targetHealthDropAt=os.clock()pending.
targetHealthDropServerTime=workspace:GetServerTimeNow()end if health<=0 and pending.targetDiedAt==nil then pending.targetDiedAt=os.clock()pending.targetDiedServerTime=workspace:GetServerTimeNow()end
end end)end local function findKnifeAnalysisTarget(aimPosition:Vector3?):Player?local selected:Player?=nil local selectedScore:number=math.huge for _,player:Player in ipairs(Players:GetPlayers())do if
player~=LocalPlayer and isPlayerAlive(player)and getPlayerRole(player)~='Murderer'and not isProtectedTarget(player)then local root:BasePart?=getTargetRoot(player)if root then local score:number=
aimPosition and(root.Position-aimPosition).Magnitude or(root.Position-(LocalPlayer.Character and LocalPlayer.Character:GetPivot().Position or Vector3.zero)).Magnitude if score<selectedScore then
selected=player selectedScore=score end end end end return selected end local function getRequestedAim():Vector3?local aim:Vector3?=nil pcall(function():()aim=playerMouse.Hit.Position end)return aim
end local function samplePendingTrajectories():()local now:number=os.clock()local function sampleQueue(queue:{any}):()for _,pending:any in ipairs(queue)do if pending.targetUserId and now-pending.
lastTrajectorySampleAt>=1/TRAJECTORY_SAMPLE_HZ then local target:Player?=Players:GetPlayerByUserId(pending.targetUserId)local point:TrajectoryPoint?=target and captureTrajectoryPoint(target,pending.
activatedAt,pending.pingMs)or nil if point then if#pending.targetTrajectory<MAX_TRAJECTORY_POINTS then table.insert(pending.targetTrajectory,point)else pending.trajectoryCapped=true end elseif pending
.samplingStoppedAt==nil then pending.samplingStoppedAt=now pending.samplingStoppedReason=target and'target-dead-or-missing'or'target-left'end if target then local root:BasePart?=getTargetRoot(target)
if root then pending.lastSeenPosition=root.Position end end pending.lastTrajectorySampleAt=now end end end sampleQueue(runtime.pendingGuns)sampleQueue(runtime.pendingKnives)for index:number=#runtime.
pendingGuns,1,-1 do local pending:PendingGun=runtime.pendingGuns[index]if pending.confirmedAt and now-pending.confirmedAt>=0.5 then table.remove(runtime.pendingGuns,index)finalizeGunCapture(pending)
end end if#runtime.pendingGuns==0 and#runtime.pendingKnives==0 and runtime.trajectorySampler then runtime.trajectorySampler:Disconnect()runtime.trajectorySampler=nil end end local function
ensureTrajectorySampler():()if runtime.trajectorySampler then return end runtime.trajectorySampler=TaskManager:Connect(function(_deltaTime:number):()compactPending(runtime.pendingGuns,os.clock())
compactPending(runtime.pendingKnives,os.clock())samplePendingTrajectories()end,nil,'MM2 shot sampler')end local function addSample(stats:RunningStats,value:number):()if not finite(value)then return
end stats.count+=1 local delta:number=value-stats.mean stats.mean+=delta/stats.count stats.m2+=delta*(value-stats.mean)stats.minimum=math.min(stats.minimum,value)stats.maximum=math.max(stats.maximum,
value)end local function statsPayload(stats:RunningStats):{[string]:number}return{count=stats.count,mean=stats.mean,m2=stats.m2,minimum=stats.count>0 and stats.minimum or 0,maximum=stats.count>0 and
stats.maximum or 0}end local function getPingMilliseconds():number local pingMs:number=0 pcall(function():()local player:any=LocalPlayer local direct:any=player and player.GetNetworkPing if type(
direct)~='function'then return end local ok:boolean,seconds:any=pcall(direct,player)if ok and finite(seconds)and(seconds::number)>0 then pingMs=(seconds::number)*1000 end end)return math.max(0,pingMs)
end local function worldPosition(value:any):Vector3?if typeof(value)=='Vector3'then return value elseif typeof(value)=='CFrame'then return value.Position elseif typeof(value)=='Instance'then if value:
IsA('Attachment')then return value.WorldPosition elseif value:IsA('BasePart')then return value.Position elseif value:IsA('Model')then return value:GetPivot().Position end end return nil end
local function projectilePosition(projectile:Instance):Vector3?local visual:Instance?=projectile:FindFirstChild('KnifeVisual',true)if visual and visual:IsA('BasePart')then return visual.Position end
local blade:Instance?=projectile:FindFirstChild('BladePosition',true)local bladePosition:Vector3?=worldPosition(blade)if bladePosition then return bladePosition end return worldPosition(projectile)end
local function toolOrigin(tool:Tool):Vector3?local handle:Instance?=tool:FindFirstChild('Handle')return worldPosition(handle)end local function appendEvent(record:CalibrationEvent):()record.sessionId=
runtime.sessionId record.buildStamp=runtime.buildStamp table.insert(runtime.events,record)runtime.revision+=1 runtime.dirty=true local states:any=type(configData)=='table'and configData.states or nil
if not states or states['MM2.Calibration.Consent']~=true then return end if runtime.saveScheduled then return end runtime.saveScheduled=true task.delay(10,function():()runtime.saveScheduled=false if
runtime.active and runtime.dirty then if#runtime.events>=MAX_EVENTS then if saveSnapshot('before-archive')and archiveEvents()then saveSnapshot('after-archive')end else saveSnapshot('autosave')end end
end)end local function captureTargetHistory(player:Player?,now:number):{HistoryPoint}local points:{HistoryPoint}={}if not player or not state.mm2Prediction or type(state.mm2Prediction.getHistory)~=
'function'then return points end local ok:boolean,history:any,age:any=pcall(state.mm2Prediction.getHistory,player,0.6)if not ok or type(history)~='table'then return points end local sampleAge:number=
predictionMath.finite(age)and age or 0 local HISTORY_LIMIT:number=36 local samples:{any}={}for _,sample:any in ipairs(history)do local position:any=sample.position local velocity:any=sample.velocity
if typeof(position)=='Vector3'and typeof(velocity)=='Vector3'then table.insert(samples,sample)end end local stride:number=1 if#samples>HISTORY_LIMIT then stride=math.ceil(#samples/(HISTORY_LIMIT-1))
end for index,sample:any in ipairs(samples)do local isLast:boolean=index==#samples if isLast or(index-1)%stride==0 then local position:any=sample.position local velocity:any=sample.velocity local move
:any=sample.moveDirection local offset:any=sample.aimOffset table.insert(points,{dt=(sample.at or 0)-sampleAge,position={position.X,position.Y,position.Z},velocity={velocity.X,velocity.Y,velocity.Z},
humanoidState=tostring(sample.state or'Running'),grounded=sample.grounded==true,moveDirection=typeof(move)=='Vector3'and{move.X,move.Y,move.Z}or nil,aimOffset=typeof(offset)=='Vector3'and{offset.X,
offset.Y,offset.Z}or nil})end end return points end local function trackConnection(connection:RBXScriptConnection):()table.insert(runtime.connections,connection)end compactPending=function(queue:{any}
,now:number):()local function drop(index:number):()local entry:any=queue[index]if entry and entry.healthConnection then pcall(function():()(entry.healthConnection::RBXScriptConnection):Disconnect()end
)entry.healthConnection=nil end table.remove(queue,index)end while#queue>0 and now-queue[1].activatedAt>MAX_PENDING_AGE do drop(1)end while#queue>12 do drop(1)end end local function vectorFromArray(
value:{number}):Vector3 return Vector3.new(value[1]or 0,value[2]or 0,value[3]or 0)end local function analyzeTargetTrajectory(trajectory:{TrajectoryPoint},pingMs:number,eventKind:string,
evaluationSeconds:number?):{[string]:any}if#trajectory<2 then return{pingBucket=pingBucketKey(pingMs),sampleCount=#trajectory}end local first:TrajectoryPoint=trajectory[1]local last:TrajectoryPoint=
trajectory[#trajectory]if evaluationSeconds then local desiredDt:number=first.dt+evaluationSeconds local closestDistance:number=math.huge for _,point:TrajectoryPoint in ipairs(trajectory)do local
distance:number=math.abs(point.dt-desiredDt)if distance<closestDistance then last=point closestDistance=distance end end end local initialPosition:Vector3=vectorFromArray(first.position)local
finalPosition:Vector3=vectorFromArray(last.position)local initialVelocity:Vector3=vectorFromArray(first.velocity)local duration:number=math.max(0.0001,last.dt-first.dt)local displacement:Vector3=
finalPosition-initialPosition local residual:Vector3=displacement-initialVelocity*duration local constantVelocityPrediction:Vector3=initialPosition+initialVelocity*duration local observedAcceleration:
Vector3=Vector3.zero if#trajectory>=3 then local second:TrajectoryPoint=trajectory[2]local accelerationDelta:number=math.max(1/240,second.dt-first.dt)observedAcceleration=(vectorFromArray(second.
velocity)-initialVelocity)/accelerationDelta if observedAcceleration.Magnitude>500 then observedAcceleration=observedAcceleration.Unit*500 end end local accelerationPrediction:Vector3=initialPosition+
initialVelocity*duration+observedAcceleration*(0.5*duration*duration)local ballisticPrediction:Vector3=initialPosition+initialVelocity*duration+Vector3.new(0,-workspace.Gravity,0)*(0.5*duration*
duration)local modelErrors:{[string]:number}={constantVelocity=(constantVelocityPrediction-finalPosition).Magnitude,observedAcceleration=(accelerationPrediction-finalPosition).Magnitude,
airborneBallistic=(ballisticPrediction-finalPosition).Magnitude}local bestMotionModel:string='constantVelocity'local bestMotionError:number=modelErrors.constantVelocity for modelName:string,modelError
:number in pairs(modelErrors)do if modelError<bestMotionError then bestMotionModel=modelName bestMotionError=modelError end end local horizontalVelocity:Vector3=Vector3.new(initialVelocity.X,0,
initialVelocity.Z)local horizontalDisplacement:Vector3=Vector3.new(displacement.X,0,displacement.Z)local horizontalLead:number?=nil if horizontalVelocity.Magnitude>0.75 then horizontalLead=math.clamp(
horizontalDisplacement:Dot(horizontalVelocity)/horizontalVelocity:Dot(horizontalVelocity),-0.15,0.6)end local verticalLead:number?=nil if math.abs(initialVelocity.Y)>0.75 then verticalLead=math.clamp(
displacement.Y/initialVelocity.Y,-0.15,0.6)end local jumped:boolean=false for _,point:TrajectoryPoint in ipairs(trajectory)do if point.jumping then jumped=true break end end local stateName:string=
jumped and'Airborne'or'Grounded'local bucketKey:string=eventKind..':'..pingBucketKey(pingMs)..':'..stateName local bucket:any=runtime.motionBuckets[bucketKey]if type(bucket)~='table'then bucket={count
=0,horizontalLeadSum=0,horizontalLeadCount=0,verticalLeadSum=0,verticalLeadCount=0,residualXSum=0,residualYSum=0,residualZSum=0,durationSum=0,constantVelocityErrorSum=0,observedAccelerationErrorSum=0,
airborneBallisticErrorSum=0}runtime.motionBuckets[bucketKey]=bucket end bucket.count+=1 bucket.durationSum+=duration bucket.residualXSum+=residual.X bucket.residualYSum+=residual.Y bucket.residualZSum
+=residual.Z bucket.constantVelocityErrorSum+=modelErrors.constantVelocity bucket.observedAccelerationErrorSum+=modelErrors.observedAcceleration bucket.airborneBallisticErrorSum+=modelErrors.
airborneBallistic if horizontalLead then bucket.horizontalLeadSum+=horizontalLead bucket.horizontalLeadCount+=1 end if verticalLead then bucket.verticalLeadSum+=verticalLead bucket.verticalLeadCount+=
1 end return{pingBucket=pingBucketKey(pingMs),sampleCount=#trajectory,initialState=first.humanoidState,finalState=last.humanoidState,jumped=jumped,durationSeconds=duration,evaluationSeconds=
evaluationSeconds or duration,displacement=vectorArray(displacement),constantVelocityResidual=vectorArray(residual),horizontalLeadSeconds=horizontalLead,verticalLeadSeconds=verticalLead,
bestMotionModel=bestMotionModel,motionModelErrors=modelErrors}end local function motionModelPayload(buckets:{[string]:any}):{[string]:any}local payload:{[string]:any}={}for key:string,bucket:any in
pairs(buckets)do if type(bucket)=='table'then local count:number=math.max(1,tonumber(bucket.count)or 1)local horizontalCount:number=math.max(1,tonumber(bucket.horizontalLeadCount)or 0)local
verticalCount:number=math.max(1,tonumber(bucket.verticalLeadCount)or 0)local meanDuration:number=tonumber(bucket.meanDurationSeconds)or(bucket.durationSum and(bucket.durationSum/count))or 0 local
meanHorizLead:number?=nil if bucket.horizontalLeadSum and horizontalCount>0 then meanHorizLead=bucket.horizontalLeadSum/horizontalCount elseif bucket.meanHorizontalLeadSeconds then meanHorizLead=
bucket.meanHorizontalLeadSeconds end local meanVertLead:number?=nil if bucket.verticalLeadSum and verticalCount>0 then meanVertLead=bucket.verticalLeadSum/verticalCount elseif bucket.
meanVerticalLeadSeconds then meanVertLead=bucket.meanVerticalLeadSeconds end local meanResidual:{number}if type(bucket.meanConstantVelocityResidual)=='table'then meanResidual=bucket.
meanConstantVelocityResidual else meanResidual={(tonumber(bucket.residualXSum)or 0)/count,(tonumber(bucket.residualYSum)or 0)/count,(tonumber(bucket.residualZSum)or 0)/count}end local meanModelErr:{[
string]:number}if type(bucket.meanModelError)=='table'then meanModelErr=bucket.meanModelError else meanModelErr={constantVelocity=(tonumber(bucket.constantVelocityErrorSum)or 0)/count,
observedAcceleration=(tonumber(bucket.observedAccelerationErrorSum)or 0)/count,airborneBallistic=(tonumber(bucket.airborneBallisticErrorSum)or 0)/count}end payload[key]={durationSum=bucket.durationSum
,horizontalLeadSum=bucket.horizontalLeadSum,horizontalLeadCount=bucket.horizontalLeadCount,verticalLeadSum=bucket.verticalLeadSum,verticalLeadCount=bucket.verticalLeadCount,residualXSum=bucket.
residualXSum,residualYSum=bucket.residualYSum,residualZSum=bucket.residualZSum,constantVelocityErrorSum=bucket.constantVelocityErrorSum,observedAccelerationErrorSum=bucket.observedAccelerationErrorSum
,airborneBallisticErrorSum=bucket.airborneBallisticErrorSum,count=tonumber(bucket.count)or 0,meanDurationSeconds=meanDuration,meanHorizontalLeadSeconds=meanHorizLead,meanVerticalLeadSeconds=
meanVertLead,meanConstantVelocityResidual=meanResidual,meanModelError=meanModelErr}end end return payload end local function median(values:{number}):number?if#values==0 then return nil end local
sorted:{number}=table.clone(values)table.sort(sorted)local middle:number=math.floor((#sorted+1)/2)if#sorted%2==1 then return sorted[middle]end return(sorted[middle]+sorted[middle+1])*0.5 end local
estimatesCache:Estimates?=nil local estimatesAt:number=-math.huge local function endpointErrorStats():(number,number?,number?)local values:{number}={}for _,ev in ipairs(runtime.events)do if ev.kind==
'gun'and predictionMath.finite(ev.endpointErrorStuds)then table.insert(values,ev.endpointErrorStuds)end end local mean:number?=nil if#values>0 then local total:number=0 for _,value in ipairs(values)do
total+=value end mean=total/#values end return#values,median(values),mean end local function estimates():Estimates local now:number=os.clock()if estimatesCache and now-estimatesAt<0.25 then return
estimatesCache end local fresh:{number}={}for _,sample in ipairs(runtime.freshGunLatency)do if now-sample.at<=60 then table.insert(fresh,sample.ms)end end local gunStats:RunningStats=runtime.
gunAcceptance local speedStats:RunningStats=runtime.knifeSpeed local spawnDelayStats:RunningStats=runtime.knifeSpawnDelay local maxStab:number?=nil for _,ev in ipairs(runtime.events)do if ev.kind==
'stab'and ev.hitTarget==true and predictionMath.finite(ev.distance)then if not maxStab or ev.distance>maxStab then maxStab=ev.distance end end end local errCount:number,errMedian:number?,errMean:
number?=endpointErrorStats()estimatesCache={predictionErrorSamples=errCount,predictionErrorMedianStuds=errMedian,predictionErrorMeanStuds=errMean,gunAcceptanceMs=gunStats.count>0 and gunStats.mean or
nil,knifeSpeed=speedStats.count>0 and speedStats.mean or nil,confirmedShots=gunStats.count,confirmedThrows=speedStats.count,maxStabDistance=maxStab,liveGunAcceptanceMs=#fresh>=6 and median(fresh)or
nil,liveGunSamples=#fresh,knifeSpawnDelayMs=spawnDelayStats.count>0 and spawnDelayStats.mean or nil,confirmedSpawnDelays=spawnDelayStats.count}estimatesAt=now return estimatesCache::Estimates end
local function preserveFailedSnapshot(encoded:string,generation:number,sessionId:string):()local states:any=type(configData)=='table'and configData.states or nil if not states or states[
'MM2.Calibration.Consent']~=true or not runtime.active or runtime.observationGeneration~=generation or type(environment.isfile)~='function'or type(environment.readfile)~='function'or type(environment.
writefile)~='function'or(type(state.ownsRuntimeData)=='function'and not state.ownsRuntimeData())then return end for _attempt:number=1,32 do runtime.recoverySequence+=1 local path:string=OUTPUT_PATH..
'.recovery-'..sessionId..'-'..tostring(runtime.recoverySequence)..'.json'local okExists:boolean,exists:any=pcall(environment.isfile,path)if not okExists or type(exists)~='boolean'then break end if not
exists then if states['MM2.Calibration.Consent']~=true or not runtime.active or runtime.observationGeneration~=generation or(type(state.ownsRuntimeData)=='function'and not state.ownsRuntimeData())then
return end local okWrite:boolean=pcall(environment.writefile,path,encoded)local okRead:boolean,recovered:any=pcall(environment.readfile,path)if okWrite and okRead and recovered==encoded then warn(
trajectoryLogPrefix..' Verified recovery snapshot: '..path)else warn(trajectoryLogPrefix..[[ Recovery write could not be verified; data remains in memory.]])end return end end warn(trajectoryLogPrefix
..' Could not reserve a recovery path; data remains in memory.')end local function writeSnapshot(reason:string):boolean local generation:number=runtime.observationGeneration local snapshotSessionId:
string=runtime.sessionId local consentStates:any=type(configData)=='table'and configData.states or nil if not consentStates or consentStates['MM2.Calibration.Consent']~=true then return false end if
runtime.persistenceBlocked or(type(state.ownsRuntimeData)=='function'and not state.ownsRuntimeData())then return false end if type(environment.writefile)~='function'then warn(trajectoryLogPrefix..
' writefile is unavailable; data was not saved.')return false end local prior:any=runtime.pendingHistoryBackup if type(prior)=='string'then local okCurrent,current=pcall(environment.readfile,
OUTPUT_PATH)if not okCurrent or current~=prior then runtime.persistenceBlocked=true warn(trajectoryLogPrefix..[[ History changed before consent; it will not be overwritten.]])return false end if not
runtime.historyBackup then local backup:string=OUTPUT_PATH..'.before-'..runtime.sessionId..'.json'local okExists:boolean,exists:any=pcall(environment.isfile,backup)if not okExists or exists~=false
then runtime.persistenceBlocked=true warn(trajectoryLogPrefix..[[ Backup path is occupied or unavailable; existing files are untouched.]])return false end if consentStates['MM2.Calibration.Consent']~=
true or not runtime.active or runtime.observationGeneration~=generation or(type(state.ownsRuntimeData)=='function'and not state.ownsRuntimeData())then return false end local okBackup:boolean=pcall(
environment.writefile,backup,prior)local okVerify,backedUp=pcall(environment.readfile,backup)if not okBackup or not okVerify or backedUp~=prior then runtime.persistenceBlocked=true warn(
trajectoryLogPrefix..[[ History backup failed; existing file will not be overwritten.]])return false end runtime.historyBackup=backup end end local savedRevision:number=runtime.revision local
mergedGun=statsPayload(runtime.gunAcceptance)local mergedKnife=statsPayload(runtime.knifeSpeed)local mergedDelay=statsPayload(runtime.knifeSpawnDelay)local payload:{[string]:any}={schema=4,kind=
'mm2-trajectory-analytics',buildStamp=runtime.buildStamp,sessionId=runtime.sessionId,historyBackup=runtime.historyBackup,settings={silentSpawn=mm2Settings.silentSpawn,silentAim=mm2Settings.silentAim,
silentSweep=mm2Settings.silentSweep,wallCheck=mm2Settings.shootWallCheck,autoLead=state.mm2Prediction and state.mm2Prediction.settings.autoLead or true,leadOffsetSeconds=state.mm2Prediction and state.
mm2Prediction.settings.extraLead or 0,replicationInterpolationSeconds=state.mm2Prediction and state.mm2Prediction.settings.interpolation or nil},placeId=game.PlaceId,savedAt=DateTime.now():ToIsoDate()
,reason=reason,aggregateScope=[[validated-history-plus-this-session; not a hit-rate denominator]],aggregate={gunAcceptanceMs=mergedGun,knifeSpeedStudsPerSecond=mergedKnife,knifeSpawnDelayMs=
mergedDelay},predictor=(function():{[string]:any}local out:{[string]:any}={}local predictorState:any=state.mm2Prediction if predictorState and type(predictorState.exportLeadState)=='function'then
local ok:boolean,value:any=pcall(predictorState.exportLeadState)if ok and type(value)=='table'then out.leadErrors=value end end if predictorState and type(predictorState.getLeadSamples)=='function'
then local ok:boolean,guns:any=pcall(predictorState.getLeadSamples,'gun')local okKnife:boolean,knives:any=pcall(predictorState.getLeadSamples,'knife')out.samples={gun=ok and guns or nil,knife=okKnife
and knives or nil}end out.readyFor=[[3+ recent samples activate a correction; each mode retains up to 8 (not proof of convergence)]]return out end)(),estimator={gunAcceptanceMs=mergedGun.count>0 and
mergedGun.mean or nil,knifeSpeed=mergedKnife.count>0 and mergedKnife.mean or nil,confirmedShots=mergedGun.count,confirmedThrows=mergedKnife.count},analytics={bucketSpec={metric='Data Ping',widthMs=5,
minimumMs=0,maximumMs=250,states={'Grounded','Airborne'}},motionModel=motionModelPayload(runtime.motionBuckets)},session={startedAt=runtime.startedAtUnix,countersScope='this-session-only',
gunConfirmationMeaning=[[matched GunFired signal; hitTarget means the ray reached the target, died means the humanoid reached 0 health]],evictedEvents=runtime.evictedEvents,archivedEvents=runtime.
archivedEvents,archiveFiles=runtime.archiveFiles,eventScope=[[events in this snapshot plus the listed verified session archive files]],healthMeaning=
[[client-observed Humanoid.Health; healthDropServerTime is GetServerTimeNow at the drop]],knifeConfirmationMeaning='own linked projectile with valid speed; not an impact',knifeObservedMeaning=
[[all ThrowingKnife projectiles visible to this client, not all server events]],knifeObserved=runtime.sessionKnifeObserved,rawKnifeActivations=runtime.rawKnifeActivations,unmatchedGunSignals=runtime.
unmatchedGunSignals,stabAttempts=runtime.sessionStabAttempts,elapsedSeconds=os.clock()-runtime.startedAt,predictionError=(function():{[string]:any}local ok:boolean,n:any,med:any,mean:any=pcall(
endpointErrorStats)return{samples=ok and n or 0,medianStuds=ok and med or nil,meanStuds=ok and mean or nil,scope=
[[own predicted point vs the torso point at the scoring instant (interpolated from the newest pre-shot sample and the root at confirmation); an estimate, not a server measurement]]}end)(),
gunByModeScope='events in the current segment only',gunByMode=(function():{[string]:any}local modes:{[string]:any}={}if type(runtime.events)=='table'then for _,ev in ipairs(runtime.events)do if ev.
kind=='gun'then if ev.phase=='authored'then local mode:string=type(ev.settings)=='table'and tostring(ev.settings.mode or'?')or'?'local row:any=modes[mode]or{attempts=0,confirmed=0,hits=0,kills=0}row.
attempts+=1 modes[mode]=row elseif ev.phase=='confirmation'then local mode:string=tostring(ev.shotMode or'?')local row:any=modes[mode]or{attempts=0,confirmed=0,hits=0,kills=0}row.confirmed+=1 if ev.
hitTarget then row.hits+=1 end if ev.died then row.kills+=1 end modes[mode]=row end end end end return modes end)(),gunAttempts=runtime.sessionGunAttempts,gunConfirmed=runtime.sessionGunConfirmed,
gunHitTarget=runtime.sessionGunHitTarget,gunKills=runtime.sessionGunKills,gunMisses=runtime.sessionGunMisses or 0,gunMissReasons=runtime.missReasons or{},knifeAttempts=runtime.sessionKnifeAttempts,
knifeConfirmed=runtime.sessionKnifeConfirmed,events=runtime.events}}local encodedOk:boolean,encoded:any=pcall(HttpService.JSONEncode,HttpService,payload)if not encodedOk or type(encoded)~='string'then
warn(trajectoryLogPrefix..' JSONEncode failed: '..tostring(encoded))return false end if type(environment.makefolder)=='function'then pcall(environment.makefolder,OUTPUT_ROOT)pcall(environment.
makefolder,OUTPUT_FOLDER)end if consentStates['MM2.Calibration.Consent']~=true or not runtime.active or runtime.observationGeneration~=generation or(type(state.ownsRuntimeData)=='function'and not
state.ownsRuntimeData())then return false end local expected:string?=if type(prior)=='string'then prior else runtime.lastSnapshotBytes if type(expected)=='string'then local okCurrent:boolean,current:
any=pcall(environment.readfile,OUTPUT_PATH)if not okCurrent or current~=expected then runtime.persistenceBlocked=true warn(trajectoryLogPrefix..
' History changed while saving; it will not be overwritten.')return false end elseif runtime.historyExpectedAbsent then local okExists:boolean,exists:any=pcall(environment.isfile,OUTPUT_PATH)if not
okExists or exists~=false then runtime.persistenceBlocked=true warn(trajectoryLogPrefix..[[ History appeared before saving; it will not be overwritten.]])return false end end if consentStates[
'MM2.Calibration.Consent']~=true or not runtime.active or runtime.observationGeneration~=generation or(type(state.ownsRuntimeData)=='function'and not state.ownsRuntimeData())then return false end
local writeOk:boolean,writeError:any=pcall(environment.writefile,OUTPUT_PATH,encoded)if not writeOk then warn(trajectoryLogPrefix..' writefile failed: '..tostring(writeError))local changed:boolean if
type(expected)=='string'then local okCurrent:boolean,current:any=pcall(environment.readfile,OUTPUT_PATH)changed=not okCurrent or current~=expected else local okExists:boolean,exists:any=pcall(
environment.isfile,OUTPUT_PATH)changed=not okExists or exists~=false end if changed then runtime.persistenceBlocked=true preserveFailedSnapshot(encoded,generation,snapshotSessionId)end return false
end local okRead:boolean,written:any=pcall(environment.readfile,OUTPUT_PATH)if not okRead or written~=encoded then runtime.persistenceBlocked=true warn(trajectoryLogPrefix..
[[ Snapshot readback failed; the previous verified backup is kept.]])preserveFailedSnapshot(encoded,generation,snapshotSessionId)return false end runtime.lastSnapshotBytes=encoded runtime.
pendingHistoryBackup=nil runtime.historyExpectedAbsent=false if consentStates['MM2.Calibration.Consent']~=true or not runtime.active or runtime.observationGeneration~=generation or(type(state.
ownsRuntimeData)=='function'and not state.ownsRuntimeData())then return true end runtime.dirty=runtime.revision~=savedRevision print(trajectoryLogPrefix..' Saved '..OUTPUT_PATH..' ('..tostring(#
encoded)..' bytes)')return true end saveSnapshot=function(reason:string):boolean if runtime.snapshotInProgress then runtime.resaveRequested=true return false end runtime.snapshotInProgress=true local
ok:boolean,saved:any=pcall(writeSnapshot,reason)runtime.snapshotInProgress=false local resave:boolean=runtime.resaveRequested runtime.resaveRequested=false if not ok then warn(trajectoryLogPrefix..
' Snapshot error: '..tostring(saved))end if resave and runtime.active and runtime.dirty and not runtime.persistenceBlocked then task.defer(function():()if runtime.active and runtime.dirty then
saveSnapshot('coalesced')end end)end return ok and saved==true end archiveEvents=function():boolean local states:any=type(configData)=='table'and configData.states or nil if not runtime.active or not
states or states['MM2.Calibration.Consent']~=true or runtime.persistenceBlocked or runtime.archiveInProgress or type(environment.writefile)~='function'or type(environment.readfile)~='function'or(type(
state.ownsRuntimeData)=='function'and not state.ownsRuntimeData())then return false end runtime.archiveInProgress=true local generation:number=runtime.observationGeneration local archiveSessionId:
string=runtime.sessionId local captured:{any}=table.clone(runtime.events)local count:number=#captured local path:string?=nil for _attempt:number=1,64 do runtime.archiveSequence+=1 local candidate:
string=OUTPUT_FOLDER..'/MM2_Trajectory_'..runtime.sessionId..'_part_'..tostring(runtime.archiveSequence)..'.json'local okExists:boolean,exists:any=false,nil if type(environment.isfile)=='function'then
okExists,exists=pcall(environment.isfile,candidate)end if not okExists or type(exists)~='boolean'then break end if not exists then path=candidate break end end if not path then runtime.
archiveInProgress=false warn(trajectoryLogPrefix..[[ Cannot verify a fresh archive path; events remain in memory.]])return false end local encodedOk:boolean,encoded:any=pcall(HttpService.JSONEncode,
HttpService,{schema=4,kind='mm2-trajectory-event-archive',sessionId=runtime.sessionId,buildStamp=runtime.buildStamp,events=captured})if not encodedOk or type(encoded)~='string'then runtime.
archiveInProgress=false warn(trajectoryLogPrefix..' Archive encode failed; events remain in memory.')return false end if not runtime.active or states['MM2.Calibration.Consent']~=true or runtime.
observationGeneration~=generation or runtime.sessionId~=archiveSessionId or(type(state.ownsRuntimeData)=='function'and not state.ownsRuntimeData())then runtime.archiveInProgress=false return false end
local okWrite:boolean=pcall(environment.writefile,path,encoded)local okRead:boolean,verified:any=pcall(environment.readfile,path)if not okWrite or not okRead or verified~=encoded or not runtime.active
or states['MM2.Calibration.Consent']~=true or runtime.observationGeneration~=generation or runtime.sessionId~=archiveSessionId or(type(state.ownsRuntimeData)=='function'and not state.ownsRuntimeData()
)then runtime.archiveInProgress=false warn(trajectoryLogPrefix..' Archive could not be verified; events remain in memory.')return false end local remaining:{any}={}for index:number=count+1,#runtime.
events do table.insert(remaining,runtime.events[index])end runtime.events=remaining runtime.archivedEvents+=count table.insert(runtime.archiveFiles,path)runtime.revision+=1 runtime.dirty=true runtime.
archiveInProgress=false return true end local function normalizeGun(value:any):Tool?if typeof(value)~='Instance'then return nil end if value:IsA('Tool')then return value end return value:
FindFirstAncestorOfClass('Tool')end local function measureLeadErrorSeconds(pending:PendingGun):number?local history:{HistoryPoint}=pending.targetHistory or{}local newest:HistoryPoint?=history[#history
]local aim:Vector3?=pending.requestedAim local latencyMs:any=pending.latencyMs local confirmation:any=pending.rootAtConfirmation if not newest or not aim or not confirmation then return nil end if not
predictionMath.finite(latencyMs)or latencyMs<=0 or latencyMs>1200 then return nil end if not predictionMath.finite(pending.activatedAt)then return nil end local beforeAt:number=pending.activatedAt+(
newest.dt or 0)local afterAt:number=pending.confirmedAt and pending.confirmedAt or(pending.activatedAt+latencyMs/1000)if not predictionMath.finite(afterAt)or afterAt<=beforeAt then return nil end
local scoringAt:number=pending.activatedAt+latencyMs/1000 local alpha:number=math.clamp((scoringAt-beforeAt)/(afterAt-beforeAt),0,1)local p0:{number}?=newest.position local p1:{number}?=confirmation.
position if type(p0)~='table'or type(p1)~='table'or#p0<3 or#p1<3 then return nil end local rootX:number=p0[1]+(p1[1]-p0[1])*alpha local rootY:number=p0[2]+(p1[2]-p0[2])*alpha local rootZ:number=p0[3]+
(p1[3]-p0[3])*alpha local offset:{number}?=newest.aimOffset if type(offset)=='table'and#offset>=3 then rootX+=offset[1]rootY+=offset[2]rootZ+=offset[3]end local velocity:{number}?=newest.velocity if
type(velocity)~='table'or#velocity<3 then return nil end local vx:number=velocity[1]local vz:number=velocity[3]local speed:number=math.sqrt(vx*vx+vz*vz)if speed<3 then return nil end local along:
number=((rootX-aim.X)*vx+(rootZ-aim.Z)*vz)/speed if math.abs(along)>6 then return nil end return along/speed end local function measureEndpointErrorStuds(pending:PendingGun):number?local history:{
HistoryPoint}=pending.targetHistory or{}local newest:HistoryPoint?=history[#history]local aim:Vector3?=pending.requestedAim local latencyMs:any=pending.latencyMs local confirmation:any=pending.
rootAtConfirmation if not newest or not aim or not confirmation then return nil end if not predictionMath.finite(latencyMs)or latencyMs<=0 or latencyMs>1200 then return nil end if not predictionMath.
finite(pending.activatedAt)then return nil end local beforeAt:number=pending.activatedAt+(newest.dt or 0)local afterAt:number=pending.confirmedAt and pending.confirmedAt or(pending.activatedAt+
latencyMs/1000)if not predictionMath.finite(afterAt)or afterAt<=beforeAt then return nil end local scoringAt:number=pending.activatedAt+latencyMs/1000 local alpha:number=math.clamp((scoringAt-beforeAt
)/(afterAt-beforeAt),0,1)local p0:{number}?=newest.position local p1:{number}?=confirmation.position if type(p0)~='table'or type(p1)~='table'or#p0<3 or#p1<3 then return nil end local rootX:number=p0[1
]+(p1[1]-p0[1])*alpha local rootY:number=p0[2]+(p1[2]-p0[2])*alpha local rootZ:number=p0[3]+(p1[3]-p0[3])*alpha local offset:{number}?=newest.aimOffset if type(offset)=='table'and#offset>=3 then rootX
+=offset[1]rootY+=offset[2]rootZ+=offset[3]end local dx:number=rootX-aim.X local dy:number=rootY-aim.Y local dz:number=rootZ-aim.Z local distance:number=math.sqrt(dx*dx+dy*dy+dz*dz)if not
predictionMath.finite(distance)or distance>30 then return nil end return distance end finalizeGunCapture=function(pending:PendingGun):()if pending.targetDiedAt then runtime.sessionGunKills+=1 end if
pending.healthConnection then pcall(function():()(pending.healthConnection::RBXScriptConnection):Disconnect()end)pending.healthConnection=nil end if state.mm2Prediction and type(state.mm2Prediction.
noteEcho)=='function'and predictionMath.finite(pending.latencyMs)and predictionMath.finite(pending.activatedAt)then state.mm2Prediction.noteEcho(pending.activatedAt,pending.activatedAt+pending.
latencyMs/1000)end local leadError:number?=measureLeadErrorSeconds(pending)if leadError and state.mm2Prediction and type(state.mm2Prediction.noteLeadError)=='function'then pcall(state.mm2Prediction.
noteLeadError,leadError,pending.shotMode)end local leadCorrection:number?=nil local leadCorrectionTarget:number?=nil local leadFeedbackSamples:number?=nil if state.mm2Prediction then if type(state.
mm2Prediction.getLeadCorrection)=='function'then local ok:boolean,value:any=pcall(state.mm2Prediction.getLeadCorrection,pending.shotMode)if ok and predictionMath.finite(value)then leadCorrection=value
end end if type(state.mm2Prediction.getLeadCorrectionTarget)=='function'then local ok:boolean,value:any=pcall(state.mm2Prediction.getLeadCorrectionTarget,pending.shotMode)if ok and predictionMath.
finite(value)then leadCorrectionTarget=value end end if type(state.mm2Prediction.getLeadSamples)=='function'then local ok:boolean,value:any=pcall(state.mm2Prediction.getLeadSamples,pending.shotMode)if
ok and type(value)=='number'then leadFeedbackSamples=value end end end local endpointError:number?=measureEndpointErrorStuds(pending)local analysis:{[string]:any}=analyzeTargetTrajectory(pending.
targetTrajectory,pending.pingMs,'gun',(pending.latencyMs or 0)/1000)appendEvent({shotId=pending.shotId,phase='confirmation',kind='gun',serverTime=workspace:GetServerTimeNow(),clock=os.clock(),pingMs=
pending.pingMs,latencyMs=pending.latencyMs,targetUserId=pending.targetUserId,targetName=pending.targetName,targetState=analysis.initialState,targetJumped=analysis.jumped,targetTrajectory=pending.
targetTrajectory,targetHistory=#(pending.targetHistory or{})>0 and pending.targetHistory or nil,observedDisplacement=analysis.displacement,constantVelocityResidual=analysis.constantVelocityResidual,
horizontalLeadSeconds=analysis.horizontalLeadSeconds,verticalLeadSeconds=analysis.verticalLeadSeconds,endpoint=pending.confirmedEndpoint and vectorArray(pending.confirmedEndpoint)or nil,hitTarget=
pending.confirmedHitTarget,shotMode=pending.shotMode,silentSpawn=pending.silentSpawn,serverEchoMs=(pending.serverConfirmedAt and pending.serverActivatedAt)and(pending.serverConfirmedAt-pending.
serverActivatedAt)*1000 or nil,serverActivatedAt=pending.serverActivatedAt,serverConfirmedAt=pending.serverConfirmedAt,serverOrigin=pending.confirmedServerOrigin and vectorArray(pending.
```

```
confirmedServerOrigin)or nil,aimOffsetAtFire=(function():{number}?local history:{HistoryPoint}=pending.targetHistory or{}local newest:HistoryPoint?=history[#history]return newest and newest.aimOffset
or nil end)(),leadErrorSeconds=leadError,leadCorrectionSeconds=leadCorrection,leadCorrectionTargetSeconds=leadCorrectionTarget,leadFeedbackSamples=leadFeedbackSamples,aimMode=pending.shotMode,
endpointErrorStuds=endpointError,trajectoryPoints=#pending.targetTrajectory,trajectorySpanSeconds=(#pending.targetTrajectory>1 and(pending.targetTrajectory[#pending.targetTrajectory].dt-pending.
targetTrajectory[1].dt))or 0,trajectoryCapped=pending.trajectoryCapped==true,samplingStoppedReason=pending.samplingStoppedReason,samplingStoppedAt=pending.samplingStoppedAt,rttEngineAtConfirmation=
pending.rttEngineAtConfirmation,rttDataAtConfirmation=pending.rttDataAtConfirmation,aim=pending.requestedAim and vectorArray(pending.requestedAim)or nil,origin=pending.origin and vectorArray(pending.
origin)or nil,hitPart=pending.confirmedHitPart,hitPartPosition=pending.confirmedHitPartPosition and vectorArray(pending.confirmedHitPartPosition)or nil,hitPartSize=pending.confirmedHitPartSize and
vectorArray(pending.confirmedHitPartSize)or nil,rootAtConfirmation=pending.rootAtConfirmation,lastSeenPosition=pending.lastSeenPosition and vectorArray(pending.lastSeenPosition)or nil,healthAtAim=
pending.targetHealthAtAim,healthAtConfirmation=pending.targetHealthAtConfirmation,healthAfter=pending.targetHealthAfter,healthLatest=pending.targetHealthLatest,healthEvidenceScope=
'client-observed health; not proof this shot caused a change',healthDropAt=pending.targetHealthDropAt,healthDropServerTime=pending.targetHealthDropServerTime,diedAt=pending.targetDiedAt,diedServerTime
=pending.targetDiedServerTime,died=pending.targetDiedAt~=nil,pingBucket=analysis.pingBucket,bestMotionModel=analysis.bestMotionModel,motionModelErrors=analysis.motionModelErrors,evaluationSeconds=
analysis.evaluationSeconds,missReason=missReason})end local function handleGunFired(eventGun:any,eventOriginValue:any,endpointValue:any,hitPartValue:any):()if not runtime.active then return end local
now:number=os.clock()compactPending(runtime.pendingGuns,now)local eventTool:Tool?=normalizeGun(eventGun)local eventOrigin:Vector3?=worldPosition(eventOriginValue)local selectedIndex:number?=nil local
fallbackIndex:number?=nil for index:number=1,#runtime.pendingGuns do local pending:PendingGun=runtime.pendingGuns[index]if not pending.confirmedAt then local toolMatches:boolean=eventTool~=nil and
eventTool==pending.tool local originMatches:boolean=eventOrigin~=nil and pending.origin~=nil and(eventOrigin-pending.origin).Magnitude<=8 if toolMatches and originMatches then selectedIndex=index
break end if fallbackIndex==nil then if toolMatches and(eventOrigin==nil or pending.origin==nil)then fallbackIndex=index elseif eventTool==nil and originMatches then fallbackIndex=index end end end
end if not selectedIndex then selectedIndex=fallbackIndex end if not selectedIndex then local character:Model?=LocalPlayer.Character local root:BasePart?=character and character:FindFirstChild(
'HumanoidRootPart')::BasePart?local isLocalShooter:boolean=false if eventTool and eventTool.Parent==character then isLocalShooter=true elseif root and eventOrigin and(eventOrigin-root.Position).
Magnitude<=20 then isLocalShooter=true end if isLocalShooter then runtime.unmatchedGunSignals+=1 appendEvent({kind='gun',phase='unmatched-signal',serverTime=workspace:GetServerTimeNow(),pingMs=
getPingMilliseconds()})end return end local pending:PendingGun=runtime.pendingGuns[selectedIndex]local latencyMs:number=math.max(0,(now-pending.activatedAt)*1000)local target:Player?=pending.
targetUserId and Players:GetPlayerByUserId(pending.targetUserId)or nil if target then local finalPoint:TrajectoryPoint?=captureTrajectoryPoint(target,pending.activatedAt,pending.pingMs)if finalPoint
then table.insert(pending.targetTrajectory,finalPoint)end end local endpoint:Vector3?=worldPosition(endpointValue)local targetCharacter:Model?=target and target.Character local hitTarget:boolean=
typeof(hitPartValue)=='Instance'and targetCharacter~=nil and hitPartValue:IsDescendantOf(targetCharacter)pending.serverConfirmedAt=workspace:GetServerTimeNow()pending.confirmedServerOrigin=eventOrigin
local engineNow:number?,dataNow:number?=readLatencyReadings()pending.rttEngineAtConfirmation=engineNow pending.rttDataAtConfirmation=dataNow if typeof(hitPartValue)=='Instance'then pcall(function():()
pending.confirmedHitPart=string.sub(hitPartValue.Name,1,80)if hitPartValue:IsA('BasePart')then pending.confirmedHitPartPosition=hitPartValue.Position pending.confirmedHitPartSize=hitPartValue.Size end
end)end if target then local root:BasePart?=getTargetRoot(target)local humanoid:Humanoid?=getTargetHumanoid(target)if root then local stateName:string?=nil if humanoid then pcall(function():()
stateName=humanoid:GetState().Name end)end pending.rootAtConfirmation={position=vectorArray(root.Position),velocity=vectorArray(root.AssemblyLinearVelocity),state=stateName}end if humanoid then
pending.targetHealthAtConfirmation=humanoid.Health end end if latencyMs>=30 and latencyMs<=450 then table.insert(runtime.freshGunLatency,{at=now,ms=latencyMs})if#runtime.freshGunLatency>24 then table.
remove(runtime.freshGunLatency,1)end estimatesAt=-math.huge end addSample(runtime.gunAcceptance,latencyMs)runtime.sessionGunConfirmed+=1 if hitTarget then runtime.sessionGunHitTarget+=1 end local
missReason:string?=nil if hitTarget then missReason='hit'elseif typeof(hitPartValue)=='Instance'then local otherOwner:any=nil pcall(function():()otherOwner=Players:GetPlayerFromCharacter(hitPartValue:
FindFirstAncestorOfClass('Model'))end)if otherOwner and otherOwner~=target then missReason='other-player'else missReason='geometry'end else missReason='no-hit-part'end pending.missReason=missReason if
missReason~='hit'then runtime.sessionGunMisses=(runtime.sessionGunMisses or 0)+1 runtime.missReasons=runtime.missReasons or{}runtime.missReasons[missReason]=(runtime.missReasons[missReason]or 0)+1 end
pending.confirmedAt=now pending.latencyMs=latencyMs pending.confirmedEndpoint=endpoint pending.confirmedHitTarget=hitTarget emitShot('shotConfirmation',{shotId=pending.shotId,kind='gun',hitTarget=
hitTarget,targetName=pending.targetName,latencyMs=latencyMs,endpoint=endpoint,evidence='GunFired.hitPart'})ensureTrajectorySampler()end local KNIFE_FEEDBACK_MIN_SPEED:number=2 local
KNIFE_FEEDBACK_MAX_MISS_STUDS:number=6 local KNIFE_FEEDBACK_SAMPLE_LIMIT:number=0.15 local function knifeLeadErrorSeconds(track:KnifeTrack):number?local aimRoot:Vector3?=track.requestedRoot local
approach:Vector3?=track.approachRoot if not aimRoot or not approach then return nil end local reference:Vector3?=track.aimVelocity if not reference then reference=track.approachVelocity end if not
reference then return nil end local horizontal:Vector3=Vector3.new(reference.X,0,reference.Z)local speed:number=horizontal.Magnitude if speed<KNIFE_FEEDBACK_MIN_SPEED then return nil end local offset:
Vector3=approach-aimRoot local flat:Vector3=Vector3.new(offset.X,0,offset.Z)if flat.Magnitude>KNIFE_FEEDBACK_MAX_MISS_STUDS then return nil end local seconds:number=flat:Dot(horizontal)/(speed*speed)
if not predictionMath.finite(seconds)or math.abs(seconds)>KNIFE_FEEDBACK_SAMPLE_LIMIT then return nil end return seconds end local finishKnifeTrack:(projectile:Instance,reason:string)->()
local function stopKnifeSamplerIfIdle():()if next(runtime.tracks)==nil then disconnectFeatureConnection('MM2KnifeSampler')end end finishKnifeTrack=function(projectile:Instance,reason:string):()local
track:any=runtime.tracks[projectile]if not track then return end runtime.tracks[projectile]=nil if track.destroyConnection then track.destroyConnection:Disconnect()end local pending:any=track.pending
if pending and pending.healthConnection then pending.healthConnection:Disconnect()pending.healthConnection=nil end local measuredSpeed:number?=median(track.speeds)local resolvedSpeed:number?=track.
rawSpeed if not resolvedSpeed or resolvedSpeed<24 or resolvedSpeed>300 then resolvedSpeed=measuredSpeed end local validSpeed:boolean=resolvedSpeed~=nil and resolvedSpeed>=24 and resolvedSpeed<=300
local leadErrorSeconds:number?=track.localOwner and knifeLeadErrorSeconds(track)or nil if leadErrorSeconds and type(state.mm2Prediction)=='table'and type(state.mm2Prediction.noteLeadError)=='function'
then pcall(state.mm2Prediction.noteLeadError,leadErrorSeconds,'knife')end if track.localOwner and validSpeed then addSample(runtime.knifeSpeed,resolvedSpeed::number)runtime.sessionKnifeConfirmed+=1
end local analysis:{[string]:any}=analyzeTargetTrajectory(track.targetTrajectory,track.pingMs,'knife',nil)appendEvent({shotId=track.shotId,projectileId=track.projectileId,phase='projectile-observed',
kind='knife',serverTime=workspace:GetServerTimeNow(),pingMs=track.pingMs,latencyMs=track.spawnDelayMs,speed=validSpeed and resolvedSpeed or nil,rawSpeed=track.rawSpeed,samples=#track.speeds,reason=
reason,ownerName=track.ownerName,ownerUserId=track.ownerUserId,localOwner=track.localOwner,source=track.source,direction=track.direction,projectileTrajectory=track.projectileTrajectory,
projectileSamplingSeconds=os.clock()-track.startedAt,targetUserId=track.targetUserId,targetName=track.targetName,targetState=analysis.initialState,targetJumped=analysis.jumped,targetTrajectory=track.
targetTrajectory,targetHistory=#track.targetHistory>0 and track.targetHistory or nil,observedDisplacement=analysis.displacement,constantVelocityResidual=analysis.constantVelocityResidual,
horizontalLeadSeconds=analysis.horizontalLeadSeconds,verticalLeadSeconds=analysis.verticalLeadSeconds,pingBucket=analysis.pingBucket,bestMotionModel=analysis.bestMotionModel,motionModelErrors=analysis
.motionModelErrors,evaluationSeconds=analysis.evaluationSeconds,closestApproachStuds=track.closestApproach,closestApproachAtMs=track.closestApproachAt,bodyRadiusStuds=track.bodyRadius,leadErrorSeconds
=leadErrorSeconds,predictedRoot=track.requestedRoot and vectorArray(track.requestedRoot)or nil,approachRoot=track.approachRoot and vectorArray(track.approachRoot)or nil,approachAtMs=track.approachAt,
aimLeadSeconds=track.aimLeadSeconds,targetHealthAtAim=pending and pending.targetHealthAtAim or nil,targetHealthAfter=pending and pending.targetHealthAfter or nil,targetHealthLatest=pending and pending
.targetHealthLatest or nil,healthEvidenceScope=[[client-observed health; not proof this projectile caused a change]],passedThroughBody=track.closestApproach~=nil and track.bodyRadius~=nil and track.
closestApproach<=track.bodyRadius or nil,geometryScope=[[client-estimated closest approach, not server-confirmed impact]]})stopKnifeSamplerIfIdle()end local function ensureKnifeSampler():()if
featureConnections.MM2KnifeSampler then return end featureConnections.MM2KnifeSampler=TaskManager:Connect(function(_deltaTime:number):()local now:number=os.clock()local completed:{Instance}={}for
projectile:Instance,track:any in pairs(runtime.tracks)do if not projectile.Parent or now-track.startedAt>=5 then table.insert(completed,projectile)continue end local position:Vector3?=
projectilePosition(projectile)if position and track.previousPosition and track.previousAt then local sampleDelta:number=now-track.previousAt local displacement:number=(position-track.previousPosition)
.Magnitude if sampleDelta>=1/240 and displacement>=0.01 then local speed:number=displacement/sampleDelta if speed>=24 and speed<=300 then table.insert(track.speeds,speed)end end end if position then
track.previousPosition=position track.previousAt=now if now-track.lastProjectileSampleAt>=1/30 and#track.projectileTrajectory<160 then table.insert(track.projectileTrajectory,{dt=now-track.startedAt,
position=vectorArray(position)})track.lastProjectileSampleAt=now end end if track.targetUserId then local target:Player?=Players:GetPlayerByUserId(track.targetUserId)local body:BasePart?=target and
getTargetRoot(target)or nil if body then local distance:number=(position-body.Position).Magnitude if not track.closestApproach or distance<track.closestApproach then track.closestApproach=distance
track.closestApproachAt=(now-track.startedAt)*1000 track.approachRoot=body.Position track.approachVelocity=body.AssemblyLinearVelocity track.approachAt=(now-track.startedAt)*1000 local size:Vector3=
body.Size track.bodyRadius=0.5*math.max(size.X,math.max(size.Y,size.Z))end end end end if track.targetUserId and now-track.lastTrajectorySampleAt>=1/30 then local target:Player?=Players:
GetPlayerByUserId(track.targetUserId)local point:TrajectoryPoint?=target and captureTrajectoryPoint(target,track.startedAt-track.spawnDelayMs/1000,track.pingMs)or nil if point and#track.
targetTrajectory<120 then table.insert(track.targetTrajectory,point)end track.lastTrajectorySampleAt=now end end for _,projectile:Instance in ipairs(completed)do finishKnifeTrack(projectile,
'sample-window-complete')end if next(runtime.tracks)==nil then disconnectFeatureConnection('MM2KnifeSampler')end end,nil,'MM2 knife sampler')end local function resolveKnifeProjectile(projectile:
Instance):()if not runtime.active or runtime.tracks[projectile]then return end local sessionId:string=runtime.sessionId local now:number=os.clock()local projectileId:string=nextShotId()local position:
Vector3?=projectilePosition(projectile)local rawSpeed:number?=nil local direction:any=nil pcall(function():()rawSpeed=tonumber(projectile:GetAttribute('ThrowSpeed'))direction=projectile:GetAttribute(
'Direction')end)if rawSpeed and not finite(rawSpeed)then rawSpeed=nil end local directionArray:any=nil if typeof(direction)=='Vector3'and finite(direction.X)and finite(direction.Y)and finite(direction
.Z)then directionArray=vectorArray(direction)end local track:any={shotId=projectileId,projectileId=projectileId,startedAt=now,previousAt=now,previousPosition=position,lastProjectileSampleAt=now,
projectileTrajectory=position and{{dt=0,position=vectorArray(position)}}or{},speeds={},rawSpeed=rawSpeed,direction=directionArray,spawnDelayMs=0,pingMs=getPingMilliseconds(),targetTrajectory={},
targetHistory={},lastTrajectorySampleAt=now}runtime.sessionKnifeObserved+=1 runtime.tracks[projectile]=track appendEvent({kind='knife',phase='projectile-spawn',shotId=projectileId,projectileId=
projectileId,serverTime=workspace:GetServerTimeNow(),pingMs=track.pingMs,origin=position and vectorArray(position)or nil,direction=directionArray,rawSpeed=rawSpeed,evidence=
'client-observed ThrowingKnife tag'})track.destroyConnection=projectile.Destroying:Connect(function():()finishKnifeTrack(projectile,'destroyed')end)ensureKnifeSampler()task.spawn(function():()local
linkedHandle:Instance?=nil for _attempt:number=1,20 do if not runtime.active or runtime.sessionId~=sessionId or runtime.tracks[projectile]~=track or not projectile.Parent then return end local link:
Instance?=projectile:FindFirstChild('HandleLink',true)if link and link:IsA('ObjectValue')and link.Value then linkedHandle=link.Value break end task.wait(0.05)end if not linkedHandle or runtime.tracks[
projectile]~=track then return end local owner:Player?=nil local linkedTool:Tool?=nil pcall(function():()linkedTool=linkedHandle:FindFirstAncestorOfClass('Tool')::Tool?local character:Model?=
linkedHandle:FindFirstAncestorOfClass('Model')::Model?if character then owner=Players:GetPlayerFromCharacter(character)end end)track.ownerName=owner and owner.Name or nil track.ownerUserId=owner and
owner.UserId or nil track.localOwner=owner==LocalPlayer or(linkedTool~=nil and runtime.observedTools[linkedTool]==true)compactPending(runtime.pendingKnives,os.clock())local selectedIndex:number?=nil
local fallbackIndex:number?=nil for index:number=1,#runtime.pendingKnives do local pending:PendingKnife=runtime.pendingKnives[index]local tool:Tool?=pending.tool if tool then local handle:Instance?=
tool:FindFirstChild('Handle')if linkedHandle==handle or linkedHandle:IsDescendantOf(tool)then if pending.attemptCounted then selectedIndex=index break end fallbackIndex=fallbackIndex or index end end
end selectedIndex=selectedIndex or fallbackIndex if selectedIndex then local pending:any=table.remove(runtime.pendingKnives,selectedIndex)track.pending=pending track.shotId=pending.shotId track.
localOwner=true track.ownerName=LocalPlayer.Name track.ownerUserId=LocalPlayer.UserId track.source=pending.source track.spawnDelayMs=math.max(0,(now-pending.activatedAt)*1000)track.pingMs=pending.
pingMs track.targetUserId=pending.targetUserId track.targetName=pending.targetName track.requestedAim=pending.requestedAim track.requestedRoot=pending.requestedRoot track.aimVelocity=pending.
aimVelocity track.aimLeadSeconds=pending.aimLeadSeconds track.activatedAt=pending.activatedAt track.targetTrajectory=pending.targetTrajectory track.targetHistory=pending.targetHistory or{}addSample(
runtime.knifeSpawnDelay,track.spawnDelayMs)if not pending.attemptCounted then runtime.sessionKnifeAttempts+=1 emitShot('authoringShot',{shotId=pending.shotId,kind='knife',toolName=pending.tool.Name,
originPos=pending.origin,aimPos=pending.requestedAim,targetName=pending.targetName,eventSource='observed-projectile'})end emitShot('shotConfirmation',{shotId=pending.shotId,kind='knife',evidence=
'projectile-observed',targetName=pending.targetName})end appendEvent({kind='knife',phase='projectile-linked',shotId=track.shotId,projectileId=projectileId,serverTime=workspace:GetServerTimeNow(),
pingMs=track.pingMs,ownerName=track.ownerName,ownerUserId=track.ownerUserId,localOwner=track.localOwner,targetName=track.targetName,source=track.source})end)end local function isGun(tool:Tool):boolean
return tool.Name=='Gun'or CollectionService:HasTag(tool,'Weapon_Gun')or tool:FindFirstChild('Shoot')~=nil end local function isKnife(tool:Tool):boolean return tool.Name=='Knife'or CollectionService:
HasTag(tool,'Weapon_Knife')or(tool:FindFirstChild('Events')~=nil and tool.Events:FindFirstChild('KnifeThrown')~=nil)end local function beginPendingShot(kind:string,entry:any,target:Player?):any local
now:number=entry.activatedAt or os.clock()entry.activatedAt=now entry.serverActivatedAt=workspace:GetServerTimeNow()entry.targetUserId=target and target.UserId or entry.targetUserId entry.targetName=
target and target.Name or entry.targetName if entry.targetTrajectory==nil then entry.targetTrajectory={}if target then local point:TrajectoryPoint?=captureTrajectoryPoint(target,now,entry.pingMs or 0)
if point then table.insert(entry.targetTrajectory,point)end end end if entry.targetHistory==nil then entry.targetHistory=captureTargetHistory(target,now)end entry.lastTrajectorySampleAt=entry.
lastTrajectorySampleAt or now watchPendingHealth(entry,target)local queue:{any}=kind=='gun'and runtime.pendingGuns or runtime.pendingKnives table.insert(queue,entry)compactPending(queue,now)
ensureTrajectorySampler()return entry end local function observeTool(tool:Tool):()if runtime.observedTools[tool]then return end local gun:boolean=isGun(tool)local knife:boolean=isKnife(tool)if not gun
and not knife then return end runtime.observedTools[tool]=true trackConnection(tool.Activated:Connect(function():()if not runtime.active then return end local now:number=os.clock()local requestedAim:
Vector3?=getRequestedAim()local pingMs:number=getPingMilliseconds()local target:Player?=gun and findMurderer()or findKnifeAnalysisTarget(requestedAim)local targetTrajectory:{TrajectoryPoint}={}if
target then local initialPoint:TrajectoryPoint?=captureTrajectoryPoint(target,now,pingMs)if initialPoint then table.insert(targetTrajectory,initialPoint)end end local shotId:string=nextShotId()
appendEvent({shotId=shotId,kind=gun and'gun'or'knifeActivation',phase=gun and'attempt'or'activation',serverTime=workspace:GetServerTimeNow(),pingMs=pingMs,targetName=target and target.Name or nil})
emitShot('authoringShot',{shotId=shotId,kind=gun and'gun'or'knifeActivation',toolName=tool.Name,originPos=toolOrigin(tool),aimPos=requestedAim,targetName=target and target.Name or nil,eventSource=
'tool-activation'})if gun then runtime.sessionGunAttempts+=1 beginPendingShot('gun',{shotId=shotId,source='native-activation',attemptCounted=true,tool=tool,activatedAt=now,origin=toolOrigin(tool),
requestedAim=requestedAim,pingMs=pingMs,targetTrajectory=targetTrajectory},target)elseif knife then runtime.rawKnifeActivations+=1 beginPendingShot('knife',{shotId=shotId,attemptCounted=false,tool=
tool,activatedAt=now,origin=toolOrigin(tool),requestedAim=requestedAim,pingMs=pingMs,targetTrajectory=targetTrajectory},target)end end))end local function observeContainer(container:Instance?):()if
not container then return end for _,child:Instance in ipairs(container:GetChildren())do if child:IsA('Tool')then observeTool(child)end end trackConnection(container.ChildAdded:Connect(function(child:
Instance):()if child:IsA('Tool')then observeTool(child)end end))end local function installGunConfirmation(generation:number):()local function current():boolean return runtime.active and runtime.
observationGeneration==generation end task.spawn(function():()local clientServices:Instance?=nil local weaponModule:Instance?=nil for _attempt:number=1,20 do if not current()then return end
clientServices=game:GetService('ReplicatedStorage'):FindFirstChild('ClientServices')weaponModule=clientServices and clientServices:FindFirstChild('WeaponService')if weaponModule then break end task.
wait(0.25)end if not weaponModule then warn(trajectoryLogPrefix..' WeaponService did not appear; Knife remains active.')return end local loaded:boolean,weaponService:any=pcall(state.requireModule,
weaponModule)if not current()then return end if not loaded or type(weaponService)~='table'then warn(trajectoryLogPrefix..' Could not observe WeaponService: '..tostring(weaponService))return end pcall(
function():()local signal:any=weaponService.KnifeThrown and weaponService.KnifeThrown.Event if not signal or type(signal.Connect)~='function'then return end trackConnection(signal:Connect(function():(
)if not current()then return end local knife:Tool?=getPlayerWeapon(LocalPlayer,'Knife',true)if not knife or knife.Parent~=LocalPlayer.Character then return end local aim:Vector3?=getRequestedAim()
controller:noteAuthoredShot({kind='knife',tool=knife,toolName=knife.Name,originPos=toolOrigin(knife),aimPos=aim,target=findKnifeAnalysisTarget(aim),source='native-throw-event',settings={mode='Manual',
aimScope='mouse at input; actual redirected aim may differ'}})end))end)local connected:boolean,connection:any=pcall(function():any return weaponService.GunFired.OnClientEvent:Connect(function(...:any)
:()if current()then handleGunFired(...)end end)end)if connected and connection then trackConnection(connection::RBXScriptConnection)print(trajectoryLogPrefix..' GunFired confirmation connected.')else
warn(trajectoryLogPrefix..' GunFired no es compatible: '..tostring(connection))end end)end controller={}::Controller function controller:prepare():()if not runtime.historyLoaded then runtime.
historyLoaded=true loadSavedHistory()end end function controller:start():()local states:any=type(configData)=='table'and configData.states or nil if not states or states['MM2.Calibration.Consent']~=
true then return end if runtime.active then return end if not GAME_CHECK.MM2Active then warn(trajectoryLogPrefix..' Inicio omitido fuera de MM2.')return end runtime.active=true runtime.
observationGeneration+=1 local generation:number=runtime.observationGeneration runtime.startedAt=os.clock()runtime.startedAtUnix=os.time()self:prepare()if type(task)=='table'and type(task.spawn)==
'function'then task.spawn(function():()for _attempt:number=1,20 do if not runtime.active or runtime.observationGeneration~=generation then return end if applyLearnedSeeds()then return end task.wait(
0.5)end end)end observeContainer(LocalPlayer:FindFirstChildOfClass('Backpack'))observeContainer(LocalPlayer.Character)trackConnection(LocalPlayer.CharacterAdded:Connect(function(character:Model):()
observeContainer(character)end))trackConnection(CollectionService:GetInstanceAddedSignal('ThrowingKnife'):Connect(resolveKnifeProjectile))pcall(function():()for _,projectile:Instance in ipairs(
CollectionService:GetTagged('ThrowingKnife'))do resolveKnifeProjectile(projectile)end end)installGunConfirmation(generation)print(trajectoryLogPrefix..' Passive calibration started in the background.'
)end local function clearHistory():()runtime.gunAcceptance=newStats()runtime.knifeSpeed=newStats()runtime.knifeSpawnDelay=newStats()runtime.motionBuckets={}runtime.events={}runtime.archivedEvents=0
runtime.archiveFiles={}runtime.archiveSequence=0 runtime.freshGunLatency={}estimatesCache=nil estimatesAt=-math.huge runtime.sessionGunAttempts=0 runtime.sessionGunConfirmed=0 runtime.
sessionKnifeAttempts=0 runtime.sessionKnifeConfirmed=0 runtime.sessionKnifeObserved=0 runtime.rawKnifeActivations=0 runtime.unmatchedGunSignals=0 runtime.sessionStabAttempts=0 runtime.startedAt=os.
clock()runtime.startedAtUnix=os.time()for _,track in pairs(runtime.tracks)do if track.destroyConnection then track.destroyConnection:Disconnect()end end for _,queue in ipairs({runtime.pendingGuns,
runtime.pendingKnives})do for _,pending in ipairs(queue)do if pending.healthConnection then pending.healthConnection:Disconnect()end end end table.clear(runtime.tracks)disconnectFeatureConnection(
'MM2TrajectorySampler')runtime.sessionId=HttpService:GenerateGUID(false)runtime.shotSequence=0 table.clear(runtime.pendingGuns)table.clear(runtime.pendingKnives)end local function consentValue():
boolean?local states:any=type(configData)=='table'and configData.states or nil if not states then return nil end local value:any=states['MM2.Calibration.Consent']if value==true then return true end if
value==false then return false end return nil end function controller:getConsent():boolean?return consentValue()end function controller:setConsent(value:boolean?):boolean local nextValue:boolean?=nil
if value==true then nextValue=true elseif value==false then nextValue=false end if type(configData)=='table'and configData.states then configData.states['MM2.Calibration.Consent']=nextValue end if
type(host.queueConfigSave)=='function'then pcall(host.queueConfigSave)end if nextValue==true then self:start()if not runtime.active then return false end runtime.dirty=true local generation:number=
runtime.observationGeneration local queued:boolean=runtime.snapshotInProgress local saved:boolean=saveSnapshot('consent')if runtime.observationGeneration~=generation then return saved or queued end if
not saved and not queued then runtime.dirty=false self:destroy()end return saved or queued end runtime.saveScheduled=false runtime.dirty=false if runtime.active then self:destroy()end return true end
function controller:save(reason:string?):boolean if not runtime.active then return false end local saved:boolean=saveSnapshot(reason or'manual')return saved end function controller:reset():()if
consentValue()~=true or not runtime.active or runtime.persistenceBlocked then return end clearHistory()runtime.revision+=1 runtime.dirty=true local queued:boolean=runtime.snapshotInProgress local
saved:boolean=saveSnapshot('reset')if saved or queued then print(trajectoryLogPrefix..' History cleared.')else warn(trajectoryLogPrefix..' Cleared in memory; the new snapshot could not be saved.')end
end function controller:getEstimates():Estimates return estimates()end function controller:noteTracerData(info:any):()if not runtime.active or type(info)~='table'then return end local function
validVector(value:any):boolean return typeof(value)=='Vector3'and predictionMath.finite(value.X)and predictionMath.finite(value.Y)and predictionMath.finite(value.Z)and value.Magnitude<1e7 end if not
validVector(info.origin)or not validVector(info.endpoint)then return end local function pack(value:Vector3):{number}return{value.X,value.Y,value.Z}end local shotId:string?=nil local now:number=os.
clock()for index:number=#runtime.pendingGuns,1,-1 do local pending:PendingGun=runtime.pendingGuns[index]if now-pending.activatedAt<=2.5 and pending.origin and(pending.origin-info.origin).Magnitude<=8
then shotId=pending.shotId break end end local target:any=info.target local targetPosition:Vector3?=validVector(info.targetPosition)and info.targetPosition or nil local path:string?=nil local
hitTarget:boolean?=nil if typeof(info.hitPart)=='Instance'then pcall(function():()path=string.sub(info.hitPart:GetFullName(),1,512)end)if target and target.Character then hitTarget=info.hitPart:
IsDescendantOf(target.Character)end end appendEvent({kind='gun',phase='tracer-data',shotId=shotId,serverTime=workspace:GetServerTimeNow(),pingMs=getPingMilliseconds(),origin=pack(info.origin),endpoint
=pack(info.endpoint),shotRange=(info.endpoint-info.origin).Magnitude,targetUserId=target and target.UserId or nil,targetName=target and target.Name or nil,observedTargetPosition=targetPosition and
pack(targetPosition)or nil,endpointToObservedTarget=targetPosition and(info.endpoint-targetPosition).Magnitude or nil,targetPositionScope=
[[client-observed-at-GunFired-receipt; not server impact-time position]],hitPartPath=path,hitTarget=hitTarget,evidence='GunFired endpoint/hitPart'})end function controller:noteBlockedShot(blocker:any)
:()if not runtime.active or type(blocker)~='table'then return end local now:number=os.clock()if now-runtime.lastBlockedAt<1 then return end runtime.lastBlockedAt=now appendEvent({kind='gun',phase=
'blocked',serverTime=workspace:GetServerTimeNow(),pingMs=getPingMilliseconds(),reason=type(blocker.reason)=='string'and blocker.reason or'wall',blockedName=type(blocker.name)=='string'and blocker.name
or nil,blockedTransparency=tonumber(blocker.transparency),blockedCanCollide=blocker.canCollide==true,blockedSkipped=blocker.skipped==true,blockedDistance=tonumber(blocker.distance)})end function
controller:noteAuthoredShot(info:any):()if not runtime.active or type(info)~='table'then return end local kind:string=tostring(info.kind or'gun')local shotId:string=nextShotId()local now:number=os.
clock()local pingMs:number=getPingMilliseconds()local target:Player?=info.target and info.target.Character~=nil and info.target or nil local targetHistory:{HistoryPoint}=captureTargetHistory(target,
now)if kind~='stab'then appendEvent({kind=kind,shotId=shotId,phase='authored',serverTime=workspace:GetServerTimeNow(),clock=now,pingMs=pingMs,settings=info.settings,source=info.source or'menu',origin=
typeof(info.originPos)=='Vector3'and vectorArray(info.originPos)or nil,endpoint=typeof(info.aimPos)=='Vector3'and vectorArray(info.aimPos)or nil,targetUserId=target and target.UserId or nil,targetName
=target and target.Name or nil,targetHistory=#targetHistory>0 and targetHistory or nil})end local targetTrajectory:{TrajectoryPoint}={}if target then local initialPoint:TrajectoryPoint?=
captureTrajectoryPoint(target,now,pingMs)if initialPoint then table.insert(targetTrajectory,initialPoint)end end if kind=='stab'then runtime.sessionStabAttempts+=1 local targetRoot:BasePart?=
getTargetRoot(target)local localRoot:BasePart?=LocalPlayer.Character and LocalPlayer.Character:FindFirstChild('HumanoidRootPart')::BasePart?local distance:number?=targetRoot and localRoot and(
targetRoot.Position-localRoot.Position).Magnitude appendEvent({kind='stab',shotId=shotId,phase='attempt',serverTime=workspace:GetServerTimeNow(),pingMs=pingMs,distance=distance and math.floor(distance
*10+0.5)/10 or nil,targetUserId=target and target.UserId or nil,targetName=target and target.Name or nil,targetState=targetTrajectory[1]and targetTrajectory[1].humanoidState or nil,targetTrajectory=#
targetTrajectory>0 and targetTrajectory or nil,targetHistory=#targetHistory>0 and targetHistory or nil})else local isGun:boolean=kind~='knife'local authoredTool:Tool?=nil if typeof(info.tool)==
'Instance'and info.tool:IsA('Tool')then authoredTool=info.tool::Tool end if isGun or authoredTool~=nil then if isGun then local countedActivation:boolean=false if info.source=='native-redirect'and
authoredTool then for index:number=#runtime.pendingGuns,1,-1 do local pending:any=runtime.pendingGuns[index]local age:number=now-(pending.activatedAt or now)if pending.source=='native-activation'and
pending.tool==authoredTool and not pending.confirmedAt and age>=0 and age<=0.3 then if pending.healthConnection then pending.healthConnection:Disconnect()end table.remove(runtime.pendingGuns,index)
countedActivation=true break end end end if not countedActivation then runtime.sessionGunAttempts+=1 end else runtime.sessionKnifeAttempts+=1 if info.source=='native-throw-event'then for index:number=
#runtime.pendingKnives,1,-1 do local pending:any=runtime.pendingKnives[index]local age:number=now-(pending.activatedAt or now)if not pending.attemptCounted and pending.tool==authoredTool and age>=0
and age<=0.3 then if pending.healthConnection then pending.healthConnection:Disconnect()end table.remove(runtime.pendingKnives,index)break end end end end beginPendingShot(isGun and'gun'or'knife',{
shotId=shotId,attemptCounted=true,source=info.source or'menu',tool=authoredTool,shotMode=type(info.settings)=='table'and info.settings.mode or(info.silent and'Silent'or nil),silentSpawn=info.variant,
activatedAt=now,origin=info.originPos,requestedAim=info.aimPos,requestedRoot=typeof(info.aimRoot)=='Vector3'and info.aimRoot or nil,aimVelocity=typeof(info.aimVelocity)=='Vector3'and info.aimVelocity
or nil,aimLeadSeconds=predictionMath.finite(info.leadSeconds)and info.leadSeconds or nil,pingMs=pingMs,targetTrajectory=targetTrajectory,targetHistory=targetHistory},target)end end if type(state.
emitGameBridgeEvent)=='function'then emitShot('authoringShot',{shotId=shotId,eventSource='authored',settings=info.settings,kind=kind,toolName=tostring(info.toolName or(kind=='knife'and'Knife'or'Gun'))
,originPos=info.originPos,aimPos=info.aimPos,targetName=target and target.Name or nil,silent=info.silent==true,variant=info.variant,offset=info.offset})end end function controller:status():string
local current:Estimates=estimates()local knifeText:string=current.knifeSpeed and string.format('%.2f studs/s',current.knifeSpeed)or'learning'local bucketCount:number=0 for _key:string in pairs(runtime
.motionBuckets)do bucketCount+=1 end local consent:boolean?=consentValue()local consentText:string=consent==true and((runtime.persistenceBlocked or type(environment.writefile)~='function')and
'saving unavailable; see console'or'giving calibration')or(consent==false and'not giving calibration')or'calibration not answered'return string.format(
'MM2 session observations \183 guns %d \183 knives %d \183 motion buckets %d \183 knife %s \183 %s',runtime.sessionGunConfirmed,runtime.sessionKnifeConfirmed,bucketCount,knifeText,consentText)end
function controller:destroy():()if not runtime.active then return end runtime.observationGeneration+=1 if consentValue()==true then local projectiles:{Instance}={}for projectile:Instance in pairs(
runtime.tracks)do table.insert(projectiles,projectile)end for _,projectile:Instance in ipairs(projectiles)do finishKnifeTrack(projectile,'menu-unload')end end if runtime.dirty then saveSnapshot(
'module-destroy')end runtime.active=false disconnectFeatureConnection('MM2TrajectorySampler')disconnectFeatureConnection('MM2KnifeSampler')if runtime.trajectorySampler then runtime.trajectorySampler:
Disconnect()runtime.trajectorySampler=nil end for projectile:Instance,track:KnifeTrack in pairs(runtime.tracks)do if track.destroyConnection then track.destroyConnection:Disconnect()end if track.
pending and track.pending.healthConnection then track.pending.healthConnection:Disconnect()end runtime.tracks[projectile]=nil end for _,connection:RBXScriptConnection in ipairs(runtime.connections)do
connection:Disconnect()end table.clear(runtime.connections)for _,queue in ipairs({runtime.pendingGuns,runtime.pendingKnives})do for _,pending in ipairs(queue)do if pending.healthConnection then
pending.healthConnection:Disconnect()end end table.clear(queue)end table.clear(runtime.observedTools)end return controller end trajectoryCalibration=createTrajectoryCalibration()state.
mm2TrajectoryCalibration=trajectoryCalibration trajectoryCalibration:prepare()local calibrationPromptLive:boolean=true local calibrationStates:any=type(configData)=='table'and configData.states or{}
local rememberedCalibration:boolean=calibrationStates['MM2.Calibration.DontShowAgain']==true and type(calibrationStates['MM2.Calibration.Consent'])=='boolean'if rememberedCalibration then if
calibrationStates['MM2.Calibration.Consent']==true then if not trajectoryCalibration:setConsent(true)then notify([[Calibration could not save; check the console and executor file support.]])end end
else calibrationStates['MM2.Calibration.Consent']=nil task.delay(3,function():()if not calibrationPromptLive or(type(state.ownsRuntimeData)=='function'and not state.ownsRuntimeData())then return end
if type(state.confirmDialog)~='function'then return end local function answer(value:boolean,remember:boolean):()if not calibrationPromptLive or(type(state.ownsRuntimeData)=='function'and not state.
ownsRuntimeData())then return end calibrationStates['MM2.Calibration.DontShowAgain']=remember==true local saved:boolean=trajectoryCalibration:setConsent(value)if value then if saved then notify(
[[Calibration ON: saving observed MM2 knife and shot data to Wurst/Telemetry.]])else notify([[Calibration could not save; check the console and executor file support.]])end else notify(
[[Calibration OFF: no new data will be recorded; previous files are kept.]])end end state.confirmDialog({title='Give your calibration?',message=
[[Would you like to save the MM2 knives and shots your client observes to your executor workspace for analysis? Previous files are kept if you choose No.]],yes='Yes',no='No',rememberLabel=
"Don't Show Again",onYes=function(remember:boolean):()answer(true,remember)end,onNo=function(remember:boolean):()answer(false,remember)end})end)end pcall(function():()if type((game::any).BindToClose)
=='function'then game:BindToClose(function():()if type(trajectoryCalibration.save)=='function'then pcall(trajectoryCalibration.save,trajectoryCalibration,'close')end end)end end)local
weaponServiceModule:any=nil type PendingShot={target:Player,targetCharacter:Model?,gun:Tool,origin:Vector3,aimPosition:Vector3?,healthBefore:number,targetedMurderer:boolean,queuedAt:number,ambiguous:
boolean?}state.mm2ShotFeedback={shootActive=false,pending=nil::PendingShot?,lastAccepted=nil::PendingShot?,token=0,defaultDuration=5,gunFiredConnected=false,theme={Background=Color3.fromRGB(14,14,18),
Accent=Color3.fromRGB(205,82,42),Outline=Color3.fromRGB(220,115,58),Text=Color3.fromRGB(255,232,208)},resolve=function(_candidate:PendingShot,_directHit:boolean):()end,showMiss=function(_gun:Tool,
_candidate:PendingShot):()end}state.mm2ShotFeedback.root=create('Frame',{Parent=ScreenGui,Name='MM2ShotCooldown',AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=state.mm2ShotFeedback.theme.
Background,BackgroundTransparency=0.38,BorderSizePixel=0,ClipsDescendants=true,Position=UDim2.fromScale(0.5,0.62),Size=UDim2.fromOffset(320,36),Visible=false,ZIndex=240})::Frame create('UICorner',{
Parent=state.mm2ShotFeedback.root,CornerRadius=UDim.new(0,7)})create('UIStroke',{Parent=state.mm2ShotFeedback.root,Color=state.mm2ShotFeedback.theme.Outline,Transparency=0.35,Thickness=1})state.
mm2ShotFeedback.fill=create('Frame',{Parent=state.mm2ShotFeedback.root,Name='Fill',BackgroundColor3=state.mm2ShotFeedback.theme.Accent,BackgroundTransparency=0.46,BorderSizePixel=0,Size=UDim2.
fromScale(0,1),ZIndex=241})::Frame create('UICorner',{Parent=state.mm2ShotFeedback.fill,CornerRadius=UDim.new(0,7)})state.mm2ShotFeedback.text=create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,
ClipsDescendants=true,Parent=state.mm2ShotFeedback.root,Name='Countdown',BackgroundTransparency=1,Font=Enum.Font.GothamBold,Size=UDim2.fromScale(1,1),Text='',TextColor3=state.mm2ShotFeedback.theme.
Text,TextSize=13,ZIndex=242})::TextLabel state.mm2ShotFeedback.hide=function():()disconnectFeatureConnection('MM2ShootCooldown')state.mm2ShotFeedback.token+=1 state.mm2ShotFeedback.root.Visible=false
state.mm2ShotFeedback.fill.Size=UDim2.fromScale(0,1)state.mm2ShotFeedback.text.Text=''end state.mm2ShotFeedback.getDuration=function(gun:Tool?):number if gun then for _,attributeName:string in ipairs(
{'Cooldown','ReloadTime','ShotCooldown'})do local value:number?=tonumber(gun:GetAttribute(attributeName))if value and value>=0.5 and value<=15 then return value end end end return state.
mm2ShotFeedback.defaultDuration end state.mm2ShotFeedback.queue=function(target:Player?,gun:Tool?,origin:CFrame?,aim:CFrame?):()if not target or not gun or not origin then return end local
targetCharacter:Model?=target.Character local targetHumanoid:Humanoid?=targetCharacter and targetCharacter:FindFirstChildOfClass('Humanoid')::Humanoid?local candidate:PendingShot={target=target,
targetCharacter=targetCharacter,gun=gun,ambiguous=state.mm2ShotFeedback.pending~=nil,origin=origin.Position,aimPosition=aim and aim.Position,healthBefore=targetHumanoid and targetHumanoid.Health or 0,
targetedMurderer=getPlayerRole(target)=='Murderer',queuedAt=os.clock()}state.mm2ShotFeedback.pending=candidate task.delay(1.8,function():()if state.mm2ShotFeedback.pending==candidate then state.
mm2ShotFeedback.pending=nil end end)if not state.mm2ShotFeedback.gunFiredConnected then task.delay(math.max(0.35,mm2Settings.predictionRtt+0.18),function():()if state.mm2ShotFeedback.pending~=
candidate or state.mm2ShotFeedback.gunFiredConnected then return end state.mm2ShotFeedback.pending=nil state.mm2ShotFeedback.resolve(candidate,false)end)end end state.mm2ShotFeedback.showMiss=function
(gun:Tool,candidate:PendingShot):()state.mm2ShotFeedback.hide()local duration:number=state.mm2ShotFeedback.getDuration(gun)local cooldownEndsAt:number=candidate.queuedAt+duration local
initialRemaining:number=cooldownEndsAt-os.clock()if initialRemaining<=0 then return end disconnectFeatureConnection('MM2ShootCooldown')state.mm2ShotFeedback.token+=1 local token:number=state.
mm2ShotFeedback.token state.mm2ShotFeedback.root.Visible=true local lastText:string=''local lastScale:number=-1 local lastRefresh:number=-math.huge local function updateCountdown():()if token~=state.
mm2ShotFeedback.token then disconnectFeatureConnection('MM2ShootCooldown')state.mm2ShotFeedback.root.Visible=false return end local now:number=os.clock()local remaining:number=math.max(0,
cooldownEndsAt-now)if remaining<=0 then state.mm2ShotFeedback.hide()return end if now-lastRefresh<0.08 then return end lastRefresh=now local progress:number=math.clamp(remaining/duration,0,1)local
scale:number=math.floor(progress*200+0.5)/200 if scale~=lastScale then lastScale=scale state.mm2ShotFeedback.fill.Size=UDim2.fromScale(scale,1)end local nextText:string=string.format(
'Miss cooldown - %.1fs',remaining)if nextText~=lastText then lastText=nextText state.mm2ShotFeedback.text.Text=nextText end end updateCountdown()featureConnections.MM2ShootCooldown=TaskManager:
Connect(updateCountdown,nil,'Miss cooldown')end state.mm2ShotFeedback.toPosition=function(value:any):Vector3?if typeof(value)=='Vector3'then return value end if typeof(value)=='CFrame'then return
value.Position end if typeof(value)=='Instance'then if value:IsA('Attachment')then return value.WorldPosition end if value:IsA('BasePart')then return value.Position end end return nil end state.
mm2ShotFeedback.resolve=function(candidate:PendingShot,directHit:boolean):()state.mm2ShotFeedback.hide()state.mm2ShotFeedback.lastAccepted=candidate local confirmationDelay:number=math.clamp(math.max(
0.18,mm2Settings.predictionRtt+0.12),0.18,0.55)task.delay(confirmationDelay,function():()if state.mm2ShotFeedback.lastAccepted~=candidate then return end state.mm2ShotFeedback.lastAccepted=nil local
targetCharacter:Model?=candidate.targetCharacter local currentHumanoid:Humanoid?=targetCharacter and targetCharacter:FindFirstChildOfClass('Humanoid')::Humanoid?local hitConfirmed:boolean=directHit or
candidate.target.Parent==nil or targetCharacter==nil or targetCharacter.Parent==nil or currentHumanoid==nil or currentHumanoid.Health<=0 or currentHumanoid.Health<candidate.healthBefore or(candidate.
targetedMurderer and getPlayerRole(candidate.target)~='Murderer')if hitConfirmed then state.mm2ShotFeedback.hide()end if not hitConfirmed and mm2Settings.showMissCooldown then state.mm2ShotFeedback.
showMiss(candidate.gun,candidate)end end)end state.mm2ShotFeedback.handleGunFired=function(firedGun:any,eventOrigin:any,endpoint:any,hitPart:any):()if not state.mm2ShotFeedback.shootActive then return
end local candidate:PendingShot?=state.mm2ShotFeedback.pending if not candidate then return end if os.clock()-candidate.queuedAt>1.6 then state.mm2ShotFeedback.pending=nil return end local eventObject
:Instance?=typeof(firedGun)=='Instance'and firedGun or nil local eventTool:Tool?=nil if eventObject then if eventObject:IsA('Tool')then eventTool=eventObject else eventTool=eventObject:
FindFirstAncestorOfClass('Tool')end end local weaponMatches:boolean=eventTool==candidate.gun or eventObject==candidate.gun or(eventObject~=nil and eventObject:IsDescendantOf(candidate.gun))local
eventOriginPosition:Vector3?=state.mm2ShotFeedback.toPosition(eventOrigin)local originMatches:boolean=eventOriginPosition~=nil and(eventOriginPosition-candidate.origin).Magnitude<=3.5 local
endpointPosition:Vector3?=state.mm2ShotFeedback.toPosition(endpoint)local directionMatches:boolean=false if eventOriginPosition and endpointPosition and candidate.aimPosition then local eventDelta:
Vector3=endpointPosition-eventOriginPosition local expectedDelta:Vector3=candidate.aimPosition-candidate.origin if eventDelta.Magnitude>0.05 and expectedDelta.Magnitude>0.05 then directionMatches=
eventDelta.Unit:Dot(expectedDelta.Unit)>=0.965 end end if eventTool and not weaponMatches then return end if not originMatches or not directionMatches or endpointPosition==nil then return end state.
mm2ShotFeedback.pending=nil if candidate.ambiguous then return end local targetCharacter:Model?=candidate.targetCharacter local hitTarget:boolean=typeof(hitPart)=='Instance'and targetCharacter~=nil
and hitPart:IsDescendantOf(targetCharacter)state.mm2ShotFeedback.resolve(candidate,hitTarget)end type KnifeSettings={aura:boolean,auraRange:number}local function getWeaponServiceModule():any if type(
weaponServiceModule)=='table'then return weaponServiceModule end local loaded:boolean,result:any=pcall(function():any return state.requireModule(game:GetService('ReplicatedStorage'):WaitForChild(
'ClientServices'):WaitForChild('WeaponService'))end)if loaded and type(result)=='table'then weaponServiceModule=result return result end return nil end local function connectGunFiredSignal(
weaponService:any):boolean if featureConnections.MM2GunFired then state.mm2ShotFeedback.gunFiredConnected=true return true end local gunFired:any=weaponService.GunFired local connected:boolean,
connection:any=pcall(function():any return gunFired.OnClientEvent:Connect(state.mm2ShotFeedback.handleGunFired)end)if connected and connection then featureConnections.MM2GunFired=connection state.
mm2ShotFeedback.gunFiredConnected=true return true end state.mm2ShotFeedback.gunFiredConnected=false return false end local function mm2RoleProvider(player:Player):string?if roundLifecycleActive==
false then return nil end return getPlayerRole(player)end local function mm2TargetProvider():any if not hasActiveRoundRoles()then return nil end local localRole:string?=getPlayerRole(LocalPlayer)local
target:Player?=nil local source:string='MM2 role priority'if localRole=='Sheriff'or localRole=='Hero'then target=findMurderer()source='Sheriff priority \183 Murderer'elseif localRole=='Murderer'then
target=findSheriff()source=target and'Murderer priority \183 Sheriff/Hero'or'Murderer priority'if not target then for _,player:Player in ipairs(Players:GetPlayers())do if player~=LocalPlayer and
isPlayerAlive(player)and not isProtectedTarget(player)then target=player break end end end else target=findMurderer()source='Threat priority \183 Murderer'end if state.mm2ShotFeedback then local
pending:any=state.mm2ShotFeedback.pending or state.mm2ShotFeedback.lastAccepted local pendingRole:string?=pending and pending.target and getPlayerRole(pending.target)or nil local preservesRolePriority
:boolean=localRole~='Sheriff'and localRole~='Hero'or pendingRole=='Murderer'local selectedRole:string?=target and getPlayerRole(target)or nil if localRole=='Murderer'and(selectedRole=='Sheriff'or
selectedRole=='Hero')then preservesRolePriority=pendingRole=='Sheriff'or pendingRole=='Hero'end if pending and pending.target and isPlayerAlive(pending.target)and preservesRolePriority then target=
pending.target source='Recent shot target'end end if not target then return nil end local role:string?=getPlayerRole(target)return{player=target,role=role,source=source,accent=role=='Murderer'and
Theme.negative or((role=='Sheriff'or role=='Hero')and Theme.accent or Theme.positive)}end state.gameRoleProvider=mm2RoleProvider state.gameTargetProvider=mm2TargetProvider moduleCleanup=function():()
unregisterMM2Roles()if state.gameRoleProvider==mm2RoleProvider then state.gameRoleProvider=nil end if state.gameTargetProvider==mm2TargetProvider then state.gameTargetProvider=nil end end
local function validateGunShot(target:Player,character:Model,targetCharacter:Model,muzzle:Vector3,origin:Vector3,endpoint:Vector3,walls:boolean):(boolean,string?,any?,any?)if LocalPlayer.Character~=
character or not target or target==LocalPlayer or target.Character~=targetCharacter or not isPlayerAlive(target)or isProtectedTarget(target)or getPlayerRole(target)~='Murderer'then return false,
'Target changed or is not the murderer.'end if roundLifecycleActive==false then return false,'The round is not active.'end local roundRole=getRoundRole(target)if roundRole and roundRole~='Murderer'
then return false,'Conflicting target role.'end if not roundRole and not getPlayerWeapon(target,'Knife')and not hasMurdererCollisionGroup(targetCharacter)then return false,
'No current murderer evidence.'end local ownHumanoid=character:FindFirstChildOfClass('Humanoid')if not ownHumanoid or not predictionMath.finite(ownHumanoid.Health)or ownHumanoid.Health<=0 then return
false,'Your character is not alive.'end if not predictionMath.finite(muzzle.Magnitude)or not predictionMath.finite(origin.Magnitude)or not predictionMath.finite(endpoint.Magnitude)or(endpoint-origin).
Magnitude<0.05 or(endpoint-muzzle).Magnitude<0.05 then return false,'Invalid shot geometry.'end if walls then local options={ignoreTarget=true}local clear,blocker=isWeaponPathClear(muzzle,endpoint,
targetCharacter,nil,options)if not clear then return false,'A wall or another body blocks the shot.',blocker end local solidClear,solidBlocker=isWeaponPathClear(muzzle,endpoint,targetCharacter,true,
options)if not solidClear then return false,'An invisible wall blocks the shot.',solidBlocker end if(origin-muzzle).Magnitude>0.05 then local originClear,originBlocker=isWeaponPathClear(origin,
endpoint,targetCharacter,nil,options)if not originClear then return false,'A wall or another body blocks the shot.',originBlocker end local originSolid,originSolidBlocker=isWeaponPathClear(origin,
endpoint,targetCharacter,true,options)if not originSolid then return false,'An invisible wall blocks the shot.',originSolidBlocker end end end local occluder:any=nil if walls then pcall(function():()
local first:any=describeRayHit(muzzle,endpoint,targetCharacter)local reach:number=(endpoint-muzzle).Magnitude if first and not first.onTarget and predictionMath.finite(first.distance)and first.
distance<reach-0.5 then occluder=first end end)end return true,nil,nil,occluder end local function disconnectGunFiredObserver():()disconnectFeatureConnection('MM2GunFired')state.mm2ShotFeedback.
gunFiredConnected=false end registerMM2Roles()state.mm2Core={settings=mm2Settings,mm2Settings=mm2Settings,MM2Effects=MM2Effects,GunEffects=GunEffects,TrapEffects=TrapEffects,CoinEffects=CoinEffects,
clearEffects=clearEffects,createMM2Marker=createMM2Marker,createMM2Cham=createMM2Cham,mm2GameplayRemotes=mm2GameplayRemotes,getRoundData=function():{[any]:any}return mm2RoundData end,
isRoundLifecycleActive=function():boolean?return roundLifecycleActive end,onRoundRoles=onRoundRoles,requestRoleRefresh=requestRoleRefresh,onRoundPhase=onRoundPhase,getRoundPhase=function():string
return roundPhase end,roundTimer=roundTimer,findRoundTimerPart=findRoundTimerPart,readRoundTimerPartSeconds=readRoundTimerPartSeconds,getPlayerWeapon=getPlayerWeapon,isPlayerAlive=isPlayerAlive,
findMM2Role=findMM2Role,playerFromRoundKey=playerFromRoundKey,getRoundRole=getRoundRole,getPlayerRole=getPlayerRole,getPlayerPerk=getPlayerPerk,invalidateRoleCaches=invalidateRoleCaches,
roundRolesKnown=roundRolesKnown,hasActiveRoundRoles=hasActiveRoundRoles,findMurderer=findMurderer,findSheriff=findSheriff,isProtectedTarget=isProtectedTarget,findMM2Map=findMM2Map,findDroppedGun=
findDroppedGun,isWeaponPathClear=isWeaponPathClear,describeRayHit=describeRayHit,validateGunShot=validateGunShot,predictionMath=predictionMath,getEstimatedLatency=getEstimatedLatency,
getLatencyReadings=readLatencyReadings,getGunOriginCFrame=getGunOriginCFrame,trajectoryCalibration=trajectoryCalibration,getWeaponServiceModule=getWeaponServiceModule,connectGunFiredSignal=
connectGunFiredSignal,disconnectGunFiredObserver=disconnectGunFiredObserver}cleanupMM2Runtime=function()calibrationPromptLive=false unregisterMM2Roles()if state.gameRoleProvider==mm2RoleProvider then
state.gameRoleProvider=nil end if state.gameTargetProvider==mm2TargetProvider then state.gameTargetProvider=nil end trajectoryCalibration:destroy()disconnectGunFiredObserver()state.mm2ShotFeedback.
hide()for player:Player,_ in pairs(roleWatchers)do unwatchPlayer(player)end table.clear(roleIndex.roles)for _,connectionName:string in ipairs({'MM2PlayerDataChanged','MM2TimerTrackerStart',
'MM2TimerTrackerEnd','MM2RoundPhase','MM2RoundClock','MM2RoleWatchAdded','MM2RoleWatchRemoving','MM2RoleTagAddedWeapon_Knife','MM2RoleTagRemovedWeapon_Knife','MM2RoleTagAddedWeapon_Gun',
'MM2RoleTagRemovedWeapon_Gun'})do disconnectFeatureConnection(connectionName)end state.isProtectedTarget=function(player:Player?):boolean return player==nil or player==LocalPlayer end table.clear(
roundRolesListeners)table.clear(roundPhaseListeners)state.mm2Core=nil end end function Module.init(runtime:any):any assert(type(runtime)=='table','MM2 requires a Runtime table')assert(type(runtime.
Menu)=='table','MM2 requires Runtime.Menu')assert(runtime.TaskManager~=nil,'MM2 requires Runtime.TaskManager')if Module.Initialized then return Module end Module.Menu=runtime.Menu Module.Runtime=
runtime local success:boolean,buildError:any=pcall(buildMM2Features)if not success then moduleCleanup()error(buildError,0)end local environment:any=getfenv()if type(environment.cleanupMM2Runtime)==
'function'then moduleCleanup=environment.cleanupMM2Runtime end Module.Events=featureConnections Module.Initialized=true return Module end function Module.destroy():()if not Module.Initialized then
return end Module.Initialized=false moduleCleanup()Module.Events={}Module.Menu=nil Module.Runtime=nil end return Module
-- src/games/universal/Blatant/Fly.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='Fly',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.
init(context:Runtime):any local host:any=context.host local framework:any=context.framework local movementInput:any=context.services.movementInput local mobileActions:any=context.services.
mobileActions local getCharacterParts:any=host.getCharacterParts local UserInputService:any=host.UserInputService local currentWorkspace:Workspace=host.workspace or workspace type FlySettings={method:
string,floatMethod:string,speed:number,verticalSpeed:number,response:number,burstInterval:number,wallCheck:boolean,faceCamera:boolean,upKey:Enum.KeyCode,downKey:Enum.KeyCode}local flySettings:
FlySettings={method='Velocity',floatMethod='Velocity',speed=70,verticalSpeed=56,response=1,burstInterval=0.2,wallCheck=true,faceCamera=true,upKey=Enum.KeyCode.Space,downKey=Enum.KeyCode.LeftControl}
local flyRuntime:any={yLevel=nil::number?,burstAt=0,platform=nil::BasePart?,walkSpeed=nil::number?,walkHumanoid=nil::Humanoid?,root=nil::BasePart?,method=nil::string?,floatMethod=nil::string?}local
flySmoothedVelocity:Vector3=Vector3.zero local flyObjects:{[Instance]:boolean}=setmetatable({},{__mode='k'})::any local flyTouchVertical:number=0 local flyRaycastParams:RaycastParams?=nil local
flyRaycastCharacter:Instance?=nil local function raycastParamsFor(root:BasePart):RaycastParams if not flyRaycastParams then flyRaycastParams=RaycastParams.new()flyRaycastParams.FilterType=Enum.
RaycastFilterType.Exclude end if root.Parent~=flyRaycastCharacter then flyRaycastCharacter=root.Parent flyRaycastParams.FilterDescendantsInstances=root.Parent and{root.Parent}or{}end return
flyRaycastParams::RaycastParams end local function removeFlyObjects():()for object:Instance in pairs(flyObjects)do if object.Parent then object:Destroy()end end flyObjects=setmetatable({},{__mode='k'}
)::any end local function restoreWalkSpeed():()local humanoid:Humanoid?=flyRuntime.walkHumanoid if humanoid and humanoid.Parent and flyRuntime.walkSpeed~=nil then humanoid.WalkSpeed=flyRuntime.
walkSpeed end flyRuntime.walkSpeed=nil flyRuntime.walkHumanoid=nil end local function clearPlatform():()if flyRuntime.platform then flyRuntime.platform:Destroy()end flyRuntime.platform=nil end
local function ensureFlyConstraints(root:BasePart):(LinearVelocity,AlignOrientation)local attachment:Attachment?=root:FindFirstChild('Wurst_FlyAttachment')::Attachment?if not attachment then local
newAttachment:Attachment=Instance.new('Attachment')newAttachment.Name='Wurst_FlyAttachment'newAttachment.Parent=root attachment=newAttachment end local resolvedAttachment:Attachment=attachment::
Attachment flyObjects[resolvedAttachment]=true local velocity:LinearVelocity?=root:FindFirstChild('Wurst_FlyVelocity')::LinearVelocity?if not velocity then local newVelocity:LinearVelocity=Instance.
new('LinearVelocity')newVelocity.Name='Wurst_FlyVelocity'newVelocity.Attachment0=resolvedAttachment newVelocity.MaxForce=math.huge newVelocity.RelativeTo=Enum.ActuatorRelativeTo.World newVelocity.
VelocityConstraintMode=Enum.VelocityConstraintMode.Vector newVelocity.Parent=root velocity=newVelocity end flyObjects[velocity::LinearVelocity]=true local orientation:AlignOrientation?=root:
FindFirstChild('Wurst_FlyOrientation')::AlignOrientation?if not orientation then local newOrientation:AlignOrientation=Instance.new('AlignOrientation')newOrientation.Name='Wurst_FlyOrientation'
newOrientation.Attachment0=resolvedAttachment newOrientation.MaxTorque=math.huge newOrientation.Mode=Enum.OrientationAlignmentMode.OneAttachment newOrientation.Responsiveness=28 newOrientation.
RigidityEnabled=false newOrientation.Parent=root orientation=newOrientation end flyObjects[orientation::AlignOrientation]=true return velocity::LinearVelocity,orientation::AlignOrientation end
local function getFlyDirection(camera:Camera):(Vector3,number,Vector3)local cameraForward:Vector3=Vector3.new(camera.CFrame.LookVector.X,0,camera.CFrame.LookVector.Z)local cameraRight:Vector3=Vector3.
new(camera.CFrame.RightVector.X,0,camera.CFrame.RightVector.Z)cameraForward=cameraForward.Magnitude>0.001 and cameraForward.Unit or Vector3.new(0,0,-1)cameraRight=cameraRight.Magnitude>0.001 and
cameraRight.Unit or Vector3.new(1,0,0)local forwardScalar:number,rightScalar:number=movementInput.getVector()local horizontal:Vector3=cameraForward*forwardScalar+cameraRight*rightScalar if horizontal.
Magnitude>1 then horizontal=horizontal.Unit end local vertical:number=0 local keyboardAvailable:boolean=UserInputService.KeyboardEnabled and UserInputService:GetFocusedTextBox()==nil if
flyTouchVertical>0 or(keyboardAvailable and UserInputService:IsKeyDown(flySettings.upKey))then vertical=vertical+1 end if flyTouchVertical<0 or(keyboardAvailable and UserInputService:IsKeyDown(
flySettings.downKey))then vertical=vertical-1 end if vertical==0 and not UserInputService.KeyboardEnabled and movementInput.isJumpHeld()then vertical=1 end return horizontal,vertical,cameraForward end
local applyFlyHorizontal:(BasePart,Humanoid,Vector3,number)->()local applyFlyVertical:(BasePart,Humanoid,number,number)->()do applyFlyHorizontal=function(root:BasePart,humanoid:Humanoid,moveDirection:
Vector3,deltaTime:number):()local method:string=flySettings.method local currentVelocity:Vector3=root.AssemblyLinearVelocity local target:Vector3=moveDirection*flySettings.speed if method=='WalkSpeed'
then if not flyRuntime.walkSpeed then flyRuntime.walkSpeed=humanoid.WalkSpeed flyRuntime.walkHumanoid=humanoid end return end if humanoid.WalkSpeed~=flySettings.speed then humanoid.WalkSpeed=flySettings.speed
end if flySettings.speed<0 then root.AssemblyLinearVelocity=Vector3.new(target.X,currentVelocity.Y,target.Z)end return end if method=='Pulse'then local period:number=math.max(flySettings.burstInterval
,0.05)*2 local burst:boolean=(os.clock()%period)<(period*0.5)target=target*(burst and 1.8 or 0.2)end if method=='CFrame'or method=='Blink'then local step:Vector3=moveDirection*flySettings.speed*
deltaTime if method=='Blink'then local interval:number=math.max(flySettings.burstInterval,0.05)if os.clock()<flyRuntime.burstAt then step=Vector3.zero else flyRuntime.burstAt=os.clock()+interval step=
moveDirection*flySettings.speed*interval end end if step.Magnitude>0.001 and flySettings.wallCheck then local wallHit:RaycastResult?=currentWorkspace:Raycast(root.Position,step+step.Unit*2,
raycastParamsFor(root))if wallHit then step=Vector3.zero end end if step.Magnitude>0.001 then root.CFrame=root.CFrame+step end root.AssemblyLinearVelocity=Vector3.new(0,currentVelocity.Y,0)return end
if method=='Impulse'then local difference:Vector3=(target-currentVelocity)*Vector3.new(1,0,1)local threshold:number=moveDirection.Magnitude<0.01 and 10 or 2 if difference.Magnitude>threshold then root
:ApplyImpulse(difference*root.AssemblyMass)end return end local alpha:number=1-math.exp(-12*flySettings.response*deltaTime)flySmoothedVelocity=flySmoothedVelocity:Lerp(target,alpha)if method==
'Constraint'then return end root.AssemblyLinearVelocity=Vector3.new(flySmoothedVelocity.X,currentVelocity.Y,flySmoothedVelocity.Z)end applyFlyVertical=function(root:BasePart,humanoid:Humanoid,vertical
:number,deltaTime:number):()local method:string=flySettings.floatMethod local currentVelocity:Vector3=root.AssemblyLinearVelocity local targetY:number=vertical*flySettings.verticalSpeed if method==
'Floor'then local platform:BasePart?=flyRuntime.platform if not platform or not platform.Parent then local newPlatform:BasePart=Instance.new('Part')newPlatform.Name='Wurst_FlyPlatform'newPlatform.
Anchored=true newPlatform.CanQuery=false newPlatform.CanTouch=false newPlatform.Size=Vector3.new(8,1,8)newPlatform.Transparency=1 newPlatform.Parent=currentWorkspace.CurrentCamera flyRuntime.platform=
newPlatform platform=newPlatform end local resolved:BasePart=platform::BasePart resolved.CFrame=vertical<0 and CFrame.new(0,-4096,0)or CFrame.new(root.Position-Vector3.new(0,humanoid.HipHeight+1.5,0))
if vertical>0 then root.AssemblyLinearVelocity=Vector3.new(currentVelocity.X,targetY,currentVelocity.Z)end return end if method=='Bypass'then local period:number=math.max(flySettings.burstInterval,
0.05)*5 if(os.clock()%period)>(period*0.8)then flyRuntime.yLevel=nil return end local held:number=flyRuntime.yLevel or root.Position.Y held=held+targetY*deltaTime flyRuntime.yLevel=held root.
AssemblyLinearVelocity=Vector3.new(currentVelocity.X,0,currentVelocity.Z)root.CFrame=root.CFrame+Vector3.new(0,held-root.Position.Y,0)return end if method=='Hover'or method=='Jump'then local level:
number=flyRuntime.yLevel or root.Position.Y level=level+targetY*deltaTime if flySettings.wallCheck then local offset:number=level-root.Position.Y if math.abs(offset)>0.001 then local ceilingHit:
RaycastResult?=currentWorkspace:Raycast(root.Position,Vector3.new(0,offset,0),raycastParamsFor(root))if ceilingHit then level=ceilingHit.Position.Y-math.sign(offset)*(humanoid.HipHeight+0.5)end end
end flyRuntime.yLevel=level if method=='Hover'then root.AssemblyLinearVelocity=Vector3.new(currentVelocity.X,0,currentVelocity.Z)root.CFrame=root.CFrame+Vector3.new(0,level-root.Position.Y,0)elseif
root.Position.Y<level-0.5 then humanoid:ChangeState(Enum.HumanoidStateType.Jumping)end return end if method=='Bounce'then local period:number=math.max(flySettings.burstInterval,0.05)*4 local rising:
boolean=(os.clock()%period)<(period*0.5)targetY=targetY+(rising and 1 or-1)*flySettings.verticalSpeed*0.45 end if method=='Impulse'then local difference:number=targetY-currentVelocity.Y if math.abs(
difference)>2 then root:ApplyImpulse(Vector3.new(0,difference,0)*root.AssemblyMass)end return end root.AssemblyLinearVelocity=Vector3.new(currentVelocity.X,targetY,currentVelocity.Z)end end local fly:
any fly=framework.Categories.Exploits:CreateModule({Name='Fly',Category='Exploits',ConfigKey='Universal.Fly',Tooltip=[[Use WASD to move, Space to rise and LeftControl to descend. ]]..
'Method picks how you travel sideways and Float picks what '..'holds you up; if a game blocks one combination, another '..'usually works.',Function=function(enabled:boolean):()removeFlyObjects()
flySmoothedVelocity=Vector3.zero flyTouchVertical=0 flyRuntime.yLevel=nil flyRuntime.burstAt=0 clearPlatform()restoreWalkSpeed()flyRuntime.root=nil flyRuntime.method=nil flyRuntime.floatMethod=nil if
not enabled then flyRaycastCharacter=nil if flyRaycastParams then flyRaycastParams.FilterDescendantsInstances={}end fly:SetStatus(nil)local _,humanoid:Humanoid?,root:BasePart?=getCharacterParts()if
root then(root::BasePart).AssemblyLinearVelocity=Vector3.zero end return end fly:SetStatus(flySettings.method)fly:Loop(function(deltaTime:number):()local _,humanoid,root=getCharacterParts()local
camera:Camera?=currentWorkspace.CurrentCamera if not humanoid or not root or not camera then removeFlyObjects()clearPlatform()restoreWalkSpeed()flyRuntime.root=nil flyRuntime.yLevel=nil return end
```

```
local method:string=flySettings.method local floatMethod:string=flySettings.floatMethod if flyRuntime.root~=root or flyRuntime.method~=method or flyRuntime.floatMethod~=floatMethod then
removeFlyObjects()clearPlatform()restoreWalkSpeed()flyRuntime.yLevel=nil flyRuntime.burstAt=0 flySmoothedVelocity=Vector3.zero flyRuntime.root=root flyRuntime.method=method flyRuntime.floatMethod=
floatMethod end local moveDirection,vertical,cameraForward=getFlyDirection(camera)if flySettings.wallCheck and moveDirection.Magnitude>0.001 then local wallHit:RaycastResult?=currentWorkspace:Raycast(
root.Position,moveDirection.Unit*2.75,raycastParamsFor(root))if wallHit then moveDirection=Vector3.zero flySmoothedVelocity=Vector3.zero end end if method=='Constraint'then local velocity,orientation=
ensureFlyConstraints(root)applyFlyHorizontal(root,humanoid,moveDirection,deltaTime)velocity.Enabled=true velocity.VectorVelocity=Vector3.new(flySmoothedVelocity.X,vertical*flySettings.verticalSpeed,
flySmoothedVelocity.Z)orientation.Enabled=flySettings.faceCamera orientation.CFrame=CFrame.lookAt(Vector3.zero,cameraForward)return end removeFlyObjects()applyFlyHorizontal(root,humanoid,moveDirection
,deltaTime)applyFlyVertical(root,humanoid,vertical,deltaTime)if floatMethod~='Floor'and flyRuntime.platform then(flyRuntime.platform::BasePart):Destroy()flyRuntime.platform=nil end if floatMethod~=
'Hover'and floatMethod~='Jump'and floatMethod~='Bypass'then flyRuntime.yLevel=nil end if flySettings.faceCamera and cameraForward.Magnitude>0.001 then root.CFrame=CFrame.lookAt(root.Position,root.
Position+cameraForward)end end)fly:CreateDropdown({Name='Method',List={'Velocity','Constraint','Impulse','CFrame','Blink','Pulse','WalkSpeed'},Index=1,Function=function(value:string):()
flySettings.method=value if fly.Enabled then fly:SetStatus(value)end end,Tooltip=[[Sideways engine. Velocity is smooth, CFrame ignores physics, ]]..'WalkSpeed just runs faster.'})fly:CreateDropdown({
Name='Float',List={'Velocity','Impulse','Hover','Jump','Bounce','Floor','Bypass'},Index=1,Function=function(value:string):()flySettings.floatMethod=value end,Tooltip=
[[Applies to non-Constraint methods; Constraint holds altitude with its own velocity. Hover locks an altitude, Floor ]]..[[stands you on an invisible part, and Bypass drops you to the ]]..
'ground for an instant every few seconds so the server sees '..'you land.'})fly:CreateSlider({Name='Horizontal speed',Min=1,Max=250,Default=flySettings.speed,Function=function(value:number):()
flySettings.speed=value end,Tooltip=[[Studs per second while holding WASD. Negative values reverse the direction.]]})fly:CreateSlider({Name='Vertical speed',Min=1,Max=250,Default=flySettings.
verticalSpeed,Function=function(value:number):()flySettings.verticalSpeed=value end,Tooltip=[[Studs per second while rising or descending. Negative values reverse the direction.]]})fly:CreateSlider({
Name='Response multiplier',Show={Option='Method',Values={'Velocity','Constraint','Pulse'}},Min=0.5,Max=2,Default=flySettings.response,Function=function(value:number):()flySettings.response=value end,
Tooltip=[[How sharply the flight reacts to input. Higher is twitchier.]]})fly:CreateSlider({Name='Burst interval (s)',Min=0.05,Max=1,Default=flySettings.burstInterval,Function=function(value:number):(
)flySettings.burstInterval=value end,Tooltip='Cycle length for Blink, Pulse, Bounce and Bypass. Shorter '..'is smoother.'})fly:CreateToggle({Name='Wall check',Default=flySettings.wallCheck,Function=
function(value:boolean):()flySettings.wallCheck=value end,Tooltip='Stop against solid geometry instead of flying through it.'})fly:CreateToggle({Name='Face camera',Default=flySettings.faceCamera,
Function=function(value:boolean):()flySettings.faceCamera=value end,Tooltip='Rotate the character to follow wherever the camera looks.'})local upKey:any=fly:CreateBind({Name='Up key',Default=
flySettings.upKey,Function=function(value:Enum.KeyCode):()flySettings.upKey=value end,Tooltip='Hold to gain altitude.'})local downKey:any=fly:CreateBind({Name='Down key',Default=flySettings.downKey,
Function=function(value:Enum.KeyCode):()flySettings.downKey=value end,Tooltip='Hold to lose altitude.'})if mobileActions.bindPlacement then mobileActions.bindPlacement(upKey.Object,'UniversalFlyUp',
'UP',function():()end,{onPress=function():()flyTouchVertical=1 end,onRelease=function():()if flyTouchVertical==1 then flyTouchVertical=0 end end})mobileActions.bindPlacement(downKey.Object,
'UniversalFlyDown','DOWN',function():()end,{onPress=function():()flyTouchVertical=-1 end,onRelease=function():()if flyTouchVertical==-1 then flyTouchVertical=0 end end})end activeCleanup=function():()
if fly.Enabled then fly:Toggle(false)end removeFlyObjects()clearPlatform()restoreWalkSpeed()end Module.Initialized=true return fly end function Module.destroy():()if activeCleanup then activeCleanup()
end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Blatant/Invisible.lua
local Module={Name='Invisibility',PlaceId=0,Initialized=false}local cleanup function Module.init(context)local host,services=context.host,context.services local player,input,run=host.LocalPlayer,host.
UserInputService,host.RunService local world=host.workspace or workspace local VOID_DEPTH_MIN,VOID_DEPTH_MAX=0,350 local VOID_DEPTH_DEFAULT=3 local settings={fly=false,flySpeed=55,walkSpeed=16,depth=
VOID_DEPTH_DEFAULT,transparency=0.5}local state={connections={},tracks={},bodyParts={},savedWalkSpeed=nil,savedPlatformStand=nil,savedAnchored=nil}local card do local stored=type(configData)=='table'
and configData.values or nil if type(stored)=='table'and stored['UI.GhostDepthDefault']~=3 then local saved=tonumber(stored['Universal.Ghost.Voiddepth'])if saved==nil or saved==120 or saved==1 or
saved==5 then stored['Universal.Ghost.Voiddepth']=VOID_DEPTH_DEFAULT elseif saved~=nil and(saved<VOID_DEPTH_MIN or saved>VOID_DEPTH_MAX)then stored['Universal.Ghost.Voiddepth']=math.clamp(saved,
VOID_DEPTH_MIN,VOID_DEPTH_MAX)end stored['UI.GhostDepthDefault']=3 if type(queueConfigSave)=='function'then queueConfigSave()end end do local stored2=type(configData)=='table'and configData.values or
nil local migrated=stored2 and tonumber(stored2['Universal.Ghost.Voiddepth'])if migrated then settings.depth=math.clamp(migrated,VOID_DEPTH_MIN,VOID_DEPTH_MAX)end end end local function parts(
character)return character and character:FindFirstChildOfClass('Humanoid'),character and character:FindFirstChild('HumanoidRootPart')end local function activity(active)if services.activity then
services.activity.set('ghost',active)end end local function cameraBusy()return services.freeCam and services.freeCam.IsActive()or services.spectate and services.spectate.IsActive()end local function
destroyHeight()local height=world.FallenPartsDestroyHeight if height~=height or math.abs(height)==math.huge then return-500 end return height end local function parkBody(character,root,humanoid)state.
bodyParts={}for _,object in ipairs(character:GetDescendants())do if object:IsA('BasePart')then local collide=object.CanCollide if type(collide)~='boolean'then collide=true end table.insert(state.
bodyParts,{part=object,collide=collide})object.CanCollide=false end end if humanoid then state.savedWalkSpeed=humanoid.WalkSpeed state.savedPlatformStand=humanoid.PlatformStand humanoid.PlatformStand=
true pcall(function()humanoid.WalkSpeed=0 end)end state.savedAnchored=root.Anchored root.AssemblyLinearVelocity,root.AssemblyAngularVelocity=Vector3.zero,Vector3.zero root.Anchored=true end
local function unparkBody()for _,entry in ipairs(state.bodyParts)do local part=entry.part if part and part.Parent then pcall(function()part.CanCollide=entry.collide end)end end table.clear(state.
bodyParts)end local function clearGhost()local camera=world.CurrentCamera if camera and camera.CameraSubject==state.humanoid then camera.CameraSubject=parts(player.Character)end for _,track in pairs(
state.tracks)do pcall(function()track:Stop(0)end)end table.clear(state.tracks)if state.model then state.model:Destroy()end state.model,state.humanoid,state.root,state.animation,state.flyMode=nil,nil,
nil,nil,nil state.syncedAt,state.camera=nil,nil end local function stop()for _,connection in ipairs(state.connections)do connection:Disconnect()end table.clear(state.connections)unparkBody()local
humanoid,root=parts(player.Character)if humanoid and humanoid.Parent then if state.savedWalkSpeed~=nil then pcall(function()humanoid.WalkSpeed=state.savedWalkSpeed end)elseif humanoid.WalkSpeed==0
then pcall(function()humanoid.WalkSpeed=settings.walkSpeed or 16 end)end if state.savedPlatformStand~=nil then humanoid.PlatformStand=state.savedPlatformStand else humanoid.PlatformStand=false end end
if root and root.Parent then if state.root and state.root.Parent and state.character==player.Character and humanoid and humanoid.Health>0 then root.CFrame=state.root.CFrame end root.
AssemblyLinearVelocity,root.AssemblyAngularVelocity=Vector3.zero,Vector3.zero if state.savedAnchored~=nil then root.Anchored=state.savedAnchored else root.Anchored=false end end state.savedWalkSpeed,
state.savedPlatformStand,state.savedAnchored=nil,nil,nil clearGhost()state.character=nil activity(false)end local function turnOff()local feature=card and card.Feature if feature and feature.enabled
and type(feature.activate)=='function'then feature.activate()elseif card and card.Enabled then card:Toggle(false)else stop()end end local function applyAppearance()if not state.model then return end
local transparency=math.clamp(settings.transparency,0,1)for _,object in ipairs(state.model:GetDescendants())do if object:IsA('BasePart')then object.Transparency=transparency end end end local function
applyMode()if not state.model then return end local ghostRoot=state.root for _,part in ipairs(state.model:GetDescendants())do if part:IsA('BasePart')then part.Anchored=false part.CanCollide=not
settings.fly and(part.Name=='Head'or part.Name=='Torso'or part.Name=='UpperTorso'or part.Name=='LowerTorso')end end state.humanoid.PlatformStand=settings.fly ghostRoot.AssemblyLinearVelocity,ghostRoot
.AssemblyAngularVelocity=Vector3.zero,Vector3.zero state.flyMode=settings.fly end local function spawnGhost(character,root)clearGhost()local archivable=character.Archivable character.Archivable=true
local ok,clone=pcall(function()return character:Clone()end)character.Archivable=archivable if not ok or not clone then return false end local humanoid,ghostRoot=parts(clone)if not humanoid or not
ghostRoot then clone:Destroy()return false end for _,object in ipairs(clone:GetDescendants())do if object:IsA('Script')or object:IsA('LocalScript')or object:IsA('Tool')then object:Destroy()elseif
object:IsA('BasePart')then object.Anchored=false object.CanTouch,object.CanQuery,object.CastShadow=false,false,false object.Transparency=math.clamp(settings.transparency,0,1)object.CanCollide=object.
Name=='Head'or object.Name=='Torso'or object.Name=='UpperTorso'or object.Name=='LowerTorso'elseif object:IsA('Highlight')or object:IsA('BillboardGui')or object:IsA('SurfaceGui')or object:IsA(
'SelectionBox')then object:Destroy()end end clone.Name='Wurst_Ghost'pcall(function()clone:SetAttribute('WurstDisguise',true)end)humanoid.AutoRotate=true humanoid.PlatformStand=false humanoid.Sit=false
humanoid.DisplayDistanceType=Enum.HumanoidDisplayDistanceType.None humanoid.WalkSpeed=math.clamp(settings.walkSpeed,0,500)clone:PivotTo(CFrame.new(root.Position+Vector3.new(0,3,0)))clone.Parent=world
state.character,state.model,state.humanoid,state.root=character,clone,humanoid,ghostRoot state.camera,state.syncedAt,state.flyMode=nil,nil,nil applyAppearance()return true end local function animate()
local humanoid=state.humanoid local kind=humanoid:GetState()local name=kind==Enum.HumanoidStateType.Jumping and'jump'or humanoid.FloorMaterial==Enum.Material.Air and'fall'or humanoid.MoveDirection.
Magnitude>0.05 and'walk'or'idle'if settings.fly then name='idle'end if state.animation==name then return end if state.animation and state.tracks[state.animation]then state.tracks[state.animation]:
Stop(0.15)end state.animation=name if state.tracks[name]==false then return end if not state.tracks[name]then local source=state.character:FindFirstChild('Animate')local folder=source and source:
FindFirstChild(name)local animation=folder and folder:FindFirstChildWhichIsA('Animation',true)local temporary if not animation then local r6=humanoid.RigType==Enum.HumanoidRigType.R6 local ids=r6 and{
idle=180435571,walk=180426354,run=180426354,jump=125750702,fall=180436148}or{idle=507766666,walk=507777826,run=507767714,jump=507765000,fall=507767968}temporary=Instance.new('Animation')temporary.
AnimationId='rbxassetid://'..ids[name]animation=temporary end local ok,track=pcall(function()local animator=humanoid:FindFirstChildOfClass('Animator')if not animator then animator=Instance.new(
'Animator')animator.Parent=humanoid end return animator:LoadAnimation(animation)end)if temporary then temporary:Destroy()end state.tracks[name]=ok and track or false end local track=state.tracks[name]
if track then track.Looped=name~='jump'track.Priority=name=='idle'and Enum.AnimationPriority.Idle or Enum.AnimationPriority.Movement track:Play(0.15)end end local function moveVector()if input:
GetFocusedTextBox()then return Vector3.zero end local movement=services.movementInput if movement and type(movement.getVector)=='function'then local ok,forward,right=pcall(movement.getVector)if ok and
type(forward)=='number'and type(right)=='number'then local camera=world.CurrentCamera local lookForward,lookRight=Vector3.new(0,0,-1),Vector3.new(1,0,0)if camera then local look=camera.CFrame.
LookVector local rightVector=camera.CFrame.RightVector lookForward=Vector3.new(look.X,0,look.Z)lookRight=Vector3.new(rightVector.X,0,rightVector.Z)lookForward=lookForward.Magnitude>0.001 and
lookForward.Unit or Vector3.new(0,0,-1)lookRight=lookRight.Magnitude>0.001 and lookRight.Unit or Vector3.new(1,0,0)end local direction=lookForward*forward+lookRight*right return direction.Magnitude>1
and direction.Unit or direction end end local humanoid=parts(player.Character)return humanoid and humanoid.MoveDirection or Vector3.zero end local function verticalIntent()if input:GetFocusedTextBox()
then return 0 end local up,down=0,0 if input:IsKeyDown(Enum.KeyCode.Space)then up=1 end if input:IsKeyDown(Enum.KeyCode.LeftControl)then down=1 end if up==0 and down==0 then local movement=services.
movementInput if movement and type(movement.isJumpHeld)=='function'then local ok,held=pcall(movement.isJumpHeld)if ok and held==true then up=1 end end end return up-down end local SYNC_INTERVAL=0.1
local function syncBody(root,ghostRoot)local now=os.clock()if state.syncedAt and now-state.syncedAt<SYNC_INTERVAL then return end state.syncedAt=now local depth=math.clamp(settings.depth or
VOID_DEPTH_MIN,0,VOID_DEPTH_MAX)local position=ghostRoot.Position root.CFrame=CFrame.new(position.X,math.max(position.Y-depth,destroyHeight()+12),position.Z)end local function step(dt)local character=
player.Character local humanoid,root=parts(character)if not humanoid or not root or humanoid.Health<=0 then return end if state.character~=character then if not spawnGhost(character,root)then turnOff(
)card:Notify('Could not create a live character.')return end parkBody(character,root,humanoid)applyMode()end if not state.model.Parent or not state.root.Parent or state.humanoid.Health<=0 then
turnOff()return end local ghostRoot,ghostHumanoid=state.root,state.humanoid if state.flyMode~=settings.fly then applyMode()end local camera=world.CurrentCamera if camera and not cameraBusy()and(state.
camera~=camera or camera.CameraSubject==humanoid)then camera.CameraSubject=ghostHumanoid state.camera=camera end if settings.fly then local direction=moveVector()local vertical=verticalIntent()local
speed=math.clamp(settings.flySpeed,0,1000)local velocity=direction*speed+Vector3.yAxis*(vertical*speed)ghostRoot.AssemblyLinearVelocity=velocity ghostRoot.AssemblyAngularVelocity=Vector3.zero
ghostHumanoid.WalkSpeed=0 else local focused=input:GetFocusedTextBox()local speed=math.clamp(settings.walkSpeed,0,500)if ghostHumanoid.WalkSpeed~=speed then ghostHumanoid.WalkSpeed=speed end
ghostHumanoid:Move(focused and Vector3.zero or moveVector(),false)end animate()syncBody(root,ghostRoot)end card=context.framework.Categories.Utilities:CreateModule({Name='Invisibility',Category=
'Utilities',Order=8,ConfigKey='Universal.Ghost',Tooltip=
[[A local clone walks under real gravity and you see through it; your real body is parked Void depth studs underneath. Fly turns the clone into a smooth velocity flight.]],Function=function(enabled)
stop()if not enabled then return end local character=player.Character local humanoid,root=parts(character)if not humanoid or not root or humanoid.Health<=0 or not spawnGhost(character,root)then card:
Toggle(false)error('Invisibility could not create a live character.',0)end activity(true)parkBody(character,root,humanoid)applyMode()table.insert(state.connections,run.PreSimulation:Connect(step))
table.insert(state.connections,input.JumpRequest:Connect(function()if not settings.fly and state.humanoid and not input:GetFocusedTextBox()and state.humanoid.FloorMaterial~=Enum.Material.Air then
state.humanoid.Jump=true end end))step(0)end})card:CreateToggle({Name='Fly',Default=false,Function=function(value)settings.fly=value applyMode()end,Tooltip=
[[OFF: native walking and jumping. ON: smooth velocity flight with WASD or the mobile stick, Space up, LeftControl down.]]})card:CreateSlider({Name='Fly speed',Show={Option='Fly'},Min=0,Max=250,
Default=settings.flySpeed,Function=function(value)settings.flySpeed=value end})card:CreateSlider({Name='Void depth',Min=VOID_DEPTH_MIN,Max=VOID_DEPTH_MAX,Default=settings.depth,Function=function(value
)settings.depth=math.clamp(value,VOID_DEPTH_MIN,VOID_DEPTH_MAX)end,Tooltip=
[[How far under the clone your real body is parked. 0-3 = same height so melee/weapons can hit, 5+ hides deeper. Default 3 (melee-friendly).]]})card:CreateSlider({Name='WalkSpeed',Min=0,Max=200,
Default=settings.walkSpeed,Function=function(value)settings.walkSpeed=math.clamp(value,0,500)if state.humanoid and not settings.fly then state.humanoid.WalkSpeed=settings.walkSpeed end end,Tooltip=
[[Walking speed of the clone (studs/s). 0 = stop, 16 = default Roblox. Works with Fly off.]]})card:CreateSlider({Name='Transparency',Min=0,Max=1,Step=0.05,Default=settings.transparency,Function=
function(value)settings.transparency=math.clamp(value,0,1)applyAppearance()end})cleanup=turnOff Module.Initialized=true return card end function Module.destroy()if cleanup then cleanup()end cleanup=
nil Module.Initialized=false end return Module
-- src/games/universal/Blatant/Noclip.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='Noclip',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.init(context:
Runtime):any local host:any=context.host local LocalPlayer:any=host.LocalPlayer local TaskManager:any=host.TaskManager local featureConnections:any=host.featureConnections local
disconnectFeatureConnection:any=host.disconnectFeatureConnection local createUniversalFeature:any=host.createUniversalFeature local addFeatureTooltip:any=host.addFeatureTooltip local originalCollision
=setmetatable({},{__mode='k'})local function restoreCollision()for part,canCollide in pairs(originalCollision)do if part and part.Parent then part.CanCollide=canCollide end end originalCollision=
setmetatable({},{__mode='k'})end local function toggleNoclip(enabled:boolean):()disconnectFeatureConnection('Noclip')disconnectFeatureConnection('NoclipDescendant')restoreCollision()if not enabled
then return end local sweptCharacter:any=nil local sweptAt:number=-math.huge local addConnection:any=nil local function noClipPart(part:any):()if part:IsA('BasePart')then if originalCollision[part]==
nil then originalCollision[part]=part.CanCollide end if part.CanCollide then part.CanCollide=false end end end local function attach(character:any):()if sweptCharacter==character then return end
sweptCharacter=character if addConnection then addConnection:Disconnect()addConnection=nil end for _,descendant in ipairs(character:GetDescendants())do noClipPart(descendant)end addConnection=
character.DescendantAdded:Connect(noClipPart)end featureConnections.Noclip=TaskManager:Connect(function():()local character=LocalPlayer.Character if not character then return end if character~=
sweptCharacter then attach(character)sweptAt=os.clock()return end if os.clock()-sweptAt<0.2 then return end sweptAt=os.clock()for _,descendant in ipairs(character:GetDescendants())do if descendant:
IsA('BasePart')and(descendant::any).CanCollide then noClipPart(descendant)end end end,nil,'Noclip')featureConnections.NoclipDescendant={Disconnect=function():()if addConnection then addConnection:
Disconnect()addConnection=nil end sweptCharacter=nil end}end local NoclipFeature=createUniversalFeature('Noclip','Disable character collisions',11,toggleNoclip,{noOptions=true,categoryName='Exploits'}
)addFeatureTooltip(NoclipFeature,'Disables collisions for every local character part. '..'Original collision states are restored when turned off.')activeCleanup=function():()toggleNoclip(false)end
Module.Initialized=true return NoclipFeature end function Module.destroy():()if activeCleanup then activeCleanup()end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Blatant/Spider.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='Wall Crawling',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.init(
context:Runtime):any local host:any=context.host local Players:any=host.Players local LocalPlayer:any=host.LocalPlayer local TaskManager:any=host.TaskManager local featureConnections:any=host.
featureConnections local disconnectFeatureConnection:any=host.disconnectFeatureConnection local getCharacterParts:any=host.getCharacterParts local createUniversalFeature:any=host.
createUniversalFeature local addToggleOption:any=host.addToggleOption local addNumberOption:any=host.addNumberOption local addCycleOption:any=host.addCycleOption local addFeatureTooltip:any=host.
addFeatureTooltip local workspace:Workspace=host.workspace or workspace type SpiderSettings={mode:string,speed:number,climbState:boolean,impulseInterval:number}local spiderSettings:SpiderSettings={
mode='Velocity',speed=30,climbState=false,impulseInterval=0.15}local SpiderFeature:any=nil local climbingHumanoid:Humanoid?=nil local function releaseClimbState()local humanoid=climbingHumanoid
climbingHumanoid=nil if humanoid and humanoid.Parent and humanoid.Health>0 and humanoid:GetState()==Enum.HumanoidStateType.Climbing then humanoid:ChangeState(humanoid.FloorMaterial==Enum.Material.Air
and Enum.HumanoidStateType.Freefall or Enum.HumanoidStateType.Running)end end local function toggleSpider(enabled:boolean):()disconnectFeatureConnection('Spider')releaseClimbState()if not enabled then
if SpiderFeature then SpiderFeature:SetStatus(nil)end return end SpiderFeature:SetStatus(spiderSettings.mode)local impulseElapsed:number=math.huge local raycastParams:RaycastParams=RaycastParams.new()
raycastParams.FilterType=Enum.RaycastFilterType.Exclude raycastParams.RespectCanCollide=true local ignoreRefreshAt:number=-math.huge local ignoredCharacter:Model?=nil local ignoreList:{Instance}={}
local function refreshIgnore(character:Model):()local now:number=os.clock()if character==ignoredCharacter and now-ignoreRefreshAt<0.5 then return end ignoredCharacter,ignoreRefreshAt=character,now
table.clear(ignoreList)table.insert(ignoreList,character)for _,player:Player in ipairs(Players:GetPlayers())do if player~=LocalPlayer and player.Character then table.insert(ignoreList,player.Character
)end end raycastParams.FilterDescendantsInstances=ignoreList end featureConnections.Spider=TaskManager:Connect(function(deltaTime:number):()impulseElapsed+=math.max(0,deltaTime)local character:Model?,
humanoidOrNil:Humanoid?,rootOrNil:BasePart?=getCharacterParts()if not character or not humanoidOrNil or not rootOrNil or humanoidOrNil.Health<=0 or humanoidOrNil.Sit or rootOrNil.Anchored then
releaseClimbState()return end local moveDir:Vector3=humanoidOrNil.MoveDirection if moveDir.Magnitude<0.05 then releaseClimbState()return end if climbingHumanoid and climbingHumanoid~=humanoidOrNil
then releaseClimbState()end local humanoid:Humanoid=humanoidOrNil::Humanoid local root:BasePart=rootOrNil::BasePart refreshIgnore(character::Model)raycastParams.CollisionGroup=root.CollisionGroup
local hipHeight:number=humanoid.HipHeight>0 and humanoid.HipHeight or 2 local rootPos:Vector3=root.Position local origin:Vector3=rootPos-Vector3.new(0,math.max(hipHeight-0.5,0),0)local wall:
RaycastResult?=workspace:Raycast(origin,moveDir.Unit*2.5,raycastParams)local velocity:Vector3=root.AssemblyLinearVelocity if not wall or math.abs((wall::RaycastResult).Normal.Y)>=0.35 then
releaseClimbState()return end if moveDir:Dot(wall.Normal)>-0.1 then releaseClimbState()return end if spiderSettings.climbState then climbingHumanoid=humanoid humanoid:ChangeState(Enum.
HumanoidStateType.Climbing)end if spiderSettings.mode=='CFrame'then root.AssemblyLinearVelocity=Vector3.new(velocity.X,0,velocity.Z)local rise=Vector3.new(0,spiderSettings.speed*math.clamp(deltaTime,0
,0.1),0)local ceiling=workspace:Raycast(rootPos,rise+Vector3.new(0,hipHeight+1,0),raycastParams)if ceiling then rise=Vector3.new(0,math.max(0,ceiling.Distance-hipHeight-1),0)end root.CFrame=root.
CFrame+rise elseif spiderSettings.mode=='Impulse'then if impulseElapsed>=spiderSettings.impulseInterval then impulseElapsed=0 local deltaVelocity:number=math.max(0,spiderSettings.speed-velocity.Y)if
deltaVelocity>0 then root:ApplyImpulse(Vector3.new(0,deltaVelocity,0)*root.AssemblyMass)end end else root.AssemblyLinearVelocity=Vector3.new(velocity.X,spiderSettings.speed,velocity.Z)end end,nil,
'Spider')end SpiderFeature=createUniversalFeature('Wall Crawling','Climb a wall while moving toward it',20,toggleSpider,{categoryName='Exploits',configKey='Universal.Spider'})local modeRows:{[string]:
any}={}local function refreshModeRows():()for name:string,row:any in pairs(modeRows)do local visible:boolean=name=='Impulse interval'and spiderSettings.mode=='Impulse'if host.setOptionVisible then
host.setOptionVisible(SpiderFeature,row,visible)elseif row then row.Visible=visible end end end addCycleOption(SpiderFeature,'Mode',{'Velocity','Impulse','CFrame'},1,function(value:string):()
spiderSettings.mode=value refreshModeRows()if SpiderFeature.enabled then toggleSpider(false)toggleSpider(true)end end,
[[Velocity: continuous climb. Impulse: timed upward boosts. CFrame: bounded vertical steps.]])addNumberOption(SpiderFeature,'Climb speed',spiderSettings.speed,1,250,function(value:number):()
spiderSettings.speed=value end)addToggleOption(SpiderFeature,'Climb state',spiderSettings.climbState,function(value:boolean):()spiderSettings.climbState=value if not value then releaseClimbState()end
end,[[Uses climbing state in classic modes (Velocity, Impulse, CFrame).]])local impulseRow:any=addNumberOption(SpiderFeature,'Impulse interval',spiderSettings.impulseInterval,0.05,0.5,function(value:
number):()spiderSettings.impulseInterval=value end,[[Seconds between bounded upward impulses. Only used by Impulse mode.]],0.05)if impulseRow then modeRows['Impulse interval']=impulseRow end
refreshModeRows()addFeatureTooltip(SpiderFeature,[[Climb only when pushing into a nearby vertical wall. No wall rotation or ceiling walking.]])activeCleanup=function():()toggleSpider(false)end Module.
Initialized=true return SpiderFeature end function Module.destroy():()if activeCleanup then activeCleanup()end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Blatant/Swim.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='Air Swim',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup function Module.init(context
:Runtime):any local host,framework=context.host,context.framework local player=host.LocalPlayer local terrain=host.workspace:FindFirstChildWhichIsA('Terrain')local lastpos:Region3?=nil local swim
local function restoreWater()if terrain and lastpos then terrain:ReplaceMaterial(lastpos,4,Enum.Material.Water,Enum.Material.Air)lastpos=nil end end swim=framework.Categories.Exploits:CreateModule({
Name='Air Swim',Category='Exploits',ConfigKey='Universal.Swim',Order=3,Tooltip=[[Swim using local terrain water. Moving or turning off clears existing water in the affected region.]],Function=function
(enabled)if not enabled then swim:SetStatus(nil)return end swim:Clean(restoreWater)swim:Clean(host.RunService.PreSimulation:Connect(function()local character=player.Character if not character then
return end local root=character:FindFirstChild('HumanoidRootPart')local humanoid=character:FindFirstChild('Humanoid')if not root or not humanoid or humanoid.Health<=0 or not terrain then return end
local moving=humanoid.MoveDirection~=Vector3.zero local space=host.UserInputService:IsKeyDown(Enum.KeyCode.Space)local factor=(moving or space)and Vector3.new(6,6,6)or Vector3.new(2,1,2)local pos=root
.Position-Vector3.new(0,1,0)local newpos=Region3.new(pos-factor,pos+factor):ExpandToGrid(4)if lastpos and lastpos.CFrame.Position==newpos.CFrame.Position and lastpos.Size==newpos.Size then return end
restoreWater()terrain:ReplaceMaterial(newpos,4,Enum.Material.Air,Enum.Material.Water)lastpos=newpos end))swim:SetStatus('Active')end})activeCleanup=function()if swim.Enabled then swim:Toggle(false)end
restoreWater()end Module.Initialized=true return swim end function Module.destroy()if activeCleanup then activeCleanup()activeCleanup=nil end Module.Initialized=false end return Module
-- src/games/universal/Blatant/WalkFling.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='WalkFling',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.init(
context:Runtime):any local host:any=context.host local LocalPlayer:any=host.LocalPlayer local RunService:any=host.RunService or game:GetService('RunService')local createUniversalFeature:any=host.
createUniversalFeature local addFeatureTooltip:any=host.addFeatureTooltip local addInformationOption:any=host.addInformationOption local configData:any=host.configData local services:any=(context::any
).services local activity:any=services and services.activity if type(configData)=='table'and type(configData.states)=='table'then configData.states['Universal.WalkFling']=false end local enabled:
boolean=false local session:number=0 local POWER:number=50000 local SPIN:number=50000 local savedCollision:any=setmetatable({},{__mode='k'})local walkFlingFeature:any=nil local function
getCharacterRoot():(Model?,Humanoid?,BasePart?)local character:Model?=LocalPlayer and LocalPlayer.Character if not character or not character.Parent then return nil,nil,nil end local humanoid:Humanoid
?=character:FindFirstChildOfClass('Humanoid')local root:BasePart?=(humanoid and humanoid.RootPart)or(character:FindFirstChild('HumanoidRootPart')::BasePart?)or(character:FindFirstChild('Torso')::
BasePart?)if not humanoid or humanoid.Health<=0 or not root then return character,humanoid,nil end return character,humanoid,root end local function rememberCollision(character:Model):()for _,part:
Instance in ipairs(character:GetChildren())do if part:IsA('BasePart')and savedCollision[part]==nil then savedCollision[part]=(part::BasePart).CanCollide end end end local function restoreCollision():(
)for part:any,canCollide:any in pairs(savedCollision)do if part and part.Parent then pcall(function():()part.CanCollide=canCollide end)end end savedCollision=setmetatable({},{__mode='k'})end
local function stopWalkFling():()local wasEnabled:boolean=enabled enabled=false session+=1 restoreCollision()local _character:Model?,_humanoid:Humanoid?,root:BasePart?=getCharacterRoot()if wasEnabled
and root and root.Parent then pcall(function():()root.Velocity=Vector3.zero root.AssemblyLinearVelocity=Vector3.zero root.AssemblyAngularVelocity=Vector3.zero end)end if activity and type(activity.set
)=='function'then activity.set('walkfling',false)end if walkFlingFeature then walkFlingFeature:SetStatus(nil)end end local function startWalkFling():()if enabled then return end enabled=true session+=
1 local mySession:number=session local movel:number=0.1 if activity and type(activity.set)=='function'then activity.set('walkfling',true)end if walkFlingFeature then walkFlingFeature:SetStatus('ON')
end task.spawn(function():()while enabled and session==mySession and Module.Initialized do RunService.Heartbeat:Wait()if not enabled or session~=mySession then break end local character:Model?,
humanoid:Humanoid?,root:BasePart?=getCharacterRoot()if character and root and root.Parent then local savedVelocity:Vector3=root.AssemblyLinearVelocity local savedAngular:Vector3=root.
AssemblyAngularVelocity or Vector3.zero if humanoid and humanoid.ChangeState then pcall(function()humanoid:ChangeState(Enum.HumanoidStateType.Physics)end)end pcall(function():()rememberCollision(
character)for _,part:Instance in ipairs(character:GetChildren())do if part:IsA('BasePart')then(part::BasePart).CanCollide=false end end end)local spike:Vector3=savedVelocity*POWER+Vector3.new(0,POWER,0)
local spin:Vector3=Vector3.new(SPIN,SPIN,SPIN)root.Velocity=spike root.AssemblyLinearVelocity=spike root.AssemblyAngularVelocity=spin pcall(function()(root::any).RotVelocity=spin end)end)RunService.
RenderStepped:Wait()local stillRunning:boolean=enabled and session==mySession if stillRunning and character.Parent and root.Parent then pcall(function():()root.Velocity=savedVelocity root.
AssemblyLinearVelocity=savedVelocity root.AssemblyAngularVelocity=savedAngular pcall(function()(root::any).RotVelocity=savedAngular end)end)end RunService.Stepped:Wait()if stillRunning and enabled and
session==mySession and character.Parent and root.Parent then pcall(function():()local oscillated:Vector3=savedVelocity+Vector3.new(0,movel,0)root.Velocity=oscillated root.AssemblyLinearVelocity=
oscillated local spinOsc:Vector3=Vector3.new(0,SPIN*0.1,0)root.AssemblyAngularVelocity=spinOsc movel=-movel end)end end end if session==mySession then restoreCollision()end end)end local function
mm2Game():boolean local shared:any=host.state if shared and shared.activeGameName=='MM2'then return true end local ok:boolean,placeId:any=pcall(function():any return game.PlaceId end)return ok and
placeId==142823291 end local function antiFlingOn():boolean local shared:any=host.state return shared~=nil and shared.antiFlingActive==true end local function toggleWalkFling(state:boolean):()if state
==true then if mm2Game()and antiFlingOn()then error('Anti Fling is ON. Turn it off first to use Walk Fling.',0)end startWalkFling()return end stopWalkFling()end walkFlingFeature=
createUniversalFeature('Walk Fling','Fling on contact while walking (Infinite Yield walkfling)',8,toggleWalkFling,{categoryName='Exploits'})if type(addInformationOption)=='function'then
addInformationOption(walkFlingFeature,[[Collisions are restored and the body parked when the card is turned off.]])end addFeatureTooltip(walkFlingFeature,
[[While ON, every physics frame the local character is noclipped and given a ]]..[[huge velocity for one frame, which launches whoever you walk into. ]]..
[[Collisions are restored when it stops. It never starts on injection. ]]..'In MM2 it refuses to start while Anti Fling is ON.')activeCleanup=function():()stopWalkFling()end Module.Initialized=true
return walkFlingFeature end function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Combat/AimAssist.lua
export type Runtime={framework:any,entity:any,host:any,libraries:{[string]:any},services:any}local Module={Name='AimAssist',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCard:any=
nil function Module.init(context:Runtime):any local framework:any=context.framework local host:any=context.host local services:any=context.services local targeting:any=context.libraries and context.
libraries.targeting local entity:any=context.entity local userInput:UserInputService=host.UserInputService local currentWorkspace:Workspace=host.workspace or workspace local localPlayer:Player?=host.
LocalPlayer local runtime:any={target=nil,lockedTarget=nil,lastFire=-math.huge,fireMode='Hold',clickPressed=false}local card:any local function camera():Camera?return currentWorkspace.CurrentCamera
end local function getHeldTool():Tool?local player:Player?=localPlayer or host.LocalPlayer if not player then return nil end local character:Model?=player.Character if not character then return nil
end for _,child:Instance in ipairs(character:GetChildren())do if child:IsA('Tool')then return child::Tool end end return nil end local function getMouseScreenPosition(cameraObject:Camera?):Vector2
local mouseOk:boolean,mousePos:any=pcall(function():any return userInput:GetMouseLocation()end)if mouseOk and typeof(mousePos)=='Vector2'then return mousePos end local vp:Vector2=cameraObject and
cameraObject.ViewportSize or Vector2.new(800,600)local insetY:number=36 pcall(function():()local guiService:any=game:GetService('GuiService')local inset:Vector2=guiService:GetGuiInset()if typeof(inset
)=='Vector2'then insetY=inset.Y end end)return Vector2.new(vp.X/2,vp.Y/2+insetY)end local function tryFireOnce(fireMode:string?,clickPressed:boolean?):boolean local tool:Tool?=getHeldTool()if tool
then if tool.Enabled==false then return false end if fireMode=='Hold'and clickPressed==true then return true end local ok:boolean=pcall(function():()(tool::Tool):Activate()end)if ok then return true
end end local mouse:Vector2=getMouseScreenPosition(camera())local vimOk:boolean=pcall(function():()local vim:any=game:GetService('VirtualInputManager')if vim and type(vim.SendMouseButtonEvent)==
'function'then vim:SendMouseButtonEvent(mouse.X,mouse.Y,0,true,game,0)vim:SendMouseButtonEvent(mouse.X,mouse.Y,0,false,game,0)else error('no vim')end end)if vimOk then return true end local env:any=
getfenv()local click:any=env.mouse1click or env.mouse1press if type(click)=='function'then local ok2:boolean=pcall(function():()click()end)if ok2 then return true end local rel:any=env.mouse1release
if type(rel)=='function'then pcall(function():()click()task.wait(0.02)rel()end)return true end end return false end local function getCPS():number local opt:any=nil if card and card.Options then opt=
card.Options['Clicks Per Second']or card.Options['Fire rate']or card.Options['CPS']end if opt and opt.Value~=nil then return math.max(1,tonumber(opt.Value)or 12)end return 12 end local function
shouldAutoFire():boolean if not card or not card.Options or not card.Options['Auto Fire']then return false end if not card.Options['Auto Fire'].Value then return false end local rate:number=getCPS()
local now:number=os.clock()if now-runtime.lastFire<1/rate then return false end local fireMode:string='Hold'if card.Options['Auto Fire mode']then fireMode=tostring(card.Options['Auto Fire mode'].Value
)end local okHeld:boolean,isPressed:any=pcall(function():any return userInput:IsMouseButtonPressed(Enum.UserInputType.MouseButton1)end)local clickPressed:boolean=okHeld and isPressed==true runtime.
fireMode=fireMode runtime.clickPressed=clickPressed if fireMode=='Always'then runtime.lastFire=now return true end if clickPressed then runtime.lastFire=now return true end return false end
local function moveCamera(target:any,deltaTime:number):boolean local cameraObject:Camera?=camera()if not cameraObject or not target then return false end local mode:string=card.Options['Mode'].Value
local cameraModeRaw:string='Auto'if card.Options['Camera']then cameraModeRaw=tostring(card.Options['Camera'].Value)end local cameraMode:string=cameraModeRaw if cameraModeRaw=='CFrame'then cameraMode=
'Camera'end local current:CFrame=cameraObject.CFrame local offset:Vector3=target.position-current.Position if offset.Magnitude<=0.001 then return false end local alpha:number=1 if mode~='Aimbot'then
local dt=math.max(deltaTime,0)local speed=math.max(0,card.Options.Speed.Value)local strength=math.max(0,card.Options.Strength.Value)/100 if speed==0 or strength==0 or dt==0 then return false end local
angle=math.acos(math.clamp(current.LookVector:Dot(offset.Unit),-1,1))if angle<math.rad(0.5)then return false end alpha=math.min(1-math.exp(-strength*10*dt),angle>1e-6 and math.rad(speed)*dt/angle or 1
)end local function isMouseLocked():boolean local okBeh:boolean,beh:any=pcall(function():any return(userInput::any).MouseBehavior end)if okBeh and beh then local okEnum:boolean,lockCenter:any=pcall(
function():any return(Enum::any).MouseBehavior.LockCenter end)local okCurr:boolean,lockCurr:any=pcall(function():any return(Enum::any).MouseBehavior.LockCurrentPosition end)if(okEnum and beh==
lockCenter)or(okCurr and beh==lockCurr)then return true end end local lp:Player?=localPlayer or host.LocalPlayer local char:Model?=lp and lp.Character local head:BasePart?=char and char:
FindFirstChild('Head')::BasePart?if head then local dist:number=(cameraObject.CFrame.Position-head.Position).Magnitude if dist<2.5 then return true end end return false end local function tryMouse():
boolean if cameraMode=='Camera'then return false end local env:any=getfenv()local moverel:any=env.mousemoverel or env.mousemove or(host and(host.mousemoverel or host.mousemove))if type(moverel)~=
'function'then return false end local ok:boolean,screenPos:any,onScreen:any=pcall(function():any return cameraObject:WorldToScreenPoint(target.position)end)if not ok or type(screenPos)~='Vector3'then
return false end if onScreen==false and mode~='Aimbot'then return false end local m:Vector2=getMouseScreenPosition(cameraObject)local delta:Vector2=Vector2.new(screenPos.X-m.X,screenPos.Y-m.Y)if mode
~='Aimbot'then delta=delta*alpha end if delta.Magnitude<0.3 then return false end local maxDelta:number=500 if math.abs(delta.X)>maxDelta or math.abs(delta.Y)>maxDelta then delta=delta.Unit*maxDelta
end local moved:boolean=false local okMove:boolean=pcall(function():()moverel(delta.X,delta.Y)moved=true end)return okMove and moved end local function tryCamera():boolean if cameraMode=='Mouse'then
return false end local currentLook:Vector3=current.LookVector local blended:Vector3=offset.Unit if alpha<1 then local dot=math.clamp(currentLook:Dot(offset.Unit),-1,1)local tangent=offset.Unit-
currentLook*dot if tangent.Magnitude<1e-6 then tangent=current.RightVector end local turn=math.acos(dot)*alpha blended=currentLook*math.cos(turn)+tangent.Unit*math.sin(turn)end if blended.Magnitude<=
0.001 then return false end local newCFrame:CFrame=CFrame.lookAt(current.Position,current.Position+blended.Unit)local ok:boolean=pcall(function():()cameraObject.CFrame=newCFrame end)return ok end if
cameraMode=='Auto'then local locked:boolean=isMouseLocked()if locked then local didMouse:boolean=tryMouse()local didCamera:boolean=tryCamera()return didMouse or didCamera else if tryMouse()then return
true end if tryCamera()then return true end return false end elseif cameraMode=='Camera'then return tryCamera()elseif cameraMode=='Mouse'then return tryMouse()end return tryCamera()end local function
anotherAimAdapterIsActive():boolean local activity:any=services and services.activity local free=services and services.freeCam local spectate=services and services.spectate if free and free.IsActive()
or spectate and spectate.IsActive()then return true end return activity~=nil and type(activity.isActive)=='function'and activity.isActive('mvsdSilentAim')==true end local function selectTarget():any
if anotherAimAdapterIsActive()or not targeting or type(targeting.List)~='function'then return nil end local options:any=card.Options local mode:string=options['Mode'].Value local candidates:{any}=
targeting:List(entity,{FOV=mode=='Aim Assist'and options['FOV'].Value or nil,MaxDistance=options['Max distance'].Value,TargetPart=options['Target part'].Value,Prediction=options['Prediction'].Value,
TeamCheck=options['Team check'].Value,VisibilityCheck=options['Visibility check'].Value,Priority=options['Priority'].Value,IncludeNPCs=options['Target NPCs'].Value,AllowOffscreen=mode=='Aimbot'})if
mode~='Aimbot'then runtime.lockedTarget=nil return candidates[1]end if options['Sticky target'].Value and runtime.lockedTarget then for _,candidate:any in ipairs(candidates)do if candidate.entity==
runtime.lockedTarget.entity then runtime.lockedTarget=candidate return candidate end end end runtime.lockedTarget=candidates[1]return runtime.lockedTarget end pcall(function():()local cfg:any=host.
configData if cfg and type(cfg.values)=='table'then local key:string='Universal.AimAssist.Camera'if cfg.values[key]=='CFrame'then cfg.values[key]='Camera'end local oldKey:string=
'Universal.AimAssist.Fire rate'local newKey:string='Universal.AimAssist.Clicks Per Second'if cfg.values[oldKey]~=nil and cfg.values[newKey]==nil then cfg.values[newKey]=cfg.values[oldKey]end end end)
card=framework.Categories.Exploits:CreateModule({Name='Aimbot',Category='Exploits',ConfigKey='Universal.AimAssist',Order=4,Tooltip=[[Eases the local camera toward a valid target; no raycast or ]]..
'remote hooks.',Function=function(enabled:boolean):()runtime.target=nil runtime.lockedTarget=nil runtime.lastFire=-math.huge if not enabled then card:SetStatus(nil)local rs:any=(host::any).RunService
if rs and type(rs.UnbindFromRenderStep)=='function'then pcall(function():()rs:UnbindFromRenderStep('WurstAimAssistCam')end)end return end card:SetStatus('searching')card:Render(function(deltaTime:
number):()if anotherAimAdapterIsActive()or services.menu.isVisible()or services.menu.isCapturingInput()then runtime.target=nil runtime.lockedTarget=nil card:SetStatus('paused')return end if userInput:
GetFocusedTextBox()then runtime.target=nil runtime.lockedTarget=nil card:SetStatus('paused')return end local selected:any=selectTarget()runtime.target=selected if not selected then card:SetStatus(
'searching')return end local moved:boolean=moveCamera(selected,deltaTime)if shouldAutoFire()then pcall(function():()tryFireOnce(runtime.fireMode,runtime.clickPressed)end)end local targetPlayer:Player?
=selected.player local displayName:string=targetPlayer and tostring(targetPlayer.DisplayName or targetPlayer.Name)or tostring(selected.entity.Name or selected.character.Name)local distance:number=math
.round(selected.distance)card:SetStatus(displayName..' \183 '..tostring(distance)..' studs'..(moved and' \183 locked'or' \183 ready'))end)card:Clean(function():()runtime.target=nil card:SetStatus(nil)
local rs:any=(host::any).RunService if rs and type(rs.UnbindFromRenderStep)=='function'then pcall(function():()rs:UnbindFromRenderStep('WurstAimAssistCam')end)end end)end})card:CreateDropdown({Name=
'Mode',List={'Aim Assist','Aimbot'},Index=1,Function=function():()runtime.lockedTarget=nil end,Tooltip=
[[Aim Assist = ayuda suave dentro de FOV usando Speed/Strength (natural). Aimbot = snap directo con Sticky opcional.]]})card:CreateDropdown({Name='Camera',List={'Auto','Camera','Mouse'},Index=1,
Tooltip='Auto: 3rd=Mouse (natural) 1st/FPS=Mouse+Camera h\237brido para que el arma siga (Mouse Hit + CFrame instant). Camera: solo CFrame (1st person). Mouse: solo mousemoverel.'})card:CreateSlider({
Name='FOV',Show={Option='Mode',Values={'Aim Assist'}},Min=15,Max=500,Step=5,Default=180,Tooltip=
'FOV en p\237xeles: radio alrededor del cursor. 15 peque\241o \u{2014} 500 pantalla completa. Default 180.'})card:CreateSlider({Name='Speed',Min=0,Max=360,Default=180,Step=5,Show={Option='Mode',Values
={'Aim Assist'}},Tooltip='Speed \176/s: rotaci\243n m\225xima. 0 sin ayuda, 180 default, 360 instant\225neo.'})card:CreateSlider({Name='Strength',Min=0,Max=100,Default=75,Step=5,Show={Option='Mode',
Values={'Aim Assist'}},Tooltip='Strength %: fuerza de arrastre. 0 nada, 50 suave, 75 default, 100 m\225xima.'})card:CreateNote(
'FOV: 15-500 px \u{2014} 180 default \u{2022} Speed: 0-360 \176/s \u{2014} 180 default \u{2022} Strength: 0-100% \u{2014} 75 default.')card:CreateDropdown({Name='Target part',List={'Closest','Head',
'Torso'},Index=1,Tooltip='Body part used as the camera target.'})card:CreateDropdown({Name='Priority',List={'Cursor','Distance'},Index=1,Tooltip=
[[Choose the target nearest the cursor or nearest in the world.]]})card:CreateSlider({Name='Prediction',Min=0,Max=0.5,Step=0.01,Default=0.06,Tooltip=
'Prediction s: anticipa movimiento. 0 sin predicci\243n, 0.06 default (60 ms).'})card:CreateSlider({Name='Max distance',Min=25,Max=2000,Step=5,Default=600,Tooltip=
'Max distance studs: ignora targets m\225s lejos. 600 default.'})card:CreateNote('Prediction: 0-0.5 s \u{2014} 0.06 s default. Max distance 600 studs default.')card:CreateToggle({Name=
'Visibility check',Default=true,Tooltip='Skip targets hidden by map geometry.'})card:CreateToggle({Name='Team check',Default=true,Tooltip='Skip teammates and Friend List entries.'})card:CreateToggle({
Name='Target NPCs',Default=false,Tooltip='Include non-player humanoid models in both modes.'})card:CreateToggle({Name='Sticky target',Show={Option='Mode',Values={'Aimbot'}},Default=true,Function=
function(value:boolean):()if not value then runtime.lockedTarget=nil end end,Tooltip=[[Hold the current Aimbot target. Turn off to release it every frame.]]})card:CreateToggle({Name='Auto Fire',
Default=false,Tooltip='Autoclick gen\233rico: si no hay Tool hace click virtual. No toca GUI.'})card:CreateDropdown({Name='Auto Fire mode',List={'Hold','Always'},Index=1,Show={Option='Auto Fire'},
Tooltip=[[Hold = solo mientras mantienes click izquierdo (para semi-auto). Always = dispara solo al tener target.]]})card:CreateSlider({Name='Clicks Per Second',Min=1,Max=20,Step=1,Default=12,Show={
Option='Auto Fire'},Tooltip='Clicks por segundo manteniendo. 12 default = pistola semi-auto. Sube para autom\225tico.'})card:CreateNote(
'Natural: Auto 3rd=Mouse, 1st=Mouse+Camera h\237brido (sincroniza Mouse Hit + CFrame) para que el arma no se quede atr\225s.')pcall(function()local feature:any=(card::any).Feature or(card::any).
_feature or card local optionsFrame:any=(feature::any).options or(card::any).options if not optionsFrame or type(optionsFrame.FindFirstChild)~='function'then return end local function patchOption(
optionName:string,minText:string,maxText:string,suffix:string)local row:any=optionsFrame:FindFirstChild('Option_'..optionName:gsub('%W',''))if not row then return end local label:any=row:
FindFirstChild('OptionLabel')if label and suffix then pcall(function()label.Text=label.Text..' '..suffix end)end local slider:any=row:FindFirstChild('Slider')if slider then local minL:any=slider:
FindFirstChild('MinLabel')local maxL:any=slider:FindFirstChild('MaxLabel')if minL then pcall(function()minL.Text=minText end)end if maxL then pcall(function()maxL.Text=maxText end)end end end
patchOption('FOV','15 px','500 px','(p\237xeles)')patchOption('Speed','0\176/s','360\176/s','(\176/s)')patchOption('Strength','0%','100%','(%)')patchOption('Prediction','0 s','0.5 s','(s)')
patchOption('Maxdistance','25 studs','2000 studs','(studs)')patchOption('ClicksPerSecond','1','20','(cps)')end)activeCard=card Module.Initialized=true return card end function Module.destroy():()if
activeCard and activeCard.Enabled then activeCard:Toggle(false)end activeCard=nil Module.Initialized=false end return Module
-- src/games/universal/Combat/KillAura.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='KillAura',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil local function
compileIgnoreList(text:string?):{[string]:boolean}local ignored:{[string]:boolean}={}for name:string in string.gmatch(text or'','[^,;%s]+')do ignored[string.lower(name)]=true end return ignored end
local function isIgnoredEntity(entity:any,ignored:{[string]:boolean}):boolean local player:any=entity and entity.Player if not player then return false end return ignored[string.lower(player.Name or''
)]==true or ignored[string.lower(player.DisplayName or'')]==true or ignored[tostring(player.UserId)]==true end local function collectTools(localPlayer:Player):{Tool}local tools:{Tool}={}local
character:Model?=localPlayer.Character if character then for _,child:Instance in ipairs(character:GetChildren())do if child:IsA('Tool')then table.insert(tools,child::Tool)end end end local backpack:
Instance?=localPlayer:FindFirstChild('Backpack')if backpack then for _,child:Instance in ipairs(backpack:GetChildren())do if child:IsA('Tool')then table.insert(tools,child::Tool)end end end return
tools end local function toolHandle(tool:Tool?):BasePart?if not tool then return nil end local resolved:Tool=tool::Tool local handle:Instance?=resolved:FindFirstChild('Handle')if handle and handle:
IsA('BasePart')then return handle::BasePart end for _,descendant:Instance in ipairs(resolved:GetDescendants())do if descendant:IsA('BasePart')then return descendant::BasePart end end return nil end
local function activateTool(tool:Tool?):boolean if not tool or tool.Enabled==false then return false end return pcall(function():()tool:Activate()end)end local function fireContact(handle:BasePart?,
target:BasePart):()if not handle then return end local environment:any=getfenv()local fire:any=environment.firetouchinterest if type(fire)~='function'then return end pcall(fire,handle,target,0)pcall(
fire,handle,target,1)end function Module.init(context:Runtime):any local host:any=context.host local framework:any=context.framework local entityLibrary:any=context.entity local localPlayer:Player=
host.LocalPlayer local userInput:UserInputService=host.UserInputService local currentWorkspace:Workspace=host.workspace or workspace local teamCheck:any=host.state and host.state.teamCheck local
nextSwingAt:number=0 local nextStatusAt:number=0 local highlights:{[any]:Highlight}={}local function clearHighlights(keep:{[any]:boolean}?):()for key:any,highlight:Highlight in pairs(highlights)do if
not keep or not keep[key]then highlight:Destroy()highlights[key]=nil end end end local function markTarget(entity:any,options:any):()if not options['Highlight targets'].Value then return end local key
:any=entity.Player or entity.Character local existing:Highlight?=highlights[key]if not existing or not existing.Parent then local created:Highlight=Instance.new('Highlight')created.Name=
'Wurst_KillAuraTarget'created.FillTransparency=0.75 created.Parent=entity.Character highlights[key]=created existing=created end local highlight:Highlight=existing::Highlight highlight.Adornee=entity.
Character highlight.DepthMode=Enum.HighlightDepthMode.AlwaysOnTop local colour:Color3=options['Highlight colour'].Value highlight.FillColor=colour highlight.OutlineColor=colour end local aura:any=nil
local weapons:any=context.weapons local function updateTargetStatus(count:number):()local now:number=os.clock()if now<nextStatusAt then return end nextStatusAt=now+0.25 aura:SetStatus(tostring(count)
..' in range')end local learnedWeapon:string?=nil local learnConnections:{RBXScriptConnection}={}local function watchTool(tool:Tool):()table.insert(learnConnections,tool.Activated:Connect(function():(
)learnedWeapon=tool.Name end))end local function clearToolWatches():()for _,connection:RBXScriptConnection in ipairs(learnConnections)do connection:Disconnect()end table.clear(learnConnections)end
local function watchAllTools():()clearToolWatches()for _,tool:Tool in ipairs(collectTools(localPlayer))do watchTool(tool)end end local function matchesWanted(tool:Tool,wanted:string):boolean return
string.find(string.lower(tool.Name),string.lower(wanted),1,true)~=nil end local function equippedTool():(Tool?,BasePart?)local character:Model?=localPlayer.Character if not character then return nil,
nil end local wanted:string=tostring(aura.Options['Weapon name'].Value or'')local held:{Tool}={}for _,child:Instance in ipairs(character:GetChildren())do if child:IsA('Tool')then table.insert(held,
child::Tool)end end if wanted~=''then for _,tool:Tool in ipairs(held)do if matchesWanted(tool,wanted)then return tool,toolHandle(tool)end end end if learnedWeapon then for _,tool:Tool in ipairs(held)
do if tool.Name==learnedWeapon then return tool,toolHandle(tool)end end end local first:Tool?=held[1]return first,toolHandle(first)end local function equipFromBackpack():()local backpack:Instance?=
localPlayer:FindFirstChild('Backpack')local humanoid:Humanoid?=localPlayer.Character and localPlayer.Character:FindFirstChildOfClass('Humanoid')::Humanoid?if not backpack or not humanoid then return
end local wanted:string=tostring(aura.Options['Weapon name'].Value or'')local chosen:Tool?=nil for _,child:Instance in ipairs(backpack:GetChildren())do if not child:IsA('Tool')then continue end local
tool:Tool=child::Tool if wanted~=''and matchesWanted(tool,wanted)then chosen=tool break end if learnedWeapon and tool.Name==learnedWeapon then chosen=tool break end if not chosen then chosen=tool end
end if chosen then pcall(function():()humanoid:EquipTool(chosen::Tool)end)end end local store:any=host.configData if store and store.values then local reachKey:string='Universal.KillAura.Reach'local
stored:any=store.values['Universal.KillAura.Swingrange']if store.values[reachKey]==nil and type(stored)=='number'then store.values[reachKey]=stored if type(host.queueConfigSave)=='function'then host.
queueConfigSave()end end end aura=framework.Categories.Exploits:CreateModule({Name='KillAura',DisplayName=host.GAME_CHECK and host.GAME_CHECK.MM2Active and'Knife Stab'or'KillAura',Category='Exploits',
ConfigKey='Universal.KillAura',Order=1,Tooltip='Swings your weapon at everyone in reach, with its own '..'range, rate, field of view and rotation.',Function=function(enabled:boolean):()if aura.
CombatAdapter then clearHighlights()aura.CombatAdapter(enabled)return end nextSwingAt=0 nextStatusAt=0 if not enabled then aura:SetStatus(nil)clearHighlights()return end aura:SetStatus('0 in range')
aura:Clean(function():()clearHighlights()end)watchAllTools()aura:Event(localPlayer.CharacterAdded,function():()task.defer(function()if aura.Enabled then watchAllTools()end end)end)local backpack:
Instance?=localPlayer:FindFirstChild('Backpack')if backpack then aura:Event(backpack.ChildAdded,function(child:Instance):()if child:IsA('Tool')then watchTool(child::Tool)end end)end aura:Clean(
clearToolWatches)aura:Loop(function():()local options:any=aura.Options local suppress:any=host.suppressTouchControls if type(suppress)=='function'and options['Activation'].Value=='Hold key'then local
hold:any=options['Hold key'].Value local isMouseHold:boolean=false pcall(function()isMouseHold=(hold==Enum.KeyCode.Unknown and false)or(typeof(hold)=='EnumItem'and string.find(tostring(hold),
'MouseButton')~=nil)or(typeof(hold)=='EnumItem'and hold.EnumType==Enum.UserInputType)end)if isMouseHold or userInput:IsKeyDown(hold)then pcall(suppress)end end if options['Activation'].Value==
'Hold key'and not userInput:IsKeyDown(options['Hold key'].Value)then updateTargetStatus(0)clearHighlights()return end if not entityLibrary:IsAlive()then updateTargetStatus(0)clearHighlights()return
end local tool:Tool?,handle:BasePart?=equippedTool()if not tool then if options['Auto equip'].Value then equipFromBackpack()end if options['Require tool'].Value then updateTargetStatus(0)
clearHighlights()return end end entityLibrary:Refresh()local targets:{any}=entityLibrary:InRange({Range=options['Reach'].Value,IgnoreTeam=not options['Team check'].Value,IgnoreWalls=not options[
'Wall check'].Value,Angle=options['Field of view'].Value,IncludeNPCs=options['Target NPCs'].Value,Sort=options['Priority'].Value})if options['Team check'].Value and teamCheck then local kept:{any}={}
for _,entity:any in ipairs(targets)do local teammate:boolean=false if entity.Player and type(teamCheck.isSameTeam)=='function'then teammate=teamCheck.isSameTeam(entity.Player)end if not teammate then
table.insert(kept,entity)end end targets=kept end local ignoreText:string=tostring(options['Ignore players'].Value or'')if ignoreText~=''then local ignored:{[string]:boolean}=compileIgnoreList(
ignoreText)local kept:{any}={}for _,entity:any in ipairs(targets)do if not isIgnoredEntity(entity,ignored)then table.insert(kept,entity)end end targets=kept end local targetLimit:number=math.max(1,
math.round(options['Max targets'].Value))while#targets>targetLimit do table.remove(targets)end updateTargetStatus(#targets)if#targets==0 then clearHighlights()return end local now:number=os.clock()
local rate:number=math.max(options['Swings per second'].Value,0.5)local jitter:number=1/rate*0.25 local canSwing:boolean=now>=nextSwingAt local keep:{[any]:boolean}={}local root:BasePart?=
entityLibrary.LocalEntity and entityLibrary.LocalEntity.RootPart local rotation:string=options['Rotation'].Value local originalCFrame:CFrame?=root and root.CFrame or nil if canSwing then nextSwingAt=
now+(1/rate)+(math.random()-0.5)*2*jitter end if rotation~='Off'and root then local first:any=targets[1]local flat:Vector3=first.RootPart.Position*Vector3.new(1,0,1)root.CFrame=CFrame.lookAt(root.
Position,Vector3.new(flat.X,root.Position.Y+0.01,flat.Z))end for _,entity:any in ipairs(targets)do keep[entity.Player or entity.Character]=true markTarget(entity,options)if not canSwing then continue
end local swung:boolean=false if weapons then for _,label:string in ipairs(aura.Options['Weapons'].Selected or{})do local candidate:any=weapons:Find(label)if candidate and weapons:Activate(candidate,
entity)then swung=true end end end if not swung and tool then swung=activateTool(tool)end if not swung and weapons then local fallback:any=weapons:Best({Melee=true})swung=fallback~=nil and weapons:
Activate(fallback,entity)end local extraHits:number=math.max(math.round(options['Hits per swing'].Value)-1,0)for _=1,extraHits do if tool then activateTool(tool)elseif weapons then weapons:Swing()end
end if not options['Contact damage'].Value then continue end if not options['Hit through walls'].Value and not entityLibrary:VisibleFrom(entity)then continue end local overlap:OverlapParams=
OverlapParams.new()overlap.FilterType=Enum.RaycastFilterType.Include overlap.FilterDescendantsInstances={entity.Character}local parts:{BasePart}=currentWorkspace:GetPartBoundsInBox(entity.RootPart.
CFrame,Vector3.new(5,6,5),overlap)for _,part:BasePart in ipairs(parts)do fireContact(handle,part)end end if rotation=='Silent'and root and originalCFrame then root.CFrame=originalCFrame::CFrame end
clearHighlights(keep)end)end})aura:CreateDropdown({Name='Activation',List={'Always','Hold key'},Index=1})aura:CreateBind({Name='Hold key',Show={Option='Activation',Values={'Hold key'}},Default=Enum.
KeyCode.Unknown})aura:CreateSlider({Name='Reach',Min=5,Max=60,Default=18,Tooltip='Distance at which the weapon swings and its touch events '..'fire. This is the only Kill Aura distance.'})aura:
CreateSlider({Name='Max targets',Min=1,Max=10,Default=3,Tooltip='How many valid players or NPCs one swing may hit. One '..'sticks to the best target.'})aura:CreateDropdown({Name='Priority',List={
'Distance','Health','Threat'},Index=1,Tooltip='Which target comes first: nearest, weakest, or the one '..'most dangerous to you.'})aura:CreateSlider({Name='Swings per second',Min=1,Max=20,Default=8,
Tooltip='A fast human clicks about 8-12 times a second. Twenty is '..'not a person, and a server that logs click rates knows it.'})aura:CreateSlider({Name='Field of view',Min=30,Max=360,Default=360,
Tooltip='Degrees around your character. 360 scans every direction; '..'this is not cursor aim.'})aura:CreateDropdown({Name='Rotation',List={'Off','Silent','Face'},Index=2,Tooltip=
'Silent turns you for the swing only and puts you back; '..'Face keeps you pointed at the target.'})aura:CreateSlider({Name='Hits per swing',Min=1,Max=8,Default=1,Tooltip=
'How many times the weapon is pressed for each swing the '..'rate allows. Some games count every press.'})aura:CreateList({Name='Weapons',Items=function():{string}if not weapons then return{}end
return weapons:Labels({All=true,Melee=true})end,EmptyText='',Tooltip='Melee weapons this module may swing: tools, view models, '..[[inventory slots and the game's own attack button. Empty means ]]..
```

```
'whatever you are holding.'})aura:CreateTextBox({Name='Weapon name',Default='',Tooltip="Part of the tool's name. Leave empty and the module uses "..'whatever you last swung by hand.'})aura:
CreateButton({Name='Bind held tool',Tooltip='Remembers whatever is in your hands right now.',Function=function():()local tool:Tool?=equippedTool()if not tool then aura:Notify('nothing equipped')return
end learnedWeapon=(tool::Tool).Name aura:SetStatus((tool::Tool).Name..' bound')end})aura:CreateToggle({Name='Auto equip',Default=false,Tooltip=
'Pull the weapon out of the backpack when empty-handed. Off '..'by default: most games this module helps in do not need '..'anything in your hands at all.',Function=function(value:boolean):()if
weapons then weapons:SetAllowEquip(value)end end})aura:CreateToggle({Name='Require tool',Default=false,Tooltip='Do nothing while your hands are empty.'})aura:CreateToggle({Name='Team check',Default=
true,Tooltip='Skip teammates, analysed by the always-on advanced team '..[[check (game teams, roles, team colours, neutral rules), plus ]]..'Friend List entries.'})aura:CreateToggle({Name='Wall check'
,Default=false,Tooltip='Off by default: the aura hits through walls and floors '..'until you ask it not to.'})aura:CreateToggle({Name='Contact damage',Default=true,Tooltip=
"Also fire the weapon's touch events. Needed by most melee "..'weapons; requires firetouchinterest.'})aura:CreateToggle({Name='Hit through walls',Default=true,Tooltip=
'Fire contact damage even when the target is behind '..'geometry. Independent of the wall check, which decides '..'whether the target is picked at all.'})aura:CreateToggle({Name='Target NPCs',Default=
false,Tooltip='Include non-player humanoid models in the reach scan.'})aura:CreateTextBox({Name='Ignore players',Default='',Tooltip='Names, display names or user ids, separated by commas. '..
[[They are skipped by targeting: never picked, rotated to or hit.]]})aura:CreateToggle({Name='Highlight targets',Default=false})aura:CreateColor({Name='Highlight colour',Show={Option=
'Highlight targets'},Default=Color3.fromRGB(235,110,110)})aura:CreateNote([[Servers may enforce their own rate and reach rules. Reach is the ]]..'only distance this card reads; there is no '..
'second, longer one. Friend List protects players from this '..'module in every game.')activeCleanup=function():()if aura.Enabled then aura:Toggle(false)else clearHighlights()end end Module.
Initialized=true return aura end function Module.destroy():()if activeCleanup then activeCleanup()end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Movement/AirWalk.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='Air Walk',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function
Module.init(context:Runtime):any local host:any=context.host local framework:any=context.framework local LocalPlayer:Player=host.LocalPlayer local UserInputService:UserInputService=host.
UserInputService local RunService:RunService=host.RunService local world:Workspace=host.workspace local create:any=host.create local platform:Part?=nil local gui:Frame?=nil local airY:number?=nil
local trackedCharacter:Model?=nil local connections:{RBXScriptConnection}={}local held:{[string]:{[any]:boolean}}={Up={},Down={}}local VERTICAL_SPEED:number=22 local function characterParts():(Model?,
Humanoid?,BasePart?)local character:Model?=LocalPlayer.Character if not character then return nil,nil,nil end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')::Humanoid?local root:
BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?return character,humanoid,root end local function footOffsetOf(humanoid:Humanoid,root:BasePart):number local hip:number=humanoid.
HipHeight if hip<=0 then hip=2 end return hip+root.Size.Y*0.5+0.5 end local function destroyPlatform():()if platform then platform:Destroy()platform=nil end end local function destroyGui():()if gui
then gui:Destroy()gui=nil end end local function clearConnections():()for _,conn:RBXScriptConnection in ipairs(connections)do conn:Disconnect()end table.clear(connections)table.clear(held.Up)table.
clear(held.Down)end local function setHold(direction:string,source:any,isDown:boolean):()local bucket:{[any]:boolean}?=held[direction]if not bucket then return end bucket[source]=isDown and true or
nil end local function isDirectionHeld(direction:string):boolean local bucket:{[any]:boolean}?=held[direction]return bucket~=nil and next(bucket)~=nil end local function ensurePlatform():Part if
platform and platform.Parent then return platform end local p:Part=Instance.new('Part')p.Name='Wurst_AirWalkPlatform'p.Anchored=true p.CanCollide=true p.CanQuery=false p.CanTouch=false p.CastShadow=
false p.Size=Vector3.new(16,1,16)p.Transparency=1 p.Parent=world platform=p return p end local airWalk:any airWalk=framework.Categories.Exploits:CreateModule({Name='Air Walk',Category='Exploits',
ConfigKey='Universal.AirWalk',Order=11,Tooltip=[[Stay in the air with every jump and walk in mid-air. Space to go up, Shift to go down, with mobile Up/Down buttons.]],Function=function(enabled:boolean
):()clearConnections()destroyGui()destroyPlatform()airY=nil trackedCharacter=nil if not enabled then airWalk:SetStatus(nil)return end airWalk:SetStatus('Active')local function connect(signal:any,
callback:any):()table.insert(connections,signal:Connect(callback))end connect(UserInputService.InputBegan,function(event:InputObject,processed:boolean):()if processed or UserInputService:
GetFocusedTextBox()then return end if event.UserInputType==Enum.UserInputType.Keyboard then if event.KeyCode==Enum.KeyCode.Space then setHold('Up','Space',true)elseif event.KeyCode==Enum.KeyCode.
LeftShift or event.KeyCode==Enum.KeyCode.RightShift then setHold('Down',event.KeyCode,true)end end end)connect(UserInputService.InputEnded,function(event:InputObject):()if event.UserInputType==Enum.
UserInputType.Keyboard then if event.KeyCode==Enum.KeyCode.Space then setHold('Up','Space',false)elseif event.KeyCode==Enum.KeyCode.LeftShift or event.KeyCode==Enum.KeyCode.RightShift then setHold(
'Down',event.KeyCode,false)end elseif event.UserInputType==Enum.UserInputType.Touch then setHold('Up',event,false)setHold('Down',event,false)end end)if UserInputService.WindowFocusReleased then
connect(UserInputService.WindowFocusReleased,function():()table.clear(held.Up)table.clear(held.Down)end)end if UserInputService.TextBoxFocused then connect(UserInputService.TextBoxFocused,function():(
)table.clear(held.Up)table.clear(held.Down)end)end if UserInputService.TouchEnabled and type(create)=='function'and host.ScreenGui then gui=create('Frame',{Name='AirWalkControls',Parent=host.ScreenGui
,Size=UDim2.fromScale(1,1),BackgroundTransparency=1,ZIndex=30})local function makeMobileButton(name:string,text:string,at:UDim2,size:UDim2):any local object:any=create('TextButton',{Name=name,Parent=
gui,Position=at,Size=size,Text=text,Font=Enum.Font.Code,TextSize=18,TextColor3=Color3.new(1,1,1),BackgroundColor3=Color3.fromRGB(25,25,25),BackgroundTransparency=0.3,BorderSizePixel=1,BorderColor3=
Color3.fromRGB(120,120,120),AutoButtonColor=true,ZIndex=31})if host.ThemeEngine then host.ThemeEngine.Bind(object,'BackgroundColor3','background')host.ThemeEngine.Bind(object,'TextColor3','text')end
return object end for _,name:string in ipairs({'Up','Down'})do local btn:any=makeMobileButton(name,if name=='Up'then'+'else'\u{2212}',UDim2.new(1,-70,1,if name=='Up'then-152 else-96),UDim2.fromOffset(
48,48))connect(btn.InputBegan,function(event:InputObject):()if event.UserInputType==Enum.UserInputType.Touch or event.UserInputType==Enum.UserInputType.MouseButton1 then setHold(name,event,true)end
end)connect(btn.InputEnded,function(event:InputObject):()if event.UserInputType==Enum.UserInputType.Touch or event.UserInputType==Enum.UserInputType.MouseButton1 then setHold(name,event,false)end end)
end end connect(RunService.Heartbeat,function(dt:number):()local character:Model?,humanoid:Humanoid?,root:BasePart?=characterParts()if not character or not humanoid or not root or humanoid.Health<=0
then destroyPlatform()airY=nil trackedCharacter=nil return end local stepDt:number=if type(dt)=='number'and dt>0 and dt<0.5 then dt else(1/60)local offset:number=footOffsetOf(humanoid,root)local
currentFootY:number=root.Position.Y-offset if trackedCharacter~=character or airY==nil then trackedCharacter=character airY=currentFootY end local keyUp:boolean=false local keyDown:boolean=false if
not UserInputService:GetFocusedTextBox()then keyUp=UserInputService:IsKeyDown(Enum.KeyCode.Space)keyDown=UserInputService:IsKeyDown(Enum.KeyCode.LeftShift)or UserInputService:IsKeyDown(Enum.KeyCode.
RightShift)end local goingUp:boolean=keyUp or isDirectionHeld('Up')local goingDown:boolean=keyDown or isDirectionHeld('Down')local part:Part=ensurePlatform()if goingDown and not goingUp then airY=
currentFootY-VERTICAL_SPEED*stepDt-1.4 local vel:Vector3=root.AssemblyLinearVelocity if vel.Y>-VERTICAL_SPEED then root.AssemblyLinearVelocity=Vector3.new(vel.X,-VERTICAL_SPEED,vel.Z)end elseif
goingUp and not goingDown then airY=math.max(airY::number,currentFootY)+VERTICAL_SPEED*stepDt local vel:Vector3=root.AssemblyLinearVelocity if vel.Y<VERTICAL_SPEED then root.AssemblyLinearVelocity=
Vector3.new(vel.X,VERTICAL_SPEED,vel.Z)end else if currentFootY>(airY::number)then airY=currentFootY end end if not part.CanCollide then part.CanCollide=true end part.CFrame=CFrame.new(root.Position.X
,airY::number,root.Position.Z)end)airWalk:Clean(function():()clearConnections()destroyGui()destroyPlatform()airY=nil trackedCharacter=nil end)end})activeCleanup=function():()clearConnections()
destroyGui()destroyPlatform()if airWalk and airWalk.Enabled then airWalk:Toggle(false)end end Module.Initialized=true return airWalk end function Module.destroy():()if activeCleanup then
activeCleanup()end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Movement/AntiFling.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='AntiFling',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function
Module.init(context:Runtime):any local host:any=context.host local services:any=context.services local activity:any=services and services.activity local framework:any=context.framework local players:
any=host.Players local localPlayer:any=host.LocalPlayer local world:any=host.workspace or workspace local featureConnections:any=host.featureConnections local disconnect:any=host.
disconnectFeatureConnection local enabled:boolean=false local playerShield:boolean=true local impactShield:boolean=true local feature:any=nil local character:Model?=nil local links:{[BasePart]:{[
BasePart]:any}}={}local hazardSeen:{[BasePart]:number}={}local dangerousUntil:{[Model]:number}={}local nextScan:number=0 local dangerUntil:number=0 local suspended:boolean=false local warnedOverlap:
boolean=false local SCAN_INTERVAL:number=0.15 local RADIUS:number=30 local HAZARD_HOLD:number=0.75 local LINEAR_LIMIT:number=80 local SURFACE_LIMIT:number=70 local MAX_LINKS:number=2048 local
MAX_PARTS:number=200 local function isBusy():boolean return activity~=nil and type(activity.isActive)=='function'and(activity.isActive('fling')or activity.isActive('walkfling'))end local function
destroyLinks():()for _,perLocal:{[BasePart]:any}in pairs(links)do for _,constraint:any in pairs(perLocal)do pcall(function():()constraint:Destroy()end)end end table.clear(links)table.clear(hazardSeen)
table.clear(dangerousUntil)dangerUntil=0 end local function isToolPart(part:Instance):boolean local ok:boolean,result:any=pcall(function():any return part:FindFirstAncestorOfClass('Tool')end)return ok
and result~=nil end local function listBodyParts(model:Model?,onlyCollidable:boolean):{BasePart}local parts:{BasePart}={}if not model or not model.Parent then return parts end local ok:boolean,
descendants:any=pcall(function():any return model:GetDescendants()end)if not ok or type(descendants)~='table'then return parts end for _,object:Instance in ipairs(descendants)do if object:IsA(
'BasePart')and not isToolPart(object)then local part:BasePart=object::BasePart if not onlyCollidable or part.CanCollide then table.insert(parts,part)end end end return parts end local function
fastPart(part:BasePart,scale:number?):boolean local ok:boolean,fast:any=pcall(function():boolean local linear:Vector3=part.AssemblyLinearVelocity local angular:Vector3=part.AssemblyAngularVelocity
local radius:number=math.min(part.Size.Magnitude*0.5,20)local multiplier:number=scale or 1 return linear.Magnitude>LINEAR_LIMIT*multiplier or angular.Magnitude*radius>SURFACE_LIMIT*multiplier end)
return ok and fast==true end local function supportAssembly(root:BasePart,humanoid:Humanoid,owner:Model):BasePart?local ok:boolean,result:any=pcall(function():any local params:RaycastParams=
RaycastParams.new()params.FilterType=Enum.RaycastFilterType.Exclude params.FilterDescendantsInstances={owner}params.IgnoreWater=true params.RespectCanCollide=true local distance:number=math.max(4,(
tonumber(humanoid.HipHeight)or 2)+root.Size.Y*0.5+2)return world:Raycast(root.Position,Vector3.new(0,-distance,0),params)end)local ground:any=ok and result and result.Instance if ground and ground:
IsA('BasePart')then return ground.AssemblyRootPart or ground end return nil end local function synchronize(owner:Model,humanoid:Humanoid,root:BasePart,now:number):()local ownParts:{BasePart}=
listBodyParts(owner,true)local desired:{[BasePart]:boolean}={}local candidates:{any}={}local characterModels:{[Model]:boolean}={}local roster:any=players:GetPlayers()for _,player:Player in ipairs(
roster)do local other:Model?=player~=localPlayer and player.Character or nil if other and other~=owner and other.Parent then characterModels[other]=true if playerShield then local otherRoot:BasePart?=
other:FindFirstChild('HumanoidRootPart')::BasePart?local distance:number=otherRoot and(otherRoot.Position-root.Position).Magnitude or math.huge if distance<=RADIUS then local parts:{BasePart}=
listBodyParts(other,false)local hazard:boolean=false for _,part:BasePart in ipairs(parts)do if part.CanCollide and fastPart(part,nil)then hazard=true break end end if hazard then dangerousUntil[other]
=now+HAZARD_HOLD dangerUntil=math.max(dangerUntil,now+HAZARD_HOLD)end end end end end if playerShield then for model:Model,expiresAt:number in pairs(dangerousUntil)do if expiresAt<=now or not model.
Parent then dangerousUntil[model]=nil end end for model:Model in pairs(dangerousUntil)do local otherRoot:BasePart?=model:FindFirstChild('HumanoidRootPart')::BasePart?local distance:number=otherRoot
and(otherRoot.Position-root.Position).Magnitude or math.huge local parts:{BasePart}=listBodyParts(model,true)table.insert(candidates,{model=model,distance=distance,parts=parts})for _,part:BasePart in
ipairs(parts)do desired[part]=true end end end table.sort(candidates,function(a:any,b:any):boolean return a.distance<b.distance end)if impactShield and not humanoid.Sit and not humanoid.SeatPart then
local overlap:any=nil pcall(function():()overlap=OverlapParams.new()overlap.FilterType=Enum.RaycastFilterType.Exclude overlap.FilterDescendantsInstances={owner}overlap.MaxParts=MAX_PARTS overlap.
RespectCanCollide=true end)local ok:boolean,nearby:any=pcall(function():any return world:GetPartBoundsInRadius(root.Position,RADIUS,overlap)end)if not ok or type(nearby)~='table'then if not
warnedOverlap and type(host.notify)=='function'then warnedOverlap=true host.notify([[Anti Fling: moving-object scan unavailable; player protection remains active.]])end else local floorAssembly:
BasePart?=supportAssembly(root,humanoid,owner)for _,instance:Instance in ipairs(nearby)do if instance:IsA('BasePart')then local part:BasePart=instance::BasePart local isCharacterPart:boolean=false
local ancestor:Instance?=part.Parent while ancestor and ancestor~=world do if characterModels[ancestor::any]then isCharacterPart=true break end ancestor=ancestor.Parent end local inOwner:boolean=part:
IsDescendantOf(owner)local isTool:boolean=isToolPart(part)local assembly:BasePart?=part.AssemblyRootPart or part if part.Parent and part.CanCollide and not part.Anchored and not inOwner and(not
isCharacterPart or isTool)and(not floorAssembly or assembly~=floorAssembly)and fastPart(part,if isTool then 3 else nil)then hazardSeen[part]=now dangerUntil=math.max(dangerUntil,now+HAZARD_HOLD)end
end end end for part:BasePart,seen:number in pairs(hazardSeen)do if not part.Parent or now-seen>HAZARD_HOLD then hazardSeen[part]=nil else desired[part]=true end end else table.clear(hazardSeen)end
local ordered:{BasePart}={}local included:{[BasePart]:boolean}={}local function include(part:BasePart):()if desired[part]and part.Parent and not included[part]then included[part]=true table.insert(
ordered,part)end end local hazards:{BasePart}={}for part:BasePart in pairs(hazardSeen)do table.insert(hazards,part)end table.sort(hazards,function(a:BasePart,b:BasePart):boolean return(a.Position-root
.Position).Magnitude<(b.Position-root.Position).Magnitude end)for _,part:BasePart in ipairs(hazards)do include(part)end if playerShield then for _,candidate:any in ipairs(candidates)do for _,part:
BasePart in ipairs(candidate.parts)do include(part)end end end local accepted:{[BasePart]:{[BasePart]:boolean}}={}local allocated:number=0 for _,other:BasePart in ipairs(ordered)do if allocated>=
MAX_LINKS then break end local perLocal:{[BasePart]:boolean}={}for _,own:BasePart in ipairs(ownParts)do if allocated>=MAX_LINKS then break end if own~=other then perLocal[own]=true allocated+=1 end
end if next(perLocal)then accepted[other]=perLocal end end for other:BasePart,perLocal:{[BasePart]:any}in pairs(links)do local wanted:{[BasePart]:boolean}?=accepted[other]for own:BasePart,constraint:
any in pairs(perLocal)do if not wanted or not wanted[own]or not own.Parent or not other.Parent then pcall(function():()constraint:Destroy()end)perLocal[own]=nil end end if next(perLocal)==nil then
links[other]=nil end end for other:BasePart,wanted:{[BasePart]:boolean}in pairs(accepted)do local perLocal:{[BasePart]:any}=links[other]or{}links[other]=perLocal for own:BasePart in pairs(wanted)do if
not perLocal[own]then local ok:boolean,constraint:any=pcall(function():any local link:NoCollisionConstraint=Instance.new('NoCollisionConstraint')link.Part0,link.Part1=own,other link.Parent=own return
link end)if ok and constraint then perLocal[own]=constraint end end end if next(perLocal)==nil then links[other]=nil end end end local function step():()if not enabled then return end local owner:
Model?,humanoid:Humanoid?,root:BasePart?=host.getCharacterParts()if owner~=character then destroyLinks()character=owner nextScan=0 end if not owner or not humanoid or not root or not owner.Parent or
humanoid.Health<=0 then destroyLinks()return end if isBusy()then if not suspended then destroyLinks()end suspended=true return end if suspended then suspended=false nextScan=0 end local now:number=os.
clock()if now>=nextScan then nextScan=now+SCAN_INTERVAL local ok:boolean,err:any=pcall(synchronize,owner,humanoid,root,now)if not ok then destroyLinks()if type(host.warn)=='function'then host.warn(
'Anti Fling: '..tostring(err))end end end if now>dangerUntil or root.Anchored or humanoid.Sit or humanoid.SeatPart then return end local walkSpeed:number=math.max(16,tonumber(humanoid.WalkSpeed)or 16)
local speed:any=framework and(framework:GetModule('Speed Settings')or framework:GetModule('Speed Hack'))if speed and speed.Enabled and speed.Options.Speed then walkSpeed=math.max(walkSpeed,tonumber(
speed.Options.Speed.Value)or 0)end local jumpPower:number=math.max(50,tonumber(humanoid.JumpPower)or 50)local velocity:Vector3=root.AssemblyLinearVelocity local planar:Vector3=Vector3.new(velocity.X,0
,velocity.Z)local limit:number=math.max(120,walkSpeed*4)local riseLimit:number=math.max(180,jumpPower*4)local nextVelocity:Vector3=velocity local changed:boolean=false if planar.Magnitude>limit then
local move:Vector3=humanoid.MoveDirection local direction:Vector3=move.Magnitude>0.05 and move.Unit*walkSpeed or Vector3.zero nextVelocity=Vector3.new(direction.X,nextVelocity.Y,direction.Z)changed=
true end if nextVelocity.Y>riseLimit then nextVelocity=Vector3.new(nextVelocity.X,riseLimit,nextVelocity.Z)changed=true end pcall(function():()if root.AssemblyAngularVelocity.Magnitude>25 then root.
AssemblyAngularVelocity=Vector3.zero end if changed then root.AssemblyLinearVelocity=nextVelocity end end)end local function toggleAntiFling(value:boolean):()enabled=value==true if host.state then
host.state.antiFlingActive=enabled end disconnect('AntiFling')destroyLinks()character=nil nextScan=0 suspended=false if enabled then featureConnections.AntiFling=host.TaskManager:Connect(step,nil,
'Anti Fling')end end feature=host.createUniversalFeature('Anti Fling',[[Collision Shield: local player and extreme-object protection]],2,toggleAntiFling,{categoryName='Utilities'})host.
addToggleOption(feature,'Player Shield',true,function(value:boolean):()playerShield=value==true nextScan=0 if not playerShield and not impactShield then destroyLinks()end end,
[[Shield yourself only from bodies that are moving or spinning at fling speeds nearby, for a moment after each hit. Calm players keep normal collisions, so you no longer walk through their bodies.]])
host.addToggleOption(feature,'Impact Shield',true,function(value:boolean):()impactShield=value==true nextScan=0 if not playerShield and not impactShield then destroyLinks()end end,
[[Temporarily avoid nearby extreme-speed loose objects; support platforms, seated vehicles and ordinary tools are left alone.]])host.addFeatureTooltip(feature,
[[Collision Shield only changes local collision pairs. Extremely fast/streamed objects and server-owned physics can still launch you.]])activeCleanup=function():()toggleAntiFling(false)if host.state
then host.state.antiFlingActive=false end end Module.Initialized=true return feature end function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.Initialized=
false end return Module
-- src/games/universal/Movement/ClickTeleport.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='ClickTeleport',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil
local function characterRoot(localPlayer:Player):(BasePart?,Humanoid?)local character:Model?=localPlayer.Character if not character then return nil,nil end return character:FindFirstChild(
'HumanoidRootPart')::BasePart?,character:FindFirstChildOfClass('Humanoid')::Humanoid?end function Module.init(context:Runtime):any local host:any=context.host local framework:any=context.framework
local localPlayer:Player=host.LocalPlayer local userInput:UserInputService=host.UserInputService local menu:any=context.services.menu local runtime:any={armedUntil=0,point=nil::string?,lastOrigin=nil
::Vector3?}local teleport:any teleport=framework.Categories.Utilities:CreateModule({Name='Teleport',ConfigKey='Universal.ClickTeleport',Category='Utilities',Order=1,Kind='action',Silent=true,Tooltip=
'Press to travel to the destination selected below.',Function=function():()Module.run(context,teleport,runtime)end})local destinations={'Tap point','Nearest player','Named player','Item','Waypoint',
'Last position'}if host.GAME_CHECK and host.GAME_CHECK.MM2Active then table.insert(destinations,1,'MM2')local config=host.configData if config then local key='Universal.ClickTeleport.Destination'if
config.values[key]==nil or string.sub(tostring(config.values[key]),1,4)=='MM2 'then config.values[key]='MM2'end end end teleport:CreateDropdown({Name='Destination',List=destinations,Index=1,Tooltip=
[[MM2 shows all game teleport buttons together. Other destinations use the main action.]],Function=function(value)runtime.armedUntil=0 if runtime.cancelTravel then runtime.cancelTravel()end local
mm2Controls={['Teleport to Map']=true,['Teleport to Lobby']=true,['Teleport to Player']=true,['Get Gun']=true,['Player name']=true,['Get Gun key']=true,['Auto Get Gun']=true}teleport.OptionFilter=
function(name)if name=='Destination'then return true end return value=='MM2'and mm2Controls[name]==true or value~='MM2'and not mm2Controls[name]end teleport:RefreshVisibility()end})teleport:
CreateTextBox({Name='Player',Show={Option='Destination',Values={'Named player'}},Default='',Tooltip='Part of a name or display name, for Named player.'})teleport:CreateTextBox({Name='Item',Show={
Option='Destination',Values={'Item'}},Default='',Tooltip='Part or model name to look for, for Item. Partial names '..'match, so "chest" finds "GoldChest".'})local pointList=teleport:CreateList({Name=
'Point',Single=true,Show={Option='Destination',Values={'Waypoint'}},EmptyText='No points. Save one in Markers.',Items=function()local points=context.services.points return points and points.Names()or{
}end,Function=function(_,names)runtime.point=names[1]end,Tooltip=[[The same saved points as Markers. Choose one, then Teleport to point.]]})teleport:CreateButton({Name='Teleport to point',Show={Option
='Destination',Values={'Waypoint'}},Function=function()Module.run(context,teleport,runtime)end})local unsubscribe if context.services.points then unsubscribe=context.services.points.Subscribe(function
()if runtime.point and not context.services.points.Get(runtime.point)then if runtime.cancelTravel then runtime.cancelTravel()end pointList:Set(runtime.point,false)end pointList:Refresh()end)end
teleport:CreateSlider({Name='Arm window',Show={Option='Destination',Values={'Tap point'}},Min=1,Max=15,Default=6,Tooltip='Seconds Tap point stays armed after the button is pressed.'})teleport:
CreateSlider({Name='Search range',Show={Option='Destination',Values={'Nearest player','Named player','Item'}},Min=50,Max=5000,Default=1500,Tooltip='How far to look for items and players.'})teleport:
CreateSlider({Name='Player offset',Show={Option='Destination',Values={'Nearest player','Named player'}},Min=0,Max=20,Default=4,Tooltip='Studs to stop short of a player, so you do not land inside '..
'them.'})teleport:CreateToggle({Name='Skip friends',Show={Option='Destination',Values={'Nearest player'}},Default=false,Tooltip='Nearest player ignores anyone on the Friend List.'})teleport:
CreateDropdown({Name='Movement',List={'Instant','Glide'},Index=1,Tooltip=
[[Instant jumps to the destination. Glide moves continuously at Glide speed (studs/sec), regardless of frame rate. A new trip cancels the previous one.]]})teleport:CreateSlider({Name='Glide speed',
Show={Option='Movement',Values={'Glide'}},Min=1,Max=500,Default=60,Tooltip=[[Travel speed in studs per second, not distance per frame. Higher = faster.]]})teleport:CreateSlider({Name='Height offset',
Min=0,Max=12,Default=3,Tooltip='Studs above whatever you land on.'})teleport:CreateToggle({Name='Keep momentum',Default=false,Tooltip='Leave your velocity alone on arrival instead of zeroing it.'})
local tapConnection:RBXScriptConnection=userInput.InputBegan:Connect(function(input:InputObject,processed:boolean):()if processed or os.clock()>=runtime.armedUntil then return end if menu.
isCapturingInput()then return end local touched:boolean=input.UserInputType==Enum.UserInputType.Touch local clicked:boolean=input.UserInputType==Enum.UserInputType.MouseButton1 if not touched and not
clicked then return end runtime.armedUntil=0 Module.travel(context,teleport,runtime,Module.pointUnder(context,input.Position),true)end)activeCleanup=function():()pcall(function():()tapConnection:
Disconnect()end)runtime.armedUntil=0 if runtime.cancelTravel then runtime.cancelTravel()end if unsubscribe then unsubscribe()end end Module.Initialized=true return teleport end function Module.
pointUnder(context:Runtime,screenPosition:Vector3):Vector3?local host:any=context.host local currentWorkspace:Workspace=host.workspace or workspace local camera:Camera?=currentWorkspace.CurrentCamera
if not camera then return nil end local resolvedCamera:Camera=camera::Camera local ray:Ray=resolvedCamera:ScreenPointToRay(screenPosition.X,screenPosition.Y)local parameters:RaycastParams=
RaycastParams.new()parameters.FilterType=Enum.RaycastFilterType.Exclude parameters.IgnoreWater=true parameters.FilterDescendantsInstances={host.LocalPlayer.Character::any,resolvedCamera}local result:
RaycastResult?=currentWorkspace:Raycast(ray.Origin,ray.Direction.Unit*4096,parameters)return result and result.Position or nil end local function matchesName(candidate:string,query:string):boolean
return string.find(string.lower(candidate),string.lower(query),1,true)~=nil end function Module.findItem(context:Runtime,query:string,origin:Vector3,range:number):Vector3?local host:any=context.host
local currentWorkspace:Workspace=host.workspace or workspace local best:Vector3?=nil local bestDistance:number=range local character:Model?=host.LocalPlayer.Character for _,descendant:Instance in
ipairs(currentWorkspace:GetDescendants())do if character and descendant:IsDescendantOf(character)then continue end if not matchesName(descendant.Name,query)then continue end local position:Vector3?=
nil if descendant:IsA('BasePart')then position=(descendant::BasePart).Position elseif descendant:IsA('Model')then local pivotOk:boolean,pivot:any=pcall(function():CFrame return(descendant::Model):
GetPivot()end)if pivotOk then position=(pivot::CFrame).Position end end if not position then continue end local distance:number=((position::Vector3)-origin).Magnitude if distance<bestDistance then
best=position bestDistance=distance end end return best end function Module.run(context:Runtime,teleport:any,runtime:any):()if runtime.cancelTravel then runtime.cancelTravel()end if teleport.Options.
Destination.Value=='MM2'then teleport:Notify('Choose a MM2 teleport button below')return end local host:any=context.host local entityLibrary:any=context.entity local localPlayer:Player=host.
LocalPlayer local options:any=teleport.Options local destination:string=options['Destination'].Value local root:BasePart?=characterRoot(localPlayer)if not root then teleport:Notify('no character')
return end local origin:Vector3=(root::BasePart).Position local range:number=options['Search range'].Value if destination=='Tap point'then runtime.armedUntil=os.clock()+options['Arm window'].Value
teleport:Notify('tap where you want to go')return end if destination=='Waypoint'then local points=context.services.points local position,surface if points and runtime.point then position,surface=
points.Get(runtime.point)end if not position then teleport:Notify('Choose a saved point from Markers first.')return end Module.travel(context,teleport,runtime,position,surface==true)return end if
destination=='Last position'then if not runtime.lastOrigin then teleport:Notify('nowhere to go back to')return end Module.travel(context,teleport,runtime,runtime.lastOrigin,false)return end if
destination=='Item'then local query:string=tostring(options['Item'].Value or'')if query==''then teleport:Notify('type an item name first')return end local found:Vector3?=Module.findItem(context,query,
origin,range)if not found then teleport:Notify('no "'..query..'" within range')return end Module.travel(context,teleport,runtime,found,true)return end entityLibrary:Refresh()local target:any=nil if
destination=='Nearest player'then local bestDistance:number=range for _,entity:any in ipairs(entityLibrary.List)do if options['Skip friends'].Value and entityLibrary:IsProtected(entity.Player)then
continue end if entity.Distance<bestDistance then target=entity bestDistance=entity.Distance end end else local query:string=tostring(options['Player'].Value or'')if query==''then teleport:Notify(
'type a player name first')return end for _,entity:any in ipairs(entityLibrary.List)do if matchesName(entity.Player.Name,query)or matchesName(entity.Player.DisplayName,query)then target=entity break
end end end if not target then teleport:Notify('no matching player')return end local targetPosition:Vector3=target.RootPart.Position local offset:number=options['Player offset'].Value local approach:
Vector3=targetPosition-origin if offset>0 and approach.Magnitude>offset then targetPosition=targetPosition-approach.Unit*offset end Module.travel(context,teleport,runtime,targetPosition,false)end
function Module.travel(context:Runtime,teleport:any,runtime:any,destination:Vector3?,liftToSurface:boolean):()local host:any=context.host local localPlayer:Player=host.LocalPlayer if runtime.
cancelTravel then runtime.cancelTravel()end if not destination then teleport:Notify('nothing there')return end local root:BasePart?,humanoid:Humanoid?=characterRoot(localPlayer)if not root then return
end local resolvedRoot:BasePart=root::BasePart local options:any=teleport.Options runtime.lastOrigin=resolvedRoot.Position local target:Vector3=destination::Vector3 if liftToSurface then local
hipHeight:number=humanoid and(humanoid::Humanoid).HipHeight or 2 target=target+Vector3.new(0,hipHeight+options['Height offset'].Value,0)end local keepMomentum:boolean=options['Keep momentum'].Value if
options['Movement'].Value=='Instant'then resolvedRoot.CFrame=resolvedRoot.CFrame+(target-resolvedRoot.Position)if not keepMomentum then resolvedRoot.AssemblyLinearVelocity=Vector3.zero resolvedRoot.
AssemblyAngularVelocity=Vector3.zero end return end local speed=math.max(0,options['Glide speed'].Value)if speed==0 then teleport:Notify('Glide speed must be greater than zero.')return end local
character=localPlayer.Character local connection local function finish()if connection then connection:Disconnect()connection=nil end runtime.cancelTravel=nil if context.services.activity then context.
services.activity.set('teleport',false)end teleport:SetStatus(nil)end if runtime.cancelTravel then runtime.cancelTravel()end runtime.cancelTravel=finish if context.services.activity then context.
services.activity.set('teleport',true)end teleport:SetStatus('Gliding')connection=host.RunService.Heartbeat:Connect(function(dt)if localPlayer.Character~=character or not resolvedRoot.Parent or
characterRoot(localPlayer)~=resolvedRoot or humanoid and humanoid.Health<=0 then finish()return end local offset=target-resolvedRoot.Position local distance=offset.Magnitude local amount=math.min(
distance,speed*math.clamp(dt,0,0.1))if distance<=amount+0.001 then resolvedRoot.CFrame=resolvedRoot.CFrame+(target-resolvedRoot.Position)if not keepMomentum then resolvedRoot.AssemblyLinearVelocity=
Vector3.zero resolvedRoot.AssemblyAngularVelocity=Vector3.zero end finish()return end if distance>0 then resolvedRoot.CFrame+=offset.Unit*amount end if not keepMomentum then resolvedRoot.
AssemblyLinearVelocity=Vector3.zero resolvedRoot.AssemblyAngularVelocity=Vector3.zero end end)end function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.
Initialized=false end return Module
-- src/games/universal/Movement/FreezeMovements.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='FreezeMovements',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.init
(context:Runtime):any local host:any=context.host local TaskManager:any=host.TaskManager local featureConnections:any=host.featureConnections local disconnectFeatureConnection:any=host.
disconnectFeatureConnection local getCharacterParts:any=host.getCharacterParts local createUniversalFeature:any=host.createUniversalFeature local addCycleOption:any=host.addCycleOption local
addToggleOption:any=host.addToggleOption local addNumberOption:any=host.addNumberOption local LOCK_MODE:string='Position Lock'local MOTION_MODE:string='Motion'local freezeSettings={mode=LOCK_MODE,
emotes=true,animation=false,anyEmote=true,lockRotation=true,returnDistance=0,keepGravity=false}local config:any=host.configData if config and config.values and config.states then local key:string=
'Movement.FreezeMovements.Freezemode'local legacy:any=config.values[key]if legacy=='Freeze Emotes'or legacy=='Glitch Animations'then config.values[key]=MOTION_MODE local prefix:string=
'Movement.FreezeMovements.'if config.states[prefix..'FreezeEmotes']==nil then config.states[prefix..'FreezeEmotes']=legacy=='Freeze Emotes'end if config.states[prefix..'FreezeAnimation']==nil then
config.states[prefix..'FreezeAnimation']=legacy=='Glitch Animations'end elseif legacy=='Anchor'or legacy=='Freeze Physics'then config.values[key]=LOCK_MODE end end local FreezeFeature:any=nil local
originalFreezeState=setmetatable({},{__mode='k'})local stickyEmoteTracks:{any}={}local emoteLastSeen:{[any]:number}=setmetatable({},{__mode='k'})local function getAnimator(humanoid:any):any local
animator:any=humanoid and humanoid:FindFirstChildOfClass('Animator')if not animator and humanoid and humanoid.Parent then local character:any=humanoid.Parent animator=character and character:
FindFirstChildWhichIsA('Animator',true)end return animator end local emoteConnection:any=nil local catalogEmoteConnection:any=nil local emoteGeneration:number=0 local watchedHumanoid:any=nil local
watchedAnimator:any=nil local watchedAnimate:any=nil local nextMetadataRefresh:number=0 local freezeEnabled:boolean=false local resolvedAssets:{[number]:boolean}={}local assetAttempts:{[number]:number
}={}local pendingEmoteTracks:{any}={}local lastMetadataRefresh:number=-math.huge local emoteNames:{[string]:boolean}={}local emoteIds:{[string]:boolean}={}local managedEmotes:{[any]:any}={}local
ownedEmotes:{[any]:boolean}=setmetatable({},{__mode='k'})local isEmoteTrack:(any)->boolean local function clearStickyEmoteTracks():()emoteGeneration+=1 if emoteConnection then emoteConnection:
Disconnect()end if catalogEmoteConnection then catalogEmoteConnection:Disconnect()end emoteConnection=nil catalogEmoteConnection=nil local previousHumanoid:any=watchedHumanoid watchedHumanoid=nil
watchedAnimator=nil watchedAnimate=nil resolvedAssets={}assetAttempts={}pendingEmoteTracks={}local stopped:{[any]:boolean}={}local function stopOnce(track:any):()if track and not stopped[track]then
stopped[track]=true pcall(function():()track:Stop(0.1)end)end end for _,source:any in ipairs(stickyEmoteTracks)do stopOnce(source)end local animator:any=getAnimator(previousHumanoid)if animator then
pcall(function():()for _,track:any in ipairs(animator:GetPlayingAnimationTracks())do if track.IsPlaying and not ownedEmotes[track]and isEmoteTrack(track)then stopOnce(track)end end end)end for _,track
:any in pairs(managedEmotes)do stopOnce(track)pcall(function():()track:Destroy()end)end managedEmotes={}ownedEmotes=setmetatable({},{__mode='k'})emoteNames={}emoteIds={}stickyEmoteTracks={}
emoteLastSeen=setmetatable({},{__mode='k'})end local CLASSIC_EMOTES={dance=true,dance1=true,dance2=true,dance3=true,wave=true,point=true,cheer=true,laugh=true}local EMOTE_KEYWORDS={'emote','dance',
'dancing','taunt','celebrat','cheer','wave','laugh','salute','groove','pose','victory','expression','emoji','flex','party','clap','bow','ridicule','point'}local NON_EMOTE_KEYWORDS={'attack','shoot',
'slash','stab','swing','punch','kick','reload','equip','unequip','tool','hit','block','parry','cast','spell','sword','gun','knife','punch','throw','fling'}local function hasKeyword(name:string,list:{
string}):boolean local lower:string=string.lower(name)for _,keyword:string in ipairs(list)do if string.find(lower,keyword,1,true)then return true end end return false end local function
looksLikeEmoteName(name:string):boolean local lower:string=string.lower(name)if emoteNames[lower]then return true end if hasKeyword(lower,NON_EMOTE_KEYWORDS)then return false end lower=lower:gsub(
'anim$','')if CLASSIC_EMOTES[lower]==true then return true end return hasKeyword(lower,EMOTE_KEYWORDS)end isEmoteTrack=function(track:any):boolean if looksLikeEmoteName(tostring(track.Name or''))then
return true end local animation:any=track.Animation if typeof(animation)=='Instance'then if looksLikeEmoteName(animation.Name)then return true end local node:any=animation.Parent for _depth:number=1,6
do if not node then break end if looksLikeEmoteName(node.Name)then return true end node=node.Parent end local id:string?=string.match(animation.AnimationId,'%d+')if id and emoteIds[id]then return true
end end return false end local function looksLikeStandingEmote(humanoid:any,root:any,track:any):boolean if not freezeSettings.anyEmote then return false end if not track or ownedEmotes[track]or
isEmoteTrack(track)then return false end local name:string=tostring(track.Name or'')if hasKeyword(name,NON_EMOTE_KEYWORDS)then return false end local animation:any=track.Animation if typeof(animation)
~='Instance'then return false end if track.Looped~=true then return false end local priority:any=track.Priority if priority~=Enum.AnimationPriority.Action and priority~=Enum.AnimationPriority.Idle
then return false end if root and root.AssemblyLinearVelocity and root.AssemblyLinearVelocity.Magnitude>2 then return false end local ok:boolean,state:any=pcall(function():any return humanoid:
GetState()end)if ok and(state==Enum.HumanoidStateType.Running or state==Enum.HumanoidStateType.Jumping or state==Enum.HumanoidStateType.Freefall)then return false end return true end local function
rememberEmote(track:any):()for source:any,managed:any in pairs(managedEmotes)do pcall(function():()managed:Stop(0.1)end)pcall(function():()managed:Destroy()end)ownedEmotes[managed]=nil managedEmotes[
source]=nil end stickyEmoteTracks={track}emoteLastSeen=setmetatable({},{__mode='k'})emoteLastSeen[track]=os.clock()end local function freezeEmotesTick(humanoid:any,root:any):()local animator:any=
getAnimator(humanoid)if not animator then if watchedAnimator then clearStickyEmoteTracks()end return end local animate=humanoid.Parent and humanoid.Parent:FindFirstChild('Animate')if watchedHumanoid~=
humanoid or watchedAnimator~=animator or watchedAnimate~=animate or os.clock()>=nextMetadataRefresh then if watchedHumanoid~=humanoid or watchedAnimator~=animator then clearStickyEmoteTracks()else if
emoteConnection then emoteConnection:Disconnect()emoteConnection=nil end if catalogEmoteConnection then catalogEmoteConnection:Disconnect()catalogEmoteConnection=nil end end watchedHumanoid=humanoid
watchedAnimator=animator watchedAnimate=animate lastMetadataRefresh=os.clock()nextMetadataRefresh=lastMetadataRefresh+2 pcall(function():()catalogEmoteConnection=humanoid.EmoteTriggered:Connect(
function(success:boolean,track:any):()if success and track and watchedHumanoid==humanoid and not ownedEmotes[track]then local animation:any=track.Animation if animation then local id:string?=string.
match(animation.AnimationId,'%d+')if id then emoteIds[id]=true end end rememberEmote(track)end end)end)local catalogAssets:{number}={}pcall(function():()local description:any=humanoid:
GetAppliedDescription()local ok:boolean,emotes:any=pcall(description.GetEmotes,description)description:Destroy()if not ok then return end local seen:{[number]:boolean}={}for name:string,ids:any in
pairs(emotes)do emoteNames[string.lower(name)]=true for _,id:any in ipairs(ids)do emoteIds[tostring(id)]=true local numberId:number?=tonumber(id)if numberId and not seen[numberId]and not
resolvedAssets[numberId]and#catalogAssets<48 then seen[numberId]=true resolvedAssets[numberId]=true assetAttempts[numberId]=(assetAttempts[numberId]or 0)+1 table.insert(catalogAssets,numberId)end end
end end)local generation:number=emoteGeneration if#catalogAssets>0 and task and game then task.spawn(function():()for _,assetId:number in ipairs(catalogAssets)do if generation~=emoteGeneration then
return end local ok:boolean,objects:any=pcall(function():any return game:GetObjects('rbxassetid://'..tostring(assetId))end)if not ok and generation==emoteGeneration and(assetAttempts[assetId]or 0)<3
then resolvedAssets[assetId]=nil end if ok and type(objects)=='table'then for _,object:any in ipairs(objects)do pcall(function():()local nodes:{any}=object:GetDescendants()table.insert(nodes,object)
for _,node:any in ipairs(nodes)do if generation==emoteGeneration and node:IsA('Animation')then local id:string?=string.match(node.AnimationId,'%d+')if id then emoteIds[id]=true end end end end)pcall(
function():()object:Destroy()end)end end end end)end pcall(function():()local animate:any=humanoid.Parent:FindFirstChild('Animate')if animate then for _,animation:any in ipairs(animate:GetDescendants(
))do if animation:IsA('Animation')then local node:any=animation while node and node~=animate do local name:string=string.lower(node.Name):gsub('anim$','')if CLASSIC_EMOTES[name]or emoteNames[name]then
local id:string?=string.match(animation.AnimationId,'%d+')if id then emoteIds[id]=true end break end node=node.Parent end end end end end)emoteConnection=animator.AnimationPlayed:Connect(function(
track:any):()if not ownedEmotes[track]and isEmoteTrack(track)and not emoteLastSeen[track]then rememberEmote(track)elseif not ownedEmotes[track]and not emoteLastSeen[track]and looksLikeStandingEmote(
humanoid,root,track)then rememberEmote(track)elseif not ownedEmotes[track]then if#pendingEmoteTracks>=8 then table.remove(pendingEmoteTracks,1)end table.insert(pendingEmoteTracks,{track=track,at=os.
clock()})if os.clock()-lastMetadataRefresh>=0.25 then nextMetadataRefresh=0 end end end)end for index=#pendingEmoteTracks,1,-1 do local candidate=pendingEmoteTracks[index]if isEmoteTrack(candidate.
track)then rememberEmote(candidate.track)table.remove(pendingEmoteTracks,index)elseif os.clock()-candidate.at>5 then table.remove(pendingEmoteTracks,index)end end local okTracks:boolean,tracks:any=
pcall(function():any return animator:GetPlayingAnimationTracks()end)if not okTracks or type(tracks)~='table'then return end local playing:{[any]:boolean}={}for _,track:any in ipairs(tracks)do if track
and track.IsPlaying then playing[track]=true if not ownedEmotes[track]and not emoteLastSeen[track]and isEmoteTrack(track)then rememberEmote(track)end if emoteLastSeen[track]then emoteLastSeen[track]=
os.clock()end end end local moving:boolean=root~=nil and root.AssemblyLinearVelocity.Magnitude>1 local now:number=os.clock()for index:number=#stickyEmoteTracks,1,-1 do local track:any=
stickyEmoteTracks[index]if not track then table.remove(stickyEmoteTracks,index)elseif playing[track]or(managedEmotes[track]and managedEmotes[track].IsPlaying)then emoteLastSeen[track]=now else local
lastSeen:number=emoteLastSeen[track]if lastSeen==nil then lastSeen=now emoteLastSeen[track]=now end if moving and now-lastSeen<1.5 then pcall(function():()local managed:any=managedEmotes[track]if not
managed then managed=animator:LoadAnimation(track.Animation)ownedEmotes[managed]=true managedEmotes[track]=managed managed.Looped=true managed.Priority=Enum.AnimationPriority.Action end if not managed
.IsPlaying then managed:Play(0.1)end end)elseif now-lastSeen>0.5 then local managed:any=managedEmotes[track]if managed then pcall(function():()managed:Stop(0.1)end)pcall(function():()managed:Destroy()
end)managedEmotes[track]=nil ownedEmotes[managed]=nil end table.remove(stickyEmoteTracks,index)emoteLastSeen[track]=nil end end end end local motionSource:any=nil local motionTrack:any=nil local
motionHumanoid:any=nil local motionAnimator:any=nil local function clearMotionTrack():()if motionTrack then pcall(function():()motionTrack:Stop(0.1)end)pcall(function():()motionTrack:Destroy()end)end
motionSource,motionTrack,motionHumanoid,motionAnimator=nil,nil,nil,nil end local LOCOMOTION={walk=true,run=true,idle=true,jump=true,fall=true,climb=true,swim=true,swimidle=true}local function
isLocomotion(track:any):boolean if not track or ownedEmotes[track]or isEmoteTrack(track)then return false end if track.Priority==Enum.AnimationPriority.Movement then return true end if track.Priority
~=Enum.AnimationPriority.Core and track.Priority~=Enum.AnimationPriority.Idle then return false end local function known(name:any):boolean local key:string=string.lower(tostring(name or'')):gsub(
'anim$','')return LOCOMOTION[key]==true end if known(track.Name)then return true end local node:any=track.Animation for _index:number=1,5 do if not node then break end if known(node.Name)then return
true end node=node.Parent end return false end local function freezeAnimationTick(humanoid:any):()local animator:any=getAnimator(humanoid)if not animator then if motionAnimator then clearMotionTrack()
end return end if motionHumanoid~=humanoid or motionAnimator~=animator then clearMotionTrack()motionHumanoid=humanoid motionAnimator=animator end if not motionSource then local ok:boolean,tracks:any=
pcall(function():any return animator:GetPlayingAnimationTracks()end)if not ok or type(tracks)~='table'then return end local weight:number=-1 local function consider(track:any):()if not track or not
track.IsPlaying or ownedEmotes[track]or isEmoteTrack(track)then return end local current:number=tonumber(track.WeightCurrent)or 1 if current>weight then motionSource,weight=track,current end end for _
,track:any in ipairs(tracks)do if isLocomotion(track)then consider(track)end end if not motionSource and freezeSettings.anyAnimation then for _,track:any in ipairs(tracks)do consider(track)end end end
if not motionSource then return end pcall(function():()if not motionTrack then motionTrack=animator:LoadAnimation(motionSource.Animation)motionTrack.Looped=true motionTrack.Priority=Enum.
AnimationPriority.Action motionTrack:Play(0.1)motionTrack.TimePosition=motionSource.TimePosition motionTrack:AdjustSpeed(motionSource.Speed or 1)end local emotePlaying:boolean=false if freezeSettings.
emotes then for _,source:any in ipairs(stickyEmoteTracks)do if source.IsPlaying or(managedEmotes[source]and managedEmotes[source].IsPlaying)then emotePlaying=true break end end end if emotePlaying
then if motionTrack.IsPlaying then motionTrack:Stop(0.1)end elseif not motionTrack.IsPlaying then motionTrack:Play(0.1)end end)end local function restoreFreezeMovements():()for root:BasePart,original:
any in pairs(originalFreezeState)do root.Anchored=original.anchored end originalFreezeState=setmetatable({},{__mode='k'})end local function lockPosition(root:any):()local original:any=
originalFreezeState[root]if not original then local cframe:any=nil local look:any=nil pcall(function():()if root.CFrame~=nil then cframe=root.CFrame look=root.CFrame.LookVector end end)original={
anchored=root.Anchored,position=root.Position,cframe=cframe,look=look}originalFreezeState[root]=original end local lockedPosition:any=original.position local gravity:boolean=freezeSettings.keepGravity
==true local lockRotation:boolean=freezeSettings.lockRotation==true root.Anchored=not gravity and lockRotation if not gravity and not lockRotation and lockedPosition~=nil then root.Position=
lockedPosition end if root.AssemblyLinearVelocity~=nil then if gravity then local vertical:number=tonumber(root.AssemblyLinearVelocity.Y)or 0 root.AssemblyLinearVelocity=Vector3.new(0,math.min(
vertical,0),0)else root.AssemblyLinearVelocity=Vector3.zero end end if freezeSettings.lockRotation and root.AssemblyAngularVelocity~=nil then root.AssemblyAngularVelocity=Vector3.zero end if
freezeSettings.lockRotation and original.look~=nil then pcall(function():()if CFrame~=nil and root.CFrame~=nil then root.CFrame=CFrame.lookAt(root.Position,root.Position+original.look)end end)end
local limit:number=freezeSettings.returnDistance or 0 if gravity and limit>0 and lockedPosition~=nil and root.Position~=nil then local dx:number=(tonumber(root.Position.X)or 0)-(tonumber(
lockedPosition.X)or 0)local dz:number=(tonumber(root.Position.Z)or 0)-(tonumber(lockedPosition.Z)or 0)if math.sqrt(dx*dx+dz*dz)>limit then root.Position=Vector3.new(lockedPosition.X or 0,root.Position
.Y,lockedPosition.Z or 0)end end end local toggleFreezeMovements:(boolean)->()toggleFreezeMovements=function(enabled:boolean):()freezeEnabled=enabled disconnectFeatureConnection('FreezeMovements')if
not enabled then disconnectFeatureConnection('FreezeCharacterAdded')disconnectFeatureConnection('FreezeCharacterRemoving')end restoreFreezeMovements()clearMotionTrack()clearStickyEmoteTracks()if not
enabled then if FreezeFeature then FreezeFeature:SetStatus(nil)end return end local player=host.LocalPlayer if player and player.CharacterAdded and not featureConnections.FreezeCharacterAdded then
featureConnections.FreezeCharacterAdded=player.CharacterAdded:Connect(function()if freezeEnabled then toggleFreezeMovements(true)end end)end if player and player.CharacterRemoving and not
featureConnections.FreezeCharacterRemoving then featureConnections.FreezeCharacterRemoving=player.CharacterRemoving:Connect(function()restoreFreezeMovements()clearMotionTrack()clearStickyEmoteTracks()
end)end FreezeFeature:SetStatus(freezeSettings.mode)local previousRoot:any=nil featureConnections.FreezeMovements=TaskManager:Connect(function():()local _,humanoid,root=getCharacterParts()if root~=
previousRoot then restoreFreezeMovements()clearMotionTrack()clearStickyEmoteTracks()previousRoot=root end if not humanoid then clearMotionTrack()clearStickyEmoteTracks()return end if humanoid:
GetState()==Enum.HumanoidStateType.Dead then restoreFreezeMovements()clearMotionTrack()clearStickyEmoteTracks()return end if freezeSettings.mode=='Motion'then if freezeSettings.emotes then
freezeEmotesTick(humanoid,root)end if freezeSettings.animation then freezeAnimationTick(humanoid)end elseif root then lockPosition(root)end end,nil,'Freeze Animations')end FreezeFeature=
createUniversalFeature('Freeze Animations',[[Pin the body where it stands, or keep any animation playing while moving]],4,toggleFreezeMovements,{categoryName='Utilities',configKey=
'Movement.FreezeMovements'})local motionRows:{any}={}local lockRows:{any}={}local function rowFor(handle:any):any if not handle then return nil end local name:string=typeof(handle)=='Instance'and
handle.Name or''if string.sub(name,1,7)=='Option_'or string.sub(name,1,8)=='Section_'then return handle end return handle.Parent or handle end local function registerRow(list:{any},handle:any):()local
row:any=rowFor(handle)if row==nil or row==FreezeFeature.options then return end table.insert(list,row)end local function refreshOptionGroups():()local motion:boolean=freezeSettings.mode==MOTION_MODE
for _,row:any in ipairs(motionRows)do if host.setOptionVisible then host.setOptionVisible(FreezeFeature,row,motion)else row.Visible=motion end end for _,row:any in ipairs(lockRows)do if host.
setOptionVisible then host.setOptionVisible(FreezeFeature,row,not motion)else row.Visible=not motion end end end local function reapply():()if FreezeFeature and FreezeFeature.enabled then
toggleFreezeMovements(true)end end addCycleOption(FreezeFeature,'Freeze mode',{LOCK_MODE,MOTION_MODE},1,function(value:string):()if value=='Anchor'or value=='Freeze Physics'then value=LOCK_MODE end
freezeSettings.mode=value refreshOptionGroups()reapply()end)local lockHeading:any=host.addSectionOption and host.addSectionOption(FreezeFeature,'Position Lock settings')registerRow(lockRows,
lockHeading)local lockRotation:any=addToggleOption(FreezeFeature,'Lock rotation',true,function(value:boolean):()freezeSettings.lockRotation=value reapply()end,
[[Also stop the body from spinning or being spun by the game. OFF keeps the position and lets the character turn.]])registerRow(lockRows,lockRotation)local keepGravity:any=addToggleOption(
FreezeFeature,'Keep gravity',false,function(value:boolean):()freezeSettings.keepGravity=value reapply()end,
[[Do not anchor the body: it can still fall and land, but pushes and knockback are erased every frame.]])registerRow(lockRows,keepGravity)local returnRow:any=addNumberOption(FreezeFeature,
'Return if pushed',0,0,40,function(value:number):()freezeSettings.returnDistance=value end,[[Studs of drift allowed before the character is snapped back. 0 disables the return.]])registerRow(lockRows,
returnRow)local heading:any=host.addSectionOption and host.addSectionOption(FreezeFeature,'Motion settings')registerRow(motionRows,heading)local emotes:any=addToggleOption(FreezeFeature,
'Freeze Emotes',true,function(value:boolean):()freezeSettings.emotes=value if not value then clearStickyEmoteTracks()end end,
[[Keep emotes playing while moving, Roblox's own and the ones a game ships itself. Takes precedence over the locked clip.]])registerRow(motionRows,emotes)local anyEmote:any=addToggleOption(
FreezeFeature,'Any emote',true,function(value:boolean):()freezeSettings.anyEmote=value if not value then clearStickyEmoteTracks()end end,
[[Hold a custom emote even when its name and asset say nothing: a looped clip started while standing still counts as one.]])registerRow(motionRows,anyEmote)local animation:any=addToggleOption(
FreezeFeature,'Freeze Animation',false,function(value:boolean):()freezeSettings.animation=value clearMotionTrack()end,
[[Keep the current clip looping through jumps and other movement. Does not freeze a single frame.]])registerRow(motionRows,animation)local anyAnimation:any=addToggleOption(FreezeFeature,
'Any animation',true,function(value:boolean):()freezeSettings.anyAnimation=value clearMotionTrack()end,[[Freeze a game's own locomotion clip too, not only Roblox's walk/run/idle set.]])registerRow(
motionRows,anyAnimation)refreshOptionGroups()activeCleanup=function():()toggleFreezeMovements(false)end Module.Initialized=true return FreezeFeature end function Module.destroy():()if activeCleanup
then activeCleanup()end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Movement/Gravity.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='Gravity',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.init(context
:Runtime):any local host:any=context.host local TaskManager:any=host.TaskManager local featureConnections:any=host.featureConnections local disconnectFeatureConnection:any=host.
disconnectFeatureConnection local createUniversalFeature:any=host.createUniversalFeature local addNumberOption:any=host.addNumberOption local workspace:any=host.workspace local gravitySettings={value=
196.2}local GravityFeature:any=nil local originalGravity:number?=nil local function restoreGravity()if originalGravity~=nil then workspace.Gravity=originalGravity originalGravity=nil end end
local function toggleGravity(enabled)disconnectFeatureConnection('Gravity')if not enabled then if GravityFeature then GravityFeature:SetStatus(nil)end restoreGravity()return end GravityFeature:
SetStatus(tostring(math.round(gravitySettings.value)))if originalGravity==nil then originalGravity=workspace.Gravity end featureConnections.Gravity=TaskManager:Connect(function()if workspace.Gravity~=
gravitySettings.value then workspace.Gravity=gravitySettings.value end end,nil,'Gravity')end GravityFeature=createUniversalFeature('Gravity','Keep workspace gravity at a custom value',5,toggleGravity,
{categoryName='World'})addNumberOption(GravityFeature,'Gravity value',gravitySettings.value,-500,500,function(value)gravitySettings.value=value if GravityFeature.enabled then GravityFeature:SetStatus(
tostring(math.round(value)))end end)activeCleanup=function():()disconnectFeatureConnection('Gravity')restoreGravity()end Module.Initialized=true return GravityFeature end function Module.destroy():()
if activeCleanup then activeCleanup()end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Movement/InfiniteJump.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='Infinite Jump',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function
Module.init(context:Runtime):any local host:any=context.host local framework:any=context.framework local services:any=context.services local LocalPlayer:Player=host.LocalPlayer local UserInputService:
UserInputService=host.UserInputService local RunService:RunService=host.RunService local movementInput:any=services and services.movementInput local lastBoostAt:number=-math.huge local pressHeld:
boolean=false local jumpTouch:any=nil local heldKey:any=nil local infiniteJump:any local function mode():string local option:any=infiniteJump and infiniteJump.Options and infiniteJump.Options.Mode
return option and option.Value=='Rise'and'Rise'or'Normal'end local function characterParts():(Model?,Humanoid?,BasePart?)local character:Model?=LocalPlayer and LocalPlayer.Character if not character
then return nil,nil,nil end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')::Humanoid?local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?return character,
humanoid,root end local function canBoostJump(humanoid:Humanoid?):boolean if not humanoid or not humanoid.Parent or humanoid.Health<=0 then return false end local currentState:Enum.HumanoidStateType=
humanoid:GetState()return currentState~=Enum.HumanoidStateType.Seated and currentState~=Enum.HumanoidStateType.Swimming and currentState~=Enum.HumanoidStateType.Climbing and currentState~=Enum.
HumanoidStateType.Physics and currentState~=Enum.HumanoidStateType.PlatformStanding and currentState~=Enum.HumanoidStateType.Ragdoll and currentState~=Enum.HumanoidStateType.Dead end infiniteJump=
framework.Categories.Exploits:CreateModule({Name='Infinite Jump',Category='Exploits',ConfigKey='Universal.InfiniteJump',Order=10,Tooltip=
[[Normal adds one jump per press; Rise ascends while the jump key or mobile jump button is held.]],Function=function(enabled:boolean):()lastBoostAt=-math.huge pressHeld=false jumpTouch=nil heldKey=nil
if not enabled then infiniteJump:SetStatus(nil)return end infiniteJump:SetStatus(mode())local function executeAirJump():()local _,humanoid:Humanoid?,root:BasePart?=characterParts()if not humanoid or
not root or root.Anchored or not canBoostJump(humanoid)then return end local power:number=infiniteJump.Options['Jump Power']and infiniteJump.Options['Jump Power'].Value or 50 if humanoid.FloorMaterial
~=Enum.Material.Air and power>=0 then return end local now:number=os.clock()local interval:number=infiniteJump.Options['Jump interval']and infiniteJump.Options['Jump interval'].Value or 0.18 if now-
lastBoostAt<interval then return end lastBoostAt=now local velocity:Vector3=root.AssemblyLinearVelocity root.AssemblyLinearVelocity=Vector3.new(velocity.X,power,velocity.Z)humanoid:ChangeState(if
power<0 then Enum.HumanoidStateType.Freefall else Enum.HumanoidStateType.Jumping)end local function onJumpRequested():()if UserInputService and type(UserInputService.GetFocusedTextBox)=='function'and
UserInputService:GetFocusedTextBox()then return end if pressHeld then return end pressHeld=true if UserInputService and UserInputService.TouchEnabled then local okTouches:boolean,touches:any=pcall(
function():any return UserInputService:GetTouches()end)if okTouches and type(touches)=='table'then if#touches==1 then jumpTouch=touches[1]else local gui:Instance?=LocalPlayer and LocalPlayer:
FindFirstChildOfClass('PlayerGui')local touchGui:Instance?=gui and gui:FindFirstChild('TouchGui')local button:any=touchGui and touchGui:FindFirstChild('JumpButton',true)if button then local at:Vector2
=button.AbsolutePosition local size:Vector2=button.AbsoluteSize for _,touch:any in ipairs(touches)do local p:Vector3=touch.Position if p.X>=at.X and p.Y>=at.Y and p.X<=at.X+size.X and p.Y<=at.Y+size.Y
then jumpTouch=touch break end end end end end end executeAirJump()end if movementInput and type(movementInput.onJumpRequest)=='function'then infiniteJump:Clean(movementInput.onJumpRequest(
onJumpRequested))elseif UserInputService and UserInputService.JumpRequest then infiniteJump:Clean(UserInputService.JumpRequest:Connect(onJumpRequested))end if UserInputService and UserInputService.
InputBegan then infiniteJump:Clean(UserInputService.InputBegan:Connect(function(input:any,gameProcessed:boolean):()if input.KeyCode==Enum.KeyCode.Space or input.KeyCode==Enum.KeyCode.ButtonA then
local typing:boolean=type(UserInputService.GetFocusedTextBox)=='function'and UserInputService:GetFocusedTextBox()~=nil if not gameProcessed and not typing then heldKey=input.KeyCode onJumpRequested()
end elseif input.UserInputType==Enum.UserInputType.Touch then local gui:Instance?=LocalPlayer and LocalPlayer:FindFirstChildOfClass('PlayerGui')local touchGui:Instance?=gui and gui:FindFirstChild(
'TouchGui')local button:any=touchGui and touchGui:FindFirstChild('JumpButton',true)if button and button.AbsolutePosition and button.AbsoluteSize and input.Position then local at:Vector2=button.
AbsolutePosition local size:Vector2=button.AbsoluteSize local p:Vector3=input.Position if p.X>=at.X and p.Y>=at.Y and p.X<=at.X+size.X and p.Y<=at.Y+size.Y then jumpTouch=input pressHeld=true
executeAirJump()end end end end))end if UserInputService and UserInputService.InputEnded then infiniteJump:Clean(UserInputService.InputEnded:Connect(function(event:any):()if(event.KeyCode==Enum.
KeyCode.Space or event.KeyCode==Enum.KeyCode.ButtonA)and(heldKey==nil or heldKey==event.KeyCode)then pressHeld=false heldKey=nil jumpTouch=nil elseif event.UserInputType==Enum.UserInputType.Touch and
not heldKey and(not jumpTouch or event==jumpTouch)then pressHeld=false jumpTouch=nil end end))end if UserInputService and UserInputService.WindowFocusReleased then infiniteJump:Clean(UserInputService.
WindowFocusReleased:Connect(function():()pressHeld=false jumpTouch=nil heldKey=nil end))end if UserInputService and UserInputService.TextBoxFocused then infiniteJump:Clean(UserInputService.
TextBoxFocused:Connect(function():()pressHeld=false jumpTouch=nil heldKey=nil end))end if RunService and RunService.PreSimulation then infiniteJump:Clean(RunService.PreSimulation:Connect(function():()
local _,humanoid:Humanoid?,root:BasePart?=characterParts()if not humanoid or not root or humanoid.Health<=0 then return end local power:number=infiniteJump.Options['Jump Power']and infiniteJump.
Options['Jump Power'].Value or 50 local typing:boolean=type(UserInputService.GetFocusedTextBox)=='function'and UserInputService:GetFocusedTextBox()~=nil if typing then pressHeld=false jumpTouch=nil
heldKey=nil return end if mode()=='Rise'and pressHeld and not root.Anchored and canBoostJump(humanoid)then local vel:Vector3=root.AssemblyLinearVelocity root.AssemblyLinearVelocity=Vector3.new(vel.X,
power,vel.Z)if power<0 and humanoid:GetState()==Enum.HumanoidStateType.Jumping then humanoid:ChangeState(Enum.HumanoidStateType.Freefall)end elseif mode()=='Normal'and power<0 and humanoid:GetState()
==Enum.HumanoidStateType.Jumping then local vel:Vector3=root.AssemblyLinearVelocity root.AssemblyLinearVelocity=Vector3.new(vel.X,power,vel.Z)humanoid:ChangeState(Enum.HumanoidStateType.Freefall)end
end))end end})infiniteJump:CreateDropdown({Name='Mode',List={'Normal','Rise'},Index=1,Function=function(value:string):()if infiniteJump.Enabled then infiniteJump:SetStatus(value)end end,Tooltip=
[[Normal = one jump per press in air, Rise = hold to ascend. Jump Power + Interval control both.]]})infiniteJump:CreateSlider({Name='Jump Power',Min=1,Max=250,Default=50,Tooltip=
[[Vertical jump velocity applied in mid-air (1-250). Negative inverts jump.]]})infiniteJump:CreateSlider({Name='Jump interval',Min=0.05,Max=0.6,Step=0.01,Default=0.18,Tooltip=
[[Minimum delay between mid-air jumps in seconds. Default 0.18 s (vanilla-like). Lower = faster spam, higher = slower rhythm.]]})infiniteJump:CreateNote(
[[Jump interval: 0.18 s is default Roblox rhythm. 0.05 s = spam fast, 0.6 s = slow.]])activeCleanup=function():()if infiniteJump.Enabled then infiniteJump:Toggle(false)end end Module.Initialized=true
return infiniteJump end function Module.destroy():()if activeCleanup then activeCleanup()end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Movement/Jump.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='Jump Settings',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function
Module.init(context:Runtime):any local host:any=context.host local framework:any=context.framework local movementInput:any=context.services and context.services.movementInput local UserInputService:
any=host.UserInputService or game:GetService('UserInputService')local LocalPlayer:Player=host.LocalPlayer or game:GetService('Players').LocalPlayer local originalJumpPower=setmetatable({},{__mode='k'}
)::{[Humanoid]:{jumpPower:number,useJumpPower:boolean}}local function characterParts():(Model?,Humanoid?,BasePart?)local character:Model?=LocalPlayer.Character if not character then return nil,nil,nil
end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?return character,humanoid,root end local function
restoreJumpPower():()for humanoid:Humanoid,original in pairs(originalJumpPower)do if humanoid and humanoid.Parent then humanoid.JumpPower=original.jumpPower humanoid.UseJumpPower=original.useJumpPower
end end originalJumpPower=setmetatable({},{__mode='k'})::any end local function canControlJump(humanoid:Humanoid):boolean if humanoid.Health<=0 or humanoid.Sit or humanoid.SeatPart then return false
end local state=humanoid:GetState()return state~=Enum.HumanoidStateType.Seated and state~=Enum.HumanoidStateType.Swimming and state~=Enum.HumanoidStateType.Climbing and state~=Enum.HumanoidStateType.
Physics and state~=Enum.HumanoidStateType.PlatformStanding and state~=Enum.HumanoidStateType.Ragdoll and state~=Enum.HumanoidStateType.FallingDown and state~=Enum.HumanoidStateType.Dead end
local function performHighJump(velocityOverride:number?):()local _,humanoid:Humanoid?,root:BasePart?=characterParts()if not humanoid or not root or root.Anchored or not canControlJump(humanoid)then
return end local velocityVal:number=velocityOverride or 72 local current:Vector3=root.AssemblyLinearVelocity root.AssemblyLinearVelocity=Vector3.new(current.X,velocityVal,current.Z)if humanoid.
FloorMaterial~=Enum.Material.Air then humanoid:ChangeState(Enum.HumanoidStateType.Freefall)end end local longJumpSpeed=50 local longJumpMode='Velocity'local function performLongJump():()local _,
humanoid:Humanoid?,root:BasePart?=characterParts()if not humanoid or not root or root.Anchored or not canControlJump(humanoid)then return end local optsMode:any=jump and jump.Options and jump.Options[
'Long jump mode']local optsSpeed:any=jump and jump.Options and jump.Options['Long jump speed']local mode:string=optsMode and optsMode.Value or longJumpMode local speed:number=optsSpeed and optsSpeed.
Value or longJumpSpeed local moveDir:Vector3=humanoid.MoveDirection local flat:Vector3=Vector3.new(moveDir.X,0,moveDir.Z)if flat.Magnitude<0.05 then local cam:any=host.workspace and host.workspace.
CurrentCamera if cam and cam.CFrame then local lv:Vector3=cam.CFrame.LookVector flat=Vector3.new(lv.X,0,lv.Z)else flat=Vector3.new(0,0,-1)end end if flat.Magnitude<0.001 then return end flat=flat.
Magnitude>1 and flat.Unit or flat local cur:Vector3=root.AssemblyLinearVelocity if mode=='Velocity'then root.AssemblyLinearVelocity=Vector3.new(flat.X*speed,cur.Y,flat.Z*speed)elseif mode=='Impulse'
then local diff:Vector3=Vector3.new(flat.X*speed-cur.X,0,flat.Z*speed-cur.Z)if diff.Magnitude>0.001 then root:ApplyImpulse(diff*root.AssemblyMass)end else local step:Vector3=flat*speed*0.25 local pos:
Vector3=root.CFrame.Position root.CFrame=CFrame.new(Vector3.new(pos.X+step.X,pos.Y+step.Y,pos.Z+step.Z))end if humanoid.FloorMaterial~=Enum.Material.Air then humanoid:ChangeState(Enum.
HumanoidStateType.Jumping)end end local jump:any jump=framework.Categories.Exploits:CreateModule({Name='Jump Settings',Category='Exploits',ConfigKey='Universal.Jump',Order=6,Tooltip=
[[Jump controls: adjust Jump Power, High Jump button, Long Jump button and Auto Jump. Infinite Jump moved to its own module.]],Function=function(enabled:boolean):()restoreJumpPower()if not enabled
then jump:SetStatus(nil)return end jump:SetStatus('active')jump:Clean(function():()restoreJumpPower()end)jump:Clean(host.RunService.PreSimulation:Connect(function():()local opts=jump.Options local _,
humanoid:Humanoid?,root:BasePart?=characterParts()if not humanoid or not root or humanoid.Health<=0 then return end local canMove=canControlJump(humanoid)and not root.Anchored if not canMove then
return end if opts['Custom jump power']and opts['Custom jump power'].Value then local wantedPower:number=opts['Jump power']and opts['Jump power'].Value or 80 if originalJumpPower[humanoid]==nil then
originalJumpPower[humanoid]={jumpPower=humanoid.JumpPower,useJumpPower=humanoid.UseJumpPower}end humanoid.UseJumpPower=true humanoid.JumpPower=wantedPower if wantedPower<0 and humanoid:GetState()==
Enum.HumanoidStateType.Jumping then local curVel:Vector3=root.AssemblyLinearVelocity root.AssemblyLinearVelocity=Vector3.new(curVel.X,wantedPower,curVel.Z)end else if originalJumpPower[humanoid]then
humanoid.JumpPower=originalJumpPower[humanoid].jumpPower humanoid.UseJumpPower=originalJumpPower[humanoid].useJumpPower originalJumpPower[humanoid]=nil end end if canMove and opts['Auto jump']and opts
['Auto jump'].Value then local moveDir:Vector3=humanoid.MoveDirection if moveDir.Magnitude>0.05 and humanoid.FloorMaterial~=Enum.Material.Air then humanoid:ChangeState(Enum.HumanoidStateType.Jumping)
end end end))end})jump:CreateToggle({Name='Custom jump power',Default=false,Tooltip='Enables custom jump height/power for ground jumps.'})jump:CreateSlider({Name='Jump power',Show={Option=
'Custom jump power',Values={true}},Min=1,Max=250,Default=80,Function=function(value:number):()if jump.Enabled and jump.Options['Custom jump power'].Value then jump:SetStatus(tostring(math.round(value)
))end end,Tooltip=[[Character jump power (default 50). Negative launches downward.]]})jump:CreateButton({Name='High jump',Function=function():()local velocityVal:number=jump.Options[
'High jump velocity']and jump.Options['High jump velocity'].Value or 72 performHighJump(velocityVal)end,Tooltip=[[Launches your character into a single massive vertical jump.]]})jump:CreateSlider({
Name='High jump velocity',Min=1,Max=250,Default=72,Tooltip=[[Vertical velocity for High Jump button. Negative = downward.]]})jump:CreateToggle({Name='Auto jump',Default=false,Tooltip=
'Continuously jumps while walking on the ground.'})jump:CreateButton({Name='Long jump',Function=function():()performLongJump()end,Tooltip=
[[Horizontal boost in your move/camera direction (button like High Jump). Uses Long jump mode/speed below.]]})jump:CreateDropdown({Name='Long jump mode',List={'Velocity','Impulse','CFrame'},Index=1,
Function=function(value:string):()longJumpMode=value end,Tooltip=[[How Long Jump is applied: Velocity (smooth), Impulse (physics), CFrame (teleport).]]})jump:CreateSlider({Name='Long jump speed',Min=1
,Max=250,Default=50,Function=function(value:number):()longJumpSpeed=value end,Tooltip=[[Horizontal speed for Long Jump button in studs/s. Negative = backwards.]]})activeCleanup=function():()if jump.
Enabled then jump:Toggle(false)end restoreJumpPower()end Module.Initialized=true return jump end function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.
Initialized=false end return Module
-- src/games/universal/Movement/Speed.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='Speed Settings',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup function Module.init(
context:Runtime):any local host,framework=context.host,context.framework local player,world=host.LocalPlayer,host.workspace or workspace local walk,speed local teleportElapsed,lastMode=0,nil local
wallParams,wallCharacter,wallGroup local function clearWallFilter()if wallParams then wallParams.FilterDescendantsInstances={}end wallCharacter=nil end local function parts()local character=player.
Character return character,character and character:FindFirstChildOfClass('Humanoid'),character and character:FindFirstChild('HumanoidRootPart')end local function restoreWalk()if walk then if walk.
humanoid.Parent and walk.humanoid.WalkSpeed==walk.applied then walk.humanoid.WalkSpeed=walk.base end walk=nil end end local function applyWalk(humanoid,value)if walk and walk.humanoid~=humanoid then
restoreWalk()end if not walk then walk={humanoid=humanoid,base=humanoid.WalkSpeed}end local current=humanoid.WalkSpeed if walk.applied~=nil and current~=walk.applied then walk.base=current end if
current~=value then humanoid.WalkSpeed=value end walk.applied=value if value<0 then local moveDir=humanoid.MoveDirection if moveDir and moveDir.Magnitude>0.01 and type(humanoid.Move)=='function'then
humanoid:Move(-moveDir,false)end end end local function direction(humanoid)local move=humanoid.MoveDirection local flat=Vector3.new(move.X,0,move.Z)return flat.Magnitude>1 and flat.Unit or flat end
local function blocked(root,step)if step.Magnitude<0.001 then return step end if not wallParams then wallParams=RaycastParams.new()wallParams.FilterType=Enum.RaycastFilterType.Exclude wallParams.
RespectCanCollide,wallParams.IgnoreWater=true,true end if wallCharacter~=player.Character then wallCharacter=player.Character wallParams.FilterDescendantsInstances=player.Character and{player.
Character}or{}end if wallGroup~=root.CollisionGroup then wallGroup=root.CollisionGroup wallParams.CollisionGroup=wallGroup end if world.Blockcast then local hit=world:Blockcast(root.CFrame,root.Size*
0.9,step,wallParams)return hit and step.Unit*math.max(0,hit.Distance-0.1)or step end local hit=world:Raycast(root.Position,step,wallParams)return hit and step.Unit*math.max(0,(hit.Position-root.
Position).Magnitude-root.Size.Magnitude*0.5-0.1)or step end local function stepSpeed(dt)local _,humanoid,root=parts()if not humanoid or not root or humanoid.Health<=0 then restoreWalk()if speed.
ClearSprint then speed.ClearSprint()end return end if not speed.Enabled and not speed.SprintHeld then restoreWalk()return end local opts=speed.Options local mode=speed.Enabled and opts.Mode.Value or
'WalkSpeed'local target=speed.Enabled and opts.Speed.Value or 0 if speed.SprintHeld then target=math.max(target,30)end if mode~=lastMode then restoreWalk()teleportElapsed=0 lastMode=mode end local
seat:any=humanoid.SeatPart local isSeated:boolean=seat~=nil or humanoid.Sit if isSeated then if speed.SprintFrame then speed.SprintFrame(speed.SprintHeld and false,root)end if mode=='WalkSpeed'then
restoreWalk()return end local vehicle:any=seat if not vehicle or not vehicle.Parent then restoreWalk()return end restoreWalk()local fly=framework.GetModule and framework:GetModule('Fly')if vehicle.
Anchored or(fly and fly.Enabled)then return end local move=direction(humanoid)local vehicleMul:number=opts['Vehicle multiplier']and opts['Vehicle multiplier'].Value or 1.2 local vehicleTarget:number=
target*vehicleMul local response:number=opts['Response']and opts['Response'].Value or 1 local keepY:boolean=opts['Keep vertical']==nil or opts['Keep vertical'].Value if mode=='Velocity'or(mode==
'WalkSpeed'and target<0)then local vel:Vector3=vehicle.AssemblyLinearVelocity local targetVel:Vector3=Vector3.new(move.X*vehicleTarget,keepY and vel.Y or 0,move.Z*vehicleTarget)if response>=1 then
vehicle.AssemblyLinearVelocity=targetVel else local lerpAlpha:number=math.clamp(response*(dt or 1/60)*12,0,1)local curXZ:Vector3=Vector3.new(vel.X,0,vel.Z)local tarXZ:Vector3=Vector3.new(targetVel.X,0
,targetVel.Z)local newXZ:Vector3=Vector3.new(curXZ.X+(tarXZ.X-curXZ.X)*lerpAlpha,0,curXZ.Z+(tarXZ.Z-curXZ.Z)*lerpAlpha)vehicle.AssemblyLinearVelocity=Vector3.new(newXZ.X,targetVel.Y,newXZ.Z)if move.
Magnitude<0.01 then local brake:number=opts['Braking']and opts['Braking'].Value or 0.8 if brake<0.99 then local brakeAlpha:number=math.clamp((1-brake)*(dt or 1/60)*10,0,1)local curV:Vector3=vehicle.
AssemblyLinearVelocity vehicle.AssemblyLinearVelocity=Vector3.new(curV.X*(1-brakeAlpha),curV.Y,curV.Z*(1-brakeAlpha))if vehicle.AssemblyLinearVelocity.Magnitude<0.5 then vehicle.AssemblyLinearVelocity
=Vector3.new(0,curV.Y,0)end end end end elseif mode=='Impulse'then local vel:Vector3=vehicle.AssemblyLinearVelocity local targetVelXZ:Vector3=Vector3.new(move.X*vehicleTarget,0,move.Z*vehicleTarget)
local diff:Vector3=Vector3.new(targetVelXZ.X-vel.X,0,targetVelXZ.Z-vel.Z)diff*=math.clamp(response,0.2,2)if diff.Magnitude>0.001 then vehicle:ApplyImpulse(diff*vehicle.AssemblyMass)end elseif move.
Magnitude>0.001 then local elapsed=math.clamp(dt or 0,0,0.1)if mode=='Teleport'then teleportElapsed+=elapsed local interval=opts['Burst delay'].Value if teleportElapsed<interval then return end
teleportElapsed%=interval elapsed=interval end local step=move*vehicleTarget*elapsed if opts['Wall check'].Value then step=blocked(vehicle,step)end vehicle.CFrame+=step if root and root.Parent then
root.CFrame=vehicle.CFrame*CFrame.new(0,2,0)end else teleportElapsed=0 end if speed.Enabled and opts['Auto jump'].Value and move.Magnitude>0.05 then end return end if speed.SprintFrame then speed.
SprintFrame(speed.SprintHeld and not isSeated,root)end local fly=framework.GetModule and framework:GetModule('Fly')local state=humanoid:GetState()if root.Anchored or(fly and fly.Enabled)or state==Enum
.HumanoidStateType.Climbing or state==Enum.HumanoidStateType.Swimming or state==Enum.HumanoidStateType.Physics or state==Enum.HumanoidStateType.PlatformStanding then restoreWalk()return end if mode==
'WalkSpeed'then applyWalk(humanoid,target)else restoreWalk()end local move=direction(humanoid)if speed.Enabled and opts['Auto jump'].Value and move.Magnitude>0.05 and humanoid.FloorMaterial~=Enum.
Material.Air then humanoid:ChangeState(Enum.HumanoidStateType.Jumping)end if mode=='WalkSpeed'and target>=0 then return end local velocity=root.AssemblyLinearVelocity local response:number=opts[
'Response']and opts['Response'].Value or 1 local keepY:boolean=opts['Keep vertical']==nil or opts['Keep vertical'].Value local braking:number=opts['Braking']and opts['Braking'].Value or 0.8 if mode==
'Velocity'or(mode=='WalkSpeed'and target<0)then local targetVel:Vector3=Vector3.new(move.X*target,keepY and velocity.Y or 0,move.Z*target)if response>=1 or move.Magnitude<0.01 and braking>=0.99 then
root.AssemblyLinearVelocity=targetVel else local lerpAlpha:number=math.clamp(response*(dt or 1/60)*12,0,1)local curXZ:Vector3=Vector3.new(velocity.X,0,velocity.Z)local tarXZ:Vector3=Vector3.new(
targetVel.X,0,targetVel.Z)local newXZ:Vector3=Vector3.new(curXZ.X+(tarXZ.X-curXZ.X)*lerpAlpha,0,curXZ.Z+(tarXZ.Z-curXZ.Z)*lerpAlpha)root.AssemblyLinearVelocity=Vector3.new(newXZ.X,keepY and velocity.Y
or targetVel.Y,newXZ.Z)if move.Magnitude<0.01 then if braking<0.99 then local brakeAlpha:number=math.clamp((1-braking)*(dt or 1/60)*10,0,1)local curV:Vector3=root.AssemblyLinearVelocity root.
AssemblyLinearVelocity=Vector3.new(curV.X*(1-brakeAlpha),curV.Y,curV.Z*(1-brakeAlpha))if root.AssemblyLinearVelocity.Magnitude<0.5 and keepY then root.AssemblyLinearVelocity=Vector3.new(0,velocity.Y,0
)elseif root.AssemblyLinearVelocity.Magnitude<0.5 then root.AssemblyLinearVelocity=Vector3.zero end end end end elseif mode=='Impulse'then local diff=Vector3.new(move.X*target-velocity.X,0,move.Z*
target-velocity.Z)diff*=math.clamp(response,0.2,2)if diff.Magnitude>0.001 then root:ApplyImpulse(diff*root.AssemblyMass)end elseif move.Magnitude>0.001 then local elapsed=math.clamp(dt or 0,0,0.1)if
mode=='Teleport'then teleportElapsed+=elapsed local interval=opts['Burst delay'].Value if teleportElapsed<interval then return end teleportElapsed%=interval elapsed=interval end local speedDelta=
target<0 and(target-humanoid.WalkSpeed)or math.max(0,target-humanoid.WalkSpeed)local step=move*speedDelta*elapsed step*=math.clamp(response,0.5,2)if opts['Wall check'].Value then step=blocked(root,
step)end root.CFrame+=step else teleportElapsed=0 end end speed=framework.Categories.Exploits:CreateModule({Name='Speed Settings',Category='Exploits',ConfigKey='Universal.Speed',Order=1,Tooltip=
[[Movement with full vehicle support. WalkSpeed on foot only; Velocity/Impulse/CFrame/Teleport work in any vehicle or object using velocity.]],Function=function(enabled)restoreWalk()teleportElapsed,
lastMode=0,nil if not enabled then clearWallFilter()if speed.ClearSprint then speed.ClearSprint()end speed:SetStatus(nil)if speed.SprintHeld then stepSpeed(0)end return end speed:Clean(function()
restoreWalk()if speed.ClearSprint then speed.ClearSprint()end end)speed:Clean(host.RunService.PreSimulation:Connect(stepSpeed))speed:SetStatus(speed.Options.Mode.Value)end})speed.SprintHeld=false
speed.SetSprintHeld=function(held)speed.SprintHeld=held stepSpeed(0)if not held and speed.ClearSprint then speed.ClearSprint()end end speed.StepSprint=function(dt)if speed.SprintHeld and not speed.
Enabled then stepSpeed(dt)end end local function changed()restoreWalk()teleportElapsed,lastMode=0,nil if speed.Enabled then speed:SetStatus(speed.Options.Mode.Value)end end speed:CreateDropdown({Name=
'Mode',List={'WalkSpeed','Velocity','Impulse','CFrame','Teleport'},Index=2,Function=changed})speed:CreateSlider({Name='Speed',Min=1,Max=250,Default=32,Tooltip=
[[Speed in studs/s. WalkSpeed sets Humanoid.WalkSpeed, Velocity/Impulse drive AssemblyLinearVelocity, CFrame/Teleport move CFrame.]]})speed:CreateSlider({Name='Response',Min=0.2,Max=2,Step=0.1,Default
=1,Show={Option='Mode',Values={'Velocity','Impulse','CFrame'}},Tooltip=[[Response: how fast velocity reaches target. 0.2 sluggish, 1 default balanced, 2 instant.]]})speed:CreateSlider({Name='Braking',
Min=0,Max=1,Step=0.05,Default=0.8,Show={Option='Mode',Values={'Velocity'}},Tooltip=[[Braking when no input: 0 = instant stop, 0.8 default smooth, 1 = keep momentum.]]})speed:CreateSlider({Name=
'Vehicle multiplier',Min=0.5,Max=3,Step=0.1,Default=1.2,Tooltip=
[[Scales Speed when in a vehicle/seat (Velocity/Impulse/CFrame/Teleport only). 1 = same as on foot, 1.2 default slightly faster in vehicle.]]})speed:CreateToggle({Name='Keep vertical',Default=true,
Show={Option='Mode',Values={'Velocity','Impulse'}},Tooltip=[[Preserve Y velocity (gravity/jump). OFF = full 3D velocity control (also affects vehicles).]]})speed:CreateToggle({Name='Wall check',
Default=true,Show={Option='Mode',Values={'CFrame','Teleport'}}})speed:CreateSlider({Name='Burst delay',Min=0.05,Max=1,Default=0.2,Show={Option='Mode',Values={'Teleport'}}})speed:CreateToggle({Name=
'Auto jump',Default=false})speed:RefreshVisibility()activeCleanup=function()speed.SprintHeld=false if speed.Enabled then speed:Toggle(false)end restoreWalk()clearWallFilter()if speed.ClearSprint then
speed.ClearSprint()end end Module.Initialized=true return speed end function Module.destroy()if activeCleanup then activeCleanup()activeCleanup=nil end Module.Initialized=false end return Module
-- src/games/universal/Other/Animations.lua
export type Runtime={framework:any,host:any}local Module={Name='Animations',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCard:any=nil local activeCleanup:(()->())?=nil function
Module.init(context:Runtime):any local host:any=context.host local framework:any=context.framework local Players:Players=host.Players or(game::any):GetService('Players')local LocalPlayer:Player=host.
LocalPlayer or Players.LocalPlayer local TaskManager:any=host.TaskManager local featureConnections:any=host.featureConnections local disconnectFeatureConnection:any=host.disconnectFeatureConnection
local configData:any=host.configData local savedKey:string='Universal.Animations.Saved'local savedAnims:{string}={}if configData and configData.values and type(configData.values[savedKey])=='table'
then local stored:any=configData.values[savedKey]for _,v:any in ipairs(stored)do if type(v)=='string'and string.match(v,'^%d+$')then table.insert(savedAnims,v)elseif type(v)=='number'then table.
insert(savedAnims,tostring(v))end end end local currentTrack:AnimationTrack?=nil local currentAnim:Animation?=nil local currentId:string=''local currentSpeed:number=1 local isPlaying:boolean=false
local function getAnimator():Animator?local character:Model?=LocalPlayer and LocalPlayer.Character if not character then return nil end local humanoid:Humanoid?=character:FindFirstChildOfClass(
'Humanoid')::Humanoid?if not humanoid then return nil end local animator:Animator?=humanoid:FindFirstChildOfClass('Animator')::Animator?if animator then return animator end local ok:boolean,created:
any=pcall(function()local a=Instance.new('Animator')a.Parent=humanoid return a end)if ok and created then return created::Animator end return nil end local function stopCurrent():()if currentTrack
then pcall(function()currentTrack:Stop(0.1)end)pcall(function()currentTrack:Destroy()end)currentTrack=nil end if currentAnim then pcall(function()currentAnim:Destroy()end)currentAnim=nil end isPlaying
=false end local function playAnimation(id:string,speed:number):()local animator:Animator?=getAnimator()if not animator then if activeCard then activeCard:Notify('No animator found (respawn?)')end
return end local numericId:string?=string.match(tostring(id or''),'%d+')if not numericId then if activeCard then activeCard:Notify('Enter a valid animation ID')end return end stopCurrent()local anim:
Animation=Instance.new('Animation')anim.Name='Wurst_Animation_'..numericId anim.AnimationId='rbxassetid://'..numericId local success:boolean,result:any=pcall(function()return animator:LoadAnimation(
anim)end)if not success or not result then pcall(function()anim:Destroy()end)if activeCard then activeCard:Notify('Failed to load anim: '..tostring(result or'invalid id'))end return end local track:
AnimationTrack=result::AnimationTrack pcall(function()track.Priority=Enum.AnimationPriority.Action track.Looped=true end)local cleanSpeed:number=tonumber(speed)or 1 if cleanSpeed~=cleanSpeed or
cleanSpeed<=0 then cleanSpeed=1 end if cleanSpeed>999 then cleanSpeed=999 end if cleanSpeed<0.1 then cleanSpeed=0.1 end track:Play()track:AdjustSpeed(cleanSpeed)local myTrack:AnimationTrack=track
local conn:RBXScriptConnection?=nil conn=track.Stopped:Connect(function()if conn then pcall(function()conn:Disconnect()end)conn=nil end if isPlaying and myTrack==currentTrack and activeCard and
activeCard.Enabled then pcall(function()myTrack:Play()myTrack:AdjustSpeed(cleanSpeed)end)end end)if activeCard then activeCard:Clean(function()if conn then pcall(function()conn:Disconnect()end)end end
)end currentTrack=track currentAnim=anim currentId=numericId currentSpeed=cleanSpeed isPlaying=true if activeCard then activeCard:SetStatus('Playing '..numericId..' \183 '..tostring(cleanSpeed)..'x')
end end local function saveCurrent():()local id:string?=string.match(tostring(currentId or activeCard and activeCard.Options['Animation ID'].Value or''),'%d+')if not id then local raw:any=activeCard
and activeCard.Options['Animation ID']and activeCard.Options['Animation ID'].Value or''id=string.match(tostring(raw or''),'%d+')end if not id then if activeCard then activeCard:Notify(
'Enter an animation ID first')end return end for _,existing:string in ipairs(savedAnims)do if existing==id then if activeCard then activeCard:Notify('Already saved: '..id)end return end end table.
insert(savedAnims,1,id)while#savedAnims>20 do table.remove(savedAnims)end if configData and configData.values then configData.values[savedKey]=savedAnims if host.queueConfigSave then pcall(host.
queueConfigSave)end end if activeCard and activeCard.Options['Saved Animations']then pcall(function()activeCard.Options['Saved Animations']:Refresh()end)end if activeCard then activeCard:Notify(
'Saved '..id)end end local function refreshSavedList(card:any):()if card and card.Options['Saved Animations']then pcall(function()card.Options['Saved Animations']:Refresh()end)end end local card:any=
framework.Categories.World:CreateModule({Name='Animations',Category='World',ConfigKey='Universal.Animations',Tooltip=
[[Plays any animation ID on your character with Animator, visible to others. Save IDs and click to replay.]],Function=function(enabled:boolean):()if not enabled then stopCurrent()if activeCard then
activeCard:SetStatus(nil)end return end local id:string=tostring(card.Options['Animation ID'].Value or'')local speed:number=tonumber(card.Options['Speed'].Value)or 1 if string.match(id,'%d+')then
playAnimation(id,speed)else if activeCard then activeCard:SetStatus('Enter an ID and press Save or Play')end end end})activeCard=card card:CreateTextBox({Name='Animation ID',Placeholder=
'rbxassetid or numbers only, e.g. 507777826',Default='',Tooltip=[[Animation asset ID (numbers only). Works with any Roblox animation. Example: 507777826 (walk).]]})card:CreateSlider({Name='Speed',Min=
1,Max=5,Step=0.1,Default=1,Tooltip=[[Playback speed: 1 = normal, 0.5 = half, 5 = 5x, up to 999 if typed manually. Roblox allows high values.]]})card:CreateButton({Name='Play Animation',Tooltip=
[[Play the ID above at the selected speed (visible to others via Animator).]],Function=function()local id:string=tostring(card.Options['Animation ID'].Value or'')local speed:number=tonumber(card.
Options['Speed'].Value)or 1 if not string.match(id,'%d+')then card:Notify('Enter an animation ID first')return end if not card.Enabled then card:Toggle(true)task.defer(function()if card.Enabled then
playAnimation(id,speed)end end)else playAnimation(id,speed)end end})card:CreateButton({Name='Save Animation',Tooltip=[[Save the current ID to the list below. Max 20, persists via config.]],Function=
saveCurrent})local savedList:any=card:CreateList({Name='Saved Animations',Items=function():{string}local out:{string}={}for _,id:string in ipairs(savedAnims)do table.insert(out,id)end return out end,
Default={},EmptyText='No saved yet. Enter an ID and press Save Animation.',Function=function(selected:string,all:{string})if type(selected)~='string'or selected==''then return end local id:string=
selected if#all>0 then if type(selected)=='table'then id=all[#all]end end if not string.match(id or'','%d+')then return end local speed:number=tonumber(card.Options['Speed'].Value)or 1 if card.Options
['Animation ID']then pcall(function()local opt:any=card.Options['Animation ID']if opt.Object then local tb:any=opt.Object:FindFirstChildOfClass('TextBox')or opt.Object:FindFirstChild('TextBox')or opt.
Object:FindFirstChild('Input',true)if tb then tb.Text=id end end opt.Value=id end)end currentId=id if not card.Enabled then card:Toggle(true)task.defer(function()if card.Enabled then playAnimation(id,
speed)end end)else playAnimation(id,speed)end task.defer(function()pcall(function()savedList:Set(id,false)end)end)end,OnRemove=function(id:string)for i:number,v:string in ipairs(savedAnims)do if v==id
then table.remove(savedAnims,i)break end end if configData and configData.values then configData.values[savedKey]=savedAnims if host.queueConfigSave then pcall(host.queueConfigSave)end end
refreshSavedList(card)end,Tooltip='Click an ID to play it. Saved IDs persist across sessions.'})local origSpeedOption:any=card.Options['Speed']if origSpeedOption and origSpeedOption.Object then local
lastSpeed:number=tonumber(origSpeedOption.Value)or 1 card:Render(function()if not card.Enabled or not currentTrack then return end local now:number=tonumber(card.Options['Speed'].Value)or lastSpeed if
now~=lastSpeed then lastSpeed=now local clean:number=math.clamp(now,0.1,999)pcall(function()currentTrack:AdjustSpeed(clean)end)currentSpeed=clean if card then card:SetStatus('Playing '..currentId..
' \183 '..tostring(clean)..'x')end end end)end if LocalPlayer and LocalPlayer.CharacterAdded then local conn:RBXScriptConnection=LocalPlayer.CharacterAdded:Connect(function()if card.Enabled and
currentId~=''then task.wait(1)if card.Enabled then playAnimation(currentId,currentSpeed)end end end)card:Clean(function()pcall(function()conn:Disconnect()end)end)end card:Clean(function()stopCurrent()
end)activeCleanup=function()stopCurrent()if card and card.Enabled then pcall(function()card:Toggle(false)end)end end Module.Initialized=true return card end function Module.destroy():()if
activeCleanup then pcall(activeCleanup)activeCleanup=nil end if activeCard and activeCard.Enabled then pcall(function()activeCard:Toggle(false)end)end activeCard=nil Module.Initialized=false end
return Module
-- src/games/universal/Other/AntiAfk.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='AntiAfk',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.init(context
:Runtime):any local host:any=context.host local LocalPlayer:any=host.LocalPlayer local featureConnections:any=host.featureConnections local disconnectFeatureConnection:any=host.
disconnectFeatureConnection local createUniversalFeature:any=host.createUniversalFeature local addFeatureTooltip:any=host.addFeatureTooltip local function pokeAfk():boolean local camera:Camera?=host.
workspace.CurrentCamera local centre:Vector2=Vector2.new(400,300)if camera then centre=camera.ViewportSize/2 end local virtualUser:any=nil pcall(function():()virtualUser=(game::any):GetService(
'VirtualUser')end)if virtualUser then local ok:boolean=pcall(function():()virtualUser:CaptureController()virtualUser:Button2Down(centre,camera and camera.CFrame)virtualUser:Button2Up(centre,camera and
camera.CFrame)end)if ok then return true end local legacyOk:boolean=pcall(function():()virtualUser:ClickButton2(centre)end)if legacyOk then return true end end local virtualInput:any=nil pcall(
function():()virtualInput=(game::any):GetService('VirtualInputManager')end)if virtualInput then local ok:boolean=pcall(function():()virtualInput:SendMouseButtonEvent(centre.X,centre.Y,0,true,game,
false)virtualInput:SendMouseButtonEvent(centre.X,centre.Y,0,false,game,false)end)if ok then return true end end return false end local function toggleAntiAfk(enabled:boolean):()
disconnectFeatureConnection('AntiAFK')disconnectFeatureConnection('AntiAFKTimer')if not enabled then return end featureConnections.AntiAFK=LocalPlayer.Idled:Connect(function():()pokeAfk()end)local
elapsed:number=0 featureConnections.AntiAFKTimer=host.TaskManager:Connect(function(deltaTime:number):()elapsed+=deltaTime if elapsed<150 then return end elapsed=0 pokeAfk()end,nil,'Anti AFK')end local
AntiAfkFeature=createUniversalFeature('Anti AFK',[[Simulate input before Roblox's idle kick (about every 2.5 minutes, and on the idle warning)]],13,toggleAntiAfk,{noOptions=true,configKey=
'Universal.AntiAFK',categoryName='World'})addFeatureTooltip(AntiAfkFeature,'Attempts input every ~150 seconds and on Idled. '..
[[Tries VirtualUser first, then VirtualInputManager; availability depends on the executor. ]]..'Disconnects immediately when disabled.')activeCleanup=function():()disconnectFeatureConnection('AntiAFK'
)disconnectFeatureConnection('AntiAFKTimer')end Module.Initialized=true return AntiAfkFeature end function Module.destroy():()if activeCleanup then activeCleanup()end activeCleanup=nil Module.
Initialized=false end return Module
-- src/games/universal/Other/Disguise.lua
export type Runtime={framework:any,host:any,services:any}type AppearanceTemplate={object:Instance,parent:Instance?}type MeshTemplate={part:MeshPart,meshId:string,textureId:string}type
AnimationTemplate={animation:Animation,animationId:string}type AppearanceSnapshot={character:Model,items:{AppearanceTemplate},meshParts:{MeshTemplate},animations:{AnimationTemplate},emotes:any,
equippedEmotes:any,scales:{[string]:number},descAnimations:{[string]:number}}local Module={Name='Disguise',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCard:any=nil local
DEFAULT_USER_ID:number=78492 local DEFAULT_ANIMATION_IDS:{[string]:{string}}={idle={'507766666','507766951','507766388'},walk={'507777826'},run={'507767714'},jump={'507765000'},fall={'507767968'},
climb={'507765644'},swim={'507784897'},swimidle={'507785072'}}local DEFAULT_ANIMATION_IDS_R6:{[string]:{string}}={idle={'180435571','180435792'},walk={'180426354'},run={'180426354'},jump={'125750702'}
,fall={'180436148'},climb={'180436334'}}local R15_DEFAULT_IDS:{[string]:boolean}={}for _,ids:{string}in pairs(DEFAULT_ANIMATION_IDS)do for _,id:string in ipairs(ids)do R15_DEFAULT_IDS[id]=true end end
local R6_DEFAULT_IDS:{[string]:boolean}={}for _,ids:{string}in pairs(DEFAULT_ANIMATION_IDS_R6)do for _,id:string in ipairs(ids)do R6_DEFAULT_IDS[id]=true end end local SCALE_FIELDS:{string}={
'HeightScale','WidthScale','DepthScale','HeadScale','BodyTypeScale','ProportionScale'}local BODY_PARTS:{[string]:boolean}={Head=true,Torso=true,UpperTorso=true,LowerTorso=true,LeftArm=true,RightArm=
true,LeftLeg=true,RightLeg=true,LeftUpperArm=true,LeftLowerArm=true,LeftHand=true,RightUpperArm=true,RightLowerArm=true,RightHand=true,LeftUpperLeg=true,LeftLowerLeg=true,LeftFoot=true,RightUpperLeg=
true,RightLowerLeg=true,RightFoot=true}local MOVEMENT_ANIMATIONS:{{descriptionField:string,scriptCategory:string}}={{descriptionField='IdleAnimation',scriptCategory='idle'},{descriptionField=
'WalkAnimation',scriptCategory='walk'},{descriptionField='RunAnimation',scriptCategory='run'},{descriptionField='JumpAnimation',scriptCategory='jump'},{descriptionField='ClimbAnimation',scriptCategory
='climb'},{descriptionField='FallAnimation',scriptCategory='fall'},{descriptionField='SwimAnimation',scriptCategory='swim'},{descriptionField='SwimIdleAnimation',scriptCategory='swimidle'},{
descriptionField='MoodAnimation',scriptCategory='mood'}}function Module.init(context:Runtime):any local framework:any=context.framework local host:any=context.host local Players:Players=host.Players
or(game::any):GetService('Players')local LocalPlayer:Player=host.LocalPlayer or Players.LocalPlayer local cloned=setmetatable({},{__mode='k'})::{[Instance]:boolean}local disguisedItems=setmetatable({}
,{__mode='k'})::{[Instance]:boolean}local snapshot:AppearanceSnapshot?=nil local filteredCharacter:Model?=nil local filterConnection:RBXScriptConnection?=nil local generation:number=0 local restoring:
boolean=false local emotesAppliedToHumanoid:boolean=false local animationOverrideApplied:boolean=false local foreignPackNotified:boolean=false local lastAppliedSlotCount:number=0 local
lastDroppedSlotCount:number=0 local stopPoseMirror:()->()=function():()end local playingAnimationTracks:(Humanoid?)->{AnimationTrack}=function():{AnimationTrack}return{}end type TargetCache={userId:
number,slots:{[string]:any},emotes:any,equippedEmotes:any,rig:string?}local targetCache:TargetCache?=nil local function savedToggle(name:string,fallback:boolean):boolean local store:any=host.
configData local states:any=store and store.states if type(states)~='table'then return fallback end local saved:any=states['Universal.Disguise.'..name:gsub('%W','')]if type(saved)=='boolean'then
return saved end return fallback end local disguiseCharDefault:boolean=savedToggle('Disguise Character',true)local settings={disguiseCharacter=disguiseCharDefault,mode=if disguiseCharDefault then
'Character'else'Animation',useTargetMovementAnimations=true,animationSpeed=1,id=tostring(DEFAULT_USER_ID),showTargetEmotes=savedToggle('Show target emotes',true),showAdvanced=false,customPackSlots=(
nil::any),customPackName=(nil::any),customSlotIds=({}::any)}local function status(message:string):()if activeCard then activeCard:SetStatus(message)end end local function notify(message:string):()
pcall(print,'[Wurst:Disguise] '..message)if activeCard then pcall(activeCard.Notify,activeCard,message)end end local function isAppearanceItem(obj:Instance):boolean return obj:IsA('Accessory')or obj:
IsA('ShirtGraphic')or obj:IsA('Shirt')or obj:IsA('Pants')or obj:IsA('BodyColors')or((obj:IsA('Decal')or obj:IsA('Texture'))and string.lower(obj.Name)=='face')end local function isInsideCharacter(obj:
Instance,character:Model):boolean local ok:boolean,result:any=pcall(function()return obj:IsDescendantOf(character)end)return ok and result==true end local function rebindAccessory(character:Model,
accessory:Accessory):()for _,descendant:Instance in ipairs(accessory:GetDescendants())do if descendant:IsA('Weld')and descendant.Part1 then local part:Instance?=character:FindFirstChild(descendant.
Part1.Name)if part and part:IsA('BasePart')then descendant.Part1=part end elseif descendant:IsA('RigidConstraint')and descendant.Attachment1 then local attachment:Instance?=character:FindFirstChild(
descendant.Attachment1.Name,true)if attachment and attachment:IsA('Attachment')then descendant.Attachment1=attachment end end end end local function destroySnapshot():()local current:
AppearanceSnapshot?=snapshot snapshot=nil emotesAppliedToHumanoid=false animationOverrideApplied=false if current then for _,entry:AppearanceTemplate in ipairs(current.items)do pcall(entry.object.
Destroy,entry.object)end end end local function getAppliedDescription(humanoid:Humanoid):(any?,boolean)local child:HumanoidDescription?=humanoid:FindFirstChildOfClass('HumanoidDescription')::
HumanoidDescription?if not child then local waitOk:boolean,waited:any=pcall(function()return humanoid:WaitForChild('HumanoidDescription',2)end)if waitOk and waited and waited:IsA('HumanoidDescription'
)then child=waited::HumanoidDescription end end if child then return child,false end local ok:boolean,result:any=pcall(function()return humanoid:GetAppliedDescription()end)if ok and result then return
result,true end return nil,false end local function cloneTable(value:any):any if type(value)~='table'then return value end local copy:any={}for key:any,entry:any in pairs(value)do copy[key]=if type(
entry)=='table'then table.clone(entry)else entry end return copy end local function readDescriptionEmotes(description:any):(any?,any?)local ok:boolean,emotes:any,equipped:any=pcall(function()return
description:GetEmotes(),description:GetEquippedEmotes()end)if not ok or type(emotes)~='table'or type(equipped)~='table'then return nil,nil end return cloneTable(emotes),cloneTable(equipped)end
local function applyDescriptionToHumanoid(humanoid:Humanoid,description:any):(boolean,string?)local humanoidObject:any=humanoid local lastError:string='ApplyDescription is unavailable.'for _,
methodName:string in ipairs({'ApplyDescriptionResetAsync','ApplyDescriptionAsync','ApplyDescriptionReset','ApplyDescription'})do local method:any=humanoidObject[methodName]if type(method)=='function'
then local ok:boolean,result:any=pcall(method,humanoidObject,description)if ok then return true,nil end lastError=tostring(result)end end return false,lastError end local function setHumanoidEmotes(
humanoid:Humanoid,emotes:any,equippedEmotes:any):(boolean,string?)if type(emotes)~='table'or type(equippedEmotes)~='table'then return false,'The avatar description did not expose emote data.'end local
description:any,isTemporary:boolean=getAppliedDescription(humanoid)if not description then return false,'Could not read the current HumanoidDescription.'end local setOk:boolean,setError:any=pcall(
function()description:SetEmotes(cloneTable(emotes))description:SetEquippedEmotes(cloneTable(equippedEmotes))end)if not setOk then if isTemporary then pcall(description.Destroy,description)end return
false,tostring(setError)end if not isTemporary then return true,nil end local applied:boolean,applyError:string?=applyDescriptionToHumanoid(humanoid,description)pcall(description.Destroy,description)
return applied,applyError end local function captureSnapshot(character:Model,humanoid:Humanoid):AppearanceSnapshot local items:{AppearanceTemplate}={}local meshParts:{MeshTemplate}={}local animations:
{AnimationTemplate}={}local scales:{[string]:number}={}local descAnimations:{[string]:number}={}local emotes:any={}local equippedEmotes:any={}for _,obj:Instance in ipairs(character:GetDescendants())do
if isAppearanceItem(obj)then local ok:boolean,copy:any=pcall(function()return obj:Clone()end)if ok and copy then copy.Parent=nil table.insert(items,{object=copy,parent=obj.Parent})end elseif obj:IsA(
'MeshPart')and BODY_PARTS[obj.Name]then table.insert(meshParts,{part=obj,meshId=obj.MeshId,textureId=obj.TextureID})elseif obj:IsA('Animation')then local animate:Instance?=character:FindFirstChild(
'Animate')if animate and obj:IsDescendantOf(animate)then table.insert(animations,{animation=obj,animationId=obj.AnimationId})end end end local description:any,isTemporary:boolean=
getAppliedDescription(humanoid)if description then for _,field:string in ipairs(SCALE_FIELDS)do local ok:boolean,value:any=pcall(function()return description[field]end)if ok and type(value)=='number'
then scales[field]=value end end for _,mapping in ipairs(MOVEMENT_ANIMATIONS)do local okField:boolean,fieldValue:any=pcall(function()return description[mapping.descriptionField]end)if okField and
type(fieldValue)=='number'then descAnimations[mapping.descriptionField]=fieldValue end end local originalEmotes:any,originalEquipped:any=readDescriptionEmotes(description)if originalEmotes and
originalEquipped then emotes=originalEmotes equippedEmotes=originalEquipped end if isTemporary then pcall(description.Destroy,description)end end return{character=character,items=items,meshParts=
meshParts,animations=animations,emotes=emotes,equippedEmotes=equippedEmotes,scales=scales,descAnimations=descAnimations}end local function ensureSnapshot(character:Model,humanoid:Humanoid):
AppearanceSnapshot if snapshot and snapshot.character==character then return snapshot end destroySnapshot()snapshot=captureSnapshot(character,humanoid)return snapshot end local function
restartAnimateScript(animate:Instance):()local scriptObject:any=animate local enabledOk:boolean,wasEnabled:any=pcall(function()return scriptObject.Enabled end)if enabledOk and type(wasEnabled)==
'boolean'then if not wasEnabled then return end local disabledOk:boolean=pcall(function()scriptObject.Enabled=false end)if disabledOk then task.wait()pcall(function()scriptObject.Enabled=wasEnabled
end)task.wait()return end end local disabledOk:boolean,wasDisabled:any=pcall(function()return scriptObject.Disabled end)if disabledOk and type(wasDisabled)=='boolean'then local pauseOk:boolean=pcall(
function()scriptObject.Disabled=true end)if pauseOk then task.wait()pcall(function()scriptObject.Disabled=wasDisabled end)task.wait()end end end local function restoreMovementAnimationsOnly(character:
Model,humanoid:Humanoid?):()stopPoseMirror()local current:AppearanceSnapshot?=snapshot if not current or current.character~=character or not animationOverrideApplied then return end if humanoid and
current.descAnimations then local description:any,isTemporary:boolean=getAppliedDescription(humanoid)if description then local working:any=description if not isTemporary then local cloneOk:boolean,
clone:any=pcall(function():any return description:Clone()end)if cloneOk and clone then working=clone isTemporary=true end end pcall(function():()for field:string,asset:number in pairs(current.
descAnimations)do working[field]=asset end end)applyDescriptionToHumanoid(humanoid,working)if isTemporary then pcall(function():()working:Destroy()end)end end end for _,entry:AnimationTemplate in
ipairs(current.animations)do if isInsideCharacter(entry.animation,character)then pcall(function()entry.animation.AnimationId=entry.animationId end)end end animationOverrideApplied=false local animate:
Instance?=character:FindFirstChild('Animate')if not animate then return end if humanoid then local animator:Animator?=humanoid:FindFirstChildOfClass('Animator')if animator then local tracksOk:boolean,
tracks:any=pcall(function()return animator:GetPlayingAnimationTracks()end)if tracksOk and type(tracks)=='table'then for _,track:AnimationTrack in ipairs(tracks)do pcall(track.Stop,track)end end end
end restartAnimateScript(animate)end local function animationIdFromValue(value:any):string?local number:number?=nil if type(value)=='number'and value>0 then number=value elseif type(value)=='string'
then local matched:string?=string.match(value,'%d+')if matched then number=tonumber(matched)end end if not number or number<=0 then return nil end return'rbxassetid://'..tostring(math.floor(number))
end local function animationSlotsFromDescription(description:any):{[string]:any}local slots:{[string]:string}={}for _,mapping in ipairs(MOVEMENT_ANIMATIONS)do local readOk:boolean,value:any=pcall(
function()return description[mapping.descriptionField]end)local assetId:any=if readOk then value else nil local chosenId:string?=animationIdFromValue(assetId)if chosenId then slots[mapping.
scriptCategory]=chosenId end end return slots end local function rigDefaultSlots(humanoid:Humanoid?):{[string]:any}local rig:any=nil if humanoid then pcall(function():()rig=(humanoid::any).RigType end
)end if rig==Enum.HumanoidRigType.R6 then return DEFAULT_ANIMATION_IDS_R6 end return DEFAULT_ANIMATION_IDS end local function rigName(humanoid:Humanoid?):string?local rig:any=nil if humanoid then
pcall(function():()rig=(humanoid::any).RigType end)end if rig==Enum.HumanoidRigType.R6 then return'R6'end if rig==Enum.HumanoidRigType.R15 then return'R15'end return nil end local function
belongsToDefaultSet(value:any,set:{[string]:boolean}):boolean local function owns(entry:any):boolean if type(entry)=='number'then entry=tostring(math.floor(entry))end if type(entry)~='string'then
return false end local digits:string?=string.match(entry,'%d+')return digits~=nil and set[digits]==true end if type(value)=='table'then if#value==0 then return false end for _,entry:any in ipairs(
value)do if not owns(entry)then return false end end return true end return owns(value)end local function belongsToR15Defaults(value:any):boolean return belongsToDefaultSet(value,R15_DEFAULT_IDS)end
local function belongsToR6Defaults(value:any):boolean return belongsToDefaultSet(value,R6_DEFAULT_IDS)end local function resolvedAnimationSlots(slots:{[string]:any},humanoid:Humanoid?,_targetRig:
string?):({[string]:any},number)local defaults:{[string]:any}=rigDefaultSlots(humanoid)local isR6:boolean=defaults==DEFAULT_ANIMATION_IDS_R6 local dropped:number=0 local resolved:{[string]:any}={}for
category:string,value:any in pairs(slots)do local owned:boolean if isR6 then owned=belongsToR6Defaults(value)else owned=not belongsToR6Defaults(value)end if owned then resolved[category]=value elseif
not belongsToR15Defaults(value)then dropped+=1 end end for category:string,ids:any in pairs(defaults)do if resolved[category]==nil then resolved[category]=ids end end return resolved,dropped end
local function writeAnimationSlots(animate:Instance,slots:{[string]:any}):number local changedCount:number=0 for _,mapping in ipairs(MOVEMENT_ANIMATIONS)do local chosen:any=slots[mapping.
scriptCategory]local ids:{string}={}if type(chosen)=='table'then for _,value:any in ipairs(chosen)do local id:string?=animationIdFromValue(value)if id then table.insert(ids,id)end end else local id:
string?=animationIdFromValue(chosen)if id then table.insert(ids,id)end end local category:Instance?=animate:FindFirstChild(mapping.scriptCategory)if#ids>0 and category then local targets:{Animation}={
}if category:IsA('Animation')then table.insert(targets,category)end for _,descendant:Instance in ipairs(category:GetDescendants())do if descendant:IsA('Animation')then table.insert(targets,descendant)
end end for index:number,animation:Animation in ipairs(targets)do local chosenId:string=#ids==1 and ids[1]or ids[math.min(index,#ids)]if animation.AnimationId~=chosenId then local changed:boolean=
pcall(function()animation.AnimationId=chosenId end)if changed then changedCount+=1 end end end end end return changedCount end local function applySlotsViaDescription(humanoid:Humanoid,resolved:{[
string]:any}):number local description:any,isTemporary:boolean=getAppliedDescription(humanoid)if not description then return 0 end local working:any=description if not isTemporary then local cloneOk:
boolean,clone:any=pcall(function():any return description:Clone()end)if cloneOk and clone then working=clone isTemporary=true end end local setCount:number=0 local setOk:boolean=pcall(function():()for
_,mapping in ipairs(MOVEMENT_ANIMATIONS)do local chosen:any=resolved[mapping.scriptCategory]if type(chosen)=='table'then chosen=chosen[1]end local digits:string?=chosen and string.match(tostring(
chosen),'%d+')or nil local asset:number?=digits and tonumber(digits)or nil if asset and asset>0 then working[mapping.descriptionField]=asset setCount+=1 end end end)if not setOk or setCount==0 then if
isTemporary then pcall(function():()working:Destroy()end)end return 0 end local applied:boolean=applyDescriptionToHumanoid(humanoid,working)if isTemporary then pcall(function():()working:Destroy()end)
end return applied and setCount or 0 end local sanitizedSlots:(humanoid:Humanoid,resolved:{[string]:any})->({[string]:any},number)=function(_humanoid:Humanoid,resolved:{[string]:any}):({[string]:any},
number)return resolved,0 end local invalidClipNotified:boolean=false local preloadedClipIds:{[string]:boolean}={}local function preloadAnimations(ids:{string}):()local pending:{string}={}for _,id:
string in ipairs(ids)do if not preloadedClipIds[id]then preloadedClipIds[id]=true table.insert(pending,id)end end if#pending==0 then return end pcall(function():()local provider:any=(host::any).
ContentProvider or game:GetService('ContentProvider')if not provider or type(provider.PreloadAsync)~='function'then return end local probes:{Instance}={}for _,id:string in ipairs(pending)do local ok:
boolean,probe:any=pcall(function():any local animation:any=Instance.new('Animation')animation.AnimationId=id return animation end)if ok and probe then table.insert(probes,probe)end end if#probes>0
then pcall(function():()provider:PreloadAsync(probes)end)for _,probe:Instance in ipairs(probes)do pcall(function():()probe:Destroy()end)end end end)end local function awaitAnimateScript(character:
Model):Instance?local animate:Instance?=character:FindFirstChild('Animate')if animate then return animate end local runService:any=(host::any).RunService if not runService then pcall(function():()
runService=game:GetService('RunService')end)end if not runService then return nil end for _=1,240 do local stepped:boolean=pcall(function():()runService.Heartbeat:Wait()end)if not stepped or character
.Parent==nil then return nil end animate=character:FindFirstChild('Animate')if animate then return animate end end return nil end local function applyAnimationSlots(character:Model,humanoid:Humanoid,
slots:{[string]:any},targetRig:string?):number local animate:Instance?=awaitAnimateScript(character)if not animate then return 0 end local resolved:{[string]:any},dropped:number=
resolvedAnimationSlots(slots,humanoid,targetRig)lastDroppedSlotCount=dropped if dropped>0 and not foreignPackNotified then foreignPackNotified=true notify("That pack's clips are not "..(rigName(
humanoid)or'your rig')..' clips; your own animations stay.')end local sanitized:{[string]:any},replacedClips:number=sanitizedSlots(humanoid,resolved)resolved=sanitized lastResolvedSlots=resolved if
replacedClips>0 and not invalidClipNotified then invalidClipNotified=true notify(tostring(replacedClips)..' of those animation slots cannot be used in '..'this game (the engine refuses the clips); '..
"your rig's own clips fill them instead.")end local clipIds:{string}={}for _,value:any in pairs(resolved)do local list:{any}=if type(value)=='table'then value else{value}for _,entry:any in ipairs(list
)do local id:string?=animationIdFromValue(entry)if id then table.insert(clipIds,id)end end end preloadAnimations(clipIds)local changedCount:number=writeAnimationSlots(animate,resolved)if changedCount
==0 then changedCount=applySlotsViaDescription(humanoid,resolved)end if changedCount==0 then return 0 end animationOverrideApplied=true for _,track:AnimationTrack in ipairs(playingAnimationTracks(
humanoid))do pcall(function():()track:Stop()end)end restartAnimateScript(animate)return changedCount end local function targetRigFor(userId:number):string?local ok:boolean,player:any=pcall(function():
any return Players:GetPlayerByUserId(userId)end)if not ok or not player then return nil end local character:any=player.Character local humanoid:any=character and character:FindFirstChildOfClass(
'Humanoid')if not humanoid then return nil end return rigName(humanoid)end local function cacheTarget(userId:number,description:any):()local emotes:any,equippedEmotes:any=readDescriptionEmotes(
description)local slots:{[string]:any}=animationSlotsFromDescription(description)foreignPackNotified=false targetCache={userId=userId,slots=slots,emotes=emotes,equippedEmotes=equippedEmotes,rig=
targetRigFor(userId)}end local function refreshMeshPartSnapshot(character:Model,current:AppearanceSnapshot):()local originalByName:{[string]:{meshId:string,textureId:string}}={}for _,entry:
MeshTemplate in ipairs(current.meshParts)do originalByName[entry.part.Name]={meshId=entry.meshId,textureId=entry.textureId}end local refreshed:{MeshTemplate}={}for _,obj:Instance in ipairs(character:
GetDescendants())do if obj:IsA('MeshPart')and BODY_PARTS[obj.Name]then local original:{meshId:string,textureId:string}?=originalByName[obj.Name]table.insert(refreshed,{part=obj,meshId=original and
original.meshId or obj.MeshId,textureId=original and original.textureId or obj.TextureID})end end if#refreshed>0 then current.meshParts=refreshed end end playingAnimationTracks=function(humanoid:
Humanoid?):{AnimationTrack}local playing:{AnimationTrack}={}if not humanoid then return playing end local animator:Animator?=humanoid:FindFirstChildOfClass('Animator')if not animator then return
playing end local ok:boolean,tracks:any=pcall(function():any return animator:GetPlayingAnimationTracks()end)if not ok or type(tracks)~='table'then return playing end for _,track:AnimationTrack in
ipairs(tracks)do local isPlaying:boolean=false pcall(function():()isPlaying=track.IsPlaying==true end)if isPlaying then table.insert(playing,track)end end return playing end local function
poseSignature(humanoid:Humanoid?):number?local character:Instance?=humanoid and humanoid.Parent if not character then return nil end local joints:number=0 local signature:number=0 for _,object:
Instance in ipairs(character:GetDescendants())do if object:IsA('Motor6D')or object:IsA('AnimationConstraint')then local ok:boolean,transform:CFrame=pcall(function():CFrame return(object::any).
Transform end)if ok and transform then joints+=1 signature+=transform.X*13.17+transform.Y*7.31+transform.Z*3.77+transform.R00*29.1+transform.R11*17.3+transform.R22*11.9 end end end if joints==0 then
return nil end return signature end local function poseDriven(humanoid:Humanoid?):boolean local character:Instance?=humanoid and humanoid.Parent if not character then return false end for _,object:
Instance in ipairs(character:GetDescendants())do if object:IsA('Motor6D')or object:IsA('AnimationConstraint')then local ok:boolean,transform:CFrame=pcall(function():CFrame return(object::any).
Transform end)if ok and transform then if math.abs(transform.X)>1e-4 or math.abs(transform.Y)>1e-4 or math.abs(transform.Z)>1e-4 or math.abs(transform.R00-1)>1e-3 or math.abs(transform.R11-1)>1e-3 or
math.abs(transform.R22-1)>1e-3 then return true end end end end return false end local function poseMoves(humanoid:Humanoid?,seconds:number):boolean?local before:number?=poseSignature(humanoid)if
before==nil then return nil end local steps:number=5 for _=1,steps do task.wait(seconds/steps)local after:number?=poseSignature(humanoid)if after==nil then return nil end if math.abs(after-before)>
1e-3 then return true end end return false end local clipValidity:{[string]:boolean}={}local function clipLoads(_humanoid:Humanoid,id:string):boolean local cached:boolean?=clipValidity[id]if cached~=
nil then return cached end local valid:boolean=true if belongsToR15Defaults(id)or belongsToR6Defaults(id)then clipValidity[id]=true return true end local digits:string?=string.match(id,'%d+')local
assetId:number?=digits and tonumber(digits)or nil if assetId then local service:any=nil pcall(function():()service=(host::any).MarketplaceService or(game::any):GetService('MarketplaceService')end)if
service and type(service.GetProductInfo)=='function'then local infoOk:boolean,info:any=pcall(function():any return service:GetProductInfo(assetId,Enum.InfoType.Asset)end)if not infoOk or type(info)~=
'table'then valid=false else local assetType:any=info.AssetTypeId if assetType~=nil then local isAnimation:boolean=false pcall(function():()if type(assetType)=='number'then isAnimation=assetType==Enum
.AssetType.Animation.Value else isAnimation=assetType==Enum.AssetType.Animation end end)if not isAnimation then valid=false end end end end end clipValidity[id]=valid return valid end sanitizedSlots=
function(humanoid:Humanoid,resolved:{[string]:any}):({[string]:any},number)local defaults:{[string]:any}=rigDefaultSlots(humanoid)local out:{[string]:any}={}local replaced:number=0 for category:string
,value:any in pairs(resolved)do local kept:{string}={}local list:{any}=type(value)=='table'and value or{value}for _,entry:any in ipairs(list)do local id:string?=animationIdFromValue(entry)if id and
clipLoads(humanoid,id)then table.insert(kept,id)end end if#kept>0 then out[category]=#kept==1 and kept[1]or kept else replaced+=1 out[category]=defaults[category]end end return out,replaced end local
mirrorAllowed:boolean=false local lastResolvedSlots:{[string]:any}?=nil local function retryAnimateScript(character:Model,humanoid:Humanoid,attempts:number):boolean local animate:Instance?=character:
FindFirstChild('Animate')if not animate then return false end for attempt:number=1,attempts do restartAnimateScript(animate)task.wait(0.2+attempt*0.15)if character.Parent==nil then return false end
local tracks:number=#playingAnimationTracks(humanoid)if tracks>0 and(poseMoves(humanoid,0.25)~=false or poseDriven(humanoid))then return true end if tracks==0 and attempt==1 and lastResolvedSlots then
writeAnimationSlots(animate,lastResolvedSlots)end end return false end local function settleMovementAnimations(character:Model,humanoid:Humanoid):boolean if not animationOverrideApplied then return
false end local animate:Instance?=character:FindFirstChild('Animate')if not animate then return false end local tracks:number=#playingAnimationTracks(humanoid)if tracks>0 then if poseMoves(humanoid,
0.3)~=false then return true end if poseDriven(humanoid)then return true end end if retryAnimateScript(character,humanoid,3)then return true end if mirrorAllowed and lastResolvedSlots then local
viaDescription:number=applySlotsViaDescription(humanoid,lastResolvedSlots)if viaDescription>0 then restartAnimateScript(animate)task.wait(0.15)if#playingAnimationTracks(humanoid)>0 and(poseMoves(
humanoid,0.3)~=false or poseDriven(humanoid))then notify("The target's animations were applied through "..'their avatar description.')return true end end end if mirrorAllowed and startPoseMirror(
character,humanoid)then return true end local defaults:{[string]:any}=rigDefaultSlots(humanoid)local isR6:boolean=defaults==DEFAULT_ANIMATION_IDS_R6 if writeAnimationSlots(animate,defaults)>0 then for
_,track:AnimationTrack in ipairs(playingAnimationTracks(humanoid))do pcall(function():()track:Stop()end)end restartAnimateScript(animate)task.wait(0.1)if#playingAnimationTracks(humanoid)>0 and
poseMoves(humanoid,0.3)~=false then notify('Those clips do not play on your '..(isR6 and'R6'or'R15').." rig; your rig's own set is back.")return true end end restoreMovementAnimationsOnly(character,
humanoid)notify("The target's animations do not play on your rig; "..'your own animations are back.')return false end local function animationAssetId(itemId:any):string?local normalizedId:string?=
animationIdFromValue(itemId)if not normalizedId then return nil end if type((game::any).GetObjects)=='function'then local ok:boolean,objects:any=pcall(function():any return(game::any):GetObjects(
normalizedId)end)if ok and type(objects)=='table'then for _,object:any in ipairs(objects)do if typeof(object)=='Instance'then if object:IsA('Animation')then local id:string?=animationIdFromValue(
object.AnimationId)if id then return id end end for _,descendant:Instance in ipairs(object:GetDescendants())do if descendant:IsA('Animation')then local id:string?=animationIdFromValue(descendant.
AnimationId)if id then return id end end end end end end end return nil end local function bundleItemCategory(rawName:any):string?local name:string=string.lower(tostring(rawName or''))local hasSwim:
boolean=string.find(name,'swim',1,true)~=nil local hasIdle:boolean=string.find(name,'idle',1,true)~=nil if hasSwim and hasIdle then return'swimidle'end if hasIdle then return'idle'end if string.find(
name,'walk',1,true)then return'walk'end if string.find(name,'run',1,true)then return'run'end if string.find(name,'jump',1,true)then return'jump'end if string.find(name,'climb',1,true)then return
'climb'end if string.find(name,'fall',1,true)then return'fall'end if hasSwim then return'swim'end return nil end local function slotsFromBundleItems(items:any):{[string]:any}local slots:{[string]:any}
={}if type(items)~='table'then return slots end for _,item:any in ipairs(items)do if type(item)=='table'then local category:string?=bundleItemCategory(item.Name or item.name or item.DisplayName or
item.displayName or item.Type or item.type)local itemId:any=item.Id or item.id or item.AssetId or item.assetId if category and itemId then local id:string?=animationAssetId(itemId)or
animationIdFromValue(itemId)if id then local existing:any=slots[category]if type(existing)=='table'then table.insert(existing,id)elseif existing~=nil then slots[category]={existing,id}else slots[
category]=id end end end end return slots end local function bundleAnimationSlots(bundleId:number):({[string]:any}?,string?)local service:any=(host::any).MarketplaceService or(game::any):
```

```
GetService('MarketplaceService')local ok:boolean,product:any=pcall(function():any return service:GetProductInfo(bundleId,Enum.InfoType.Bundle)end)if ok and type(product)=='table'and(product.BundleType
==nil or product.BundleType=='AvatarAnimations')and type(product.Items)=='table'then local slots:{[string]:any}=slotsFromBundleItems(product.Items)if next(slots)~=nil then return slots,nil end end
local httpOk:boolean,body:any=pcall(function():any return(game::any):HttpGet('https://catalog.roblox.com/v1/bundles/'..tostring(math.floor(bundleId))..'/details')end)if httpOk and type(body)=='string'
then local httpService:any=(game::any):GetService('HttpService')local decodeOk:boolean,decoded:any=pcall(function():any return httpService:JSONDecode(body)end)if decodeOk and type(decoded)=='table'and
type(decoded.items)=='table'then local slots:{[string]:any}=slotsFromBundleItems(decoded.items)if next(slots)~=nil then return slots,nil end end end if ok and type(product)=='table'and product.
BundleType~=nil and product.BundleType~='AvatarAnimations'then return nil,'That bundle is not an animation pack.'end return nil,[[That ID is not a user and no animation pack items were found for it.]]
end local function targetAnimateSlots(userId:number):{[string]:any}?local player:any=nil pcall(function():()player=Players:GetPlayerByUserId(userId)end)local targetCharacter:any=player and player.
Character local animate:any=targetCharacter and targetCharacter:FindFirstChild('Animate')if not animate then return nil end local slots:{[string]:any}={}for _,mapping in ipairs(MOVEMENT_ANIMATIONS)do
local container:any=animate:FindFirstChild(mapping.scriptCategory)local animation:any=nil if container then if container:IsA('Animation')then animation=container else for _,descendant:any in ipairs(
container:GetDescendants())do if descendant:IsA('Animation')then animation=descendant break end end end end if animation then local id:string?=animationIdFromValue(animation.AnimationId)if id then
slots[mapping.scriptCategory]=id end end end if next(slots)==nil then return nil end return slots end local function applyCachedMovementAnimations(character:Model,humanoid:Humanoid):number local
cached:TargetCache?=targetCache if not cached then return 0 end local slots:{[string]:any}=targetAnimateSlots(cached.userId)or cached.slots local changedCount:number=applyAnimationSlots(character,
humanoid,slots,cached.rig)if changedCount>0 or animationOverrideApplied then mirrorAllowed=true settleMovementAnimations(character,humanoid)mirrorAllowed=false end return changedCount end
local function selectedPackSlots():{[string]:any}?return settings.customPackSlots end local function packDisplayName():string return settings.customPackName or'Custom'end local function
applyAnimationPack(character:Model,humanoid:Humanoid):number stopPoseMirror()local slots:{[string]:any}?=selectedPackSlots()if not slots then return 0 end local changedCount:number=
applyAnimationSlots(character,humanoid,slots,nil)if changedCount>0 or animationOverrideApplied then settleMovementAnimations(character,humanoid)end return changedCount end local function
applyCachedEmotes(character:Model,humanoid:Humanoid):boolean local cached:TargetCache?=targetCache local current:AppearanceSnapshot?=snapshot if not cached or not current or current.character~=
character then return false end if type(cached.emotes)~='table'or type(cached.equippedEmotes)~='table'then return false end local applied:boolean=setHumanoidEmotes(humanoid,cached.emotes,cached.
equippedEmotes)if not applied then return false end emotesAppliedToHumanoid=true refreshMeshPartSnapshot(character,current)return true end local function restoreEmotesOnly(character:Model,humanoid:
Humanoid?):()local current:AppearanceSnapshot?=snapshot if not humanoid or not current or current.character~=character or not emotesAppliedToHumanoid then return end local restored:boolean=
setHumanoidEmotes(humanoid,current.emotes,current.equippedEmotes)if restored then emotesAppliedToHumanoid=false refreshMeshPartSnapshot(character,current)end end local function
restoreOriginalAppearance(character:Model,humanoid:Humanoid?):()local current:AppearanceSnapshot?=snapshot if not current or current.character~=character then return end restoring=true if
emotesAppliedToHumanoid and humanoid then local emotesRestored:boolean=setHumanoidEmotes(humanoid,current.emotes,current.equippedEmotes)if emotesRestored then emotesAppliedToHumanoid=false
refreshMeshPartSnapshot(character,current)end end local removals:{Instance}={}table.clear(disguisedItems)table.clear(cloned)for _,obj:Instance in ipairs(character:GetDescendants())do if
isAppearanceItem(obj)then table.insert(removals,obj)end end for _,obj:Instance in ipairs(removals)do pcall(obj.Destroy,obj)end for _,entry:MeshTemplate in ipairs(current.meshParts)do if
isInsideCharacter(entry.part,character)then pcall(function()entry.part.MeshId=entry.meshId entry.part.TextureID=entry.textureId end)end end for _,entry:AnimationTemplate in ipairs(current.animations)
do if isInsideCharacter(entry.animation,character)then pcall(function()entry.animation.AnimationId=entry.animationId end)end end local restartRestoredAnimations:boolean=animationOverrideApplied
animationOverrideApplied=false for _,entry:AppearanceTemplate in ipairs(current.items)do local ok:boolean,copy:any=pcall(function()return entry.object:Clone()end)if ok and copy then local parent:
Instance=entry.parent or character if parent~=character and not isInsideCharacter(parent,character)then if(copy:IsA('Decal')or copy:IsA('Texture'))and string.lower(copy.Name)=='face'then parent=
character:FindFirstChild('Head')or character else parent=character end end if copy:IsA('Accessory')then rebindAccessory(character,copy)end pcall(function()copy.Parent=parent end)end end restoring=
false if restartRestoredAnimations then local animate:Instance?=character:FindFirstChild('Animate')if animate then if humanoid then local animator:Animator?=humanoid:FindFirstChildOfClass('Animator')
if animator then local tracksOk:boolean,tracks:any=pcall(function()return animator:GetPlayingAnimationTracks()end)if tracksOk and type(tracks)=='table'then for _,track:AnimationTrack in ipairs(tracks)
do pcall(track.Stop,track)end end end end restartAnimateScript(animate)end end end local function stopAppearanceFilter():()if filterConnection then pcall(filterConnection.Disconnect,filterConnection)
filterConnection=nil end filteredCharacter=nil end local function removeUnexpectedAppearance(obj:Instance):()if restoring or not isAppearanceItem(obj)or cloned[obj]or disguisedItems[obj]or not
activeCard or not activeCard.Enabled or not settings.disguiseCharacter then return end task.defer(function():()if restoring or not activeCard or not activeCard.Enabled or not settings.
disguiseCharacter or cloned[obj]or disguisedItems[obj]then return end pcall(obj.Destroy,obj)end)end local function installAppearanceFilter(character:Model):()if filteredCharacter==character and
filterConnection and filterConnection.Connected then return end stopAppearanceFilter()filteredCharacter=character filterConnection=character.DescendantAdded:Connect(removeUnexpectedAppearance)if
activeCard then activeCard:Clean(filterConnection)end end local function parseId():(number?,string?)local trimmed:string=tostring(settings.id):gsub('^%s+',''):gsub('%s+$','')if trimmed==''then return
nil,'Enter a positive integer ID or username first.'end local value:number?=tonumber(trimmed)if value and value>0 and value%1==0 then return value,nil end for _,p:Player in ipairs(Players:GetPlayers()
)do if string.lower(p.Name)==string.lower(trimmed)or string.lower(p.DisplayName)==string.lower(trimmed)then return p.UserId,nil end end local ok:boolean,resolvedId:any=pcall(function()return Players:
GetUserIdFromNameAsync(trimmed)end)if ok and type(resolvedId)=='number'and resolvedId>0 then return resolvedId,nil end return nil,'Enter a positive integer ID or valid username.'end local function
getUserDescription(userId:number):(any?,string?)local playersObject:any=Players local asyncGetter:any=playersObject.GetHumanoidDescriptionFromUserIdAsync local syncGetter:any=playersObject.
GetHumanoidDescriptionFromUserId if type(asyncGetter)=='function'then local ok:boolean,result:any=pcall(asyncGetter,playersObject,userId)if ok and result then return result,nil end if not ok then
local asyncError:string=tostring(result)if type(syncGetter)~='function'then return nil,asyncError end end end if type(syncGetter)=='function'then local ok:boolean,result:any=pcall(syncGetter,
playersObject,userId)if ok and result then return result,nil end return nil,tostring(result)end return nil,[[This client does not expose a HumanoidDescription lookup API.]]end local function
stillCurrent(token:number,character:Model):boolean return generation==token and activeCard~=nil and activeCard.Enabled and LocalPlayer.Character==character end local function setDescriptionScales(
description:any,current:AppearanceSnapshot):()for _,field:string in ipairs(SCALE_FIELDS)do local value:number?=current.scales[field]if value~=nil then pcall(function()description[field]=value end)end
end end local function copyCharacterAppearance(character:Model,clone:Model):()local realHead:Instance?=character:FindFirstChild('Head')for _,source:Instance in ipairs(clone:GetChildren())do if
isAppearanceItem(source)then cloned[source]=true disguisedItems[source]=true if source:IsA('Accessory')then rebindAccessory(character,source)end source.Parent=character end end local targetFace:
Instance?=clone:FindFirstChild('face',true)if targetFace and isAppearanceItem(targetFace)and realHead then cloned[targetFace]=true disguisedItems[targetFace]=true targetFace.Parent=realHead end end
local function destroyInstance(instance:Instance?):()if instance then pcall(instance.Destroy,instance)end end local function applyCharacterDisguise(token:number,character:Model,humanoid:Humanoid,
userId:number):()lastDroppedSlotCount=0 local description:any,fetchError:string?=getUserDescription(userId)if not description then if stillCurrent(token,character)then notify(
'Could not load that avatar: '..tostring(fetchError))end return end if not stillCurrent(token,character)then destroyInstance(description)return end cacheTarget(userId,description)local cached:
TargetCache?=targetCache local current:AppearanceSnapshot=ensureSnapshot(character,humanoid)restoreOriginalAppearance(character,humanoid)setDescriptionScales(description,current)local oldArchivable:
boolean=character.Archivable character.Archivable=true local cloneOk:boolean,cloneResult:any=pcall(function()return character:Clone()end)character.Archivable=oldArchivable if not cloneOk or not
cloneResult then destroyInstance(description)notify('Could not stage the disguise character.')return end local clone:Model=cloneResult::Model local parentOk:boolean=pcall(function()clone.Parent=game
end)if not parentOk then destroyInstance(description)destroyInstance(clone)notify('Could not stage the disguise character in this client.')return end local cloneHumanoid:Humanoid?=clone:
FindFirstChildOfClass('Humanoid')if not cloneHumanoid then destroyInstance(description)destroyInstance(clone)notify('The staged character has no Humanoid.')return end local applied:boolean,applyError:
string?=applyDescriptionToHumanoid(cloneHumanoid,description)if not applied then destroyInstance(description)destroyInstance(clone)notify('Could not apply the avatar description: '..tostring(
applyError))return end if not stillCurrent(token,character)then destroyInstance(description)destroyInstance(clone)return end local emoteWarning:string?=nil if settings.showTargetEmotes then if cached
and type(cached.emotes)=='table'and type(cached.equippedEmotes)=='table'then local emotesApplied:boolean,emoteError:string?=setHumanoidEmotes(humanoid,cached.emotes,cached.equippedEmotes)if
emotesApplied then emotesAppliedToHumanoid=true refreshMeshPartSnapshot(character,current)else emoteWarning=tostring(emoteError)local rolledBack:boolean=setHumanoidEmotes(humanoid,current.emotes,
current.equippedEmotes)if not rolledBack then emotesAppliedToHumanoid=true emoteWarning..=' The original emotes could not be restored yet.'end end else emoteWarning=
'The target avatar description did not expose emotes.'end end if not stillCurrent(token,character)then if emotesAppliedToHumanoid then local rolledBack:boolean=setHumanoidEmotes(humanoid,current.
emotes,current.equippedEmotes)if rolledBack then emotesAppliedToHumanoid=false end end destroyInstance(description)destroyInstance(clone)return end installAppearanceFilter(character)restoring=true
local oldItems:{Instance}={}for _,obj:Instance in ipairs(character:GetDescendants())do if isAppearanceItem(obj)then table.insert(oldItems,obj)end end for _,obj:Instance in ipairs(oldItems)do pcall(obj
.Destroy,obj)end restoring=false local copied:boolean,copyError:any=pcall(function()copyCharacterAppearance(character,clone)end)destroyInstance(clone)if not copied then destroyInstance(description)
notify([[The avatar was loaded, but some appearance parts could not be copied: ]]..tostring(copyError))return end if not stillCurrent(token,character)then destroyInstance(description)return end
activeCard:SetStatus('Character')if emoteWarning and settings.showTargetEmotes then notify('Disguise emotes could not be applied: '..emoteWarning)end do local movementOk:boolean,movementResult:any=
pcall(applyCachedMovementAnimations,character,humanoid)lastAppliedSlotCount=movementOk and(tonumber(movementResult)or 0)or 0 if not movementOk then notify(
[[The appearance was applied, but movement animations failed: ]]..tostring(movementResult))end end if settings.customPackSlots~=nil then local packOk:boolean,packResult:any=pcall(applyAnimationPack,
character,humanoid)if packOk then lastAppliedSlotCount=tonumber(packResult)or 0 end end destroyInstance(description)local applied:string=string.format('%s \183 %s \183 %d slots',if settings.
disguiseCharacter then'Character'else'Emotes only',rigName(humanoid)or'rig?',lastAppliedSlotCount)if(lastDroppedSlotCount or 0)>0 then applied..=string.format(' \183 %d pack clips skipped',
lastDroppedSlotCount or 0)end if settings.customPackSlots~=nil then applied..=' \183 pack '..packDisplayName()end status(applied)end local function applyAnimationDisguise(token:number,character:Model,
humanoid:Humanoid,userId:number):()lastDroppedSlotCount=0 local bundleTarget:boolean=false local description:any,fetchError:string?=getUserDescription(userId)if description then if not stillCurrent(
token,character)then destroyInstance(description)return end cacheTarget(userId,description)else local slots:{[string]:any}?,bundleError:string?=bundleAnimationSlots(userId)if not slots then if
stillCurrent(token,character)then notify('Could not load that ID: '..tostring(fetchError)..' '..tostring(bundleError))end return end if not stillCurrent(token,character)then return end settings.
customPackSlots=slots settings.customPackName='Bundle '..tostring(userId)bundleTarget=true targetCache={userId=userId,slots=slots::{[string]:any},emotes=nil,equippedEmotes=nil}end local cached:
TargetCache?=targetCache local current:AppearanceSnapshot=ensureSnapshot(character,humanoid)restoreOriginalAppearance(character,humanoid)stopAppearanceFilter()local emotesApplied:boolean=false if
cached and type(cached.emotes)=='table'and type(cached.equippedEmotes)=='table'then local applied:boolean,emoteError:string?=setHumanoidEmotes(humanoid,cached.emotes,cached.equippedEmotes)if applied
then emotesApplied=true emotesAppliedToHumanoid=true refreshMeshPartSnapshot(character,current)else notify('Disguise emotes could not be applied: '..tostring(emoteError))end end local changedCount:
number=applyCachedMovementAnimations(character,humanoid)if not bundleTarget and settings.customPackSlots~=nil then local packCount:number=applyAnimationPack(character,humanoid)if packCount>0 then
changedCount=packCount end end destroyInstance(description)if not stillCurrent(token,character)then if emotesApplied then local rolledBack:boolean=setHumanoidEmotes(humanoid,current.emotes,current.
equippedEmotes)if rolledBack then emotesAppliedToHumanoid=false end end return end if not emotesApplied and changedCount==0 then notify('That avatar did not expose emotes or movement animations.')
return end activeCard:SetStatus('Animation')local animationLine:string=string.format('Animation \183 %s \183 emotes %s \183 %d slots',rigName(humanoid)or'rig?',emotesApplied and'applied'or
'not available',changedCount)if(lastDroppedSlotCount or 0)>0 then animationLine..=string.format(' \183 %d pack clips skipped',lastDroppedSlotCount or 0)end if settings.customPackSlots~=nil then
animationLine..=' \183 pack '..packDisplayName()end status(animationLine..'. Your own avatar stays visible.')end local function applyDisguise():()lastDroppedSlotCount=0 generation+=1 local token:
number=generation local character:Model?=LocalPlayer.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')::Humanoid?if not character or not humanoid or humanoid
.Health<=0 then notify('Your character is not ready yet.')return end local numericId:number?,idError:string?=parseId()if not numericId then notify(tostring(idError))return end local disguiseChar:
boolean=settings.disguiseCharacter local repairControls:any=host.suppressTouchControls if type(repairControls)=='function'then task.delay(0.6,function():()pcall(repairControls)end)end targetCache=nil
task.spawn(function():()if disguiseChar then applyCharacterDisguise(token,character::Model,humanoid::Humanoid,numericId::number)else applyAnimationDisguise(token,character::Model,humanoid::Humanoid,
numericId::number)end end)end local function activeCharacter():(any,any)local character:any=LocalPlayer.Character local humanoid:any=character and character:FindFirstChildOfClass('Humanoid')if not
character or not humanoid then return nil,nil end return character,humanoid end local function applyAnimationSpeedValue(speed:number):()local _character:any,humanoid:any=activeCharacter()if not
humanoid then return end for _,track:AnimationTrack in ipairs(playingAnimationTracks(humanoid))do pcall(function():()track:AdjustSpeed(speed)end)end end local function applyAnimationSpeed():()
applyAnimationSpeedValue(settings.animationSpeed)end local function setMovementAnimationsEnabled(enabled:boolean):()settings.useTargetMovementAnimations=enabled if not activeCard or not activeCard.
Enabled or settings.mode=='Animation'or not settings.disguiseCharacter then return end local character:any,humanoid:any=activeCharacter()if not character then return end if enabled then if not
targetCache then applyDisguise()return end local appliedCount:number=applyCachedMovementAnimations(character,humanoid)if appliedCount==0 and not animationOverrideApplied then applyDisguise()return end
else restoreMovementAnimationsOnly(character,humanoid)end status(if settings.disguiseCharacter then'Character'else'Emotes only')end local function setTargetEmotesEnabled(enabled:boolean):()settings.
showTargetEmotes=enabled if settings.mode=='Animation'and settings.disguiseCharacter~=false then return end if not activeCard or not activeCard.Enabled then return end local character:any,humanoid:any
=activeCharacter()if not character then return end if enabled then if not targetCache then applyDisguise()return end if type(targetCache.emotes)~='table'or type(targetCache.equippedEmotes)~='table'
then notify('That avatar did not expose emotes.')return end applyCachedEmotes(character,humanoid)else restoreEmotesOnly(character,humanoid)end status(if settings.disguiseCharacter then'Character'else
'Emotes only')end local function restoreAndRelease():()generation+=1 applyAnimationSpeedValue(1)stopAppearanceFilter()local character:Model?=LocalPlayer.Character local humanoid:Humanoid?=character
and character:FindFirstChildOfClass('Humanoid')::Humanoid?if character then restoreOriginalAppearance(character,humanoid)end destroySnapshot()targetCache=nil table.clear(cloned)table.clear(
disguisedItems)end local card:any card=framework.Categories.World:CreateModule({Name='Disguise',Category='World',ConfigKey='Universal.Disguise',Order=26,Tooltip=
[[Disguise Character ON (default): copies accessories, clothing, face, body colors and animations/emotes onto your preserved body/rig. OFF: only animations/emotes are shown without hiding it, your character stays as is. Default ID 78492.]]
,Function=function(enabled:boolean):()if enabled then card:SetStatus(if settings.disguiseCharacter then'Character'else'Emotes only')local runService:any=host.RunService if not runService then local ok
:boolean,service:any=pcall(function():any return(game::any):GetService('RunService')end)if ok then runService=service end end if runService and runService.Heartbeat then local lastSpeedAt:number=0
card:Clean(runService.Heartbeat:Connect(function():()local now:number=os.clock()if now-lastSpeedAt<0.2 then return end lastSpeedAt=now applyAnimationSpeed()end))end card:Clean(LocalPlayer.
CharacterAdded:Connect(function():()generation+=1 stopAppearanceFilter()destroySnapshot()restoring=false emotesAppliedToHumanoid=false animationOverrideApplied=false table.clear(cloned)table.clear(
disguisedItems)task.wait(0.5)if card.Enabled then applyDisguise()end end))applyDisguise()else restoreAndRelease()card:SetStatus(nil)end end})card:CreateToggle({Name='Disguise Character',Default=
settings.disguiseCharacter,Function=function(value:boolean):()settings.disguiseCharacter=value settings.mode=if value then'Character'else'Animation'card:SetStatus(if value then'Character'else
'Emotes only')if card.Enabled then if not value then local character:Model?=LocalPlayer.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')::Humanoid?if
character then restoreOriginalAppearance(character,humanoid)end stopAppearanceFilter()end applyDisguise()end end,Tooltip=
[[ON (default): full character disguise (appearance + animations/emotes). OFF: only animations/emotes are shown, your character stays as is.]]})card:CreateDropdown({Name='Mode',List={'Character',
'Animation'},Index=if settings.disguiseCharacter then 1 else 2,Show={Option='Disguise Character',Values={'__hidden__'}},Function=function(value:string):()settings.mode=value settings.disguiseCharacter
=(value=='Character')card:SetStatus(value)if card.Enabled then if value=='Animation'then local character:Model?=LocalPlayer.Character local humanoid:Humanoid?=character and character:
FindFirstChildOfClass('Humanoid')::Humanoid?if character then restoreOriginalAppearance(character,humanoid)end stopAppearanceFilter()end applyDisguise()end end,Tooltip=
'Legacy: use Disguise Character instead.'})card:CreateSlider({Name='Animation Speed',Min=0,Max=2,Step=0.05,Default=settings.animationSpeed,Function=function(value:number):()settings.animationSpeed=
math.clamp(value,0,2)if card.Enabled then applyAnimationSpeed()end end,Tooltip=
[[Playback speed of your character's animations while Disguise is on: 0 freezes them, 1 is normal, 2 is double speed. Turning Disguise off restores speed 1.]]})card:CreateToggle({Name='Show Advanced',
Default=settings.showAdvanced,Function=function(value:boolean):()settings.showAdvanced=value end,Tooltip=
[[Shows bundle inputs for a complete pack or individual slots. Opening or closing these inputs does not change the animations already applied.]]})card:CreateTextBox({Name='Pack Bundle ID',Show={Option
='Show Advanced',Values={true}},Default='',Function=function(value:string):()local trimmed:string=tostring(value or''):gsub('^%s+',''):gsub('%s+$','')local digits:string?=string.match(trimmed,'%d+')
local bundleId:number?=digits and tonumber(digits)or nil if not bundleId then notify([[Enter an animation bundle ID first (the number in its catalog link).]])return end local slots:{[string]:any}?,
bundleError:string?=bundleAnimationSlots(bundleId)if not slots then notify('Could not load that pack: '..tostring(bundleError))return end settings.customPackSlots=slots settings.customPackName=
'Bundle '..tostring(bundleId)table.clear(settings.customSlotIds)if not card.Enabled then notify('Pack '..tostring(bundleId)..' loaded; it applies while Disguise is on.')return end local character:
Model?=LocalPlayer.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')::Humanoid?if character and humanoid then applyAnimationPack(character,humanoid)status((
if settings.disguiseCharacter then'Character'else'Emotes only')..' \183 pack Bundle '..tostring(bundleId))end end,Tooltip=
[[Any animation pack bundle ID from its catalog link (roblox.com/bundles/<id>/...), official or UGC -- for example 4899847411546. The bundle's items are resolved when you enter it (nothing is downloaded; the engine streams the clips) and the pack is kept for the session. It wins over the target's movement animations. R15-only clips are not forced onto an incompatible R6 body.]]
})local SLOT_INSERTS:{{string}}={{'Idle','idle'},{'Walk','walk'},{'Run','run'},{'Jump','jump'},{'Fall','fall'},{'Climb','climb'},{'Swim','swim'},{'Swim Idle','swimidle'}}for _,slotInsert:{string}in
ipairs(SLOT_INSERTS)do local label:string=slotInsert[1]local category:string=slotInsert[2]card:CreateTextBox({Name=label..' Bundle ID',Show={Option='Show Advanced',Values={true}},Default='',Function=
function(value:string):()local digits:string?=string.match(tostring(value or''),'%d+')local bundleId:number?=digits and tonumber(digits)or nil if not bundleId then notify('Enter a bundle ID for the '
..label..' slot first.')return end local slots:{[string]:any}?,bundleError:string?=bundleAnimationSlots(bundleId)if not slots then notify('Could not load that pack: '..tostring(bundleError))return end
local clip:any=slots[category]if type(clip)=='table'then clip=clip[1]end if not clip then notify('That bundle has no '..label..' clip to take.')return end settings.customSlotIds[category]=clip local
merged:{[string]:any}={}for mergedCategory:string,mergedId:any in pairs(settings.customSlotIds)do merged[mergedCategory]=mergedId end settings.customPackSlots=merged settings.customPackName=
'Custom slots'if not card.Enabled then notify(label..' slot set; it applies while Disguise is on.')return end local character:Model?=LocalPlayer.Character local humanoid:Humanoid?=character and
character:FindFirstChildOfClass('Humanoid')::Humanoid?if character and humanoid then applyAnimationPack(character,humanoid)status((if settings.disguiseCharacter then'Character'else'Emotes only')..
' \183 pack Custom slots')end end,Tooltip='Set the '..label..[[ slot from any animation pack: paste the pack's bundle ID (official or UGC) and this slot takes that pack's ]]..label..
[[ clip. Mix slots from different packs freely; empty slots use your rig's own set.]]})end card:CreateTextBox({Name='Target ID',Default=settings.id,Function=function(value:string):()local trimmed:
string=tostring(value or''):gsub('^%s+',''):gsub('%s+$','')if trimmed==''then notify('Enter a user ID or animation bundle ID first.')return end settings.id=trimmed if card.Enabled then applyDisguise()
end end,Tooltip=
[[User ID: when Disguise Character is ON copies appearance + animations/emotes onto your preserved body/rig; when OFF only animations/emotes are applied and your character stays as is. Default 78492. Missing slots fall back to your rig's own set.]]
})card:CreateToggle({Name='Use target movement animations if able',Show={Option='Disguise Character',Values={'__hidden__'}},Default=settings.useTargetMovementAnimations,Function=function(value:boolean
):()setMovementAnimationsEnabled(value)end,Tooltip=
[[Character mode only. Shows the target user's movement animations (idle, walk, run, jump, climb, fall, swim, and Roblox's default set for any slot the avatar leaves empty) on the disguised avatar. They are already loaded, so this is instant; turning it off restores your own animations. Emotes are independent and stay untouched.]]
})card:CreateToggle({Name='Show target emotes',Default=settings.showTargetEmotes,Function=function(value:boolean):()setTargetEmotesEnabled(value)end,Tooltip=
[[Shows the target user's emotes and equipped emote slots on the disguised avatar. Works with or without Disguise Character.]]})activeCard=card Module.Initialized=true return card end function Module.
destroy():()if activeCard then if activeCard.Enabled then pcall(activeCard.Toggle,activeCard,false)end activeCard=nil end Module.Initialized=false end return Module
-- src/games/universal/Other/ImproveFps.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='ImproveFps',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.init(
context:Runtime):any local host:any=context.host local featureConnections:any=host.featureConnections local disconnectFeatureConnection:any=host.disconnectFeatureConnection local
createUniversalFeature:any=host.createUniversalFeature local addToggleOption:any=host.addToggleOption local fpsSettings:{[string]:boolean}={textures=true,particles=true,shadows=true,materials=false,
lighting=false,post=false,guiEffects=false,worldGui=false,gameGui=false,coreGui=false,geometry=false,scenery=false,quality=false,terrain=false,lights=false}local fpsModeEnabled:boolean=false local
generation:number=0 local ImproveFpsFeature:any=nil local function selectedEffectCount():number local count:number=0 for _name:string,selected:boolean in pairs(fpsSettings)do if selected then count+=1
end end return count end local fpsVisualCache:any={}local propertyWatches:{[any]:any}={}local function cacheFpsProperty(object:any,property:string):()local data:any=fpsVisualCache[object]if not data
then data={}fpsVisualCache[object]=data local ok,connection=pcall(function()return object.Destroying:Connect(function()fpsVisualCache[object]=nil local watch=propertyWatches[object]propertyWatches[
object]=nil if watch then watch:Disconnect()end end)end)if ok then propertyWatches[object]=connection end end if data[property]==nil then data[property]=object[property]end end local lighting:any=host
.Lighting local playerGui:any=host.PlayerGui local scenery:{[any]:boolean}=setmetatable({},{__mode='k'})local coreGuiCache:{[any]:boolean}={}local function setProperty(object:any,property:string,value
:any):()pcall(function()cacheFpsProperty(object,property)object[property]=value end)end local function protectedGui(object:any):boolean local current=object while current do if current==host.ScreenGui
or current.Name=='TouchGui'or current.Name=='Chat'or current.Name=='BubbleChat'then return true end current=current.Parent if type(current)~='table'and typeof(current)~='Instance'then break end end
return false end local function updateScenery(part:any):()local character=host.LocalPlayer and host.LocalPlayer.Character local root=character and character:FindFirstChild('HumanoidRootPart')local
cached=fpsVisualCache[part]local function restorePart()if cached and cached.LocalTransparencyModifier~=nil then pcall(function()part.LocalTransparencyModifier=cached.LocalTransparencyModifier end)end
end if not root or not part.Parent or not part.Anchored or not part.CanCollide then restorePart()return end local current=part.Parent while current and current~=workspace do if current:IsA('Tool')or
current:IsA('Accessory')or current:FindFirstChildOfClass('Humanoid')then restorePart()return end current=current.Parent end local far=(part.Position-root.Position).Magnitude>350 if far then
setProperty(part,'LocalTransparencyModifier',1)elseif cached and cached.LocalTransparencyModifier~=nil then pcall(function()part.LocalTransparencyModifier=cached.LocalTransparencyModifier end)end end
local function applyFpsObject(object:Instance):()if object~=workspace and object~=lighting and object~=playerGui and object.Parent==nil then return end local target:any=object if lighting and object==
lighting and fpsSettings.lighting then setProperty(object,'EnvironmentDiffuseScale',0)setProperty(object,'EnvironmentSpecularScale',0)end if fpsSettings.shadows and object==lighting then setProperty(
object,'GlobalShadows',false)end if fpsSettings.lights and object:IsA('Light')then setProperty(object,'Enabled',false)end if fpsSettings.terrain and object:IsA('Terrain')then setProperty(object,
'Decoration',false)setProperty(object,'WaterWaveSize',0)setProperty(object,'WaterWaveSpeed',0)setProperty(object,'WaterReflectance',0)end if fpsSettings.post and object:IsA('PostEffect')then
setProperty(object,'Enabled',false)end if not protectedGui(object)then if fpsSettings.guiEffects and(object:IsA('UIStroke')or object:IsA('UIGradient'))then setProperty(object,'Enabled',false)elseif
```

```
fpsSettings.guiEffects and object:IsA('ViewportFrame')then setProperty(object,'Visible',false)end if fpsSettings.worldGui and(object:IsA('BillboardGui')or object:IsA('SurfaceGui'))then setProperty(
object,'Enabled',false)end if fpsSettings.gameGui and object:IsA('ScreenGui')then setProperty(object,'Enabled',false)end end if fpsSettings.geometry and(object:IsA('MeshPart')or object:IsA(
'PartOperation'))then setProperty(object,'RenderFidelity',Enum.RenderFidelity.Performance)end if fpsSettings.scenery and object:IsA('BasePart')and object.Anchored and object.CanCollide then scenery[
object]=true updateScenery(object)end if fpsSettings.textures and(object:IsA('Decal')or object:IsA('Texture'))then cacheFpsProperty(object,'Transparency')target.Transparency=1 elseif fpsSettings.
textures and object:IsA('MeshPart')then setProperty(object,'TextureID','')elseif fpsSettings.textures and object:IsA('SpecialMesh')then setProperty(object,'TextureId','')end if fpsSettings.particles
and(object:IsA('ParticleEmitter')or object:IsA('Trail')or object:IsA('Beam')or object:IsA('Smoke')or object:IsA('Fire')or object:IsA('Sparkles'))then cacheFpsProperty(object,'Enabled')target.Enabled=
false end if object:IsA('BasePart')then if fpsSettings.shadows then cacheFpsProperty(object,'CastShadow')object.CastShadow=false end if fpsSettings.materials then cacheFpsProperty(object,'Material')
cacheFpsProperty(object,'Reflectance')object.Material=Enum.Material.SmoothPlastic object.Reflectance=0 end end end local function restoreFpsObjects():()for _,connection in pairs(propertyWatches)do
pcall(function()connection:Disconnect()end)end propertyWatches={}for object:any,properties:any in pairs(fpsVisualCache)do if object then for property:string,value:any in pairs(properties)do pcall(
function():()object[property]=value end)end end end fpsVisualCache={}scenery=setmetatable({},{__mode='k'})local starterGui=host.StarterGui if starterGui then for kind,value in pairs(coreGuiCache)do
pcall(function()starterGui:SetCoreGuiEnabled(kind,value)end)end end coreGuiCache={}end local function toggleImproveFps(enabled:boolean):()generation+=1 local token:number=generation fpsModeEnabled=
enabled for _,name in ipairs({'ImproveFPS','ImproveFPSLighting','ImproveFPSGui','ImproveFPSDistance'})do disconnectFeatureConnection(name)end restoreFpsObjects()if not enabled then if
ImproveFpsFeature then ImproveFpsFeature:SetStatus(nil)end return end ImproveFpsFeature:SetStatus(tostring(selectedEffectCount())..' options')if fpsSettings.quality then pcall(function()setProperty(
settings().Rendering,'QualityLevel',Enum.QualityLevel.Level01)end)end local containers={workspace}if lighting then table.insert(containers,lighting)end if playerGui then table.insert(containers,
playerGui)end for index,container in ipairs(containers)do local name=({'ImproveFPS','ImproveFPSLighting','ImproveFPSGui'})[index]featureConnections[name]=container.DescendantAdded:Connect(function(
object)if fpsModeEnabled and token==generation then pcall(applyFpsObject,object)end end)end if fpsSettings.coreGui and host.StarterGui then for _,kind in ipairs({Enum.CoreGuiType.PlayerList,Enum.
CoreGuiType.Backpack,Enum.CoreGuiType.Chat,Enum.CoreGuiType.EmotesMenu})do pcall(function()local value=host.StarterGui:GetCoreGuiEnabled(kind)if type(value)=='boolean'then coreGuiCache[kind]=value
host.StarterGui:SetCoreGuiEnabled(kind,false)end end)end end if fpsSettings.scenery and host.TaskManager then local elapsed,cursor=0,1 local pending:{BasePart}={}featureConnections.ImproveFPSDistance=
host.TaskManager:Connect(function(dt)elapsed+=dt if cursor>#pending then if elapsed<1 then return end elapsed=0 table.clear(pending)for part in pairs(scenery)do if part.Parent then table.insert(
pending,part)else scenery[part]=nil end end cursor=1 end local last=math.min(#pending,cursor+199)while cursor<=last do if not fpsModeEnabled or token~=generation then return end pcall(updateScenery,
pending[cursor])cursor+=1 end end,nil,'Improve FPS')end task.spawn(function():()local scanned=0 for _,container in ipairs(containers)do if not fpsModeEnabled or token~=generation then return end
pcall(applyFpsObject,container)for _,object in ipairs(container:GetDescendants())do if not fpsModeEnabled or token~=generation then return end pcall(applyFpsObject,object)scanned+=1 if scanned%250==0
then task.wait()end end end end)end local function refreshFps():()if fpsModeEnabled then toggleImproveFps(true)end end ImproveFpsFeature=createUniversalFeature('Improve FPS',
'Apply only the selected reversible optimizations',26,toggleImproveFps,{categoryName='World'})addToggleOption(ImproveFpsFeature,'Remove textures',true,function(value:boolean):()fpsSettings.textures=
value refreshFps()end)addToggleOption(ImproveFpsFeature,'Disable particles',true,function(value:boolean):()fpsSettings.particles=value refreshFps()end)addToggleOption(ImproveFpsFeature,
'Disable shadows',true,function(value:boolean):()fpsSettings.shadows=value refreshFps()end)addToggleOption(ImproveFpsFeature,'Simple materials',false,function(value:boolean):()fpsSettings.materials=
value refreshFps()end)for _,definition in ipairs({{'Lowest render quality','quality',[[Requests render quality Level01 where the client permits it; restores the original level when OFF.]]},{
'Simple lighting','lighting','Disables environment diffuse/specular reflections.'},{'Disable local lights','lights',[[Turns off PointLight, SpotLight and SurfaceLight instances. May darken the map.]]}
,{'Simplify terrain','terrain',[[Disables terrain decoration, water waves and water reflections where writable. Physics is unchanged.]]},{'Disable post effects','post',
[[Disables bloom, blur (including menu blur), depth of field, color correction and sun rays.]]},{'Simplify game UI','guiEffects',
[[Disables game UI strokes, gradients and 3D viewport previews. Menu and touch controls are protected.]]},{'Hide world UI','worldGui',
[[Hides floating labels and SurfaceGui displays; may hide game information.]]},{'Hide game UI','gameGui',
[[Aggressive: hides game ScreenGuis, including HUD/objectives. Keeps this menu, chat and TouchGui.]]},{'Hide Roblox panels','coreGui',
[[Hides player list, backpack, chat and emotes panel. Health and mobile movement remain available.]]},{'Low detail meshes','geometry',
[[Requests Performance render fidelity for meshes/unions; does not change collisions.]]},{'Hide distant scenery','scenery',
[[Aggressive: locally hides static map parts beyond 350 studs, updating periodically. May hide distant objectives; collisions remain.]]}})do local key:string=definition[2]addToggleOption(
ImproveFpsFeature,definition[1],false,function(value:boolean)fpsSettings[key]=value refreshFps()end,definition[3])end activeCleanup=function():()toggleImproveFps(false)end Module.Initialized=true
return ImproveFpsFeature end function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Other/ProximityPrompts.lua
export type Runtime={framework:any,entity:any,host:any}local Module={Name='ProximityPrompts',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function Module.
init(context:Runtime):any local host:any=context.host local TaskManager:any=host.TaskManager local featureConnections:any=host.featureConnections local disconnectFeatureConnection:any=host.
disconnectFeatureConnection local createUniversalFeature:any=host.createUniversalFeature local addToggleOption:any=host.addToggleOption local addNumberOption:any=host.addNumberOption type
InteractionSettings={distance:number,instantPrompts:boolean,proximityPrompts:boolean,clickDetectors:boolean}type PromptState={distance:number,holdDuration:number}local interactionSettings:
InteractionSettings={distance=32,instantPrompts=false,proximityPrompts=false,clickDetectors=false}local originalPrompts:{[ProximityPrompt]:PromptState}=setmetatable({},{__mode='k'})::any local
originalClickDetectors:{[ClickDetector]:number}=setmetatable({},{__mode='k'})::any local interactionExtenderEnabled:boolean=false local InteractFeature:any=nil local interactionRefreshClock:number=0
local function promptsChannel():boolean return interactionSettings.proximityPrompts or interactionSettings.instantPrompts end local function detectorsChannel():boolean return interactionSettings.
clickDetectors end local function anyChannel():boolean return promptsChannel()or detectorsChannel()end local function updatePrompt(prompt:ProximityPrompt,original:PromptState):()local distance:number=
promptsChannel()and interactionSettings.distance or original.distance local hold:number=promptsChannel()and interactionSettings.instantPrompts and 0 or original.holdDuration if prompt.
MaxActivationDistance~=distance then prompt.MaxActivationDistance=distance end if prompt.HoldDuration~=hold then prompt.HoldDuration=hold end end local function updateDetector(detector:ClickDetector,
original:number):()local distance:number=detectorsChannel()and interactionSettings.distance or original if detector.MaxActivationDistance~=distance then detector.MaxActivationDistance=distance end end
local function refreshInteractionExtender():()for prompt:ProximityPrompt,original:PromptState in pairs(originalPrompts)do if prompt.Parent then updatePrompt(prompt,original)end end for detector:
ClickDetector,original:number in pairs(originalClickDetectors)do if detector.Parent then updateDetector(detector,original)end end end local function applyInteractionExtension(instance:Instance):()if
not anyChannel()then return end if instance:IsA('ProximityPrompt')then if originalPrompts[instance]==nil then originalPrompts[instance]={distance=instance.MaxActivationDistance,holdDuration=instance.
HoldDuration}end updatePrompt(instance::ProximityPrompt,originalPrompts[instance])elseif instance:IsA('ClickDetector')then if originalClickDetectors[instance]==nil then originalClickDetectors[instance
]=instance.MaxActivationDistance end updateDetector(instance::ClickDetector,originalClickDetectors[instance])end end local function restoreInteractionExtender():()for prompt:ProximityPrompt,original:
PromptState in pairs(originalPrompts)do if prompt.Parent then if prompt.MaxActivationDistance~=original.distance then prompt.MaxActivationDistance=original.distance end if prompt.HoldDuration~=
original.holdDuration then prompt.HoldDuration=original.holdDuration end end end for detector:ClickDetector,original:number in pairs(originalClickDetectors)do if detector.Parent and detector.
MaxActivationDistance~=original then detector.MaxActivationDistance=original end end originalPrompts=setmetatable({},{__mode='k'})::any originalClickDetectors=setmetatable({},{__mode='k'})::any end
local scanning:boolean=false local function startScan():()if scanning then return end scanning=true local scanConnection:any=featureConnections.InteractExtender task.spawn(function():()for index:
number,instance:Instance in ipairs(workspace:GetDescendants())do if featureConnections.InteractExtender~=scanConnection then scanning=false return end applyInteractionExtension(instance)if index%240==
0 then task.wait()end end scanning=false end)end local function syncInteractions():()if not anyChannel()then disconnectFeatureConnection('InteractExtender')disconnectFeatureConnection(
'InteractExtenderRefresh')scanning=false restoreInteractionExtender()interactionRefreshClock=0 if InteractFeature then InteractFeature:SetStatus(nil)end return end if not featureConnections.
InteractExtender then featureConnections.InteractExtender=workspace.DescendantAdded:Connect(applyInteractionExtension)end if not featureConnections.InteractExtenderRefresh then featureConnections.
InteractExtenderRefresh=TaskManager:Connect(function(deltaTime:number):()interactionRefreshClock+=deltaTime if interactionRefreshClock>=0.5 then interactionRefreshClock=0 refreshInteractionExtender()
end end,nil,'Proximity Prompts')startScan()else refreshInteractionExtender()end if InteractFeature and interactionExtenderEnabled then InteractFeature:SetStatus(tostring(math.round(interactionSettings
.distance))..' studs')end end local function setRow(name:string,value:boolean):()local controls:any=nil pcall(function():()if state and type(state.toggleControls)=='table'then controls=state.
toggleControls end end)local control:any=controls and controls[InteractFeature.configKey..'.'..name]if control and type(control.setChecked)=='function'then pcall(control.setChecked,value)end end
local function toggleInteractionExtender(enabled:boolean):()interactionExtenderEnabled=enabled if enabled then interactionSettings.proximityPrompts=true interactionSettings.clickDetectors=true setRow(
'Proximityprompts',true)setRow('Clickdetectors',true)syncInteractions()startScan()return else interactionSettings.instantPrompts=false interactionSettings.proximityPrompts=false interactionSettings.
clickDetectors=false setRow('Instantprompts',false)setRow('Proximityprompts',false)setRow('Clickdetectors',false)end syncInteractions()end InteractFeature=createUniversalFeature('Proximity Prompts',
[[Extend prompt and click interaction range without auto-firing]],24,toggleInteractionExtender,{categoryName='Exploits',configKey='Universal.InteractExtender'})addNumberOption(InteractFeature,
'Interaction distance',interactionSettings.distance,8,250,function(value:number):()interactionSettings.distance=value if InteractFeature.enabled then InteractFeature:SetStatus(tostring(math.round(
value))..' studs')end syncInteractions()end)addToggleOption(InteractFeature,'Instant prompts',interactionSettings.instantPrompts,function(value:boolean):()interactionSettings.instantPrompts=value
syncInteractions()end)addToggleOption(InteractFeature,'Proximity prompts',interactionSettings.proximityPrompts,function(value:boolean):()interactionSettings.proximityPrompts=value syncInteractions()
end)addToggleOption(InteractFeature,'Click detectors',interactionSettings.clickDetectors,function(value:boolean):()interactionSettings.clickDetectors=value syncInteractions()end)activeCleanup=function
():()interactionExtenderEnabled=false disconnectFeatureConnection('InteractExtender')disconnectFeatureConnection('InteractExtenderRefresh')restoreInteractionExtender()interactionRefreshClock=0 if
InteractFeature then InteractFeature:SetStatus(nil)end end Module.Initialized=true return InteractFeature end function Module.destroy():()if activeCleanup then activeCleanup()end activeCleanup=nil
Module.Initialized=false end return Module
-- src/games/universal/Other/RejoinServer.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='RejoinServer',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function
Module.init(context:Runtime):any local host:any=context.host local LocalPlayer:any=host.LocalPlayer local createUniversalFeature:any=host.createUniversalFeature local cloneReference:any=host.
cloneReference local notify:(string)->()=host.notify local function rejoinCurrentServer():()local teleportService:TeleportService=cloneReference(game:GetService('TeleportService'))local placeId:number
=game.PlaceId local jobId:string=tostring(game.JobId or'')local rejoined:boolean=false if jobId~=''then rejoined=pcall(function():()teleportService:TeleportToPlaceInstance(placeId,jobId,LocalPlayer)
end)end if not rejoined then local ok:boolean,teleportError:any=pcall(function():()teleportService:Teleport(placeId,LocalPlayer)end)if not ok then notify('Rejoin failed: '..tostring(teleportError))end
end end local actionId:string='RejoinServer'local registries:any=context.services.registries local wurstOptions:any=registries.wurstOptions()if wurstOptions and type(wurstOptions.RegisterAction)==
'function'then local registered:boolean,registration:any=pcall(wurstOptions.RegisterAction,actionId,'Rejoin Server',rejoinCurrentServer,{description='Reconnect to the current server instance',
configKey='Universal.RejoinServer'})if not registered then error('Wurst Options refused Rejoin Server: '..tostring(registration))end local search:any=registries.moduleSearch()local searchFeature:any={
name='Rejoin Server',configKey='Universal.RejoinServer',kind='action',searchable=false,enabled=false,activate=rejoinCurrentServer}if search and type(search.Register)=='function'then search.Register(
searchFeature)end activeCleanup=function():()if search and type(search.Unregister)=='function'then search.Unregister(searchFeature)end if type(registration)=='table'and type(registration.Unregister)==
'function'then registration.Unregister()elseif type(wurstOptions.UnregisterAction)=='function'then pcall(wurstOptions.UnregisterAction,actionId)end end Module.Initialized=true return registration end
local feature:any=createUniversalFeature('Rejoin Server','Reconnect to the current server instance',22,rejoinCurrentServer,{action=true,noOptions=true,categoryName='Other',searchable=false})
activeCleanup=function():()end Module.Initialized=true return feature end function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.Initialized=false end
return Module
-- src/games/universal/Other/ServerSettings.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='ServerSettings',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCard:any=nil local
disconnectTicker:((string)->())?=nil local HOP_COOLDOWN:number=3 local STAT_INTERVAL:number=1 local MAX_PAGES:number=2 function Module.init(context:Runtime):any local host:any=context.host local
LocalPlayer:any=host.LocalPlayer local createUniversalFeature:any=host.createUniversalFeature local addActionOption:any=host.addActionOption local addInformationOption:any=host.addInformationOption
local notify:any=host.notify local cloneReference:any=host.cloneReference local services:any=context.services or{}local players:any=host.Players or game:GetService('Players')local httpService:any=
cloneReference and cloneReference(game:GetService('HttpService'))or game:GetService('HttpService')local teleportService:any=cloneReference and cloneReference(game:GetService('TeleportService'))or game
:GetService('TeleportService')local tick:any=if host.TaskManager and host.TaskManager.Connect then host.TaskManager else nil local featureConnections:any=host.featureConnections local
disconnectFeatureConnection:any=host.disconnectFeatureConnection if type(disconnectFeatureConnection)=='function'then disconnectTicker=disconnectFeatureConnection end local settings={showInfo=true}
local lastHopAt:number=-math.huge local elapsed:number=0 local hopping:boolean=false local card:any=nil local statRows:{[string]:any}={}local function say(message:string,kind:string?):()if type(notify
)=='function'then pcall(notify,message,kind)end end local function jobId():string local ok:boolean,value:any=pcall(function():any return game.JobId end)if not ok or type(value)~='string'or value==''
then return''end return value end local function playerList():{any}local ok:boolean,list:any=pcall(function():any return players:GetPlayers()end)if not ok or type(list)~='table'then return{}end return
list end local function maxPlayers():number local ok:boolean,value:any=pcall(function():any return players.MaxPlayers end)if ok and type(value)=='number'then return value end return 0 end
local function uptimeSeconds():number local ok:boolean,value:any=pcall(function():any return workspace.DistributedGameTime end)if ok and type(value)=='number'and value>=0 then return value end return
0 end local function clockText():string local ok:boolean,value:any=pcall(function():any return workspace:GetServerTimeNow()end)local stamp:number=if ok and type(value)=='number'then math.floor(value)
else os.time()local date:any=os.date('!*t',stamp)if type(date)~='table'then return'--:--:--'end return string.format('%02d:%02d:%02d',date.hour,date.min,date.sec)end local function durationText(
seconds:number):string local total:number=math.max(0,math.floor(seconds))local hours:number=math.floor(total/3600)local minutes:number=math.floor((total%3600)/60)local secs:number=total%60 return
string.format('%d:%02d:%02d',hours,minutes,secs)end local function placeText():string local placeOk:boolean,placeId:any=pcall(function():any return game.PlaceId end)local gameOk:boolean,gameId:any=
pcall(function():any return game.GameId end)local label:string='Unknown place'local marketOk:boolean,marketName:any=pcall(function():any local ms:any=game:GetService('MarketplaceService')local info:
any=ms:GetProductInfo(placeId)return info and info.Name end)if marketOk and type(marketName)=='string'and marketName~=''then label=marketName else local nameOk:boolean,name:any=pcall(function():any
return game.Name end)if nameOk and type(name)=='string'and name~=''then label=name elseif placeOk and type(placeId)=='number'then label='Place '..tostring(placeId)end end if placeOk and gameOk and
type(placeId)=='number'and type(gameId)=='number'and placeId~=gameId then return label..' \183 subplace'end return label end local function namesText(limit:number):string local list:{any}=playerList()
local names:{string}={}for _,player:any in ipairs(list)do local ok:boolean,name:any=pcall(function():any return player.Name end)if ok and type(name)=='string'then table.insert(names,name)end end if#
names==0 then return'\u{2014}'end local joined:string=table.concat(names,', ')if#joined>limit then joined=string.sub(joined,1,math.max(1,limit-1))..'\u{2026}'end return joined end local function
shortJobId():string local id:string=jobId()if id==''then return'private server'end if#id>8 then return string.sub(id,1,8)end return id end local function infoText():string local limit:number=
maxPlayers()return string.format('%s \183 %d/%d \183 up %s',shortJobId(),#playerList(),if limit>0 then limit else#playerList(),durationText(uptimeSeconds()))end local statRowsDefinition:{{key:string,
label:string,read:()->string}}={{key='server',label='Server',read=shortJobId},{key='players',label='Players',read=function():string local limit:number=maxPlayers()return string.format('%d / %d',#
playerList(),if limit>0 then limit else#playerList())end},{key='inServer',label='In server',read=function():string return namesText(34)end},{key='uptime',label='Uptime',read=function():string return
durationText(uptimeSeconds())end},{key='serverTime',label='Server time',read=function():string return clockText()end},{key='place',label='Place',read=placeText}}local function ensureStatRows():()if
not card or next(statRows)~=nil then return end if type(state.addSectionOption)=='function'then pcall(state.addSectionOption,card,'Overlay Info')end if type(state.addStatOption)~='function'then return
end for _,definition:any in ipairs(statRowsDefinition)do local ok:boolean,row:any=pcall(state.addStatOption,card,definition.label,'\u{2014}')if ok and row then statRows[definition.key]=row end end end
local function refreshStats():()ensureStatRows()for _,definition:any in ipairs(statRowsDefinition)do local row:any=statRows[definition.key]if row and type(row.set)=='function'then local ok:boolean,
value:any=pcall(definition.read)local text:string=if ok and type(value)=='string'and value~=''then value else'\u{2014}'pcall(row.set,text)end end if card and type(card.SetStatus)=='function'then if
settings.showInfo then pcall(function():()card:SetStatus(infoText())end)else pcall(function():()card:SetStatus(nil)end)end end end local function startTicker():()if not(tick and featureConnections)
then return end if featureConnections.ServerSettingsTick then return end featureConnections.ServerSettingsTick=tick:Connect(function(deltaTime:number):()elapsed+=math.max(0,deltaTime or 0)if elapsed<
STAT_INTERVAL then return end elapsed=0 refreshStats()end,nil,'Server Settings')end local function copyJobId():()local id:string=jobId()if id==''then say('This server has no Job ID to copy.')return
end local clipboard:any=host.setclipboard or setclipboard local copied:boolean=false if type(clipboard)=='function'then copied=pcall(clipboard,id)end if copied then say('Job ID copied: '..id,'good')
else say('Job ID: '..id)end end local function fetchPage(sortOrder:string,cursor:string?):({any}?,string?,string?)local endpoint:string=string.format(
[[https://games.roblox.com/v1/games/%d/servers/Public?sortOrder=%s&limit=100]],game.PlaceId,sortOrder)if type(cursor)=='string'and cursor~=''then endpoint=endpoint..'&cursor='..cursor end local ok:
boolean,body:any=pcall(function():any return(game::any):HttpGet(endpoint)end)if not ok or type(body)~='string'or body==''then return nil,nil,
'the server list request failed (executor HTTP is blocked?)'end local decodeOk:boolean,decoded:any=pcall(function():any return httpService:JSONDecode(body)end)if not decodeOk or type(decoded)~='table'
or type(decoded.data)~='table'then return nil,nil,'the server list answer could not be read'end local nextCursor:any=decoded.nextPageCursor return decoded.data,if type(nextCursor)=='string'then
nextCursor else nil,nil end local function collectPool(orders:{string}):({any},number,string?)local here:string=jobId()local pool:{any}={}local seen:{[string]:boolean}={}local total:number=0 local
lastError:string?=nil for _,order:string in ipairs(orders)do local cursor:string?=nil for _=1,MAX_PAGES do local page:{any}?,nextCursor:string?,errorText:string?=fetchPage(order,cursor)if not page
then lastError=errorText break end for _,entry:any in ipairs(page)do total+=1 if type(entry)=='table'and type(entry.id)=='string'and entry.id~=''and entry.id~=here and not seen[entry.id]then seen[
entry.id]=true table.insert(pool,entry)end end if not nextCursor then break end cursor=nextCursor end end return pool,total,lastError end local function slotFree(entry:any):boolean local playing:
number?=tonumber(entry and entry.playing)if not playing or playing<0 then return false end local limit:number?=tonumber(entry and entry.maxPlayers)if limit and limit>0 then return playing<limit end
return playing<100 end local function playersIn(entry:any):number return tonumber(entry and entry.playing)or 0 end local function joinablePool(entries:{any}):{any}local pool:{any}={}for _,entry:any in
ipairs(entries)do if slotFree(entry)then table.insert(pool,entry)end end return pool end local function pick(pool:{any},mode:string):any if#pool==0 then return nil end local populated:{any}={}for _,
entry:any in ipairs(pool)do if playersIn(entry)>=1 then table.insert(populated,entry)end end local candidates:{any}=if#populated>0 then populated else pool if mode=='random'then return candidates[math
.random(1,#candidates)]end local chosen:any=candidates[1]for index:number=2,#candidates do local entry:any=candidates[index]if mode=='largest'then if playersIn(entry)>playersIn(chosen)then chosen=
entry end elseif playersIn(entry)<playersIn(chosen)then chosen=entry end end return chosen end local function hop(mode:string):()if hopping then return end if os.clock()-lastHopAt<HOP_COOLDOWN then
say('Slow down: wait a moment before hopping again.')return end hopping=true task.spawn(function():()local ok:boolean,failure:any=pcall(function():()local orders:{string}if mode=='largest'then orders=
{'Desc'}elseif mode=='random'then orders={'Asc','Desc'}else orders={'Asc'}end local entries:{any},total:number,errorText:string?=collectPool(orders)local pool:{any}=joinablePool(entries)local chosen:
any=pick(pool,mode)lastHopAt=os.clock()if not chosen then local reason:string=if total==0 then tostring(errorText or'the server list is empty')else string.format('all %d listed servers are full',total
)say('Server hop: '..reason..' \183 joining any server instead\u{2026}','danger')local teleported:boolean=pcall(function():()teleportService:Teleport(game.PlaceId,LocalPlayer)end)if not teleported
then say('Server hop was refused by TeleportService.','danger')end return end local playing:number=playersIn(chosen)local limit:number=tonumber(chosen.maxPlayers)or 0 say(string.format(
'Hopping to the %s server: %d/%d players\u{2026}',mode,playing,if limit>0 then limit else playing+1))local teleported:boolean=pcall(function():()teleportService:TeleportToPlaceInstance(game.PlaceId,
chosen.id,LocalPlayer)end)if not teleported then say('Server hop was refused by TeleportService.','danger')end end)hopping=false if not ok then say('Server hop failed: '..tostring(failure),'danger')
end end)end card=createUniversalFeature('Server Settings',[[Live server readout (Job ID, players, uptime, place) plus hops that only land on a server with a free slot.]],8,function(enabled:boolean):()
if enabled then elapsed=STAT_INTERVAL refreshStats()end end,{categoryName='Utilities'})activeCard=card addActionOption(card,'Copy Job ID',copyJobId)addActionOption(card,'Hop: smallest server',function
():()hop('smallest')end)addActionOption(card,'Hop: largest server',function():()hop('largest')end)addActionOption(card,'Hop: random server',function():()hop('random')end)addInformationOption(card,
[[A Roblox client cannot change another server's settings. This card ]]..[[shows YOUR server, live, and moves you between public ones. ]]..
[[A hop only lands where a slot is free; "smallest" skips empty ]]..[[instances and "largest" is the fullest server you can still ]]..
[[join (e.g. 11/12). If every listed server is full, the hop joins ]]..'a random one instead of failing.')ensureStatRows()refreshStats()startTicker()state.serverSettingsCard=card state.serverSettings=
settings state.serverSettingsStats=statRows state.serverSettingsPick=function(entries:{any},mode:string):string?local here:string=jobId()local filtered:{any}={}for _,entry:any in ipairs(entries or{})
do if type(entry)=='table'and type(entry.id)=='string'and entry.id~=here and slotFree(entry)then table.insert(filtered,entry)end end local chosen:any=pick(filtered,mode)return if chosen then tostring(
chosen.id)else nil end Module.Initialized=true return card end function Module.destroy():()if not Module.Initialized then return end Module.Initialized=false if type(disconnectTicker)=='function'then
pcall(disconnectTicker,'ServerSettingsTick')end state.serverSettingsCard=nil state.serverSettings=nil state.serverSettingsStats=nil state.serverSettingsPick=nil activeCard=nil end return Module
-- src/games/universal/Render/Arrows.lua
export type Runtime={framework:any,entity:any,render:any,host:any,services:any}local Module={Name='Player Pointers',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCard:any=nil
function Module.init(context:Runtime):any local framework:any=context.framework local entity:any=context.entity local render:any=context.render local protectedTargets:any=context.services.
protectedTargets local gameBridge:any=context.services.gameBridge local teamCheck:any=context.host.state and context.host.state.teamCheck local currentWorkspace:Workspace=context.host.workspace or
workspace local layer:Frame=render:Layer('ArrowsLayer')local shown:{[Player]:boolean}={}local function roleColour(player:Player):Color3?if gameBridge and type(gameBridge.playerRoleColor)=='function'
then local ok:boolean,colour:any=pcall(gameBridge.playerRoleColor,player)if ok and typeof(colour)=='Color3'then return colour end end return nil end local card:any local function clear():()render:
ReleaseAll(layer)table.clear(shown)layer.Visible=false end card=framework.Categories.Visuals:CreateModule({Name='Player Pointers',Category='Render',ConfigKey='Universal.Arrows',Order=3,Tooltip=
'Points toward players outside the camera view.',Function=function(enabled:boolean):()layer.Visible=enabled if not enabled then clear()card:SetStatus(nil)return end card:Render(function():()local
camera:Camera?=currentWorkspace.CurrentCamera if not camera then render:ReleaseAll(layer)table.clear(shown)return end local viewport:Vector2=camera.ViewportSize local centre:Vector2=Vector2.new(
viewport.X/2,viewport.Y/2)local radius:number=math.max(40,math.min(viewport.X,viewport.Y)*0.4)local seen:{[Player]:boolean}={}local count:number=0 for _,target:any in ipairs(entity:Refresh())do local
player:Player=target.Player local friendly:boolean=target.IsFriendly if not friendly and teamCheck and type(teamCheck.isSameTeam)=='function'then friendly=teamCheck.isSameTeam(player)end if(not card.
Options['Teammates'].Value and friendly)or protectedTargets.isProtected(player)or not target.Humanoid or target.Humanoid.Health<=0 or not target.RootPart or target.Distance>card.Options['Max distance'
].Value then render:Release(layer,player)continue end local projected:Vector3,onScreen:boolean=camera:WorldToViewportPoint(target.RootPart.Position)if onScreen then render:Release(layer,player)
continue end local drawing:any=render:Set(layer,player)local delta:Vector2=Vector2.new(projected.X,projected.Y)-centre if projected.Z<0 then delta=Vector2.new(-delta.X,-delta.Y)end if delta.Magnitude<
0.001 then delta=Vector2.new(0,-1)end local direction:Vector2=delta.Unit local perpendicular:Vector2=Vector2.new(-direction.Y,direction.X)local tip:Vector2=centre+direction*radius local back:Vector2=
tip-direction*16 local roleColourForTarget:Color3?=roleColour(player)local colour:Color3=roleColourForTarget or(card.Options['Team colours'].Value and friendly and player.Team and player.Team.
TeamColor.Color or card.Options['Colour'].Value)drawing:Line('Arrow',1,tip,back+perpendicular*8,2,colour)drawing:Line('Arrow',2,tip,back-perpendicular*8,2,colour)drawing:HideLines('Arrow',3,8)if card.
Options['Distance'].Value then drawing:Label('Distance','['..tostring(math.round(target.Distance))..']',tip-direction*26,0.5,12,colour,true)else drawing:HideLabel('Distance')end drawing:Show(true)seen
[player]=true count+=1 end for player:Player in pairs(shown)do if not seen[player]then render:Release(layer,player)end end shown=seen card:SetStatus(tostring(count))end)card:Clean(clear)end})card:
CreateColor({Name='Colour',Default=Color3.fromRGB(85,255,85),Tooltip=[[Base arrow colour. Arrows use role colours instead whenever the game knows roles.]]})card:CreateToggle({Name='Distance',Default=
true})card:CreateSlider({Name='Max distance',Min=25,Max=2000,Step=25,Default=500})card:CreateToggle({Name='Teammates',Default=false,Tooltip=
[[Also point at teammates. The advanced team check decides who counts.]]})card:CreateToggle({Name='Team colours',Default=false,Tooltip=
[[Teammates point in their Roblox team colour; everyone else keeps the base colour. Role colours always win.]]})activeCard=card Module.Initialized=true return card end function Module.destroy():()if
activeCard and activeCard.Enabled then activeCard:Toggle(false)end activeCard=nil Module.Initialized=false end return Module
-- src/games/universal/Render/Chams.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='Chams',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCard:any=nil local activeBridgeCleanup:
(()->())?=nil function Module.init(context:Runtime):any local framework:any=context.framework local entity:any=context.entity local host:any=context.host local protectedTargets:any=context.services.
protectedTargets local gameBridge:any=context.services.gameBridge local teamCheck:any=host.state and host.state.teamCheck local currentWorkspace:Workspace=host.workspace or workspace local highlights:
{[Player]:Highlight}={}local lastUpdate:number=-math.huge local roles:any=nil local function playerPerk(player:Player):string?if type(gameBridge)~='table'or type(gameBridge.playerPerk)~='function'then
return nil end local ok:boolean,perk:any=pcall(gameBridge.playerPerk,player)if ok and type(perk)=='string'then return perk end return nil end local PERK_ICONS:{[string]:string}={['Footsteps']=
'rbxassetid://336404773',['Sleight']='rbxassetid://340309581',['Ninja']='rbxassetid://416832054',['Haste']='rbxassetid://416942717',['FakeGun']='rbxassetid://17899774181',['Fake Gun']=
'rbxassetid://17899774181',['Ghost']='rbxassetid://335206856',['Xray']='rbxassetid://340313896',['X-Ray']='rbxassetid://340313896',['Trap']='rbxassetid://4966828196',['Sprint']=
'rbxassetid://4970219155',['VampireBat']='rbxassetid://129078122681639',['Bat Form']='rbxassetid://129078122681639',['Decoy']='rbxassetid://113722008817201'}local perkTag:BillboardGui?=nil
local function destroyPerkTag():()if perkTag then pcall(perkTag.Destroy,perkTag)perkTag=nil end end local function clear():()for player:Player,highlight:Highlight in pairs(highlights)do highlights[
player]=nil highlight:Destroy()end destroyPerkTag()end if type(gameBridge)=='table'and type(gameBridge.onEvent)=='function'then activeBridgeCleanup=gameBridge.onEvent(function(kind:string):()if kind==
'roleUpdate'then lastUpdate=-math.huge end end)end local store:any=host.configData if store and store.values and store.states then local oldMode:any=store.values['Universal.PlayerESP.Chams']if store.
states['Universal.Chams']==nil and(oldMode=='Overlay'or oldMode=='Outline')then store.states['Universal.Chams']=true local fillKey:string='Universal.Chams.Filltransparency'if store.values[fillKey]==
nil then store.values[fillKey]=oldMode=='Outline'and 1 or 0.55 end if type(host.queueConfigSave)=='function'then host.queueConfigSave()end end local teamKey:string='Universal.Chams.Teamcolours'if
store.states[teamKey]==true and store.values['Universal.Chams.Fillcolour']==nil and store.values['Universal.Chams.Outlinecolour']==nil then store.states[teamKey]=false if type(host.queueConfigSave)==
'function'then host.queueConfigSave()end end end local card:any card=framework.Categories.Visuals:CreateModule({Name='Chams',Category='Render',ConfigKey='Universal.Chams',Order=2,Tooltip=
[[Player highlights with independent role palettes and filters. Names and health have their own modules.]],Function=function(enabled:boolean):()if not enabled then clear()card:SetStatus(nil)return end
card:SetStatus('0')card:Render(function():()local now:number=os.clock()if now-lastUpdate<0.2 then return end lastUpdate=now local seen:{[Player]:boolean}={}local count:number=0 local roleMode=roles
and roles.provider~=nil local teamsReplaced=roleMode and roles.provider.ReplacesTeams==true local useTeamColours=not teamsReplaced and card.Options['Team colours'].Value local allowTeammates=not
teamsReplaced and card.Options['Teammates'].Value local showPerk=card.Options['Show Perk']and card.Options['Show Perk'].Value local perkTarget:Player?=nil local targets:any=entity:Refresh()if roleMode
and type(roles.provider.DeadRoles)=='table'and host.Players then targets=table.clone(targets)local included={}for _,target in ipairs(targets)do included[target.Player]=true end for _,player in ipairs(
host.Players:GetPlayers())do if player==host.LocalPlayer or included[player]then continue end local character=player.Character local humanoid=character and character:FindFirstChildOfClass('Humanoid')
if character and character.Parent and humanoid and humanoid.Health<=0 and table.find(roles.provider.DeadRoles,roles:Role(player))then table.insert(targets,{Player=player,Character=character,Humanoid=
humanoid,IsFriendly=entity.IsFriendly and entity:IsFriendly(player)or false})end end end for _,target:any in ipairs(targets)do local player:Player=target.Player local allowedTeam:boolean=true if not
teamsReplaced then local friendly:boolean=target.IsFriendly if not friendly and teamCheck and type(teamCheck.isSameTeam)=='function'then friendly=teamCheck.isSameTeam(player)end allowedTeam=
allowTeammates or not friendly end if not(allowedTeam and not protectedTargets.isProtected(player)and target.Character and target.Humanoid and(target.Humanoid.Health>0 or roleMode and type(roles.
provider.DeadRoles)=='table'and table.find(roles.provider.DeadRoles,roles:Role(player))))then continue end local roleColour:Color3?=nil if roleMode then if not roles:Allows(player)then continue end
local role roleColour,role=roles:Resolve(player,nil)if role=='Murderer'then perkTarget=player end end local teamColour:Color3?=nil if not roleColour and useTeamColours and target.IsFriendly then local
team:Team?=player.Team if team then teamColour=team.TeamColor.Color end end local fillColour:Color3=roleColour or teamColour or card.Options['Fill colour'].Value local outlineColour:Color3=roleColour
or teamColour or card.Options['Outline colour'].Value local fillTransparency:number=card.Options['Fill transparency'].Value local outlineTransparency:number=card.Options['Outline transparency'].Value
local depthMode:Enum.HighlightDepthMode=card.Options['Through walls'].Value and Enum.HighlightDepthMode.AlwaysOnTop or Enum.HighlightDepthMode.Occluded seen[player]=true count+=1 local highlight:
Highlight?=highlights[player]if not highlight or not highlight.Parent then highlight=Instance.new('Highlight')highlight.Name='Wurst_Chams'highlight.Parent=currentWorkspace highlights[player]=highlight
end local resolved:Highlight=highlight::Highlight if resolved.Adornee~=target.Character then resolved.Adornee=target.Character end if resolved.FillColor~=fillColour then resolved.FillColor=fillColour
end if resolved.OutlineColor~=outlineColour then resolved.OutlineColor=outlineColour end if resolved.FillTransparency~=fillTransparency then resolved.FillTransparency=fillTransparency end if resolved.
OutlineTransparency~=outlineTransparency then resolved.OutlineTransparency=outlineTransparency end if resolved.DepthMode~=depthMode then resolved.DepthMode=depthMode end end for player:Player,
highlight:Highlight in pairs(highlights)do if not seen[player]then highlights[player]=nil highlight:Destroy()end end if roleMode and showPerk and perkTarget and perkTarget.Character then local head:
BasePart?=perkTarget.Character:FindFirstChild('Head')or perkTarget.Character:FindFirstChild('HumanoidRootPart')::BasePart?local perk:string?=head and playerPerk(perkTarget)or nil if head and perk then
local iconId:string?=PERK_ICONS[perk]or PERK_ICONS[string.gsub(perk,'%s+','')]if iconId then if not perkTag or not perkTag.Parent or perkTag.Adornee~=head then destroyPerkTag()local billboard:
BillboardGui=Instance.new('BillboardGui')billboard.Name='Wurst_ChamPerk'billboard.AlwaysOnTop=true billboard.LightInfluence=0 billboard.Size=UDim2.fromOffset(26,26)billboard.StudsOffset=Vector3.new(0,
2.9,0)billboard.MaxDistance=1500 billboard.Adornee=head billboard.Parent=head local icon:ImageLabel=Instance.new('ImageLabel')icon.Name='Perk'icon.BackgroundTransparency=1 icon.Size=UDim2.fromScale(1,
1)icon.Image=iconId icon.Parent=billboard perkTag=billboard end local icon:any=perkTag and perkTag:FindFirstChild('Perk')if icon and icon.Image~=iconId then icon.Image=iconId end else destroyPerkTag()
end else destroyPerkTag()end else destroyPerkTag()end card:SetStatus(tostring(count))end)card:Clean(clear)end})card:CreateColor({Name='Fill colour',Default=Color3.fromRGB(85,255,85)})card:CreateColor(
{Name='Outline colour',Default=Color3.fromRGB(85,255,85)})card:CreateSlider({Name='Fill transparency',Min=0,Max=1,Step=0.05,Default=0.55})card:CreateSlider({Name='Outline transparency',Min=0,Max=1,
Step=0.05,Default=0})card:CreateToggle({Name='Through walls',Default=true})card:CreateToggle({Name='Teammates',Default=false})card:CreateToggle({Name='Team colours',Default=false})card:CreateToggle({
Name='Show Perk',Default=true,Tooltip='Perk icon above the murderer.'})roles=context.services.gameRoles:Bind(card,{filters=true})card.OptionFilter=function(name)if roles.provider and roles.provider.
ReplacesTeams and(name=='Teammates'or name=='Team colours')then return false end if name=='Show Perk'then return roles.provider~=nil and roles.provider.Name=='MM2'end return true end card:
RefreshVisibility()Module.roles=roles activeCard=card Module.Initialized=true return card end function Module.destroy():()if activeCard and activeCard.Enabled then activeCard:Toggle(false)end if
activeBridgeCleanup then pcall(activeBridgeCleanup)activeBridgeCleanup=nil end if Module.roles then Module.roles:Destroy()Module.roles=nil end activeCard=nil Module.Initialized=false end return Module
-- src/games/universal/Render/FieldOfView.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='FieldOfView',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil function
Module.init(context:Runtime):any local host:any=context.host local TaskManager:any=host.TaskManager local featureConnections:any=host.featureConnections local disconnectFeatureConnection:any=host.
disconnectFeatureConnection local fovOwnership:any=context.services.fovOwnership local createUniversalFeature:any=host.createUniversalFeature local addNumberOption:any=host.addNumberOption local
workspace:any=host.workspace local fovSettings={value=90}local FovFeature:any=nil local function toggleFov(enabled)disconnectFeatureConnection('FOV')fovOwnership.set('Universal',enabled and
fovSettings.value or nil)if not enabled then if FovFeature then FovFeature:SetStatus(nil)end return end FovFeature:SetStatus(tostring(math.round(fovSettings.value)))featureConnections.FOV=TaskManager:
Connect(function()fovOwnership.set('Universal',fovSettings.value)fovOwnership.apply(workspace.CurrentCamera)end,nil,'FOV')end FovFeature=createUniversalFeature('FOV',
'Keep the camera field of view fixed',10,toggleFov,{categoryName='Render'})addNumberOption(FovFeature,'Field of view',fovSettings.value,20,120,function(value)fovSettings.value=value if FovFeature.
enabled then FovFeature:SetStatus(tostring(math.round(value)))end end)activeCleanup=function():()disconnectFeatureConnection('FOV')fovOwnership.set('Universal',nil)end Module.Initialized=true return
FovFeature end function Module.destroy():()if activeCleanup then activeCleanup()end activeCleanup=nil Module.Initialized=false end return Module
-- src/games/universal/Render/FreeCam.lua
local Module={Name='FreeCam',Initialized=false}local card,services function Module.init(context)local host=context.host services=context.services local input,run=host.UserInputService,host.RunService
local actions=game:GetService('ContextActionService')local access=context.services.cameraAccess local world=host.workspace or workspace local create=host.create local id='ARandomMenu.FreeCam.'..
tostring({})local cameraState={IsActive=function()return card~=nil and card.Enabled end}context.services.freeCam=cameraState local function spectating()return context.services.spectate and context.
services.spectate.IsActive()end card=context.framework.Categories.Visuals:CreateModule({Name='FreeCam',Category='World',ConfigKey='Universal.FreeCam',Order=6,Tooltip=
[[Move the camera without moving your character. WASD, Q/E, right-drag; touch controls on mobile.]],Function=function(enabled)cameraState.Session=enabled and{}or nil if not enabled then card:
SetStatus(nil)return end local keys,held,connections={},{},{}local camera,saved,position,pitch,yaw local writtenFrame,writtenFocus local controls,controlsEnabled local touchGui,touchGuiEnabled local
moveTouch,lookTouch,dragging local stick=Vector2.zero local mouseMode,mouseIcon local gui,pad,knob local function connect(signal,callback)table.insert(connections,signal:Connect(callback))end
local function releaseMouse()if not dragging then return end dragging=false if input.MouseBehavior==Enum.MouseBehavior.LockCurrentPosition then input.MouseBehavior=mouseMode end if input.
MouseIconEnabled==false then input.MouseIconEnabled=mouseIcon end end local function resetInput()table.clear(keys)table.clear(held)moveTouch,lookTouch=nil,nil stick=Vector2.zero if knob then knob.
Position=UDim2.fromScale(0.5,0.5)end releaseMouse()end local function restoreCamera()if camera and saved then pcall(function()if camera.CameraType~=Enum.CameraType.Scriptable then return end camera.
CameraType=saved.kind if camera.CFrame==writtenFrame then camera.CFrame=saved.frame end if camera.Focus==writtenFocus then camera.Focus=saved.focus end end)end camera,saved,writtenFrame,writtenFocus=
nil,nil,nil,nil end local function releaseControls()if controls and controlsEnabled then pcall(controls.Enable,controls)end controls,controlsEnabled=nil,nil if touchGui then pcall(function()touchGui.
Enabled=touchGuiEnabled end)end touchGui=nil end card:Clean(function()run:UnbindFromRenderStep(id)table.clear(keys)table.clear(held)for _,name in ipairs({'Forward','Back','Left','Right','Up','Down'})
do actions:UnbindAction(id..name)end for _,connection in ipairs(connections)do connection:Disconnect()end releaseMouse()restoreCamera()releaseControls()if gui then gui:Destroy()end end)local function
hold(name,source,down)held[name]=held[name]or{}held[name][source]=down and true or nil keys[name]=next(held[name])~=nil end for _,definition in ipairs({{'Forward',Enum.KeyCode.W,Enum.KeyCode.Up},{
'Back',Enum.KeyCode.S,Enum.KeyCode.Down},{'Left',Enum.KeyCode.A,Enum.KeyCode.Left},{'Right',Enum.KeyCode.D,Enum.KeyCode.Right},{'Up',Enum.KeyCode.E,Enum.KeyCode.Space},{'Down',Enum.KeyCode.Q,Enum.
KeyCode.LeftControl}})do local name=definition[1]actions:BindActionAtPriority(id..name,function(_,state,event)if spectating()or input:GetFocusedTextBox()then held[name]=nil keys[name]=false return
Enum.ContextActionResult.Pass end if state==Enum.UserInputState.Cancel then held[name]=nil keys[name]=false else hold(name,event.KeyCode,state==Enum.UserInputState.Begin or state==Enum.UserInputState.
Change)end return Enum.ContextActionResult.Sink end,false,Enum.ContextActionPriority.High.Value+1,definition[2],definition[3])end local function pan(delta)if spectating()then return end local
sensitivity=math.rad(card.Options.Sensitivity.Value)yaw-=delta.X*sensitivity pitch=math.clamp(pitch-delta.Y*sensitivity,-math.rad(89),math.rad(89))end connect(input.InputBegan,function(event,processed
)if processed or spectating()or input:GetFocusedTextBox()or not position then return end if event.UserInputType==Enum.UserInputType.MouseButton2 then mouseMode,mouseIcon=input.MouseBehavior,input.
MouseIconEnabled dragging=true input.MouseBehavior=Enum.MouseBehavior.LockCurrentPosition input.MouseIconEnabled=false end end)connect(input.InputChanged,function(event)if dragging and event.
UserInputType==Enum.UserInputType.MouseMovement and not input:GetFocusedTextBox()then pan(event.Delta)end end)connect(input.InputEnded,function(event)if event.UserInputType==Enum.UserInputType.
MouseButton2 then releaseMouse()end end)connect(input.WindowFocusReleased,resetInput)connect(input.TextBoxFocused,resetInput)if input.TouchEnabled then gui=create('Frame',{Name='FreeCamControls',
Parent=host.ScreenGui,Size=UDim2.fromScale(1,1),BackgroundTransparency=1,ZIndex=30})local function button(name,text,at,size)local object=create('TextButton',{Name=name,Parent=gui,Position=at,Size=size
,Text=text,Font=Enum.Font.Code,TextSize=18,TextColor3=Color3.new(1,1,1),BackgroundColor3=Color3.fromRGB(25,25,25),BackgroundTransparency=0.3,BorderSizePixel=1,BorderColor3=Color3.fromRGB(120,120,120),
AutoButtonColor=true,ZIndex=31})if host.ThemeEngine then host.ThemeEngine.Bind(object,'BackgroundColor3','background')host.ThemeEngine.Bind(object,'TextColor3','text')end return object end pad=button(
'Move','',UDim2.new(0,22,1,-152),UDim2.fromOffset(104,104))knob=create('Frame',{Parent=pad,Name='Stick',AnchorPoint=Vector2.new(0.5,0.5),Position=UDim2.fromScale(0.5,0.5),Size=UDim2.fromOffset(26,26),
BackgroundColor3=Color3.fromRGB(220,220,220),BorderSizePixel=0,ZIndex=32})local function moveStick(event)local centre=pad.AbsolutePosition+pad.AbsoluteSize*0.5 local delta=(Vector2.new(event.Position.
X,event.Position.Y)-centre)/42 stick=delta.Magnitude>1 and delta.Unit or delta if stick.Magnitude<0.12 then stick=Vector2.zero end knob.Position=UDim2.new(0.5,stick.X*36,0.5,stick.Y*36)end connect(pad
.InputBegan,function(event)if event.UserInputType==Enum.UserInputType.Touch and not moveTouch then moveTouch=event moveStick(event)end end)for _,name in ipairs({'Up','Down'})do local object=button(
name,name=='Up'and'+'or'\u{2212}',UDim2.new(1,-70,1,name=='Up'and-152 or-96),UDim2.fromOffset(48,48))connect(object.InputBegan,function(event)if event.UserInputType==Enum.UserInputType.Touch then
hold(name,event,true)end end)connect(object.InputEnded,function(event)if event.UserInputType==Enum.UserInputType.Touch then hold(name,event,false)end end)end local exit=button('Exit','Exit FreeCam',
UDim2.new(0.5,-66,0,48),UDim2.fromOffset(132,30))connect(exit.Activated,function()if card.Feature and card.Feature.enabled and card.Feature.activate then card.Feature.activate()else card:Toggle(false)
end end)connect(input.TouchStarted,function(event,processed)local current=world.CurrentCamera if not processed and not spectating()and not lookTouch and event~=moveTouch and current and event.Position
.X>current.ViewportSize.X*0.45 and not input:GetFocusedTextBox()then lookTouch=event end end)connect(input.TouchMoved,function(event)if event==moveTouch then moveStick(event)elseif event==lookTouch
and position then pan(event.Delta)end end)connect(input.TouchEnded,function(event)hold('Up',event,false)hold('Down',event,false)if event==moveTouch then moveTouch=nil stick=Vector2.zero knob.Position=
UDim2.fromScale(0.5,0.5)end if event==lookTouch then lookTouch=nil end end)end card:SetStatus('WASD / touch')run:BindToRenderStep(id,Enum.RenderPriority.Camera.Value+1,function(dt)if spectating()then
resetInput()if gui then gui.Visible=false end return end if gui then gui.Visible=true end local nextCamera=world.CurrentCamera if camera~=nextCamera then restoreCamera()camera=nextCamera if camera
then saved={kind=camera.CameraType,frame=camera.CFrame,focus=camera.Focus}if not position then position=camera.CFrame.Position local look=camera.CFrame.LookVector pitch,yaw=math.asin(math.clamp(look.Y
,-1,1)),math.atan2(-look.X,-look.Z)end end end if not camera then return end if writtenFrame and camera.CameraType~=Enum.CameraType.Scriptable then saved.kind,saved.frame,saved.focus=camera.CameraType
,camera.CFrame,camera.Focus elseif writtenFrame and camera.CFrame~=writtenFrame then saved.frame,saved.focus=camera.CFrame,camera.Focus end local nextControls=access:GetControls()if type(nextControls)
~='table'or type(nextControls.Disable)~='function'or type(nextControls.Enable)~='function'then nextControls=nil end if nextControls~=controls then releaseControls()controls=nextControls if controls
then controlsEnabled=controls.controlsEnabled~=false local ok=pcall(controls.Disable,controls)if not ok then controls,controlsEnabled=nil,nil end end end local playerGui=host.LocalPlayer:
FindFirstChildOfClass('PlayerGui')local nativeTouch=playerGui and playerGui:FindFirstChild('TouchGui')if nativeTouch~=touchGui then if touchGui then pcall(function()touchGui.Enabled=touchGuiEnabled
end)end touchGui=nativeTouch if touchGui then touchGuiEnabled=touchGui.Enabled end end if touchGui then touchGui.Enabled=false end if not controls then local character=host.LocalPlayer.Character local
humanoid=character and character:FindFirstChildOfClass('Humanoid')if humanoid then humanoid:Move(Vector3.zero,false)end end local rotation=CFrame.fromOrientation(pitch,yaw,0)if not input:
GetFocusedTextBox()then local move=Vector3.new((keys.Right and 1 or 0)-(keys.Left and 1 or 0)+stick.X,(keys.Up and 1 or 0)-(keys.Down and 1 or 0),(keys.Back and 1 or 0)-(keys.Forward and 1 or 0)+stick
.Y)local direction=rotation:VectorToWorldSpace(Vector3.new(move.X,0,move.Z))+Vector3.yAxis*move.Y if direction.Magnitude>1 then direction=direction.Unit end local speed=card.Options.Speed.Value*(input
:IsKeyDown(Enum.KeyCode.LeftShift)and 3 or 1)position+=direction*speed*math.clamp(dt,0,0.1)end camera.CameraType=Enum.CameraType.Scriptable camera.CFrame=CFrame.new(position)*rotation camera.Focus=
camera.CFrame*CFrame.new(0,0,-16)writtenFrame,writtenFocus=camera.CFrame,camera.Focus end)end})card:CreateSlider({Name='Speed',Min=1,Max=250,Default=48})card:CreateSlider({Name='Sensitivity',Min=0.05,
Max=0.6,Default=0.2})Module.Initialized=true return card end function Module.destroy()if card and card.Enabled then card:Toggle(false)end if services then services.freeCam=nil end card,services=nil,
nil Module.Initialized=false end return Module
-- src/games/universal/Render/Fullbright.lua
local Module={Name='Fullbright',PlaceId=0,Initialized=false}local card function Module.init(context)local lighting=context.host.Lighting local original,applied={},{}local originalTime,appliedTime,
elapsed local function write(key,value)local current=lighting[key]if applied[key]~=nil and current~=applied[key]then original[key]=current end if original[key]==nil then original[key]=current end if
current~=value then lighting[key]=value end applied[key]=value end local function restoreTime()if appliedTime~=nil and lighting.ClockTime==appliedTime then lighting.ClockTime=originalTime end
originalTime,appliedTime=nil,nil end local function restore()restoreTime()for key,value in pairs(original)do if lighting[key]==applied[key]then lighting[key]=value end end table.clear(original)table.
clear(applied)end local function tick(dt)local brightness=card.Options.Brightness.Value if brightness==math.huge then brightness=1000 end write('Brightness',brightness)write('GlobalShadows',false)
write('FogEnd',100000)write('Ambient',Color3.fromRGB(178,178,178))write('OutdoorAmbient',Color3.fromRGB(178,178,178))if card.Options['Time Changer'].Value then if appliedTime~=nil and lighting.
ClockTime~=appliedTime then originalTime=lighting.ClockTime end if originalTime==nil then originalTime=lighting.ClockTime end elapsed=(elapsed+math.max(0,dt or 0))%4 appliedTime=(card.Options[
'Fast time'].Value and elapsed*6 or card.Options.Time.Value)%24 if lighting.ClockTime~=appliedTime then lighting.ClockTime=appliedTime end else restoreTime()end end card=context.framework.Categories.
Render:CreateModule({Name='Fullbright',Category='Render',ConfigKey='Universal.Fullbright',Tooltip='Bright lighting; optionally control the time of day.',Function=function(enabled)if not enabled then
restore()return end elapsed=0 card:Clean(restore)tick(0)card:Loop(tick)end})card:CreateSlider({Name='Brightness',Min=-10,Max=math.huge,Default=3})card:CreateToggle({Name='Time Changer',Default=false,
Function=function(value)if not value then restoreTime()end end})card:CreateSlider({Name='Time',Min=0,Max=24,Default=14,Show={Option='Time Changer',Values={true}}})card:CreateToggle({Name='Fast time',
Default=false,Show={Option='Time Changer',Values={true}},Tooltip='Loop a full day every four seconds.'})Module.Initialized=true return card end function Module.destroy()if card and card.Enabled then
card:Toggle(false)end card=nil Module.Initialized=false end return Module
-- src/games/universal/Render/HP.lua
local Module={Name='HP',Initialized=false}local activeCard,cleanup,hud,roleBinding local LOW_HEALTH=Color3.fromRGB(255,70,70)local MID_HEALTH=Color3.fromRGB(255,220,60)local HIGH_HEALTH=Color3.
fromRGB(85,255,85)local function healthValues(health,maximum)if type(health)~='number'or health~=health or math.abs(health)==math.huge then health=0 end if type(maximum)~='number'or maximum~=maximum
or maximum<=0 or maximum==math.huge then maximum=100 end health=math.max(0,health)return health,maximum,math.clamp(health/maximum,0,1)end local function healthColour(fraction)return fraction<0.5 and
LOW_HEALTH:Lerp(MID_HEALTH,fraction*2)or MID_HEALTH:Lerp(HIGH_HEALTH,(fraction-0.5)*2)end function Module.init(context:any):any local host,render,entity=context.host,context.render,context.entity
local create=host.create local layer,tracked=render:Layer('HPLayer'),{}local store=host.configData if store and store.states['Universal.HP']==nil then store.states['Universal.HP']=store.states[
'Universal.PlayerESP']==true and(store.states['Universal.PlayerESP.Healthbar']~=false or store.states['Universal.PlayerESP.Healthtext']==true)end hud=create('TextLabel',{Name='SelfHP',Parent=host.
ScreenGui,AnchorPoint=Vector2.new(0.5,0),Position=UDim2.new(0.5,6,0.5,30),Size=UDim2.fromOffset(140,24),BackgroundTransparency=1,BorderSizePixel=0,Visible=false,ZIndex=20,Text='',Font=Enum.Font.Arial,
TextSize=18,TextStrokeTransparency=1,TextColor3=healthColour(1)})local scale=create('UIScale',{Parent=hud,Scale=1})local centerPosition,bottomPosition=UDim2.new(0.5,6,0.5,30),UDim2.new(0.5,0,1,-110)
local card local function clear()render:ReleaseAll(layer)table.clear(tracked)layer.Visible=false hud.Visible=false hud.Text=''end card=context.framework.Categories.Visuals:CreateModule({Name='HP',
Category='Render',ConfigKey='Universal.HP',Order=4,Tooltip=[[Independent player health bars and optional heart-style Self HP display.]],Function=function(enabled)if not enabled then clear()card:
SetStatus(nil)return end card:Clean(clear)card:Render(function(dt)local options=card.Options local targets=entity:Refresh()local camera=(host.workspace or workspace).CurrentCamera layer.Visible=true
local seen,count={},0 if camera and options.Players.Value then for _,target in ipairs(targets)do if not target.Player or target.Health<=0 or target.Distance>options['Max distance'].Value or not(
roleBinding.provider and roleBinding.provider.ReplacesTeams)and not options.Teammates.Value and target.IsFriendly then continue end local rect=render:ModelRect(camera,target.Character)if not rect or
not options['Through walls'].Value and not entity:VisibleFrom(target)then continue end local health,maximum,fraction=healthValues(target.Health,target.MaxHealth)local drawing=render:Set(layer,target.
Player)drawing:Show(true)drawing:Bar(Vector2.new(rect.left-2,rect.top),rect.height,fraction)if options['Health text'].Value then drawing:Label('HP',tostring(math.ceil(health)),Vector2.new(rect.left-14
,rect.top+(1-fraction)*rect.height),0.5,12,healthColour(fraction),false)else drawing:HideLabel('HP')end tracked[target.Player]=true seen[target.Player]=true count+=1 end end for player in pairs(
tracked)do if not seen[player]then render:Release(layer,player)tracked[player]=nil end end local showSelf=options['Self HP'].Value if hud.Visible~=showSelf then hud.Visible=showSelf end if showSelf
then local character=host.LocalPlayer.Character local humanoid=character and character:FindFirstChildOfClass('Humanoid')local health,maximum,fraction=healthValues(humanoid and humanoid.Health,humanoid
and humanoid.MaxHealth)local wanted=humanoid and humanoid.Health>0 and tostring(math.ceil(health))..' \u{2764}\u{fe0f}'or''if hud.Text~=wanted then hud.Text=wanted end local colour=healthColour(
fraction)if hud.TextColor3~=colour then hud.TextColor3=colour end local position=options['Self HP position'].Value=='Bottom'and bottomPosition or centerPosition if hud.Position~=position then hud.
Position=position end local wantedScale=options['Self HP scale'].Value if camera then wantedScale=math.min(wantedScale,math.max(0.5,(camera.ViewportSize.X-24)/140))end if scale.Scale~=wantedScale then
scale.Scale=wantedScale end end card:SetStatus(options['Self HP'].Value and tostring(count)..' + self'or tostring(count))end)end})card:CreateToggle({Name='Players',Default=true})card:CreateToggle({
Name='Health text',Default=false,Show={Option='Players'}})card:CreateSlider({Name='Max distance',Min=50,Max=5000,Default=2000,Show={Option='Players'}})card:CreateToggle({Name='Teammates',Default=true,
Show={Option='Players'}})card:CreateToggle({Name='Through walls',Default=true,Show={Option='Players'}})card:CreateToggle({Name='Self HP',Default=false})card:CreateDropdown({Name='Self HP position',
List={'Center','Bottom'},Index=1,Show={Option='Self HP'}})card:CreateSlider({Name='Self HP scale',Min=0.8,Max=1.6,Decimal=1,Default=1,Show={Option='Self HP'}})roleBinding=context.services.gameRoles:
Bind(card,{palette=false})card.OptionFilter=function(name)return not(name=='Teammates'and roleBinding.provider and roleBinding.provider.ReplacesTeams)end card:RefreshVisibility()activeCard,cleanup=
card,clear Module.Initialized=true return card end function Module.destroy()if activeCard and activeCard.Enabled then activeCard:Toggle(false)end if cleanup then cleanup()end if hud then hud:Destroy()
end if roleBinding then roleBinding:Destroy()end activeCard,cleanup,hud,roleBinding=nil,nil,nil,nil Module.Initialized=false end return Module
```

```
-- src/games/universal/Render/ItemRender.lua
export type Runtime={framework:any,entity:any,host:any,render:any,services:any}local Module={Name='ItemRender',PlaceId=0,Events={}::{[string]:any},Initialized=false,layer=nil::Frame?}local
activeCleanup:(()->())?=nil local function setProviderEnabled(render:any,providerStates:{[string]:boolean},provider:any,value:boolean):()providerStates[provider.name]=value if render.Enabled then
pcall(provider.setEnabled,value)return end if not value then return end local feature:any=render.Feature if feature and type(feature.activate)=='function'and feature.enabled~=true then feature.
activate()else render:Toggle(true)end end function Module.appearanceSignature(object:Instance):string?local parts:{BasePart}={}local queue:{Instance}={object}local cursor:number=1 while cursor<=#queue
do local node:Instance=queue[cursor]cursor+=1 if node:IsA('BasePart')then table.insert(parts,node::BasePart)end if#parts>32 then return nil end for _,child:Instance in ipairs(node:GetChildren())do if#
queue>=96 then return nil end table.insert(queue,child)end end if#parts==0 then return nil end local center:Vector3=Vector3.zero for _,part:BasePart in ipairs(parts)do center+=part.Position end center
/=#parts local function numberKey(value:number):string return tostring(math.round(value*20))end local function vectorKey(value:Vector3):string return numberKey(value.X)..','..numberKey(value.Y)..','..
numberKey(value.Z)end local tokens:{string}={}for _,part:BasePart in ipairs(parts)do local meshId:string=''local meshShape:string=''if part:IsA('MeshPart')then meshId=(part::MeshPart).MeshId end local
mesh:SpecialMesh?=part:FindFirstChildOfClass('SpecialMesh')::SpecialMesh?if mesh then meshId=mesh.MeshId meshShape=tostring(mesh.MeshType)..':'..vectorKey(mesh.Scale)..':'..vectorKey(mesh.Offset)end
local shape:string=part:IsA('Part')and tostring((part::Part).Shape)or part.ClassName local distances:{string}={}for _,other:BasePart in ipairs(parts)do table.insert(distances,numberKey((part.Position-
other.Position).Magnitude))end table.sort(distances)table.insert(tokens,table.concat({part.ClassName,shape,vectorKey(part.Size),meshId,meshShape,numberKey((part.Position-center).Magnitude),table.
concat(distances,',')},':'))end table.sort(tokens)return table.concat(tokens,'|')end function Module.init(context:Runtime):any local host:any=context.host local framework:any=context.framework local
localPlayer:Player=host.LocalPlayer local userInput:UserInputService=host.UserInputService local currentWorkspace:Workspace=host.workspace or workspace local menu:any=context.services.menu local
tracked:{[Instance]:boolean}={}local index:{Instance}={}local indexConnections:{RBXScriptConnection}={}local indexDirty:boolean=true local known:{string}={}local wanted:{string}={}local picking:
boolean=false local touchMatching:string='Appearance'local appearances:{[string]:string}={}local signatureCache:{[Instance]:any}=setmetatable({},{__mode='k'})local profileSequence:number=0 local
nextIndexAt:number=0 local nextShapeRefresh:number=0 local function signature(instance:Instance):string?if signatureCache[instance]==nil then signatureCache[instance]=Module.appearanceSignature(
instance)or false end return signatureCache[instance]or nil end local function release(instance:Instance):()if not tracked[instance]then return end tracked[instance]=nil local renderLib:any=context.
render if renderLib and Module.layer then renderLib:Release(Module.layer,instance)end end local function releaseAll():()for instance:Instance in pairs(tracked)do release(instance)end end
local function matches(instance:Instance):boolean local lowered:string=string.lower(instance.Name)for _,entry:string in ipairs(wanted)do local appearance:string?=appearances[entry]if appearance then
if signature(instance)==appearance then return true end elseif not string.match(entry,'^Look #%d+ ')and string.find(lowered,string.lower(entry),1,true)then return true end end return false end
local function rebuildIndex():()table.clear(index)indexDirty=false if#wanted==0 then return end for _,descendant:Instance in ipairs(currentWorkspace:GetDescendants())do if descendant:IsA('BasePart')or
descendant:IsA('Model')or descendant:IsA('Tool')then if matches(descendant)then table.insert(index,descendant)end end end end local function watchWorkspace():()local function changed(descendant:
Instance):()signatureCache[descendant]=nil local ancestor:Instance?=descendant.Parent while ancestor and ancestor~=currentWorkspace do signatureCache[ancestor]=nil ancestor=ancestor.Parent end if#
wanted>0 then indexDirty=true end end table.insert(indexConnections,currentWorkspace.DescendantAdded:Connect(changed))table.insert(indexConnections,currentWorkspace.DescendantRemoving:Connect(changed)
)end local renderLib:any=context.render local layer:Frame?=renderLib and renderLib:Layer('ItemRenderLayer')or nil Module.layer=layer local itemEsp:any=context.services and context.services.itemEsp
local providerStates:{[string]:boolean}={}local function forEachProvider(callback:(any)->()):()if not itemEsp or type(itemEsp.providers)~='function'then return end for _,provider:any in ipairs(itemEsp
.providers())do pcall(callback,provider)end end local render:any render=framework.Categories.Visuals:CreateModule({Name='ItemESP',Category='Render',ConfigKey='Universal.ItemRender',Order=2,Tooltip=
[[Names and outlines every object on the list, wherever it is.]],Function=function(enabled:boolean):()if layer then layer.Visible=enabled end if not enabled then forEachProvider(function(provider:any)
:()if providerStates[provider.name]then pcall(provider.setEnabled,false)end end)render:SetStatus(nil)releaseAll()picking=false return end forEachProvider(function(provider:any):()if providerStates[
provider.name]then pcall(provider.setEnabled,true)end end)render:SetStatus('0')render:Clean(function():()releaseAll()picking=false if layer then layer.Visible=false end end)indexDirty=true
watchWorkspace()render:Clean(function():()for _,connection:RBXScriptConnection in ipairs(indexConnections)do pcall(function():()connection:Disconnect()end)end indexConnections={}table.clear(index)end)
render:Render(function():()if next(appearances)~=nil and os.clock()>=nextShapeRefresh then nextShapeRefresh=os.clock()+5 signatureCache=setmetatable({},{__mode='k'})indexDirty=true end if indexDirty
and os.clock()>=nextIndexAt then nextIndexAt=os.clock()+0.25 rebuildIndex()end Module.render(context,render,tracked,wanted,release,index,matches)end)end})local objectList:any objectList=render:
CreateList({Name='Objects',Items=function():{string}return known end,Tooltip='Every object the module knows about. Tick the ones to draw.',Function=function(_selected:any,names:{string}):()wanted=
names indexDirty=true releaseAll()end})local function learn(name:string):()if name==''then return end if not table.find(known,name)then table.insert(known,name)table.sort(known)end objectList:Set(name
,true)end local function findBestMatchingName(query:string):string?local lowered:string=string.lower(query)local candidates:{string}={}for _,descendant:Instance in ipairs(currentWorkspace:
GetDescendants())do if descendant:IsA('BasePart')or descendant:IsA('Model')or descendant:IsA('Tool')then local dName:string=descendant.Name local dLower:string=string.lower(dName)if dLower==lowered
then return dName end if string.find(dLower,lowered,1,true)then table.insert(candidates,dName)end end end for _,c:string in ipairs(candidates)do if string.find(string.lower(c),'^'..lowered)then return
c end end if#candidates>0 then return candidates[1]end return nil end local pendingInput:string=''local addBox:any local function commitAddByName(rawText:string?):()local text:string=rawText or
pendingInput local trimmed:string=string.match(text,'^%s*(.-)%s*$')or text if trimmed==''or#trimmed<2 then return end local best:string=findBestMatchingName(trimmed)or trimmed learn(best)render:
SetStatus('Added: '..best)pendingInput=''if addBox and addBox.Object and typeof(addBox.Object)=='Instance'and addBox.Object:IsA('TextBox')then addBox.Object.Text=''end end addBox=render:CreateTextBox(
{Name='Add by name',Default='',Tooltip=[[Type a name and press Enter (or click Add Object). Finds closest matching items.]],Function=function(value:string):()pendingInput=value end})if addBox and
addBox.Object and typeof(addBox.Object)=='Instance'and addBox.Object:IsA('TextBox')then local box:TextBox=addBox.Object box.FocusLost:Connect(function(enterPressed:boolean)if enterPressed then
commitAddByName(box.Text)end end)end render:CreateButton({Name='Add object',Tooltip='Adds the name typed above to the list.',Function=function():()local text:string=(addBox.Object and addBox.Object.
Text)or pendingInput commitAddByName(text)end})local function resolveItemInstance(part:BasePart):Instance local tool:Tool?=part:FindFirstAncestorOfClass('Tool')if tool then return tool end local
parent:Instance?=part.Parent if parent and parent:IsA('Model')and parent~=currentWorkspace then local parentNameLower:string=string.lower(parent.Name)local isGenericMap:boolean=parentNameLower=='map'
or parentNameLower=='workspace'or parentNameLower=='geometry'or parentNameLower=='spawns'or parentNameLower=='terrain'or parentNameLower=='baseplate'local childCount:number=#parent:GetChildren()if not
isGenericMap and childCount<=15 then return parent end end return part end render:CreateDropdown({Name='Touch matching',List={'Appearance','Name'},Index=1,Function=function(value:string):()
touchMatching=value end,Tooltip=[[Appearance matches a small geometry/mesh template, not every Model. ]]..[[Templates last this session; reselect after rejoining. Name keeps the old search.]]})render:
CreateButton({Name='Touch part',Tooltip='Hides the menu, then adds whatever you tap next.',Function=function():()if picking then picking=false render:SetStatus('cancelled')return end picking=true menu
.setVisible(false)render:Notify('tap the object you want rendered')end})render:CreateButton({Name='Clear list',Function=function():()for _,name:string in ipairs(known)do objectList:Set(name,false)end
table.clear(known)table.clear(appearances)signatureCache=setmetatable({},{__mode='k'})wanted={}objectList:Refresh()releaseAll()render:SetStatus('list cleared')end})render:CreateToggle({Name=
'Show distance',Default=true})render:CreateToggle({Name='Outline',Default=true,Tooltip="Draw the object's silhouette through walls."})render:CreateSlider({Name='Max distance',Min=50,Max=5000,Default=
1200})render:CreateSlider({Name='Text size',Min=8,Max=22,Default=12})render:CreateColor({Name='Colour',Default=Color3.fromRGB(85,255,85)})local pickConnection:RBXScriptConnection=userInput.InputBegan:
Connect(function(input:InputObject,processed:boolean):()if not picking or processed then return end if input.UserInputType~=Enum.UserInputType.Touch and input.UserInputType~=Enum.UserInputType.
MouseButton1 then return end local camera:Camera?=currentWorkspace.CurrentCamera if not camera then return end local ray:Ray=(camera::Camera):ScreenPointToRay(input.Position.X,input.Position.Y)local
parameters:RaycastParams=RaycastParams.new()parameters.FilterType=Enum.RaycastFilterType.Exclude parameters.IgnoreWater=true parameters.FilterDescendantsInstances={localPlayer.Character::any,camera}
local result:RaycastResult?=currentWorkspace:Raycast(ray.Origin,ray.Direction.Unit*4096,parameters)if not result then render:Notify('nothing under that tap')return end picking=false local hit:Instance
=(result::RaycastResult).Instance if not hit:IsA('BasePart')then render:Notify('invalid part tapped')menu.setVisible(true)return end local targetItem:Instance=resolveItemInstance(hit::BasePart)local
name:string=targetItem.Name if touchMatching=='Appearance'then local fingerprint:string?=Module.appearanceSignature(targetItem)if not fingerprint then menu.setVisible(true)render:Notify(
'Object too large/complex; select a smaller part or use Name')return end profileSequence+=1 name='Look #'..tostring(profileSequence)..' ('..name..')'appearances[name]=fingerprint end learn(name)menu.
setVisible(true)render:SetStatus(name..' added')end)render:CreateNote([[Touch matching: Appearance selects matching geometry in this session; Name selects by name.]])local providerOptions:{[string]:
any}={}local activeProviders:{[string]:any}={}local function refreshProviders():()local previous:{[string]:any}=activeProviders local current:{[string]:any}={}forEachProvider(function(provider:any):()
current[provider.name]=provider end)activeProviders=current for name:string,provider:any in pairs(previous)do if current[name]~=provider and providerStates[name]then pcall(provider.setEnabled,false)
end end for name:string,provider:any in pairs(current)do if not providerOptions[name]then if render.Feature then render.Feature.optionEntersFeature=true end providerOptions[name]=render:CreateToggle({
Name=name,Default=false,Tooltip=provider.tooltip,Function=function(value:boolean):()providerStates[name]=value local selected:any=activeProviders[name]if selected then setProviderEnabled(render,
providerStates,selected,value)end end})elseif previous[name]~=provider and render.Enabled and providerStates[name]then pcall(provider.setEnabled,true)end end for name:string,option:any in pairs(
providerOptions)do option:SetVisible(current[name]~=nil)end end refreshProviders()local providerWatch:(()->())?=nil if itemEsp and type(itemEsp.onChanged)=='function'then providerWatch=itemEsp.
onChanged(refreshProviders)end activeCleanup=function():()if render.Enabled then render:Toggle(false)end picking=false pcall(function():()pickConnection:Disconnect()end)if providerWatch then pcall(
providerWatch)providerWatch=nil end for _,connection:RBXScriptConnection in ipairs(indexConnections)do connection:Disconnect()end table.clear(indexConnections)releaseAll()end Module.Initialized=true
return render end function Module.render(context:Runtime,render:any,tracked:{[Instance]:any},wanted:{string},release:(Instance)->(),index:{Instance},matches:(Instance)->boolean):()local host:any=
context.host local renderLib:any=context.render local layer:Frame?=Module.layer local localPlayer:Player=host.LocalPlayer local options:any=render.Options local currentWorkspace:Workspace=host.
workspace or workspace local camera:Camera?=currentWorkspace.CurrentCamera if#wanted==0 or not renderLib or not layer or not camera then for instance:Instance in pairs(tracked)do release(instance)end
return end local resolvedCamera:Camera=camera::Camera local resolvedLayer:Frame=layer::Frame local character:Model?=localPlayer.Character local root:BasePart?=character and character:FindFirstChild(
'HumanoidRootPart')::BasePart?local origin:Vector3=root and(root::BasePart).Position or Vector3.zero local maxDistance:number=options['Max distance'].Value local colour:Color3=options['Colour'].Value
local textSize:number=math.round(options['Text size'].Value)local seen:{[Instance]:boolean}={}for _,descendant:Instance in ipairs(index)do if not descendant.Parent then continue end local isPart:
boolean=descendant:IsA('BasePart')local isModel:boolean=descendant:IsA('Model')or descendant:IsA('Tool')if not isPart and not isModel then continue end if character and descendant:IsDescendantOf(
character)then continue end if not matches(descendant)then continue end local duplicate:boolean=false local ancestor:Instance?=descendant.Parent while ancestor and ancestor~=currentWorkspace do if(
ancestor:IsA('Model')or ancestor:IsA('Tool'))and matches(ancestor)then duplicate=true break end ancestor=ancestor.Parent end if duplicate then continue end local anchor:BasePart?=nil if isPart then
anchor=descendant::BasePart else anchor=descendant:IsA('Model')and(descendant::Model).PrimaryPart or nil anchor=anchor or descendant:FindFirstChildWhichIsA('BasePart',true)::BasePart?end if not anchor
then continue end local resolvedAnchor:BasePart=anchor::BasePart local distance:number=(resolvedAnchor.Position-origin).Magnitude if distance>maxDistance then continue end local projected:Vector2?=
renderLib:Project(resolvedCamera,resolvedAnchor.Position)if not projected then continue end seen[descendant]=true tracked[descendant]=true local drawings:any=renderLib:Set(resolvedLayer,descendant)
drawings:Show(true)drawings:Label('NameTag',options['Show distance'].Value and('['..tostring(math.round(distance))..'] '..descendant.Name)or descendant.Name,projected::Vector2,0.5,textSize,colour,true
)drawings:Highlight(descendant,colour,1,options['Outline'].Value and'Outline'or'Off')end local count:number=0 for instance:Instance in pairs(tracked)do if not seen[instance]or not instance.Parent then
release(instance)else count+=1 end end local status:string=tostring(count)if render.Status~=status then render:SetStatus(status)end end function Module.destroy():()if activeCleanup then pcall(
activeCleanup)end activeCleanup=nil Module.layer=nil Module.Initialized=false end return Module
-- src/games/universal/Render/Markers.lua
local Module={Name='Markers',Initialized=false}local card,clear,inputConnection,services function Module.init(context)local host,render=context.host,context.render services=context.services local menu
=services.menu local layer=render:Layer('MarkersLayer')local key='Universal.Markers.Places.'..tostring(game.PlaceId)local config=host.configData local points={}local stored=config and config.values[
key]if type(stored)~='table'then stored={}end local loaded=0 for name,position in pairs(stored)do if type(name)=='string'and type(position)=='table'and loaded<128 then local x,y,z=tonumber(position[1]
),tonumber(position[2]),tonumber(position[3])if x and y and z and x==x and y==y and z==z and math.max(math.abs(x),math.abs(y),math.abs(z))<1e7 then points[name]={x,y,z,position[4]=='surface'and
'surface'or nil}loaded+=1 end end end local listeners={}local function pointNames()local names={}for name in pairs(points)do table.insert(names,name)end table.sort(names)return names end services.
points={Names=pointNames,Get=function(name)local data=points[name]if data then return Vector3.new(data[1],data[2],data[3]),data[4]=='surface'end end,Subscribe=function(callback)listeners[callback]=
true return function()listeners[callback]=nil end end}local list,tracked=nil,{}local picking,pickName,pickMenuVisible=false,nil,false local function finishPick(status,reopen)local wasPicking=picking
picking=false if card then card:SetStatus(status)end if wasPicking and pickMenuVisible and reopen~=false then menu.setVisible(true)end pickMenuVisible=false end local function save()if config then
config.values[key]=points if host.queueConfigSave then host.queueConfigSave()end end for callback in pairs(listeners)do callback()end end clear=function(reopen)render:ReleaseAll(layer)table.clear(
tracked)layer.Visible=false finishPick(nil,reopen)end card=context.framework.Categories.Render:CreateModule({Name='Markers',Category='Render',ConfigKey='Universal.Markers',Tooltip=
[[Save your standing coordinates or mark a surface. Points are shared with Teleport and kept per game.]],Function=function(enabled)if not enabled then clear()return end layer.Visible=true card:Clean(
clear)card:Render(function()local camera=(host.workspace or workspace).CurrentCamera if not camera then clear()return end layer.Visible=true local seen={}for name,data in pairs(points)do if not list:
IsSelected(name)then continue end local position=Vector3.new(data[1],data[2],data[3])local at=render:Project(camera,position)if not at or at.X<0 or at.Y<0 or at.X>camera.ViewportSize.X or at.Y>camera.
ViewportSize.Y then continue end local text=name if card.Options.Distance.Value then text..=' ['..math.round((camera.CFrame.Position-position).Magnitude)..']'end local drawing=render:Set(layer,name)
drawing:Show(true)drawing:Label('Marker',text,Vector2.new(at.X,at.Y),1,14,card.Options.Colour.Value,true)seen[name],tracked[name]=true,true end for name in pairs(tracked)do if not seen[name]then
render:Release(layer,name)tracked[name]=nil end end end)end})local function pointName()return string.sub(string.match(card.Options.Name.Value or'','^%s*(.-)%s*$')or'',1,48)end local function
nextPointName()local base=pointName()local index=1 while true do local suffix=base==''and tostring(index)or' #'..index local name=(base==''and'Point 'or string.sub(base,1,48-#suffix))..suffix if not
points[name]then return name end index+=1 end end local function savedName()local name=pointName()return name~=''and name or nextPointName()end local function storePoint(position,name,surface)if name
==''then card:Notify('Enter a point name.')return false end if not position or position.X~=position.X or position.Y~=position.Y or position.Z~=position.Z or math.max(math.abs(position.X),math.abs(
position.Y),math.abs(position.Z))>=1e7 then card:Notify('Position unavailable. Use coordinates in the form x, y, z.')return false end local count=0 for _ in pairs(points)do count+=1 end if count>=128
and not points[name]then card:Notify('Marker limit reached (128). Delete a point first.')return false end points[name]={position.X,position.Y,position.Z,surface and'surface'or nil}save()list:Refresh()
list:Set(name,true)card:SetStatus(name)return true end card:CreateTextBox({Name='Name',Default='',Placeholder='Home, exit, meeting spot...'})card:CreateButton({Name='Save here',Tooltip=
[[Save your character standing x, y, z here, not the camera position.]],Function=function()local character=host.LocalPlayer.Character local root=character and character:FindFirstChild(
'HumanoidRootPart')storePoint(root and root.Position,savedName())end})card:CreateButton({Name='Touch',Tooltip=
[[Hides the menu to pick a surface. Escape cancels; new points are numbered, never overwritten. Works while OFF too.]],Function=function()if picking then finishPick()return end pickName=nextPointName(
)picking=true pickMenuVisible=menu.isVisible()card:SetStatus('Click/tap a surface \183 Esc cancels')menu.setVisible(false)end})if config and config.values['Universal.Markers.Point']==nil then config.
values['Universal.Markers.Point']=config.values['Universal.Markers.Places']end list=card:CreateList({Name='Point',Items=pointNames,Default=pointNames(),EmptyText='No points. Use Save here or Touch.',
OnRemove=function(name)if not points[name]then return end list:Set(name,false)points[name]=nil save()list:Refresh()render:Release(layer,name)tracked[name]=nil end})card:CreateToggle({Name=
'Manual coordinates',Default=false})card:CreateTextBox({Name='Coordinates',Default='',Placeholder='x, y, z',Show={Option='Manual coordinates',Values={true}}})card:CreateButton({Name='Save coordinates'
,Show={Option='Manual coordinates',Values={true}},Function=function()local x,y,z=string.match(card.Options.Coordinates.Value or'',
[[^%s*([%+%-%.%deE]+)%s*,%s*([%+%-%.%deE]+)%s*,%s*([%+%-%.%deE]+)%s*$]])x,y,z=tonumber(x),tonumber(y),tonumber(z)storePoint(x and y and z and Vector3.new(x,y,z)or nil,savedName())end})card:
CreateToggle({Name='Distance',Default=true})card:CreateColor({Name='Colour',Default=Color3.fromRGB(85,255,85)})inputConnection=host.UserInputService.InputBegan:Connect(function(event,processed)if not
picking then return end if event.KeyCode==Enum.KeyCode.Escape then finishPick()return end if processed or host.UserInputService:GetFocusedTextBox()then return end if event.UserInputType~=Enum.
UserInputType.MouseButton1 and event.UserInputType~=Enum.UserInputType.Touch then return end local camera=(host.workspace or workspace).CurrentCamera if not camera then return end local params=
RaycastParams.new()params.FilterType=Enum.RaycastFilterType.Exclude params.FilterDescendantsInstances=host.LocalPlayer.Character and{host.LocalPlayer.Character}or{}params.RespectCanCollide=true params
.IgnoreWater=true local ray=camera:ScreenPointToRay(event.Position.X,event.Position.Y)local hit=(host.workspace or workspace):Raycast(ray.Origin,ray.Direction*5000,params)if hit and storePoint(hit.
Position,pickName,true)then finishPick(pickName)else card:SetStatus('No surface \183 tap again')end end)Module.Initialized=true return card end function Module.destroy()if clear then clear(false)end
if card and card.Enabled then card:Toggle(false)end if inputConnection then inputConnection:Disconnect()inputConnection=nil end if services then services.points=nil end card,clear,services=nil,nil,nil
Module.Initialized=false end return Module
-- src/games/universal/Render/NPCESP.lua
export type Runtime={framework:any,entity:any,render:any,host:any,services:any}local Module={Name='NPCESP',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCard:any=nil local
activeInputConnection:RBXScriptConnection?=nil local STANDARD_PART_NAMES:{[string]:boolean}={head=true,torso=true,uppertorso=true,lowertorso=true,humanoidrootpart=true,leftarm=true,rightarm=true,
leftleg=true,rightleg=true,leftupperarm=true,leftlowerarm=true,lefthand=true,rightupperarm=true,rightlowerarm=true,righthand=true,leftupperleg=true,leftlowerleg=true,leftfoot=true,rightupperleg=true,
rightlowerleg=true,rightfoot=true}local VERDICT_COOLDOWN:number=0.5 local function normalizeName(value:string):string local normalized:string=string.lower(value):gsub('[%s_%-]','')return normalized
end function Module.init(context:Runtime):any local framework:any=context.framework local render:any=context.render local host:any=context.host local players:Players=host.Players local
currentWorkspace:Workspace=host.workspace or workspace local layer:Frame=render:Layer('NpcEspLayer')local lastStatusAt:number=-math.huge local lastSweepAt:number=-math.huge local bots:{[Model]:boolean
}={}local rigInfo:any=setmetatable({},{__mode='k'})local verdicts:any=setmetatable({},{__mode='k'})local forcedNames:{string}={}local function modelFor(instance:Instance):Model?if instance:IsA('Model'
)then return instance::Model end return instance:FindFirstAncestorOfClass('Model')::Model?end local function isPlayerCharacterModel(model:Model):boolean if players:GetPlayerFromCharacter(model)~=nil
then return true end for _,player:Player in ipairs(players:GetPlayers())do local character:Model?=player.Character if character and model:IsDescendantOf(character)then return true end end return false
end local function matchesForcedName(model:Model):boolean local name:string=string.lower(model.Name)for _,needle:string in ipairs(forcedNames)do if needle~=''and string.find(name,needle,1,true)then
return true end end return false end local function classify(model:Model):string if model:GetAttribute('WurstDisguise')==true then return'reject'end if matchesForcedName(model)then return'rig'end
local humanoid:Humanoid?=model:FindFirstChildOfClass('Humanoid')::Humanoid?local standardParts:number=0 local motors:number=0 for _,child:Instance in ipairs(model:GetChildren())do if child:IsA(
'BasePart')then if STANDARD_PART_NAMES[normalizeName(child.Name)]then standardParts+=1 end elseif child:IsA('Motor6D')then motors+=1 end end if humanoid~=nil then if standardParts>=3 then return'rig'
end return'creature'end if standardParts>=4 or(standardParts>=3 and motors>=2)then return'rig'end if model:FindFirstChildWhichIsA('Humanoid',true)~=nil then return'creature'end return'reject'end
local function verdictFor(model:Model):string local entry:any=verdicts[model]if entry==nil then entry={verdict=isPlayerCharacterModel(model)and'reject'or classify(model),at=os.clock(),dirty=false}
verdicts[model]=entry return entry.verdict end if entry.dirty then local now:number=os.clock()if now-entry.at>=VERDICT_COOLDOWN then entry.verdict=isPlayerCharacterModel(model)and'reject'or classify(
model)entry.at=now entry.dirty=false end end return entry.verdict end local function markDirty(model:Model):()local entry:any=verdicts[model]if entry and not entry.dirty then entry.dirty=true end
rigInfo[model]=nil end local function invalidate(instance:Instance):()local parent:Instance?=instance.Parent if parent and parent:IsA('Model')then markDirty(parent::Model)end if instance:IsA('Model')
then markDirty(instance::Model)end end local function add(model:Model,verdict:string):()if verdict~='rig'then bots[model]=true return end for existing:Model in pairs(bots)do if existing:
IsDescendantOf(model)then return end if model:IsDescendantOf(existing)then bots[existing]=nil render:Release(layer,existing)end end bots[model]=true end local function inspect(instance:Instance):()if
not(instance:IsA('Model')or instance:IsA('Humanoid')or instance:IsA('BasePart')or instance:IsA('Motor6D'))then return end invalidate(instance)local model:Model?=modelFor(instance)local creature:Model?
=nil while model and model~=currentWorkspace do local verdict:string=verdictFor(model)if verdict=='rig'then add(model,'rig')return end if verdict=='creature'and creature==nil then creature=model end
local parent:Instance?=model.Parent if not parent then break end model=parent:IsA('Model')and parent::Model or parent:FindFirstAncestorOfClass('Model')::Model?end if creature then add(creature,
'creature')end end local function sweep(now:number):()for model:any,entry:any in pairs(verdicts)do if entry.dirty and now-entry.at>=VERDICT_COOLDOWN then local resolved:Model=model::Model local
verdict:string=isPlayerCharacterModel(resolved)and'reject'or classify(resolved)entry.verdict=verdict entry.at=now entry.dirty=false if verdict=='rig'then add(resolved,'rig')end end end end
local function scan():()render:ReleaseAll(layer)table.clear(bots)for _,object:Instance in ipairs(currentWorkspace:GetDescendants())do if object:IsA('Model')then inspect(object)end end end
local function clear():()render:ReleaseAll(layer)table.clear(bots)layer.Visible=false end local card:any card=framework.Categories.Visuals:CreateModule({Name='NPCESP',Category='Render',ConfigKey=
'Universal.NPCESP',Order=4,Tooltip='Displays character-shaped NPC models - bots, shopkeepers '..[[and display statues like the creator's rig - whether or not ]]..'their Humanoid is alive.',Function=
function(enabled:boolean):()layer.Visible=enabled if not enabled then clear()card:SetStatus(nil)return end scan()card:Event(currentWorkspace.DescendantAdded,inspect)card:Event(currentWorkspace.
DescendantRemoving,invalidate)card:Render(function():()local camera:Camera?=currentWorkspace.CurrentCamera if not camera then return end local resolvedCamera:Camera=camera::Camera local now:number=os.
clock()if now-lastSweepAt>=1 then lastSweepAt=now sweep(now)end local maxDistance:number=card.Options['Max distance'].Value local includeCreatures:boolean=card.Options['All humanoids'].Value local
localCharacter:Model?=host.LocalPlayer.Character local localRoot:BasePart?=localCharacter and(localCharacter:FindFirstChild('HumanoidRootPart')::BasePart?or localCharacter.PrimaryPart or
localCharacter:FindFirstChildWhichIsA('BasePart',true)::BasePart?)or nil local origin:Vector3=localRoot and localRoot.Position or resolvedCamera.CFrame.Position local count:number=0 for model:Model in
pairs(bots)do if not model:IsDescendantOf(currentWorkspace)or players:GetPlayerFromCharacter(model)~=nil then bots[model]=nil render:Release(layer,model)verdicts[model]={verdict='reject',at=now,dirty=
false}continue end local verdict:string=verdictFor(model)if verdict=='reject'then bots[model]=nil render:Release(layer,model)continue end if verdict=='creature'and not includeCreatures then render:
Release(layer,model)continue end local drawing:any=render:Set(layer,model)local info:any=rigInfo[model]if not(info and info.root and info.root.Parent)then info={root=model:FindFirstChild(
'HumanoidRootPart')::BasePart?or model.PrimaryPart or model:FindFirstChildWhichIsA('BasePart',true)::BasePart?,humanoid=model:FindFirstChildOfClass('Humanoid')::Humanoid?or model:
FindFirstChildWhichIsA('Humanoid',true)::Humanoid?}rigInfo[model]=info end local root:BasePart?=info.root if not root or(root.Position-origin).Magnitude>maxDistance then drawing:Show(false)continue
end if verdict=='creature'and info.humanoid and info.humanoid.Health<=0 then drawing:Show(false)continue end local rect:any=render:ModelRect(resolvedCamera,model)if not rect then drawing:Show(false)
continue end local colour:Color3=card.Options['Colour'].Value drawing:Box(rect,card.Options['Box'].Value and'Full'or'Off',1,colour)if card.Options['Name'].Value then local text:string=card.Options[
'Model name'].Value and model.Name~=''and model.Name or'BOT'local humanoid:Humanoid?=info.humanoid if humanoid and card.Options['Health'].Value then text..=' ['..tostring(math.round(humanoid.Health))
..']'end drawing:Label('NameTag',text,Vector2.new(rect.centreX,rect.top-4),1,13,colour,true)else drawing:HideLabel('NameTag')end drawing:Highlight(model,colour,0.55,not card.Options['Highlight'].Value
and'Off'or(card.Options['Through walls'].Value and'AlwaysOnTop'or'Occluded'))drawing:Show(true)count+=1 end if now-lastStatusAt>=0.25 then lastStatusAt=now card:SetStatus(tostring(count))end end)card:
Clean(clear)end})card:CreateToggle({Name='Box',Default=true})card:CreateToggle({Name='Highlight',Default=true})card:CreateToggle({Name='Through walls',Default=true,Show={Option='Highlight'}})card:
CreateToggle({Name='Name',Default=true})card:CreateToggle({Name='Model name',Show={Option='Name'},Default=false,Tooltip='Label each model with its own name instead of BOT.'})card:CreateToggle({Name=
'Health',Show={Option='Name'},Default=true,Tooltip="Append the Humanoid's hit points to the label when the "..'model has one.'})card:CreateToggle({Name='All humanoids',Default=false,Tooltip=
'Also flag non-character models that merely carry a living '..[[Humanoid (creatures, monsters). Off by default: NPCESP tracks ]]..'character-shaped rigs.'})card:CreateColor({Name='Colour',Default=
Color3.fromRGB(85,255,85)})card:CreateSlider({Name='Max distance',Min=25,Max=2000,Step=25,Default=500})local pendingName:string=''local addBox:any local function addForcedName():()local trimmed:string
=string.match(pendingName,'^%s*(.-)%s*$')or pendingName if trimmed==''then return end local needle:string=string.lower(trimmed)if not table.find(forcedNames,needle)then table.insert(forcedNames,needle
)end table.clear(verdicts)table.clear(rigInfo)if card.Enabled then scan()end addBox:Set('')end addBox=card:CreateTextBox({Name='Force by name',Default='',Tooltip=
[[Type a model name and press Enter or Add forced name. Partial names match, so statue catches CreatorStatue.]],Function=function(value:string):()pendingName=value end})if pendingName~=''then
addForcedName()end if addBox.Object and typeof(addBox.Object)=='Instance'and addBox.Object:IsA('TextBox')then activeInputConnection=addBox.Object.FocusLost:Connect(function(enterPressed:boolean):()if
enterPressed then addForcedName()end end)end card:CreateButton({Name='Add forced name',Tooltip='Add the name typed above, including on touch devices.',Function=addForcedName})card:CreateButton({Name=
'Clear forced names',Function=function():()table.clear(forcedNames)table.clear(verdicts)table.clear(rigInfo)if card.Enabled then scan()end end})activeCard=card Module.Initialized=true return card end
function Module.destroy():()if activeCard and activeCard.Enabled then activeCard:Toggle(false)end if activeInputConnection then activeInputConnection:Disconnect()activeInputConnection=nil end
activeCard=nil Module.Initialized=false end return Module
-- src/games/universal/Render/Nametags.lua
local Module={Name='Nametags',Initialized=false}local activeCard,cleanup,roleBinding local BLACK=Color3.new(0,0,0)function Module.init(context:any):any local host,entity,render=context.host,context.
entity,context.render local layer=render:Layer('NametagsLayer')local tracked={}local store=host.configData if store and store.states['Universal.Nametags']==nil then local states,values=store.states,
store.values states['Universal.Nametags']=states['Universal.Chams']==true and states['Universal.Chams.Nametags']==true or states['Universal.PlayerESP']==true and states['Universal.PlayerESP.Nametags']
~=false if values['Universal.Nametags.Textsize']==nil then values['Universal.Nametags.Textsize']=values['Universal.PlayerESP.Textsize']end for new,old in pairs({Background='Nameplate',Displayname=
'Displayname',Heldtool='Heldtool',Showrole='Showrole',Rolecolours='Rolecolours'})do local key='Universal.Nametags.'..new if states[key]==nil then states[key]=states['Universal.PlayerESP.'..old]end end
for key,colour in pairs(table.clone(values))do if string.sub(key,1,20)=='Universal.PlayerESP.'and string.lower(string.sub(key,-6))=='colour'then local newKey='Universal.Nametags.'..string.sub(key,21)
if values[newKey]==nil then values[newKey]=colour end end end end local textService=game:GetService('TextService')local card local function clear()render:ReleaseAll(layer)table.clear(tracked)layer.
Visible=false end local function fit(drawing,slot,text,size,viewport)local label=drawing.labels[slot]if not label then return end local key=text..':'..tostring(size)..':'..tostring(viewport.X)drawing.
textMetrics=drawing.textMetrics or{}if not drawing.textMetrics[slot]or drawing.textMetrics[slot].key~=key then local width=#text*size*0.65+8 pcall(function()width=textService:GetTextSize(text,size,
Enum.Font.Arial,Vector2.new(10000,100)).X+8 end)width=math.max(1,math.min(width,viewport.X-24))drawing.textMetrics[slot]={key=key,width=width}label.AutomaticSize=Enum.AutomaticSize.None label.Size=
UDim2.fromOffset(width,size+7)label.TextTruncate=Enum.TextTruncate.AtEnd end local half=drawing.textMetrics[slot].width*0.5 local clampedX=math.clamp(label.Position.X.Offset,half+8,math.max(half+8,
viewport.X-half-8))if math.abs(clampedX-label.Position.X.Offset)>=0.5 then label.Position=UDim2.fromOffset(clampedX,label.Position.Y.Offset)end end card=context.framework.Categories.Visuals:
CreateModule({Name='Nametags',Category='Render',ConfigKey='Universal.Nametags',Order=3,Tooltip=[[Readable player nameplates, account names, distance and game roles. Independent of ESP and Chams.]],
Function=function(enabled)if not enabled then clear()card:SetStatus(nil)return end layer.Visible=true card:Clean(clear)card:Render(function()local camera=(host.workspace or workspace).CurrentCamera if
not camera then clear()return end layer.Visible=true local seen,count={},0 local options=card.Options for _,target in ipairs(entity:Refresh())do local player=target.Player if not player or target.
Health<=0 or target.Distance>options['Max distance'].Value or not(roleBinding.provider and roleBinding.provider.ReplacesTeams)and not options.Teammates.Value and target.IsFriendly then continue end
local head=target.Head or target.RootPart local point=head and render:Project(camera,head.Position+Vector3.new(0,head.Size.Y*0.5+0.7,0))if not point or point.X<0 or point.Y<0 or point.X>camera.
ViewportSize.X or point.Y>camera.ViewportSize.Y or not options['Through walls'].Value and not entity:VisibleFrom(target)then continue end local colour,role=roleBinding:Resolve(player,options.Colour.
Value)local shown=options['Display name'].Value and player.DisplayName or player.Name local details={}if options['Account name'].Value and shown~=player.Name then table.insert(details,'@'..player.Name
)end if options['Show role'].Value and role then table.insert(details,role)end if options['Held tool'].Value then local tool=target.Character:FindFirstChildOfClass('Tool')if tool then table.insert(
details,tool.Name)end end seen[player]=true tracked[player]=true count+=1 local drawing=render:Set(layer,player)drawing:Show(true)local size=math.round(options['Text size'].Value)if#details>0 then
shown..=' ['..table.concat(details,'] [')..']'end if options.Distance.Value then shown='['..tostring(math.round(target.Distance))..'] '..shown end local y=math.max(point.Y,size+16)drawing:Label('Name'
,shown,Vector2.new(point.X,y),1,size,colour,options.Background.Value)local label=drawing.labels.Name if label then if not drawing.flatNametag then for _,child in ipairs(label:GetChildren())do if child
:IsA('UICorner')or child:IsA('UIStroke')then child:Destroy()elseif child:IsA('UIPadding')then child.PaddingLeft,child.PaddingRight=UDim.new(0,4),UDim.new(0,4)end end drawing.flatNametag=true end if
label.Font~=Enum.Font.Arial then label.Font=Enum.Font.Arial end if label.BackgroundColor3~=BLACK then label.BackgroundColor3=BLACK end local plate:number=options.Background.Value and 0.5 or 1 if label
.BackgroundTransparency~=plate then label.BackgroundTransparency=plate end if label.TextStrokeTransparency~=1 then label.TextStrokeTransparency=1 end end fit(drawing,'Name',shown,size,camera.
ViewportSize)drawing:HideLabel('Details')end for player in pairs(tracked)do if not seen[player]then render:Release(layer,player)tracked[player]=nil end end card:SetStatus(tostring(count))end end)card
:CreateToggle({Name='Display name',Default=true})card:CreateToggle({Name='Account name',Default=false,Show={Option='Display name'}})card:CreateToggle({Name='Distance',Default=false})card:CreateToggle(
{Name='Show role',Default=false})card:CreateToggle({Name='Held tool',Default=false})card:CreateToggle({Name='Background',Default=true})card:CreateSlider({Name='Text size',Min=10,Max=24,Default=14})
card:CreateColor({Name='Colour',Default=Color3.fromRGB(85,255,85)})card:CreateSlider({Name='Max distance',Min=50,Max=5000,Default=1000})card:CreateToggle({Name='Through walls',Default=true})card:
CreateToggle({Name='Teammates',Default=true})roleBinding=context.services.gameRoles:Bind(card)card.OptionFilter=function(name)if name=='Show role'then return roleBinding.provider~=nil end return not(
name=='Teammates'and roleBinding.provider and roleBinding.provider.ReplacesTeams)end card:RefreshVisibility()activeCard,cleanup=card,clear Module.Initialized=true return card end function Module.
destroy()if activeCard and activeCard.Enabled then activeCard:Toggle(false)end if cleanup then cleanup()end if roleBinding then roleBinding:Destroy()end activeCard,cleanup,roleBinding=nil,nil,nil
Module.Initialized=false end return Module
-- src/games/universal/Render/PlayerESP.lua
export type Runtime={framework:any,entity:any,host:any,services:any}local Module={Name='PlayerESP',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil local
activeCard:any=nil function Module.init(context:Runtime):any local host:any=context.host local framework:any=context.framework local players:Players=host.Players local render:any=context.render local
container:Frame=render:Layer('PlayerEspLayer')local pool:{[Player]:boolean}={}local lastStatusAt:number=-math.huge local visibilityCache:any=setmetatable({},{__mode='k'})local esp:any esp=framework.
Categories.Visuals:CreateModule({Name='PlayerESP',Category='Render',ConfigKey='Universal.PlayerESP',Order=1,Tooltip=[[Outlined boxes, skeletons and tracers. Nametags and HP are independent modules.]],
Function=function(enabled:boolean):()container.Visible=enabled if not enabled then esp:SetStatus(nil)render:ReleaseAll(container)table.clear(pool)return end lastStatusAt=-math.huge esp:SetStatus('0')
esp:Render(function():()Module.render(context,container,pool,esp,visibilityCache)local now:number=os.clock()if now-lastStatusAt>=0.25 then lastStatusAt=now local count:number=0 for _player:Player in
pairs(pool)do count+=1 end esp:SetStatus(tostring(count))end end)esp:Event(players.PlayerRemoving,function(player:Player):()render:Release(container,player)pool[player]=nil end)esp:Clean(function():()
render:ReleaseAll(container)table.clear(pool)container.Visible=false end)end})esp:CreateDropdown({Name='Boxes',List={'Corner','Full','Off'},Index=1,Tooltip=
'Corner draws four brackets, Full draws the whole rectangle.'})esp:CreateSlider({Name='Box thickness',Show={Option='Boxes',Values={'Corner','Full'}},Min=1,Max=5,Default=1})esp:CreateToggle({Name=
'Filled box',Show={Option='Boxes',Values={'Corner','Full'}},Default=false,Tooltip="Wash the inside of the box in the target's colour. Reads "..'at a glance across a map; noisy in a crowd.'})esp:
CreateToggle({Name='Skeleton',Default=false,Tooltip="Draw the rig's bones. Works on both R6 and R15."})esp:CreateDropdown({Name='Tracers',List={'Off','Bottom','Centre','Cursor'},Index=1,Tooltip=
'Where the line to each target starts.'})esp:CreateToggle({Name='Team check',Default=false,Tooltip='Skip players on your own team.'})esp:CreateToggle({Name='Visibility check',Default=false,Tooltip=
'Recolour targets that are behind geometry.'})esp:CreateSlider({Name='Max distance',Min=50,Max=5000,Default=2000})esp:CreateColor({Name='Colour',Default=Color3.fromRGB(85,255,85),Tooltip=
'Used for every target the game gives no colour of its own.'})esp:CreateToggle({Name='Team colours',Default=true,Tooltip="Draw each player in their own team's colour where the game "..'has teams.'})
local roles=context.services.gameRoles:Bind(esp)Module.roles=roles esp.OptionFilter=function(name)return not(roles.provider and roles.provider.ReplacesTeams and(name=='Team check'or name==
'Team colours'))end esp:RefreshVisibility()local function addExtraRows(extra:any):()if type(extra)~='table'or type(extra.Name)~='string'or esp.Options[extra.Name]then return end esp:CreateToggle({Name
=extra.Name,Default=extra.Default==true,Tooltip=extra.Tooltip,Function=function(value:boolean):()if type(extra.Toggle)=='function'then pcall(extra.Toggle,value)end end})if extra.Color then esp:
CreateColor({Name=extra.Name..' colour',Show={Option=extra.Name},Default=extra.Color,Function=function(value:Color3):()if type(extra.SetColor)=='function'then pcall(extra.SetColor,value)end end})end
for _,definition:any in ipairs(extra.Options or{})do local rowName:string=extra.Name..' '..definition.Name if definition.Kind=='toggle'then esp:CreateToggle({Name=rowName,Show={Option=extra.Name},
Default=definition.Default==true,Tooltip=definition.Tooltip,Function=function(value:boolean):()if type(definition.Set)=='function'then pcall(definition.Set,value)end end})else esp:CreateSlider({Name=
rowName,Show={Option=extra.Name},Min=definition.Min or 0,Max=definition.Max or 100,Default=definition.Default or definition.Min or 0,Tooltip=definition.Tooltip,Function=function(value:number):()if
type(definition.Set)=='function'then pcall(definition.Set,value)end end})end end end local gameBridge:any=context.services.gameBridge if type(gameBridge.onEvent)=='function'then Module.bridgeCleanup=
gameBridge.onEvent(function(kind:string,payload:any):()if kind=='extra'then addExtraRows(payload)end end)end activeCleanup=function():()render:ReleaseAll(container)table.clear(pool)container.Visible=
false end activeCard=esp Module.Initialized=true return esp end function Module.render(context:Runtime,container:Frame,pool:{[Player]:any},esp:any,visibilityCache:any):()local host:any=context.host
local entityLibrary:any=context.entity local render:any=context.render local gameBridge:any=context.services.gameBridge local currentWorkspace:Workspace=host.workspace or workspace local camera:Camera
?=currentWorkspace.CurrentCamera if not camera then container.Visible=false return end container.Visible=true local resolvedCamera:Camera=camera::Camera local options:any=esp.Options local viewport:
Vector2=resolvedCamera.ViewportSize entityLibrary:Refresh()local seen:{[Player]:boolean}={}for _,entity:any in ipairs(entityLibrary.List)do local player:Player=entity.Player if not(Module.roles and
Module.roles.provider and Module.roles.provider.ReplacesTeams)and options['Team check'].Value and entity.IsFriendly then continue end if entity.Distance>options['Max distance'].Value then continue end
local rect:any=render:ModelRect(resolvedCamera,entity.Character)seen[player]=true pool[player]=true local drawings:any=render:Set(container,player)if not rect then drawings:Show(false)continue end
local visible:boolean=true if options['Visibility check'].Value then local cache:any=visibilityCache[player]local now:number=os.clock()if not cache or now-cache.at>0.12 then cache={at=now,value=
entityLibrary:VisibleFrom(entity)}visibilityCache[player]=cache end visible=cache.value end local colour=options['Colour'].Value local roles=Module.roles local roleColour=roles and roles:Resolve(
player,nil)if roleColour then colour=roleColour elseif not(roles and roles.provider and roles.provider.ReplacesTeams)and options['Team colours'].Value and player.Team then colour=player.Team.TeamColor
.Color end if not visible then colour=colour:Lerp(Color3.new(0,0,0),0.45)end local left:number=rect.left local top:number=rect.top local bottom:number=rect.bottom local height:number=rect.height local
centreX:number=rect.centreX drawings:Show(true)local thickness:number=math.round(options['Box thickness'].Value)local boxMode:string=options['Boxes'].Value drawings:Fill(options['Filled box'].Value
and boxMode~='Off'and rect or nil,colour,0.82)drawings:Box(rect,boxMode,thickness,colour)if options['Skeleton'].Value then local bones:{{BasePart}}=entityLibrary:Rig(entity)for index:number=1,14 do
local pair:{BasePart}?=bones[index]local first:Vector2?=pair and render:Project(resolvedCamera,pair[1].Position)local second:Vector2?=pair and render:Project(resolvedCamera,pair[2].Position)if first
and second then drawings:Line('Bone',index,first::Vector2,second::Vector2,1,colour)else drawings:HideLines('Bone',index,index)end end else drawings:HideLines('Bone',1,14)end local tracerMode:string=
options['Tracers'].Value if tracerMode=='Off'then drawings:HideLines('Tracer',1,1)else local origin:Vector2 if tracerMode=='Bottom'then origin=Vector2.new(viewport.X*0.5,viewport.Y)elseif tracerMode==
'Centre'then origin=viewport*0.5 else origin=host.UserInputService:GetMouseLocation()end drawings:Line('Tracer',1,origin,Vector2.new(centreX,bottom),1,colour)end end for player:Player in pairs(pool)do
if not seen[player]then render:Release(container,player)pool[player]=nil end end end function Module.destroy():()if activeCard and activeCard.Enabled then activeCard:Toggle(false)end if activeCleanup
then pcall(activeCleanup)end if Module.roles then Module.roles:Destroy()Module.roles=nil end if Module.bridgeCleanup then Module.bridgeCleanup()Module.bridgeCleanup=nil end activeCleanup=nil
activeCard=nil Module.Initialized=false end return Module
-- src/games/universal/Render/Spectate.lua
local Module={Name='Spectate',Initialized=false}local card,services,stop function Module.init(context)local host=context.host services=context.services local world=host.workspace or workspace local
chosen,camera,saved,writtenSubject local enabled=false local config=host.configData if config then if config.states['Universal.FreeCam.Spectate']==nil then config.states['Universal.FreeCam.Spectate']=
config.states['Universal.Spectate']end if config.values['Universal.FreeCam.SpectatePlayer']==nil then config.values['Universal.FreeCam.SpectatePlayer']=config.values['Universal.Spectate.Player']end
end card=assert(context.framework:GetModule('FreeCam'),'Spectate requires FreeCam options')local playerList local id='ARandomMenu.Spectate.'..tostring({})local function localHumanoid()local character=
host.LocalPlayer.Character return character and character:FindFirstChildOfClass('Humanoid')end local function freeCamSession()local free=services.freeCam return free and free.IsActive()and free.
Session or nil end local function restore()if camera and saved then pcall(function()if camera.CameraType~=Enum.CameraType.Custom or camera.CameraSubject~=writtenSubject then return end local
returnedToFreeCam=saved.freeCam and saved.freeCam==freeCamSession()if saved.freeCam and not returnedToFreeCam then camera.CameraType=Enum.CameraType.Custom camera.CameraSubject=localHumanoid()else
camera.CameraType=saved.kind local subject=saved.subject if subject==saved.localHumanoid or typeof(subject)=='Instance'and subject.Parent==nil then subject=localHumanoid()end camera.CameraSubject=
subject camera.CFrame,camera.Focus=saved.frame,saved.focus end end)end camera,saved,writtenSubject=nil,nil,nil end local function targetHumanoid()for _,player in ipairs(host.Players:GetPlayers())do if
player~=host.LocalPlayer and player.Name==chosen then local character=player.Character local humanoid=character and character:FindFirstChildOfClass('Humanoid')return humanoid and humanoid.Health>0 and
humanoid or nil,true end end return nil,false end local function step()local humanoid,present=targetHumanoid()if chosen and not present then local departed=chosen chosen=nil if playerList then
playerList:Set(departed,false)playerList:Refresh()end end if not humanoid then restore()card:SetStatus(chosen and'Waiting for respawn'or'Choose a player')return end local nextCamera=world.
CurrentCamera if camera~=nextCamera then restore()end if not nextCamera then return end if not camera then camera=nextCamera saved={kind=camera.CameraType,subject=camera.CameraSubject,frame=camera.
CFrame,focus=camera.Focus,freeCam=freeCamSession(),localHumanoid=localHumanoid()}elseif camera.CameraType~=Enum.CameraType.Custom or camera.CameraSubject~=writtenSubject then saved={kind=camera.
CameraType,subject=camera.CameraSubject,frame=camera.CFrame,focus=camera.Focus,freeCam=false,localHumanoid=localHumanoid()}end camera.CameraType=Enum.CameraType.Custom camera.CameraSubject=humanoid
writtenSubject=humanoid card:SetStatus(chosen)end local state={IsActive=function()if not enabled or not chosen then return false end local humanoid=targetHumanoid()return humanoid~=nil end}services.
spectate=state stop=function()enabled=false host.RunService:UnbindFromRenderStep(id)restore()end card:CreateToggle({Name='Spectate',Default=false,Tooltip=
[[Follow a player independently: FreeCam may stay OFF. Turn this off or deselect to return.]],Function=function(value)stop()enabled=value if enabled then host.RunService:BindToRenderStep(id,Enum.
RenderPriority.Camera.Value-1,step)step()end end})playerList=card:CreateList({Name='Spectate Player',Single=true,Show={Option='Spectate'},Items=function()local names={}for _,player in ipairs(host.
Players:GetPlayers())do if player~=host.LocalPlayer then table.insert(names,player.Name)end end table.sort(names)return names end,EmptyText='No other players',Tooltip=
[[Select one player; click the selected player again to stop spectating.]],Function=function(_,names)local nextName=names[1]if chosen~=nextName then restore()chosen=nextName end if enabled then step()
end end})Module.Initialized=true return Module end function Module.destroy()if stop then stop()end if services then services.spectate=nil end card,services,stop=nil,nil,nil Module.Initialized=false
end return Module
-- src/games/universal/Render/XRay.lua
local Module={Name='XRay',PlaceId=0,Initialized=false}local card function Module.init(context)local host,world=context.host,context.host.workspace or workspace local records=setmetatable({},{__mode=
'k'})local indexed=setmetatable({},{__mode='k'})local candidates,cursor,pause={},1,0 local named,kept={},{}local function names(text)local result={}for name in string.gmatch(string.lower(text or''),
'[^,]+')do name=string.match(name,'^%s*(.-)%s*$')if name~=''then result[name]=true end end return result end local function restore(part)local record=records[part]if not record then return end pcall(
function()if part.LocalTransparencyModifier==record.applied then part.LocalTransparencyModifier=record.base end end)records[part]=nil end local function restoreAll()for part in pairs(records)do
restore(part)end end local function changed()restoreAll()cursor=1 pause=0 if card and card.Options['Part names']and card.Options['Keep visible']then named=names(card.Options['Part names'].Value)kept=
names(card.Options['Keep visible'].Value)end end local function add(part)if part:IsA('BasePart')and not indexed[part]then indexed[part]=true table.insert(candidates,part)end end local function
eligible(part,origin,root)local opts=card.Options if part.Transparency>=0.98 or kept[string.lower(part.Name)]then return false end if not opts['Include unanchored'].Value and not part.Anchored then
return false end if opts['Collidable only'].Value and not part.CanCollide then return false end if opts['Limit range'].Value and origin and(part.Position-origin).Magnitude>opts.Range.Value+part.Size.
Magnitude*0.5 then return false end if part.Name=='Wurst_AntiVoidPlatform'or host.isMenuOwned and host.isMenuOwned(part)then return false end local ancestor=part.Parent while ancestor and ancestor~=
world do if ancestor:IsA('Tool')or ancestor:IsA('Model')and ancestor:FindFirstChildOfClass('Humanoid')then return false end ancestor=ancestor.Parent end local thinFloor=part.Size.Y<=math.min(part.Size
.X,part.Size.Z)*0.35 if opts['Keep floors'].Value and thinFloor and root and part.Position.Y<=root.Position.Y then return false end local mode=opts.Scope.Value if mode=='Named parts'then return named[
string.lower(part.Name)]==true end if mode=='Walls'then return not thinFloor and part.Size.Y>math.min(part.Size.X,part.Size.Z)*1.5 end return true end local function step(dt)pause=math.max(0,pause-(dt
or 0))if pause>0 or#candidates==0 then return end local camera=world.CurrentCamera local character=host.LocalPlayer.Character local root=character and character:FindFirstChild('HumanoidRootPart')local
origin=camera and camera.CFrame.Position or root and root.Position for _=1,200 do local part=candidates[cursor]if not part then cursor=1 pause=0.2 break end if not part.Parent or not part:
IsDescendantOf(world)then restore(part)indexed[part]=nil candidates[cursor]=candidates[#candidates]candidates[#candidates]=nil else if eligible(part,origin,root)then local current=part.
LocalTransparencyModifier local record=records[part]if not record then record={base=current}records[part]=record elseif current~=record.applied then record.base=current end record.applied=math.max(
record.base,card.Options.Transparency.Value)if current~=record.applied then part.LocalTransparencyModifier=record.applied end else restore(part)end cursor+=1 end end end card=context.framework.
Categories.Render:CreateModule({Name='X-Ray',Category='Render',ConfigKey='Universal.XRay',Tooltip=[[Fade map parts locally. Keep named objects visible; never fade characters or held tools.]],Function=
function(enabled)restoreAll()table.clear(candidates)table.clear(indexed)cursor=1 pause=0 if not enabled then card:SetStatus(nil)return end changed()for _,part in ipairs(world:GetDescendants())do add(
part)end card:Clean(restoreAll)card:Clean(world.DescendantAdded:Connect(add))card:Loop(step)step(0)end})card:CreateDropdown({Name='Scope',List={'All map','Walls','Named parts'},Index=1,Function=
changed})card:CreateSlider({Name='Transparency',Min=0,Max=1,Default=0.5,Step=0.05,Decimal=2,Function=changed})card:CreateTextBox({Name='Part names',Default='',Placeholder='Wall, Window',Show={Option=
'Scope',Values={'Named parts'}},Function=changed,Tooltip=[[Exact part names, separated by commas. Only these names fade in Named parts scope.]]})card:CreateTextBox({Name='Keep visible',Default='',
Placeholder='Spawn, Objective',Function=changed,Tooltip=[[Vape-style kept-part list. Exact names separated by commas; these parts never fade.]]})card:CreateToggle({Name='Keep floors',Default=true,
Function=changed})card:CreateToggle({Name='Collidable only',Default=true,Function=changed})card:CreateToggle({Name='Include unanchored',Default=false,Function=changed})card:CreateToggle({Name=
'Limit range',Default=true,Function=changed})card:CreateSlider({Name='Range',Min=25,Max=5000,Default=250,Show={Option='Limit range',Values={true}},Function=changed})Module.Initialized=true return card
end function Module.destroy()if card and card.Enabled then card:Toggle(false)end card=nil Module.Initialized=false end return Module
-- src/games/universal/Render/ZoomUnlocker.lua
local Module={Name='ZoomUnlocker',Initialized=false}local cleanup function Module.init(context)local host,player=context.host,context.host.LocalPlayer local access=context.services.cameraAccess local
settings={minimum=0.5,maximum=400,firstPerson=true,walls=false}local original,feature local function bounds(minimum,maximum)if player.CameraMinZoomDistance>maximum then player.CameraMinZoomDistance=
minimum end if player.CameraMaxZoomDistance~=maximum then player.CameraMaxZoomDistance=maximum end if player.CameraMinZoomDistance~=minimum then player.CameraMinZoomDistance=minimum end end
local function apply()bounds(math.min(settings.minimum,settings.maximum),math.max(settings.minimum,settings.maximum))local mode=settings.firstPerson and Enum.CameraMode.Classic or original.mode if
player.CameraMode~=mode then player.CameraMode=mode end local walls=access:ThroughWalls(settings.walls)feature:SetStatus(settings.walls and(walls and tostring(math.round(settings.maximum))..' / walls'
or'walls unavailable')or tostring(math.round(settings.maximum))..' studs')end cleanup=function()host.disconnectFeatureConnection('ZoomUnlocker')access:ThroughWalls(false)if original then bounds(
original.minimum,original.maximum)player.CameraMode=original.mode original=nil end end feature=host.createUniversalFeature('Zoom','Extend zoom; optionally let the camera pass through walls.',23,
function(enabled)cleanup()if not enabled then feature:SetStatus(nil)return end original={minimum=player.CameraMinZoomDistance,maximum=player.CameraMaxZoomDistance,mode=player.CameraMode}apply()host.
featureConnections.ZoomUnlocker=host.TaskManager:Connect(apply,nil,'Zoom Unlocker')end,{configKey='Universal.ZoomUnlocker',categoryName='Render'})local function update(key,value)settings[key]=value if
feature.enabled then apply()end end host.addNumberOption(feature,'Maximum distance',settings.maximum,25,2000,function(value)update('maximum',value)end)host.addNumberOption(feature,'Minimum distance',
settings.minimum,0.5,50,function(value)update('minimum',value)end)host.addToggleOption(feature,'Unlock forced first person',settings.firstPerson,function(value)update('firstPerson',value)end)host.
addToggleOption(feature,'Through walls',settings.walls,function(value)update('walls',value)end)Module.Initialized=true return feature end function Module.destroy()if cleanup then cleanup()cleanup=nil
end Module.Initialized=false end return Module
-- src/libraries/CameraAccess.lua
local Module={Name='CameraAccess',Initialized=false}local active function Module.init(host)local api={playerModule=nil,source=nil,pending=false,checkedAt=-math.huge}local patch function api:Refresh()
if self.destroyed then return end local scripts=host.LocalPlayer:FindFirstChild('PlayerScripts')local source=scripts and scripts:FindFirstChild('PlayerModule')if source~=self.source then self.
playerModule,self.poppercam=nil,nil end if self.pending or source==self.source and self.playerModule then return end if os.clock()-self.checkedAt<0.5 then return end self.checkedAt=os.clock()if not
source then return end self.source,self.pending=source,true task.spawn(function()local ok,result=pcall(require,source)local cameraModule=source:FindFirstChild('CameraModule')local popper=cameraModule
and cameraModule:FindFirstChild('Poppercam')local cameraOk,cameraClass if popper then cameraOk,cameraClass=pcall(require,popper)end if not self.destroyed and self.source==source then self.playerModule
=ok and type(result)=='table'and result or nil self.poppercam=cameraOk and type(cameraClass)=='table'and cameraClass or nil end self.pending=false end)end local function get(method)api:Refresh()local
playerModule=api.playerModule if playerModule and type(playerModule[method])=='function'then local ok,result=pcall(playerModule[method],playerModule)if ok then return result end end end function api:
GetControls()return get('GetControls')end function api:GetCameras()local result=get('GetCameras')return type(result)=='table'and result or nil end local function restore()if not patch then return end
local old=patch patch=nil if old.module.Update==old.replacement then local ok=pcall(function()old.module.Update=old.update end)if ok and old.enabled then pcall(old.module.Enable,old.module,true)end
end end function api:ThroughWalls(enabled)if not enabled or self.destroyed then restore()return false end local cameras=self:GetCameras()local occlusion=cameras and cameras.activeOcclusionModule or
self.poppercam if patch and patch.module~=occlusion then restore()end if patch then return patch.module.Update==patch.replacement end if type(occlusion)~='table'or type(occlusion.Update)~='function'
then return false end local wasEnabled=false if type(occlusion.GetEnabled)=='function'and type(occlusion.Enable)=='function'then local ok,value=pcall(occlusion.GetEnabled,occlusion)wasEnabled=ok and
value==true if wasEnabled then pcall(occlusion.Enable,occlusion,false)end end local cleaned=setmetatable({},{__mode='k'})local replacement=function(instance,_,cameraFrame,focus)if not cleaned[instance
]then cleaned[instance]=true if type(instance.Cleanup)=='function'then pcall(instance.Cleanup,instance)end end local extrapolator=instance.focusExtrapolator if extrapolator and type(extrapolator.Reset
)=='function'then extrapolator:Reset()end return cameraFrame,focus end local update=occlusion.Update local ok=pcall(function()occlusion.Update=replacement end)if not ok then if wasEnabled then pcall(
occlusion.Enable,occlusion,true)end return false end patch={module=occlusion,update=update,replacement=replacement,enabled=wasEnabled}return true end function api:Destroy()restore()self.destroyed=true
self.playerModule,self.poppercam=nil,nil end host.services.cameraAccess=api active=api Module.Initialized=true return api end function Module.destroy()if active then active:Destroy()active=nil end
Module.Initialized=false end return Module
-- src/libraries/Entity.lua
export type Entity={Player:Player?,Character:Model,Humanoid:Humanoid,RootPart:BasePart,Head:BasePart?,Health:number,MaxHealth:number,Distance:number,Team:Team?,IsFriendly:boolean,IsNPC:boolean,Name:
string}export type RangeQuery={Range:number,Limit:number?,IgnoreTeam:boolean?,IgnoreWalls:boolean?,IgnoreFriends:boolean?,Angle:number?,Sort:string?,IncludeNPCs:boolean?}export type EntityQuery={
Radius:number?,MaxDistance:number?,IgnoreTeam:boolean?,IgnoreWalls:boolean?,IgnoreFriends:boolean?,Part:string?,IncludeNPCs:boolean?}export type EntityLibrary={List:{Entity},ByPlayer:{[Player]:Entity}
,NPCList:{Entity},ByModel:{[Model]:Entity},LocalEntity:Entity?,IsAlive:(self:EntityLibrary)->boolean,Refresh:(self:EntityLibrary)->{Entity},Get:(self:EntityLibrary,player:Player)->Entity?,GetNpc:(self
:EntityLibrary,model:Model)->Entity?,ForEach:(self:EntityLibrary,callback:(Entity)->())->(),IsFriendly:(self:EntityLibrary,player:Player?)->boolean,IsFriendlyModel:(self:EntityLibrary,model:Model)->
boolean,IsEnemy:(self:EntityLibrary,player:Player?)->boolean?,IsProtected:(self:EntityLibrary,player:Player?)->boolean,InRange:(self:EntityLibrary,query:RangeQuery)->{Entity},ClosestToRay:(self:
EntityLibrary,origin:Vector3,direction:Vector3,query:EntityQuery?)->Entity?,ClosestToCursor:(self:EntityLibrary,query:EntityQuery?)->Entity?,VisibleFrom:(self:EntityLibrary,entity:Entity)->boolean,Rig
:(self:EntityLibrary,entity:Entity)->{{BasePart}},Destroy:(self:EntityLibrary)->()}local Module={Name='Entity',PlaceId=0,Events={}::{[string]:any},Initialized=false}local R15_BONES:{{string}}={{'Head'
,'UpperTorso'},{'UpperTorso','LowerTorso'},{'UpperTorso','LeftUpperArm'},{'LeftUpperArm','LeftLowerArm'},{'LeftLowerArm','LeftHand'},{'UpperTorso','RightUpperArm'},{'RightUpperArm','RightLowerArm'},{
'RightLowerArm','RightHand'},{'LowerTorso','LeftUpperLeg'},{'LeftUpperLeg','LeftLowerLeg'},{'LeftLowerLeg','LeftFoot'},{'LowerTorso','RightUpperLeg'},{'RightUpperLeg','RightLowerLeg'},{'RightLowerLeg'
,'RightFoot'}}local R6_BONES:{{string}}={{'Head','Torso'},{'Torso','Left Arm'},{'Torso','Right Arm'},{'Torso','Left Leg'},{'Torso','Right Leg'}}local activeLibrary:EntityLibrary?=nil function Module.
init(context:any):EntityLibrary local host:any=context local players:Players=host.Players local localPlayer:Player=host.LocalPlayer local currentWorkspace:Workspace=host.workspace or workspace local
library:any={List={},ByPlayer={},NPCList={},ByModel={},LocalEntity=nil,Connections={},lastRefresh=-1,pool=setmetatable({},{__mode='k'}),npcPool=setmetatable({},{__mode='k'}),npcModels=setmetatable({},
{__mode='k'}),rigs=setmetatable({},{__mode='k'})}local function characterParts(player:Player):(Model?,Humanoid?,BasePart?)local character:Model?=player.Character if not character or not character.
Parent then for _,containerName:string in ipairs({'PlayersContainer','Characters','Live','Players'})do local container:Instance?=currentWorkspace:FindFirstChild(containerName)local candidate:Instance?
=container and container:FindFirstChild(player.Name)if candidate and candidate:IsA('Model')then character=candidate::Model break end end end if not character or not character.Parent then return nil,
nil,nil end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')::Humanoid?local root:BasePart?=character:FindFirstChild('HumanoidRootPart')::BasePart?or character.PrimaryPart or
character:FindFirstChildWhichIsA('BasePart')::BasePart?return character,humanoid,root end local function localRoot():BasePart?local _,_,root=characterParts(localPlayer)return root end local
TEAM_ATTRIBUTE_NAMES:{string}={'Team','TeamId','TeamID','Faction','FactionId','FactionID'}local function normalizeTeamValue(value:any):string? if value==nil then return nil end if typeof(value)==
'Instance'then return'name:'..string.lower((value::Instance).Name)end if type(value)=='table'and type(value.Name)=='string'then return'name:'..string.lower(value.Name)end if type(value)=='table'and
type(value.R)=='number'and type(value.G)=='number'and type(value.B)=='number'then return string.format('colour:%d:%d:%d',math.round(value.R*255),math.round(value.G*255),math.round(value.B*255))end if
type(value)=='table'and value.Color~=nil then local nested:string?=normalizeTeamValue(value.Color)if nested then return nested end end if type(value)=='string'then return'name:'..string.lower(value)
end local text:string=tostring(value)if text==''or text=='nil'then return nil end return'value:'..string.lower(text)end local function teamToken(instance:Instance):string?for _,attributeName:string in
ipairs(TEAM_ATTRIBUTE_NAMES)do local ok:boolean,value:any=pcall(instance.GetAttribute,instance,attributeName)if ok then local normalized:string?=normalizeTeamValue(value)if normalized then return
normalized end end end local directOk:boolean,direct:any=pcall(function():any return(instance::any).Team end)if directOk then local normalizedDirect:string?=normalizeTeamValue(direct)if
normalizedDirect then return normalizedDirect end end local colourOk:boolean,colour:any=pcall(function():any return(instance::any).TeamColor end)if colourOk then local normalizedColour:string?=
normalizeTeamValue(colour)if normalizedColour then return normalizedColour end end local child:Instance?=instance:FindFirstChild('Team')if child then local valueOk:boolean,childValue:any=pcall(
function():any return(child::any).Value end)return normalizeTeamValue(valueOk and childValue or child.Name)end return nil end local function localTeamToken():string?return teamToken(localPlayer)end
local function teamResolver():any local hostState:any=host.state if hostState and hostState.gameBridge then return hostState.gameBridge.teamProvider end return nil end local function providerTeamToken
(instance:Instance):string?local provider:any=teamResolver()if not provider or type(provider.Get)~='function'then return nil end local ok:boolean,token:any=pcall(provider.Get,instance)if ok and type(
token)=='string'and token~=''then return token end return nil end local function providerRelation(instance:Instance):(boolean,string?)local provider:any=teamResolver()if not provider or type(provider.
GetRelation)~='function'then return false,nil end local ok:boolean,relation:any=pcall(provider.GetRelation,instance)if ok and type(relation)=='string'then return true,relation end return true,nil end
local function advancedVerdict(player:Player):any local checker:any=host.state and host.state.teamCheck if checker and type(checker.analyze)=='function'then local ok:boolean,verdict:any=pcall(checker.
analyze,player)if ok and type(verdict)=='table'then return verdict end end return nil end function library:IsFriendly(player:Player?):boolean if not player or player==localPlayer then return false end
local verdict:any=advancedVerdict(player)if verdict and verdict.method~='unknown'then return verdict.sameTeam==true end local handled:boolean,relation:string?=providerRelation(player)if handled then
return relation=='Teammate'end local provider:any=teamResolver()if provider and type(provider.IsFriendly)=='function'then local ok:boolean,friendly:any=pcall(provider.IsFriendly,player)if ok and type(
friendly)=='boolean'then return friendly end end local providerToken:string?=providerTeamToken(player)if providerToken~=nil then local localProviderToken:string?=providerTeamToken(localPlayer)if
localProviderToken~=nil then return localProviderToken==providerToken end end if provider==nil and(player.Neutral==true or localPlayer.Neutral==true)then return false end local localToken:string?=
localTeamToken()local targetToken:string?=teamToken(player)return localToken~=nil and targetToken~=nil and localToken==targetToken end function library:IsFriendlyModel(model:Model):boolean if model==
localPlayer.Character then return false end local handled:boolean,relation:string?=providerRelation(model)if handled then return relation=='Teammate'end local providerToken:string?=providerTeamToken(
model)if providerToken~=nil then local localProviderToken:string?=providerTeamToken(localPlayer)if localProviderToken~=nil then return localProviderToken==providerToken end end local localToken:string
?=localTeamToken()local targetToken:string?=teamToken(model)return localToken~=nil and targetToken~=nil and localToken==targetToken end function library:IsEnemy(player:Player?):boolean?if not player
or player==localPlayer then return false end local verdict:any=advancedVerdict(player)if verdict then if type(verdict.hostile)=='boolean'then return verdict.hostile end if verdict.method==
'provider-unknown'or verdict.method=='roles-unknown'or verdict.method=='attributes-unknown'then return false end return nil end local handled:boolean,relation:string?=providerRelation(player)if
handled then return relation=='Enemy'end return nil end local function passesTeamFilter(entity:Entity,ignoreTeam:boolean?):boolean if ignoreTeam then return true end if entity.Player then local
hostile:boolean?=library:IsEnemy(entity.Player)if hostile~=nil then return hostile end end return not entity.IsFriendly end local function npcParts(model:Model):(Humanoid?,BasePart?)local humanoid:
Humanoid?=model:FindFirstChildOfClass('Humanoid')::Humanoid?local root:BasePart?=model:FindFirstChild('HumanoidRootPart')::BasePart?or model.PrimaryPart or model:FindFirstChildWhichIsA('BasePart')::
BasePart?return humanoid,root end local function isNpcModel(model:Model):boolean if not model.Parent or model==localPlayer.Character then return false end if players:GetPlayerFromCharacter(model)~=nil
then return false end if model:GetAttribute('WurstDisguise')==true then return false end local humanoid:Humanoid?,root:BasePart?=npcParts(model)return humanoid~=nil and root~=nil end local function
modelFor(instance:Instance):Model?if instance:IsA('Model')then return instance::Model end return instance:FindFirstAncestorOfClass('Model')::Model?end local function indexNpc(instance:Instance):()
local model:Model?=modelFor(instance)if model and isNpcModel(model)then library.npcModels[model]=true end end local function unindexNpc(instance:Instance):()local model:Model?=modelFor(instance)if
model and(instance==model or not model.Parent)then library.npcModels[model]=nil library.ByModel[model]=nil end end local function refreshNpcs(originPosition:Vector3):()local list:{Entity}=library.
NPCList local byModel:{[Model]:Entity}=library.ByModel table.clear(list)table.clear(byModel)for model:Model in pairs(library.npcModels)do if not isNpcModel(model)then library.npcModels[model]=nil
continue end local humanoid:Humanoid?,root:BasePart?=npcParts(model)if not humanoid or not root or humanoid.Health<=0 then continue end local entity:any=library.npcPool[model]if not entity then entity
={}library.npcPool[model]=entity end entity.Player=nil entity.Character=model entity.Humanoid=humanoid entity.RootPart=root entity.Head=model:FindFirstChild('Head')::BasePart?entity.Health=humanoid.
Health entity.MaxHealth=math.max(humanoid.MaxHealth,1)entity.Distance=(originPosition-root.Position).Magnitude entity.Team=nil entity.IsFriendly=library:IsFriendlyModel(model)entity.IsNPC=true entity.
Name=model.Name byModel[model]=entity table.insert(list,entity)end end function library:Refresh(force:boolean?):{Entity}local now:number=os.clock()if not force and now-library.lastRefresh<1/30 then
return library.List end library.lastRefresh=now local origin:BasePart?=localRoot()local originPosition:Vector3=origin and origin.Position or Vector3.zero local list:{Entity}=library.List table.clear(
list)local byPlayer:{[Player]:Entity}=library.ByPlayer table.clear(byPlayer)for _,player:Player in ipairs(players:GetPlayers())do local character,humanoid,root=characterParts(player)if not character
or not humanoid or not root then continue end if humanoid.Health<=0 then continue end local entity:any=library.pool[player]if not entity then entity={}library.pool[player]=entity end entity.Player=
player entity.Character=character entity.Humanoid=humanoid entity.RootPart=root entity.Head=character:FindFirstChild('Head')::BasePart?entity.Health=humanoid.Health entity.MaxHealth=math.max(humanoid.
MaxHealth,1)entity.Distance=(originPosition-root.Position).Magnitude entity.Team=player.Team entity.IsFriendly=library:IsFriendly(player)entity.IsNPC=false entity.Name=player.Name byPlayer[player]=
entity if player==localPlayer then library.LocalEntity=entity else table.insert(list,entity)end end if not byPlayer[localPlayer]then library.LocalEntity=nil end refreshNpcs(originPosition)return list
end function library:IsAlive():boolean local _,humanoid,root=characterParts(localPlayer)return humanoid~=nil and root~=nil and humanoid.Health>0 end function library:Get(player:Player):Entity?return
library.ByPlayer[player]end function library:GetNpc(model:Model):Entity?return library.ByModel[model]end function library:ForEach(callback:(Entity)->()):()for _,entity:Entity in ipairs(library.List)do
callback(entity)end end function library:VisibleFrom(entity:Entity):boolean local camera:Camera?=currentWorkspace.CurrentCamera if not camera then return false end local parameters:RaycastParams=
RaycastParams.new()parameters.FilterType=Enum.RaycastFilterType.Exclude parameters.IgnoreWater=true parameters.FilterDescendantsInstances={entity.Character,localPlayer.Character::any,camera}local
origin:Vector3=camera.CFrame.Position local target:Vector3=entity.RootPart.Position local result:RaycastResult?=currentWorkspace:Raycast(origin,target-origin,parameters)return result==nil end function
library:Rig(entity:Entity):{{BasePart}}local character:Model=entity.Character local cached:any=library.rigs[character]if cached then return cached end local bones:{{BasePart}}={}local definitions:{{
string}}=character:FindFirstChild('UpperTorso')and R15_BONES or R6_BONES for _,pair:{string}in ipairs(definitions)do local first:BasePart?=character:FindFirstChild(pair[1])::BasePart?local second:
BasePart?=character:FindFirstChild(pair[2])::BasePart?if first and second then table.insert(bones,{first,second})end end library.rigs[character]=bones return bones end function library:IsProtected(
player:Player?):boolean if not player then return false end local targets:any=host.services and host.services.protectedTargets if targets and type(targets.isProtected)=='function'then return targets.
isProtected(player)==true end return false end function library:InRange(query:RangeQuery):{Entity}local origin:BasePart?=localRoot()if not origin then return{}end local resolvedOrigin:BasePart=origin
::BasePart local facing:Vector3=resolvedOrigin.CFrame.LookVector*Vector3.new(1,0,1)local halfAngle:number?=query.Angle and math.rad(query.Angle)/2 or nil local candidates:{Entity}={}local function
consider(entity:Entity):()if entity.Distance>query.Range then return end if not passesTeamFilter(entity,query.IgnoreTeam)then return end if not query.IgnoreFriends and library:IsProtected(entity.
Player)then return end if halfAngle then local delta:Vector3=(entity.RootPart.Position-resolvedOrigin.Position)*Vector3.new(1,0,1)if delta.Magnitude>0.001 and facing.Magnitude>0.001 then local dot:
number=facing.Unit:Dot(delta.Unit)if math.acos(math.clamp(dot,-1,1))>halfAngle then return end end end if not query.IgnoreWalls and not library:VisibleFrom(entity)then return end table.insert(
candidates,entity)end for _,entity:Entity in ipairs(library.List)do consider(entity)end if query.IncludeNPCs then for _,entity:Entity in ipairs(library.NPCList)do consider(entity)end end local
sortMode:string=query.Sort or'Distance'table.sort(candidates,function(first:Entity,second:Entity):boolean if sortMode=='Health'then return first.Health<second.Health end if sortMode=='Threat'then
return first.Distance/math.max(first.Health,1)>second.Distance/math.max(second.Health,1)end return first.Distance<second.Distance end)local limit:number=query.Limit or#candidates while#candidates>
limit do table.remove(candidates)end return candidates end function library:ClosestToRay(origin:Vector3,direction:Vector3,query:EntityQuery?):Entity?local options:EntityQuery=query or{}local unit:
Vector3=direction.Magnitude>0 and direction.Unit or Vector3.new(0,0,-1)local maxDistance:number=options.MaxDistance or math.huge local best:Entity?=nil local bestScore:number=math.huge local function
consider(entity:Entity):()if not passesTeamFilter(entity,options.IgnoreTeam)then return end if entity.Distance>maxDistance then return end if not options.IgnoreFriends and library:IsProtected(entity.
Player)then return end local part:BasePart=entity.RootPart if options.Part=='Head'and entity.Head then part=entity.Head::BasePart end local offset:Vector3=part.Position-origin local along:number=
offset:Dot(unit)if along<=0 then return end local perpendicular:number=(offset-unit*along).Magnitude if perpendicular<bestScore and(options.IgnoreWalls or library:VisibleFrom(entity))then best=entity
bestScore=perpendicular end end for _,entity:Entity in ipairs(library.List)do consider(entity)end if options.IncludeNPCs then for _,entity:Entity in ipairs(library.NPCList)do consider(entity)end end
local radius:number?=options.Radius if radius and bestScore>radius then return nil end return best end function library:ClosestToCursor(query:EntityQuery?):Entity?local camera:Camera?=currentWorkspace
.CurrentCamera if not camera then return nil end local ray:Ray?=host.services and host.services.aim.getRay()local origin:Vector3=ray and ray.Origin or camera.CFrame.Position local direction:Vector3=
ray and ray.Direction or camera.CFrame.LookVector return library:ClosestToRay(origin,direction,query)end function library:Destroy():()for _,connection:RBXScriptConnection in ipairs(library.Connections
)do pcall(function():()connection:Disconnect()end)end library.Connections={}library.List={}library.ByPlayer={}library.NPCList={}library.ByModel={}library.npcModels=setmetatable({},{__mode='k'})library
.LocalEntity=nil if activeLibrary==library then activeLibrary=nil end end for _,descendant:Instance in ipairs(currentWorkspace:GetDescendants())do indexNpc(descendant)end table.insert(library.
Connections,currentWorkspace.DescendantAdded:Connect(indexNpc))table.insert(library.Connections,currentWorkspace.DescendantRemoving:Connect(unindexNpc))table.insert(library.Connections,players.
PlayerAdded:Connect(function(player:Player):()table.insert(library.Connections,player.CharacterAdded:Connect(function():()library:Refresh(true)end))library:Refresh(true)end))table.insert(library.
Connections,players.PlayerRemoving:Connect(function(player:Player):()library.ByPlayer[player]=nil library:Refresh(true)end))for _,player:Player in ipairs(players:GetPlayers())do table.insert(library.
Connections,player.CharacterAdded:Connect(function():()library:Refresh(true)end))end library:Refresh(true)activeLibrary=(library::any)::EntityLibrary Module.Initialized=true return(library::any)::
EntityLibrary end function Module.destroy():()local library:EntityLibrary?=activeLibrary if library then library:Destroy()end activeLibrary=nil Module.Initialized=false end return Module
-- src/libraries/Framework.lua
export type CleanupItem=any export type MovementInputService={getVector:()->(number,number),isJumpHeld:()->boolean,onJumpRequest:((()->()))->RBXScriptConnection}export type OwnershipService<T> ={set:(
ownerId:string,value:T?)->(),apply:(instance:Instance?)->()}export type ModuleServices={movementInput:MovementInputService,aim:{getRay:()->Ray?},screenCapture:{isAvailable:()->boolean,capture:(string
)->(boolean,string?),writeText:(string,string)->boolean},menu:{isVisible:()->boolean,isCapturingInput:()->boolean,setVisible:(boolean)->()},mobileActions:{bindPlacement:(...any)->any},protectedTargets
:{isProtected:(Player?)->boolean,setProvider:(((Player?)->boolean)?)->()},fovOwnership:OwnershipService<number>,platformStandOwnership:OwnershipService<boolean>,activity:{set:(string,boolean)->(),
isActive:(string)->boolean},spoofAvatar:any,projectileCalibration:any,gameBridge:any,shortcuts:any,registries:any}export type ModuleContext={framework:Framework,entity:any,weapons:any,render:any,
libraries:{[string]:any},host:any,services:ModuleServices}export type Option={Name:string,Value:any,Object:any,SetVisible:(self:Option,visible:boolean)->()}export type Module={Name:string,Category:
string,Enabled:boolean,Feature:any,Options:{[string]:Option},OptionMetadata:{any},CreateToggle:(self:Module,definition:any)->Option,CreateSlider:(self:Module,definition:any)->Option,CreateTwoSlider:(
self:Module,definition:any)->Option,CreateDropdown:(self:Module,definition:any)->Option,CreateBind:(self:Module,definition:any)->Option,CreateTextBox:(self:Module,definition:any)->Option,CreateColor:(
self:Module,definition:any)->Option,CreateList:(self:Module,definition:any)->Option,CreateButton:(self:Module,definition:any)->Option,CreateSection:(self:Module,name:string)->Option,CreateNote:(self:
Module,text:string)->(),Clean:(self:Module,item:CleanupItem)->CleanupItem,Loop:(self:Module,callback:(number)->())->any,Render:(self:Module,callback:(number)->())->any,Event:(self:Module,signal:any,
callback:(...any)->())->any,CleanAll:(self:Module)->(),Toggle:(self:Module,value:boolean?)->(),Notify:(self:Module,message:string)->(),SetStatus:(self:Module,status:string?)->()}export type Category={
Name:string,Modules:{Module},CreateModule:(self:Category,definition:any)->Module}export type Framework={Categories:{[string]:Category},Modules:{[string]:Module},Order:{string},Host:any,GetCategory:(
self:Framework,name:string)->Category,GetModule:(self:Framework,name:string)->Module?,DisableAll:(self:Framework)->(),Destroy:(self:Framework)->()}local Module={Name='Framework',PlaceId=0,Events={}::{
[string]:any},Initialized=false}local KNOWN_CATEGORIES:{string}={'Exploits','Utilities','World','Render','Blocks','Items'}local activeFramework:Framework?=nil local function isKnownCategory(name:
string):boolean return table.find(KNOWN_CATEGORIES,name)~=nil end local CATEGORY_ALIASES:{[string]:string}={Combat='Exploits',Blatant='Exploits',Movement='World',Other='Utilities',Fun='Utilities',
Visuals='Render',Protection='Utilities',Utility='Utilities',Spoof='Utilities',General='Utilities',Player='World'}local function canonicalCategory(name:string):string if isKnownCategory(name)then
return name end return CATEGORY_ALIASES[name]or'Utilities'end local function disposeItem(item:CleanupItem):()if item==nil then return end local itemType:string=typeof(item)if itemType=='function'then
pcall(item::any)return end if itemType=='RBXScriptConnection'then pcall(function():()(item::any):Disconnect()end)return end if itemType=='Instance'then pcall(function():()(item::any):Destroy()end)
return end if itemType=='table'then local candidate:any=item if type(candidate.Disconnect)=='function'then pcall(candidate.Disconnect,candidate)return end if type(candidate.Destroy)=='function'then
pcall(candidate.Destroy,candidate)return end if type(candidate.disconnect)=='function'then pcall(candidate.disconnect,candidate)return end end end function Module.init(context:any):Framework local
host:any=context local framework:any={Categories={},Modules={},Order={},Host=host}local function normalizeSearchName(value:any):string return string.lower(tostring(value or'')):gsub('[%s%-%_]','')end
local moduleSearch:any={entries={},aliases={esp='playeresp',playeresp='playeresp',itemesp='itemesp',autoclicker='autoclicker',aimassist='aimbot',killaura='killaura',fly='flight',speed='speedhack',
nofall='nofall'}}function moduleSearch.Register(feature:any):()if type(feature)~='table'then return end local normalized:string=normalizeSearchName(feature.name)local binding:any=host.state.
shortcutBindings and host.state.shortcutBindings[feature.configKey]local entry:any={normalized=normalized,displayName=feature.name,configKey=feature.configKey,actionId=feature.configKey,aliases={},
kind=feature.kind,searchable=feature.searchable~=false,feature=feature,activate=feature.activate,invoke=feature.activate}function entry.currentKey():any return binding and binding.key or Enum.KeyCode.
Unknown end function entry.conflict():boolean local key:any=entry.currentKey()if key==nil or key==Enum.KeyCode.Unknown then return false end local uses:number=0 for _,candidate:any in pairs(host.state
.shortcutBindings or{})do if candidate.key==key then uses+=1 end end return uses>1 end function entry.capture(key:Enum.KeyCode):boolean return binding~=nil and binding.assign(key)==true end function
entry.remove():()if binding and binding.key~=Enum.KeyCode.Unknown then binding.assign(binding.key)end end for alias:string,target:string in pairs(moduleSearch.aliases)do if target==normalized then
table.insert(entry.aliases,alias)end end table.sort(entry.aliases)function entry.enabled():boolean return feature.enabled==true end table.insert(moduleSearch.entries,entry)end function moduleSearch.
Unregister(feature:any):()for index:number=#moduleSearch.entries,1,-1 do if moduleSearch.entries[index].feature==feature then table.remove(moduleSearch.entries,index)end end end function moduleSearch.
Resolve(query:any):any local normalized:string=normalizeSearchName(query)if normalized==''then return{status='not_found'}end normalized=moduleSearch.aliases[normalized]or normalized local matches:{any
}={}for _,entry:any in ipairs(moduleSearch.entries)do if entry.searchable and entry.normalized==normalized then table.insert(matches,entry)end end if#matches==0 then return{status='not_found'}end if#
matches>1 then return{status='ambiguous',matches=matches}end return{status='success',entry=matches[1]}end function moduleSearch.Execute(query:any):any local result:any=moduleSearch.Resolve(query)if
result.status~='success'then return result end local entry:any=result.entry if entry.kind~='toggle'and entry.kind~='action'then return{status='not_toggleable',entry=entry}end if type(entry.activate)~=
'function'then return{status='not_toggleable',entry=entry}end entry.activate()return{status='success',entry=entry,enabled=entry.enabled()}end host.state.moduleSearch=moduleSearch local function
newOption(name:string,initialValue:any,optionType:string,definition:any?):any local source:any=definition or{}local defaultValue:any=initialValue if type(initialValue)=='table'then defaultValue=table.
clone(initialValue)end local step:number?=source.Step if step==nil and(optionType=='slider'or optionType=='range')then local integerOnly:boolean=type(source.Min)=='number'and type(source.Max)==
'number'and source.Min%1==0 and source.Max%1==0 and((type(initialValue)=='number'and initialValue%1==0)or(type(initialValue)=='table'and initialValue.Min%1==0 and initialValue.Max%1==0))step=
integerOnly and 1 or 0 end local option:any={Name=name,Value=initialValue,Object=nil,Metadata={id='',configKey='',label=name,type=optionType,default=defaultValue,current=initialValue,min=source.Min,
max=source.Max,step=step,values=source.List and table.clone(source.List)or nil,callback=source.Function,show=source.Show,visible=source.Show==nil,action=optionType=='action',bind=optionType=='bind',
color=optionType=='color',rangeLow=source.DefaultMin,rangeHigh=source.DefaultMax}}function option:SetValue(value:any):()self.Value=value self.Metadata.current=value if type(value)=='table'and value.
Min~=nil and value.Max~=nil then self.Metadata.rangeLow=value.Min self.Metadata.rangeHigh=value.Max end end function option:SetVisible(visible:boolean):()self.Metadata.visible=visible local row:any=
self.Object if typeof(row)=='Instance'and row.ClassName~='Frame'then row=row.Parent end if host.setOptionVisible and row then host.setOptionVisible(self.Feature,row,visible)end end return option end
local function createModule(category:any,definition:any):Module local moduleName:string=definition.Name local displayName:string=definition.DisplayName or moduleName local categoryName:string=
definition.Category or category.Name local created:any={Name=moduleName,Category=categoryName,Enabled=false,Options={},OptionMetadata={},Cleanup={}}local function onToggle(enabled:boolean?):()local
nextState:boolean=enabled==true created.Enabled=nextState if not nextState then created:CleanAll()end local handler:any=definition.Function if type(handler)=='function'then handler(nextState)end if
not nextState then created:CleanAll()end end local kind:string=definition.Kind or(definition.Action==true and'action')or'toggle'local feature:any=host.createUniversalFeature(displayName,definition.
Tooltip or definition.Description or'',definition.Order or(#category.Modules+1),kind=='action'and function():()if type(definition.Function)=='function'then definition.Function(true)end end or onToggle
,{categoryName=categoryName,kind=kind,silentAction=definition.Silent==true,searchable=definition.Searchable~=false,registry=definition.Registry,parent=definition.Parent,configKey=definition.ConfigKey}
)created.Feature=feature feature.frameworkModule=true local function register(option:any,definition2:any?):Option option.Feature=feature local optionId:string=option.Name:gsub('%W','')option.Metadata.
id=optionId option.Metadata.configKey=feature.configKey..'.'..optionId local gated:boolean=false if definition2 and type(definition2.Show)=='table'then option.Show=definition2.Show gated=true end
created.Options[option.Name]=option table.insert(created.OptionMetadata,option.Metadata)feature.optionMetadata=created.OptionMetadata if gated then created:RefreshVisibility()end return option end
local function ruleHolds(rule:any):boolean local source:any=created.Options[rule.Option]local holds:boolean=false if source~=nil then if type(rule.Values)=='table'then for _,wanted:any in ipairs(rule.
Values)do if source.Value==wanted then holds=true break end end else holds=source.Value==true end end if rule.Invert==true then return not holds end return holds end function created:RefreshVisibility
():()for _,option:any in pairs(created.Options)do local rule:any=option.Show local available:boolean=created.OptionFilter==nil or created.OptionFilter(option.Name)if type(rule)~='table'and available
and not option.FilteredOut then continue end option.FilteredOut=not available local visible:boolean=available if type(rule)~='table'then elseif rule.Option~=nil then visible=visible and ruleHolds(rule
)else for _,nested:any in ipairs(rule)do if not ruleHolds(nested)then visible=false break end end end if option.Visible~=visible then option.Visible=visible option:SetVisible(visible)end end end
local function valueCallback(option:any,definition2:any):(any)->()local function changed(value:any):()option:SetValue(value)if type(definition2.Function)=='function'then definition2.Function(value)end
created:RefreshVisibility()end option.Metadata.callback=changed return changed end function created:CreateToggle(definition2:any):Option local option:any=newOption(definition2.Name,definition2.Default
==true,'toggle',definition2)option.Object=host.addToggleOption(feature,definition2.Name,definition2.Default==true,valueCallback(option,definition2),definition2.Tooltip,definition2.Bindable==true)
return register(option,definition2)end function created:CreateSlider(definition2:any):Option local default:number=definition2.Default or definition2.Min or 0 local option:any=newOption(definition2.
Name,default,'slider',definition2)option.Object=host.addNumberOption(feature,definition2.Name,default,definition2.Min or 0,definition2.Max or 100,valueCallback(option,definition2),definition2.Tooltip,
definition2.Step)return register(option,definition2)end function created:CreateTwoSlider(definition2:any):Option local minimum:number=definition2.Min or 0 local maximum:number=definition2.Max or 100
local low:number=definition2.DefaultMin or minimum local high:number=definition2.DefaultMax or maximum local option:any=newOption(definition2.Name,{Min=low,Max=high},'range',definition2)local builder:
any=host.addRangeOption or(host.state and host.state.addRangeOption)option.GetRandomValue=function(_self:any):number local band:any=option.Value local lowValue:number=band.Min local highValue:number=
math.max(band.Max,lowValue)return lowValue+math.random()*(highValue-lowValue)end if type(builder)~='function'then return register(option)end local function rangeChanged(newLow:number,newHigh:number):(
)option:SetValue({Min=newLow,Max=newHigh})if type(definition2.Function)=='function'then definition2.Function(newLow,newHigh)end created:RefreshVisibility()end option.Metadata.callback=rangeChanged
option.Object=builder(feature,definition2.Name,{min=minimum,max=maximum,step=definition2.Step,defaultLow=low,defaultHigh=high,integer=definition2.Decimal==nil,hint=definition2.Tooltip,onChanged=
rangeChanged})return register(option,definition2)end function created:CreateDropdown(definition2:any):Option local list:{string}=definition2.List or{}local index:number=definition2.Index or 1 local
option:any=newOption(definition2.Name,list[index],'dropdown',definition2)option.Object=host.addCycleOption(feature,definition2.Name,list,index,valueCallback(option,definition2),definition2.Tooltip)
return register(option,definition2)end function created:CreateBind(definition2:any):Option local default:Enum.KeyCode=definition2.Default or Enum.KeyCode.Unknown local option:any=newOption(definition2
.Name,default,'bind',definition2)option.Object=host.addKeyOption(feature,definition2.Name,default,valueCallback(option,definition2),definition2.Tooltip)return register(option,definition2)end function
created:CreateTextBox(definition2:any):Option local default:string=definition2.Default or''local option:any=newOption(definition2.Name,default,'text',definition2)option.Object=host.addTextOption(
feature,definition2.Name,default,valueCallback(option,definition2),definition2.Persist~=false,definition2.Tooltip)if option.Object and definition2.Placeholder then option.Object.PlaceholderText=
definition2.Placeholder end option.Set=function(self:any,value:string):()self.Metadata.callback(value)if self.Object then self.Object.Text=value end end return register(option,definition2)end function
created:CreateColor(definition2:any):Option local default:Color3=definition2.Default or Color3.new(1,1,1)local option:any=newOption(definition2.Name,default,'color',definition2)option.Object=host.
addColorOption(feature,definition2.Name,default,valueCallback(option,definition2),definition2.Tooltip)return register(option,definition2)end function created:CreateList(definition2:any):Option local
option:any=newOption(definition2.Name,{},'list',definition2)option.Selected={}local builder:any=host.addListOption or(host.state and host.state.addListOption)if type(builder)~='function'then return
register(option)end local optionKey:string=feature.configKey..'.'..definition2.Name:gsub('%W','')local function listChanged(selected:any,names:{string}):()option:SetValue(selected)option.Selected=
names if type(definition2.Function)=='function'then definition2.Function(selected,names)end created:RefreshVisibility()end option.Metadata.callback=listChanged option.Object=builder(feature,
definition2.Name,{items=definition2.Items,single=definition2.Single==true,onRemove=definition2.OnRemove,default=definition2.Default,hint=definition2.Tooltip,emptyText=definition2.EmptyText,onChanged=
listChanged})local controls:any=host.state and host.state.listControls and host.state.listControls[optionKey]option.Set=function(_self:any,item:string,on:boolean):()if controls then controls.set(item,
on)end end option.Refresh=function(_self:any):()if controls then controls.refresh()end end option.IsSelected=function(_self:any,item:string):boolean if controls then return controls.selected[item]==
true end return false end return register(option,definition2)end function created:CreateButton(definition2:any):Option if definition2.Hold then assert(type(definition2.OnRelease)=='function',
'A held action requires OnRelease')end local option:any=newOption(definition2.Name,nil,'action',definition2)option.Metadata.hold=definition2.Hold==true local function run():()if option.Metadata.
visible==false then return end if type(definition2.Function)=='function'then definition2.Function()end end option.Metadata.callback=run option.Object=host.addActionOption(feature,definition2.Name,run,
definition2.Tooltip,definition2.Hold and{hold=true,onRelease=definition2.OnRelease}or nil)return register(option,definition2)end function created:CreateSection(name:string):Option local option:any=
newOption(name,nil,'section',nil)option.Object=host.addSectionOption and host.addSectionOption(feature,name)or nil return register(option)end function created:CreateNote(text:string):()if host.
addInformationOption then host.addInformationOption(feature,text)end end function created:Clean(item:CleanupItem):CleanupItem table.insert(created.Cleanup,item)return item end function created:Loop(
callback:(number)->()):any return created:Clean(host.TaskManager:Connect(callback,nil,moduleName))end function created:Render(callback:(number)->()):any return created:Clean(host.TaskManager:Connect(
callback,'render',moduleName))end function created:Event(signal:any,callback:(...any)->()):any return created:Clean(signal:Connect(callback))end function created:CleanAll():()local items:{CleanupItem}
=created.Cleanup created.Cleanup={}for index:number=#items,1,-1 do disposeItem(items[index])end end function created:Toggle(value:boolean?):()if value~=nil and value==created.Enabled then return end
onToggle(value==nil and not created.Enabled or value==true)end function created:Reveal():()if feature and type(feature.activate)=='function'then feature.activate()end end function created:SetStatus(
status:string?):()created.Status=status if feature and type(feature.SetStatus)=='function'then feature:SetStatus(status)end end function created:Notify(message:string):()if host.notify then host.
notify(moduleName..': '..message)end end table.insert(category.Modules,created)framework.Modules[moduleName]=created table.insert(framework.Order,moduleName)return(created::any)::Module end function
framework:GetCategory(name:string):Category local existing:any=framework.Categories[name]if existing then return existing end local canonical:string=canonicalCategory(name)local category:any=framework
.Categories[canonical]if not category then category={Name=canonical,Modules={}}function category:CreateModule(definition:any):Module return createModule(category,definition)end framework.Categories[
canonical]=category end if name~=canonical then framework.Categories[name]=category end return(category::any)::Category end function framework:GetModule(name:string):Module?return framework.Modules[
name]end function framework:DisableAll():()for _,moduleName:string in ipairs(framework.Order)do local entry:any=framework.Modules[moduleName]if entry and entry.Enabled then pcall(function():()entry:
Toggle(false)end)end if entry then pcall(function():()entry:CleanAll()end)end end end function framework:Destroy():()framework:DisableAll()framework.Categories={}framework.Modules={}framework.Order={}
if host.state.moduleSearch==moduleSearch then host.state.moduleSearch=nil end table.clear(moduleSearch.entries)if activeFramework==framework then activeFramework=nil end end for _,name:string in
ipairs(KNOWN_CATEGORIES)do framework:GetCategory(name)end for oldName:string in pairs(CATEGORY_ALIASES)do framework:GetCategory(oldName)end activeFramework=(framework::any)::Framework Module.
Initialized=true return(framework::any)::Framework end function Module.destroy():()local framework:Framework?=activeFramework if framework then framework:Destroy()end activeFramework=nil Module.
Initialized=false end return Module
-- src/libraries/GameRoles.lua
local Module={Name='GameRoles',Initialized=false}local active:any=nil function Module.init(host:any):any local bridge=host.services.gameBridge local service={bindings={}}function service:Register(
provider:any):()->()assert(type(provider)=='table'and type(provider.Get)=='function','GameRoles needs Get(player)')assert(type(provider.Roles)=='table','GameRoles needs a Roles array')bridge.
registerRoles(provider)return function()bridge.unregisterRoles(provider)end end function service:Bind(card:any,settings:any?):any settings=settings or{}local binding={provider=nil,rows={},roles={},
dead=false}local function visibility(option,visible)if visible then option.Show=nil else option.Show={Option='__GameRolesAbsent',Values={true}}end if option.SetVisible then option:SetVisible(visible)
end end local function update(provider)if binding.dead then return end binding.provider=type(provider)=='table'and provider or nil table.clear(binding.roles)if binding.provider then for _,role in
ipairs(type(provider.Roles)=='table'and provider.Roles or{})do if type(role)=='string'and role~=''and#binding.roles<32 and not table.find(binding.roles,role)then table.insert(binding.roles,role)end
end end if settings.palette~=false and#binding.roles>0 and not card.Options['Role colours']then binding.rows['Role colours']=card:CreateToggle({Name='Role colours',Default=true})end for _,role in
ipairs(binding.roles)do local colourName=role..' colour'if settings.palette~=false and not binding.rows[colourName]then local colour=provider.Colors and provider.Colors[role]if type(provider.GetColor)
=='function'then local ok,value=pcall(provider.GetColor,role)if ok and typeof(value)=='Color3'then colour=value end end binding.rows[colourName]=card:CreateColor({Name=colourName,Default=typeof(colour
)=='Color3'and colour or Color3.fromRGB(236,236,240),Show={Option='Role colours'}})end if settings.filters and not binding.rows[role..' ESP']then binding.rows[role..' ESP']=card:CreateToggle({Name=
role..' ESP',Default=true})end end for name,option in pairs(binding.rows)do local present=name=='Role colours'and#binding.roles>0 for _,role in ipairs(binding.roles)do if name==role..' colour'or name
==role..' ESP'then present=true break end end visibility(option,present)if present and name~='Role colours'and string.sub(name,-7)==' colour'then option.Show={Option='Role colours'}end end card:
RefreshVisibility()end function binding:Role(player)if not self.provider then return nil end local getter=bridge.playerRole or self.provider.Get local ok,role=pcall(getter,player)return ok and type(
role)=='string'and table.find(self.roles,role)and role or nil end function binding:Resolve(player,fallback)local role=self:Role(player)local option=role and card.Options[role..' colour']local enabled=
card.Options['Role colours']return option and enabled and enabled.Value and option.Value or fallback,role end function binding:Allows(player)local role=self:Role(player)local option=role and card.
Options[role..' ESP']return not option or option.Value~=false end local disconnect=bridge.onEvent(function(kind,provider)if kind=='roles'then update(provider)end end)function binding:Destroy()if self.
dead then return end self.dead=true if disconnect then disconnect()disconnect=nil end self.provider=nil local index=table.find(service.bindings,self)if index then table.remove(service.bindings,index)
end end table.insert(service.bindings,binding)return binding end function service:Destroy()for _,binding in ipairs(table.clone(self.bindings))do binding:Destroy()end end host.services.gameRoles=
service active=service Module.Initialized=true return service end function Module.destroy()if active then active:Destroy()active=nil end Module.Initialized=false end return Module
-- src/libraries/Manifest.lua
export type ModuleEntry={path:string,name:string,category:string}export type GamePlaceEntry={placeId:number,module:string,displayName:string}export type CategoryCatalog={order:{string},aliases:{[
string]:string},default:string}export type Manifest={version:number,categories:CategoryCatalog,cards:{[string]:string},widgets:string,core:{string},libraries:{string},modules:{ModuleEntry},gamePlaces:
{GamePlaceEntry},gameModules:{[string]:{string}}}local Manifest:Manifest={version=1,categories={order={'Exploits','Utilities','World','Render','Blocks','Items'},aliases={['Combat']='Exploits',[
'Blatant']='Exploits',['Movement']='World',['Other']='Utilities',['Visuals']='Render',['Protection']='Utilities',['Utility']='Utilities',['Spoof']='Utilities',['Fun']='Utilities',['General']=
'Utilities',['Player']='World'},default='Utilities'},cards={['Aimbot']='Exploits',['Anti AFK']='World',['Anti Fling']='Utilities',['Anti Void']='Utilities',['Anti-AFK']='Other',['Anti-Fling']=
'Movement',['AntiAFK']='Other',['Auto Buy']='Other',['Auto Dribble']='Exploits',['Auto Generator']='Other',['Auto Header']='Exploits',['Auto Parry / Auto Block']='Movement',['Auto Pickup']='Other',[
'Auto Shoot']='Utilities',['Auto Swing']='Exploits',['Auto Tackle']='Exploits',['Ball Trajectory']='Render',['Bed Nuker']='Exploits',['Best Goal Visual']='Render',['Blurt Roles']='World',[
'Cooldown Modifier']='Exploits',['Damage Boost']='Exploits',['Animations']='World',['Disguise']='World',['FOV']='Render',['Fast Vault Optimizer']='Movement',['Fling']='Exploits',['Fly']='Exploits',[
'Freeze Animations']='Utilities',['Fullbright']='Render',['Goalkeeper Assist']='Exploits',['Gravity']='World',['Gun Aimbot']='Exploits',['Improve FPS']='World',['Invisibility']='Utilities',[
'Invisible']='Exploits',['Jump Settings']='Exploits',['KillAura']='Exploits',['Killer Hit Aura']='Exploits',['Killer Power Cooldowns']='Render',['Killer Proximity Visualizer']='Render',['Knife Stab']=
'Exploits',['Knife Throw']='Exploits',['Lag Switch']='Exploits',['Large Shoot']='Exploits',['Line of Sight ESP']='Render',['Locker & Hideout Detector']='Render',['Noclip']='Exploits',[
'Noclip (Vaults & Pallets)']='Exploits',['Notifications']='Utilities',['Optimize Blind / Flash Protection']='Movement',['Pass Assist']='Exploits',['Player ESP']='Render',['Power Shot']='Exploits',[
'Proximity Prompts']='Exploits',['Quick Actions']='Other',['Rejoin Server']='Other',['Remote Logger']='Other',['Round Info']='Render',['Scaffold']='Movement',['Server Aura']='Exploits',['Server Miner'
]='Other',['Server Scaffold']='Movement',['Server Settings']='Utilities',['Shoot Assist']='Exploits',['Silent Aim']='Exploits',['Silent Steps / Anti-Sound']='Movement',['SilentAIM']='Exploits',[
'Speed Settings']='Exploits',['Spider']='Exploits',['Teleport']='Utilities',['Time Changer']='Render',['VD Field of View']='Render',['VD Fullbright']='Render',['Walk Speed']='World',['WalkSpeed Hack']
='Exploits',['X-Ray']='Render',['Zoom Unlocker']='Render'},widgets='src/guis/Wurst/Code/Widgets.lua',core={'src/libraries/Framework.lua'},libraries={'src/libraries/Entity.lua',
'src/libraries/Targeting.lua','src/libraries/Weapons.lua','src/libraries/Render.lua','src/libraries/GameRoles.lua','src/libraries/CameraAccess.lua','src/guis/Wurst/Code/Cards.lua',
'src/guis/Wurst/Code/WindowManager.lua','src/guis/Wurst/Code/ClickGui.lua','src/guis/Wurst/Code/FloatingWindows.lua','src/guis/Wurst/Code/SettingsPage.lua','src/guis/Wurst/Code/MobileActions.lua',
'src/guis/Wurst/Code/Furniture.lua'},modules={{path='src/games/universal/Combat/AimAssist.lua',name='Aimbot',category='Exploits'},{path='src/games/universal/Blatant/Swim.lua',name='Air Swim',category=
'Exploits'},{path='src/games/universal/Movement/AirWalk.lua',name='Air Walk',category='Exploits'},{path='src/games/universal/Other/AntiAfk.lua',name='Anti AFK',category='World'},{path=
'src/games/universal/Movement/AntiFling.lua',name='Anti Fling',category='Utilities'},{path='src/games/universal/Render/Chams.lua',name='Chams',category='Render'},{path=
'src/games/universal/Other/Animations.lua',name='Animations',category='World'},{path='src/games/universal/Other/Disguise.lua',name='Disguise',category='World'},{path=
'src/games/universal/Blatant/Fly.lua',name='Fly',category='Exploits'},{path='src/games/universal/Render/FieldOfView.lua',name='FOV',category='Render'},{path='src/games/universal/Render/FreeCam.lua',
name='FreeCam',category='World'},{path='src/games/universal/Movement/FreezeMovements.lua',name='Freeze Animations',category='Utilities'},{path='src/games/universal/Render/Fullbright.lua',name=
'Fullbright',category='Render'},{path='src/games/universal/Movement/Gravity.lua',name='Gravity',category='World'},{path='src/games/universal/Render/HP.lua',name='HP',category='Render'},{path=
'src/games/universal/Other/ImproveFps.lua',name='Improve FPS',category='World'},{path='src/games/universal/Movement/InfiniteJump.lua',name='Infinite Jump',category='Exploits'},{path=
'src/games/universal/Blatant/Invisible.lua',name='Invisibility',category='Utilities'},{path='src/games/universal/Render/ItemRender.lua',name='ItemESP',category='Render'},{path=
'src/games/universal/Movement/Jump.lua',name='Jump Settings',category='Exploits'},{path='src/games/universal/Combat/KillAura.lua',name='KillAura',category='Exploits'},{path=
'src/games/universal/Render/Markers.lua',name='Markers',category='Render'},{path='src/games/universal/Render/Nametags.lua',name='Nametags',category='Render'},{path=
'src/games/universal/Blatant/Noclip.lua',name='Noclip',category='Exploits'},{path='src/games/universal/Render/NPCESP.lua',name='NPCESP',category='Render'},{path='src/games/universal/Render/Arrows.lua'
,name='Player Pointers',category='Render'},{path='src/games/universal/Render/PlayerESP.lua',name='PlayerESP',category='Render'},{path='src/games/universal/Other/ProximityPrompts.lua',name=
'Proximity Prompts',category='Exploits'},{path='src/games/universal/Other/RejoinServer.lua',name='Rejoin Server',category='Utilities'},{path='src/games/universal/Other/ServerSettings.lua',name=
'Server Settings',category='Utilities'},{path='src/games/universal/Render/Spectate.lua',name='Spectate',category='World'},{path='src/games/universal/Movement/Speed.lua',name='Speed Settings',category=
'Exploits'},{path='src/games/universal/Movement/ClickTeleport.lua',name='Teleport',category='Utilities'},{path='src/games/universal/Blatant/WalkFling.lua',name='Walk Fling',category='Exploits'},{path=
'src/games/universal/Blatant/Spider.lua',name='Wall Crawling',category='Exploits'},{path='src/games/universal/Render/XRay.lua',name='X-Ray',category='Render'},{path=
'src/games/universal/Render/ZoomUnlocker.lua',name='Zoom',category='Render'}},gamePlaces={{placeId=71480482338212,module='BedFight',displayName='Bed Fight'},{placeId=142823291,module='MM2',displayName
='Murder Mystery 2'},{placeId=14315258385,module='TRS',displayName='The Regular Show'},{placeId=93978595733734,module='VD',displayName='Virtual Dealer'},{placeId=92876937625630,module='X2Duels',
displayName='X2 Duels (Legacy)'},{placeId=135856908115931,module='X2Duels',displayName='X2 Duels'}},gameModules={MM2={'src/games/MM2/base.lua','src/games/MM2/Prediction.lua',
'src/games/MM2/Blatant/Shoot.lua','src/games/MM2/Blatant/KnifeAura.lua','src/games/MM2/Combat/AutoShoot.lua','src/games/MM2/Combat/KnifeThrow.lua','src/games/MM2/Render/RoundEsp.lua',
'src/games/MM2/Render/CoinEsp.lua','src/games/MM2/Movement/Sprint.lua','src/games/MM2/Movement/Teleport.lua','src/games/MM2/Fun/BlurtRoles.lua','src/games/MM2/Other/Notifications.lua',
'src/games/MM2/Other/AutoUnbox.lua','src/games/MM2/Other/FlingEngine.lua','src/games/MM2/Other/Fling.lua','src/games/MM2/Other/AutoFarm.lua','src/games/MM2/Other/AutoVote.lua',
'src/games/MM2/Other/CoinsAura.lua','src/games/MM2/Other/GunAura.lua','src/games/MM2/Other/AutoWin.lua','src/games/MM2/Other/GetAllEmotes.lua'},VD={'src/games/VD/base.lua'},BedFight={
'src/games/BedFight/base.lua','src/games/BedFight/Combat/AutoSwing.lua','src/games/BedFight/Render/RoundInfo.lua'},X2Duels={'src/games/X2Duels/Rules.lua','src/games/X2Duels/base.lua',
'src/games/X2Duels/Other/AutoFarm.lua'},TRS={'src/games/TRS/base.lua'}}return Manifest
-- src/libraries/Render.lua
export type Rect={left:number,right:number,top:number,bottom:number,width:number,height:number,centreX:number,centreY:number}export type DrawingSet={root:Frame,Line:(self:DrawingSet,group:string,index
:number,from:Vector2,to:Vector2,thickness:number,colour:Color3)->(),Box:(self:DrawingSet,rect:Rect,mode:string,thickness:number,colour:Color3)->(),Fill:(self:DrawingSet,rect:Rect?,colour:Color3,
transparency:number)->(),Label:(self:DrawingSet,slot:string,text:string,at:Vector2,anchorY:number,size:number,colour:Color3,plate:boolean)->(),HideLabel:(self:DrawingSet,slot:string)->(),HideLines:(
self:DrawingSet,group:string,first:number,last:number)->(),Bar:(self:DrawingSet,at:Vector2,height:number,fraction:number)->(),HideBar:(self:DrawingSet)->(),Show:(self:DrawingSet,visible:boolean)->(),
Highlight:(self:DrawingSet,adornee:Instance?,colour:Color3,fill:number,mode:string)->(),Destroy:(self:DrawingSet)->()}export type RenderLibrary={Layer:(self:RenderLibrary,name:string)->Frame,Set:(self
:RenderLibrary,layer:Frame,key:any)->DrawingSet,Release:(self:RenderLibrary,layer:Frame,key:any)->(),ReleaseAll:(self:RenderLibrary,layer:Frame)->(),Project:(self:RenderLibrary,camera:Camera,position:
Vector3)->Vector2?,ModelRect:(self:RenderLibrary,camera:Camera,model:Model)->Rect?,Destroy:(self:RenderLibrary)->()}local Module={Name='Render',PlaceId=0,Events={}::{[string]:any},Initialized=false}
local activeLibrary:RenderLibrary?=nil function Module.init(context:any):RenderLibrary local host:any=context local create:any=host.create local screenGui:Instance=host.ScreenGui local library:any={}
local pools:any=setmetatable({},{__mode='k'})function library:Layer(name:string):Frame local existing:Instance?=screenGui:FindFirstChild(name)if existing and existing:IsA('Frame')then return existing
::Frame end local layer:Frame=create('Frame',{Parent=screenGui,Name=name,BackgroundTransparency=1,BorderSizePixel=0,Size=UDim2.fromScale(1,1),Visible=false,ZIndex=0})::Frame pools[layer]={}return
layer end local function newSet(layer:Frame,key:any):DrawingSet local root:Frame=create('Frame',{Parent=layer,Name=typeof(key)=='Instance'and(key::Instance).Name or tostring(key),
BackgroundTransparency=1,BorderSizePixel=0,Size=UDim2.fromScale(1,1),ZIndex=0})::Frame local set:any={root=root,lines={}::{[string]:{Frame}},labels={}::{[string]:TextLabel},fill=nil::Frame?,highlight=
nil::Highlight?}local function line(group:string,index:number):Frame local bucket:{Frame}=set.lines[group]if not bucket then bucket={}set.lines[group]=bucket end local existing:Frame?=bucket[index]if
existing then return existing::Frame end local frame:Frame=create('Frame',{Parent=root,Name=group..tostring(index),AnchorPoint=Vector2.new(0.5,0.5),BorderSizePixel=0,Size=UDim2.fromOffset(0,1),Visible
=false,ZIndex=1})::Frame bucket[index]=frame return frame end function set:Line(group:string,index:number,from:Vector2,to:Vector2,thickness:number,colour:Color3):()local frame:Frame=line(group,index)
local delta:Vector2=to-from local length:number=delta.Magnitude if length<0.5 then frame.Visible=false return end frame.Visible=true frame.BackgroundColor3=colour frame.Position=UDim2.fromOffset(math.
round((from.X+to.X)*0.5),math.round((from.Y+to.Y)*0.5))frame.Size=UDim2.fromOffset(math.round(length),thickness)frame.Rotation=math.deg(math.atan2(delta.Y,delta.X))end function set:HideLines(group:
string,first:number,last:number):()local bucket:{Frame}?=set.lines[group]if not bucket then return end for index:number=first,last do local frame:Frame?=(bucket::{Frame})[index]if frame then(frame::
Frame).Visible=false end end end function set:Box(rect:Rect,mode:string,thickness:number,colour:Color3):()if mode=='Off'or rect.width<=0 or rect.height<=0 then set:HideLines('Box',1,8)set:HideLines(
'BoxOutline',1,8)return end local function edge(index,from,to)set:Line('BoxOutline',index,from,to,thickness+2,Color3.fromRGB(8,10,15))set:Line('Box',index,from,to,thickness,colour)set.lines.BoxOutline
[index].ZIndex=1 set.lines.Box[index].ZIndex=2 end local corners:{Vector2}={Vector2.new(rect.left,rect.top),Vector2.new(rect.right,rect.top),Vector2.new(rect.right,rect.bottom),Vector2.new(rect.left,
rect.bottom)}if mode=='Full'then for index:number=1,4 do edge(index,corners[index],corners[index%4+1])end set:HideLines('Box',5,8)set:HideLines('BoxOutline',5,8)return end local segment:number=math.
min(math.max(math.min(rect.width,rect.height)*0.28,2),rect.width*0.5,rect.height*0.5,18)local index:number=1 for corner:number=1,4 do local from:Vector2=corners[corner]local nextCorner:Vector2=corners
[corner%4+1]local previousCorner:Vector2=corners[(corner+2)%4+1]edge(index,from,from+(nextCorner-from).Unit*segment)edge(index+1,from,from+(previousCorner-from).Unit*segment)index+=2 end end function
set:Fill(rect:Rect?,colour:Color3,transparency:number):()if not rect and not set.fill then return end if not set.fill then set.fill=create('Frame',{Parent=root,Name='BoxFill',BackgroundColor3=colour,
BackgroundTransparency=transparency,BorderSizePixel=0,Visible=false,ZIndex=0})::Frame end local frame:Frame=set.fill::Frame if not rect then frame.Visible=false return end local resolved:Rect=rect::
Rect frame.Visible=true frame.BackgroundColor3=colour frame.BackgroundTransparency=transparency frame.Position=UDim2.fromOffset(math.round(resolved.left),math.round(resolved.top))frame.Size=UDim2.
fromOffset(math.round(resolved.width),math.round(resolved.height))end function set:Label(slot:string,text:string,at:Vector2,anchorY:number,size:number,colour:Color3,plate:boolean):()local label:
TextLabel?=set.labels[slot]if not label then label=create('TextLabel',{Parent=root,Name=slot,AnchorPoint=Vector2.new(0.5,anchorY),AutomaticSize=Enum.AutomaticSize.X,BackgroundColor3=Color3.fromRGB(0,0
,0),BackgroundTransparency=1,BorderSizePixel=0,Text='',Font=Enum.Font.Arial,RichText=false,TextStrokeColor3=Color3.fromRGB(0,0,0),TextStrokeTransparency=1,TextColor3=Color3.fromRGB(236,236,240),
TextSize=size,Size=UDim2.fromOffset(0,size+6),TextTruncate=Enum.TextTruncate.None,TextXAlignment=Enum.TextXAlignment.Center,TextYAlignment=Enum.TextYAlignment.Center,Visible=false,ZIndex=2})::
TextLabel create('UIPadding',{Parent=label,PaddingLeft=UDim.new(0,4),PaddingRight=UDim.new(0,4)})set.labels[slot]=label end local resolved:TextLabel=label::TextLabel resolved.Visible=true if resolved.
Text~=text then resolved.Text=text end if resolved.TextColor3~=colour then resolved.TextColor3=colour end if resolved.TextSize~=size then resolved.TextSize=size resolved.Size=UDim2.fromOffset(resolved
.Size.X.Offset,size+6)end local plateTransparency:number=plate and 0.5 or 1 if resolved.BackgroundTransparency~=plateTransparency then resolved.BackgroundTransparency=plateTransparency end local x:
number=math.round(at.X)local y:number=math.round(at.Y)local position:UDim2=resolved.Position if position.X.Offset~=x or position.Y.Offset~=y then resolved.Position=UDim2.fromOffset(x,y)end end
function set:HideLabel(slot:string):()local label:TextLabel?=set.labels[slot]if label then(label::TextLabel).Visible=false end end function set:Bar(at:Vector2,height:number,fraction:number):()if not
set.barTrack then set.barTrack=create('Frame',{Parent=root,Name='HealthTrack',AnchorPoint=Vector2.new(1,0),BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=0.35,BorderSizePixel=0,Visible=
false,ZIndex=1})::Frame set.barFill=create('Frame',{Parent=set.barTrack,Name='HealthFill',AnchorPoint=Vector2.new(0,1),BackgroundColor3=Color3.fromRGB(120,220,140),BorderSizePixel=0,Position=UDim2.
new(0,1,1,0),Size=UDim2.new(0,1,1,0),ZIndex=2})::Frame end local track:Frame=set.barTrack::Frame local fill:Frame=set.barFill::Frame local clamped:number=fraction==fraction and math.clamp(fraction,0,1
)or 0 track.Visible=true track.Position=UDim2.fromOffset(math.round(at.X),math.round(at.Y))track.Size=UDim2.fromOffset(3,math.max(1,math.round(height)))fill.Size=UDim2.new(0,1,clamped,0)local low,mid,
high=Color3.fromRGB(255,70,70),Color3.fromRGB(255,220,60),Color3.fromRGB(85,255,85)fill.BackgroundColor3=clamped<0.5 and low:Lerp(mid,clamped*2)or mid:Lerp(high,(clamped-0.5)*2)end function set:
HideBar():()if set.barTrack then(set.barTrack::Frame).Visible=false end end function set:Show(visible:boolean):()root.Visible=visible if set.highlight then(set.highlight::Highlight).Enabled=visible
end end function set:Highlight(adornee:Instance?,colour:Color3,fill:number,mode:string):()if not adornee or mode=='Off'then if set.highlight then(set.highlight::Highlight):Destroy()set.highlight=nil
end return end local highlight:Highlight?=set.highlight if not highlight or not(highlight::Highlight).Parent then highlight=create('Highlight',{Parent=adornee,Name='Wurst_Chams',DepthMode=Enum.
HighlightDepthMode.AlwaysOnTop})::Highlight set.highlight=highlight end local resolved:Highlight=highlight::Highlight resolved.Adornee=adornee resolved.Parent=adornee resolved.Enabled=root.Visible
resolved.DepthMode=mode=='Occluded'and Enum.HighlightDepthMode.Occluded or Enum.HighlightDepthMode.AlwaysOnTop resolved.FillColor=colour resolved.OutlineColor=colour resolved.FillTransparency=mode==
'Outline'and 1 or fill resolved.OutlineTransparency=0 end function set:Destroy():()if set.highlight then pcall(function():()(set.highlight::Highlight):Destroy()end)end pcall(function():()root:Destroy(
)end)end return(set::any)::DrawingSet end function library:Set(layer:Frame,key:any):DrawingSet local pool:any=pools[layer]if not pool then pool={}pools[layer]=pool end local existing:any=pool[key]if
existing then return existing end local set:DrawingSet=newSet(layer,key)pool[key]=set return set end function library:Release(layer:Frame,key:any):()local pool:any=pools[layer]local set:any=pool and
pool[key]if not set then return end pool[key]=nil set:Destroy()end function library:ReleaseAll(layer:Frame):()local pool:any=pools[layer]if not pool then return end for key:any,set:any in pairs(pool)
do pool[key]=nil set:Destroy()end end function library:Project(camera:Camera,position:Vector3):Vector2?local point:Vector3=camera:WorldToViewportPoint(position)if point.Z<=0 then return nil end return
Vector2.new(point.X,point.Y)end local function rectFromPoints(camera:Camera,viewport:Vector2,points:{Vector3},minimumAspect:number?):Rect?local minimumX:number,minimumY:number=math.huge,math.huge
local maximumX:number,maximumY:number=-math.huge,-math.huge local inFront:number=0 for _,position:Vector3 in ipairs(points)do local point:Vector3=camera:WorldToViewportPoint(position)if point.Z>0 then
inFront+=1 if point.X<minimumX then minimumX=point.X end if point.Y<minimumY then minimumY=point.Y end if point.X>maximumX then maximumX=point.X end if point.Y>maximumY then maximumY=point.Y end end
end if inFront<2 then return nil end minimumX-=2 maximumX+=2 minimumY-=2 maximumY+=2 local width:number=maximumX-minimumX local height:number=maximumY-minimumY if minimumAspect and height>0 and height
*minimumAspect>width then width=height*minimumAspect local centreX:number=(minimumX+maximumX)*0.5 minimumX=centreX-width*0.5 maximumX=centreX+width*0.5 end if maximumX<0 or minimumX>viewport.X or
maximumY<0 or minimumY>viewport.Y then return nil end local left:number=math.max(minimumX,0)local right:number=math.min(maximumX,viewport.X)local top:number=math.max(minimumY,0)local bottom:number=
math.min(maximumY,viewport.Y)if right-left<1 or bottom-top<1 then return nil end return{left=left,right=right,top=top,bottom=bottom,width=right-left,height=bottom-top,centreX=(left+right)*0.5,centreY=
(top+bottom)*0.5}end function library:ModelRect(camera:Camera,model:Model):Rect?if not model.Parent then return nil end local viewport:Vector2=camera.ViewportSize local humanoid:Humanoid?=model:
FindFirstChildOfClass('Humanoid')::Humanoid?local root:BasePart?=model:FindFirstChild('HumanoidRootPart')::BasePart?or model.PrimaryPart if humanoid and root then local head:BasePart?=model:
FindFirstChild('Head')::BasePart?local rootPosition:Vector3=root.Position local topPosition:Vector3=head and(head.Position+Vector3.new(0,head.Size.Y*0.5+0.1,0))or(rootPosition+Vector3.new(0,2.3,0))
local leftFoot=model:FindFirstChild('LeftFoot')or model:FindFirstChild('Left Leg')local rightFoot=model:FindFirstChild('RightFoot')or model:FindFirstChild('Right Leg')local lowestY:number=rootPosition
.Y-math.max(humanoid.HipHeight+root.Size.Y*0.5,2.0)if leftFoot and leftFoot:IsA('BasePart')then lowestY=math.min(lowestY,leftFoot.Position.Y-leftFoot.Size.Y*0.5)end if rightFoot and rightFoot:IsA(
'BasePart')then lowestY=math.min(lowestY,rightFoot.Position.Y-rightFoot.Size.Y*0.5)end local bottomPosition:Vector3=Vector3.new(rootPosition.X,lowestY,rootPosition.Z)local rootCFrame:CFrame=root.CFrame return rectFromPoints(camera,viewport,{topPosition,bottomPosition,rootCFrame*Vector3.new(1.35,0,0),rootCFrame*Vector3.new(-1.35,0,0),rootCFrame*Vector3.new(0,0,0.75),rootCFrame*
Vector3.new(0,0,-0.75)},0.38)end local boundsOk:boolean,boundsCFrame:any,boundsSize:any=pcall(model.GetBoundingBox,model)if not boundsOk or typeof(boundsSize)~='Vector3'or boundsSize.Magnitude<=0 then
return nil end local half:Vector3=boundsSize*0.5 return rectFromPoints(camera,viewport,{boundsCFrame*Vector3.new(half.X,half.Y,half.Z),boundsCFrame*Vector3.new(-half.X,half.Y,half.Z),boundsCFrame*
Vector3.new(half.X,-half.Y,half.Z),boundsCFrame*Vector3.new(-half.X,-half.Y,half.Z),boundsCFrame*Vector3.new(half.X,half.Y,-half.Z),boundsCFrame*Vector3.new(-half.X,half.Y,-half.Z),boundsCFrame*
Vector3.new(half.X,-half.Y,-half.Z),boundsCFrame*Vector3.new(-half.X,-half.Y,-half.Z)})end function library:Destroy():()for layer:any,pool:any in pairs(pools)do for key:any,set:any in pairs(pool)do
pool[key]=nil pcall(function():()set:Destroy()end)end pcall(function():()(layer::Frame):Destroy()end)end if activeLibrary==library then activeLibrary=nil end end activeLibrary=(library::any)::
RenderLibrary Module.Initialized=true return(library::any)::RenderLibrary end function Module.destroy():()local library:RenderLibrary?=activeLibrary if library then library:Destroy()end activeLibrary=
nil Module.Initialized=false end return Module
-- src/libraries/Targeting.lua
export type Query={FOV:number?,MaxDistance:number?,TargetPart:string?,Prediction:number?,TeamCheck:boolean?,VisibilityCheck:boolean?,Priority:string?,IncludeNPCs:boolean?,AllowOffscreen:boolean?}
export type Target={player:Player?,entity:any,character:Model,part:BasePart,position:Vector3,screenPosition:Vector2,distance:number,cursorDistance:number}export type Library={List:(self:Library,
entities:any,query:Query?)->{Target},Select:(self:Library,entities:any,query:Query?)->Target?,Destroy:(self:Library)->()}local Module={Name='Targeting',PlaceId=0,Events={}::{[string]:any},Initialized=
false}local activeLibrary:Library?=nil local function asBasePart(instance:Instance?):BasePart?if instance and instance:IsA('BasePart')then return instance::BasePart end return nil end function Module.
init(context:any):Library local host:any=context.host or context local userInput:UserInputService?=host.UserInputService local currentWorkspace:Workspace=host.workspace or workspace local localPlayer:
Player?=host.LocalPlayer local services:any=context.services or host.services local library:any={}local function camera():Camera?return currentWorkspace.CurrentCamera end local function cursor(
cameraObject:Camera):Vector2 if userInput and userInput.MouseEnabled then local ok:boolean,location:any=pcall((userInput::UserInputService).GetMouseLocation,userInput)if ok and typeof(location)==
'Vector2'then return location::Vector2 end end local viewport:Vector2=cameraObject.ViewportSize return Vector2.new(viewport.X/2,viewport.Y/2)end local function targetParts(entity:any,requested:string?
):{BasePart}local character:Model=entity.Character local wanted:string=requested or'Closest'if wanted=='Head'then return{asBasePart(character:FindFirstChild('Head'))or entity.RootPart}end if wanted==
'Torso'then return{asBasePart(character:FindFirstChild('UpperTorso'))or asBasePart(character:FindFirstChild('Torso'))or entity.RootPart}end if wanted~='Closest'then return{entity.RootPart}end local
head:BasePart?=asBasePart(character:FindFirstChild('Head'))if head then return{head,entity.RootPart}end return{entity.RootPart}end local function protected(player:Player?):boolean if not player then
return false end local policy:any=services and services.protectedTargets if policy and type(policy.isProtected)=='function'then return policy.isProtected(player)==true end return false end
local function visibleFrom(cameraObject:Camera,entity:any,position:Vector3):boolean local origin:Vector3=cameraObject.CFrame.Position local parameters:RaycastParams=RaycastParams.new()parameters.
FilterType=Enum.RaycastFilterType.Exclude local ignored:{Instance}={entity.Character,cameraObject}if localPlayer and localPlayer.Character then table.insert(ignored,localPlayer.Character)end
parameters.FilterDescendantsInstances=ignored parameters.IgnoreWater=true local ok:boolean,result:any=pcall(currentWorkspace.Raycast,currentWorkspace,origin,position-origin,parameters)return ok and
result==nil end local function candidateForPart(cameraObject:Camera,pointer:Vector2,entity:any,part:BasePart,prediction:number,query:Query):Target?local velocity:Vector3=part.AssemblyLinearVelocity or
Vector3.zero local predicted:Vector3=part.Position+velocity*prediction local projected:Vector3,onScreen:boolean=cameraObject:WorldToViewportPoint(predicted)if(not onScreen or projected.Z<=0)and not
query.AllowOffscreen then return nil end local screenPosition:Vector2=Vector2.new(projected.X,projected.Y)local cursorDistance:number=onScreen and(screenPosition-pointer).Magnitude or math.huge if
query.FOV~=nil and cursorDistance>(query.FOV::number)then return nil end if query.VisibilityCheck==true and not visibleFrom(cameraObject,entity,predicted)then return nil end return{player=entity.
Player,entity=entity,character=entity.Character,part=part,position=predicted,screenPosition=screenPosition,distance=entity.Distance,cursorDistance=cursorDistance}end function library:List(entities:any
,query:Query?):{Target}local options:Query=query or{}local cameraObject:Camera?=camera()if not cameraObject or not entities then return{}end if type(entities.Refresh)=='function'then entities:Refresh(
)end local pointer:Vector2=cursor(cameraObject::Camera)local found:{Target}={}local maxDistance:number=options.MaxDistance or math.huge local prediction:number=math.max(options.Prediction or 0,0)
local function consider(entity:any):()if entity.Distance>maxDistance then return end if options.TeamCheck~=false then local hostile:boolean?=nil if entity.Player and type(entities.IsEnemy)=='function'
then hostile=entities:IsEnemy(entity.Player)end if hostile==false or hostile==nil and entity.IsFriendly then return end end if protected(entity.Player)then return end local bestForEntity:Target?=nil
for _,part:BasePart in ipairs(targetParts(entity,options.TargetPart))do local candidate:Target?=candidateForPart(cameraObject::Camera,pointer,entity,part,prediction,options)if candidate and(not
bestForEntity or candidate.cursorDistance<(bestForEntity::Target).cursorDistance)then bestForEntity=candidate end end if bestForEntity then table.insert(found,bestForEntity::Target)end end for _,
entity:any in ipairs(entities.List or{})do consider(entity)end if options.IncludeNPCs then for _,entity:any in ipairs(entities.NPCList or{})do consider(entity)end end local priority:string=options.
Priority or'Cursor'table.sort(found,function(first:Target,second:Target):boolean if priority=='Distance'then if first.distance==second.distance then return first.cursorDistance<second.cursorDistance
end return first.distance<second.distance end if first.cursorDistance==second.cursorDistance then return first.distance<second.distance end return first.cursorDistance<second.cursorDistance end)return
found end function library:Select(entities:any,query:Query?):Target?return self:List(entities,query)[1]end function library:Destroy():()if activeLibrary==library then activeLibrary=nil end end
activeLibrary=(library::any)::Library Module.Initialized=true return(library::any)::Library end function Module.destroy():()local library:Library?=activeLibrary if library then library:Destroy()end
activeLibrary=nil Module.Initialized=false end return Module
-- src/libraries/Weapons.lua
export type Candidate={id:string,label:string,kind:string,melee:boolean,instance:Instance}export type GameSource={scan:()->{{string}},press:(label:string,target:any?)->boolean}export type
WeaponLibrary={Scan:(self:WeaponLibrary,query:any?)->{Candidate},Labels:(self:WeaponLibrary,query:any?)->{string},Find:(self:WeaponLibrary,label:string)->Candidate?,IsMelee:(self:WeaponLibrary,name:
string)->boolean,Activate:(self:WeaponLibrary,candidate:Candidate,target:any?)->boolean,Best:(self:WeaponLibrary,query:any?)->Candidate?,Invalidate:(self:WeaponLibrary)->(),SetAllowEquip:(self:
WeaponLibrary,allowed:boolean)->(),Swing:(self:WeaponLibrary)->boolean,RegisterGameSource:(self:WeaponLibrary,source:GameSource?)->(),PressButton:(self:WeaponLibrary,button:GuiButton)->boolean,Destroy
:(self:WeaponLibrary)->()}local Module={Name='Weapons',PlaceId=0,Events={}::{[string]:any},Initialized=false}local WEAPON_WORDS:{string}={'sword','katana','dagger','blade','knife','axe','mace','spear'
,'scythe','sabre','saber','machete','hammer','club','bat','staff','pickaxe','fist','punch','attack','swing','hit','melee','weapon','combat'}local MELEE_WORDS:{string}={'sword','katana','dagger',
'blade','knife','axe','mace','spear','scythe','sickle','sabre','saber','rapier','cutlass','machete','glaive','halberd','hammer','club','baton','bat','pickaxe','fist','knuckle'}local RANGED_WORDS:{
string}={'bow','crossbow','arrow','gun','rifle','pistol','revolver','shotgun','sniper','launcher','rocket','grenade','snowball','projectile','throw','wand','staff','gauntlet','fireball'}local
RIG_WORDS:{string}={'hand','arm','leg','foot','torso','head','upper','lower','root','limb','joint','motor','weld','attachment','rig','bone','hitbox','collider','humanoid','camera','viewmodel','idle',
'walk','run','sprint','sneak','crouch','jump','fall','land','swim','pose','track','sequence'}local NOT_WEAPON_WORDS:{string}={'emote','animation','anim','dance','cosmetic','cape','skin','kit','shop',
'setting','gift','trade','coin','gem','wool','block','plank','brick','glass','ladder','wood ','sand','dirt','obsidian','potion','food','apple','bed','chest','music','sound','menu','close','back',
'shopkeeper'}local INVENTORY_WORDS:{string}={'hotbar','backpack','inventory','toolbar','slots','items'}local TOUCH_WORDS:{string}={'mobile','touch','buttons','controls'}local activeLibrary:
WeaponLibrary?=nil local function containsWord(text:string,words:{string}):boolean local lowered:string=string.lower(text)for _,word:string in ipairs(words)do if string.find(lowered,word,1,true)then
return true end end return false end local function isMeleeName(name:string):boolean if containsWord(name,NOT_WEAPON_WORDS)or containsWord(name,RANGED_WORDS)or containsWord(name,RIG_WORDS)then return
false end return containsWord(name,MELEE_WORDS)end local function ancestryMatches(instance:Instance,words:{string},depth:number):boolean local current:Instance?=instance local steps:number=0 while
current and steps<depth do if containsWord(current.Name,words)then return true end current=current.Parent steps+=1 end return false end function Module.init(context:any):WeaponLibrary local host:any=
context local localPlayer:Player=host.LocalPlayer local currentWorkspace:Workspace=host.workspace or workspace local menuGui:Instance?=host.ScreenGui local library:any={cache={},cachedAt=-1,cachedAll=
false,cachedMelee=false,allowEquip=false}function library:PressButton(button:GuiButton):boolean local environment:any=getfenv()local connectionsOf:any=environment.getconnections local fireSignal:any=
environment.firesignal local pressed:boolean=false if type(connectionsOf)=='function'then for _,signal:any in ipairs({button.MouseButton1Down,button.MouseButton1Click,(button::any).Activated,button.
MouseButton1Up})do local ok:boolean,connections:any=pcall(connectionsOf,signal)if not ok or type(connections)~='table'then continue end for _,connection:any in ipairs(connections)do local fired:
boolean=pcall(function():()connection:Fire()end)pressed=pressed or fired end end end if not pressed and type(fireSignal)=='function'then pressed=pcall(fireSignal,(button::any).Activated)==true end
return pressed end local function isMenuOwned(instance:Instance):boolean return menuGui~=nil and instance:IsDescendantOf(menuGui::Instance)end function library:Invalidate():()library.cachedAt=-1 end
function library:RegisterGameSource(source:any):()library.gameSource=source library.cachedAt=-1 end function library:SetAllowEquip(allowed:boolean):()library.allowEquip=allowed==true end function
library:Scan(query:any?):{Candidate}local options:any=query or{}local everything:boolean=options.All==true local now:number=os.clock()if now-library.cachedAt<1 and library.cachedAll==everything and
library.cachedMelee==(options.Melee==true)then return library.cache end local found:{Candidate}={}local seen:{[string]:boolean}={}local meleeOnly:boolean=options.Melee==true local function add(kind:
string,label:string,instance:Instance):()local id:string=kind..':'..label if seen[id]then return end local melee:boolean=isMeleeName(label)if meleeOnly and not melee then return end seen[id]=true
table.insert(found,{id=id,label=label,kind=kind,melee=melee,instance=instance})end if library.gameSource and type(library.gameSource.scan)=='function'then local ok:boolean,listed:any=pcall(library.
gameSource.scan)if ok and type(listed)=='table'then for _,entry:any in ipairs(listed)do if type(entry)=='table'and entry[1]and entry[2]then add('Game',tostring(entry[1]),entry[2])end end end end local
character:Model?=localPlayer.Character if character then for _,child:Instance in ipairs(character:GetChildren())do if child:IsA('Tool')then add('Tool',child.Name,child)end end end local backpack:
Instance?=localPlayer:FindFirstChild('Backpack')if backpack then for _,child:Instance in ipairs(backpack:GetChildren())do if child:IsA('Tool')then add('Tool',child.Name,child)end end end local
viewRoots:{Instance}={}local camera:Camera?=currentWorkspace.CurrentCamera if camera then table.insert(viewRoots,camera)end if character then table.insert(viewRoots,character)end for _,root:Instance
in ipairs(viewRoots)do for _,child:Instance in ipairs(root:GetChildren())do if not child:IsA('Model')then continue end if containsWord(child.Name,WEAPON_WORDS)and not containsWord(child.Name,
NOT_WEAPON_WORDS)then add('ViewModel',child.Name,child)end if string.find(string.lower(child.Name),'viewmodel',1,true)then for _,held:Instance in ipairs(child:GetChildren())do if not held:IsA('Model')
and not held:IsA('BasePart')then continue end if not isMeleeName(held.Name)then continue end add('ViewModel',held.Name,held)end end end end local playerGui:Instance?=localPlayer:FindFirstChild(
'PlayerGui')if playerGui then for _,descendant:Instance in ipairs(playerGui:GetDescendants())do if not descendant:IsA('GuiButton')then continue end if isMenuOwned(descendant)then continue end if
containsWord(descendant.Name,NOT_WEAPON_WORDS)then continue end local named:boolean=containsWord(descendant.Name,WEAPON_WORDS)if not named then if not everything then continue end if not
ancestryMatches(descendant,TOUCH_WORDS,4)then continue end add('Button',descendant.Name,descendant)continue end local inInventory:boolean=ancestryMatches(descendant,INVENTORY_WORDS,4)add(inInventory
and'Slot'or'Button',descendant.Name,descendant)end end library.cache=found library.cachedAt=now library.cachedAll=everything library.cachedMelee=meleeOnly return found end function library:Labels(
query:any?):{string}local labels:{string}={}for _,candidate:Candidate in ipairs(library:Scan(query))do table.insert(labels,candidate.label..'  \183  '..candidate.kind)end table.sort(labels)return
labels end function library:IsMelee(name:string):boolean return isMeleeName(name)end function library:Find(label:string):Candidate?local wanted:string=string.lower(label)for _,candidate:Candidate in
ipairs(library:Scan({All=true}))do if string.lower(candidate.label)==wanted then return candidate end if string.lower(candidate.label..'  \183  '..candidate.kind)==wanted then return candidate end end
return nil end function library:Best(query:any?):Candidate?local found:{Candidate}=library:Scan(query or{All=false})local order:{string}={'Game','Tool','Button','Slot','ViewModel'}local character:
Model?=localPlayer.Character for _,kind:string in ipairs(order)do for _,candidate:Candidate in ipairs(found)do if candidate.kind~=kind then continue end if kind=='Tool'and character and candidate.
instance.Parent~=character and not library.allowEquip then continue end return candidate end end return nil end function library:Swing():boolean local candidate:Candidate?=library:Best({Melee=true})or
library:Best()if not candidate then return false end return library:Activate(candidate::Candidate)end function library:Activate(candidate:Candidate,target:any?):boolean local instance:Instance=
candidate.instance if candidate.kind=='Game'then if not library.gameSource or type(library.gameSource.press)~='function'then return false end local ok:boolean,pressed:any=pcall(library.gameSource.
press,candidate.label,target)return ok and pressed==true end if not instance.Parent then return false end if candidate.kind=='Tool'then local tool:Tool=instance::Tool if library.allowEquip and tool.
Parent~=localPlayer.Character then local humanoid:Humanoid?=localPlayer.Character and localPlayer.Character:FindFirstChildOfClass('Humanoid')::Humanoid?if humanoid then pcall(function():()(humanoid::
Humanoid):EquipTool(tool)end)end end if tool.Parent~=localPlayer.Character or tool.Enabled==false then return false end return pcall(function():()tool:Activate()end)end if candidate.kind=='Button'or
candidate.kind=='Slot'then return library:PressButton(instance::GuiButton)end if candidate.kind=='ViewModel'then for _,other:Candidate in ipairs(library:Scan({All=true}))do if other.kind=='Button'then
return library:PressButton(other.instance::GuiButton)end end return false end return false end function library:Destroy():()if activeLibrary==library then activeLibrary=nil end end activeLibrary=(
library::any)::WeaponLibrary Module.Initialized=true return(library::any)::WeaponLibrary end function Module.destroy():()local library:WeaponLibrary?=activeLibrary if library then library:Destroy()end
activeLibrary=nil Module.Initialized=false end return Module
-- src/guis/Wurst/Code/Cards.lua
local Module={Name='Cards',PlaceId=0,Events={}::{[string]:any},Initialized=false}local moduleCleanup:()->()=function():()end function Module.init(context:any):any local host:any=context local state:
any=host.state local configData:any=host.configData local queueConfigSave:any=host.queueConfigSave local flushConfigSave:any=host.flushConfigSave or queueConfigSave local create:any=host.create local
makeButton:any=host.makeButton local makeTextLabel:any=host.makeTextLabel local notify:any=host.notify local Theme:any=host.Theme local CONTROL_FONT:any=host.CONTROL_FONT local trackUiConnection:any=
host.trackUiConnection local beginKeyCapture:any=host.beginKeyCapture local UserInputService:any=host.UserInputService local applyKeySlotStyle:any=host.applyKeySlotStyle local bindActionShortcut:any=
host.bindActionShortcut local getFeatureCardParent:any=host.getFeatureCardParent local FEATURE_CATEGORIES:any=host.FEATURE_CATEGORIES local DEFAULT_FEATURE_CATEGORY:any=host.DEFAULT_FEATURE_CATEGORY
local normalizeFeatureCategory:any=host.normalizeFeatureCategory local universalFeatures:any=host.universalFeatures local movementFeatures:any=host.movementFeatures local mm2Features:any=host.
mm2Features local allFeatures:any=host.allFeatures local registeredSearchFeatures:{any}={}local applyFeaturePresentation:any=host.applyFeaturePresentation local refreshFeatureToggle:any=host.
refreshFeatureToggle local addFeatureTooltip:any=host.addFeatureTooltip local drawPixelIcon:any=host.drawPixelIcon or function(parent:Instance,_kind:string,_color:Color3?):Frame local existing:
Instance?=(parent::any):FindFirstChild('PixelIcon')if existing then existing:Destroy()end return create('Frame',{Parent=parent,Name='PixelIcon',BackgroundTransparency=1})::Frame end local function
createUniversalFeature(name:string,description:string,layoutOrder:number,onToggle:any,configuration:any?):any configuration=configuration or{}local registry:any=configuration.registry or
universalFeatures local sectionName:string=configuration.section or(registry==mm2Features and'MM2')or(registry==state.trsFeatures and'TRS')or(registry==state.vdFeatures and'VD')or(registry==state.
bedFightFeatures and'BedFight')or(registry==state.bedWarsFeatures and'BedWars')or(registry==state.mvsdFeatures and'MVSD')or(registry==movementFeatures and'Movement')or'Universal'local configKey:string
=configuration.configKey or(sectionName..'.'..name:gsub('%W',''))local rawCategory:any=type(configuration.categoryName)=='string'and configuration.categoryName or FEATURE_CATEGORIES[name]or
DEFAULT_FEATURE_CATEGORY local categoryName:string=type(normalizeFeatureCategory)=='function'and normalizeFeatureCategory(rawCategory)or tostring(rawCategory)local feature:any={name=name,description=
description,categoryName=categoryName,enabled=false,expanded=false,kind=configuration.kind or(configuration.action==true and'action')or(configuration.holdAction==true and'hold')or(configuration.
category==true and'group')or'toggle',silentAction=configuration.silentAction==true,searchable=configuration.searchable~=false,onHold=configuration.onHold,optionCount=0,optionRows={},collapsedHeight=
state.layout.cardHeight,onToggle=onToggle,configKey=configKey,registry=registry,sectionName=sectionName,searchText=string.lower(name..' '..description..' '..sectionName..' '..categoryName),tooltipText
=description,tooltipBound=false}feature.isAction=feature.kind=='action'feature.isCategory=feature.kind=='group'feature.isHoldAction=feature.kind=='hold'feature.isAlwaysOn=feature.kind=='hold'or
feature.kind=='group'local requestedParent:Instance?=configuration.parent::any local fallback:Instance=(host::any).ScreenGui or(host::any).PopupLayer or({}::any)local rowParent:Instance=
getFeatureCardParent(requestedParent or fallback)if type(rowParent)~='table'or(rowParent::any).Children==nil then rowParent=fallback end local row:Frame=(create('Frame',{Parent=rowParent,Name=name:
gsub('%W','')..'Feature',Active=true,BackgroundColor3=state.guiStyle=='Default'and Theme.surface or Theme.background,BackgroundTransparency=state.guiStyle=='Default'and 0.82 or 1,BorderSizePixel=0,
ClipsDescendants=true,LayoutOrder=layoutOrder,Size=UDim2.new(1,0,0,feature.collapsedHeight)})::any)::Frame feature.row=row row:SetAttribute('FeatureCategory',categoryName)row:SetAttribute(
'FeatureSortName',string.lower(name))local bitmapText:any=state.bitmapText local title:TextLabel?=nil if bitmapText and type(bitmapText.draw)=='function'then feature.drawTitle=function(color:Color3?,
transparency:number?):()if color~=nil then feature.titleColor=color end if transparency~=nil then feature.titleTransparency=transparency end bitmapText.draw(row,{{text=feature.name,color=feature.
titleColor or Theme.text}},{Name='Title',TextSize=state.layout.titleFontSize,ZIndex=7,align='Center',transparency=feature.titleTransparency or 0,maxWidth=140})end feature.drawTitle(Theme.text,0)else
title=(makeTextLabel(row,name,state.layout.titleFontSize)::any)::TextLabel local label:TextLabel=title::TextLabel label.FontFace=CONTROL_FONT label.Position=UDim2.fromOffset(28,0)label.Size=UDim2.new(
1,-56,0,feature.collapsedHeight)label.TextXAlignment=Enum.TextXAlignment.Center label.TextTruncate=Enum.TextTruncate.AtEnd label.TextColor3=state.guiStyle=='Default'and Theme.text or Theme.text
feature.title=label end local arrow:TextButton?=nil local toggleExpansion:()->()=function():()end function feature.hasOptions():boolean if configuration.noOptions==true then return false end for _,
optionRow:GuiObject in ipairs(feature.optionRows)do if optionRow.Visible then return true end end return false end local function syncArrow():()local expandable:boolean=feature.hasOptions()feature.
hasSettings=expandable if expandable and not arrow then local built:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,Name='Expand',Active=true,AnchorPoint=Vector2.new(1
,0),AutoButtonColor=false,BackgroundTransparency=1,BorderSizePixel=0,FontFace=CONTROL_FONT,Position=UDim2.new(1,0,0,0),Size=UDim2.fromOffset(22,feature.collapsedHeight),Text='',TextSize=10,ZIndex=9})
::any)::TextButton trackUiConnection(built.MouseButton1Click:Connect(function():()toggleExpansion()end))arrow=built feature.arrow=built end if arrow then(arrow::TextButton).Visible=expandable
drawPixelIcon(arrow::TextButton,feature.expanded and'up'or'down',nil,'arrows')end end feature.syncArrow=syncArrow if state and type(state.registerThemeRefresh)=='function'then state.
registerThemeRefresh(function():()if row.Parent and arrow then syncArrow()end end,{'arrows'})end local activationTarget:TextButton=create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,
ClipsDescendants=true,Parent=row,Name='ActivationTarget',Active=true,AutoButtonColor=false,BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.fromOffset(0,0),Size=UDim2.new(1,0,0,feature.
collapsedHeight),Text='',ZIndex=6})::TextButton feature.activationTarget=activationTarget if title then(title::TextLabel).ZIndex=7 end trackUiConnection(row.MouseEnter:Connect(function():()state.
applyCardSkin(feature,true)end))trackUiConnection(row.MouseLeave:Connect(function():()state.applyCardSkin(feature,false)end))local options:Frame=(create('Frame',{Parent=row,Name='Options',Active=true,
BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.fromOffset(4,feature.collapsedHeight),Size=UDim2.new(1,-8,0,0),Visible=false,ZIndex=row.ZIndex+10})::any)::Frame feature.options=options
create('UIPadding',{Parent=options,PaddingBottom=UDim.new(0,4),PaddingLeft=UDim.new(0,0),PaddingRight=UDim.new(0,0),PaddingTop=UDim.new(0,4)})local optionsLayout:UIListLayout=(create('UIListLayout',{
Parent=options,FillDirection=Enum.FillDirection.Vertical,HorizontalAlignment=Enum.HorizontalAlignment.Center,Padding=UDim.new(0,4),SortOrder=Enum.SortOrder.LayoutOrder})::any)::UIListLayout
local function updateExpandedSize(_animate:boolean?):()local visibleOptions:number=0 local declaredHeight:number=0 for _,optionRow:GuiObject in ipairs(feature.optionRows)do if optionRow.Visible then
visibleOptions+=1 declaredHeight+=optionRow.Size.Y.Offset end end local expectedHeight:number=if visibleOptions>0 then declaredHeight+math.max(0,visibleOptions-1)*4 else 0 local measuredHeight:number=
math.max(optionsLayout.AbsoluteContentSize.Y,expectedHeight)local singleOptionExtra:number=if visibleOptions==1 then 8 else 0 local optionsHeight:number=measuredHeight+8+singleOptionExtra feature.
optionsHeight=optionsHeight options.Size=UDim2.new(1,0,0,optionsHeight)syncArrow()local settingsWindow:any=feature.settingsWindow if settingsWindow and settingsWindow.resizeToOptions then
settingsWindow.resizeToOptions()end end feature.updateExpandedSize=updateExpandedSize trackUiConnection(optionsLayout:GetPropertyChangedSignal('AbsoluteContentSize'):Connect(function():()
updateExpandedSize(false)end))toggleExpansion=function():()if not feature.hasOptions()then syncArrow()return end local windows:any=state.windows if windows and type(windows.OpenFeatureSettings)==
'function'then local window:any=feature.settingsWindow if window and window.root and window.root.Visible==true then window.root.Visible=false feature.expanded=false if feature.syncArrow then feature.
syncArrow()end return end windows.OpenFeatureSettings(feature)end end feature.toggleExpansion=toggleExpansion function feature.setHeld(held:boolean):()if feature.held==held then return end feature.
held=held local handler:any=feature.onHold if type(handler)~='function'then return end local ok:boolean,failure:any=pcall(handler,held)if not ok then warn('[Random Testing Menu] '..name..': '..
tostring(failure))return end feature.active=held refreshFeatureToggle(feature)end local activationGeneration:number=0 local function paintToggle():()local ok:boolean,failure:any=pcall(
refreshFeatureToggle,feature)if not ok then warn('[Random Testing Menu] '..name..' UI: '..tostring(failure))end end local function activateFeature():()if feature.isHoldAction then return end if
feature.isCategory then if not feature.hasOptions()then return end feature.expanded=not feature.expanded updateExpandedSize(true)return end if feature.isAction then if feature.actionRunning then
return end feature.actionRunning=true task.spawn(function():()local success:boolean,errorMessage:any=pcall(feature.onToggle)feature.actionRunning=false if not success then warn(
'[Random Testing Menu] '..name..': '..tostring(errorMessage))notify('Could not run '..name)end end)return end local requestedState:boolean=not feature.enabled activationGeneration+=1 local generation:
number=activationGeneration feature.enabled=requestedState if requestedState then feature.enabledAt=os.clock()end paintToggle()local success:boolean,errorMessage:any=pcall(feature.onToggle,
requestedState)if generation~=activationGeneration or feature.enabled~=requestedState then return end if success then configData.states[feature.configKey]=requestedState flushConfigSave()else warn(
'[Random Testing Menu] '..name..': '..tostring(errorMessage))feature.enabled=false paintToggle()pcall(feature.onToggle,false)if generation~=activationGeneration then return end local reason:string=
tostring(errorMessage)if#reason>90 then reason=string.sub(reason,1,90)end notify('Could not toggle '..name..': '..reason)end end feature.activate=activateFeature state.applyCardSkin(feature,false)if
feature.isAction then trackUiConnection(activationTarget.MouseButton1Down:Connect(function():()if feature.row then(feature.row::Frame).BackgroundTransparency=0.46 end end))trackUiConnection(
activationTarget.MouseButton1Up:Connect(function():()state.applyCardSkin(feature,true)end))end trackUiConnection(activationTarget.MouseButton1Click:Connect(activateFeature))trackUiConnection(
activationTarget.InputBegan:Connect(function(input:InputObject):()if input.UserInputType==Enum.UserInputType.MouseButton2 then toggleExpansion()end end))local shortcutButton:TextButton?=nil if state.
isMobile then local slot:TextButton=(makeButton(row,'KEY',9)::any)::TextButton slot.Name='Shortcut'slot.AnchorPoint=Vector2.new(0,0.5)slot.Position=UDim2.new(0,4,0,feature.collapsedHeight/2)slot.Size=
UDim2.fromOffset(22,22)slot.ZIndex=9 applyKeySlotStyle(slot)slot.TextScaled=true create('UITextSizeConstraint',{Parent=slot,MaxTextSize=11,MinTextSize=7})shortcutButton=slot end feature.shortcutButton
=shortcutButton if feature.kind=='group'then if shortcutButton then(shortcutButton::TextButton).Visible=false end elseif feature.kind=='hold'then bindActionShortcut(shortcutButton,feature.configKey,
name,function():()end,{hold=true,onPress=function():()feature.setHeld(true)end,onRelease=function():()feature.setHeld(false)end})else bindActionShortcut(shortcutButton,feature.configKey,name,
activateFeature,{feedback=function():string return name..(feature.enabled==true and' on'or' off')end})end if state.isMobile and shortcutButton and state.mobileUi then local slot:TextButton=
shortcutButton::TextButton local mobileUi:any=state.mobileUi local actionId:string=mobileUi.actionId(feature.configKey)local function syncMobileSlot():()local action:any=mobileUi.actions[actionId]
local corner:UICorner?=slot:FindFirstChildOfClass('UICorner')if action then slot.Text=''slot.BackgroundColor3=mobileUi.colorFromHex(action.color)or Theme.surfaceRaised slot.BackgroundTransparency=0.12
if corner then corner.CornerRadius=action.shape=='Circle'and UDim.new(1,0)or action.shape=='Rounded'and UDim.new(0,10)or UDim.new(0,2)end else slot.Text='+'slot.BackgroundColor3=Theme.surfaceRaised
slot.BackgroundTransparency=0.35 if corner then corner.CornerRadius=UDim.new(0,6)end end end syncMobileSlot()if type(mobileUi.onChanged)=='function'then mobileUi.onChanged(syncMobileSlot)end local
binding:any=state.shortcutBindings and state.shortcutBindings[feature.configKey]if binding then local previous:any=binding.refresh binding.refresh=function():()if type(previous)=='function'then
previous()end syncMobileSlot()end end end if(feature.kind=='toggle'or feature.kind=='action'or feature.kind=='hold')and not state.isMobile and type(beginKeyCapture)=='function'and type(state.
shortcutBindings)=='table'and state.shortcutBindings[feature.configKey]~=nil then local binding:any=state.shortcutBindings[feature.configKey]local UNBOUND_BG:Color3=Color3.fromRGB(0,0,0)local square:
TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,Name='BindSquare',Active=true,AnchorPoint=Vector2.new(0,0.5),AutoButtonColor=false,BackgroundColor3=UNBOUND_BG,
BackgroundTransparency=0.55,BorderSizePixel=0,FontFace=CONTROL_FONT,Position=UDim2.new(0,4,0.5,0),Size=UDim2.fromOffset(18,18),Text='',TextColor3=Theme.text,TextTransparency=0,TextSize=8,ZIndex=8})::
any)::TextButton square:SetAttribute('CaptureIdleText','')local edge:UIStroke=(create('UIStroke',{Parent=square,Name='StyleStroke',Color=Theme.outline,Thickness=1,Transparency=0.6})::any)::UIStroke
```

```
local COMPACT_KEY_NAMES:{[string]:string}={LeftShift='LS',RightShift='RS',LeftControl='LC',RightControl='RC',LeftAlt='LA',RightAlt='RA',CapsLock='CL',Backspace='Bk',Return='En',Escape='Es',Space='Sp',
PageUp='PU',PageDown='PD',Insert='Ins',Delete='Del',Home='Hm',End='End'}local function keyCaption():string if binding.key==nil or binding.key==Enum.KeyCode.Unknown then return''end local name:string=
tostring(binding.key.Name or'')local compact:string?=COMPACT_KEY_NAMES[name]if compact then return compact end local display:string=state.keyDisplayName and tostring(state.keyDisplayName(name))or name
return string.sub(display,1,3)end local capturing:boolean=false local function conflictWith():any if binding.key==nil or binding.key==Enum.KeyCode.Unknown then return nil end for _,other:any in pairs(
state.shortcutBindings)do if other~=binding and other.key==binding.key then return other end end return nil end local function syncSquare():()if capturing then return end local caption:string=
keyCaption()square.Text=caption square:SetAttribute('CaptureIdleText',caption)local bound:boolean=binding.key~=nil and binding.key~=Enum.KeyCode.Unknown local conflict:any=bound and conflictWith()or
nil if conflict then square.BackgroundColor3=Theme.negative square.BackgroundTransparency=0.35 edge.Color=Theme.negative edge.Transparency=0.15 elseif bound and feature.enabled==true then square.
BackgroundColor3=Theme.enabled square.BackgroundTransparency=0.35 edge.Color=Theme.enabled edge.Transparency=0.25 else square.BackgroundColor3=UNBOUND_BG square.BackgroundTransparency=0.55 edge.Color=
Theme.outline edge.Transparency=if bound then 0.25 else 0.6 end end feature.bindSquare=square feature.syncBindSquare=syncSquare if state.registerThemeRefresh then state.registerThemeRefresh(function()
:()if square.Parent then syncSquare()end end,{'enabled','outline'})end local refreshBinding:any=binding.refresh binding.refresh=function():()if refreshBinding then refreshBinding()end syncSquare()end
syncSquare()local function syncKeyboard():()square.Visible=UserInputService==nil or UserInputService.KeyboardEnabled~=false end syncKeyboard()pcall(function():()trackUiConnection(UserInputService:
GetPropertyChangedSignal('KeyboardEnabled'):Connect(syncKeyboard))end)local function tipText():string local bound:boolean=binding.key~=nil and binding.key~=Enum.KeyCode.Unknown if not bound then
return'Bind: click, then press a key'end local conflict:any=conflictWith()if conflict then return binding.key.Name..' \u{2014} also bound to '..tostring(conflict.id)end return'Bound: '..binding.key.
Name end trackUiConnection(square.MouseEnter:Connect(function():()local tooltip:any=state.featureTooltip if not tooltip then return end state.tooltipToken=(state.tooltipToken or 0)+1 local token:
number=state.tooltipToken task.delay(0.4,function():()if state.tooltipToken~=token or not square.Parent then return end local rawText:string=tipText()local bitmap:any=state.bitmapText local measured:
number=bitmap and type(bitmap.measure)=='function'and tonumber(bitmap.measure({{text=rawText}},16))or#rawText*8 local width:number=math.clamp(measured+24,140,360)local wrapped:string=rawText local
lines:number=1 if bitmap and type(bitmap.wrap)=='function'then wrapped,lines=bitmap.wrap(rawText,width-24,16)end tooltip.Text=wrapped tooltip.TextTransparency=1 tooltip.Size=UDim2.fromOffset(width,
math.max(30,lines*18+12))tooltip.Position=UDim2.fromOffset(square.AbsolutePosition.X,square.AbsolutePosition.Y+square.AbsoluteSize.Y+8)if state.bitmapText and type(state.bitmapText.draw)=='function'
then state.bitmapText.draw(tooltip,{{text=tooltip.Text}},{Name='Glyphs',TextSize=16,maxWidth=width-24,ellipsis=true,ZIndex=251})end tooltip.Visible=true end)end))trackUiConnection(square.MouseLeave:
Connect(function():()state.tooltipToken=(state.tooltipToken or 0)+1 if state.featureTooltip then state.featureTooltip.Visible=false end end))local outsideConnection:RBXScriptConnection?=nil
local function cleanupCapture():()capturing=false if outsideConnection then pcall(function():()(outsideConnection::RBXScriptConnection):Disconnect()end)outsideConnection=nil end syncSquare()task.
delay(0.2,function():()if square.Parent and not capturing then syncSquare()end end)end trackUiConnection(square.MouseButton1Click:Connect(function():()if capturing then if state.keyCaptureCancel then
state.keyCaptureCancel()else cleanupCapture()end return end capturing=true beginKeyCapture(square,function(keyCode:Enum.KeyCode):()local backspaceKey:any=(Enum.KeyCode::any).Backspace local deleteKey:
any=(Enum.KeyCode::any).Delete if keyCode==backspaceKey or keyCode==deleteKey then if binding.key~=Enum.KeyCode.Unknown and type(binding.assign)=='function'then binding.assign(binding.key)end elseif
type(binding.assign)=='function'then binding.assign(keyCode)end cleanupCapture()end)local baseCancel:any=state.keyCaptureCancel state.keyCaptureCancel=function():()if baseCancel then baseCancel()end
cleanupCapture()end pcall(function():()outsideConnection=UserInputService.InputBegan:Connect(function(input:InputObject):()if input.UserInputType~=Enum.UserInputType.MouseButton1 and input.
UserInputType~=Enum.UserInputType.Touch then return end local px:number=input.Position.X local py:number=input.Position.Y local ax:number=square.AbsolutePosition.X local ay:number=square.
AbsolutePosition.Y if px<ax or px>ax+square.AbsoluteSize.X or py<ay or py>ay+square.AbsoluteSize.Y then if state.keyCaptureCancel then state.keyCaptureCancel()end end end)end))end feature.
applyLayout=function():()local metrics:LayoutMetrics=state.layout feature.collapsedHeight=metrics.cardHeight if title then state.setTextMetric(title,metrics.titleFontSize)elseif feature.drawTitle then
feature.drawTitle(nil,nil)end if arrow then(arrow::TextButton).Size=UDim2.fromOffset(22,metrics.cardHeight)end if shortcutButton then(shortcutButton::TextButton).Size=UDim2.fromOffset(22,22);(
shortcutButton::TextButton).Position=UDim2.new(0,4,0,metrics.cardHeight/2)end activationTarget.Size=UDim2.new(1,0,0,metrics.cardHeight)options.Position=UDim2.fromOffset(4,metrics.cardHeight)syncArrow(
)updateExpandedSize(false)end feature.applyLayout()function feature:SetStatus(status:string?):()feature.status=status end function feature:SetName(nextName:string):()if type(nextName)~='string'or
nextName==''then return end local previous:string=feature.name if previous~=nextName and registry[previous]==feature then registry[previous]=nil registry[nextName]=feature end feature.name=nextName
feature.searchText=string.lower(nextName..' '..(feature.description or'')..' '..sectionName..' '..categoryName)row:SetAttribute('FeatureSortName',string.lower(nextName))if feature.title then feature.
title.Text=nextName end if feature.drawTitle then feature.drawTitle()end local search:any=state.moduleSearch if search and type(search.Unregister)=='function'and type(search.Register)=='function'then
search.Unregister(feature)search.Register(feature)end if state.clickGui and type(state.clickGui.Resort)=='function'then state.clickGui.Resort(feature)end if state.floatingUi and type(state.floatingUi.
registerFeature)=='function'then state.floatingUi.registerFeature(feature)end end registry[name]=feature table.insert(allFeatures,feature)if state.floatingUi then state.floatingUi.registerFeature(
feature)end if state.clickGui then state.clickGui.place(feature)end applyFeaturePresentation(feature,state.guiStyle=='Default')if type(addFeatureTooltip)=='function'and description~=''then
addFeatureTooltip(feature,description)end local search:any=state.moduleSearch if search and type(search.Register)=='function'then search.Register(feature)table.insert(registeredSearchFeatures,feature)
end if feature.isAlwaysOn then task.defer(function():()local ok:boolean,failure:any=pcall(feature.onToggle,true)if not ok then warn('[Random Testing Menu] '..name..': '..tostring(failure))end end)end
if not feature.isAction and not feature.isCategory and not feature.isHoldAction and configuration.restore~=false and configData.states[feature.configKey]==true then task.defer(function():()if
activationGeneration~=0 or configData.states[feature.configKey]~=true then return end activationGeneration+=1 local generation:number=activationGeneration feature.enabled=true feature.enabledAt=os.
clock()paintToggle()local success:boolean,errorMessage:any=pcall(feature.onToggle,true)if generation~=activationGeneration or feature.enabled~=true then return end if not success then feature.enabled=
false paintToggle()configData.states[feature.configKey]=false queueConfigSave()warn('[Random Testing Menu] Could not restore '..name..': '..tostring(errorMessage))end end)end return feature end
moduleCleanup=function():()local search:any=state.moduleSearch if search and type(search.Unregister)=='function'then for _,feature:any in ipairs(registeredSearchFeatures)do search.Unregister(feature)
end end table.clear(registeredSearchFeatures)end Module.Initialized=true return{createUniversalFeature=createUniversalFeature}end function Module.destroy():()moduleCleanup()moduleCleanup=function():()
end Module.Initialized=false end return Module
-- src/guis/Wurst/Code/ClickGui.lua
export type CategoryWindow={name:string,window:any,layout:UIListLayout,count:number}local Module={Name='ClickGui',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=
nil local WINDOW_WIDTH:number=200 local WINDOW_MAX_HEIGHT:number=300 local SCALE_MINIMUM:number=0.7 local SCALE_MAXIMUM:number=1.6 local CALIBRATION_WIDTH:number=1600 local CALIBRATION_HEIGHT:number=
900 local GRID_MARGIN:number=10 local WINDOW_GAP:number=2 local GRID_GAP:number=10 local TITLE_HEIGHT:number=26 local DEFAULT_CATEGORY_ORDER:{string}={'Exploits','Utilities','World','Render','Blocks',
'Items'}local categories:any={order=DEFAULT_CATEGORY_ORDER,aliases={Combat='Exploits'},default='Utilities'}local function useCatalog(hostState:any):()local catalog:any=hostState and hostState.
categories if type(catalog)=='table'and type(catalog.order)=='table'and#catalog.order>0 then categories=catalog end end local function order():{string}return categories.order or DEFAULT_CATEGORY_ORDER
end local function fallbackNormalize(raw:any):string local name:string=tostring(raw or categories.default or'Utilities')if table.find(order(),name)~=nil then return name end local aliases:any=
categories.aliases if type(aliases)=='table'and type(aliases[name])=='string'then return aliases[name]end return categories.default or'Utilities'end function Module.init(context:any):any local host:
any=context local state:any=host.state useCatalog(state)local create:any=host.create local allFeatures:{any}=host.allFeatures local normalizeFeatureCategory:any=host.normalizeFeatureCategory
local function viewportSize():Vector2 local saved:any=state.layoutViewport if saved and saved.X>1 and saved.Y>1 then return saved end local camera:Camera?=workspace.CurrentCamera local current:any=
camera and camera.ViewportSize if current and current.X>1 and current.Y>1 then return current end return Vector2.new(CALIBRATION_WIDTH,CALIBRATION_HEIGHT)end local function viewportScale():number
local shape:any=host.ThemeEngine and host.ThemeEngine.shape local wanted:number=shape and tonumber(shape.scale)or 1 local viewport:Vector2=viewportSize()local fit:number=math.clamp(math.min(viewport.X
/CALIBRATION_WIDTH,viewport.Y/CALIBRATION_HEIGHT),SCALE_MINIMUM,SCALE_MAXIMUM)if wanted~=wanted then wanted=1 end return fit*math.clamp(wanted,0.7,1.3)end local function logicalViewport():Vector2
local viewport:Vector2=viewportSize()local factor:number=viewportScale()return Vector2.new(viewport.X/factor,viewport.Y/factor)end local windows:any=state.windows if not windows then error(
'ClickGui requires the window manager')end local ClickGui:any={categories={}::{[string]:CategoryWindow},order={}::{string},connections={}::{RBXScriptConnection}}local productVersion:string=tostring((
host.PRODUCT and host.PRODUCT.version)or((host.BUILD_LABEL or'0.0')..' Release'))local LOGO_CHIP_WIDTH:number=142+10+(8+#productVersion*15)+4 local LOGO_CHIP_HEIGHT:number=44 local TOP_ROW_X:number=14
+LOGO_CHIP_WIDTH+GRID_GAP local TOP_ROW_Y:number=10 local nextX:number=TOP_ROW_X local nextY:number=TOP_ROW_Y local rowHeight:number=LOGO_CHIP_HEIGHT local function nextSlot(height:number):Vector2
local viewport:Vector2=logicalViewport()if nextX+WINDOW_WIDTH>viewport.X-GRID_MARGIN then nextX=GRID_MARGIN nextY+=rowHeight+GRID_GAP rowHeight=0 end local slot:Vector2=Vector2.new(nextX,nextY)nextX+=
WINDOW_WIDTH+GRID_GAP rowHeight=math.max(rowHeight,height)return slot end local function categoryWindow(name:string):CategoryWindow local existing:CategoryWindow?=ClickGui.categories[name]if existing
then return existing::CategoryWindow end local height:number=TITLE_HEIGHT+8 local window:any=windows.Create({id='ClickGui_'..name,title=name,size=Vector2.new(WINDOW_WIDTH,height),position=nextSlot(
WINDOW_MAX_HEIGHT*0.5),maxHeight=WINDOW_MAX_HEIGHT,closable=false})local layout:UIListLayout=create('UIListLayout',{Parent=window.body,Padding=UDim.new(0,WINDOW_GAP),SortOrder=Enum.SortOrder.
LayoutOrder})::UIListLayout create('UIPadding',{Parent=window.body,PaddingTop=UDim.new(0,WINDOW_GAP),PaddingBottom=UDim.new(0,WINDOW_GAP),PaddingLeft=UDim.new(0,WINDOW_GAP),PaddingRight=UDim.new(0,
WINDOW_GAP)})local record:CategoryWindow={name=name,window=window,layout=layout,count=0}ClickGui.categories[name]=record table.insert(ClickGui.order,name)table.sort(ClickGui.order,function(left:string
,right:string):boolean local leftRank:number=table.find(order(),left)or math.huge local rightRank:number=table.find(order(),right)or math.huge if leftRank==rightRank then return left<right end return
leftRank<rightRank end)local function resize():()local limit:number=tonumber(state.uiMaxHeight)or WINDOW_MAX_HEIGHT if limit<=0 then limit=math.huge end limit=math.min(limit,logicalViewport().Y-12)
local declaredHeight:number=0 local visibleRows:number=0 for _,child:Instance in ipairs(window.body:GetChildren())do if child:IsA('GuiObject')and(child::GuiObject).Visible then visibleRows+=1
declaredHeight+=(child::GuiObject).Size.Y.Offset end end declaredHeight+=math.max(visibleRows-1,0)*WINDOW_GAP local contentHeight:number=math.max(layout.AbsoluteContentSize.Y,declaredHeight)local
wanted:number=math.min(TITLE_HEIGHT+contentHeight+4,limit)window.fullHeight=wanted if not window.collapsed then window.root.Size=UDim2.fromOffset(WINDOW_WIDTH,wanted)end if windows.ClampWindow then
windows.ClampWindow(window)end end table.insert(ClickGui.connections,layout:GetPropertyChangedSignal('AbsoluteContentSize'):Connect(resize))resize()record.resize=resize::any return record end
local function sortCategoryRows(record:CategoryWindow):()local rows:{GuiObject}={}for _,child:Instance in ipairs(record.window.body:GetChildren())do if not child:IsA('GuiObject')then continue end
local key:any=child:GetAttribute('FeatureSortName')if type(key)=='string'then table.insert(rows,child::GuiObject)end end table.sort(rows,function(left:GuiObject,right:GuiObject):boolean local leftKey:
string=tostring(left:GetAttribute('FeatureSortName')or'')local rightKey:string=tostring(right:GetAttribute('FeatureSortName')or'')if leftKey~=rightKey then return leftKey<rightKey end return tostring(
left.Name)<tostring(right.Name)end)for index:number,row:GuiObject in ipairs(rows)do row.LayoutOrder=index end end function ClickGui.Resort(feature:any):()local record:CategoryWindow?=feature and
ClickGui.categories[feature.categoryName]if record then sortCategoryRows(record::CategoryWindow)end end function ClickGui.place(feature:any):()local row:Frame?=feature and feature.row if not row then
return end local raw:any=(row::Frame):GetAttribute('FeatureCategory')local name:string=type(normalizeFeatureCategory)=='function'and normalizeFeatureCategory(raw)or fallbackNormalize(raw)local record:
CategoryWindow=categoryWindow(name)record.count+=1(row::Frame).Parent=record.window.body;(row::Frame).Size=UDim2.new(1,0,0,feature.collapsedHeight or 22)sortCategoryRows(record)if record.resize then(
record.resize::()->())()end local userMoved:boolean=false for _,category:string in ipairs(ClickGui.order)do local candidate:CategoryWindow?=ClickGui.categories[category]if candidate and candidate.
window.userMoved==true then userMoved=true break end end if not userMoved and type(ClickGui.Retile)=='function'then ClickGui.Retile()end end function ClickGui.Retile():()nextX=TOP_ROW_X nextY=
TOP_ROW_Y rowHeight=LOGO_CHIP_HEIGHT for _,name:string in ipairs(ClickGui.order)do local record:CategoryWindow=ClickGui.categories[name]if record then local slot:Vector2=nextSlot(record.window.
fullHeight or WINDOW_MAX_HEIGHT*0.5)record.window.root.Position=UDim2.fromOffset(slot.X,slot.Y)if type(windows.SetHome)=='function'then windows.SetHome(record.window,slot)end if type(windows.
ClampWindow)=='function'then windows.ClampWindow(record.window)end end end end function ClickGui.RefreshHeights():()for _,name:string in ipairs(ClickGui.order)do local record:CategoryWindow=ClickGui.
categories[name]if record and record.resize then(record.resize::()->())()end end end function ClickGui.SetVisible(visible:boolean):()for _,name:string in ipairs(ClickGui.order)do local record:
CategoryWindow=ClickGui.categories[name]if not visible and record.window.pinned then continue end record.window:SetVisible(visible and record.window.userVisible~=false)end end local function
resolvedCategory(feature:any):string local row:any=feature and feature.row local raw:any=row and row:GetAttribute('FeatureCategory')if type(normalizeFeatureCategory)=='function'then return
normalizeFeatureCategory(raw)end return fallbackNormalize(raw)end for _,name:string in ipairs(order())do for _,feature:any in ipairs(allFeatures)do if resolvedCategory(feature)==name then
categoryWindow(name)break end end end for _,feature:any in ipairs(allFeatures)do ClickGui.place(feature)end local scale:UIScale=create('UIScale',{Parent=host.PopupLayer,Name='ClickGuiScale',Scale=
viewportScale()})::UIScale state.onLayout(function():()scale.Scale=viewportScale()ClickGui.RefreshHeights()local userMoved:boolean=false for _,name:string in ipairs(ClickGui.order)do local record:
CategoryWindow?=ClickGui.categories[name]if record and record.window.userMoved==true then userMoved=true break end end if not userMoved then ClickGui.Retile()end if type(windows.Reclamp)=='function'
then windows.Reclamp()end end)ClickGui.scale=scale ClickGui.ApplyScale=function():()if ClickGui.scale then ClickGui.scale.Scale=viewportScale()end end state.clickGui=ClickGui state.
addMenuVisibilityListener(function(visible:boolean):()ClickGui.SetVisible(visible)end)ClickGui.SetVisible(state.visible==true)activeCleanup=function():()for _,name:string in ipairs(ClickGui.order)do
local record:CategoryWindow?=ClickGui.categories[name]if record then pcall(function():()(record::CategoryWindow).window:Destroy()end)end end for _,connection:RBXScriptConnection in ipairs(ClickGui.
connections)do pcall(function():()connection:Disconnect()end)end ClickGui.connections={}ClickGui.categories={}ClickGui.order={}state.clickGui=nil end Module.Initialized=true return ClickGui end
function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.Initialized=false end return Module
-- src/guis/Wurst/Code/FloatingWindows.lua
export type FloatingWindowRecord={id:string,root:Frame,body:Frame,closeButton:TextButton}local activeCleanup:(()->())?=nil local Module={Name='FloatingWindows',PlaceId=0,Events={}::{[string]:any},
Initialized=false}function Module.init(context:any):any local host:any=context local state:any=host.state local configData:any=host.configData local queueConfigSave:any=host.queueConfigSave local
create:any=host.create local makeButton:any=host.makeButton local makeTextBox:any=host.makeTextBox local makeTextLabel:any=host.makeTextLabel local Theme:any=host.Theme local ThemeEngine:any=host.
ThemeEngine local CONTROL_FONT:any=host.CONTROL_FONT local THIN_FONT:any=host.THIN_FONT local VALVE_FONT:any=host.VALVE_FONT local PopupLayer:any=host.PopupLayer local UI_RADIUS:any=host.UI_RADIUS
local cornerRadius:any=host.cornerRadius local trackUiConnection:any=host.trackUiConnection local createToggleSwitch:any=host.createToggleSwitch local setToggleSwitch:any=host.setToggleSwitch local
TweenService:any=host.TweenService local UserInputService:any=host.UserInputService local Players:any=host.Players local LocalPlayer:any=host.LocalPlayer local Stats:any=host.Stats local TaskManager:
any=host.TaskManager local allFeatures:any=host.allFeatures local function bind(instance:Instance,property:string,token:string):()if ThemeEngine and type(ThemeEngine.Bind)=='function'then ThemeEngine.
Bind(instance,property,token)else(instance::any)[property]=Theme[token]end end local ownedTasks:{any}={}local FloatingUi:any={favorites={},favoriteRows={},overlays={},featureRows={},windows={}}state.
floatingUi=FloatingUi if type(configData.ui.favorites)~='table'then configData.ui.favorites={}end if type(configData.ui.overlays)~='table'then configData.ui.overlays={}end if type(configData.ui.
floatingPositions)~='table'then configData.ui.floatingPositions={}end if type(configData.ui.overlaySettings)~='table'then configData.ui.overlaySettings={}end local overlaySettings:any=configData.ui.
overlaySettings overlaySettings.TextGUI=type(overlaySettings.TextGUI)=='table'and overlaySettings.TextGUI or{}overlaySettings.SessionInfo=type(overlaySettings.SessionInfo)=='table'and overlaySettings.
SessionInfo or{}overlaySettings.TargetInfo=type(overlaySettings.TargetInfo)=='table'and overlaySettings.TargetInfo or{}overlaySettings.Radar=type(overlaySettings.Radar)=='table'and overlaySettings.
Radar or{}local textGuiSettings:any=overlaySettings.TextGUI textGuiSettings.sort=textGuiSettings.sort or'Alphabetical'textGuiSettings.font=textGuiSettings.font or'Builder Sans'textGuiSettings.
colorMode=textGuiSettings.colorMode or'Match GUI'textGuiSettings.scale=math.clamp(tonumber(textGuiSettings.scale)or 1,0.75,2)textGuiSettings.shadow=textGuiSettings.shadow~=false textGuiSettings.
gradient=textGuiSettings.gradient==true textGuiSettings.v4Gradient=textGuiSettings.v4Gradient==true textGuiSettings.animations=textGuiSettings.animations~=false textGuiSettings.watermark=
textGuiSettings.watermark==true textGuiSettings.background=textGuiSettings.background==true textGuiSettings.hideModules=textGuiSettings.hideModules==true textGuiSettings.hideRender=textGuiSettings.
hideRender==true textGuiSettings.customText=tostring(textGuiSettings.customText or'')local sessionSettings:any=overlaySettings.SessionInfo sessionSettings.showPerformance=sessionSettings.
showPerformance~=false sessionSettings.showServer=sessionSettings.showServer~=false sessionSettings.showRole=sessionSettings.showRole~=false sessionSettings.showDeaths=sessionSettings.showDeaths~=
false sessionSettings.showWins=sessionSettings.showWins~=false local targetSettings:any=overlaySettings.TargetInfo targetSettings.mode=targetSettings.mode or'Smart Priority'targetSettings.showAvatar=
targetSettings.showAvatar~=false targetSettings.showHealth=targetSettings.showHealth~=false targetSettings.showDistance=targetSettings.showDistance~=false local radarSettings:any=overlaySettings.Radar
radarSettings.range=math.clamp(tonumber(radarSettings.range)or 250,25,2000)radarSettings.size=math.clamp(tonumber(radarSettings.size)or 190,140,280)radarSettings.rotation=radarSettings.rotation or
'Camera'radarSettings.colorMode=radarSettings.colorMode or'Smart'radarSettings.outOfRange=radarSettings.outOfRange or'Clamp'radarSettings.showNames=radarSettings.showNames==true radarSettings.
showDistance=radarSettings.showDistance==true radarSettings.showTeam=radarSettings.showTeam~=false radarSettings.aliveOnly=radarSettings.aliveOnly~=false radarSettings.grid=radarSettings.grid~=false
radarSettings.rings=radarSettings.rings~=false local function getFloatingViewport():Vector2 local stable:any=state.layoutViewport if stable and stable.X>1 and stable.Y>1 then return stable end local
camera:Camera?=workspace.CurrentCamera local current:any=camera and camera.ViewportSize if current and current.X>1 and current.Y>1 then return current end return Vector2.new(1280,720)end
local function clampFloatingPosition(position:Vector2,size:Vector2):Vector2 local viewport:Vector2=getFloatingViewport()return Vector2.new(math.clamp(position.X,6,math.max(6,viewport.X-size.X-6)),math
.clamp(position.Y,6,math.max(6,viewport.Y-size.Y-6)))end local function bringFloatingToFront(record:FloatingWindowRecord):()state.windows.Raise(record.id)end local function createFloatingWindow(id:
string,titleText:string,size:Vector2,defaultPosition:Vector2,onClose:(()->())?,chromeless:boolean?):FloatingWindowRecord local bare:boolean=chromeless==true local headerHeight:number=bare and 0 or 30
local saved:any=configData.ui.floatingPositions[id]local position:Vector2=type(saved)=='table'and Vector2.new(tonumber(saved[1])or defaultPosition.X,tonumber(saved[2])or defaultPosition.Y)or
defaultPosition position=clampFloatingPosition(position,size)local root:Frame=create('Frame',{Parent=PopupLayer,Name='Floating_'..id,Active=true,BackgroundColor3=Theme.background,
BackgroundTransparency=bare and 1 or 0.08,BorderSizePixel=0,ClipsDescendants=not bare,Position=UDim2.fromOffset(position.X,position.Y),Size=UDim2.fromOffset(size.X,size.Y),Visible=false,ZIndex=150})::
Frame bind(root,'BackgroundColor3','background')if not bare then create('UICorner',{Parent=root,CornerRadius=cornerRadius(UI_RADIUS.panel)})local edge:UIStroke=create('UIStroke',{Parent=root,Name=
'WindowOutline',Color=Theme.outline,Transparency=0.08,Thickness=1})::UIStroke bind(edge,'Color','outline')local shade:Frame=create('Frame',{Parent=root,Name='Readability',BackgroundColor3=Theme.
background,BackgroundTransparency=0.28,BorderSizePixel=0,Size=UDim2.fromScale(1,1)})::Frame bind(shade,'BackgroundColor3','background')shade:SetAttribute('FloatingDepth',2)end local header:Frame=
create('Frame',{Parent=root,Name='Header',Active=not bare,BackgroundColor3=Theme.surface,BackgroundTransparency=0.24,BorderSizePixel=0,Size=UDim2.new(1,0,0,headerHeight),Visible=not bare})::Frame
bind(header,'BackgroundColor3','surface')header:SetAttribute('FloatingDepth',3)local closeButton:TextButton if bare then closeButton=create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,
ClipsDescendants=true,Parent=header,Name='Close',BackgroundTransparency=1,Text='',Visible=false})::TextButton else local accent:Frame=create('Frame',{Parent=header,BackgroundColor3=Theme.outline,
BackgroundTransparency=0.62,BorderSizePixel=0,Position=UDim2.fromOffset(0,headerHeight-1),Size=UDim2.new(1,0,0,1)})::Frame bind(accent,'BackgroundColor3','outline')accent:SetAttribute('FloatingDepth',
1)local title:TextLabel=makeTextLabel(header,titleText,10)title.FontFace=CONTROL_FONT title.Position=UDim2.fromOffset(10,0)title.Size=UDim2.new(1,-40,1,0)title.TextColor3=Theme.text bind(title,
'TextColor3','text')title:SetAttribute('FloatingDepth',2)closeButton=makeButton(header,'x',9)closeButton.Name='Close'closeButton.AnchorPoint=Vector2.new(1,0.5)closeButton.BackgroundTransparency=1
closeButton.Position=UDim2.new(1,-6,0.5,0)closeButton.Size=UDim2.fromOffset(22,22)closeButton.TextColor3=Theme.textMuted closeButton:SetAttribute('ThemeRole','Icon')closeButton:SetAttribute(
'FloatingDepth',3)local closeStroke:UIStroke?=closeButton:FindFirstChild('StyleStroke')::UIStroke?if closeStroke then closeStroke.Enabled=false end end local body:Frame=create('Frame',{Parent=root,
Name='Body',Active=false,BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.fromOffset(0,headerHeight),Size=UDim2.new(1,0,1,-headerHeight)})::Frame body:SetAttribute('FloatingDepth',4)if bare
then root.Active=state.visible==true create('UICorner',{Parent=root,CornerRadius=cornerRadius(UI_RADIUS.control)})local hint:UIStroke=create('UIStroke',{Parent=root,Name='MoveHint',Color=Theme.outline
,Thickness=1,Transparency=state.visible and 0.72 or 1})::UIStroke bind(hint,'Color','outline')state.addMenuVisibilityListener(function(visible:boolean):()if not root.Parent then return end root.Active
=visible hint.Transparency=visible and 0.72 or 1 end)end local record:FloatingWindowRecord={id=id,root=root,body=body,closeButton=closeButton}state.windows.Adopt(root,{id=id,body=body,header=header,
closeButton=closeButton,headerHeight=headerHeight,dragHandle=bare and root or header,requiresMenuOpen=bare,layered=true,onClose=onClose})return record end function FloatingUi.reclamp():()state.windows
.Reclamp()end state.onLayout(function():()FloatingUi.reclamp()end)local FavoritesWindow:FloatingWindowRecord=createFloatingWindow('Favorites','Favorites',Vector2.new(state.layout.compact and 240 or
270,236),Vector2.new(40,90),function():()configData.ui.favoritesWindowOpen=false queueConfigSave()end)local FavoritesList:ScrollingFrame=create('ScrollingFrame',{Parent=FavoritesWindow.body,Name=
'FavoriteModules',Active=true,AutomaticCanvasSize=Enum.AutomaticSize.Y,BackgroundTransparency=1,BorderSizePixel=0,CanvasSize=UDim2.fromOffset(0,0),Position=UDim2.fromOffset(8,8),ScrollBarImageColor3=
Theme.outline,ScrollBarThickness=3,Size=UDim2.new(1,-16,1,-16)})::ScrollingFrame FavoritesList:SetAttribute('FloatingDepth',2)create('UIListLayout',{Parent=FavoritesList,Padding=UDim.new(0,5),
SortOrder=Enum.SortOrder.LayoutOrder})local function syncFavoriteButton(feature:any):()if not feature.favoriteButton then return end local selected:boolean=configData.ui.favorites[feature.configKey]==
true feature.favoriteButton.Text=selected and'\u{2605}'or'\u{2606}'feature.favoriteButton.TextColor3=selected and Theme.accent or Theme.textMuted feature.favoriteButton.TextTransparency=selected and 0
or 0.34 end local function refreshFavorites():()for _,row:GuiObject in ipairs(FloatingUi.favoriteRows)do row:Destroy()end table.clear(FloatingUi.favoriteRows)local order:number=0 for _,feature:any in
ipairs(allFeatures)do if configData.ui.favorites[feature.configKey]==true then order+=1 local row:TextButton=create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,ClipsDescendants=true,Parent=
FavoritesList,Name=feature.configKey:gsub('%W',''),AutoButtonColor=false,BackgroundColor3=Theme.surface,BackgroundTransparency=0.38,BorderSizePixel=0,FontFace=THIN_FONT,LayoutOrder=order,Size=UDim2.
new(1,-4,0,36),Text=feature.name,TextColor3=feature.enabled and Theme.accent or Theme.text,TextSize=13,TextXAlignment=Enum.TextXAlignment.Left})::TextButton create('UICorner',{Parent=row,CornerRadius=
cornerRadius(8)})row:SetAttribute('FloatingDepth',3)create('UIPadding',{Parent=row,PaddingLeft=UDim.new(0,12),PaddingRight=UDim.new(0,38)})local status:Frame=create('Frame',{Parent=row,Name='Status',
AnchorPoint=Vector2.new(1,0.5),BackgroundColor3=feature.enabled and Theme.accent or Theme.textMuted,BackgroundTransparency=feature.enabled and 0 or 0.58,BorderSizePixel=0,Position=UDim2.new(1,-14,0.5,
0),Size=UDim2.fromOffset(feature.enabled and 9 or 7,feature.enabled and 9 or 7)})::Frame create('UICorner',{Parent=status,CornerRadius=UDim.new(1,0)})status:SetAttribute('FloatingDepth',2)row.
MouseButton1Click:Connect(function():()if feature.isCategory then state.setMenuVisible(true)end feature.activate()end)table.insert(FloatingUi.favoriteRows,row)end end if order==0 then local empty:
TextLabel=create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,ClipsDescendants=true,Parent=FavoritesList,BackgroundTransparency=1,FontFace=THIN_FONT,LayoutOrder=1,Size=UDim2.new(1,-4,0,52),Text=
'Star a module to keep it here',TextColor3=Theme.textMuted,TextSize=13,TextTransparency=0.28,TextXAlignment=Enum.TextXAlignment.Center})::TextLabel empty:SetAttribute('FloatingDepth',3)table.insert(
FloatingUi.favoriteRows,empty)end end function FloatingUi.toggleFavorite(feature:any):()local nextValue:boolean=configData.ui.favorites[feature.configKey]~=true configData.ui.favorites[feature.
configKey]=nextValue and true or nil syncFavoriteButton(feature)refreshFavorites()queueConfigSave()end function FloatingUi.registerFeature(feature:any):()FloatingUi.favorites[feature.configKey]=
feature syncFavoriteButton(feature)if configData.ui.favorites[feature.configKey]==true then refreshFavorites()end end function FloatingUi.unregisterFeature(feature:any):()FloatingUi.favorites[feature.
configKey]=nil if configData.ui.favorites[feature.configKey]==true then refreshFavorites()end end function FloatingUi.refreshFeature(feature:any):()if configData.ui.favorites[feature.configKey]==true
then refreshFavorites()end end local OverlayMenu:FloatingWindowRecord=createFloatingWindow('OverlayMenu','Overlays',Vector2.new(state.layout.compact and 232 or 252,220),Vector2.new(340,90),function():
()configData.ui.overlayMenuOpen=false queueConfigSave()end)local OverlayList:Frame=create('Frame',{Parent=OverlayMenu.body,BackgroundTransparency=1,Position=UDim2.fromOffset(8,8),Size=UDim2.new(1,-16,
1,-16)})::Frame OverlayList:SetAttribute('FloatingDepth',2)create('UIListLayout',{Parent=OverlayList,Padding=UDim.new(0,5),SortOrder=Enum.SortOrder.LayoutOrder})local overlayDefinitions:{any}={{id=
'SessionInfo',title='Session Info',size=Vector2.new(210,118),position=Vector2.new(24,300)},{id='TextGUI',title='Text GUI',size=Vector2.new(240,220),position=Vector2.new(250,300)},{id='TargetInfo',
title='Target Info',size=Vector2.new(260,86),position=Vector2.new(510,300)},{id='Radar',title='Radar',size=Vector2.new(radarSettings.size,radarSettings.size),position=Vector2.new(820,280)}}
local function setOverlayEnabled(id:string,enabled:boolean):()local overlay:any=FloatingUi.overlays[id]if not overlay then return end overlay.enabled=enabled overlay.window.root.Visible=enabled
setToggleSwitch(overlay.toggle,enabled,true)configData.ui.overlays[id]=enabled queueConfigSave()if enabled then bringFloatingToFront(overlay.window)end end FloatingUi.settingsWindows={}FloatingUi.
openSettings=function(id:string):()local window:FloatingWindowRecord?=FloatingUi.settingsWindows[id]if window then window.root.Visible=not window.root.Visible if window.root.Visible then
bringFloatingToFront(window)end end end for index:number,definition:any in ipairs(overlayDefinitions)do local overlayId:string=definition.id local overlayWindow:FloatingWindowRecord overlayWindow=
createFloatingWindow(overlayId,definition.title,definition.size,definition.position,function():()setOverlayEnabled(overlayId,false)end,true)local managerRow:Frame=create('Frame',{Parent=OverlayList,
BackgroundColor3=Theme.surface,BackgroundTransparency=0.48,BorderSizePixel=0,LayoutOrder=index,Size=UDim2.new(1,0,0,40)})::Frame managerRow:SetAttribute('FloatingDepth',3)create('UICorner',{Parent=
managerRow,CornerRadius=cornerRadius(8)})local managerLabel:TextLabel=makeTextLabel(managerRow,definition.title,10)managerLabel.Position=UDim2.fromOffset(10,0)managerLabel.Size=UDim2.new(1,-112,1,0)
managerLabel.TextColor3=Theme.textMuted managerLabel:SetAttribute('FloatingDepth',1)local settingsButton:TextButton=makeButton(managerRow,'',9)settingsButton.Name='Settings'state.applyLinesGlyph(
settingsButton,false)settingsButton.AnchorPoint=Vector2.new(1,0.5)settingsButton.BackgroundTransparency=1 settingsButton.Position=UDim2.new(1,-58,0.5,0)settingsButton.Size=UDim2.fromOffset(28,28)
settingsButton:SetAttribute('FloatingDepth',2)local settingsStroke:UIStroke?=settingsButton:FindFirstChild('StyleStroke')::UIStroke?if settingsStroke then settingsStroke.Enabled=false end local toggle
:TextButton=createToggleSwitch(managerRow,false,42)toggle.AnchorPoint=Vector2.new(1,0.5)toggle.Position=UDim2.new(1,-7,0.5,0)toggle:SetAttribute('FloatingDepth',2)FloatingUi.overlays[overlayId]={
window=overlayWindow,toggle=toggle,enabled=false}trackUiConnection(toggle.MouseButton1Click:Connect(function():()setOverlayEnabled(overlayId,not FloatingUi.overlays[overlayId].enabled)end))
trackUiConnection(settingsButton.MouseButton1Click:Connect(function():()FloatingUi.openSettings(overlayId)end))end local function createOverlaySettingsWindow(id:string,title:string,height:number,
position:Vector2):(FloatingWindowRecord,ScrollingFrame)local window:FloatingWindowRecord=createFloatingWindow(id..'Settings',title,Vector2.new(state.layout.compact and 268 or 292,math.min(height,math.
max(180,getFloatingViewport().Y-24))),position,nil)local scroll:ScrollingFrame=create('ScrollingFrame',{Parent=window.body,Active=true,AutomaticCanvasSize=Enum.AutomaticSize.Y,BackgroundTransparency=1
,BorderSizePixel=0,CanvasSize=UDim2.fromOffset(0,0),Position=UDim2.fromOffset(8,8),ScrollBarImageColor3=Theme.accentDim,ScrollBarThickness=3,Size=UDim2.new(1,-16,1,-16)})::ScrollingFrame scroll:
SetAttribute('FloatingDepth',2)create('UIListLayout',{Parent=scroll,Padding=UDim.new(0,5),SortOrder=Enum.SortOrder.LayoutOrder})FloatingUi.settingsWindows[id]=window return window,scroll end
local function createCompactSettingRow(parent:Instance,labelText:string):Frame local row:Frame=create('Frame',{Parent=parent,BackgroundColor3=Theme.surface,BackgroundTransparency=0.44,BorderSizePixel=
0,Size=UDim2.new(1,-4,0,38)})::Frame create('UICorner',{Parent=row,CornerRadius=cornerRadius(8)})local label:TextLabel=makeTextLabel(row,labelText,9)label.Position=UDim2.fromOffset(10,0)label.Size=
UDim2.new(0.48,-10,1,0)label.TextColor3=Theme.textMuted return row end local function addCompactCycle(parent:Instance,labelText:string,values:{string},currentValue:string,callback:(string)->()):()
local row:Frame=createCompactSettingRow(parent,labelText)local index:number=table.find(values,currentValue)or 1 local button:TextButton=makeButton(row,values[index],8)button.Position=UDim2.new(0.48,4,
0,4)button.Size=UDim2.new(0.52,-8,1,-8)trackUiConnection(button.MouseButton1Click:Connect(function():()index=index%#values+1 button.Text=values[index]callback(values[index])queueConfigSave()end))end
local function addCompactToggle(parent:Instance,labelText:string,initial:boolean,callback:(boolean)->()):()local row:Frame=createCompactSettingRow(parent,labelText)local enabled:boolean=initial local
toggle:TextButton=createToggleSwitch(row,enabled,42)toggle.AnchorPoint=Vector2.new(1,0.5)toggle.Position=UDim2.new(1,-8,0.5,0)trackUiConnection(toggle.MouseButton1Click:Connect(function():()enabled=
not enabled setToggleSwitch(toggle,enabled,true)callback(enabled)queueConfigSave()end))end local function addCompactText(parent:Instance,labelText:string,initial:string,callback:(string)->()):()local
row:Frame=createCompactSettingRow(parent,labelText)local box:TextBox=makeTextBox(row,initial)box.Position=UDim2.new(0.48,4,0,4)box.Size=UDim2.new(0.52,-8,1,-8)box.PlaceholderText='Custom text'box.
TextXAlignment=Enum.TextXAlignment.Left trackUiConnection(box.FocusLost:Connect(function():()callback(string.sub(box.Text,1,80))queueConfigSave()end))end do local textSettingsScroll:ScrollingFrame
local _textSettingsWindow:FloatingWindowRecord _textSettingsWindow,textSettingsScroll=createOverlaySettingsWindow('TextGUI','Text GUI settings',state.layout.compact and 500 or 540,Vector2.new(300,70))
addCompactCycle(textSettingsScroll,'Sort',{'Alphabetical','Length','Enabled order','Section'},textGuiSettings.sort,function(value:string):()textGuiSettings.sort=value end)addCompactCycle(
textSettingsScroll,'Font',{'Builder Sans','Source Sans','Gotham','Code','Valve'},textGuiSettings.font,function(value:string):()textGuiSettings.font=value end)addCompactCycle(textSettingsScroll,
'Color mode',{'Match GUI','Solid white','Greyscale','Rainbow','By section'},textGuiSettings.colorMode,function(value:string):()textGuiSettings.colorMode=value end)addCompactCycle(textSettingsScroll,
'Scale',{'0.75','1','1.25','1.5','2'},tostring(textGuiSettings.scale),function(value:string):()textGuiSettings.scale=tonumber(value)or 1 end)addCompactToggle(textSettingsScroll,'Shadow',
textGuiSettings.shadow,function(value:boolean):()textGuiSettings.shadow=value end)addCompactToggle(textSettingsScroll,'Gradient',textGuiSettings.gradient,function(value:boolean):()textGuiSettings.
gradient=value end)addCompactToggle(textSettingsScroll,'V4 gradient',textGuiSettings.v4Gradient,function(value:boolean):()textGuiSettings.v4Gradient=value end)addCompactToggle(textSettingsScroll,
'Animations',textGuiSettings.animations,function(value:boolean):()textGuiSettings.animations=value end)addCompactToggle(textSettingsScroll,'Watermark',textGuiSettings.watermark,function(value:boolean)
:()textGuiSettings.watermark=value end)addCompactToggle(textSettingsScroll,'Render background',textGuiSettings.background,function(value:boolean):()textGuiSettings.background=value end)
addCompactToggle(textSettingsScroll,'Hide modules',textGuiSettings.hideModules,function(value:boolean):()textGuiSettings.hideModules=value end)addCompactToggle(textSettingsScroll,'Hide render',
textGuiSettings.hideRender,function(value:boolean):()textGuiSettings.hideRender=value end)addCompactText(textSettingsScroll,'Custom text',textGuiSettings.customText,function(value:string):()
textGuiSettings.customText=value end)end do local radarSettingsScroll:ScrollingFrame local _radarSettingsWindow:FloatingWindowRecord _radarSettingsWindow,radarSettingsScroll=
createOverlaySettingsWindow('Radar','Radar settings',500,Vector2.new(630,70))addCompactCycle(radarSettingsScroll,'Range',{'75','150','250','500','1000','2000'},tostring(radarSettings.range),function(
value:string):()radarSettings.range=tonumber(value)or 250 end)addCompactCycle(radarSettingsScroll,'Radar size',{'160','190','220','250','280'},tostring(radarSettings.size),function(value:string):()
radarSettings.size=tonumber(value)or 190 end)addCompactCycle(radarSettingsScroll,'Rotation',{'Camera','Player','North'},radarSettings.rotation,function(value:string):()radarSettings.rotation=value end
)addCompactCycle(radarSettingsScroll,'Color mode',{'Smart','Team','Health','MM2 role','Spectrum'},radarSettings.colorMode,function(value:string):()radarSettings.colorMode=value end)addCompactCycle(
radarSettingsScroll,'Out of range',{'Clamp','Hide'},radarSettings.outOfRange,function(value:string):()radarSettings.outOfRange=value end)addCompactToggle(radarSettingsScroll,'Alive only',radarSettings
.aliveOnly,function(value:boolean):()radarSettings.aliveOnly=value end)addCompactToggle(radarSettingsScroll,'Show team',radarSettings.showTeam,function(value:boolean):()radarSettings.showTeam=value
end)addCompactToggle(radarSettingsScroll,'Show names',radarSettings.showNames,function(value:boolean):()radarSettings.showNames=value end)addCompactToggle(radarSettingsScroll,'Show distance',
radarSettings.showDistance,function(value:boolean):()radarSettings.showDistance=value end)addCompactToggle(radarSettingsScroll,'Grid',radarSettings.grid,function(value:boolean):()radarSettings.grid=
value end)addCompactToggle(radarSettingsScroll,'Range rings',radarSettings.rings,function(value:boolean):()radarSettings.rings=value end)end do local sessionSettingsScroll:ScrollingFrame local
_sessionSettingsWindow:FloatingWindowRecord _sessionSettingsWindow,sessionSettingsScroll=createOverlaySettingsWindow('SessionInfo','Session Info settings',310,Vector2.new(960,70))addCompactToggle(
sessionSettingsScroll,'Performance',sessionSettings.showPerformance,function(value:boolean):()sessionSettings.showPerformance=value end)addCompactToggle(sessionSettingsScroll,'Server data',
sessionSettings.showServer,function(value:boolean):()sessionSettings.showServer=value end)addCompactToggle(sessionSettingsScroll,'Role and team',sessionSettings.showRole,function(value:boolean):()
sessionSettings.showRole=value end)addCompactToggle(sessionSettingsScroll,'Deaths',sessionSettings.showDeaths,function(value:boolean):()sessionSettings.showDeaths=value end)addCompactToggle(
sessionSettingsScroll,'Wins / rounds',sessionSettings.showWins,function(value:boolean):()sessionSettings.showWins=value end)end do local targetSettingsScroll:ScrollingFrame local _targetSettingsWindow
:FloatingWindowRecord _targetSettingsWindow,targetSettingsScroll=createOverlaySettingsWindow('TargetInfo','Target Priority',280,Vector2.new(960,395))addCompactCycle(targetSettingsScroll,'Selection',{
'Smart Priority','Nearest cursor','Nearest distance'},targetSettings.mode,function(value:string):()targetSettings.mode=value end)addCompactToggle(targetSettingsScroll,'Avatar portrait',targetSettings.
showAvatar,function(value:boolean):()targetSettings.showAvatar=value end)addCompactToggle(targetSettingsScroll,'Health bar',targetSettings.showHealth,function(value:boolean):()targetSettings.
showHealth=value end)addCompactToggle(targetSettingsScroll,'Distance',targetSettings.showDistance,function(value:boolean):()targetSettings.showDistance=value end)end local SessionShadow:TextLabel=
makeTextLabel(FloatingUi.overlays.SessionInfo.window.body,'Session initializing...',9)SessionShadow.Name='SessionShadow'SessionShadow.Position=UDim2.fromOffset(1,1)SessionShadow.Size=UDim2.fromScale(1
,1)SessionShadow.TextColor3=Color3.fromRGB(0,0,0)SessionShadow.TextTransparency=0.25 SessionShadow.TextYAlignment=Enum.TextYAlignment.Top SessionShadow:SetAttribute('FloatingDepth',1)local SessionText
:TextLabel=makeTextLabel(FloatingUi.overlays.SessionInfo.window.body,'Session initializing...',9)SessionText.Position=UDim2.fromOffset(0,0)SessionText.Size=UDim2.fromScale(1,1)SessionText.TextColor3=
Theme.text SessionText.TextYAlignment=Enum.TextYAlignment.Top SessionText:SetAttribute('FloatingDepth',2)local TextGuiBody:Frame=FloatingUi.overlays.TextGUI.window.body local TextGuiShadow:TextLabel=
makeTextLabel(TextGuiBody,'No active modules',10)TextGuiShadow.Position=UDim2.fromOffset(1,1)TextGuiShadow.Size=UDim2.fromScale(1,1)TextGuiShadow.TextColor3=Color3.fromRGB(0,0,0)TextGuiShadow.
TextTransparency=0.16 TextGuiShadow.TextWrapped=true TextGuiShadow.TextYAlignment=Enum.TextYAlignment.Top TextGuiShadow:SetAttribute('FloatingDepth',1)local TextGuiLabel:TextLabel=makeTextLabel(
FloatingUi.overlays.TextGUI.window.body,'No active modules',10)TextGuiLabel.Position=UDim2.fromOffset(0,0)TextGuiLabel.Size=UDim2.fromScale(1,1)TextGuiLabel.TextColor3=Theme.accent TextGuiLabel.
TextWrapped=true TextGuiLabel.TextYAlignment=Enum.TextYAlignment.Top TextGuiLabel:SetAttribute('FloatingDepth',2)local TextGuiGradient:UIGradient=create('UIGradient',{Parent=TextGuiLabel,Color=
ColorSequence.new({ColorSequenceKeypoint.new(0,Theme.accent),ColorSequenceKeypoint.new(0.5,Theme.accentDim),ColorSequenceKeypoint.new(1,Theme.accentFaint)}),Rotation=90,Enabled=false})::UIGradient
local TargetBody:Frame=FloatingUi.overlays.TargetInfo.window.body local TargetPortraitFrame:Frame=create('Frame',{Parent=TargetBody,BackgroundColor3=Theme.surface,BackgroundTransparency=0.18,
BorderSizePixel=0,Position=UDim2.fromOffset(0,0),Size=UDim2.fromOffset(56,56)})::Frame TargetPortraitFrame:SetAttribute('FloatingDepth',2)create('UICorner',{Parent=TargetPortraitFrame,CornerRadius=
cornerRadius(UI_RADIUS.panel)})create('UIStroke',{Parent=TargetPortraitFrame,Color=Theme.outline,Transparency=0.55,Thickness=1})local TargetPortrait:ImageLabel=create('ImageLabel',{Parent=
TargetPortraitFrame,BackgroundTransparency=1,BorderSizePixel=0,Image='',Position=UDim2.fromOffset(4,4),ScaleType=Enum.ScaleType.Crop,Size=UDim2.new(1,-8,1,-8),ZIndex=3})::ImageLabel TargetPortrait:
SetAttribute('FloatingDepth',3)create('UICorner',{Parent=TargetPortrait,CornerRadius=cornerRadius(10)})local TargetName:TextLabel=makeTextLabel(TargetBody,'No target',11)TargetName.FontFace=
CONTROL_FONT TargetName.Position=UDim2.fromOffset(64,0)TargetName.Size=UDim2.new(1,-64,0,22)TargetName:SetAttribute('FloatingDepth',2)local TargetRole:TextLabel=makeTextLabel(TargetBody,
'No priority target',8)TargetRole.Position=UDim2.fromOffset(64,21)TargetRole.Size=UDim2.new(1,-64,0,18)TargetRole.TextColor3=Theme.accentDim TargetRole:SetAttribute('FloatingDepth',2)local
TargetHealthTrack:Frame=create('Frame',{Parent=TargetBody,BackgroundColor3=Theme.background,BackgroundTransparency=0.18,BorderSizePixel=0,Position=UDim2.fromOffset(64,42),Size=UDim2.new(1,-64,0,6)})::
Frame TargetHealthTrack:SetAttribute('FloatingDepth',2)create('UICorner',{Parent=TargetHealthTrack,CornerRadius=UDim.new(1,0)})local TargetHealthFill:Frame=create('Frame',{Parent=TargetHealthTrack,
BackgroundColor3=Theme.accent,BorderSizePixel=0,Size=UDim2.fromScale(0,1)})::Frame TargetHealthFill:SetAttribute('FloatingDepth',1)create('UICorner',{Parent=TargetHealthFill,CornerRadius=UDim.new(1,0)
})local TargetDetails:TextLabel=makeTextLabel(TargetBody,'Waiting for a nearby player',9)TargetDetails.Position=UDim2.fromOffset(0,58)TargetDetails.Size=UDim2.new(1,0,0,28)TargetDetails.TextColor3=
Theme.textMuted TargetDetails:SetAttribute('FloatingDepth',2)local RadarSurface:Frame=create('Frame',{Parent=FloatingUi.overlays.Radar.window.body,Name='Surface',AnchorPoint=Vector2.new(0.5,0.5),
BackgroundColor3=Theme.surface,BackgroundTransparency=0.34,BorderSizePixel=0,Position=UDim2.fromScale(0.5,0.5),Size=UDim2.fromOffset(radarSettings.size,radarSettings.size)})::Frame RadarSurface:
SetAttribute('FloatingDepth',2)create('UICorner',{Parent=RadarSurface,CornerRadius=UDim.new(1,0)})create('UIStroke',{Parent=RadarSurface,Color=Theme.outline,Transparency=0.55})local radarAxes:{Frame}=
{}for _,rotation:number in ipairs({0,90})do local axis:Frame=create('Frame',{Parent=RadarSurface,AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.outline,BackgroundTransparency=0.72,
BorderSizePixel=0,Position=UDim2.fromScale(0.5,0.5),Rotation=rotation,Size=UDim2.new(1,-16,0,1)})::Frame axis:SetAttribute('FloatingDepth',1)table.insert(radarAxes,axis)end local radarRings:{Frame}={}
for _,scale:number in ipairs({0.34,0.67})do local ring:Frame=create('Frame',{Parent=RadarSurface,AnchorPoint=Vector2.new(0.5,0.5),BackgroundTransparency=1,Position=UDim2.fromScale(0.5,0.5),Size=UDim2.
fromScale(scale,scale)})::Frame ring:SetAttribute('FloatingDepth',1)create('UICorner',{Parent=ring,CornerRadius=UDim.new(1,0)})create('UIStroke',{Parent=ring,Color=Theme.outline,Transparency=0.7,
Thickness=1})table.insert(radarRings,ring)end local RadarNorth:TextLabel=makeTextLabel(RadarSurface,'N',8)RadarNorth.AnchorPoint=Vector2.new(0.5,0)RadarNorth.Position=UDim2.new(0.5,0,0,5)RadarNorth.
Size=UDim2.fromOffset(20,18)RadarNorth.TextColor3=Theme.accent RadarNorth.TextXAlignment=Enum.TextXAlignment.Center RadarNorth:SetAttribute('FloatingDepth',3)local RadarCenter:Frame=create('Frame',{
Parent=RadarSurface,AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.accent,BorderSizePixel=0,Position=UDim2.fromScale(0.5,0.5),Size=UDim2.fromOffset(7,7)})::Frame RadarCenter:SetAttribute(
'FloatingDepth',3)create('UICorner',{Parent=RadarCenter,CornerRadius=UDim.new(1,0)})local radarBlips:{[Player]:any}={}local sessionStartedAt:number=os.clock()local overlayElapsed:number=1 local
sessionFps:number=60 local sessionStats:any={deaths=0,wins=0,rounds=0,winsAvailable=false,roundsAvailable=false,role=nil}state.sessionInfo={setRole=function(role:string?):()sessionStats.role=role end,
addWin=function():()sessionStats.winsAvailable=true sessionStats.wins+=1 end,addRound=function():()sessionStats.roundsAvailable=true sessionStats.rounds+=1 end}local deathConnection:
RBXScriptConnection?=nil local sessionBindGeneration:number=0 local sessionTrackingActive:boolean=true local function bindSessionCharacter(character:Model):()sessionBindGeneration+=1 local generation:
number=sessionBindGeneration if deathConnection then deathConnection:Disconnect()deathConnection=nil end local humanoid:Humanoid?=character:FindFirstChildOfClass('Humanoid')or character:WaitForChild(
'Humanoid',8)::Humanoid?if humanoid and sessionTrackingActive and generation==sessionBindGeneration and character==LocalPlayer.Character then deathConnection=humanoid.Died:Connect(function():()
sessionStats.deaths+=1 end)end end if LocalPlayer.Character then task.defer(bindSessionCharacter,LocalPlayer.Character)end trackUiConnection(LocalPlayer.CharacterAdded:Connect(bindSessionCharacter))
state.destroySessionTracking=function():()sessionTrackingActive=false sessionBindGeneration+=1 if deathConnection then deathConnection:Disconnect()deathConnection=nil end end local thumbnailCache:{[
number]:string}={}local thumbnailGeneration:number=0 local displayedTargetUserId:number=0 local currentOverlayTarget:Player?=nil local function requestTargetThumbnail(player:Player?):()local userId:
number=player and player.UserId or 0 if userId==displayedTargetUserId then return end displayedTargetUserId=userId thumbnailGeneration+=1 local generation:number=thumbnailGeneration TargetPortrait.
Image=userId>0 and(thumbnailCache[userId]or'')or''if userId<=0 or thumbnailCache[userId]then return end task.spawn(function():()local success:boolean,image:any,ready:any=pcall(Players.
GetUserThumbnailAsync,Players,userId,Enum.ThumbnailType.HeadShot,Enum.ThumbnailSize.Size180x180)if success and ready==true and type(image)=='string'and image~=''then thumbnailCache[userId]=image if
generation==thumbnailGeneration and TargetPortrait.Parent then TargetPortrait.Image=image end elseif generation==thumbnailGeneration then task.delay(1,function():()if generation==thumbnailGeneration
then displayedTargetUserId=0 end end)end end)end local function getTextGuiFont():Font if textGuiSettings.font=='Source Sans'then return Font.fromEnum(Enum.Font.SourceSans)elseif textGuiSettings.font==
'Gotham'then return Font.fromEnum(Enum.Font.Gotham)elseif textGuiSettings.font=='Code'then return Font.fromEnum(Enum.Font.Code)elseif textGuiSettings.font=='Valve'then return VALVE_FONT end return
CONTROL_FONT end local function getSessionPing():number?local success:boolean,value:any=pcall(function():number local player:any=LocalPlayer or Players.LocalPlayer local direct:any=player and player.
GetNetworkPing if type(direct)~='function'then return 0 end local seconds:any=direct(player)return if type(seconds)=='number'then seconds*1000 else 0 end)if not success then return nil end local
milliseconds:number?=tonumber(value)if not milliseconds or milliseconds<=0 then return nil end return milliseconds end local lastTextGuiText:string=''table.insert(ownedTasks,TaskManager:Connect(
function(deltaTime:number):()local performance:any=state.performance if performance and type(performance.fps)=='number'then sessionFps=performance.fps else sessionFps+=((1/math.max(deltaTime,1/240))-
sessionFps)*0.12 end if not(FloatingUi.overlays.SessionInfo.enabled or FloatingUi.overlays.TextGUI.enabled or FloatingUi.overlays.TargetInfo.enabled or FloatingUi.overlays.Radar.enabled)then
overlayElapsed=0 return end overlayElapsed+=deltaTime if overlayElapsed<0.12 then return end overlayElapsed=0 local now:number=os.clock()local elapsed:number=math.max(0,math.floor(now-sessionStartedAt
))if FloatingUi.overlays.SessionInfo.enabled then local role:string?=sessionStats.role if state.gameRoleProvider then local success:boolean,resolved:any=pcall(state.gameRoleProvider,LocalPlayer)if
success and type(resolved)=='string'then role=resolved end end local lines:{string}={string.format('Injected  %02d:%02d:%02d',math.floor(elapsed/3600),math.floor(elapsed/60)%60,elapsed%60)}if
sessionSettings.showDeaths then table.insert(lines,'Deaths    '..tostring(sessionStats.deaths))end if sessionSettings.showWins then table.insert(lines,'Wins      '..(sessionStats.winsAvailable and
tostring(sessionStats.wins)or'N/A'))table.insert(lines,'Rounds    '..(sessionStats.roundsAvailable and tostring(sessionStats.rounds)or'N/A'))end if sessionSettings.showRole then table.insert(lines,
'Role      '..tostring(role or'N/A'))table.insert(lines,'Team      '..tostring(LocalPlayer.Team and LocalPlayer.Team.Name or'None'))end if sessionSettings.showPerformance then local ping:number?=
getSessionPing()table.insert(lines,string.format('FPS       %.0f',math.clamp(sessionFps,0,999)))table.insert(lines,'Ping      '..(ping and string.format('%.0f ms',ping)or'N/A'))local cost:any=state.
performance if cost and type(cost.tasksMs)=='number'then table.insert(lines,string.format('Menu sample %.2f+%.2f ms',cost.tasksMs,cost.renderMs))end if cost and(cost.spikeCount or 0)>0 then table.
insert(lines,string.format('Spikes    %d \183 worst %.0f ms',cost.spikeCount,cost.worstSpikeMs or 0))end if cost and cost.peakTask then table.insert(lines,string.format('Task peak %s \183 %.2f ms',
tostring(cost.peakTask),tonumber(cost.peakTaskMs)or 0))end end if sessionSettings.showServer then table.insert(lines,string.format('Players   %d/%d',#Players:GetPlayers(),Players.MaxPlayers))table.
insert(lines,'Game      '..tostring(state.activeGameName or'Universal'))end SessionText.Text=table.concat(lines,'\n')end if FloatingUi.overlays.TextGUI.enabled then local enabledFeatures:{any}={}if
not textGuiSettings.hideModules then for _,feature:any in ipairs(allFeatures)do local lowerName:string=string.lower(feature.name)local renderLike:boolean=string.find(lowerName,'esp',1,true)~=nil or
string.find(lowerName,'visual',1,true)~=nil or string.find(lowerName,'x-ray',1,true)~=nil or string.find(lowerName,'fov',1,true)~=nil or string.find(lowerName,'fullbright',1,true)~=nil if feature.
enabled and not feature.isCategory and not feature.isAction and(not textGuiSettings.hideRender or not renderLike)then table.insert(enabledFeatures,feature)end end end table.sort(enabledFeatures,
function(left:any,right:any):boolean if textGuiSettings.sort=='Length'then return#left.name>#right.name elseif textGuiSettings.sort=='Enabled order'then return(left.enabledAt or 0)<(right.enabledAt or
0)elseif textGuiSettings.sort=='Section'then local leftKey:string=left.sectionName..left.name local rightKey:string=right.sectionName..right.name return leftKey<rightKey end return string.lower(left.
name)<string.lower(right.name)end)local textLines:{string}={}if textGuiSettings.watermark then table.insert(textLines,'Wurst')end if textGuiSettings.customText~=''then table.insert(textLines,
textGuiSettings.customText)end for _,feature:any in ipairs(enabledFeatures)do table.insert(textLines,feature.name)end if#textLines==0 then table.insert(textLines,'No active modules')end local nextText
:string=table.concat(textLines,'\n')if nextText~=lastTextGuiText then lastTextGuiText=nextText TextGuiLabel.Text=nextText TextGuiShadow.Text=nextText if textGuiSettings.animations then TextGuiLabel.
TextTransparency=0.5 TweenService:Create(TextGuiLabel,TweenInfo.new(0.2),{TextTransparency=0}):Play()end end TextGuiLabel.FontFace=getTextGuiFont()TextGuiShadow.FontFace=TextGuiLabel.FontFace
TextGuiLabel.TextSize=math.round(14*textGuiSettings.scale)TextGuiShadow.TextSize=TextGuiLabel.TextSize TextGuiShadow.Visible=textGuiSettings.shadow TextGuiBody.BackgroundColor3=Theme.background
TextGuiBody.BackgroundTransparency=textGuiSettings.background and 0.42 or 1 local textWindowRoot:Frame=FloatingUi.overlays.TextGUI.window.root local textHeader:Frame?=textWindowRoot:FindFirstChild(
'Header')::Frame?local textShade:Frame?=textWindowRoot:FindFirstChild('Readability')::Frame?local textStroke:UIStroke?=textWindowRoot:FindFirstChildOfClass('UIStroke')if textHeader then textHeader.
Visible=state.visible end if textShade then textShade.Visible=state.visible or textGuiSettings.background end if textStroke then textStroke.Transparency=state.visible and 0.38 or 1 end textWindowRoot.
Active=state.visible textWindowRoot.BackgroundTransparency=state.visible and 0.08 or 1 TextGuiBody.Position=state.visible and UDim2.fromOffset(0,36)or UDim2.fromOffset(0,0)TextGuiBody.Size=state.
visible and UDim2.new(1,0,1,-36)or UDim2.fromScale(1,1)local gradientEnabled:boolean=textGuiSettings.gradient or textGuiSettings.v4Gradient or textGuiSettings.colorMode=='Greyscale'or textGuiSettings.
colorMode=='Rainbow'TextGuiGradient.Enabled=gradientEnabled if textGuiSettings.v4Gradient then TextGuiGradient.Color=ColorSequence.new({ColorSequenceKeypoint.new(0,Theme.accent),ColorSequenceKeypoint.
new(0.34,Theme.accentDim),ColorSequenceKeypoint.new(0.68,Theme.accentFaint),ColorSequenceKeypoint.new(1,Theme.accent)})TextGuiGradient.Rotation=(now*45)%360 elseif textGuiSettings.colorMode=='Rainbow'
then local hue:number=(now*0.12)%1 TextGuiLabel.TextColor3=Color3.fromHSV(hue,0.72,1)TextGuiGradient.Enabled=false elseif textGuiSettings.colorMode=='Greyscale'then local pulse:number=0.72+0.28*math.
sin(now*1.6)TextGuiLabel.TextColor3=Color3.fromHSV(0,0,pulse)TextGuiGradient.Enabled=false elseif textGuiSettings.colorMode=='Solid white'then TextGuiLabel.TextColor3=Theme.accent elseif
textGuiSettings.colorMode=='By section'then TextGuiLabel.TextColor3=Theme.accentDim else TextGuiLabel.TextColor3=Theme.accent TextGuiGradient.Rotation=90 end end local targetInfoEnabled:boolean=
FloatingUi.overlays.TargetInfo.enabled local localCharacter:Model?=LocalPlayer.Character local localRoot:BasePart?=localCharacter and localCharacter:FindFirstChild('HumanoidRootPart')::BasePart?local
camera:Camera?=workspace.CurrentCamera local selectedTarget:Player?=nil local targetRole:string?=nil local targetSource:string='Nearest player'local targetAccent:Color3=Theme.accentDim if
targetInfoEnabled and targetSettings.mode=='Smart Priority'and state.gameTargetProvider then local success:boolean,snapshot:any=pcall(state.gameTargetProvider)if success then if typeof(snapshot)==
'Instance'and snapshot:IsA('Player')then selectedTarget=snapshot elseif type(snapshot)=='table'and typeof(snapshot.player)=='Instance'and snapshot.player:IsA('Player')then selectedTarget=snapshot.
player targetRole=snapshot.role targetSource=tostring(snapshot.source or'Game priority')if typeof(snapshot.accent)=='Color3'then targetAccent=snapshot.accent end end end end if targetInfoEnabled and
not selectedTarget and localRoot then local bestScore:number=math.huge local pointer:Vector2=state.isMobile and getFloatingViewport()/2 or UserInputService:GetMouseLocation()for _,player:Player in
ipairs(Players:GetPlayers())do if player~=LocalPlayer then local character:Model?=player.Character local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')::Humanoid?local
root:BasePart?=character and character:FindFirstChild('HumanoidRootPart')::BasePart?if humanoid and humanoid.Health>0 and root then local distance:number=(root.Position-localRoot.Position).Magnitude
local score:number=distance if targetSettings.mode~='Nearest distance'and camera then local screenPoint:Vector3,visible:boolean=camera:WorldToViewportPoint(root.Position)score=visible and(Vector2.new(
screenPoint.X,screenPoint.Y)-pointer).Magnitude or math.huge end if score<bestScore then bestScore=score selectedTarget=player end end end end targetSource=targetSettings.mode end currentOverlayTarget
=targetInfoEnabled and selectedTarget or nil if selectedTarget and state.gameRoleProvider and not targetRole then local success:boolean,role:any=pcall(state.gameRoleProvider,selectedTarget)targetRole=
success and type(role)=='string'and role or nil end if targetRole then targetAccent=targetRole=='Murderer'and Theme.negative or((targetRole=='Sheriff'or targetRole=='Hero')and Theme.accent or Theme.
positive)end if targetInfoEnabled then local targetCharacter:Model?=selectedTarget and selectedTarget.Character local targetHumanoid:Humanoid?=targetCharacter and targetCharacter:
FindFirstChildOfClass('Humanoid')::Humanoid?local targetRoot:BasePart?=targetCharacter and targetCharacter:FindFirstChild('HumanoidRootPart')::BasePart?if selectedTarget and targetHumanoid then local
ratio:number=math.clamp(targetHumanoid.Health/math.max(1,targetHumanoid.MaxHealth),0,1)local distance:number?=localRoot and targetRoot and(targetRoot.Position-localRoot.Position).Magnitude or nil
TargetName.Text=selectedTarget.DisplayName..'  @'..selectedTarget.Name TargetRole.Text=tostring(targetRole or'Player')..'  \183  '..targetSource TargetRole.TextColor3=targetAccent TargetDetails.Text=
string.format('Health %.0f / %.0f%s',targetHumanoid.Health,targetHumanoid.MaxHealth,targetSettings.showDistance and distance and string.format('  \183  %.0f studs',distance)or'')TargetHealthTrack.
Visible=targetSettings.showHealth TargetHealthFill.Size=UDim2.fromScale(ratio,1)TargetHealthFill.BackgroundColor3=ratio>0.5 and Theme.accent or(ratio>0.25 and Theme.accentDim or Theme.accentFaint)
TargetPortraitFrame.Visible=targetSettings.showAvatar requestTargetThumbnail(selectedTarget)else TargetName.Text='No target'TargetRole.Text='No priority target'TargetDetails.Text=
'Waiting for a valid player'TargetHealthFill.Size=UDim2.fromScale(0,1)TargetHealthTrack.Visible=targetSettings.showHealth TargetPortraitFrame.Visible=targetSettings.showAvatar requestTargetThumbnail(
nil)end end if FloatingUi.overlays.Radar.enabled then local radarSize:number=radarSettings.size RadarSurface.Size=UDim2.fromOffset(radarSize,radarSize)local radarWindow:FloatingWindowRecord=FloatingUi
.overlays.Radar.window local nextRadarSize:UDim2=UDim2.fromOffset(radarSize,radarSize)if radarWindow.root.Size~=nextRadarSize then radarWindow.root.Size=nextRadarSize FloatingUi.reclamp()end for _,
axis:Frame in ipairs(radarAxes)do axis.Visible=radarSettings.grid end for _,ring:Frame in ipairs(radarRings)do ring.Visible=radarSettings.rings end RadarNorth.Visible=radarSettings.rotation=='North'
local activePlayers:{[Player]:boolean}={}local basis:CFrame?=localRoot and localRoot.CFrame or nil if localRoot and radarSettings.rotation=='Camera'and camera then local look:Vector3=Vector3.new(
camera.CFrame.LookVector.X,0,camera.CFrame.LookVector.Z)if look.Magnitude>0.01 then basis=CFrame.lookAt(localRoot.Position,localRoot.Position+look.Unit)end elseif localRoot and radarSettings.rotation
=='North'then basis=CFrame.new(localRoot.Position)end for _,player:Player in ipairs(Players:GetPlayers())do if player~=LocalPlayer and localRoot and basis then local character:Model?=player.Character
local humanoid:Humanoid?=character and character:FindFirstChildOfClass('Humanoid')::Humanoid?local targetRoot:BasePart?=character and character:FindFirstChild('HumanoidRootPart')::BasePart?local
sameTeam:boolean=LocalPlayer.Team~=nil and player.Team==LocalPlayer.Team local allowed:boolean=targetRoot~=nil and(not radarSettings.aliveOnly or(humanoid~=nil and humanoid.Health>0))and(radarSettings
.showTeam or not sameTeam)if allowed and targetRoot then local relative:Vector3=(basis::CFrame):PointToObjectSpace(targetRoot.Position)local distance:number=Vector2.new(relative.X,relative.Z).
Magnitude if distance<=radarSettings.range or radarSettings.outOfRange=='Clamp'then activePlayers[player]=true local record:any=radarBlips[player]if not record then local blip:Frame=create('Frame',{
Parent=RadarSurface,AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.accentFaint,BorderSizePixel=0,Size=UDim2.fromOffset(7,7)})::Frame blip:SetAttribute('FloatingDepth',4)create('UICorner',{
Parent=blip,CornerRadius=UDim.new(1,0)})local targetRing:UIStroke=create('UIStroke',{Parent=blip,Color=Theme.accent,Transparency=1,Thickness=2})::UIStroke local label:TextLabel=makeTextLabel(blip,'',7
)label.AnchorPoint=Vector2.new(0.5,0)label.Position=UDim2.new(0.5,0,1,3)label.Size=UDim2.fromOffset(100,18)label.TextColor3=Theme.text label.TextXAlignment=Enum.TextXAlignment.Center record={blip=blip
,label=label,ring=targetRing}radarBlips[player]=record end local radius:number=radarSize/2-12 local offset:Vector2=Vector2.new(relative.X,relative.Z)*(radius/radarSettings.range)if offset.Magnitude>
radius then offset=offset.Unit*radius end record.blip.Position=UDim2.fromOffset(radarSize/2+offset.X,radarSize/2+offset.Y)local role:string?=nil if state.gameRoleProvider then local roleOk:boolean,
roleValue:any=pcall(state.gameRoleProvider,player)role=roleOk and type(roleValue)=='string'and roleValue or nil end local color:Color3=Theme.accentFaint if radarSettings.colorMode=='Health'and
humanoid then local healthRatio:number=humanoid.Health/math.max(1,humanoid.MaxHealth)color=Color3.fromHSV(healthRatio*0.34,0.8,1)elseif radarSettings.colorMode=='Team'or radarSettings.colorMode==
'Smart'then color=sameTeam and Theme.positive or Theme.negative end if(radarSettings.colorMode=='MM2 role'or radarSettings.colorMode=='Smart')and role then color=role=='Murderer'and Theme.negative or(
(role=='Sheriff'or role=='Hero')and Theme.accent or Theme.positive)elseif radarSettings.colorMode=='Spectrum'then color=Color3.fromHSV((now*0.1+player.UserId%10/10)%1,0.7,1)end record.blip.
BackgroundColor3=color record.ring.Transparency=player==currentOverlayTarget and 0 or 1 record.blip.Size=UDim2.fromOffset(player==currentOverlayTarget and 10 or 7,player==currentOverlayTarget and 10
or 7)record.label.Visible=radarSettings.showNames or radarSettings.showDistance record.label.Text=(radarSettings.showNames and player.DisplayName or'')..(radarSettings.showDistance and string.format(
' %.0f',distance)or'')end end end end for player:Player,record:any in pairs(radarBlips)do if not activePlayers[player]then record.blip:Destroy()radarBlips[player]=nil end end end end,nil,
'Floating overlays'))FavoritesWindow.root.Visible=configData.ui.favoritesWindowOpen==true OverlayMenu.root.Visible=configData.ui.overlayMenuOpen==true for overlayId:string,overlay:any in pairs(
FloatingUi.overlays)do setOverlayEnabled(overlayId,configData.ui.overlays[overlayId]==true)end refreshFavorites()activeCleanup=function():()for _,handle:any in ipairs(ownedTasks)do if handle and
handle.Disconnect then handle:Disconnect()end end table.clear(ownedTasks)if type(state.destroySessionTracking)=='function'then state.destroySessionTracking()end end Module.Initialized=true return
state.floatingUi end function Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.Initialized=false end return Module
-- src/guis/Wurst/Code/Furniture.lua
local Module={Name='Furniture',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil local WURSTLOGO_BACKGROUND:Color3=Color3.fromRGB(255,255,255)local WURSTLOGO_TEXT
:Color3=Color3.fromRGB(0,0,0)local FURNITURE_MARGIN_X:number=14 local FURNITURE_MARGIN_Y:number=10 local LOGO_HEIGHT:number=36 local LOGO_WIDTH:number=142 local HACKLIST_MODE:string='Auto'local
HACKLIST_POSITION:string='Left'local HACKLIST_COLOR:Color3=Color3.fromRGB(255,255,255)local HACKLIST_SORT_BY:string='Name'local HACKLIST_REVERSE:boolean=false local HACKLIST_ANIMATIONS:boolean=true
local HACKLIST_FONT_SIZE:number=16 local HACKLIST_LINE_HEIGHT:number=18 local HACKLIST_TOP:number=56 local HACKLIST_SLIDE:number=24 local WURSTLOGO_VISIBILITY:string='Always'local PILL_BUTTON:number=
32 local PILL_PADDING_X:number=7 local PILL_PADDING_Y:number=5 local PILL_GAP:number=2 local TOOLTIP_DELAY_MS:number=400 function Module.init(context:any):any local host:any=context local state:any=
host.state local create:any=host.create local PRODUCT:any=host.PRODUCT local Theme:any=host.Theme local ThemeEngine:any=host.ThemeEngine local TaskManager:any=host.TaskManager local PopupLayer:any=
host.PopupLayer local TITLE_FONT:any=host.TITLE_FONT local CONTROL_FONT:any=host.CONTROL_FONT local TweenService:any=host.TweenService local allFeatures:{any}=host.allFeatures local configData:any=
host.configData local queueConfigSave:any=host.queueConfigSave local UserInputService:any=host.UserInputService local colorFromConfig:any=host.colorFromConfig local ScreenGui:any=host.ScreenGui local
makeTextBox:any=host.makeTextBox local notify:any=host.notify or function(_message:any):()end local Furniture:any={handles={}::{any},connections={}::{RBXScriptConnection}}for _,ghostName:string in
ipairs({'WurstLogo','HackList','Navigator','Pill'})do local ghost:Instance?=PopupLayer:FindFirstChild(ghostName)while ghost do ghost:Destroy()ghost=PopupLayer:FindFirstChild(ghostName)end if ScreenGui
then local screenGhost:Instance?=ScreenGui:FindFirstChild(ghostName)while screenGhost do screenGhost:Destroy()screenGhost=ScreenGui:FindFirstChild(ghostName)end end end local versionText:string=
tostring(PRODUCT.version)local versionWidth:number=8+#versionText*15 local chipWidth:number=LOGO_WIDTH+10+versionWidth+4 local chip:Frame=(create('Frame',{Parent=PopupLayer,Name='WurstLogo',Active=
false,BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.fromOffset(FURNITURE_MARGIN_X,FURNITURE_MARGIN_Y),Size=UDim2.fromOffset(chipWidth,LOGO_HEIGHT+8),ZIndex=20})::any)::Frame local
logoStripe:Frame=(create('Frame',{Parent=chip,Name='BackgroundStripe',BackgroundColor3=WURSTLOGO_BACKGROUND,BackgroundTransparency=0.5,BorderSizePixel=0,Position=UDim2.fromOffset(0,12),Size=UDim2.
fromOffset(chipWidth,22),ZIndex=20})::any)::Frame local logoImage:ImageLabel=(create('ImageLabel',{Parent=chip,Name='Logo',BackgroundTransparency=1,BorderSizePixel=0,Image='',Position=UDim2.
fromOffset(4,4),ScaleType=Enum.ScaleType.Fit,Size=UDim2.fromOffset(LOGO_WIDTH,LOGO_HEIGHT),Visible=false,ZIndex=21})::any)::ImageLabel local logoFallback:TextLabel=(create('TextLabel',{TextTruncate=
Enum.TextTruncate.AtEnd,Parent=chip,Name='LogoFallback',BackgroundTransparency=1,FontFace=TITLE_FONT,Position=UDim2.fromOffset(4,4),Size=UDim2.fromOffset(LOGO_WIDTH,LOGO_HEIGHT),Text=string.upper(
tostring(PRODUCT.name)),TextColor3=WURSTLOGO_TEXT,TextSize=14,TextXAlignment=Enum.TextXAlignment.Center,ZIndex=21})::any)::TextLabel local versionChip:Frame=(create('Frame',{Parent=chip,Name=
'VersionChip',BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.fromOffset(LOGO_WIDTH+10,12),Size=UDim2.fromOffset(versionWidth,22),ZIndex=20})::any)::Frame local bitmapText:any=state.
bitmapText local drawsVersion:boolean=bitmapText~=nil and type(bitmapText.draw)=='function'local versionColor:Color3=WURSTLOGO_TEXT local versionLabel:TextLabel?=nil local function drawVersion():()if
drawsVersion then bitmapText.draw(versionChip,{{text=versionText,color=versionColor}},{Name='Version',TextSize=20,ZIndex=21,align='Left'})end end if drawsVersion then drawVersion()else versionLabel=(
create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=versionChip,Name='Version',BackgroundTransparency=1,FontFace=TITLE_FONT,Position=UDim2.fromOffset(0,0),Size=UDim2.fromScale(1,1),Text=
versionText,TextColor3=WURSTLOGO_TEXT,TextSize=20,TextXAlignment=Enum.TextXAlignment.Left})::any)::TextLabel end local function fitLogoBand():()local measured:number=0 if drawsVersion and type(
bitmapText.measure)=='function'then measured=tonumber(bitmapText.measure({{text=versionText}},20))or 0 elseif versionLabel then local bounds:any=(versionLabel::any).TextBounds measured=bounds and
tonumber(bounds.X)or 0 end if measured<=0 then return end local fittedVersion:number=8+measured local fittedChip:number=LOGO_WIDTH+10+fittedVersion+4 versionChip.Size=UDim2.fromOffset(fittedVersion,22
)chip.Size=UDim2.fromOffset(fittedChip,LOGO_HEIGHT+8)logoStripe.Size=UDim2.fromOffset(fittedChip,22)end if versionLabel then table.insert(Furniture.connections,(versionLabel::TextLabel):
GetPropertyChangedSignal('TextBounds'):Connect(fitLogoBand))end if drawsVersion and type(bitmapText.onReady)=='table'then table.insert(bitmapText.onReady,fitLogoBand)end task.defer(fitLogoBand)if
type(state.applyAssetImage)=='function'then state.applyAssetImage('wurstLogo',logoImage,function():()logoImage.Visible=true logoFallback.Visible=false end)end local logoAlways:boolean=true
local function refreshLogoVisibility():()chip.Visible=logoAlways and state.visible==true end local function setLogoBackground(color:Color3):()logoStripe.BackgroundColor3=color end local function
setLogoTextColor(color:Color3):()versionColor=color if drawsVersion then drawVersion()elseif versionLabel then(versionLabel::TextLabel).TextColor3=color end logoFallback.TextColor3=color end
local function setLogoVisibility(value:string):()logoAlways=value=='Always'refreshLogoVisibility()end setLogoBackground(colorFromConfig(configData.values['WurstLogo.Background'],WURSTLOGO_BACKGROUND))
setLogoTextColor(colorFromConfig(configData.values['WurstLogo.Text'],WURSTLOGO_TEXT))local storedLogoVisibility:any=configData.values['WurstLogo.Visibility']setLogoVisibility(type(storedLogoVisibility
)=='string'and storedLogoVisibility or WURSTLOGO_VISIBILITY)state.wurstLogo={root=chip,SetBackground=setLogoBackground,SetTextColor=setLogoTextColor,SetVisibility=setLogoVisibility}local hudFrame:
Frame=(create('Frame',{Parent=PopupLayer,Name='HackList',Active=false,AnchorPoint=Vector2.new(1,0),BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.new(1,-12,0,HACKLIST_TOP),Size=UDim2.new(0,
260,1,-(HACKLIST_TOP+12)),ZIndex=20})::any)::Frame local hudLayout:UIListLayout=(create('UIListLayout',{Parent=hudFrame,FillDirection=Enum.FillDirection.Vertical,HorizontalAlignment=Enum.
HorizontalAlignment.Right,Padding=UDim.new(0,1),SortOrder=Enum.SortOrder.LayoutOrder})::any)::UIListLayout local hudMode:string=type(configData.values['HackList.Mode'])=='string'and configData.values[
'HackList.Mode']or HACKLIST_MODE local hudPosition:string=type(configData.values['HackList.Position'])=='string'and configData.values['HackList.Position']or HACKLIST_POSITION local hudColor:Color3=
colorFromConfig(configData.values['HackList.Color'],HACKLIST_COLOR)local hudSortBy:string=type(configData.values['HackList.Sortby'])=='string'and configData.values['HackList.Sortby']or
HACKLIST_SORT_BY local hudReverse:boolean=HACKLIST_REVERSE if configData.states['HackList.Reversesorting']~=nil then hudReverse=configData.states['HackList.Reversesorting']==true end local
hudAnimations:boolean=HACKLIST_ANIMATIONS if configData.states['HackList.Animations']~=nil then hudAnimations=configData.states['HackList.Animations']==true end local hudSize:string=type(configData.
values['HackList.Size'])=='string'and configData.values['HackList.Size']or'Normal'local hudRainbow:boolean=configData.states['HackList.Rainbow']==true local HUD_SIZES:{[string]:{font:number,line:
number}}={Small={font=12,line=14},Normal={font=HACKLIST_FONT_SIZE,line=HACKLIST_LINE_HEIGHT},Large={font=20,line=23}}local function hudMetrics():{font:number,line:number}return HUD_SIZES[hudSize]or
HUD_SIZES.Normal end local function hudOnLeft():boolean return hudPosition=='Left'or hudPosition=='Custom'end local navigatorBottom:()->number=function():number return FURNITURE_MARGIN_Y+LOGO_HEIGHT+8
end local hudCustomX:number?=tonumber(configData.values['HackList.CustomX'])local hudCustomY:number?=tonumber(configData.values['HackList.CustomY'])local function logicalViewport():Vector2 local
windows:any=state.windows if windows and type(windows.LogicalViewport)=='function'then local ok:boolean,viewport:any=pcall(windows.LogicalViewport)if ok and viewport then return viewport end end local
raw:Vector2=ScreenGui and ScreenGui.AbsoluteSize or Vector2.new(1280,720)return raw end local function clampHudCustom(x:number,y:number):(number,number)local viewport:Vector2=logicalViewport()return
math.clamp(x,0,math.max(viewport.X-260,0)),math.clamp(y,0,math.max(viewport.Y-32,0))end local function placeHudList():()if hudPosition=='Custom'then local x:number=hudCustomX or FURNITURE_MARGIN_X
local y:number=hudCustomY or navigatorBottom()x,y=clampHudCustom(x,y)hudCustomX,hudCustomY=x,y hudFrame.AnchorPoint=Vector2.new(0,0)hudFrame.Position=UDim2.fromOffset(x,y)hudLayout.HorizontalAlignment
=Enum.HorizontalAlignment.Left elseif hudOnLeft()then hudFrame.AnchorPoint=Vector2.new(0,0)hudFrame.Position=UDim2.new(0,FURNITURE_MARGIN_X,0,navigatorBottom())hudLayout.HorizontalAlignment=Enum.
HorizontalAlignment.Left else local windows:any=state.windows local logicalWidth:number?=nil if windows and type(windows.LogicalViewport)=='function'then local ok:boolean,logical:any=pcall(windows.
LogicalViewport)if ok and logical then logicalWidth=logical.X end end hudFrame.AnchorPoint=Vector2.new(1,0)if logicalWidth then hudFrame.Position=UDim2.new(0,logicalWidth-12,0,HACKLIST_TOP)else
hudFrame.Position=UDim2.new(1,-12,0,HACKLIST_TOP)end hudLayout.HorizontalAlignment=Enum.HorizontalAlignment.Right end end placeHudList()state.onLayout(placeHudList)local hudRows:{[string]:any}={}local
hudDragInput:InputObject?=nil local hudDragStart:Vector2=Vector2.zero local hudDragOrigin:Vector2=Vector2.zero local function hudHitTest(point:Vector2):boolean if hudPosition~='Custom'or state.visible
~=true then return false end for _,row:any in pairs(hudRows)do local target:GuiObject?=row.holder or row.container if target and target.Parent and target.Visible then local at:Vector2=target.
AbsolutePosition local size:Vector2=target.AbsoluteSize if size.X>0 and size.Y>0 and point.X>=at.X and point.X<=at.X+size.X and point.Y>=at.Y and point.Y<=at.Y+size.Y then return true end end end
return false end state.surfaceDragHitTest=hudHitTest hudFrame.Active=false hudFrame.ZIndex=20 table.insert(Furniture.connections,UserInputService.InputBegan:Connect(function(input:InputObject):()if
state.surfaceDragOwner~=nil or(input.UserInputType~=Enum.UserInputType.MouseButton1 and input.UserInputType~=Enum.UserInputType.Touch)or not hudHitTest(Vector2.new(input.Position.X,input.Position.Y))
then return end hudDragInput=input state.surfaceDragOwner=hudFrame hudDragStart=Vector2.new(input.Position.X,input.Position.Y)hudDragOrigin=Vector2.new(hudFrame.Position.X.Offset,hudFrame.Position.Y.
Offset)end))table.insert(Furniture.connections,UserInputService.InputChanged:Connect(function(input:InputObject):()if not hudDragInput or(input~=hudDragInput and not(hudDragInput.UserInputType==Enum.
UserInputType.MouseButton1 and input.UserInputType==Enum.UserInputType.MouseMovement))then return end local pointer:Vector2=Vector2.new(input.Position.X,input.Position.Y)local delta:Vector2=pointer-
hudDragStart local windows:any=state.windows if windows and type(windows.ScreenToLogical)=='function'then delta=windows.ScreenToLogical(delta)end local x:number,y:number=clampHudCustom(hudDragOrigin.X
+delta.X,hudDragOrigin.Y+delta.Y)hudCustomX,hudCustomY=x,y hudFrame.Position=UDim2.fromOffset(x,y)end))table.insert(Furniture.connections,UserInputService.InputEnded:Connect(function(input:InputObject
):()if not hudDragInput or(input~=hudDragInput and input.UserInputType~=Enum.UserInputType.MouseButton1)then return end hudDragInput=nil if state.surfaceDragOwner==hudFrame then state.surfaceDragOwner
=nil end if hudCustomX and hudCustomY then configData.values['HackList.CustomX']=hudCustomX configData.values['HackList.CustomY']=hudCustomY queueConfigSave()end end))local function animationsOn():
boolean return hudAnimations and state.animationsEnabled~=false end local hudRainbowClock:number=0 local function hudEntryColor(order:number?):Color3 if hudRainbow then local hue:number=(
hudRainbowClock*0.35+(order or 0)*0.05)%1 return Color3.fromHSV(hue,0.85,1)end if animationsOn()then local wave:number=0.5+0.5*math.sin(hudRainbowClock*2.4-(order or 0)*0.65)local brightness:number=
0.45+0.55*wave return Color3.new(hudColor.R*brightness,hudColor.G*brightness,hudColor.B*brightness)end return hudColor end local function drawHudEntry(row:any,name:string,status:string?):()row.name=
name row.status=status local segments:{any}={{text=name,color=hudEntryColor(row.order)}}if status and status~=''then table.insert(segments,{text=' ['..tostring(status)..']',color=Theme.textMuted})end
local bitmapText:any=state.bitmapText if bitmapText and type(bitmapText.draw)=='function'then row.holder=bitmapText.draw(row.container,segments,{Name='Entry',TextSize=hudMetrics().font,ZIndex=20,align
=hudOnLeft()and'Left'or'Right'})end end local function newHudRow(name:string):any local container:Frame=(create('Frame',{Parent=hudFrame,Name=name,BackgroundTransparency=1,BorderSizePixel=0,Size=UDim2
.new(1,0,0,hudMetrics().line),ZIndex=20})::any)::Frame return{container=container}end local function slideHudRow(row:any):()if not animationsOn()or not row.holder then return end local holder:Frame=
row.holder::Frame local resting:UDim2=holder.Position holder.Position=UDim2.new(resting.X.Scale,resting.X.Offset+HACKLIST_SLIDE,resting.Y.Scale,resting.Y.Offset)TweenService:Create(holder,TweenInfo.
new(0.25,Enum.EasingStyle.Back,Enum.EasingDirection.Out),{Position=resting}):Play()end local function dropHudRow(row:any):()if animationsOn()and row.holder then local holder:Frame=row.holder::Frame
local slid:UDim2=UDim2.new(holder.Position.X.Scale,holder.Position.X.Offset+HACKLIST_SLIDE,holder.Position.Y.Scale,holder.Position.Y.Offset)local tween:Tween=TweenService:Create(holder,TweenInfo.new(
0.15,Enum.EasingStyle.Quad,Enum.EasingDirection.In),{Position=slid})tween.Completed:Connect(function():()row.container:Destroy()end)tween:Play()else row.container:Destroy()end end local hudSignature:
string=''local hudElapsed:number=0 local hudSide:string=hudOnLeft()and'left'or'right'local hudAlways:boolean=configData.states['HackList.Show']==true local hudVisible:boolean=hudAlways==true
local function refreshHudList():()if not hudVisible then return end local entries:{any}={}for _,feature:any in ipairs(allFeatures)do if feature.enabled==true and not feature.isCategory then table.
insert(entries,{name=tostring(feature.name),status=feature.status})end end local function before(left:any,right:any):boolean if hudSortBy=='Width'then local leftWidth:number=#left.name+(left.status
and#tostring(left.status)+3 or 0)local rightWidth:number=#right.name+(right.status and#tostring(right.status)+3 or 0)if leftWidth~=rightWidth then return leftWidth>rightWidth end end return left.name<
right.name end table.sort(entries,function(left:any,right:any):boolean if hudReverse then return before(right,left)end return before(left,right)end)if hudMode=='Hidden'then entries={}elseif#entries>0
then local screenHeight:number local manager:any=state.windows if manager and type(manager.LogicalViewport)=='function'then screenHeight=manager.LogicalViewport().Y else local camera:Camera?=workspace
.CurrentCamera screenHeight=camera and(camera::Camera).ViewportSize.Y or 720 end local available:number=screenHeight-HACKLIST_TOP-12 local overflows:boolean=#entries*(hudMetrics().line+1)>available if
hudMode=='Count'or(hudMode=='Auto'and overflows)then entries={{name=tostring(#entries)..' hacks active',status=nil}}end end local signature:{string}={}for _,entry:any in ipairs(entries)do table.
insert(signature,entry.name..'\1'..tostring(entry.status or''))end local onLeft:boolean=hudOnLeft()local composed:string=table.concat(signature,'\2')..(onLeft and'\3left'or'\3right')if composed==
hudSignature then return end hudSignature=composed local side:string=onLeft and'left'or'right'local redrawAll:boolean=side~=hudSide hudSide=side local wanted:{[string]:boolean}={}for order:number,
entry:any in ipairs(entries)do wanted[entry.name]=true local row:any=hudRows[entry.name]local isNew:boolean=row==nil if not row then row=newHudRow(entry.name)hudRows[entry.name]=row end row.container.
LayoutOrder=order row.order=order if isNew or redrawAll or row.name~=entry.name or row.status~=entry.status then drawHudEntry(row,entry.name,entry.status)end if isNew then slideHudRow(row)end end for
name:string,row:any in pairs(hudRows)do if not wanted[name]then hudRows[name]=nil dropHudRow(row)end end end local hudRainbowSince:number=0 local function hudReanimate():()local bitmap:any=state.
bitmapText local canRecolor=bitmap~=nil and bitmap.recolor~=nil for _,row:any in pairs(hudRows)do if row.name and row.holder then local colors:{Color3}=row.colors if not colors then colors={hudColor,
Theme.textMuted}row.colors=colors end colors[1]=hudEntryColor(row.order)colors[2]=Theme.textMuted if not canRecolor or not bitmap.recolor(row.holder,colors)then drawHudEntry(row,row.name,row.status)
end end end end local function hudAnimating():boolean return hudVisible and(hudRainbow or animationsOn())end table.insert(Furniture.handles,TaskManager:Connect(function(deltaTime:number):()hudElapsed
+=deltaTime if hudElapsed>=0.25 then hudElapsed%=0.25 refreshHudList()end if not hudAnimating()then return end hudRainbowClock+=deltaTime hudRainbowSince+=deltaTime local interval:number=if hudRainbow
then 0.05 else 0.1 if hudRainbowSince>=interval then hudRainbowSince%=interval hudReanimate()end end,nil,'HUD animations'))refreshHudList()local function forceHudRefresh():()hudSignature='\0stale'
refreshHudList()end if state.bitmapText and type(state.bitmapText.onReady)=='table'then table.insert(state.bitmapText.onReady,forceHudRefresh)end hudFrame.Visible=hudVisible local function
setHackListVisible(value:boolean):()hudVisible=value~=false hudFrame.Visible=hudVisible if hudVisible then forceHudRefresh()for _,row:any in pairs(hudRows)do slideHudRow(row)end end end Furniture.
SetHackListVisible=setHackListVisible local function applyHudVisibility():()setHackListVisible(hudAlways==true)end if type(state.addMenuVisibilityListener)=='function'then state.
addMenuVisibilityListener(function(_visible:boolean):()if not hudAlways then setHackListVisible(false)end end)end state.hudList={frame=hudFrame,refresh=refreshHudList,SetVisible=setHackListVisible,
SetAlways=function(value:boolean):()hudAlways=value==true applyHudVisibility()end,SetMode=function(value:string):()hudMode=value forceHudRefresh()end,SetPosition=function(value:string):()hudPosition=
value placeHudList()if value=='Custom'then notify('Drag the HackList while the menu is open')end forceHudRefresh()end,SetColor=function(color:Color3):()hudColor=color forceHudRefresh()end,SetSortBy=
function(value:string):()hudSortBy=value forceHudRefresh()end,SetReverse=function(value:boolean):()hudReverse=value forceHudRefresh()end,SetAnimations=function(value:boolean):()hudAnimations=value
forceHudRefresh()end,SetSize=function(value:string):()hudSize=value for _,row:any in pairs(hudRows)do row.container.Size=UDim2.new(1,0,0,hudMetrics().line)end forceHudRefresh()end,SetRainbow=function(
value:boolean):()hudRainbow=value==true if not hudRainbow then forceHudRefresh()end end}local NAV_COLUMNS:number=3 local NAV_CELL_WIDTH:number=150 local NAV_CELL_HEIGHT:number=22 local NAV_CELL_GAP:
number=3 local NAV_WIDTH:number=NAV_COLUMNS*NAV_CELL_WIDTH+(NAV_COLUMNS-1)*NAV_CELL_GAP local navScreen:Frame=(create('Frame',{Parent=ScreenGui or PopupLayer,Name='Navigator',Active=false,
BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=1,BorderSizePixel=0,Size=UDim2.fromScale(1,1),Visible=false,ZIndex=246})::any)::Frame do local bitmap:any=state.bitmapText if bitmap and
type(bitmap.adoptTree)=='function'then table.insert(Furniture.connections,bitmap.adoptTree(navScreen))end end local navColumn:Frame=(create('Frame',{Parent=navScreen,Name='Column',AnchorPoint=Vector2.
new(0.5,0),BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=0.35,BorderSizePixel=0,Position=UDim2.new(0.5,0,0,48),Size=UDim2.fromOffset(NAV_WIDTH,32),ZIndex=247})::any)::Frame local
navStroke:UIStroke=(create('UIStroke',{Parent=navColumn,Name='NavigatorStroke',Color=Theme.outline,Thickness=1,Transparency=0.35})::any)::UIStroke local navSearch:TextBox=(create('TextBox',{
TextTruncate=Enum.TextTruncate.AtEnd,Parent=navColumn,Name='Search',BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=0.35,BorderSizePixel=0,ClearTextOnFocus=false,FontFace=CONTROL_FONT,
PlaceholderText='Search\u{2026}',Position=UDim2.fromOffset(0,0),Size=UDim2.new(1,0,0,32),Text='',TextColor3=Theme.text,TextSize=14,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=248})::any)::TextBox
create('UIPadding',{Parent=navSearch,PaddingLeft=UDim.new(0,6),PaddingRight=UDim.new(0,6)})local navResults:ScrollingFrame=(create('ScrollingFrame',{Parent=navColumn,Name='Results',Active=true,
AutomaticCanvasSize=Enum.AutomaticSize.None,BackgroundTransparency=1,BorderSizePixel=0,CanvasSize=UDim2.new(),Position=UDim2.fromOffset(0,38),ScrollBarThickness=3,ScrollingDirection=Enum.
ScrollingDirection.Y,Size=UDim2.new(1,0,1,-38),ZIndex=247})::any)::ScrollingFrame create('UIGridLayout',{Parent=navResults,CellPadding=UDim2.fromOffset(NAV_CELL_GAP,NAV_CELL_GAP),CellSize=UDim2.
fromOffset(NAV_CELL_WIDTH,NAV_CELL_HEIGHT),SortOrder=Enum.SortOrder.LayoutOrder})local function navPrefs():any if type(configData.values.NavigatorPrefs)~='table'then configData.values.NavigatorPrefs={
}end return configData.values.NavigatorPrefs end local navOrdered:{any}={}local navEntries:{TextButton}={}local navIndex:{[any]:number}={}local navSelected:number=1 local navCellConnections:{
RBXScriptConnection}={}local rebuildNavigator:()->()=function():()end local function paintNavEntry(index:number):()local entry:TextButton?=navEntries[index]local feature:any=navOrdered[index]if not
entry or not feature then return end local on:boolean=feature.enabled==true(entry::TextButton).BackgroundColor3=on and Theme.enabled or Color3.fromRGB(0,0,0);(entry::TextButton).BackgroundTransparency
=on and 0 or 0.35(entry::TextButton).TextColor3=on and Theme.enabledText or Theme.text local ring:UIStroke?=(entry::TextButton):FindFirstChild('SelectionRing')::UIStroke?if ring then(ring::UIStroke).
Transparency=index==navSelected and 0 or 1 end end local function paintNavSelection():()for index:number in ipairs(navEntries)do paintNavEntry(index)end end local function navActivate(index:number):()
local feature:any=navOrdered[index]if not feature then return end if type(feature.activate)=='function'then feature.activate()end navPrefs()[tostring(feature.name)]=(tonumber(navPrefs()[tostring(
feature.name)])or 0)+1 queueConfigSave()paintNavEntry(index)refreshHudList()end local closeNavigator:()->()=function():()end local function navOpenSettings(index:number):()local feature:any=navOrdered
[index]if not feature or type(feature.hasOptions)~='function'or not feature.hasOptions()then return end closeNavigator()if state.visible~=true then state.setMenuVisible(true)end if not feature.
expanded and type(feature.toggleExpansion)=='function'then feature.toggleExpansion()end end rebuildNavigator=function():()for _,connection:RBXScriptConnection in ipairs(navCellConnections)do pcall(
function():()connection:Disconnect()end)end table.clear(navCellConnections)for _,child:Instance in ipairs(navResults:GetChildren())do if child:IsA('GuiObject')then child:Destroy()end end local query:
string=string.lower(navSearch.Text or'')local matches:{any}={}for _,feature:any in ipairs(allFeatures)do if query==''or string.find(tostring(feature.searchText),query,1,true)then table.insert(matches,
feature)end end local prefs:any=navPrefs()table.sort(matches,function(left:any,right:any):boolean local leftUses:number=tonumber(prefs[left.name])or 0 local rightUses:number=tonumber(prefs[right.name]
)or 0 if leftUses~=rightUses then return leftUses>rightUses end return tostring(left.name)<tostring(right.name)end)navOrdered={}navEntries={}navIndex={}for index:number,feature:any in ipairs(matches)
do table.insert(navOrdered,feature)navIndex[feature]=index local entry:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=navResults,Name='Nav_'..tostring(feature.name):gsub(
'%W',''),Active=true,AutoButtonColor=false,BorderSizePixel=0,FontFace=CONTROL_FONT,LayoutOrder=index,Text=tostring(feature.name),TextSize=14,ZIndex=248})::any)::TextButton create('UIStroke',{Parent=
entry,Name='SelectionRing',Color=Theme.enabled,Thickness=1,Transparency=1})navEntries[index]=entry table.insert(navCellConnections,entry.MouseButton1Click:Connect(function():()navSelected=index
navActivate(index)paintNavSelection()end))end navSelected=math.clamp(navSelected,1,math.max(#navOrdered,1))local rows:number=math.ceil(#navOrdered/NAV_COLUMNS)local resultHeight:number=rows*
NAV_CELL_HEIGHT+math.max(rows-1,0)*NAV_CELL_GAP local screenHeight:number=ScreenGui and tonumber(ScreenGui.AbsoluteSize.Y)or 0 if screenHeight<=0 then screenHeight=480 end local columnHeight:number=
math.min(38+resultHeight,math.max(160,screenHeight-96))navColumn.Size=UDim2.fromOffset(NAV_WIDTH,columnHeight)navResults.Size=UDim2.fromOffset(NAV_WIDTH,columnHeight-38)navResults.CanvasSize=UDim2.
fromOffset(0,resultHeight)paintNavSelection()end local navigatorHidMenu:boolean=false local function openNavigator():()if state.visible==true and type(state.setMenuVisible)=='function'then
navigatorHidMenu=true state.setMenuVisible(false)else navigatorHidMenu=false end navScreen.Visible=true navSelected=1 rebuildNavigator()end closeNavigator=function():()navScreen.Visible=false
navSearch:ReleaseFocus()navSearch.Text=''if navigatorHidMenu then navigatorHidMenu=false if type(state.setMenuVisible)=='function'then state.setMenuVisible(true)end end end table.insert(Furniture.
connections,navSearch:GetPropertyChangedSignal('Text'):Connect(function():()if navScreen.Visible then rebuildNavigator()end end))table.insert(Furniture.connections,navSearch.FocusLost:Connect(function
(enterPressed:boolean):()if enterPressed and navScreen.Visible then navActivate(navSelected)end end))table.insert(Furniture.connections,UserInputService.InputBegan:Connect(function(input:InputObject,
_gameProcessed:boolean):()if not navScreen.Visible then return end local key:Enum.KeyCode=input.KeyCode local step:number=0 if key==Enum.KeyCode.Escape then closeNavigator()return elseif key==Enum.
KeyCode.Backspace then if navSearch.Text==''then closeNavigator()end return elseif key==Enum.KeyCode.Down then step=NAV_COLUMNS elseif key==Enum.KeyCode.Up then step=-NAV_COLUMNS elseif key==Enum.
KeyCode.Right or key==Enum.KeyCode.Tab then step=1 elseif key==Enum.KeyCode.Left then step=-1 elseif key==Enum.KeyCode.Space then if navSearch:IsFocused()then return end navOpenSettings(navSelected)
return else return end if#navOrdered>0 then navSelected=((navSelected-1+step)%#navOrdered)+1 paintNavSelection()end end))state.navigator={root=navScreen,refresh=rebuildNavigator,refreshFeature=
function(feature:any):()local index:number?=navIndex[feature]if navScreen.Visible and index then paintNavEntry(index)end end,Open=openNavigator,Close=closeNavigator}local touchPrimary:boolean=state.
isMobile==true local DOCK_SEARCH_WIDTH:number=190 local DOCK_BOTTOM_MARGIN:number=70 local dockSearchWidth:number=touchPrimary and 0 or DOCK_SEARCH_WIDTH local dockSearchGap:number=touchPrimary and 0
or PILL_GAP local menuButtonX:number=PILL_PADDING_X+dockSearchWidth+dockSearchGap local navButtonX:number=menuButtonX+PILL_BUTTON+PILL_GAP local pillWidth:number=PILL_PADDING_X*2+dockSearchWidth+
dockSearchGap+PILL_BUTTON+PILL_GAP+PILL_BUTTON local pillFrame:Frame=(create('Frame',{Parent=ScreenGui or PopupLayer,Name='Pill',Active=true,AnchorPoint=Vector2.new(0.5,1),BackgroundColor3=Color3.
fromRGB(16,18,28),BackgroundTransparency=0.28,BorderSizePixel=0,Position=UDim2.new(0.5,0,1,-DOCK_BOTTOM_MARGIN),Size=UDim2.fromOffset(pillWidth,PILL_BUTTON+PILL_PADDING_Y*2),ZIndex=249})::any)::Frame
create('UICorner',{Parent=pillFrame,CornerRadius=UDim.new(1,0)})create('UIStroke',{Parent=pillFrame,Color=Color3.fromRGB(255,255,255),Thickness=1,Transparency=0.91})local PILL_IDLE_TEXT:Color3=Color3.
fromRGB(232,232,238)local function showPillTip(anchor:GuiObject,text:string):()local tooltip:TextLabel?=state.featureTooltip if not tooltip then return end state.tooltipToken=state.tooltipToken+1
local bitmap:any=state.bitmapText local measured:number=bitmap and type(bitmap.measure)=='function'and tonumber(bitmap.measure({{text=text}},16))or#text*8 local width:number=math.clamp(measured+24,120
,320)local wrapped:string=text local lines:number=1 if bitmap and type(bitmap.wrap)=='function'then wrapped,lines=bitmap.wrap(text,width-24,16)end local resolvedTooltip:TextLabel=tooltip::TextLabel
resolvedTooltip.Text=wrapped resolvedTooltip.TextTransparency=1 resolvedTooltip.Size=UDim2.fromOffset(width,math.max(30,lines*18+12))resolvedTooltip.Position=UDim2.fromOffset(anchor.AbsolutePosition.X
,anchor.AbsolutePosition.Y-(math.max(30,lines*18+12)+8))if bitmap and type(bitmap.draw)=='function'then bitmap.draw(tooltip,{{text=wrapped}},{Name='Glyphs',TextSize=16,maxWidth=width-24,ellipsis=true,
ZIndex=251})else(tooltip::TextLabel).TextTransparency=0 end(tooltip::TextLabel).Visible=true end local function hidePillTip():()local tooltip:TextLabel?=state.featureTooltip if not tooltip then return
end state.tooltipToken=state.tooltipToken+1(tooltip::TextLabel).Visible=false end local function pillButton(order:number,offsetX:number,glyph:string,textSize:number,tip:string,onClick:()->()):
TextButton local button:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=pillFrame,Name='PillButton'..tostring(order),AutoButtonColor=false,BackgroundColor3=Color3.fromRGB(
255,255,255),BackgroundTransparency=1,BorderSizePixel=0,FontFace=CONTROL_FONT,Position=UDim2.fromOffset(offsetX,PILL_PADDING_Y),Size=UDim2.fromOffset(PILL_BUTTON,PILL_BUTTON),Text=glyph,TextColor3=
PILL_IDLE_TEXT,TextSize=textSize,ZIndex=241})::any)::TextButton create('UICorner',{Parent=button,CornerRadius=UDim.new(1,0)})table.insert(Furniture.connections,button.MouseEnter:Connect(function():()
if not UserInputService.MouseEnabled then return end state.tooltipToken=state.tooltipToken+1 local token:number=state.tooltipToken task.delay(TOOLTIP_DELAY_MS/1000,function():()if state.tooltipToken==
token and button.Parent then showPillTip(button,tip)end end)end))table.insert(Furniture.connections,button.MouseLeave:Connect(function():()hidePillTip()end))table.insert(Furniture.connections,button.
MouseButton1Click:Connect(function():()hidePillTip()onClick()end))return button end local menuButton:TextButton=pillButton(1,menuButtonX,'\u{2261}',16,'Show or hide the menu',function():()if navScreen
.Visible and type(state.SetMenuStyle)=='function'then state.SetMenuStyle('Wurst',true)return end if type(state.openMenuSurface)=='function'then state.openMenuSurface()else state.setMenuVisible(not
state.visible)end end)local navButton:TextButton=pillButton(2,navButtonX,'',16,'Navigator',function():()local navigator:any=state.navigator if not navigator then return end if navigator.root and
navigator.root.Visible then if type(navigator.Close)=='function'then navigator.Close()end elseif type(navigator.Open)=='function'then navigator.Open()end end)do local ring:Frame=(create('Frame',{
Parent=navButton,Name='LupaRing',AnchorPoint=Vector2.new(0.5,0.5),BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.new(0.5,-2,0.5,-2),Size=UDim2.fromOffset(11,11),ZIndex=navButton.ZIndex+1})
::any)::Frame create('UICorner',{Parent=ring,CornerRadius=UDim.new(1,0)})create('UIStroke',{Parent=ring,Color=PILL_IDLE_TEXT,Thickness=1.6,Transparency=0.1})local handle:Frame=(create('Frame',{Parent=
navButton,Name='LupaHandle',AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=PILL_IDLE_TEXT,BorderSizePixel=0,Position=UDim2.new(0.5,3,0.5,5),Rotation=45,Size=UDim2.fromOffset(2,7),ZIndex=navButton.
ZIndex+1})::any)::Frame local _=handle end local searchShell:Frame=(create('Frame',{Parent=pillFrame,Name='DockSearch',BackgroundColor3=Color3.fromRGB(255,255,255),BackgroundTransparency=0.94,
BorderSizePixel=0,Position=UDim2.fromOffset(PILL_PADDING_X,PILL_PADDING_Y),Size=UDim2.fromOffset(dockSearchWidth,PILL_BUTTON),Visible=not touchPrimary,ZIndex=241})::any)::Frame create('UICorner',{
Parent=searchShell,CornerRadius=UDim.new(1,0)})if not touchPrimary and type(makeTextBox)=='function'then local searchBox:TextBox=(makeTextBox(searchShell,'')::any)::TextBox searchBox.Name=
'DockSearchBox'searchBox.BackgroundTransparency=1 searchBox.ClearTextOnFocus=false searchBox.PlaceholderText='Search module...'searchBox.Position=UDim2.fromOffset(12,0)searchBox.Size=UDim2.new(1,-18,1
,0)searchBox.TextColor3=PILL_IDLE_TEXT searchBox.TextSize=14 searchBox.TextXAlignment=Enum.TextXAlignment.Left searchBox.ZIndex=242 local boxCorner:UICorner?=searchBox:FindFirstChild('StyleCorner')::
UICorner?if boxCorner then boxCorner:Destroy()end local boxStroke:UIStroke?=searchBox:FindFirstChild('StyleStroke')::UIStroke?if boxStroke then boxStroke:Destroy()end local function submitSearch():()
local query:string=tostring(searchBox.Text or'')if query:gsub('%s','')==''then return end searchBox.Text=''local search:any=state.moduleSearch if type(search)~='table'or type(search.Execute)~=
'function'then notify('module search unavailable')return end local result:any=search.Execute(query)local status:string=tostring(result and result.status)if status=='success'then searchBox.Text=''local
entryName:string=tostring(result.entry and result.entry.displayName or query)notify(entryName..': '..(result.enabled and'enabled'or'disabled'))elseif status=='ambiguous'then local names:{string}={}for
index:number,match:any in ipairs(result.matches or{})do if index>4 then table.insert(names,'\u{2026}')break end table.insert(names,tostring(match.displayName))end notify('ambiguous: '..table.concat(
names,', '))elseif status=='not_toggleable'then notify(tostring(result.entry and result.entry.displayName or query)..' has nothing to toggle')else notify('no module matches "'..query..'"')end end
table.insert(Furniture.connections,searchBox.FocusLost:Connect(function(enterPressed:boolean):()if enterPressed==true then submitSearch()end end))pcall(function():()table.insert(Furniture.connections,
(searchBox::any).ReturnPressedFromOnScreenKeyboard:Connect(submitSearch))end)end local function setDockVisible(value:boolean):()if touchPrimary then pillFrame.Visible=true return end pillFrame.Visible
=value~=false end if state.uiShowDock==nil then state.uiShowDock=touchPrimary end setDockVisible(state.uiShowDock==true)Furniture.SetDockVisible=setDockVisible state.dock={root=pillFrame,SetVisible=
setDockVisible}local moving=false local moveConnections,moveHandle,moveHint={},nil,nil local moveOrigin,dragInput,dragStart,dragAnchor local function dockAt(x,y)local viewport,size=ScreenGui.
AbsoluteSize,pillFrame.AbsoluteSize if viewport.X<1 or viewport.Y<1 then return end if size.X<1 or size.Y<1 then size=Vector2.new(pillWidth,PILL_BUTTON+PILL_PADDING_Y*2)end x=math.clamp(x,size.X*0.5,
math.max(size.X*0.5,viewport.X-size.X*0.5))y=math.clamp(y,size.Y,math.max(size.Y,viewport.Y))pillFrame.Position=UDim2.fromScale(x/math.max(1,viewport.X),y/math.max(1,viewport.Y))end local savedDock=
configData.ui.dockPosition if type(savedDock)=='table'and type(savedDock.x)=='number'and type(savedDock.y)=='number'and savedDock.x==savedDock.x and savedDock.y==savedDock.y then dockAt(math.clamp(
savedDock.x,0,1)*ScreenGui.AbsoluteSize.X,math.clamp(savedDock.y,0,1)*ScreenGui.AbsoluteSize.Y)end if type(savedDock)=='table'and type(savedDock.x)=='number'and type(savedDock.y)=='number'and
savedDock.x==savedDock.x and savedDock.y==savedDock.y then task.defer(function()if pillFrame.Parent then dockAt(math.clamp(savedDock.x,0,1)*ScreenGui.AbsoluteSize.X,math.clamp(savedDock.y,0,1)*
ScreenGui.AbsoluteSize.Y)end end)end local function finishDockMove(save)if not moving then return end moving=false dragInput=nil for _,connection in ipairs(moveConnections)do connection:Disconnect()
end table.clear(moveConnections)if moveHandle then moveHandle:Destroy()moveHandle=nil end if moveHint then moveHint:Destroy()moveHint=nil end if save then configData.ui.dockPosition={x=pillFrame.
Position.X.Scale,y=pillFrame.Position.Y.Scale}queueConfigSave()else pillFrame.Position=moveOrigin end setDockVisible(state.uiShowDock==true)end local function beginDockMove()if moving then
finishDockMove(true)return end moving=true moveOrigin=pillFrame.Position setDockVisible(true)moveHandle=create('TextButton',{Name='DockMoveHandle',Parent=pillFrame,Size=UDim2.fromScale(1,1),Text=
'Drag dock',FontFace=CONTROL_FONT,TextSize=14,TextColor3=Theme.text,BackgroundColor3=Theme.surface,BackgroundTransparency=0.15,BorderSizePixel=0,Active=true,ZIndex=450})moveHint=create('TextLabel',{
Name='DockMoveHint',Parent=ScreenGui,AnchorPoint=Vector2.new(0.5,0),Position=UDim2.new(0.5,0,0,48),Size=UDim2.fromOffset(330,26),Text='Drag and release \183 Enter saves \183 Esc cancels',FontFace=
CONTROL_FONT,TextSize=13,TextColor3=Theme.text,BackgroundColor3=Theme.surface,BackgroundTransparency=0.22,BorderSizePixel=0,ZIndex=450})local function connect(signal,fn)table.insert(moveConnections,
signal:Connect(fn))end connect(moveHandle.InputBegan,function(input)if input.UserInputType~=Enum.UserInputType.MouseButton1 and input.UserInputType~=Enum.UserInputType.Touch then return end dragInput=
input dragStart=Vector2.new(input.Position.X,input.Position.Y)dragAnchor=pillFrame.AbsolutePosition+Vector2.new(pillFrame.AbsoluteSize.X*0.5,pillFrame.AbsoluteSize.Y)end)connect(UserInputService.
InputChanged,function(input)if not dragInput then return end if input~=dragInput and not(dragInput.UserInputType==Enum.UserInputType.MouseButton1 and input.UserInputType==Enum.UserInputType.
MouseMovement)then return end local delta=Vector2.new(input.Position.X,input.Position.Y)-dragStart dockAt(dragAnchor.X+delta.X,dragAnchor.Y+delta.Y)end)connect(UserInputService.InputEnded,function(
input)if input==dragInput or dragInput and dragInput.UserInputType==Enum.UserInputType.MouseButton1 and input.UserInputType==Enum.UserInputType.MouseButton1 then finishDockMove(true)end end)connect(
UserInputService.InputBegan,function(input)if input.KeyCode==Enum.KeyCode.Escape then finishDockMove(false)elseif input.KeyCode==Enum.KeyCode.Return then finishDockMove(true)end end)connect(
UserInputService.WindowFocusReleased,function()finishDockMove(false)end)end Furniture.BeginDockMove=beginDockMove state.dock.BeginMove=beginDockMove state.dock.CancelMove=function()finishDockMove(
false)end table.insert(Furniture.connections,ScreenGui:GetPropertyChangedSignal('AbsoluteSize'):Connect(function()dockAt(pillFrame.Position.X.Scale*ScreenGui.AbsoluteSize.X+pillFrame.Position.X.Offset
,pillFrame.Position.Y.Scale*ScreenGui.AbsoluteSize.Y+pillFrame.Position.Y.Offset)end))state.addMenuVisibilityListener(function(visible)if not visible then finishDockMove(false)end end)local function
restyleMenuButton():()if not pillFrame.Parent then return end local open:boolean=state.visible==true menuButton.BackgroundColor3=Theme.accent menuButton.BackgroundTransparency=open and 0 or 1
menuButton.TextColor3=open and Theme.accentText or PILL_IDLE_TEXT end restyleMenuButton()state.addMenuVisibilityListener(function(visible:boolean):()restyleMenuButton()refreshLogoVisibility()
placeHudList()if visible then task.defer(function():()local manager:any=state.windows if not manager or type(manager.ReflowWindow)~='function'then return end for _,window:any in ipairs(manager.list or
{})do if(window.featureSettings or window.managementWindow)and window.root.Visible and not window.userMoved then manager.ReflowWindow(window)end end end)end end)refreshLogoVisibility()ThemeEngine.
OnChange(function():()restyleMenuButton()navSearch.TextColor3=Theme.text navStroke.Color=Theme.outline for _,entry:TextButton in ipairs(navEntries)do local ring:UIStroke?=entry:FindFirstChild(
'SelectionRing')::UIStroke?if ring then ring.Color=Theme.enabled end end paintNavSelection()end)state.pill={root=pillFrame,restyle=restyleMenuButton}if type(state.refreshLauncher)=='function'then
state.refreshLauncher()end activeCleanup=function():()finishDockMove(false)state.dock=nil for _,handle:any in ipairs(Furniture.handles)do pcall(function():()handle:Disconnect()end)end Furniture.
handles={}for _,connection:RBXScriptConnection in ipairs(Furniture.connections)do pcall(function():()connection:Disconnect()end)end Furniture.connections={}for _,connection:RBXScriptConnection in
ipairs(navCellConnections)do pcall(function():()connection:Disconnect()end)end table.clear(navCellConnections)if state.surfaceDragOwner==hudFrame then state.surfaceDragOwner=nil end if state.
surfaceDragHitTest==hudHitTest then state.surfaceDragHitTest=nil end chip:Destroy()hudFrame:Destroy()navScreen:Destroy()pillFrame:Destroy()state.hudList=nil state.wurstLogo=nil state.navigator=nil
state.pill=nil if type(state.refreshLauncher)=='function'then pcall(state.refreshLauncher)end end Module.Initialized=true return Furniture end function Module.destroy():()if activeCleanup then pcall(
activeCleanup)end activeCleanup=nil Module.Initialized=false end return Module
-- src/guis/Wurst/Code/MobileActions.lua
export type MobileHoldCallbacks={isActive:(()->boolean)?,onPress:(()->())?,onRelease:(()->())?}export type MobileActionRecord={id:string,label:string,button:TextButton,callback:()->(),hold:
MobileHoldCallbacks?,holding:boolean,activeInput:InputObject?,startPointer:Vector2,startPosition:Vector2,moved:boolean,holdToken:number,size:number,opacity:number,shape:string,color:string,connections
:{RBXScriptConnection}}export type PendingMobilePlacement={id:string,label:string,callback:()->(),hold:MobileHoldCallbacks?,restoreMenu:boolean}local Module={Name='MobileActions',PlaceId=0,Events={}::
{[string]:any},Initialized=false}function Module.init(context:any):any local host:any=context local state:any=host.state local configData:any=host.configData local queueConfigSave:any=host.
queueConfigSave local create:any=host.create local makeTextLabel:any=host.makeTextLabel local notify:any=host.notify local Theme:any=host.Theme local ThemeEngine:any=host.ThemeEngine local
CONTROL_FONT:any=host.CONTROL_FONT local ScreenGui:any=host.ScreenGui local PopupLayer:any=host.PopupLayer local TweenService:any=host.TweenService local UserInputService:any=host.UserInputService
local UI_MOTION:any=host.UI_MOTION local cornerRadius:any=host.cornerRadius local trackUiConnection:any=host.trackUiConnection local initialViewport:any=host.initialViewport or Vector2.new(1280,720)
type MobileActionRecord={id:string,label:string,button:TextButton,callback:()->(),hold:MobileHoldCallbacks?,holding:boolean,activeInput:InputObject?,startPointer:Vector2,startPosition:Vector2,moved:
boolean,holdToken:number,size:number,opacity:number,shape:string,color:string,connections:{RBXScriptConnection}}type PendingMobilePlacement={id:string,label:string,callback:()->(),hold:
MobileHoldCallbacks?,restoreMenu:boolean}local mobileActions:{[string]:MobileActionRecord}={}local pendingMobilePlacement:PendingMobilePlacement?=nil local MobilePlacementOverlay:Frame=(create('Frame'
,{Parent=PopupLayer,Name='MobilePlacementOverlay',Active=true,BackgroundColor3=Theme.background,BackgroundTransparency=0.2,BorderSizePixel=0,Size=UDim2.fromScale(1,1),Visible=false,ZIndex=200})::any)
::Frame local MobilePlacementLabel=makeTextLabel(MobilePlacementOverlay,'TOUCH WHERE THE ACTION BUTTON SHOULD GO',16)MobilePlacementLabel.FontFace=CONTROL_FONT MobilePlacementLabel.TextXAlignment=Enum
.TextXAlignment.Center MobilePlacementLabel.Position=UDim2.new(0.08,0,0.08,0)MobilePlacementLabel.Size=UDim2.new(0.84,0,0,54)MobilePlacementLabel.ZIndex=201 local MOBILE_SHAPES:{string}={'Square',
'Rounded','Circle'}local MOBILE_PALETTE:{string}={'#F2F2F2','#2B2B2B','#0A84FF','#30D158','#FF9F0A','#FF375F','#BF5AF2','#64D2FF'}local SHAPE_CORNERS:{[string]:UDim}={Square=UDim.new(0,2),Rounded=UDim
.new(0,12),Circle=UDim.new(1,0)}local changeListeners:{()->()}={}local function colorToHex(color:Color3):string local ok:boolean,hex:any=pcall(function():any return(Color3::any).toHex(color)end)if ok
and type(hex)=='string'then if string.sub(hex,1,1)~='#'then hex='#'..hex end return string.upper(hex)end return string.format('#%02X%02X%02X',math.floor(color.R*255+0.5),math.floor(color.G*255+0.5),
math.floor(color.B*255+0.5))end local function colorFromHex(value:string):Color3?if type(value)~='string'or value==''then return nil end if string.sub(value,1,1)~='#'then value='#'..value end local ok
:boolean,color:any=pcall(function():any return(Color3::any).fromHex(value)end)if ok and typeof(color)=='Color3'then return color end return nil end local function readableTextColor(background:Color3):
Color3 local luminance:number=0.2126*background.R+0.7152*background.G+0.0722*background.B return luminance>0.55 and Color3.fromRGB(18,18,18)or Color3.fromRGB(242,242,242)end local function
ensureMobileConfig():{[string]:any}if type(configData.ui.mobileActions)~='table'then configData.ui.mobileActions={}end return configData.ui.mobileActions end local function ensureMobileDefaults():{[
string]:any}if type(configData.ui.mobileButtons)~='table'then configData.ui.mobileButtons={}end local defaults:{[string]:any}=configData.ui.mobileButtons if type(defaults.shape)~='string'or not table.
find(MOBILE_SHAPES,defaults.shape)then defaults.shape='Square'end if type(defaults.color)~='string'or not colorFromHex(defaults.color)then defaults.color=colorToHex(Theme.surfaceRaised)end return
defaults end local function applyMobileStyle(record:MobileActionRecord):()local corner:UICorner?=record.button:FindFirstChildOfClass('UICorner')if corner then corner.CornerRadius=SHAPE_CORNERS[record.
shape]or SHAPE_CORNERS.Rounded end local color:Color3?=colorFromHex(record.color)if color then record.button.BackgroundColor3=color record.button.TextColor3=readableTextColor(color)end end
local function notifyMobileChanged():()for _,listener:()->()in ipairs(changeListeners)do pcall(listener)end end local function clampMobileButtonPosition(position:Vector2,size:number):Vector2 local
camera:Camera?=workspace.CurrentCamera local viewport:Vector2=camera and camera.ViewportSize or Vector2.new(1280,720)local radius:number=size/2 return Vector2.new(math.clamp(position.X,radius+6,math.
max(radius+6,viewport.X-radius-6)),math.clamp(position.Y,radius+6,math.max(radius+6,viewport.Y-radius-6)))end local function saveMobileActionPosition(record:MobileActionRecord):()local mobileConfig:{[
string]:any}=ensureMobileConfig()mobileConfig[record.id]={record.button.Position.X.Offset,record.button.Position.Y.Offset,record.size,record.opacity,record.shape,record.color}queueConfigSave()
notifyMobileChanged()end local function removeMobileAction(id:string):()local record:MobileActionRecord?=mobileActions[id]if not record then return end mobileActions[id]=nil ensureMobileConfig()[id]=
nil for _,connection:RBXScriptConnection in ipairs(record.connections)do connection:Disconnect()end table.clear(record.connections)if record.button.Parent then record.button:Destroy()end
queueConfigSave()notifyMobileChanged()notify(record.label..' mobile button removed.')end local editingRecord:MobileActionRecord?=nil local editorOverlay:Frame=(create('Frame',{Parent=ScreenGui,Name=
'MobileActionEditor',Active=true,BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=0.45,BorderSizePixel=0,Size=UDim2.fromScale(1,1),Visible=false,ZIndex=270})::any)::Frame local
editorPanel:Frame=(create('Frame',{Parent=editorOverlay,Name='Panel',AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.surfaceRaised,BackgroundTransparency=0.04,BorderSizePixel=0,Position=UDim2.
fromScale(0.5,0.5),Size=UDim2.fromOffset(300,300),ZIndex=271})::any)::Frame create('UICorner',{Parent=editorPanel,CornerRadius=UDim.new(0,12)})local editorStroke:UIStroke=(create('UIStroke',{Parent=
editorPanel,Name='WindowOutline',Color=Theme.outline,Thickness=1,Transparency=0.08})::any)::UIStroke if ThemeEngine and type(ThemeEngine.Bind)=='function'then ThemeEngine.Bind(editorStroke,'Color',
'outline')end local editorTitle:TextLabel=(makeTextLabel(editorPanel,'Mobile button',16)::any)::TextLabel editorTitle.Position=UDim2.fromOffset(14,8)editorTitle.Size=UDim2.new(1,-28,0,28)editorTitle.
TextXAlignment=Enum.TextXAlignment.Left editorTitle.ZIndex=272 local function editorLabel(text:string,y:number):TextLabel local label:TextLabel=(makeTextLabel(editorPanel,text,14)::any)::TextLabel
label.Position=UDim2.fromOffset(14,y)label.Size=UDim2.fromOffset(76,36)label.TextXAlignment=Enum.TextXAlignment.Left label.ZIndex=272 return label end editorLabel('Size',46)editorLabel('Opacity',88)
editorLabel('Shape',130)editorLabel('Colour',172)local function editorButton(name:string,text:string,x:number,y:number,width:number):TextButton local button:TextButton=(create('TextButton',{
TextTruncate=Enum.TextTruncate.AtEnd,Parent=editorPanel,Name=name,Active=true,AutoButtonColor=false,BackgroundColor3=Theme.surface,BackgroundTransparency=0.08,BorderSizePixel=0,FontFace=CONTROL_FONT,
Position=UDim2.fromOffset(x,y),Size=UDim2.fromOffset(width,34),Text=text,TextColor3=Theme.text,TextSize=14,ZIndex=272})::any)::TextButton create('UICorner',{Parent=button,CornerRadius=UDim.new(0,7)})
return button end local sizeDown:TextButton=editorButton('SizeDown','-',92,47,40)local sizeValue:TextButton=editorButton('SizeValue','52 px',138,47,72)local sizeUp:TextButton=editorButton('SizeUp','+'
,216,47,40)sizeValue.Active=false local opacityDown:TextButton=editorButton('OpacityDown','-',92,89,40)local opacityValue:TextButton=editorButton('OpacityValue','92%',138,89,72)local opacityUp:
TextButton=editorButton('OpacityUp','+',216,89,40)opacityValue.Active=false local shapeButton:TextButton=editorButton('Shape','Square',92,131,164)shapeButton.TextXAlignment=Enum.TextXAlignment.Center
local removeButton:TextButton=editorButton('Remove','Remove',14,214,78)removeButton.BackgroundColor3=Color3.fromRGB(120,28,28)local resetButton:TextButton=editorButton('Reset','Reset',101,214,78)local
doneButton:TextButton=editorButton('Done','Done',188,214,78)local applyAllButton:TextButton=editorButton('ApplyAll','Apply to all buttons',14,256,272)doneButton.BackgroundColor3=Theme.enabled
doneButton.TextColor3=Theme.enabledText if ThemeEngine and type(ThemeEngine.Bind)=='function'then ThemeEngine.Bind(doneButton,'BackgroundColor3','enabled')ThemeEngine.Bind(doneButton,'TextColor3',
'enabledText')end local function closeEditor():()editingRecord=nil editorOverlay.Visible=false end local refreshEditor:()->()=function():()end local swatches:{[string]:TextButton}={}for index:number,
hex:string in ipairs(MOBILE_PALETTE)do local color:Color3=colorFromHex(hex)or Theme.surfaceRaised local swatch:TextButton=(create('TextButton',{Parent=editorPanel,Name='Colour'..tostring(index),Active
=true,AutoButtonColor=false,BackgroundColor3=color,BorderSizePixel=0,Position=UDim2.fromOffset(92+(index-1)*25,168),Size=UDim2.fromOffset(22,22),Text='',ZIndex=272})::any)::TextButton create(
'UICorner',{Parent=swatch,CornerRadius=UDim.new(1,0)})local ring:UIStroke=(create('UIStroke',{Parent=swatch,Name='SelectedRing',Color=Theme.text,Thickness=1,Transparency=0.85})::any)::UIStroke swatch:
SetAttribute('Hex',hex)swatches[hex]=swatch trackUiConnection(swatch.MouseButton1Click:Connect(function():()local record:MobileActionRecord?=editingRecord if not record then return end record.color=
hex refreshEditor()end))local _=ring end local function refreshSwatches(selected:string):()for hex:string,swatch:TextButton in pairs(swatches)do local ring:UIStroke?=swatch:FindFirstChild(
'SelectedRing')::UIStroke?if ring then local active:boolean=string.upper(hex)==string.upper(selected)ring.Thickness=active and 2 or 1 ring.Transparency=active and 0 or 0.85 ring.Color=active and Theme
.text or Theme.outline end end end refreshEditor=function():()local record:MobileActionRecord?=editingRecord if not record then return end editorTitle.Text=record.label sizeValue.Text=tostring(math.
round(record.size))..' px'opacityValue.Text=tostring(math.round(record.opacity*100))..'%'shapeButton.Text=record.shape record.button.Size=UDim2.fromOffset(record.size,record.size)record.button.
BackgroundTransparency=1-record.opacity applyMobileStyle(record)refreshSwatches(record.color)local clamped:Vector2=clampMobileButtonPosition(Vector2.new(record.button.Position.X.Offset,record.button.
Position.Y.Offset),record.size)record.button.Position=UDim2.fromOffset(clamped.X,clamped.Y)saveMobileActionPosition(record)end local function openEditor(record:MobileActionRecord):()editingRecord=
record editorOverlay.Visible=true refreshEditor()end trackUiConnection(sizeDown.MouseButton1Click:Connect(function():()if editingRecord then editingRecord.size=math.clamp(editingRecord.size-8,36,120)
refreshEditor()end end))trackUiConnection(sizeUp.MouseButton1Click:Connect(function():()if editingRecord then editingRecord.size=math.clamp(editingRecord.size+8,36,120)refreshEditor()end end))
trackUiConnection(opacityDown.MouseButton1Click:Connect(function():()if editingRecord then editingRecord.opacity=math.clamp(editingRecord.opacity-0.1,0.2,1)refreshEditor()end end))trackUiConnection(
opacityUp.MouseButton1Click:Connect(function():()if editingRecord then editingRecord.opacity=math.clamp(editingRecord.opacity+0.1,0.2,1)refreshEditor()end end))trackUiConnection(shapeButton.
MouseButton1Click:Connect(function():()local record:MobileActionRecord?=editingRecord if not record then return end local index:number=table.find(MOBILE_SHAPES,record.shape)or 1 record.shape=
MOBILE_SHAPES[index%#MOBILE_SHAPES+1]refreshEditor()end))trackUiConnection(resetButton.MouseButton1Click:Connect(function():()local record:MobileActionRecord?=editingRecord if record then local
defaults:{[string]:any}=ensureMobileDefaults()record.size=52 record.opacity=0.92 record.shape=defaults.shape record.color=defaults.color refreshEditor()end end))trackUiConnection(applyAllButton.
MouseButton1Click:Connect(function():()local source:MobileActionRecord?=editingRecord if not source then return end for _,record:MobileActionRecord in pairs(mobileActions)do record.size=source.size
record.opacity=source.opacity record.shape=source.shape record.color=source.color record.button.Size=UDim2.fromOffset(record.size,record.size)record.button.BackgroundTransparency=1-record.opacity
applyMobileStyle(record)saveMobileActionPosition(record)end notify('mobile button style applied to all.')end))trackUiConnection(removeButton.MouseButton1Click:Connect(function():()local record:
MobileActionRecord?=editingRecord closeEditor()if record then removeMobileAction(record.id)end end))trackUiConnection(doneButton.MouseButton1Click:Connect(closeEditor))local function
setMobileActionVisibility(menuVisible:boolean):()if menuVisible then closeEditor()end for _,record:MobileActionRecord in pairs(mobileActions)do record.button.Visible=state.isMobile and not menuVisible
end end local function createMobileAction(id:string,label:string,callback:()->(),requestedPosition:Vector2?,hold:MobileHoldCallbacks?):MobileActionRecord?if not state.isMobile then return nil end
local existing:MobileActionRecord?=mobileActions[id]if existing then existing.callback=callback existing.label=label existing.hold=hold return existing end local stored:any=ensureMobileConfig()[id]
local size:number=math.clamp(type(stored)=='table'and tonumber(stored[3])or tonumber(state.mobileButtonSize)or 52,36,120)local opacity:number=math.clamp(type(stored)=='table'and tonumber(stored[4])or
0.92,0.2,1)local defaults:{[string]:any}=ensureMobileDefaults()local shape:string=type(stored)=='table'and type(stored[5])=='string'and table.find(MOBILE_SHAPES,stored[5])and stored[5]or defaults.
shape local color:string=type(stored)=='table'and type(stored[6])=='string'and colorFromHex(stored[6])and stored[6]or defaults.color local actionCount:number=0 for _ in pairs(mobileActions)do
actionCount+=1 end local defaultPosition:Vector2=Vector2.new(74+(actionCount*(size+10)),math.max(size,initialViewport.Y-size-70))local selectedPosition:Vector2=requestedPosition or(type(stored)==
'table'and Vector2.new(tonumber(stored[1])or defaultPosition.X,tonumber(stored[2])or defaultPosition.Y)or defaultPosition)selectedPosition=clampMobileButtonPosition(selectedPosition,size)local button=
create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,ClipsDescendants=true,Parent=ScreenGui,Name='MobileAction_'..id:gsub('%W',''),Active=true,AutoButtonColor=false,BackgroundColor3=Theme.
surfaceRaised,BackgroundTransparency=1-opacity,BorderSizePixel=0,FontFace=CONTROL_FONT,Position=UDim2.fromOffset(selectedPosition.X,selectedPosition.Y),AnchorPoint=Vector2.new(0.5,0.5),Size=UDim2.
fromOffset(size,size),Text=label,TextColor3=Theme.text,TextScaled=true,TextWrapped=true,Visible=not state.visible,ZIndex=180})::TextButton create('UIPadding',{Parent=button,PaddingBottom=UDim.new(0,4)
,PaddingLeft=UDim.new(0,4),PaddingRight=UDim.new(0,4),PaddingTop=UDim.new(0,4)})create('UICorner',{Parent=button,CornerRadius=cornerRadius(12)})local buttonOutline:UIStroke=(create('UIStroke',{Parent=
button,Name='MobileActionOutline',Color=Theme.outline,Transparency=0.35,Thickness=1})::any)::UIStroke if ThemeEngine and type(ThemeEngine.Bind)=='function'then ThemeEngine.Bind(buttonOutline,'Color',
'outline')end create('UITextSizeConstraint',{Parent=button,MaxTextSize=12,MinTextSize=6})local record:MobileActionRecord={id=id,label=label,button=button,callback=callback,hold=hold,holding=false,
activeInput=nil,startPointer=Vector2.zero,startPosition=selectedPosition,moved=false,holdToken=0,size=size,opacity=opacity,shape=shape,color=color,connections={}}mobileActions[id]=record
applyMobileStyle(record)saveMobileActionPosition(record)table.insert(record.connections,trackUiConnection(button.InputBegan:Connect(function(input:InputObject):()if input.UserInputType~=Enum.
UserInputType.Touch then return end record.activeInput=input record.startPointer=Vector2.new(input.Position.X,input.Position.Y)record.startPosition=Vector2.new(button.Position.X.Offset,button.Position
.Y.Offset)record.moved=false local hold:MobileHoldCallbacks?=record.hold local holdActive:boolean=hold~=nil and hold.onPress~=nil and(hold.isActive==nil or(hold.isActive::()->boolean)())if holdActive
and not record.holding then record.holding=true button.BackgroundTransparency=math.max(0,1-record.opacity-0.1)task.spawn((hold::MobileHoldCallbacks).onPress::()->())end record.holdToken+=1 local token
:number=record.holdToken task.delay(0.75,function():()if mobileActions[id]~=record or record.activeInput~=input or record.moved or record.holdToken~=token then return end record.activeInput=nil record
.holdToken+=1 if record.holding then record.holding=false if record.hold and record.hold.onRelease then task.spawn(record.hold.onRelease)end end button.BackgroundTransparency=1-record.opacity
openEditor(record)end)end)))table.insert(record.connections,trackUiConnection(UserInputService.InputEnded:Connect(function(input:InputObject):()if input.UserInputType~=Enum.UserInputType.Touch or
record.activeInput~=input then return end record.activeInput=nil record.holdToken+=1 if record.holding then record.holding=false button.BackgroundTransparency=1-record.opacity if record.hold and
record.hold.onRelease then task.spawn(record.hold.onRelease)end if record.moved then saveMobileActionPosition(record)end elseif record.moved then saveMobileActionPosition(record)else TweenService:
Create(button,TweenInfo.new(0.08,Enum.EasingStyle.Quad),{BackgroundTransparency=math.max(0,1-record.opacity-0.1)}):Play()task.delay(0.09,function():()if button.Parent then TweenService:Create(button,
TweenInfo.new(UI_MOTION.fast,Enum.EasingStyle.Quad),{BackgroundTransparency=1-record.opacity}):Play()end end)task.spawn(record.callback)end end)))return record end trackUiConnection(UserInputService.
InputChanged:Connect(function(input:InputObject):()if input.UserInputType~=Enum.UserInputType.Touch then return end for _,record:MobileActionRecord in pairs(mobileActions)do if record.activeInput==
input then local pointer:Vector2=Vector2.new(input.Position.X,input.Position.Y)local delta:Vector2=pointer-record.startPointer if delta.Magnitude>=7 then record.moved=true if record.holding then
record.holding=false record.button.BackgroundTransparency=1-record.opacity if record.hold and record.hold.onRelease then task.spawn(record.hold.onRelease)end end end if record.moved then local
position:Vector2=clampMobileButtonPosition(record.startPosition+delta,record.button.AbsoluteSize.X)record.button.Position=UDim2.fromOffset(position.X,position.Y)end end end end))local function
beginMobileActionPlacement(id:string,label:string,callback:()->(),hold:MobileHoldCallbacks?):()if not state.isMobile then return end if mobileActions[id]then openEditor(mobileActions[id])return end
pendingMobilePlacement={id=id,label=label,callback=callback,hold=hold,restoreMenu=state.visible}MobilePlacementLabel.Text='TOUCH TO PLACE '..string.upper(label)..(hold and
'\nHOLD THE BUTTON TO KEEP IT ACTIVE'or'\nHOLD THE BUTTON LATER TO CUSTOMIZE IT')MobilePlacementOverlay.Visible=true state.setMenuVisible(false)local placement:PendingMobilePlacement=
pendingMobilePlacement::PendingMobilePlacement task.delay(8,function():()if pendingMobilePlacement==placement then pendingMobilePlacement=nil MobilePlacementOverlay.Visible=false if placement.
restoreMenu then state.setMenuVisible(true)end notify('Mobile button placement cancelled.')end end)end trackUiConnection(MobilePlacementOverlay.InputBegan:Connect(function(input:InputObject):()local
placement:PendingMobilePlacement?=pendingMobilePlacement if not placement or input.UserInputType~=Enum.UserInputType.Touch then return end pendingMobilePlacement=nil MobilePlacementOverlay.Visible=
false createMobileAction(placement.id,placement.label,placement.callback,Vector2.new(input.Position.X,input.Position.Y),placement.hold)if placement.restoreMenu then state.setMenuVisible(true)end
notify(placement.label..' mobile button placed.')end))local function bindMobileActionPlacement(source:TextButton,id:string,label:string,callback:()->(),hold:MobileHoldCallbacks?):()if not state.
```

```
isMobile then return end trackUiConnection(source.MouseButton1Click:Connect(function():()local existing:MobileActionRecord?=mobileActions[id]if existing then openEditor(existing)else
beginMobileActionPlacement(id,label,callback,hold)end end))if ensureMobileConfig()[id]~=nil then createMobileAction(id,label,callback,nil,hold)end end state.bindMobileActionPlacement=
bindMobileActionPlacement local function resetMobileActions():()local ids:{string}={}for id:string in pairs(mobileActions)do table.insert(ids,id)end for _,id:string in ipairs(ids)do
removeMobileAction(id)end configData.ui.mobileActions={}queueConfigSave()end state.mobileUi={actions=mobileActions,setActionVisibility=setMobileActionVisibility,resetActions=resetMobileActions,
clampButtonPosition=clampMobileButtonPosition,saveActionPosition=saveMobileActionPosition,editor=editorOverlay,openEditor=openEditor,closeEditor=closeEditor,place=beginMobileActionPlacement,defaults=
ensureMobileDefaults,shapes=MOBILE_SHAPES,palette=MOBILE_PALETTE,style=applyMobileStyle,colorFromHex=colorFromHex,colorToHex=colorToHex,notifyChanged=notifyMobileChanged,onChanged=function(listener:(
)->()):()table.insert(changeListeners,listener)end,actionId=function(shortcutId:string):string return'Shortcut'..tostring(shortcutId):gsub('%W','')end}state.addMenuVisibilityListener(
setMobileActionVisibility)Module.Initialized=true return state.mobileUi end function Module.destroy():()Module.Initialized=false end return Module
-- src/guis/Wurst/Code/SettingsPage.lua
local Module={Name='SettingsPage',PlaceId=0,Events={}::{[string]:any},Initialized=false}local activeCleanup:(()->())?=nil local WINDOW_WIDTH:number=214 local ROW_HEIGHT:number=22 local WINDOW_GAP:
number=2 local TITLE_HEIGHT:number=26 function Module.init(context:any):any local OPACITY:number=0.85 local TOOLTIP_OPACITY:number=0.75 local MAX_HEIGHT:number=200 local MAX_SETTINGS_HEIGHT:number=200
local HACKLIST_MODE:string='Auto'local HACKLIST_POSITION:string='Left'local HACKLIST_COLOR:Color3=Color3.fromRGB(255,255,255)local HACKLIST_SORT_BY:string='Name'local HACKLIST_REVERSE:boolean=false
local HACKLIST_ANIMATIONS:boolean=true local HACKLIST_SIZE:string='Normal'local HACKLIST_RAINBOW:boolean=false local host:any=context local state:any=host.state local configData:any=host.configData
local queueConfigSave:any=host.queueConfigSave local create:any=host.create local makeButton:any=host.makeButton local makeTextBox:any=host.makeTextBox local notify:any=host.notify local Theme:any=
host.Theme local ThemeEngine:any=host.ThemeEngine local CONTROL_FONT:any=host.CONTROL_FONT local UserInputService:any=host.UserInputService local trackUiConnection:any=host.trackUiConnection local
beginKeyCapture:any=host.beginKeyCapture local setKeySlotCapture:any=host.setKeySlotCapture local bindActionShortcut:any=host.bindActionShortcut local GAME_CHECK:any=host.GAME_CHECK local ScreenGui:
any=host.ScreenGui local allFeatures:{any}=host.allFeatures or{}local REPOSITORY_RAW_BASE:any=host.REPOSITORY_RAW_BASE local RUNTIME_SAFETY_SOURCE_URL:any=host.RUNTIME_SAFETY_SOURCE_URL local
RUNTIME_COMPATIBILITY_MARKER:any=host.RUNTIME_COMPATIBILITY_MARKER local SOURCE_STAMP:any=host.SOURCE_STAMP local addToggleOption:any=host.addToggleOption local addNumberOption:any=host.
addNumberOption local addCycleOption:any=host.addCycleOption local addColorOption:any=host.addColorOption local drawPixelIcon:any=host.drawPixelIcon or function(parent:Instance,_kind:string,_color:
Color3?):Frame local existing:Instance?=(parent::any):FindFirstChild('PixelIcon')if existing then existing:Destroy()end return create('Frame',{Parent=parent,Name='PixelIcon',BackgroundTransparency=1})
::Frame end local windows:any=state.windows if not windows then error('SettingsPage requires the window manager')end local Settings:any={records={}::{any},connections={}::{RBXScriptConnection}}
local function viewport():Vector2 if type(windows.LogicalViewport)=='function'then return windows.LogicalViewport()end local camera:Camera?=workspace.CurrentCamera return camera and(camera::Camera).
ViewportSize or Vector2.new(1280,720)end local maxSettingsHeight:number=MAX_SETTINGS_HEIGHT state.uiMaxHeight=MAX_HEIGHT local function textWidth(text:any):number local bitmapText:any=state.bitmapText
if bitmapText and type(bitmapText.measure)=='function'then local measured:number=tonumber(bitmapText.measure({{text=tostring(text)}},16))or 0 if measured>0 then return measured end end return#
tostring(text)*12 end local function packedWidth(rows:{{any}}):number local widest:number=150 for _,row:{any}in ipairs(rows)do local labelWidth:number=textWidth(row[1])local valueWidth:number=row[2]~=
nil and textWidth(row[2])or 0 local extra:number=tonumber(row[3])or 0 widest=math.max(widest,8+labelWidth+24+valueWidth+extra)end return widest end local function configWindow(id:string,title:string,
definition:any):any local packRows:{{any}}?=definition.packRows local width:number=packRows and packedWidth(packRows)or tonumber(definition.width)or WINDOW_WIDTH local window:any=windows.Create({id=id
,title=title,size=Vector2.new(width,30),position=definition.position,closable=definition.closable~=false})window.managementWindow=true if definition.anchor~=nil then window.placementAnchor=definition.
anchor end do local bitmapText:any=state.bitmapText if bitmapText and type(bitmapText.adoptTree)=='function'then table.insert(window.connections,bitmapText.adoptTree(window.body))end end local layout:
UIListLayout=(create('UIListLayout',{Parent=window.body,Padding=UDim.new(0,WINDOW_GAP),SortOrder=Enum.SortOrder.LayoutOrder})::any)::UIListLayout create('UIPadding',{Parent=window.body,PaddingTop=UDim
.new(0,WINDOW_GAP),PaddingBottom=UDim.new(0,WINDOW_GAP),PaddingLeft=UDim.new(0,WINDOW_GAP),PaddingRight=UDim.new(0,WINDOW_GAP)})local record:any={window=window,layout=layout,open=false,width=width,
packRows=packRows}local function resize():()local limit:number if definition.settingsSized then limit=maxSettingsHeight else limit=tonumber(state.uiMaxHeight)or MAX_HEIGHT end if limit<=0 then limit=
math.huge end if type(windows.LogicalViewport)=='function'then limit=math.min(limit,windows.LogicalViewport().Y-12)end local wanted:number=math.min(TITLE_HEIGHT+layout.AbsoluteContentSize.Y+WINDOW_GAP
*2,limit)window.fullHeight=wanted if not window.collapsed then window.root.Size=UDim2.fromOffset(record.width,wanted)end if windows.ReflowWindow then windows.ReflowWindow(window)elseif windows.
ClampWindow then windows.ClampWindow(window)end end record.resize=resize record.repack=function(rowsOverride:{{any}}?):()if rowsOverride~=nil then record.packRows=rowsOverride end if record.packRows==
nil then return end local packed:number=packedWidth(record.packRows)if packed~=record.width then record.width=packed resize()end end table.insert(Settings.connections,layout:GetPropertyChangedSignal(
'AbsoluteContentSize'):Connect(resize))resize()table.insert(Settings.records,record)return record end local function panelFor(record:any,configKey:string):any local panel:any={options=record.window.
body,optionCount=0,optionRows={},configKey=configKey,searchText=''}panel.updateExpandedSize=function():()record.resize()end return panel end local function actionRow(record:any,order:number,labelText:
string,onClick:()->()):TextButton local row:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=record.window.body,Name='Row_'..labelText:gsub('%W',''),Active=true,
AutoButtonColor=false,BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=0.5,BorderSizePixel=0,FontFace=CONTROL_FONT,LayoutOrder=order,Size=UDim2.new(1,0,0,ROW_HEIGHT),Text='',ZIndex=record
.window.body.ZIndex+1})::any)::TextButton create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,Name='Label',BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.fromOffset(6,0)
,Size=UDim2.new(1,-28,1,0),Text=labelText,TextColor3=Theme.text,TextSize=16,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=row.ZIndex+1})local arrowBox:Frame=(create('Frame',{Parent=row,Name='Arrow',
AnchorPoint=Vector2.new(1,0),BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.new(1,-4,0,0),Size=UDim2.fromOffset(22,ROW_HEIGHT),ZIndex=row.ZIndex+1})::any)::Frame drawPixelIcon(arrowBox,
'right',nil,'arrows')trackUiConnection(row.MouseEnter:Connect(function():()row.BackgroundTransparency=0.3 end))trackUiConnection(row.MouseLeave:Connect(function():()row.BackgroundTransparency=0.5 end)
)trackUiConnection(row.MouseButton1Click:Connect(onClick))record.resize()return row end local function raiseOpen(record:any):()record.open=true if not record.placed or record.window.stalePlacement
then record.placed=true record.window.stalePlacement=nil if windows.PlaceWindow then windows.PlaceWindow(record.window)end end record.window:SetVisible(true)end local uiRecord:any=nil local
hackListRecord:any=nil local function openHackList():()if hackListRecord then raiseOpen(hackListRecord)return end local screen:Vector2=viewport()hackListRecord=configWindow('HackList','HackList',{
position=Vector2.new(screen.X-WINDOW_WIDTH*2-32,180),anchor=uiRecord and uiRecord.window,settingsSized=true,packRows={{'Show',nil,30},{'Mode','Hidden',26},{'Position','Right',26},{'Color','#FFFFFF'},{
'Sort by','Width',26},{'Reverse sorting',nil,30},{'Animations',nil,30},{'Size','Normal',26},{'Rainbow',nil,30}}})local panel:any=panelFor(hackListRecord,'HackList')local showStored:boolean=configData.
states['HackList.Show']==true addToggleOption(panel,'Show',showStored,function(value:boolean):()state.hudListAlways=value configData.states['HackList.Show']=value queueConfigSave()if state.hudList and
type(state.hudList.SetAlways)=='function'then state.hudList.SetAlways(value)end if state.hudList and type(state.hudList.SetVisible)=='function'then state.hudList.SetVisible(value)end end,
[[Show the hack list. OFF keeps it off the screen, even with the menu open.]])state.hudListAlways=showStored if state.hudList and type(state.hudList.SetAlways)=='function'then state.hudList.SetAlways(
showStored)end local modes:{string}={'Auto','Count','Hidden'}addCycleOption(panel,'Mode',modes,table.find(modes,HACKLIST_MODE)or 1,function(value:string):()if state.hudList then state.hudList.SetMode(
value)end end)local positions:{string}={'Left','Right','Custom'}addCycleOption(panel,'Position',positions,table.find(positions,HACKLIST_POSITION)or 1,function(value:string):()if state.hudList then
state.hudList.SetPosition(value)end end)addColorOption(panel,'Color',HACKLIST_COLOR,function(color:Color3):()if state.hudList then state.hudList.SetColor(color)end end)local sorts:{string}={'Name',
'Width'}addCycleOption(panel,'Sort by',sorts,table.find(sorts,HACKLIST_SORT_BY)or 1,function(value:string):()if state.hudList then state.hudList.SetSortBy(value)end end)addToggleOption(panel,
'Reverse sorting',HACKLIST_REVERSE,function(value:boolean):()if state.hudList then state.hudList.SetReverse(value)end end)addToggleOption(panel,'Animations',HACKLIST_ANIMATIONS,function(value:boolean)
:()if state.hudList then state.hudList.SetAnimations(value)end end)local sizes:{string}={'Small','Normal','Large'}addCycleOption(panel,'Size',sizes,table.find(sizes,HACKLIST_SIZE)or 2,function(value:
string):()if state.hudList and type(state.hudList.SetSize)=='function'then state.hudList.SetSize(value)end end)addToggleOption(panel,'Rainbow',HACKLIST_RAINBOW,function(value:boolean):()if state.
hudList and type(state.hudList.SetRainbow)=='function'then state.hudList.SetRainbow(value)end end)raiseOpen(hackListRecord)end local keybindsRecord:any=nil local profilesRecord:any=nil local
keybindsList:Frame?=nil local function featureLabel(bindingId:string):string for _,feature:any in ipairs(allFeatures)do if feature.configKey==bindingId then return tostring(feature.name)end end return
bindingId end local function keyLabel(name:any):string if type(state.keyDisplayName)=='function'then return state.keyDisplayName(name)end return tostring(name or'')end local function persistBindingKey
(binding:any):()local optionKey:string='Shortcut.'..tostring(binding.id):gsub('%W','')if binding.key==Enum.KeyCode.Unknown then configData.values[optionKey]=nil else configData.values[optionKey]=
binding.key.Name end queueConfigSave()end local rebuildKeybinds:()->()=function():()end local function sortedBindings():{any}local list:{any}={}for _,binding:any in pairs(state.shortcutBindings)do
table.insert(list,binding)end table.sort(list,function(left:any,right:any):boolean return featureLabel(tostring(left.id))<featureLabel(tostring(right.id))end)return list end rebuildKeybinds=function()
:()local container:Frame?=keybindsList if not container then return end for _,child:Instance in ipairs((container::Frame):GetChildren())do if child:IsA('GuiObject')then child:Destroy()end end local
keyUses:{[string]:number}={}for _,binding:any in pairs(state.shortcutBindings)do if binding.key and binding.key~=Enum.KeyCode.Unknown then local name:string=binding.key.Name keyUses[name]=(keyUses[
name]or 0)+1 end end local order:number=0 local function keybindRow(labelText:string,slotText:string,conflicted:boolean,onSlot:(TextButton)->(),onRemove:(()->())?):()order+=1 local row:Frame=(create(
'Frame',{Parent=container,Name='Bind_'..labelText:gsub('%W',''),BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=order,Size=UDim2.new(1,0,0,ROW_HEIGHT),ZIndex=(container::Frame).ZIndex+1})::any)
::Frame create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,Name='Label',BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.fromOffset(6,0),Size=UDim2.new(1,-160,1,0),Text=
labelText,TextColor3=Theme.text,TextSize=16,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=row.ZIndex+1})local slot:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,
Name='Key',Active=true,AnchorPoint=Vector2.new(1,0),AutoButtonColor=false,BackgroundTransparency=1,BorderSizePixel=0,FontFace=CONTROL_FONT,Position=UDim2.new(1,-6,0,0),Size=UDim2.fromOffset(150,
ROW_HEIGHT),Text=slotText,TextColor3=conflicted and Theme.negative or Theme.textMuted,TextSize=16,TextXAlignment=Enum.TextXAlignment.Right,ZIndex=row.ZIndex+1})::any)::TextButton slot:SetAttribute(
'CaptureIdleText',slotText)trackUiConnection(slot.MouseButton1Click:Connect(function():()onSlot(slot)end))if onRemove then trackUiConnection(slot.InputBegan:Connect(function(input:InputObject):()if
input.UserInputType==Enum.UserInputType.MouseButton2 then(onRemove::()->())()end end))end end if not UserInputService.TouchEnabled then keybindRow('Open menu',keyLabel(state.toggleKey.Name),false,
function(slot:TextButton):()if state.keyCaptureCancel then state.keyCaptureCancel()end state.keybindButton=slot state.waitingForKey=true slot.Text='...'setKeySlotCapture(slot,true)end)end for _,
binding:any in ipairs(sortedBindings())do local bound:boolean=binding.key~=nil and binding.key~=Enum.KeyCode.Unknown if not bound then continue end local slotText:string=keyLabel(binding.key.Name)
local conflicted:boolean=(keyUses[binding.key.Name]or 0)>1 keybindRow(featureLabel(tostring(binding.id)),slotText,conflicted,function(slot:TextButton):()beginKeyCapture(slot,function(keyCode:Enum.
KeyCode):()binding.key=keyCode==binding.key and Enum.KeyCode.Unknown or keyCode persistBindingKey(binding)if binding.refresh then binding.refresh()end rebuildKeybinds()end)end,function():()binding.key
=Enum.KeyCode.Unknown persistBindingKey(binding)if binding.refresh then binding.refresh()end rebuildKeybinds()notify(featureLabel(tostring(binding.id))..' unbound')end)end if keybindsRecord then local
rows:{{any}}={{'Open menu','RShift',12},{'Reset keybinds',nil,26},{'Add',nil,26},{'Profiles\u{2026}',nil,26}}for _,binding:any in ipairs(sortedBindings())do if binding.key and binding.key~=Enum.
KeyCode.Unknown then table.insert(rows,{featureLabel(tostring(binding.id)),keyLabel(binding.key.Name),12})end end if keybindsRecord.repack then keybindsRecord.repack(rows)end keybindsRecord.resize()
end end local function profileStore():any if type(configData.values.Profiles)~='table'then configData.values.Profiles={}end return configData.values.Profiles end local PROFILE_STORE_KEYS:{[string]:
boolean}={Profiles=true,KeybindProfiles=true}local function captureProfile():any local states:{[string]:any}={}for key:string,value:any in pairs(configData.states)do states[key]=value end local values
:{[string]:any}={}for key:string,value:any in pairs(configData.values)do if not PROFILE_STORE_KEYS[key]then values[key]=value end end local shortcuts:{[string]:string}={}for id:string,binding:any in
pairs(state.shortcutBindings)do if binding.key and binding.key~=Enum.KeyCode.Unknown then shortcuts[id]=binding.key.Name end end local enabledModules:number=0 for _,feature:any in ipairs(allFeatures)
do if configData.states[feature.configKey]==true then enabledModules+=1 end end local optionCount:number=0 for _ in pairs(values)do optionCount+=1 end return{savedAt=os.time(),states=states,values=
values,shortcuts=shortcuts,enabledModules=enabledModules,optionCount=optionCount}end local function applyProfile(name:string):boolean local snapshot:any=profileStore()[name]if type(snapshot)~='table'
then return false end for key:string in pairs(configData.states)do configData.states[key]=nil end for key:string,value:any in pairs(snapshot.states)do configData.states[key]=value end for key:string,
value:any in pairs(snapshot.values)do if not PROFILE_STORE_KEYS[key]then configData.values[key]=value end end for _,binding:any in pairs(state.shortcutBindings)do local savedName:any=snapshot.
shortcuts and snapshot.shortcuts[tostring(binding.id)]local savedKey:any=Enum.KeyCode.Unknown if type(savedName)=='string'and savedName~=''then local resolved:boolean,candidate:any=pcall(function():
any return(Enum.KeyCode::any)[savedName]end)if resolved then savedKey=candidate end end if savedKey==Enum.KeyCode.Unknown then if binding.key~=Enum.KeyCode.Unknown and type(binding.assign)=='function'
then binding.assign(binding.key)end elseif binding.key~=savedKey and type(binding.assign)=='function'then binding.assign(savedKey)elseif type(binding.refresh)=='function'then binding.refresh()end end
for _,feature:any in ipairs(allFeatures)do local wanted:boolean=snapshot.states[feature.configKey]==true if feature.enabled~=wanted and type(feature.activate)=='function'then pcall(feature.activate)
end end for key:string,control:any in pairs(state.toggleControls or{})do local saved:any=snapshot.states[key]if saved~=nil and type(control.setChecked)=='function'then pcall(control.setChecked,saved==
true)end end for key:string,control:any in pairs(state.valueControls or{})do local saved:any=snapshot.values[key]if saved~=nil and type(control.apply)=='function'then pcall(control.apply,saved)end end
queueConfigSave()return true end local function panicReset():()for _,feature:any in ipairs(allFeatures)do if feature.enabled and type(feature.activate)=='function'then pcall(feature.activate)end end
for _,control:any in pairs(state.toggleControls or{})do if type(control.setChecked)=='function'then pcall(control.setChecked,false)end end for _,control:any in pairs(state.valueControls or{})do if
control.default~=nil and type(control.apply)=='function'then pcall(control.apply,control.default)end end for key:string in pairs(configData.states)do configData.states[key]=nil end for key:string in
pairs(configData.values)do if not PROFILE_STORE_KEYS[key]and not string.match(key,'^Shortcut%.')then configData.values[key]=nil end end queueConfigSave()notify(
'panic: every module and option is back to defaults')end local rebuildProfiles:()->()=function():()end local profilesList:Frame?=nil local function openProfiles():()if profilesRecord then raiseOpen(
profilesRecord)rebuildProfiles()return end local screen:Vector2=viewport()profilesRecord=configWindow('Profiles','Profiles',{position=Vector2.new(screen.X-WINDOW_WIDTH*2-32,260),anchor=keybindsRecord
and keybindsRecord.window,settingsSized=true})local entryRow:Frame=(create('Frame',{Parent=profilesRecord.window.body,Name='NewProfile',BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=1,Size=
UDim2.new(1,0,0,ROW_HEIGHT),ZIndex=profilesRecord.window.body.ZIndex+1})::any)::Frame local nameBox:TextBox=(makeTextBox(entryRow,'')::any)::TextBox nameBox.PlaceholderText='New profile name'nameBox.
Position=UDim2.fromOffset(0,0)nameBox.Size=UDim2.new(1,-170,1,0)nameBox.TextSize=14 nameBox.ZIndex=entryRow.ZIndex+1 local saveButton:TextButton=(makeButton(entryRow,'SAVE',10)::any)::TextButton
saveButton.AnchorPoint=Vector2.new(1,0.5)saveButton.Position=UDim2.new(1,-66,0.5,0)saveButton.Size=UDim2.fromOffset(56,20)saveButton.ZIndex=entryRow.ZIndex+1 local panicButton:TextButton=(makeButton(
entryRow,'PANIC',10)::any)::TextButton panicButton.AnchorPoint=Vector2.new(1,0.5)panicButton.Position=UDim2.new(1,0,0.5,0)panicButton.Size=UDim2.fromOffset(56,20)panicButton.ZIndex=entryRow.ZIndex+1
trackUiConnection(panicButton.MouseButton1Click:Connect(function():()panicReset()rebuildProfiles()end))profilesList=(create('Frame',{Parent=profilesRecord.window.body,Name='ProfileList',AutomaticSize=
Enum.AutomaticSize.Y,BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=2,Size=UDim2.new(1,0,0,0),ZIndex=profilesRecord.window.body.ZIndex+1})::any)::Frame create('UIListLayout',{Parent=
profilesList,Padding=UDim.new(0,WINDOW_GAP),SortOrder=Enum.SortOrder.LayoutOrder})rebuildProfiles=function():()local container:Frame?=profilesList if not container then return end for _,child:Instance
in ipairs((container::Frame):GetChildren())do if child:IsA('GuiObject')then child:Destroy()end end local names:{string}={}for name:string in pairs(profileStore())do table.insert(names,name)end table.
sort(names,function(a:string,b:string):boolean local left:any=profileStore()[a]local right:any=profileStore()[b]local leftAt:number=type(left)=='table'and tonumber(left.savedAt)or 0 local rightAt:
number=type(right)=='table'and tonumber(right.savedAt)or 0 if leftAt~=rightAt then return leftAt>rightAt end return a<b end)for index:number,name:string in ipairs(names)do local snapshot:any=
profileStore()[name]local row:Frame=(create('Frame',{Parent=container,Name='Profile_'..name:gsub('%W',''),BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=index,Size=UDim2.new(1,0,0,44),ZIndex=(
container::Frame).ZIndex+1})::any)::Frame create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,Name='Label',BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.fromOffset(6,2)
,Size=UDim2.new(1,-180,0,18),Text=name,TextColor3=Theme.text,TextSize=14,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=row.ZIndex+1})local detail:string='profile'if type(snapshot)=='table'then local
savedAt:number=tonumber(snapshot.savedAt)or 0 local when:string=savedAt>0 and os.date('%Y-%m-%d %H:%M',savedAt)or'unknown date'detail=when..' \183 '..tostring(snapshot.enabledModules or 0)..
' modules on \183 '..tostring(snapshot.optionCount or 0)..' options'end create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,Name='Detail',BackgroundTransparency=1,FontFace=CONTROL_FONT
,Position=UDim2.fromOffset(6,22),Size=UDim2.new(1,-180,0,16),Text=detail,TextColor3=Theme.textMuted,TextSize=12,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=row.ZIndex+1})local loadButton:TextButton
=(makeButton(row,'LOAD',10)::any)::TextButton loadButton.AnchorPoint=Vector2.new(1,0.5)loadButton.Position=UDim2.new(1,-56,0.5,0)loadButton.Size=UDim2.fromOffset(48,18)loadButton.ZIndex=row.ZIndex+1
trackUiConnection(loadButton.MouseButton1Click:Connect(function():()if applyProfile(name)then notify('loaded profile '..name)rebuildProfiles()else notify('profile '..name..' is corrupted')end end))
local updateButton:TextButton=(makeButton(row,'UPD',10)::any)::TextButton updateButton.AnchorPoint=Vector2.new(1,0.5)updateButton.Position=UDim2.new(1,-108,0.5,0)updateButton.Size=UDim2.fromOffset(44,
18)updateButton.ZIndex=row.ZIndex+1 trackUiConnection(updateButton.MouseButton1Click:Connect(function():()profileStore()[name]=captureProfile()queueConfigSave()notify('updated profile '..name)
rebuildProfiles()end))local deleteButton:TextButton=(create('TextButton',{Parent=row,Name='Delete',AutoButtonColor=false,BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=0.5,
BorderSizePixel=0,FontFace=CONTROL_FONT,AnchorPoint=Vector2.new(1,0.5),Position=UDim2.new(1,-6,0.5,0),Size=UDim2.fromOffset(18,18),Text='x',TextColor3=Theme.negative,TextSize=12,ZIndex=row.ZIndex+1})
::any)::TextButton trackUiConnection(deleteButton.MouseButton1Click:Connect(function():()profileStore()[name]=nil queueConfigSave()notify('deleted profile '..name)rebuildProfiles()end))end if
profilesRecord then if profilesRecord.repack then local rows:{{any}}={{'New profile name',nil,60}}for _,name:string in ipairs(names)do table.insert(rows,{name,nil,64})end profilesRecord.repack(rows)
end profilesRecord.resize()end end trackUiConnection(saveButton.MouseButton1Click:Connect(function():()local name:string=nameBox.Text:gsub('^%s+',''):gsub('%s+$','')if name==''then notify(
'enter a profile name')return end profileStore()[name]=captureProfile()queueConfigSave()nameBox.Text=''rebuildProfiles()notify('saved profile '..name)end))rebuildProfiles()raiseOpen(profilesRecord)end
local addRecord:any=nil local addList:Frame?=nil local pendingConflict:any=nil local rebuildAddList:()->()=function():()end local function applyBinding(binding:any,keyCode:Enum.KeyCode,clearOther:any)
:()binding.key=keyCode persistBindingKey(binding)if binding.refresh then binding.refresh()end if clearOther then clearOther.key=Enum.KeyCode.Unknown persistBindingKey(clearOther)if clearOther.refresh
then clearOther.refresh()end end pendingConflict=nil rebuildKeybinds()rebuildAddList()notify(featureLabel(tostring(binding.id))..' bound to '..keyCode.Name)end rebuildAddList=function():()local
container:Frame?=addList if not container then return end for _,child:Instance in ipairs((container::Frame):GetChildren())do if child:IsA('GuiObject')then child:Destroy()end end local order:number=0
local function addRow(labelText:string,valueText:string,warning:boolean,onClick:((TextButton)->())?):()order+=1 local row:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=
container,Name='Add_'..labelText:gsub('%W',''),Active=onClick~=nil,AutoButtonColor=false,BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=onClick~=nil and 0.5 or 1,BorderSizePixel=0,
FontFace=CONTROL_FONT,LayoutOrder=order,Size=UDim2.new(1,0,0,ROW_HEIGHT),Text='',ZIndex=(container::Frame).ZIndex+1})::any)::TextButton create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=
row,Name='Label',BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.fromOffset(6,0),Size=UDim2.new(1,-120,1,0),Text=labelText,TextColor3=warning and Theme.negative or Theme.text,TextSize=16
,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=row.ZIndex+1})local slot:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,Name='Key',Active=onClick~=nil,AnchorPoint=
Vector2.new(1,0),AutoButtonColor=false,BackgroundTransparency=1,BorderSizePixel=0,FontFace=CONTROL_FONT,Position=UDim2.new(1,-6,0,0),Size=UDim2.fromOffset(110,ROW_HEIGHT),Text=valueText,TextColor3=
Theme.textMuted,TextSize=16,TextXAlignment=Enum.TextXAlignment.Right,ZIndex=row.ZIndex+1})::any)::TextButton slot:SetAttribute('CaptureIdleText',valueText)if onClick then trackUiConnection(row.
MouseButton1Click:Connect(function():()(onClick::(TextButton)->())(slot)end))trackUiConnection(slot.MouseButton1Click:Connect(function():()(onClick::(TextButton)->())(slot)end))trackUiConnection(row.
MouseEnter:Connect(function():()row.BackgroundTransparency=0.3 end))trackUiConnection(row.MouseLeave:Connect(function():()row.BackgroundTransparency=0.5 end))end end if pendingConflict then local
conflict:any=pendingConflict addRow(conflict.key.Name..' already binds '..featureLabel(tostring(conflict.other.id)),'',true,nil)addRow('Replace it','',false,function(_slot:TextButton):()applyBinding(
conflict.binding,conflict.key,conflict.other)end)addRow('Keep both','',false,function(_slot:TextButton):()applyBinding(conflict.binding,conflict.key,nil)end)addRow('Cancel','',false,function(_slot:
TextButton):()pendingConflict=nil rebuildAddList()end)else addRow('Click an action, then press a key','',false,nil)for _,binding:any in ipairs(sortedBindings())do local bound:boolean=binding.key~=nil
and binding.key~=Enum.KeyCode.Unknown addRow(featureLabel(tostring(binding.id)),bound and keyLabel(binding.key.Name)or'\u{2014}',false,function(slot:TextButton):()beginKeyCapture(slot,function(keyCode
:Enum.KeyCode):()local other:any=nil for _,candidate:any in pairs(state.shortcutBindings)do if candidate~=binding and candidate.key==keyCode then other=candidate break end end if other then
pendingConflict={binding=binding,key=keyCode,other=other}rebuildAddList()else applyBinding(binding,keyCode,nil)end end)end)end end if addRecord then if addRecord.repack then local rows:{{any}}={{
'Click an action, then press a key',nil,12}}for _,binding:any in ipairs(sortedBindings())do table.insert(rows,{featureLabel(tostring(binding.id)),binding.key and binding.key~=Enum.KeyCode.Unknown and
binding.key.Name or'\u{2014}',12})end addRecord.repack(rows)end addRecord.resize()end end local function openAddKeybind():()pendingConflict=nil if addRecord then rebuildAddList()raiseOpen(addRecord)
return end local screen:Vector2=viewport()addRecord=configWindow('KeybindAdd','Add Keybind',{position=Vector2.new(screen.X-WINDOW_WIDTH*2-32,200),anchor=keybindsRecord and keybindsRecord.window,
settingsSized=true,packRows={{'Click an action, then press a key',nil,12}}})addList=(create('Frame',{Parent=addRecord.window.body,Name='AddList',AutomaticSize=Enum.AutomaticSize.Y,
BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=1,Size=UDim2.new(1,0,0,0),ZIndex=addRecord.window.body.ZIndex+1})::any)::Frame create('UIListLayout',{Parent=addList,Padding=UDim.new(0,
WINDOW_GAP),SortOrder=Enum.SortOrder.LayoutOrder})rebuildAddList()raiseOpen(addRecord)end local function openKeybinds():()if keybindsRecord then raiseOpen(keybindsRecord)rebuildKeybinds()return end
local screen:Vector2=viewport()local keybindRows:{{any}}={{'Reset keybinds',nil,26},{'Open menu',keyLabel(state.toggleKey.Name),12}}for _,binding:any in pairs(state.shortcutBindings)do if binding.key
and binding.key~=Enum.KeyCode.Unknown then table.insert(keybindRows,{featureLabel(tostring(binding.id)),binding.key.Name,12})end end keybindsRecord=configWindow('Keybinds','Keybinds',{position=Vector2
.new(screen.X-WINDOW_WIDTH*2-32,140),anchor=uiRecord and uiRecord.window,settingsSized=true,packRows=keybindRows})keybindsList=(create('Frame',{Parent=keybindsRecord.window.body,Name='KeybindList',
AutomaticSize=Enum.AutomaticSize.Y,BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=1,Size=UDim2.new(1,0,0,0),ZIndex=keybindsRecord.window.body.ZIndex+1})::any)::Frame create('UIListLayout',{
Parent=keybindsList,Padding=UDim.new(0,WINDOW_GAP),SortOrder=Enum.SortOrder.LayoutOrder})actionRow(keybindsRecord,99,'Add',openAddKeybind)actionRow(keybindsRecord,100,'Reset keybinds',function():()for
_,binding:any in pairs(state.shortcutBindings)do binding.key=Enum.KeyCode.Unknown persistBindingKey(binding)if binding.refresh then binding.refresh()end end rebuildKeybinds()notify('keybinds reset')
end)actionRow(keybindsRecord,101,'Profiles\u{2026}',openProfiles)rebuildKeybinds()raiseOpen(keybindsRecord)end local buttonsRecord:any=nil local buttonsList:Frame?=nil local buttonsAddRecord:any=nil
local buttonsAddList:Frame?=nil local function mobileCallbacks(binding:any):(()->(),any)local hold:boolean=binding.hold==true local function press():()if type(binding.onPress)=='function'then binding.
onPress()end end local function release():()if type(binding.onRelease)=='function'then binding.onRelease()end end local callback:()->()=function():()if not hold then press()end end if not hold then
return callback,nil end return callback,{isActive=function():boolean return true end,onPress=press,onRelease=release}end local function mobileUiOrNil():any return state.mobileUi end local function
buttonRow(parent:Frame,order:number,labelText:string,valueText:string,onClick:()->()):TextButton local row:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=parent,Name=
'ButtonRow_'..labelText:gsub('%W',''),Active=true,AutoButtonColor=false,BackgroundColor3=Color3.fromRGB(0,0,0),BackgroundTransparency=0.5,BorderSizePixel=0,FontFace=CONTROL_FONT,LayoutOrder=order,Size
=UDim2.new(1,0,0,22),Text='',ZIndex=parent.ZIndex+1})::any)::TextButton create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=row,Name='Label',BackgroundTransparency=1,FontFace=CONTROL_FONT,
Position=UDim2.fromOffset(6,0),Size=UDim2.new(0.6,-6,1,0),Text=labelText,TextColor3=Theme.text,TextSize=14,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=row.ZIndex+1})create('TextLabel',{TextTruncate
=Enum.TextTruncate.AtEnd,Parent=row,Name='Value',BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.new(0.6,0,0,0),Size=UDim2.new(0.4,-8,1,0),Text=valueText,TextColor3=Theme.textMuted,
TextSize=13,TextXAlignment=Enum.TextXAlignment.Right,ZIndex=row.ZIndex+1})trackUiConnection(row.MouseButton1Click:Connect(onClick))return row end local rebuildButtons:()->()=function():()end local
openAddMobileButton:()->()=function():()end rebuildButtons=function():()local container:Frame?=buttonsList local ui:any=mobileUiOrNil()if not container or not ui then return end container:
ClearAllChildren()local order:number=1 local placed:number=0 local ids:{string}={}for id:string in pairs(ui.actions)do table.insert(ids,id)end table.sort(ids)for _,id:string in ipairs(ids)do local
record:any=ui.actions[id]if record then placed+=1 local shape:string=tostring(record.shape or'Square')buttonRow(container,order,tostring(record.label),tostring(record.shape),function():()ui.
openEditor(record)end)order+=1 end end if placed==0 then buttonRow(container,order,'No buttons yet','tap Add button',function():()end)order+=1 end local defaults:any=ui.defaults and ui.defaults()if
defaults then buttonRow(container,order,'Default shape',tostring(defaults.shape),function():()local shapes:{string}=ui.shapes local index:number=table.find(shapes,defaults.shape)or 1 defaults.shape=
shapes[index%#shapes+1]queueConfigSave()rebuildButtons()notify('new buttons: '..defaults.shape)end)order+=1 buttonRow(container,order,'Default colour',tostring(defaults.color),function():()local
palette:{string}=ui.palette local index:number=table.find(palette,string.upper(defaults.color))or 1 defaults.color=palette[index%#palette+1]queueConfigSave()rebuildButtons()notify('new buttons: '..
defaults.color)end)order+=1 end if buttonsRecord then buttonsRecord.resize()end end local function openButtons():()local ui:any=mobileUiOrNil()if not ui then notify('touch controls are not ready yet')
return end if buttonsRecord then raiseOpen(buttonsRecord)rebuildButtons()return end local screen:Vector2=viewport()buttonsRecord=configWindow('Buttons','Buttons',{position=Vector2.new(screen.X-
WINDOW_WIDTH*2-32,140),anchor=uiRecord and uiRecord.window,settingsSized=true,packRows={{'Add button',nil,26},{'Reset buttons',nil,26}}})buttonsList=(create('Frame',{Parent=buttonsRecord.window.body,
Name='ButtonList',AutomaticSize=Enum.AutomaticSize.Y,BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=1,Size=UDim2.new(1,0,0,0),ZIndex=buttonsRecord.window.body.ZIndex+1})::any)::Frame create(
'UIListLayout',{Parent=buttonsList,Padding=UDim.new(0,WINDOW_GAP),SortOrder=Enum.SortOrder.LayoutOrder})actionRow(buttonsRecord,90,'Add button',openAddMobileButton)actionRow(buttonsRecord,91,
'Reset buttons',function():()ui.resetActions()rebuildButtons()notify('mobile buttons reset')end)rebuildButtons()raiseOpen(buttonsRecord)end openAddMobileButton=function():()local ui:any=mobileUiOrNil(
)if not ui then return end if buttonsAddRecord then raiseOpen(buttonsAddRecord)return end local screen:Vector2=viewport()buttonsAddRecord=configWindow('ButtonAdd','Add button',{position=Vector2.new(
screen.X-WINDOW_WIDTH*2-32,200),anchor=buttonsRecord and buttonsRecord.window,settingsSized=true,packRows={{'Tap an action to place its button',nil,12}}})buttonsAddList=(create('Frame',{Parent=
buttonsAddRecord.window.body,Name='ButtonAddList',AutomaticSize=Enum.AutomaticSize.Y,BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=1,Size=UDim2.new(1,0,0,0),ZIndex=buttonsAddRecord.window.
body.ZIndex+1})::any)::Frame create('UIListLayout',{Parent=buttonsAddList,Padding=UDim.new(0,WINDOW_GAP),SortOrder=Enum.SortOrder.LayoutOrder})local order:number=1 for id:string,binding:any in pairs(
state.shortcutBindings)do local actionId:string=ui.actionId(id)if not ui.actions[actionId]then local label:string=featureLabel(tostring(id))buttonRow(buttonsAddList::Frame,order,label,'place',function
():()local callback:()->(),hold:any=mobileCallbacks(binding)ui.place(actionId,label,callback,hold)end)order+=1 end end if order==1 then buttonRow(buttonsAddList::Frame,order,
'Every action already has a button','',function():()end)end buttonsAddRecord.resize()raiseOpen(buttonsAddRecord)end local windowsRecord:any=nil local windowsPanel:any=nil local windowOptions:{[string]
:boolean}={}local function openWindows():()local clickGui:any=state.clickGui if not windowsRecord then local screen:Vector2=viewport()windowsRecord=configWindow('Windows','Windows',{position=Vector2.
new(screen.X-WINDOW_WIDTH*3-44,140),anchor=uiRecord and uiRecord.window,settingsSized=true,packRows={{'Combat','ON',20}}})windowsPanel=panelFor(windowsRecord,'ClickGUI.Windows')end if clickGui and
windowsPanel then for _,name:string in ipairs(clickGui.order or{})do if not windowOptions[name]then windowOptions[name]=true local category:any=clickGui.categories[name]addToggleOption(windowsPanel,
name,category.window.userVisible~=false,function(value:boolean):()category.window.userVisible=value category.window:SetVisible(value and state.visible==true)end)end end end raiseOpen(windowsRecord)end
local function applySavedWindowVisibility():boolean local clickGui:any=state.clickGui if not clickGui or type(clickGui.order)~='table'then return false end for _,name:string in ipairs(clickGui.order)
do local category:any=clickGui.categories and clickGui.categories[name]if category and category.window then local optionKey:string='ClickGUI.Windows.'..tostring(name):gsub('%W','')if configData.states
[optionKey]==false then category.window.userVisible=false pcall(function():()category.window:SetVisible(false)end)end end end return true end task.spawn(function():()for _=1,15 do
applySavedWindowVisibility()task.wait(1)end end)local optionsRecord:any=nil local actionEntries:{any}={}local actionIndex:{[string]:any}={}local actionSequence:number=0 local refreshActionRows:()->()=
function():()end state.wurstOptions={RegisterAction=function(id:any,label:any,callback:any,options:any?):any local key:string=tostring(id)local configuration:any=options or{}local entry:any=
actionIndex[key]if entry==nil then actionSequence+=1 entry={id=key,order=actionSequence}actionIndex[key]=entry table.insert(actionEntries,entry)end entry.label=tostring(label)entry.callback=callback
entry.enabled=configuration.enabled~=false entry.visible=configuration.visible~=false entry.static=configuration.static==true if entry.handle==nil then local handle:any={}function handle.SetLabel(text
:any):()entry.label=tostring(text)if entry.rowLabel and entry.rowLabel.Parent then entry.rowLabel.Text=entry.label end end function handle.SetEnabled(value:boolean):()entry.enabled=value==true
refreshActionRows()end function handle.SetVisible(value:boolean):()entry.visible=value==true refreshActionRows()end function handle.Unregister():()state.wurstOptions.UnregisterAction(entry.id)end
entry.handle=handle end if configuration.shortcut==true and entry.shortcutBound~=true and type(bindActionShortcut)=='function'then entry.shortcutBound=true bindActionShortcut(nil,'Action '..key,entry.
label,function():()state.wurstOptions.InvokeAction(key)end)end refreshActionRows()return entry.handle end,UnregisterAction=function(id:any):boolean local key:string=tostring(id)local entry:any=
actionIndex[key]if entry==nil then return false end actionIndex[key]=nil local at:number?=table.find(actionEntries,entry)if at then table.remove(actionEntries,at)end refreshActionRows()return true end
,InvokeAction=function(id:any):boolean local entry:any=actionIndex[tostring(id)]if entry==nil or entry.enabled~=true or entry.visible~=true or type(entry.callback)~='function'then return false end
local ok:boolean,failure:any=pcall(entry.callback)if not ok then warn('[Wurst] Action '..tostring(id)..': '..tostring(failure))end return ok end,List=function():{string}local ids:{string}={}for _,
entry:any in ipairs(actionEntries)do table.insert(ids,entry.id)end return ids end}refreshActionRows=function():()if not optionsRecord then return end local body:any=optionsRecord.window.body for _,
child:Instance in ipairs(body:GetChildren())do if child:IsA('GuiObject')and child:GetAttribute('RegisteredAction')==true then child:Destroy()end end local packRows:{{any}}={}for _,entry:any in ipairs(
actionEntries)do if entry.visible then table.insert(packRows,{entry.label,nil,26})if entry.static then local label:TextLabel=(create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=body,Name=
'Static_'..entry.id:gsub('%W',''),BackgroundTransparency=1,FontFace=CONTROL_FONT,LayoutOrder=entry.order,Size=UDim2.new(1,0,0,ROW_HEIGHT),Text='  '..entry.label,TextColor3=Theme.textMuted,TextSize=14,
TextXAlignment=Enum.TextXAlignment.Left,ZIndex=body.ZIndex+1})::any)::TextLabel label:SetAttribute('RegisteredAction',true)entry.rowLabel=label else local row:TextButton=actionRow(optionsRecord,entry.
order,entry.label,function():()state.wurstOptions.InvokeAction(entry.id)end)row:SetAttribute('RegisteredAction',true)local rowLabel:any=row:FindFirstChild('Label')entry.rowLabel=rowLabel if not entry.
enabled then row.Active=false if rowLabel then rowLabel.TextColor3=Theme.textMuted end end end end end if optionsRecord.repack then optionsRecord.repack(packRows)end optionsRecord.resize()end local
reinjectBusy:boolean=false local reinjectHandle:any=nil local function runReinject():()local function setReinjectLabel(text:string):()if reinjectHandle then reinjectHandle.SetLabel(text)end end if
reinjectBusy then return end local compiler:any=(getfenv()::any).loadstring if type(compiler)~='function'then notify('reinject unavailable')return end reinjectBusy=true setReinjectLabel(
'Downloading\u{2026}')local sourceUrls:{string}={REPOSITORY_RAW_BASE..'src/Wurst.lua',RUNTIME_SAFETY_SOURCE_URL}local sourceOrError:any=nil local downloaded:boolean=false local downloadFailures:{
string}={}for _,candidateUrl:string in ipairs(sourceUrls)do local requestUrl:string=candidateUrl..'?reinject=v3&v='..SOURCE_STAMP local requestOk:boolean,bodyOrError:any=pcall(function():string return
(game::any):HttpGet(requestUrl,true)end)local current:boolean=requestOk and type(bodyOrError)=='string'and#bodyOrError>=1024 and bodyOrError:find(RUNTIME_COMPATIBILITY_MARKER,1,true)~=nil if current
then downloaded=true sourceOrError=bodyOrError break end table.insert(downloadFailures,tostring(bodyOrError))end if not downloaded then reinjectBusy=false setReinjectLabel('Reinject latest')notify(
'download failed')warn('[Wurst] Reinject download: '..table.concat(downloadFailures,' | '))return end local compiled:any=nil local compileError:any=nil local compiledOk:boolean,compileResult:any,
returnedError:any=pcall(compiler,sourceOrError,'@src/Wurst.lua')if compiledOk then compiled=compileResult compileError=returnedError else compileError=compileResult end if type(compiled)~='function'
then reinjectBusy=false setReinjectLabel('Reinject latest')notify('compile failed')warn('[Wurst] Reinject compile: '..tostring(compileError))return end setReinjectLabel('Restarting\u{2026}')task.
defer(function():()local executed:boolean,executionError:any=xpcall(compiled,debug.traceback)if not executed then reinjectBusy=false setReinjectLabel('Reinject latest')warn(
'[Wurst] Reinject runtime: '..tostring(executionError))end end)end local destructHandle:any=nil local function runDestruct():()if destructHandle then destructHandle.SetLabel('Removing\u{2026}')end
task.defer(function():()local destruct:any=state.destruct if type(destruct)~='function'then pcall(function():()ScreenGui:Destroy()end)return end local ok:boolean,failure:any=pcall(destruct)if not ok
then warn('[Wurst] Destruct: '..tostring(failure))pcall(function():()ScreenGui:Destroy()end)end end)end local function runResetLayout():()configData.ui.windows={}queueConfigSave()for _,window:any in
ipairs(windows.list or{})do window.userMoved=nil end if state.clickGui and state.clickGui.Retile then state.clickGui.Retile()end for _,window:any in ipairs(windows.list or{})do if window.
featureSettings or window.managementWindow then if window.root.Visible and windows.PlaceWindow then windows.PlaceWindow(window,{force=true})else window.stalePlacement=true end end end for _,record:any
in ipairs(Settings.records)do if not record.window.root.Visible then record.placed=false end end notify('layout reset')end local recognised:string=GAME_CHECK.MM2Active and'MM2'or GAME_CHECK.TRSActive
and'TRS'or GAME_CHECK.VDActive and'VD'or GAME_CHECK.BedFightActive and'BedFight'or'Universal only'reinjectHandle=state.wurstOptions.RegisterAction('Reinject','Reinject latest',runReinject)
destructHandle=state.wurstOptions.RegisterAction('Destruct','Destruct',runDestruct)state.wurstOptions.RegisterAction('ResetLayout','Reset layout',runResetLayout)local _=recognised local function
openWurstOptions():()if optionsRecord then raiseOpen(optionsRecord)return end local screen:Vector2=viewport()optionsRecord=configWindow('WurstOptions','Wurst Options',{position=Vector2.new(screen.X-
WINDOW_WIDTH*2-32,220),anchor=uiRecord and uiRecord.window,settingsSized=true,packRows={{'Reinject latest',nil,26}}})refreshActionRows()raiseOpen(optionsRecord)end local screen:Vector2=viewport()local
versionTag:string=tostring(host.BUILD_LABEL or(function():string local buildNumber:string?=string.match(tostring(SOURCE_STAMP or''),'^v(%d+)')return if buildNumber then'0.'..buildNumber else'0.0'end)(
))uiRecord=configWindow('UISettings','UI Settings \183 '..versionTag,{position=Vector2.new(screen.X-WINDOW_WIDTH-12,64),closable=false,settingsSized=false,packRows={{'HackList',nil,26},{state.isMobile
and'Buttons'or'Keybinds',nil,26},{'Windows',nil,26},{'WurstOptions',nil,26},{'Background','#404040'},{'Accent','#101010'},{'Text','#F0F0F0'},{'Opacity','0.85'}}})uiRecord.open=true actionRow(uiRecord,
1,'HackList',openHackList)if state.isMobile then actionRow(uiRecord,2,'Buttons',openButtons)else actionRow(uiRecord,2,'Keybinds',openKeybinds)end actionRow(uiRecord,3,'Windows',openWindows)actionRow(
uiRecord,4,'WurstOptions',openWurstOptions)local uiPanel:any=panelFor(uiRecord,'ClickGUI')uiPanel.optionCount=4 local uiSettingsValues:{[string]:any}={}state.uiSettings=uiSettingsValues local
CLICKGUI_ENABLED:Color3=Theme.enabled local CLICKGUI_ARROWS:Color3=Theme.arrows or Theme.enabled or Color3.fromRGB(0,204,0)local uiPaletteRows:{[string]:Frame}={}local function paintPalette(labelText:
string,color:Color3):()local option:Frame?=uiPaletteRows[labelText]if not option then return end local bar:Frame?=option:FindFirstChild('ColorBar')::Frame?local hex:TextLabel?=option:FindFirstChild(
'HexValue')::TextLabel?if bar then bar.BackgroundColor3=color end if hex then hex.Text=string.format('#%02X%02X%02X',math.floor(color.R*255+0.5),math.floor(color.G*255+0.5),math.floor(color.B*255+0.5)
)end end local PALETTE_ROWS:{any}={{label='Background',default=Theme.background,tokens={'surface','background'}},{label='Accent',default=Theme.accent,tokens={'accent','outline'}},{label='Text',default
=Theme.text,tokens={'text','accentText'}},{label='Enabled',default=Theme.enabled,tokens={'enabled'}},{label='Buttons',key='Arrows',default=Theme.arrows,tokens={'arrows'}}}if ThemeEngine and type(
ThemeEngine.SetMany)=='function'then for _,row:any in ipairs(PALETTE_ROWS)do local saved:Color3?=colorFromConfig(configData.values[uiPanel.configKey..'.'..(row.key or row.label):gsub('%W','')],nil)if
saved then local updates:{[string]:Color3}={}for _,token:string in ipairs(row.tokens)do updates[token]=saved end ThemeEngine.SetMany(updates)end end end local paletteOk:boolean,paletteError:string?=
pcall(function():()for _,row:any in ipairs(PALETTE_ROWS)do local tokens:{string}=row.tokens local labelText:string=row.label local swatch:TextButton?=addColorOption(uiPanel,labelText,row.default or
Theme.text,function(color:Color3):()uiSettingsValues[labelText]=color if ThemeEngine and type(ThemeEngine.SetMany)=='function'then local updates:{[string]:Color3}={}for _,token:string in ipairs(tokens
)do updates[token]=color end ThemeEngine.SetMany(updates)elseif ThemeEngine and type(ThemeEngine.Set)=='function'then for _,token:string in ipairs(tokens)do ThemeEngine.Set(token,color)end end end,row
.key)if swatch and swatch.Parent and swatch.Parent:IsA('Frame')then uiPaletteRows[labelText]=swatch.Parent::Frame end end end)if not paletteOk then warn('[Wurst:Settings] palette rows skipped: '..
tostring(paletteError))end addNumberOption(uiPanel,'Opacity',OPACITY,0.15,1,function(value:number):()local wanted:number=math.clamp(tonumber(value)or OPACITY,0.15,1)uiSettingsValues['Opacity']=wanted
windows.SetOpacity(wanted)end,[[How see-through the menu BACKGROUND is. Text and controls stay fully readable at any value.]],0.01,true)uiSettingsValues['Tooltip opacity']=TOOLTIP_OPACITY if state.
featureTooltip then state.featureTooltip.BackgroundTransparency=1-TOOLTIP_OPACITY end uiSettingsValues['Max height']=MAX_HEIGHT state.uiMaxHeight=MAX_HEIGHT uiSettingsValues['Max settings height']=
MAX_SETTINGS_HEIGHT maxSettingsHeight=MAX_SETTINGS_HEIGHT state.uiMaxSettingsHeight=MAX_SETTINGS_HEIGHT uiSettingsValues['Menu style']='Wurst'local themeShape:any=ThemeEngine and ThemeEngine.shape
local initialScale:number=themeShape and tonumber(themeShape.scale)or 1 uiSettingsValues['Menu scale']=initialScale local scaleReflowToken:number=0 addNumberOption(uiPanel,'Menu scale',initialScale,
0.7,1.3,function(value:number):()local shape:any=ThemeEngine and ThemeEngine.shape if not shape then return end local wanted:number=math.clamp(tonumber(value)or 1,0.7,1.3)shape.scale=wanted
uiSettingsValues['Menu scale']=wanted configData.values['UI.MenuScale']=wanted queueConfigSave()if state.clickGui and type(state.clickGui.ApplyScale)=='function'then state.clickGui.ApplyScale()end
scaleReflowToken+=1 local token:number=scaleReflowToken task.delay(0.15,function():()if token~=scaleReflowToken then return end if type(state.refreshLayout)=='function'then state.refreshLayout(true)
end end)end,'Adjust the size of menu windows on this device.',0.01,true)addToggleOption(uiPanel,'Keep after teleport',true,function(value:boolean):()uiSettingsValues['Keep after teleport']=value local
persist:any=state.teleportPersist if persist and type(persist.SetEnabled)=='function'then persist.SetEnabled(value)if value and persist.supported~=true then notify(
'queue on teleport unavailable \u{2014} '..'the menu cannot follow teleports here')end end end)local touchPrimary:boolean=state.isMobile==true if not touchPrimary then addToggleOption(uiPanel,
'Show dock',false,function(value:boolean):()uiSettingsValues['Show dock']=value state.uiShowDock=value if state.dock and type(state.dock.SetVisible)=='function'then state.dock.SetVisible(value)end end
)end if touchPrimary then host.addActionOption(uiPanel,'Move dock',function():()if state.dock and state.dock.BeginMove then state.dock.BeginMove()end end,
'Drag and release the dock to save its position.')end uiRecord.placed=true if windows.ReflowWindow then windows.ReflowWindow(uiRecord.window)end if state.bitmapText and type(state.bitmapText.onReady)
=='table'then table.insert(state.bitmapText.onReady,function():()for _,record:any in ipairs(Settings.records)do if record.repack then record.repack()end end end)end state.addMenuVisibilityListener(
function(visible:boolean):()for _,record:any in ipairs(Settings.records)do if not visible and record.window.pinned then continue end if visible then record.window:SetVisible(record.open)else record.
open=record.window.root.Visible record.window:SetVisible(false)end end end)for _,record:any in ipairs(Settings.records)do record.window:SetVisible(state.visible==true and record.open)end Settings.ui=
uiRecord Settings.openKeybinds=openKeybinds Settings.openProfiles=openProfiles Settings.openHackList=openHackList Settings.openWurstOptions=openWurstOptions activeCleanup=function():()for _,record:any
in ipairs(Settings.records)do pcall(function():()record.window:Destroy()end)end Settings.records={}for _,connection:RBXScriptConnection in ipairs(Settings.connections)do pcall(function():()connection:
Disconnect()end)end Settings.connections={}state.uiSettings=nil state.wurstOptions=nil end Module.Initialized=true return Settings end function Module.destroy():()if activeCleanup then pcall(
activeCleanup)end activeCleanup=nil Module.Initialized=false end return Module
-- src/guis/Wurst/Code/Widgets.lua
export type OptionDefinition={kind:string,label:string,default:any?,minimum:number?,maximum:number?,values:{string}?,callback:any,persist:boolean?}export type FeatureDefinition={name:string,
description:string,order:number,onToggle:any,configuration:any?,options:{OptionDefinition}?}export type WidgetLibrary={addToggleOption:any,addActionOption:any,addNumberOption:any,addRangeOption:any,
addCycleOption:any,addKeyOption:any,addTextOption:any,addColorOption:any,addListOption:any,addSectionOption:any,addInformationOption:any,setOptionVisible:any}local Module={Name='Widgets',PlaceId=0,
Events={}::{[string]:any},Initialized=false}local boundState:any=nil local toggleThemeRefreshers:{()->()}={}local function runToggleThemeRefreshers(changed:any):()for _,refresh:any in ipairs(
toggleThemeRefreshers)do pcall(refresh,changed)end end local LABEL_FONT_SIZE:number=16 local SLIDER_BAR_HEIGHT:number=6 local SLIDER_KNOB_WIDTH:number=16 local SLIDER_KNOB_HEIGHT:number=16 local
CHILD_INSET:number=4 local CHILD_GAP:number=4 function Module.init(context:any):WidgetLibrary local WURST_GREEN:Color3=Color3.fromRGB(0,204,0)local host:any=context local UserInputService:any=host.
UserInputService local state:any=host.state boundState=state table.clear(toggleThemeRefreshers)if state then state.optionThemeRefresh=runToggleThemeRefreshers state.registerThemeRefresh=function(
refresh:any,tokens:{string}?):()table.insert(toggleThemeRefreshers,function(changed)if tokens and changed then local relevant=false for _,token in ipairs(tokens)do if changed[token]then relevant=true
break end end if not relevant then return end end refresh()end)end state.valueControls={}end local configData:any=host.configData local queueConfigSave:any=host.queueConfigSave local create:any=host.
create local makeButton:any=host.makeButton local makeTextBox:any=host.makeTextBox local function flatTextBox(parent:Instance,initial:any):TextBox local box:TextBox=(makeTextBox(parent,initial)::any)
::TextBox local corner:UICorner?=box:FindFirstChild('StyleCorner')::UICorner?if corner then corner:Destroy()end return box end local makeTextLabel:any=host.makeTextLabel local notify:any=host.notify
local Theme:any=host.Theme local ThemeEngine:any=host.ThemeEngine local CONTROL_FONT:any=host.CONTROL_FONT local ScreenGui:any=host.ScreenGui local PopupLayer:any=host.PopupLayer local
trackUiConnection:any=host.trackUiConnection local applyKeySlotStyle:any=host.applyKeySlotStyle local beginKeyCapture:any=host.beginKeyCapture local bindActionShortcut:any=host.bindActionShortcut
local runActionButton:any=host.runActionButton local colorFromConfig:any=host.colorFromConfig local colorToConfig:any=host.colorToConfig local function drawPixelIcon(parent:Instance,kind:string,color:
Color3?,token:string?):Frame local themed:boolean=token~=nil and ThemeEngine~=nil and Theme[token or'']~=nil local paint:Color3=(themed and Theme[token::string])or color or WURST_GREEN local existing:
Instance?=(parent::any):FindFirstChild('PixelIcon')if existing and existing:GetAttribute('PixelKind')==kind and existing:GetAttribute('PixelToken')==(token or'')then if not themed then for _,pixel in
ipairs(existing:GetChildren())do if pixel:IsA('Frame')then pixel.BackgroundColor3=paint end end end return existing::Frame end if existing then existing:Destroy()end local sizes:{[string]:{number}}={
down={14,8},up={14,8},right={8,14},cross={10,10},pin={10,10},check={12,10}}local size:{number}=sizes[kind]or{10,10}local container:Frame=(create('Frame',{Parent=parent,Name='PixelIcon',AnchorPoint=
Vector2.new(0.5,0.5),BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.fromScale(0.5,0.5),Size=UDim2.fromOffset(size[1],size[2]),ZIndex=(tonumber((parent::any).ZIndex)or 1)+1})::any)::Frame
local function px(x:number,y:number,w:number,h:number):()local pixel:any=create('Frame',{Parent=container,BackgroundColor3=paint,BorderSizePixel=0,Position=UDim2.fromOffset(x*2,y*2),Size=UDim2.
fromOffset(w*2,h*2),ZIndex=container.ZIndex+1})if themed then ThemeEngine.Bind(pixel,'BackgroundColor3',token::string)end end container:SetAttribute('PixelKind',kind)container:SetAttribute(
'PixelToken',token or'')if kind=='down'then px(0,0,7,1)px(1,1,5,1)px(2,2,3,1)px(3,3,1,1)elseif kind=='up'then px(3,0,1,1)px(2,1,3,1)px(1,2,5,1)px(0,3,7,1)elseif kind=='right'then px(0,0,1,7)px(1,1,1,5
)px(2,2,1,3)px(3,3,1,1)elseif kind=='cross'then px(0,0,1,1)px(4,0,1,1)px(1,1,1,1)px(3,1,1,1)px(2,2,1,1)px(1,3,1,1)px(3,3,1,1)px(0,4,1,1)px(4,4,1,1)elseif kind=='pin'then px(1,0,3,3)px(2,3,1,2)elseif
kind=='check'then px(0,2,1,1)px(1,3,1,1)px(2,4,1,1)px(3,3,1,1)px(4,2,1,1)px(5,1,1,1)end return container end state.layoutOptionControl=function(option:Frame,control:GuiObject,widthScale:number?):()
local top:number=tonumber(option:GetAttribute('ControlTop'))or 0 local height:number=tonumber(option:GetAttribute('ControlHeight'))or option.Size.Y.Offset local scale:number=widthScale or 0.5 control.
AnchorPoint=Vector2.new(1,0)control.Position=UDim2.new(1,-10,0,top)control.Size=UDim2.new(scale,-10,0,height)control.ZIndex=math.max(control.ZIndex,option.ZIndex+2)end local function createOptionRow(
feature:any,labelText:string,_hintText:string?):Frame feature.optionCount+=1 feature.searchText..=' '..string.lower(labelText)local rowHeight:number=state.layout.optionRowHeight local option:Frame=(
create('Frame',{Parent=feature.options,Name='Option_'..labelText:gsub('%W',''),BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=feature.optionCount,Size=UDim2.new(1,0,0,rowHeight),ZIndex=feature
.options.ZIndex+1})::any)::Frame table.insert(feature.optionRows,option)local label:TextLabel=(create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=option,Name='OptionLabel',
BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.fromOffset(CHILD_INSET,0),Size=UDim2.new(0.55,-CHILD_INSET,0,rowHeight),Text=labelText,TextColor3=Theme.text,TextSize=LABEL_FONT_SIZE,
TextXAlignment=Enum.TextXAlignment.Left,ZIndex=option.ZIndex+1})::any)::TextLabel label.Visible=true option:SetAttribute('ControlTop',3)option:SetAttribute('ControlHeight',rowHeight-6)feature.
updateExpandedSize()return option end state.addSectionOption=function(feature:any,titleText:string):Frame feature.optionCount+=1 local section:Frame=(create('Frame',{Parent=feature.options,Name=
'Section_'..titleText:gsub('%W',''),BackgroundTransparency=1,BorderSizePixel=0,LayoutOrder=feature.optionCount,Size=UDim2.new(1,0,0,state.layout.sectionRowHeight or 22),Visible=true,ZIndex=feature.
options.ZIndex+1})::any)::Frame local caption:TextLabel=(create('TextLabel',{Parent=section,Name='SectionLabel',BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.fromOffset(CHILD_INSET,0),
Size=UDim2.new(1,-CHILD_INSET*2,0,section.Size.Y.Offset),Text=string.upper(titleText),TextColor3=Theme.textMuted,TextSize=10,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=section.ZIndex+1})::any)::
TextLabel caption:SetAttribute('SectionTitle',titleText)create('Frame',{Parent=section,Name='SectionLine',AnchorPoint=Vector2.new(0,0.5),BackgroundColor3=Theme.outline,BackgroundTransparency=0.55,
BorderSizePixel=0,Position=UDim2.new(0,4+math.clamp(#string.upper(titleText)*7+8,24,220),0.5,0),Size=UDim2.new(1,-(4+math.clamp(#string.upper(titleText)*7+8,24,220))-4,0,1),ZIndex=section.ZIndex})
section:SetAttribute('SectionTitle',titleText)table.insert(feature.optionRows,section)feature.updateExpandedSize()return section end state.addStatOption=function(feature:any,labelText:string,
initialValue:string?):any local option:Frame=createOptionRow(feature,labelText,nil)local optionLabel:TextLabel?=option:FindFirstChild('OptionLabel')::TextLabel?if optionLabel then(optionLabel::
TextLabel).Size=UDim2.new(0.62,-CHILD_INSET,1,0);(optionLabel::TextLabel).TextColor3=Theme.textMuted end local valueLabel:TextLabel=(create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=
option,Name='StatValue',AnchorPoint=Vector2.new(1,0),BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.new(1,-CHILD_INSET,0,0),Size=UDim2.new(0.38,-CHILD_INSET,1,0),Text=tostring(
initialValue or'0'),TextColor3=Theme.text,TextSize=LABEL_FONT_SIZE,TextXAlignment=Enum.TextXAlignment.Right,ZIndex=option.ZIndex+2})::any)::TextLabel local lastText:string=tostring(initialValue or'0')
return{row=option,label=valueLabel,set=function(newText:string):()local nextText:string=tostring(newText)if nextText==lastText then return end lastText=nextText if valueLabel and valueLabel.Parent
then valueLabel.Text=nextText end end}end local function setOptionVisible(feature:any,option:GuiObject,visible:boolean):()if option==nil or option==feature.options then warn(
[[[Random Testing Menu] setOptionVisible refused the options container]])return end local row:GuiObject=option local parent:any=option.Parent local container:any=feature.options if container~=nil and
parent~=nil and parent~=container and parent.Parent~=nil and parent.Parent==container then row=parent end row.Visible=visible feature.updateExpandedSize()end local function addToggleOption(feature:any
,labelText:string,defaultValue:boolean,callback:(boolean)->(),hintText:string?,bindable:boolean?):TextButton local optionKey:string=feature.configKey..'.'..labelText:gsub('%W','')local enabled:boolean
?=configData.states[optionKey]if enabled==nil then enabled=defaultValue==true end local option:Frame=createOptionRow(feature,labelText,hintText)local button:TextButton=create('TextButton',{
TextTruncate=Enum.TextTruncate.AtEnd,ClipsDescendants=true,Parent=option,Name='OptionClickTarget',Active=true,AutoButtonColor=false,BackgroundTransparency=1,BorderSizePixel=0,Size=UDim2.fromScale(1,1)
,Text='',ZIndex=option.ZIndex+2})::TextButton local box:Frame=(create('Frame',{Parent=option,Name='OptionCheckbox',AnchorPoint=Vector2.new(0,0.5),BackgroundColor3=Color3.fromRGB(0,0,0),
BackgroundTransparency=0.5,BorderSizePixel=0,Position=UDim2.new(0,CHILD_INSET,0.5,0),Size=UDim2.fromOffset(22,22),ZIndex=option.ZIndex+3})::any)::Frame local boxStroke:UIStroke=(create('UIStroke',{
Parent=box,ApplyStrokeMode=Enum.ApplyStrokeMode.Border,Color=Theme.outline,Thickness=1,Transparency=0.35})::any)::UIStroke local function markContrast(color:Color3):Color3 local luminance:number=0.299
*color.R+0.587*color.G+0.114*color.B if luminance>0.45 then return Color3.fromRGB(12,12,12)end return Color3.fromRGB(245,245,247)end local mark:Frame=drawPixelIcon(box,'check',markContrast(Theme.
enabled or WURST_GREEN))mark.Visible=false local function paintMark():()local paint:Color3=markContrast(Theme.enabled or WURST_GREEN)for _,pixel:any in ipairs(mark:GetChildren())do pcall(function():()
pixel.BackgroundColor3=paint end)end end local optionLabel:TextLabel?=option:FindFirstChild('OptionLabel')::TextLabel?if optionLabel then(optionLabel::TextLabel).Position=UDim2.fromOffset(CHILD_INSET+
22+CHILD_GAP,0);(optionLabel::TextLabel).Size=UDim2.new(1,-(CHILD_INSET+22+CHILD_GAP),1,0)end local function refreshOptionState():()local on:boolean=enabled::boolean box.BackgroundColor3=on and Theme.
enabled or Theme.surface box.BackgroundTransparency=on and 0 or 0.5 boxStroke.Color=on and Theme.enabled or Theme.outline paintMark()mark.Visible=on end table.insert(toggleThemeRefreshers,
refreshOptionState)local function toggleOption():()if not option.Visible then return end enabled=not enabled refreshOptionState()configData.states[optionKey]=enabled callback(enabled::boolean)
queueConfigSave()notify(labelText..((enabled::boolean)and' enabled'or' disabled'))end if bindable then button.Size=UDim2.new(0.72,0,1,0)if optionLabel then optionLabel.Size=UDim2.new(0.72,-(
CHILD_INSET+22+CHILD_GAP),1,0)end local keyButton:TextButton=makeButton(option,'KEY',9)keyButton.Name='OptionKeybind'state.layoutOptionControl(option,keyButton,0.24)keyButton.Position=UDim2.new(0.74,-
16,keyButton.Position.Y.Scale,keyButton.Position.Y.Offset)keyButton.ZIndex=option.ZIndex+5 applyKeySlotStyle(keyButton)keyButton.TextScaled=true create('UITextSizeConstraint',{Parent=keyButton,
MaxTextSize=14,MinTextSize=8})bindActionShortcut(keyButton,feature.configKey..labelText,labelText,toggleOption)end trackUiConnection(button.MouseButton1Click:Connect(toggleOption))state.toggleControls
=state.toggleControls or{}state.toggleControls[optionKey]={getChecked=function():boolean return enabled::boolean end,setChecked=function(checked:boolean):()if not button.Parent then return end local
wanted:boolean=checked==true if wanted==(enabled::boolean)then return end enabled=wanted refreshOptionState()configData.states[optionKey]=enabled callback(enabled::boolean)queueConfigSave()end}
refreshOptionState()callback(enabled::boolean)return button end local function addActionOption(feature:any,labelText:string,callback:()->(),hintText:string?,interaction:any?):TextButton local option=
createOptionRow(feature,labelText,hintText)local button=makeButton(option,interaction and interaction.hold and'HOLD'or'RUN',11)button.BackgroundColor3=state.guiStyle=='Default'and Theme.buttonHover or
Theme.accentSoft state.layoutOptionControl(option,button,0.26)local function executeAction():()if not option.Visible then return end runActionButton(button,'RUN',callback,labelText..' completado',
'[Random Testing Menu] '..labelText)end local hold:boolean=interaction~=nil and interaction.hold==true local held={button=false,shortcut=false}local active:boolean=false local buttonInput:InputObject?
=nil local function setHeld(source:string,value:boolean):()if value and not option.Visible then return end held[source]=value local wanted:boolean=held.button or held.shortcut if active==wanted then
return end active=wanted button.Text=wanted and'HELD'or'HOLD'if wanted then callback()elseif interaction and interaction.onRelease then interaction.onRelease()end end if hold then trackUiConnection(
button.InputBegan:Connect(function(input:InputObject):()if input.UserInputType==Enum.UserInputType.MouseButton1 or input.UserInputType==Enum.UserInputType.Touch then if buttonInput==nil then
buttonInput=input setHeld('button',true)end end))trackUiConnection(UserInputService.InputEnded:Connect(function(input:InputObject):()if input==buttonInput or(buttonInput and buttonInput.
UserInputType==Enum.UserInputType.MouseButton1 and input.UserInputType==Enum.UserInputType.MouseButton1)then buttonInput=nil setHeld('button',false)end end))trackUiConnection(UserInputService.
WindowFocusReleased:Connect(function():()held.button=false buttonInput=nil setHeld('shortcut',false)end))else trackUiConnection(button.MouseButton1Click:Connect(executeAction))end local shortcutButton
:TextButton=(makeButton(option,'KEY',9)::any)::TextButton shortcutButton.BackgroundColor3=state.guiStyle=='Default'and Theme.button or Theme.surfaceRaised state.layoutOptionControl(option,
shortcutButton,0.24)shortcutButton.Position=UDim2.new(0.74,-16,shortcutButton.Position.Y.Scale,shortcutButton.Position.Y.Offset)applyKeySlotStyle(shortcutButton)shortcutButton.TextScaled=true create(
'UITextSizeConstraint',{Parent=shortcutButton,MaxTextSize=14,MinTextSize=8})bindActionShortcut(shortcutButton,feature.configKey..labelText,labelText,hold and function():()setHeld('shortcut',true)end
or executeAction,hold and{hold=true,onRelease=function():()setHeld('shortcut',false)end}or nil)return button::TextButton end type ActiveSlider={update:(number)->(),finish:()->()}local activeSlider:
ActiveSlider?=nil trackUiConnection(UserInputService.InputChanged:Connect(function(input:InputObject):()if activeSlider and(input.UserInputType==Enum.UserInputType.MouseMovement or input.UserInputType
==Enum.UserInputType.Touch)then activeSlider.update(input.Position.X)end end))trackUiConnection(UserInputService.InputEnded:Connect(function(input:InputObject):()if activeSlider and(input.
UserInputType==Enum.UserInputType.MouseButton1 or input.UserInputType==Enum.UserInputType.Touch)then local slider:ActiveSlider=activeSlider activeSlider=nil slider.finish()end end))local function
isNumber(value:any):boolean return type(value)=='number'and value==value end local function formatCompactNumber(value:number):string if value==math.huge then return'inf'end if value==-math.huge then
return'-inf'end local magnitude:number=math.abs(value)if magnitude>=1e9 or(magnitude>0 and magnitude<1e-4)then return string.format('%.3e',value)end return string.format('%.12g',value)end
local function effectiveMax(minimum:number,maximum:number):number if maximum~=math.huge then return maximum end return math.max(100,math.abs(minimum)*10)end local function parseNumericText(text:string
):number?local cleaned=string.lower(text:gsub('%s+',''))if cleaned=='inf'or cleaned=='infinity'or cleaned=='+inf'then return math.huge end if cleaned=='-inf'or cleaned=='-infinity'then return-math.
huge end return tonumber(text)end local function addNumberOption(feature:any,labelText:string,defaultValue:number,minimum:number,maximum:number,callback:(number)->(),hintText:string?,stepValue:number?
,boundTyped:boolean?):Frame local optionKey:string=feature.configKey..'.'..labelText:gsub('%W','')local storedValue:number?=tonumber(configData.values[optionKey])local value:number=isNumber(
storedValue)and(storedValue::number)or defaultValue if boundTyped then if value==math.huge or value==-math.huge then value=defaultValue end value=math.clamp(value,minimum,maximum)end local integerOnly
:boolean=defaultValue%1==0 and minimum%1==0 and maximum%1==0 local option:Frame=createOptionRow(feature,labelText,hintText)option.Size=UDim2.new(1,0,0,44)feature.updateExpandedSize()local headerLabel:
TextLabel?=option:FindFirstChild('OptionLabel')::TextLabel?if headerLabel then(headerLabel::TextLabel).Size=UDim2.new(0.55,-CHILD_INSET,0,22)end local requestedStep:number?=tonumber(stepValue)if
requestedStep~=nil and requestedStep<=0 then requestedStep=nil end local step:number=requestedStep or(integerOnly and 1 or 0.01)local precisionFactor:number=1 for _=1,6 do if math.abs(step*
precisionFactor-math.round(step*precisionFactor))<1e-7 then break end precisionFactor*=10 end option:SetAttribute('Min',minimum)option:SetAttribute('Max',maximum)option:SetAttribute('Step',step)option
:SetAttribute('FreeNumeric',true)local slider:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=option,Name='Slider',Active=true,AutoButtonColor=false,BackgroundTransparency
=1,BorderSizePixel=0,Position=UDim2.new(0,0,0,22),Size=UDim2.new(1,0,0,22),Text='',ZIndex=option.ZIndex+1})::any)::TextButton local minText:string=formatCompactNumber(minimum)local maxText:string=
formatCompactNumber(maximum)local bitmapContract:any=state.bitmapText local function endLabelWidth(endText:string):number if bitmapContract and type(bitmapContract.measure)=='function'then return
bitmapContract.measure({{text=endText}},14)+4 end return#endText*11+4 end local minWidth:number=endLabelWidth(minText)local maxWidth:number=endLabelWidth(maxText)create('TextLabel',{TextTruncate=Enum.
TextTruncate.AtEnd,Parent=slider,Name='MinLabel',BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.new(0,CHILD_INSET,0,0),Size=UDim2.new(0,minWidth,1,0),Text=minText,TextColor3=Theme.
textMuted,TextSize=14,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=option.ZIndex+1})create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=slider,Name='MaxLabel',AnchorPoint=Vector2.new(1,0
),BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.new(1,-CHILD_INSET,0,0),Size=UDim2.new(0,maxWidth,1,0),Text=maxText,TextColor3=Theme.textMuted,TextSize=14,TextXAlignment=Enum.
TextXAlignment.Right,ZIndex=option.ZIndex+1})local track:Frame=(create('Frame',{Parent=slider,Name='Track',BackgroundColor3=Color3.fromRGB(255,255,255),BackgroundTransparency=0.9,BorderSizePixel=0,
Position=UDim2.new(0,CHILD_INSET+minWidth+CHILD_GAP,0,(22-SLIDER_BAR_HEIGHT)//2),Size=UDim2.new(1,-(CHILD_INSET*2+minWidth+maxWidth+CHILD_GAP*2),0,SLIDER_BAR_HEIGHT),ZIndex=option.ZIndex+1})::any)::
Frame local fill:Frame=(create('Frame',{Parent=track,Name='Fill',BackgroundColor3=Theme.accent,BorderSizePixel=0,Size=UDim2.fromScale(0,1),ZIndex=track.ZIndex+1})::any)::Frame local knob:Frame=(
create('Frame',{Parent=track,Name='Knob',AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.text,BorderSizePixel=0,Position=UDim2.fromScale(0,0.5),Size=UDim2.fromOffset(SLIDER_KNOB_WIDTH,
SLIDER_KNOB_HEIGHT),ZIndex=track.ZIndex+2})::any)::Frame local valueInput:TextBox=(flatTextBox(slider,'')::any)::TextBox valueInput.Name='NumericValue'valueInput.AnchorPoint=Vector2.new(1,0)valueInput
.Position=UDim2.new(1,-CHILD_INSET,0,-22)valueInput.Size=UDim2.new(0.42,0,0,22)valueInput.BackgroundTransparency=1 valueInput.TextColor3=Theme.textMuted valueInput.TextSize=LABEL_FONT_SIZE valueInput.
TextXAlignment=Enum.TextXAlignment.Right valueInput.TextTruncate=Enum.TextTruncate.AtEnd valueInput.ZIndex=option.ZIndex+2 local valueStroke:UIStroke?=valueInput:FindFirstChild('StyleStroke')::
UIStroke?if valueStroke then(valueStroke::UIStroke).Transparency=1 end valueInput.PlaceholderText=boundTyped and(formatCompactNumber(minimum)..'\u{2013}'..formatCompactNumber(maximum))or
'Any number (type inf for infinity)'local editing:boolean=false local function quantize(raw:number,fromSlider:boolean):number if not fromSlider then return boundTyped and math.clamp(raw,minimum,
maximum)or raw end if maximum==math.huge then if raw<minimum then raw=minimum end return raw end raw=math.clamp(raw,minimum,maximum)local stepped:number=minimum+math.round((raw-minimum)/step)*step
stepped=math.round(stepped*precisionFactor)/precisionFactor return math.clamp(stepped,minimum,maximum)end local function setValue(nextValue:number,persist:boolean,_fromSlider:boolean):()if not
isNumber(nextValue)then return end if(nextValue==math.huge or nextValue==-math.huge)and maximum~=math.huge then return end value=quantize(nextValue,_fromSlider)local sliderMax:number=effectiveMax(
minimum,maximum)local alpha:number=sliderMax==minimum and 0 or math.clamp((value-minimum)/(sliderMax-minimum),0,1)fill.Size=UDim2.fromScale(alpha,1)knob.Position=UDim2.fromScale(alpha,0.5)if not
editing then valueInput.Text=formatCompactNumber(value)end callback(value)if persist then configData.values[optionKey]=value queueConfigSave()end end local function updateFromScreenX(screenX:number,
origin:number,width:number):()local alpha:number=math.clamp((screenX-origin)/width,0,1)setValue(minimum+(effectiveMax(minimum,maximum)-minimum)*alpha,true,true)end if state.valueControls then state.
valueControls[optionKey]={apply=function(stored:any):()setValue(tonumber(stored)or defaultValue,true)end,default=defaultValue}end trackUiConnection(slider.InputBegan:Connect(function(input:InputObject
):()if input.UserInputType~=Enum.UserInputType.MouseButton1 and input.UserInputType~=Enum.UserInputType.Touch then return end if input.Position.X>track.AbsolutePosition.X+track.AbsoluteSize.X then
return end local dragOrigin:number=track.AbsolutePosition.X local dragWidth:number=math.max(track.AbsoluteSize.X,1)local function updateDrag(screenX:number):()updateFromScreenX(screenX,dragOrigin,
dragWidth)end updateDrag(input.Position.X)activeSlider={update=updateDrag,finish=function():()notify(labelText..': '..formatCompactNumber(value))end}end))trackUiConnection(valueInput.Focused:Connect(
function():()editing=true valueInput.Text=tostring(value)valueInput.CursorPosition=#valueInput.Text+1 end))trackUiConnection(valueInput.FocusLost:Connect(function():()local parsed:number?=
parseNumericText(valueInput.Text)editing=false if not isNumber(parsed)then valueInput.Text=formatCompactNumber(value)notify(labelText..': invalid numeric value')return end setValue(parsed::number,true
,false)notify(labelText..': '..formatCompactNumber(value))end))task.defer(function():()if option.Parent then local corrected:boolean=boundTyped==true and storedValue~=nil and storedValue~=value
setValue(value,corrected,false)end end)return option end state.addRangeOption=function(feature:any,labelText:string,definition:any):Frame local minimum:number=tonumber(definition.min)or 0 local
maximum:number=tonumber(definition.max)or 100 local baseKey:string=feature.configKey..'.'..labelText:gsub('%W','')local storedLow:number?=tonumber(configData.values[baseKey..'.Low'])local storedHigh:
number?=tonumber(configData.values[baseKey..'.High'])local low:number=isNumber(storedLow)and(storedLow::number)or tonumber(definition.defaultLow)or minimum local high:number=isNumber(storedHigh)and(
storedHigh::number)or tonumber(definition.defaultHigh)or maximum if low>high then low,high=high,low end local integerOnly:boolean=definition.integer~=false local option:Frame=createOptionRow(feature,
labelText,definition.hint)option.Size=UDim2.new(1,0,0,44)feature.updateExpandedSize()local rangeLabel:TextLabel?=option:FindFirstChild('OptionLabel')::TextLabel?if rangeLabel then(rangeLabel::
TextLabel).Size=UDim2.new(0.55,-CHILD_INSET,0,22)end local slider:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=option,Name='RangeSlider',Active=true,AutoButtonColor=
false,BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.new(0,0,0,22),Size=UDim2.new(1,0,0,22),Text='',ZIndex=option.ZIndex+1})::any)::TextButton local track:Frame=(create('Frame',{Parent=
option,Name='Track',BackgroundColor3=Color3.fromRGB(255,255,255),BackgroundTransparency=0.9,BorderSizePixel=0,Position=UDim2.new(0,CHILD_INSET,0,22+(22-SLIDER_BAR_HEIGHT)//2),Size=UDim2.new(1,-
CHILD_INSET*2,0,SLIDER_BAR_HEIGHT),ZIndex=option.ZIndex+1})::any)::Frame local span:Frame=(create('Frame',{Parent=track,Name='Span',BackgroundColor3=Theme.accent,BorderSizePixel=0,Position=UDim2.
fromScale(0,0),Size=UDim2.fromScale(1,1),ZIndex=track.ZIndex+1})::any)::Frame local function makeKnob(name:string):Frame local knob:Frame=(create('Frame',{Parent=track,Name=name,AnchorPoint=Vector2.
new(0.5,0.5),BackgroundColor3=Theme.text,BorderSizePixel=0,Position=UDim2.fromScale(0,0.5),Size=UDim2.fromOffset(SLIDER_KNOB_WIDTH,SLIDER_KNOB_HEIGHT),ZIndex=track.ZIndex+2})::any)::Frame return knob
end local lowKnob:Frame=makeKnob('Low')local highKnob:Frame=makeKnob('High')local step:number?=tonumber(definition.step)if step~=nil and step<=0 then step=nil end option:SetAttribute('Min',minimum)
option:SetAttribute('Max',maximum)option:SetAttribute('Step',step or(integerOnly and 1 or 0))option:SetAttribute('FreeNumeric',true)local highBox:TextBox=(flatTextBox(option,'')::any)::TextBox highBox
.Name='HighValue'highBox.AnchorPoint=Vector2.new(1,0)highBox.Position=UDim2.new(1,-CHILD_INSET,0,0)highBox.Size=UDim2.fromOffset(52,22)highBox.BackgroundTransparency=1 highBox.TextColor3=Theme.
textMuted highBox.TextSize=LABEL_FONT_SIZE highBox.TextXAlignment=Enum.TextXAlignment.Right highBox.TextTruncate=Enum.TextTruncate.AtEnd highBox.ZIndex=option.ZIndex+2 local _dash:TextLabel=(create(
'TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=option,Name='RangeDash',AnchorPoint=Vector2.new(1,0),BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.new(1,-CHILD_INSET-52,0,0),
Size=UDim2.fromOffset(14,22),Text=' \u{2013} ',TextColor3=Theme.textMuted,TextSize=LABEL_FONT_SIZE,TextXAlignment=Enum.TextXAlignment.Center,ZIndex=option.ZIndex+2})::any)::TextLabel local lowBox:
TextBox=(flatTextBox(option,'')::any)::TextBox lowBox.Name='LowValue'lowBox.AnchorPoint=Vector2.new(1,0)lowBox.Position=UDim2.new(1,-CHILD_INSET-52-14,0,0)lowBox.Size=UDim2.fromOffset(52,22)lowBox.
BackgroundTransparency=1 lowBox.TextColor3=Theme.textMuted lowBox.TextSize=LABEL_FONT_SIZE lowBox.TextXAlignment=Enum.TextXAlignment.Right lowBox.TextTruncate=Enum.TextTruncate.AtEnd lowBox.ZIndex=
option.ZIndex+2 for _,box:TextBox in ipairs({lowBox,highBox})do local stroke:UIStroke?=box:FindFirstChild('StyleStroke')::UIStroke?if stroke then(stroke::UIStroke).Transparency=1 end end local
editingLow:boolean=false local editingHigh:boolean=false local function alphaOf(value:number):number if maximum==minimum then return 0 end return math.clamp((value-minimum)/(maximum-minimum),0,1)end
local function refreshRange(persist:boolean):()local lowAlpha:number=alphaOf(low)local highAlpha:number=alphaOf(high)span.Position=UDim2.fromScale(lowAlpha,0)span.Size=UDim2.fromScale(math.max(
highAlpha-lowAlpha,0),1)lowKnob.Position=UDim2.fromScale(lowAlpha,0.5)highKnob.Position=UDim2.fromScale(highAlpha,0.5)if not editingLow then lowBox.Text=formatCompactNumber(low)end if not editingHigh
then highBox.Text=formatCompactNumber(high)end if type(definition.onChanged)=='function'then definition.onChanged(low,high)end if persist then configData.values[baseKey..'.Low']=low configData.values[
baseKey..'.High']=high queueConfigSave()end end local dragging:string='High'local function valueAtScreenX(screenX:number):number local width:number=math.max(track.AbsoluteSize.X,1)local alpha:number=
math.clamp((screenX-track.AbsolutePosition.X)/width,0,1)local value:number=minimum+(maximum-minimum)*alpha if step~=nil then value=minimum+math.round((value-minimum)/(step::number))*(step::number)
elseif integerOnly then value=math.round(value)end return math.clamp(value,minimum,maximum)end local function quantizeTyped(raw:number):number return raw end local function updateFromScreenX(screenX:
number):()local value:number=valueAtScreenX(screenX)if dragging=='Low'then low=math.min(value,high)else high=math.max(value,low)end refreshRange(true)end trackUiConnection(slider.InputBegan:Connect(
function(input:InputObject):()if input.UserInputType~=Enum.UserInputType.MouseButton1 and input.UserInputType~=Enum.UserInputType.Touch then return end if input.Position.X>track.AbsolutePosition.X+
track.AbsoluteSize.X then return end local value:number=valueAtScreenX(input.Position.X)dragging=math.abs(value-low)<=math.abs(value-high)and'Low'or'High'updateFromScreenX(input.Position.X)
activeSlider={update=updateFromScreenX,finish=function():()notify(labelText..': '..formatCompactNumber(low)..' \u{2013} '..formatCompactNumber(high))end}end))trackUiConnection(lowBox.Focused:Connect(
function():()editingLow=true lowBox.Text=tostring(low)lowBox.CursorPosition=#lowBox.Text+1 end))trackUiConnection(lowBox.FocusLost:Connect(function():()local parsed:number?=parseNumericText(lowBox.
Text)editingLow=false if not isNumber(parsed)then lowBox.Text=formatCompactNumber(low)notify(labelText..': invalid numeric value')return end low=math.min(quantizeTyped(parsed::number),high)
refreshRange(true)notify(labelText..': '..formatCompactNumber(low)..' \u{2013} '..formatCompactNumber(high))end))trackUiConnection(highBox.Focused:Connect(function():()editingHigh=true highBox.Text=
tostring(high)highBox.CursorPosition=#highBox.Text+1 end))trackUiConnection(highBox.FocusLost:Connect(function():()local parsed:number?=parseNumericText(highBox.Text)editingHigh=false if not isNumber(
parsed)then highBox.Text=formatCompactNumber(high)notify(labelText..': invalid numeric value')return end high=math.max(quantizeTyped(parsed::number),low)refreshRange(true)notify(labelText..': '..
formatCompactNumber(low)..' \u{2013} '..formatCompactNumber(high))end))task.defer(function():()if option.Parent then refreshRange(false)end end)refreshRange(false)return option end state.
closeChoiceList=function():()end state.choiceListAnchor=nil::TextButton?local function watchPopupOwner(owner:any,onInvalid:()->()):{RBXScriptConnection}local connections:{RBXScriptConnection}={}if
owner==nil then return connections end local scroller:any=nil local windowRoot:any=nil local node:any=owner while node~=nil do if scroller==nil and type(node.IsA)=='function'and node:IsA(
'ScrollingFrame')then scroller=node windowRoot=node.Parent break end node=node.Parent end local function watch(instance:any,property:string,test:()->boolean):()pcall(function():()table.insert(
connections,instance:GetPropertyChangedSignal(property):Connect(function():()if test()then onInvalid()end end))end)end watch(owner,'AbsolutePosition',function()return true end)watch(owner,
'AbsoluteSize',function()return true end)watch(ScreenGui,'AbsoluteSize',function()return true end)if windowRoot then watch(windowRoot,'Visible',function():boolean return windowRoot.Visible~=true end)
pcall(function():()table.insert(connections,windowRoot.AncestryChanged:Connect(function():()if windowRoot.Parent==nil then onInvalid()end end))end)end if scroller then watch(scroller,'Visible',
function():boolean return scroller.Visible~=true end)watch(scroller,'CanvasPosition',function():boolean local ok:boolean,clippedOut:any=pcall(function():boolean local top:number=scroller.
AbsolutePosition.Y local bottom:number=top+scroller.AbsoluteSize.Y local rowTop:number=owner.AbsolutePosition.Y local rowBottom:number=rowTop+owner.AbsoluteSize.Y return rowBottom<top or rowTop>bottom
end)return ok and clippedOut==true end)end return connections end local function selectorGeometry(anchor,wantedHeight)local viewport=ScreenGui.AbsoluteSize local camera=workspace.CurrentCamera if(
viewport.X<1 or viewport.Y<1)and camera then viewport=camera.ViewportSize end local row=anchor.Parent local owner=row while owner and owner~=ScreenGui do if owner:GetAttribute('SelectorOwner')then
break end owner=owner.Parent end if not owner or owner==ScreenGui then owner=row end local width=math.min(math.max(anchor.AbsoluteSize.X,160),math.max(1,viewport.X-16))local height=math.min(
wantedHeight,math.max(24,viewport.Y-16))local bottom=math.max(anchor.AbsolutePosition.Y+anchor.AbsoluteSize.Y,owner.AbsolutePosition.Y+owner.AbsoluteSize.Y)local x=math.clamp(anchor.AbsolutePosition.X
,8,math.max(8,viewport.X-width-8))local y=bottom+6 if y+height>viewport.Y-8 then local right=owner.AbsolutePosition.X+owner.AbsoluteSize.X+6 local left=owner.AbsolutePosition.X-width-6 if right+width
<=viewport.X-8 then x=right elseif left>=8 then x=left else y=owner.AbsolutePosition.Y-height-6 end if x==right or x==left then y=anchor.AbsolutePosition.Y end end y=math.clamp(y,8,math.max(8,viewport
.Y-height-8))return UDim2.fromOffset(math.round(x),math.round(y)),UDim2.fromOffset(math.round(width),math.round(height))end local function openChoiceList(anchor:TextButton,values:{string},currentIndex
:number,onPick:(number)->()):()if state.choiceListAnchor==anchor then state.closeChoiceList()return end state.closeChoiceList()local rowHeight:number=24 local searchable:boolean=#values>8 local
searchHeight:number=searchable and 24 or 0 local visibleRows:number=math.min(#values,6)local listHeight:number=visibleRows*rowHeight+8+searchHeight local listPosition,listSize=selectorGeometry(anchor,
listHeight)local list:Frame=create('Frame',{Parent=ScreenGui,Name='ChoiceList',BackgroundColor3=Theme.surfaceRaised,BackgroundTransparency=0.22,BorderSizePixel=0,Position=listPosition,Size=listSize,
ZIndex=400})::Frame create('UIStroke',{Parent=list,Color=Theme.outline,Transparency=0.6,Thickness=1})do local bitmap:any=state.bitmapText if bitmap and type(bitmap.adoptTree)=='function'then bitmap.
adoptTree(list)end end local filterBox:TextBox?=nil if searchable then local filterFrame:Frame=create('Frame',{Parent=list,Name='Filter',BackgroundColor3=Theme.background,BackgroundTransparency=0.25,
BorderSizePixel=0,ClipsDescendants=true,Position=UDim2.fromOffset(4,4),Size=UDim2.new(1,-8,0,searchHeight-4),ZIndex=402})::Frame local box:TextBox=flatTextBox(filterFrame,'')box.Name='FilterBox'box.
BackgroundTransparency=1 box.PlaceholderText='Search'box.Position=UDim2.fromOffset(8,0)box.Size=UDim2.new(1,-12,1,0)box.TextXAlignment=Enum.TextXAlignment.Left box.TextSize=14 box.ZIndex=403 local
boxCorner:UICorner?=box:FindFirstChild('StyleCorner')::UICorner?if boxCorner then boxCorner:Destroy()end local boxStroke:UIStroke?=box:FindFirstChild('StyleStroke')::UIStroke?if boxStroke then
boxStroke:Destroy()end filterBox=box end local scroller:ScrollingFrame=create('ScrollingFrame',{Parent=list,Name='Items',Active=true,AutomaticCanvasSize=Enum.AutomaticSize.Y,BackgroundTransparency=1,
BorderSizePixel=0,CanvasSize=UDim2.new(),Position=UDim2.fromOffset(4,4+searchHeight),ScrollBarThickness=2,ScrollBarImageTransparency=0.6,Size=UDim2.new(1,-8,1,-(8+searchHeight)),ZIndex=401})::
ScrollingFrame create('UIListLayout',{Parent=scroller,Padding=UDim.new(0,1),SortOrder=Enum.SortOrder.LayoutOrder})local closed:boolean=false local outsideConnection:RBXScriptConnection?=nil local
ownerConnections:{RBXScriptConnection}={}local function close():()if closed then return end closed=true if outsideConnection then outsideConnection:Disconnect()outsideConnection=nil end for _,
connection:RBXScriptConnection in ipairs(ownerConnections)do pcall(function():()connection:Disconnect()end)end ownerConnections={}list:Destroy()if state.choiceListAnchor==anchor then state.
choiceListAnchor=nil end if state.setChoiceBlur then state.setChoiceBlur(false)end state.closeChoiceList=function():()end end state.closeChoiceList=close state.choiceListAnchor=anchor if state.
setChoiceBlur then state.setChoiceBlur(true)end ownerConnections=watchPopupOwner(anchor,close)local rowsByValue:{[string]:TextButton}={}for itemIndex:number,value:string in ipairs(values)do local item
:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,ClipsDescendants=true,Parent=scroller,Name='Choice'..tostring(itemIndex),AutoButtonColor=false,BackgroundColor3=Theme.
surfaceHover,BackgroundTransparency=itemIndex==currentIndex and 0.25 or 1,BorderSizePixel=0,FontFace=CONTROL_FONT,LayoutOrder=itemIndex,Size=UDim2.new(1,0,0,rowHeight-1),Text=value,TextColor3=
itemIndex==currentIndex and Theme.text or Theme.textMuted,TextSize=14,ZIndex=402})::any)::TextButton create('UIPadding',{Parent=item,PaddingLeft=UDim.new(0,10),PaddingRight=UDim.new(0,8)})if itemIndex
==currentIndex then create('Frame',{Parent=item,Name='PickedBar',BackgroundColor3=Theme.enabled,BorderSizePixel=0,Position=UDim2.fromOffset(-10,0),Size=UDim2.new(0,3,1,0),ZIndex=403})end local
itemStroke:UIStroke=(create('UIStroke',{Parent=item,Name='EntryStroke',Color=Theme.outline,Thickness=1,Transparency=itemIndex==currentIndex and 0.35 or 0.82})::any)::UIStroke item.TextXAlignment=Enum.
TextXAlignment.Left item.MouseEnter:Connect(function():()if itemIndex~=currentIndex then item.BackgroundTransparency=0.6 item.TextColor3=Theme.text end itemStroke.Transparency=0.35 end)item.MouseLeave
:Connect(function():()item.BackgroundTransparency=itemIndex==currentIndex and 0.25 or 1 item.TextColor3=itemIndex==currentIndex and Theme.text or Theme.textMuted itemStroke.Transparency=itemIndex==
currentIndex and 0.35 or 0.82 end)item.MouseButton1Click:Connect(function():()close()onPick(itemIndex)end)rowsByValue[value]=item end if filterBox then local box:TextBox=filterBox::TextBox
trackUiConnection(box:GetPropertyChangedSignal('Text'):Connect(function():()local query:string=string.lower(box.Text)for value:string,row:TextButton in pairs(rowsByValue)do row.Visible=query==''or
string.find(string.lower(value),query,1,true)~=nil end end))task.defer(function():()if not closed then box:CaptureFocus()end end)end outsideConnection=UserInputService.InputBegan:Connect(function(
input:InputObject):()if input.KeyCode==Enum.KeyCode.Escape then close()return end if input.UserInputType~=Enum.UserInputType.MouseButton1 and input.UserInputType~=Enum.UserInputType.Touch then return
end task.defer(function():()if closed then return end local pointer:Vector2=Vector2.new(input.Position.X,input.Position.Y)local origin:Vector2=list.AbsolutePosition local size:Vector2=list.
AbsoluteSize local inside:boolean=pointer.X>=origin.X and pointer.X<=origin.X+size.X and pointer.Y>=origin.Y and pointer.Y<=origin.Y+size.Y if not inside then close()end end)end)trackUiConnection(
outsideConnection::RBXScriptConnection)end state.openChoiceList=openChoiceList state.addMenuVisibilityListener(function(visible:boolean):()if not visible then state.closeChoiceList()end end)state.
openMultiList=function(anchor:TextButton,items:{string},selected:{[string]:boolean},onToggle:(string,boolean)->(),onBulk:((boolean)->())?,onRemove:((string)->())?):()if state.choiceListAnchor==anchor
then state.closeChoiceList()return end state.closeChoiceList()local rowHeight:number=24 local rows:number=math.min(#items+(onBulk and 1 or 0),7)local listHeight:number=math.max(rows,1)*rowHeight+8
local listPosition,listSize=selectorGeometry(anchor,listHeight)local list:Frame=create('Frame',{Parent=ScreenGui,Name='MultiList',BackgroundColor3=Theme.surface,BackgroundTransparency=0.22,
BorderSizePixel=0,Position=listPosition,Size=listSize,ZIndex=400})::Frame create('UIStroke',{Parent=list,Color=Theme.outline,Transparency=0.6,Thickness=1})local scroller:ScrollingFrame=create(
'ScrollingFrame',{Parent=list,Name='Items',Active=true,AutomaticCanvasSize=Enum.AutomaticSize.Y,BackgroundTransparency=1,BorderSizePixel=0,CanvasSize=UDim2.new(),Position=UDim2.fromOffset(4,4),
ScrollBarThickness=2,ScrollBarImageTransparency=0.6,Size=UDim2.new(1,-8,1,-8),ZIndex=401})::ScrollingFrame create('UIListLayout',{Parent=scroller,Padding=UDim.new(0,1),SortOrder=Enum.SortOrder.
LayoutOrder})local closed:boolean=false local outsideConnection:RBXScriptConnection?=nil local ownerConnections:{RBXScriptConnection}={}local function close():()if closed then return end closed=true
if outsideConnection then outsideConnection:Disconnect()outsideConnection=nil end for _,connection:RBXScriptConnection in ipairs(ownerConnections)do pcall(function():()connection:Disconnect()end)end
ownerConnections={}list:Destroy()if state.choiceListAnchor==anchor then state.choiceListAnchor=nil end if state.setChoiceBlur then state.setChoiceBlur(false)end state.closeChoiceList=function():()end
end state.closeChoiceList=close state.choiceListAnchor=anchor if state.setChoiceBlur then state.setChoiceBlur(true)end ownerConnections=watchPopupOwner(anchor,close)local function makeRow(text:string,
order:number):TextButton local row:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,ClipsDescendants=true,Parent=scroller,Name='Row'..tostring(order),AutoButtonColor=false,
BackgroundColor3=Theme.surfaceHover,BackgroundTransparency=1,BorderSizePixel=0,FontFace=CONTROL_FONT,LayoutOrder=order,Size=UDim2.new(1,0,0,rowHeight-1),Text=text,TextColor3=Theme.textMuted,TextSize=
14,ZIndex=402})::any)::TextButton create('UIPadding',{Parent=row,PaddingLeft=UDim.new(0,26),PaddingRight=UDim.new(0,8)})row.TextXAlignment=Enum.TextXAlignment.Left return row end if onBulk then local
bulk:TextButton=makeRow('All / none',0)bulk.TextColor3=Theme.text local anyOn:boolean=false for _,item:string in ipairs(items)do if selected[item]then anyOn=true break end end bulk.MouseButton1Click:
Connect(function():()close();(onBulk::(boolean)->())(not anyOn)end)end for index:number,item:string in ipairs(items)do local row:TextButton=makeRow(item,index)if onRemove then local wrapper=create(
'Frame',{Name='Row'..index,Parent=scroller,BackgroundTransparency=1,Size=UDim2.new(1,0,0,rowHeight-1),LayoutOrder=index,ZIndex=402})row.Name='Select'row.Parent=wrapper row.Size=UDim2.new(1,-24,1,0)row
.TextTruncate=Enum.TextTruncate.AtEnd local remove=create('TextButton',{Name='Remove',Parent=wrapper,AnchorPoint=Vector2.new(1,0),Position=UDim2.fromScale(1,0),Size=UDim2.new(0,24,1,0),
BackgroundTransparency=1,BorderSizePixel=0,Text='\215',FontFace=CONTROL_FONT,TextSize=16,TextColor3=Theme.textMuted,AutoButtonColor=false,ZIndex=403})remove.MouseButton1Click:Connect(function()close()
onRemove(item)end)end local mark:Frame=create('Frame',{Parent=row,Name='Mark',AnchorPoint=Vector2.new(0,0.5),BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.new(0,-18,0.5,0),Size=UDim2.
fromOffset(11,11),ZIndex=403})::Frame local markStroke:UIStroke=(create('UIStroke',{Parent=mark,Color=Theme.textMuted,Transparency=0.35,Thickness=1})::any)::UIStroke local fill:Frame=create('Frame',{
Parent=mark,Name='Fill',AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.text,BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.fromScale(0.5,0.5),Size=UDim2.fromScale(0,0),ZIndex=404})
::Frame local function paint():()local on:boolean=selected[item]==true fill.Size=on and UDim2.fromScale(0.6,0.6)or UDim2.fromScale(0,0)fill.BackgroundTransparency=on and 0.05 or 1 markStroke.
Transparency=on and 0.1 or 0.35 row.TextColor3=on and Theme.text or Theme.textMuted end paint()row.MouseEnter:Connect(function():()row.BackgroundTransparency=0.72 end)row.MouseLeave:Connect(function()
:()row.BackgroundTransparency=1 end)row.MouseButton1Click:Connect(function():()selected[item]=not selected[item]paint()onToggle(item,selected[item]==true)end)end outsideConnection=UserInputService.
InputBegan:Connect(function(input:InputObject):()if input.KeyCode==Enum.KeyCode.Escape then close()return end if input.UserInputType~=Enum.UserInputType.MouseButton1 and input.UserInputType~=Enum.
UserInputType.Touch then return end task.defer(function():()if closed then return end local pointer:Vector2=Vector2.new(input.Position.X,input.Position.Y)local origin:Vector2=list.AbsolutePosition
local size:Vector2=list.AbsoluteSize if pointer.X<origin.X or pointer.X>origin.X+size.X or pointer.Y<origin.Y or pointer.Y>origin.Y+size.Y then close()end end)end)trackUiConnection(outsideConnection::
RBXScriptConnection)end state.addListOption=function(feature:any,labelText:string,definition:any):TextButton local optionKey:string=feature.configKey..'.'..labelText:gsub('%W','')local selected:{[
string]:boolean}={}local stored:any=configData.values[optionKey]if type(stored)=='string'and stored~=''then for entry:string in string.gmatch(stored,'[^,]+')do local trimmed:string=string.match(entry,
'^%s*(.-)%s*$')or entry if trimmed~=''then selected[trimmed]=true end end elseif type(definition.default)=='table'then for _,entry:string in ipairs(definition.default)do selected[entry]=true end end
local option:Frame=createOptionRow(feature,labelText,definition.hint)local button:TextButton=makeButton(option,'',11)state.layoutOptionControl(option,button,0.5)button.TextXAlignment=Enum.
TextXAlignment.Left create('UIPadding',{Parent=button,Name='ListPadding',PaddingLeft=UDim.new(0,8),PaddingRight=UDim.new(0,10)})local function currentItems():{string}local provider:any=definition.
items if type(provider)=='function'then local ok:boolean,result:any=pcall(provider)if ok and type(result)=='table'then return result end return{}end return provider or{}end local function describe():(
)local items:{string}=currentItems()local total:number=#items local count:number=0 local last:string=''for _,item:string in ipairs(items)do if selected[item]then count+=1 last=item end end if total==0
then button.Text=definition.emptyText or''elseif count==0 then button.Text=definition.single and'Choose...'or''elseif count==1 then button.Text=last elseif count==total then button.Text='all '..
tostring(total)else button.Text=tostring(count)..' of '..tostring(total)end end local function publish():()local names:{string}={}for name:string,on:boolean in pairs(selected)do if on then table.
insert(names,name)end end table.sort(names)configData.values[optionKey]=table.concat(names,',')queueConfigSave()describe()if type(definition.onChanged)=='function'then definition.onChanged(selected,
names)end end local function setItem(item:string,on:boolean):()if definition.single and on then table.clear(selected)end selected[item]=on or nil publish()end local function openItems():()local items:
{string}=currentItems()local bulk=nil if not definition.single then bulk=function(enableAll:boolean):()for _,item:string in ipairs(items)do selected[item]=enableAll or nil end publish()end end state.
openMultiList(button,items,selected,function(item,on)setItem(item,on)if definition.single then state.closeChoiceList()end end,bulk,definition.onRemove and function(item)selected[item]=nil definition.
onRemove(item)publish()if#currentItems()>0 then openItems()end end or nil)end button.MouseButton1Click:Connect(openItems)local initialNames={}if definition.single then for name,on in pairs(selected)do
if on then table.insert(initialNames,name)end end table.sort(initialNames)if#initialNames>1 then local first=initialNames[1]table.clear(selected)selected[first]=true initialNames={first}end end
describe()if type(definition.onChanged)=='function'then definition.onChanged(selected,initialNames)end button:SetAttribute('ListOption',optionKey)state.listSelections=state.listSelections or{}state.
listSelections[optionKey]=selected state.listControls=state.listControls or{}state.listControls[optionKey]={set=function(item:string,on:boolean):()setItem(item,on)end,refresh=function()if state.
choiceListAnchor==button then state.closeChoiceList()end publish()end,selected=selected}button:SetAttribute('ListKey',optionKey)return button end local function addCycleOption(feature:any,labelText:
string,values:{string},initialIndex:number?,callback:(string)->(),hintText:string?):TextButton local optionKey=feature.configKey..'.'..labelText:gsub('%W','')local storedValue=configData.values[
optionKey]local option=createOptionRow(feature,labelText,hintText)local index=initialIndex or 1 if storedValue~=nil then for valueIndex,value in ipairs(values)do if value==storedValue then index=
valueIndex break end end end local button:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=option,Name='ComboValue',Active=true,AutoButtonColor=false,BackgroundTransparency
=1,BorderSizePixel=0,FontFace=CONTROL_FONT,Position=UDim2.new(0.55,4,0,0),Size=UDim2.new(0.45,-(26+CHILD_INSET),1,0),ClipsDescendants=true,Text=values[index],TextColor3=Theme.text,TextSize=
LABEL_FONT_SIZE,TextXAlignment=Enum.TextXAlignment.Right,ZIndex=option.ZIndex+2})::any)::TextButton local arrowBox:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=option,
Name='ComboArrow',Active=true,AnchorPoint=Vector2.new(1,0),AutoButtonColor=false,BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.new(1,-CHILD_INSET,0,0),Size=UDim2.fromOffset(22,22),Text='',
ZIndex=option.ZIndex+2})::any)::TextButton drawPixelIcon(arrowBox,'down',nil,'arrows')if state and type(state.registerThemeRefresh)=='function'then state.registerThemeRefresh(function():()if arrowBox.
Parent then drawPixelIcon(arrowBox,'down',nil,'arrows')end end,{'arrows'})end local function refreshChoiceVisual():()if#values>0 then index=math.clamp(index,1,#values)end button.Text=values[index]or''
end refreshChoiceVisual()local function openList():()local control:any=state.valueControls and state.valueControls[optionKey]if type(control)=='table'and type(control.onBeforeOpen)=='function'then
pcall(control.onBeforeOpen)refreshChoiceVisual()end openChoiceList(button,values,index,function(picked:number):()index=picked refreshChoiceVisual()callback(values[index])configData.values[optionKey]=
values[index]queueConfigSave()end)end button.MouseButton1Click:Connect(openList)arrowBox.MouseButton1Click:Connect(openList)if state.valueControls then state.valueControls[optionKey]={apply=function(
stored:any):()local picked:number?=table.find(values,tostring(stored))if picked==nil then return end index=picked::number refreshChoiceVisual()callback(values[index])configData.values[optionKey]=
values[index]queueConfigSave()end,setIndex=function(newIndex:number):()index=math.clamp(newIndex,1,math.max(1,#values))refreshChoiceVisual()end,refresh=refreshChoiceVisual,default=values[initialIndex
or 1]}end callback(values[index])return button::TextButton end local function addKeyOption(feature:any,labelText:string,defaultKey:Enum.KeyCode,callback:(Enum.KeyCode)->(),hintText:string?):TextButton
feature.hasKeyOption=true local optionKey=feature.configKey..'.'..labelText:gsub('%W','')local savedKey=nil local savedName=configData.values[optionKey]if type(savedName)=='string'and savedName~=''
then pcall(function()savedKey=Enum.KeyCode[savedName]end)end local initialKey=savedKey or defaultKey local currentKey:Enum.KeyCode=initialKey local function keyLabel(name:any):string if type(state.
keyDisplayName)=='function'then return state.keyDisplayName(name)end return tostring(name or'')end local option=createOptionRow(feature,labelText,hintText)if state.isMobile then option.Visible=false
feature.updateExpandedSize()end local button=makeButton(option,currentKey==Enum.KeyCode.Unknown and''or keyLabel(currentKey.Name),11)state.layoutOptionControl(option,button,0.5)applyKeySlotStyle(
button)button:SetAttribute('CaptureIdleText',button.Text)local function assignKey(keyCode:Enum.KeyCode):boolean local cleared:boolean=keyCode==currentKey currentKey=cleared and Enum.KeyCode.Unknown or
keyCode button.Text=currentKey==Enum.KeyCode.Unknown and''or keyLabel(currentKey.Name)button:SetAttribute('CaptureIdleText',button.Text)callback(currentKey)configData.values[optionKey]=currentKey.Name
queueConfigSave()return not cleared end button.MouseButton1Click:Connect(function()if state.isMobile then return end beginKeyCapture(button,function(keyCode:Enum.KeyCode):()assignKey(keyCode)end)end)
callback(currentKey)return button::TextButton end state.addFeatureShortcutOption=function(feature:any,label:string,hint:string?):any local binding:any=state.shortcutBindings and state.shortcutBindings
[feature.configKey]if not binding or state.isMobile then return nil end local option:any=createOptionRow(feature,label,hint)local button:TextButton=(makeButton(option,'',11)::any)::TextButton state.
layoutOptionControl(option,button,0.5)applyKeySlotStyle(button)local previousRefresh:any=binding.refresh binding.refresh=function():()if type(previousRefresh)=='function'then previousRefresh()end
local key:any=binding.key button.Text=key and key~=Enum.KeyCode.Unknown and state.keyDisplayName(key.Name)or''button:SetAttribute('CaptureIdleText',button.Text)end binding.refresh()trackUiConnection(
button.MouseButton1Click:Connect(function():()beginKeyCapture(button,function(keyCode:Enum.KeyCode):()binding.assign(keyCode)end)end))return button end local function addTextOption(feature:any,
labelText:string,defaultText:string,callback:(string)->(),persist:boolean?,hintText:string?):TextBox local optionKey=feature.configKey..'.'..labelText:gsub('%W','')local shouldPersist=persist~=false
local initialText=shouldPersist and configData.values[optionKey]or nil if initialText==nil then initialText=defaultText or''end local option=createOptionRow(feature,labelText,hintText)local box=
flatTextBox(option,initialText)box.PlaceholderText='Enter'box.TextXAlignment=Enum.TextXAlignment.Center state.layoutOptionControl(option,box,0.5)box.FocusLost:Connect(function(enterPressed)if
enterPressed then callback(box.Text)end end)box:GetPropertyChangedSignal('Text'):Connect(function()callback(box.Text)if shouldPersist then configData.values[optionKey]=box.Text queueConfigSave()end
end)if not shouldPersist and configData.values[optionKey]~=nil then configData.values[optionKey]=nil queueConfigSave()end if state.valueControls then state.valueControls[optionKey]={apply=function(
stored:any):()box.Text=tostring(stored or'')if shouldPersist then configData.values[optionKey]=box.Text queueConfigSave()end end,default=defaultText or''}end callback(initialText)return box::TextBox
end local colorPicker={}colorPicker.open=function(initialColor,onApply,ownerRow)if not colorPicker.panel then local panel=create('Frame',{Parent=PopupLayer,Name='ColorPicker',Active=true,
BackgroundColor3=Theme.surface,BorderColor3=Theme.outline,BorderSizePixel=1,Position=UDim2.new(0.5,-170,0.5,-150),Size=UDim2.fromOffset(340,300),Visible=false,ZIndex=460})do local bitmap:any=state.
bitmapText if bitmap and type(bitmap.adoptTree)=='function'then bitmap.adoptTree(panel)end end local title=makeTextLabel(panel,'COLOR',14)title.FontFace=CONTROL_FONT title.Position=UDim2.fromOffset(15
,8)title.Size=UDim2.new(1,-60,0,28)title.ZIndex=81 local close=makeButton(panel,'X',12)close.AutoButtonColor=false close.Position=UDim2.new(1,-42,0,8)close.Size=UDim2.fromOffset(28,28)close.ZIndex=82
local sv=makeButton(panel,'',10)sv.AutoButtonColor=false sv:SetAttribute('NoHover',true)sv.Position=UDim2.fromOffset(15,44)sv.Size=UDim2.fromOffset(205,165)sv.ZIndex=81 local saturationGradient=
create('UIGradient',{Parent=sv,Color=ColorSequence.new(Color3.fromRGB(255,255,255),Color3.fromRGB(255,0,0))})local shade=create('Frame',{Parent=sv,BackgroundColor3=Theme.background,BorderSizePixel=0,
Size=UDim2.fromScale(1,1),ZIndex=462})create('UIGradient',{Parent=shade,Rotation=90,Transparency=NumberSequence.new({NumberSequenceKeypoint.new(0,1),NumberSequenceKeypoint.new(1,0)})})local svCursor=
create('Frame',{Parent=sv,AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Color3.fromRGB(255,255,255),BorderColor3=Theme.surface,BorderSizePixel=2,Size=UDim2.fromOffset(12,12),ZIndex=463})create(
'UICorner',{Parent=svCursor,CornerRadius=UDim.new(1,0)})local hue=makeButton(panel,'',10)hue.AutoButtonColor=false hue:SetAttribute('NoHover',true)hue.Position=UDim2.fromOffset(232,44)hue.Size=UDim2.
fromOffset(22,165)hue.ZIndex=81 create('UIGradient',{Parent=hue,Rotation=90,Color=ColorSequence.new({ColorSequenceKeypoint.new(0,Color3.fromHSV(0,1,1)),ColorSequenceKeypoint.new(0.17,Color3.fromHSV(
0.17,1,1)),ColorSequenceKeypoint.new(0.33,Color3.fromHSV(0.33,1,1)),ColorSequenceKeypoint.new(0.5,Color3.fromHSV(0.5,1,1)),ColorSequenceKeypoint.new(0.67,Color3.fromHSV(0.67,1,1)),
ColorSequenceKeypoint.new(0.83,Color3.fromHSV(0.83,1,1)),ColorSequenceKeypoint.new(1,Color3.fromHSV(1,1,1))})})local hueCursor=create('Frame',{Parent=hue,AnchorPoint=Vector2.new(0.5,0.5),
BackgroundColor3=Color3.fromRGB(245,245,245),BorderColor3=Theme.surface,BorderSizePixel=1,Position=UDim2.new(0.5,0,0,0),Size=UDim2.new(1,6,0,5),ZIndex=463})local preview=create('Frame',{Parent=panel,
BackgroundColor3=Color3.fromRGB(255,255,255),BorderColor3=Color3.fromRGB(85,85,85),BorderSizePixel=1,Position=UDim2.fromOffset(273,45),Size=UDim2.fromOffset(48,48),ZIndex=461})local apply=makeButton(
panel,'APPLY',11)apply.AutoButtonColor=false apply.Position=UDim2.fromOffset(268,105)apply.Size=UDim2.fromOffset(58,34)apply.ZIndex=81 local hex=flatTextBox(panel,'#ffffff')hex.Name='HexField'hex.
Position=UDim2.fromOffset(15,218)hex.Size=UDim2.fromOffset(150,30)hex.PlaceholderText='#rrggbb'hex.TextXAlignment=Enum.TextXAlignment.Center hex.ZIndex=81 local rgb=flatTextBox(panel,'255, 255, 255')
rgb.Name='RgbField'rgb.Position=UDim2.fromOffset(174,218)rgb.Size=UDim2.fromOffset(151,30)rgb.PlaceholderText='r, g, b'rgb.TextXAlignment=Enum.TextXAlignment.Center rgb.ZIndex=81 local hint=
makeTextLabel(panel,[[Drag the square for saturation and brightness, the strip for hue]],10)hint.TextColor3=Color3.fromRGB(145,145,145)hint.Position=UDim2.fromOffset(15,256)hint.Size=UDim2.new(1,-30,0
,24)hint.ZIndex=81 colorPicker.panel=panel colorPicker.sv=sv colorPicker.saturationGradient=saturationGradient colorPicker.svCursor=svCursor colorPicker.hue=hue colorPicker.hueCursor=hueCursor
colorPicker.preview=preview colorPicker.hex=hex colorPicker.rgb=rgb colorPicker.apply=apply colorPicker.drag=nil local function updateColor()local color=Color3.fromHSV(colorPicker.h,colorPicker.s,
colorPicker.v)colorPicker.current=color colorPicker.preview.BackgroundColor3=color colorPicker.saturationGradient.Color=ColorSequence.new(Color3.fromRGB(255,255,255),Color3.fromHSV(colorPicker.h,1,1))
colorPicker.svCursor.Position=UDim2.fromScale(colorPicker.s,1-colorPicker.v)colorPicker.hueCursor.Position=UDim2.new(0.5,0,colorPicker.h,0)local channels=colorToConfig(color)colorPicker.hex.Text=
string.format('#%02x%02x%02x',channels[1],channels[2],channels[3])colorPicker.rgb.Text=string.format('%d, %d, %d',channels[1],channels[2],channels[3])end colorPicker.update=updateColor local function
updateFromPointer(kind,position)local object=kind=='hue'and hue or sv local relative=Vector2.new(position.X,position.Y)-object.AbsolutePosition if kind=='hue'then colorPicker.h=math.clamp(relative.Y/
object.AbsoluteSize.Y,0,1)else colorPicker.s=math.clamp(relative.X/object.AbsoluteSize.X,0,1)colorPicker.v=1-math.clamp(relative.Y/object.AbsoluteSize.Y,0,1)end updateColor()end sv.InputBegan:Connect(
function(input)if input.UserInputType==Enum.UserInputType.MouseButton1 or input.UserInputType==Enum.UserInputType.Touch then colorPicker.drag='sv'updateFromPointer('sv',input.Position)end end)hue.
InputBegan:Connect(function(input)if input.UserInputType==Enum.UserInputType.MouseButton1 or input.UserInputType==Enum.UserInputType.Touch then colorPicker.drag='hue'updateFromPointer('hue',input.
Position)end end)colorPicker.inputChanged=UserInputService.InputChanged:Connect(function(input)if colorPicker.panel.Visible and colorPicker.drag and(input.UserInputType==Enum.UserInputType.
MouseMovement or input.UserInputType==Enum.UserInputType.Touch)then updateFromPointer(colorPicker.drag,input.Position)end end)colorPicker.inputEnded=UserInputService.InputEnded:Connect(function(input)
if input.UserInputType==Enum.UserInputType.MouseButton1 or input.UserInputType==Enum.UserInputType.Touch then colorPicker.drag=nil end end)hex.FocusLost:Connect(function()local value=string.match(hex.
Text,'^#?(%x%x%x%x%x%x)$')if value then local r=tonumber(string.sub(value,1,2),16)local g=tonumber(string.sub(value,3,4),16)local b=tonumber(string.sub(value,5,6),16)colorPicker.h,colorPicker.s,
colorPicker.v=Color3.fromRGB(r,g,b):ToHSV()end updateColor()end)rgb.FocusLost:Connect(function()local r,g,b=string.match(rgb.Text,'^%s*(%d+)%s*[, ]%s*(%d+)%s*[, ]%s*(%d+)%s*$')if r and g and b then
colorPicker.h,colorPicker.s,colorPicker.v=Color3.fromRGB(math.clamp(tonumber(r)or 0,0,255),math.clamp(tonumber(g)or 0,0,255),math.clamp(tonumber(b)or 0,0,255)):ToHSV()end updateColor()end)close.
MouseButton1Click:Connect(function()panel.Visible=false end)apply.MouseButton1Click:Connect(function()if colorPicker.callback then colorPicker.callback(colorPicker.current)end panel.Visible=false end)
end colorPicker.h,colorPicker.s,colorPicker.v=initialColor:ToHSV()colorPicker.callback=onApply colorPicker.panel.Visible=true colorPicker.update()for _,connection in ipairs(colorPicker.
ownerConnections or{})do pcall(function()connection:Disconnect()end)end colorPicker.ownerConnections=watchPopupOwner(ownerRow,function()colorPicker.panel.Visible=false for _,connection in ipairs(
colorPicker.ownerConnections or{})do pcall(function()connection:Disconnect()end)end colorPicker.ownerConnections={}end)end state.colorPicker=colorPicker local function addColorOption(feature,labelText
,defaultColor,callback,hintText,keyText)local optionKey=feature.configKey..'.'..(keyText or labelText):gsub('%W','')local color=colorFromConfig(configData.values[optionKey],defaultColor)if color==nil
then color=Color3.fromRGB(255,255,255)end local option=createOptionRow(feature,labelText,hintText)option.Size=UDim2.new(1,0,0,44)feature.updateExpandedSize()local colorLabel:TextLabel?=option:
FindFirstChild('OptionLabel')::TextLabel?if colorLabel then(colorLabel::TextLabel).Size=UDim2.new(0.55,-CHILD_INSET,0,22)end local hexValue:TextLabel=(create('TextLabel',{TextTruncate=Enum.
TextTruncate.AtEnd,Parent=option,Name='HexValue',AnchorPoint=Vector2.new(1,0),BackgroundTransparency=1,FontFace=CONTROL_FONT,Position=UDim2.new(1,-CHILD_INSET,0,0),Size=UDim2.new(0.42,0,0,22),Text='',
TextColor3=Theme.textMuted,TextSize=LABEL_FONT_SIZE,TextXAlignment=Enum.TextXAlignment.Right,ZIndex=option.ZIndex+2})::any)::TextLabel local bar:Frame=(create('Frame',{Parent=option,Name='ColorBar',
BackgroundColor3=color,BorderSizePixel=0,Position=UDim2.new(0,CHILD_INSET,0,22),Size=UDim2.new(1,-CHILD_INSET*2,0,22),ZIndex=option.ZIndex+1})::any)::Frame local swatch=create('TextButton',{
TextTruncate=Enum.TextTruncate.AtEnd,Parent=option,Name='ColorClickTarget',Active=true,AutoButtonColor=false,BackgroundTransparency=1,BorderSizePixel=0,Size=UDim2.fromScale(1,1),Text='',ZIndex=option.
ZIndex+3})local function hexOf(value:Color3):string return string.format('#%02X%02X%02X',math.floor(value.R*255+0.5),math.floor(value.G*255+0.5),math.floor(value.B*255+0.5))end hexValue.Text=hexOf(
color)swatch.MouseButton1Click:Connect(function()colorPicker.open(color,function(newColor)color=newColor bar.BackgroundColor3=color hexValue.Text=hexOf(color)configData.values[optionKey]=
colorToConfig(color)callback(color)queueConfigSave()end,swatch)end)if state.valueControls then state.valueControls[optionKey]={apply=function(stored:any):()local restored:Color3=colorFromConfig(stored
,defaultColor)color=restored bar.BackgroundColor3=color hexValue.Text=hexOf(color)configData.values[optionKey]=colorToConfig(color)callback(color)queueConfigSave()end,default=colorToConfig(
defaultColor)}end callback(color)return swatch end local function addInformationOption(feature:any,text:string):()local current:string=feature.tooltipText or feature.description or''local combined:
string=current~=''and(current..'\n'..text)or text feature.tooltipText=combined feature.description=combined end type OptionDefinition={kind:string,label:string,default:any?,minimum:number?,maximum:
number?,values:{string}?,callback:any,persist:boolean?}type FeatureDefinition={name:string,description:string,order:number,onToggle:any,configuration:any?,options:{OptionDefinition}?}local function
addConfiguredOptions(feature:any,definitions:{OptionDefinition}):()for _,definition:OptionDefinition in ipairs(definitions)do if definition.kind=='toggle'then addToggleOption(feature,definition.label,
definition.default==true,definition.callback)elseif definition.kind=='slider'then addNumberOption(feature,definition.label,tonumber(definition.default)or 0,definition.minimum or 0,definition.maximum
or 100,definition.callback)elseif definition.kind=='color'then addColorOption(feature,definition.label,definition.default::Color3,definition.callback)elseif definition.kind=='cycle'then
addCycleOption(feature,definition.label,definition.values or{},tonumber(definition.default)or 1,definition.callback)elseif definition.kind=='keybind'then addKeyOption(feature,definition.label,
definition.default::Enum.KeyCode,definition.callback)elseif definition.kind=='text'then addTextOption(feature,definition.label,tostring(definition.default or''),definition.callback,definition.persist)
elseif definition.kind=='action'then addActionOption(feature,definition.label,definition.callback)elseif definition.kind=='info'then addInformationOption(feature,definition.label)end end end
local function createFeatureFromDefinition(definition:FeatureDefinition):any local feature:any=host.createUniversalFeature(definition.name,definition.description,definition.order,definition.onToggle,
definition.configuration)if definition.options then addConfiguredOptions(feature,definition.options)end return feature end local function createFeaturesFromDefinitions(definitions:{FeatureDefinition})
:{any}local createdFeatures:{any}={}for _,definition:FeatureDefinition in ipairs(definitions)do table.insert(createdFeatures,createFeatureFromDefinition(definition))end return createdFeatures end
state.components={createFeature=createFeatureFromDefinition,createFeatures=createFeaturesFromDefinitions,addOptions=addConfiguredOptions}state.setToggleChecked=function(optionKey:string,checked:
boolean):boolean local control:any=state.toggleControls and state.toggleControls[optionKey]if type(control)=='table'and type(control.setChecked)=='function'then control.setChecked(checked)return true
end return false end state.beginCardPanel=function(feature:any,name:string,pinToFront:boolean?):()->()local panels:any=feature.cardPanels if not panels then panels={selected=name,values={},groups={}}
feature.cardPanels=panels function panels:refresh():()for group:any,rows:any in pairs(self.groups)do for _,row:any in ipairs(rows)do row.Visible=group==self.selected end end feature.
updateExpandedSize()end function panels:buildSelector():()if self.selector then if state.closeChoiceList then state.closeChoiceList()end local oldRow:any=self.selector.Parent local index:number?=table
.find(feature.optionRows,oldRow)if index then table.remove(feature.optionRows,index::number)end oldRow:Destroy()end self.selector=addCycleOption(feature,'Mode',self.values,table.find(self.values,self.
selected)or 1,function(value:string):()self.selected=value self:refresh()end,'Select controls to display. Recording toggles stay '..'independent; switching views never discards '..
'data or stops an enabled observer.')if self.selector then local restored:any=(self.selector::any).Text if type(restored)=='string'and table.find(self.values,restored)then self.selected=restored end(
self.selector::any).Parent.LayoutOrder=-1 end self:refresh()end end if not table.find(panels.values,name)then if pinToFront then table.insert(panels.values,1,name)panels.selected=name else table.
insert(panels.values,name)end panels:buildSelector()end local first:number=#feature.optionRows+1 return function():()local rows:{any}={}for index:number=first,#feature.optionRows do table.insert(rows,
feature.optionRows[index])end panels.groups[name]=rows panels:refresh()end end Module.Initialized=true return{addToggleOption=addToggleOption,addActionOption=addActionOption,addNumberOption=
addNumberOption,addRangeOption=state.addRangeOption,addCycleOption=addCycleOption,addKeyOption=addKeyOption,addTextOption=addTextOption,addColorOption=addColorOption,addListOption=state.addListOption,
addSectionOption=state.addSectionOption,addInformationOption=addInformationOption,setOptionVisible=setOptionVisible,drawPixelIcon=drawPixelIcon}end function Module.destroy():()Module.Initialized=false
table.clear(toggleThemeRefreshers)if boundState and type(boundState.valueControls)=='table'then table.clear(boundState.valueControls)end if boundState and boundState.optionThemeRefresh==
runToggleThemeRefreshers then boundState.optionThemeRefresh=nil end boundState=nil end return Module
-- src/guis/Wurst/Code/WindowManager.lua
export type WindowDefinition={id:string,title:string?,size:Vector2,position:Vector2,chromeless:boolean?,closable:boolean?,collapsible:boolean?,pinnable:boolean?,maxHeight:number?,onClose:(()->())?}
export type Window={id:string,root:GuiObject,body:GuiObject,header:Frame?,closeButton:TextButton?,collapsed:boolean,pinned:boolean,SetVisible:(self:Window,visible:boolean)->(),SetCollapsed:(self:
Window,collapsed:boolean)->(),SetPinned:(self:Window,pinned:boolean)->(),Raise:(self:Window)->(),Destroy:(self:Window)->()}local Module={Name='WindowManager',PlaceId=0,Events={}::{[string]:any},
Initialized=false}local activeCleanup:(()->())?=nil local SNAP_DISTANCE:number=8 local BASE_Z:number=150 local MARGIN:number=6 local PLACEMENT_GAP:number=10 local TITLE_HEIGHT:number=26 local
TITLE_FONT_SIZE:number=16 function Module.init(context:any):any local host:any=context local state:any=host.state local configData:any=host.configData local queueConfigSave:any=host.queueConfigSave
local create:any=host.create local Theme:any=host.Theme local ThemeEngine:any=host.ThemeEngine local CONTROL_FONT:any=host.CONTROL_FONT local PopupLayer:any=host.PopupLayer local trackUiConnection:any
=host.trackUiConnection local UserInputService:any=host.UserInputService local drawPixelIcon:any=host.drawPixelIcon or function(parent:Instance,_kind:string,_color:Color3?):Frame local existing:
Instance?=(parent::any):FindFirstChild('PixelIcon')if existing then existing:Destroy()end return create('Frame',{Parent=parent,Name='PixelIcon',BackgroundTransparency=1})::Frame end configData.ui=
configData.ui or{}configData.ui.windows=configData.ui.windows or{}local manager:any={list={}::{any},byId={}::{[string]:any}}local function screenViewport():Vector2 local stable:any=state.
layoutViewport if stable and stable.X>1 and stable.Y>1 then return stable end local camera:Camera?=workspace.CurrentCamera local current:any=camera and camera.ViewportSize if current and current.X>1
and current.Y>1 then return current end return Vector2.new(1280,720)end local function layerScale():number local instance:any=PopupLayer and(PopupLayer::Instance):FindFirstChild('ClickGuiScale')local
value:number=instance and tonumber(instance.Scale)or 1 if value~=value or value<=0 then value=1 end return value end local function logicalViewport():Vector2 local screen:Vector2=screenViewport()local
factor:number=layerScale()return Vector2.new(screen.X/factor,screen.Y/factor)end local function screenToLogical(point:Vector2):Vector2 local factor:number=layerScale()return Vector2.new(point.X/factor
,point.Y/factor)end local function logicalToScreen(point:Vector2):Vector2 local factor:number=layerScale()return Vector2.new(point.X*factor,point.Y*factor)end local function windowRect(window:any):(
number,number,number,number)local root:any=window.root return root.Position.X.Offset,root.Position.Y.Offset,root.Size.X.Offset,root.Size.Y.Offset end local function snapPixel(value:number):number
local factor:number=layerScale()return math.round(value*factor)/factor end local MIN_GRAB_X:number=140 local MIN_GRAB_Y:number=40 local function clamp(position:Vector2,size:Vector2):Vector2 local
screen:Vector2=logicalViewport()local x:number if size.X<=screen.X-MARGIN*2 then x=math.clamp(position.X,MARGIN,screen.X-size.X-MARGIN)else x=math.clamp(position.X,MARGIN-(size.X-MIN_GRAB_X),screen.X-
MIN_GRAB_X)end local y:number if size.Y<=screen.Y-MARGIN*2 then y=math.clamp(position.Y,MARGIN,screen.Y-size.Y-MARGIN)else y=math.clamp(position.Y,MARGIN,screen.Y-MIN_GRAB_Y)end return Vector2.new(
snapPixel(x),snapPixel(y))end function manager.ClampWindow(window:any):()local root:any=window and window.root if not root then return end local landed:Vector2=clamp(Vector2.new(root.Position.X.Offset
,root.Position.Y.Offset),Vector2.new(root.Size.X.Offset,root.Size.Y.Offset))root.Position=UDim2.fromOffset(landed.X,landed.Y)end local function rectsOverlap(aPos:Vector2,aSize:Vector2,bPos:Vector2,
bSize:Vector2):boolean local overlapX:number=math.min(aPos.X+aSize.X,bPos.X+bSize.X)-math.max(aPos.X,bPos.X)local overlapY:number=math.min(aPos.Y+aSize.Y,bPos.Y+bSize.Y)-math.max(aPos.Y,bPos.Y)if
overlapX<=0 or overlapY<=0 then return false end return overlapX*overlapY>0.3*aSize.X*aSize.Y end function manager.Deoverlap(window:any):()local root:any=window and window.root if not root or window.
userMoved then return end local size:Vector2=Vector2.new(root.Size.X.Offset,root.Size.Y.Offset)local position:Vector2=Vector2.new(root.Position.X.Offset,root.Position.Y.Offset)local initial:Vector2=
position for _=1,10 do local stacked:boolean=false for _,other:any in ipairs(manager.list)do if other~=window and other.root and other.root.Visible then if rectsOverlap(position,size,Vector2.new(other
.root.Position.X.Offset,other.root.Position.Y.Offset),Vector2.new(other.root.Size.X.Offset,other.root.Size.Y.Offset))then stacked=true break end end end if not stacked then break end position=clamp(
position+Vector2.new(30,30),size)end root.Position=UDim2.fromOffset(position.X,position.Y)if(position.X~=initial.X or position.Y~=initial.Y)and type(manager.SetHome)=='function'then manager.SetHome(
window,position)end end function manager.MeasureWindow(window:any):Vector2 local root:any=window and window.root if not root then return Vector2.zero end local width:number=root.Size.X.Offset local
height:number if window.collapsed then height=window.headerHeight or root.Size.Y.Offset else height=math.max(root.Size.Y.Offset,tonumber(window.fullHeight)or 0)end return Vector2.new(width,height)end
function manager.AbsoluteRect(window:any):(number,number,number,number)local x:number,y:number,w:number,h:number=windowRect(window)local origin:Vector2=logicalToScreen(Vector2.new(x,y))local extent:
Vector2=logicalToScreen(Vector2.new(w,h))return origin.X,origin.Y,extent.X,extent.Y end manager.LogicalViewport=logicalViewport manager.ScreenToLogical=screenToLogical manager.LogicalToScreen=
logicalToScreen local function bind(instance:Instance,property:string,token:string):()if ThemeEngine and type(ThemeEngine.Bind)=='function'then ThemeEngine.Bind(instance,property,token)else(instance::
any)[property]=Theme[token]end end local homePositions:{[string]:Vector2}={}function manager.SetHome(window:any,position:Vector2):()if window and window.id and position then homePositions[window.id]=
Vector2.new(position.X,position.Y)end end local function persist(window:any):()local home:Vector2?=homePositions[window.id]configData.ui.windows[window.id]={x=home and home.X or window.root.Position.X
.Offset,y=home and home.Y or window.root.Position.Y.Offset,collapsed=window.collapsed,pinned=window.pinned}queueConfigSave()end local function snap(window:any,position:Vector2,size:Vector2):Vector2
local screen:Vector2=logicalViewport()local xLines:{number}={MARGIN,screen.X-size.X-MARGIN}local yLines:{number}={MARGIN,screen.Y-size.Y-MARGIN}for _,other:any in ipairs(manager.list)do if other~=
window and other.root.Visible then local otherX:number,otherY:number,otherW:number,otherH:number=windowRect(other)table.insert(xLines,otherX)table.insert(xLines,otherX+otherW)table.insert(xLines,
otherX-size.X)table.insert(yLines,otherY)table.insert(yLines,otherY+otherH)table.insert(yLines,otherY-size.Y)end end local x:number,y:number=position.X,position.Y for _,line:number in ipairs(xLines)do
if math.abs(x-line)<=SNAP_DISTANCE then x=line break end end for _,line:number in ipairs(yLines)do if math.abs(y-line)<=SNAP_DISTANCE then y=line break end end return Vector2.new(snapPixel(x),
snapPixel(y))end local function raise(window:any):()local index:number?=table.find(manager.list,window)if index then table.remove(manager.list,index)end table.insert(manager.list,window)for order:
number,entry:any in ipairs(manager.list)do local base:number=math.min(BASE_Z+order,239)entry.root.ZIndex=base if entry.layered then for _,descendant:Instance in ipairs(entry.root:GetDescendants())do
if descendant:IsA('GuiObject')then local relative:number=tonumber(descendant:GetAttribute('FloatingDepth'))or 1 descendant.ZIndex=base+relative end end end end end function manager.Raise(id:string):()
local window:any=manager.byId[id]if window then raise(window)end end function manager.Adopt(root:GuiObject,definition:any):any root:SetAttribute('SelectorOwner',true)local id:string=definition.id
local saved:any=configData.ui.windows[id]if type(saved)~='table'then local legacy:any=(configData.ui.floatingPositions or{})[id]if type(legacy)=='table'then saved={x=tonumber(legacy[1]),y=tonumber(
legacy[2])}end end local window:any={id=id,root=root,body=definition.body,header=definition.header,closeButton=definition.closeButton,collapsed=type(saved)=='table'and saved.collapsed==true,pinned=
type(saved)=='table'and saved.pinned==true,fullHeight=root.Size.Y.Offset,headerHeight=definition.headerHeight or 0,layered=definition.layered==true,connections={}::{RBXScriptConnection}}local function
own(connection:RBXScriptConnection):RBXScriptConnection table.insert(window.connections,connection)trackUiConnection(connection)return connection end local savedHome:Vector2?=nil if type(saved)==
'table'and saved.x and saved.y then local savedX:number?=tonumber(saved.x)local savedY:number?=tonumber(saved.y)if savedX and savedY then savedHome=Vector2.new(savedX,savedY)local landed:Vector2=
clamp(savedHome,Vector2.new(root.Size.X.Offset,root.Size.Y.Offset))root.Position=UDim2.fromOffset(landed.X,landed.Y)window.userMoved=true end end homePositions[id]=savedHome or Vector2.new(root.
Position.X.Offset,root.Position.Y.Offset)function window:SetVisible(visible:boolean):()root.Visible=visible if visible then raise(window)manager.Deoverlap(window)end end function window:SetCollapsed(
collapsed:boolean):()window.collapsed=collapsed if window.body then(window.body::any).Visible=not collapsed end root.Size=collapsed and UDim2.fromOffset(root.Size.X.Offset,window.headerHeight)or UDim2
.fromOffset(root.Size.X.Offset,window.fullHeight)persist(window)end function window:SetPinned(pinned:boolean):()window.pinned=pinned if not pinned and state.visible~=true then root.Visible=false end
persist(window)end function window:Raise():()raise(window)end function window:Destroy():()if state.surfaceDragOwner==root then state.surfaceDragOwner=nil end local index:number?=table.find(manager.
list,window)if index then table.remove(manager.list,index)end manager.byId[id]=nil for _,connection:RBXScriptConnection in ipairs(window.connections)do pcall(function():()connection:Disconnect()end)
end window.connections={}root:Destroy()end local dragHandle:GuiObject=definition.dragHandle or root local requiresMenuOpen:boolean=definition.requiresMenuOpen==true local dragging:boolean=false local
dragInput:InputObject?=nil local dragStart:Vector2=Vector2.zero local startPosition:Vector2=Vector2.new(root.Position.X.Offset,root.Position.Y.Offset)own(root.InputBegan:Connect(function():()raise(
window)end))own(dragHandle.InputBegan:Connect(function(input:InputObject):()if requiresMenuOpen and not state.visible then return end local pointer:Vector2=Vector2.new(input.Position.X,input.Position.
Y)if state.surfaceDragHitTest and state.surfaceDragHitTest(pointer)then return end if state.surfaceDragOwner~=nil and state.surfaceDragOwner~=root then return end if input.UserInputType~=Enum.
UserInputType.MouseButton1 and input.UserInputType~=Enum.UserInputType.Touch then return end local absoluteSize:Vector2=dragHandle.AbsoluteSize if absoluteSize.X>0 and absoluteSize.Y>0 then local
absolutePosition:Vector2=dragHandle.AbsolutePosition if input.Position.X<absolutePosition.X or input.Position.X>absolutePosition.X+absoluteSize.X or input.Position.Y<absolutePosition.Y or input.
Position.Y>absolutePosition.Y+absoluteSize.Y then return end end dragging=true dragInput=input state.surfaceDragOwner=root dragStart=Vector2.new(input.Position.X,input.Position.Y)startPosition=Vector2
.new(root.Position.X.Offset,root.Position.Y.Offset)raise(window)end))own(UserInputService.InputChanged:Connect(function(input:InputObject):()if not dragging then return end if input==dragInput or(
dragInput and dragInput.UserInputType==Enum.UserInputType.MouseButton1 and input.UserInputType==Enum.UserInputType.MouseMovement)then local pointer:Vector2=Vector2.new(input.Position.X,input.Position.
Y)local delta:Vector2=screenToLogical(pointer-dragStart)local size:Vector2=Vector2.new(root.Size.X.Offset,root.Size.Y.Offset)local wanted:Vector2=startPosition+delta local landed:Vector2=snap(window,
clamp(wanted,size),size)root.Position=UDim2.fromOffset(landed.X,landed.Y)end end))own(UserInputService.InputEnded:Connect(function(input:InputObject):()if dragging and(input==dragInput or input.
UserInputType==Enum.UserInputType.MouseButton1)then dragging=false dragInput=nil if state.surfaceDragOwner==root then state.surfaceDragOwner=nil end window.userMoved=true homePositions[window.id]=
Vector2.new(root.Position.X.Offset,root.Position.Y.Offset)persist(window)end end))if window.closeButton then own((window.closeButton::TextButton).MouseButton1Click:Connect(function():()root.Visible=
false if definition.onClose then definition.onClose()end end))end manager.byId[id]=window raise(window)return window end function manager.Create(definition:WindowDefinition):any local id:string=
definition.id local bare:boolean=definition.chromeless==true local headerHeight:number=bare and 0 or TITLE_HEIGHT local saved:any=configData.ui.windows[id]local position:Vector2=definition.position if
type(saved)=='table'then position=Vector2.new(tonumber(saved.x)or position.X,tonumber(saved.y)or position.Y)end position=clamp(position,definition.size)homePositions[id]=position local root:any=
create(bare and'Frame'or'CanvasGroup',{Parent=PopupLayer,Name='Window_'..id,Active=true,BackgroundTransparency=bare and 1 or 0,BorderSizePixel=0,ClipsDescendants=not bare,Position=UDim2.fromOffset(
position.X,position.Y),Size=UDim2.fromOffset(definition.size.X,definition.size.Y),Visible=false,ZIndex=BASE_Z})if not bare then root.GroupTransparency=0 root.BackgroundTransparency=1-(manager.opacity
or 0.85)root:SetAttribute('WurstPanelSurface',true)end bind(root,'BackgroundColor3','background')local window:any=nil local paintChrome:{()->()}={}local chromeConnections:{RBXScriptConnection}={}local
header:Frame?=nil local closeButton:TextButton?=nil if not bare then local edge:UIStroke=(create('UIStroke',{Parent=root,Name='WindowOutline',Color=Theme.outline,Transparency=0.08,Thickness=1})::any)
::UIStroke bind(edge,'Color','outline')local bar:Frame=create('Frame',{Parent=root,Name='Header',BackgroundTransparency=0,BorderSizePixel=0,Size=UDim2.new(1,0,0,headerHeight),ZIndex=2})::Frame bind(
bar,'BackgroundColor3','accent')local bitmapText:any=state.bitmapText if bitmapText and type(bitmapText.draw)=='function'then local function paintTitle():()if root.Parent==nil then return end
bitmapText.draw(bar,{{text=definition.title or id,color=Theme.accentText}},{Name='Title',TextSize=TITLE_FONT_SIZE,ZIndex=3,AnchorPoint=Vector2.new(0,0.5),Position=UDim2.new(0,8,0.5,0),maxWidth=math.
max(40,root.Size.X.Offset-68)})end paintTitle()if ThemeEngine and type(ThemeEngine.OnChange)=='function'then ThemeEngine.OnChange(paintTitle,{'accentText'})end pcall(function():()table.insert(
chromeConnections,root:GetPropertyChangedSignal('Size'):Connect(paintTitle))end)else local title:TextLabel=(create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=bar,Name='Title',
BackgroundTransparency=1,FontFace=CONTROL_FONT,Text=definition.title or id,TextSize=TITLE_FONT_SIZE})::any)::TextLabel title.AnchorPoint=Vector2.new(0,0.5)title.Position=UDim2.new(0,8,0.5,0)title.Size
=UDim2.new(1,-60,1,0)title.TextXAlignment=Enum.TextXAlignment.Left title.ZIndex=3 bind(title,'TextColor3','accentText')end local buttonX:number=-5 local function chromeButton(kind:string,paint:Color3,
token:string?):TextButton local button:TextButton=(create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,Parent=bar,Active=true,AutoButtonColor=false,Name=kind,BackgroundColor3=Theme.surface,
BackgroundTransparency=0.72,BorderSizePixel=0,FontFace=CONTROL_FONT,Text='',TextSize=11})::any)::TextButton local stroke=create('UIStroke',{Parent=button,Color=Theme.outline,Thickness=1,Transparency=
0.55})bind(button,'BackgroundColor3','surface')bind(stroke,'Color','outline')button.AnchorPoint=Vector2.new(1,0.5)button.Position=UDim2.new(1,buttonX,0.5,0)button.Size=UDim2.fromOffset(22,22)button.
ZIndex=3 drawPixelIcon(button,kind,paint,token)table.insert(chromeConnections,button.MouseEnter:Connect(function():()button.BackgroundTransparency=0.45 end))table.insert(chromeConnections,button.
MouseLeave:Connect(function():()button.BackgroundTransparency=0.72 end))buttonX-=25 return button end if definition.closable~=false then closeButton=chromeButton('cross',Theme.negative,'negative')end
if definition.pinnable~=false then local pin:TextButton=chromeButton('pin',Theme.arrows,'arrows')local pinTint:Frame=(create('Frame',{Parent=pin,Name='PinTint',BackgroundColor3=Theme.arrows,
BackgroundTransparency=1,BorderSizePixel=0,Size=UDim2.fromScale(1,1),ZIndex=pin.ZIndex})::any)::Frame bind(pinTint,'BackgroundColor3','arrows')local function paintPin():()pin.BackgroundTransparency=
window.pinned and 0.35 or 0.72 pinTint.BackgroundTransparency=window.pinned and 0.55 or 1 end table.insert(paintChrome,paintPin)table.insert(chromeConnections,pin.MouseButton1Click:Connect(function():
()window:SetPinned(not window.pinned)paintPin()end))end if definition.collapsible~=false then local chevron:TextButton=chromeButton('up',Theme.arrows,'arrows')local function paintChevron():()
drawPixelIcon(chevron,window.collapsed and'down'or'up',Theme.arrows,'arrows')end table.insert(chromeConnections,chevron.MouseButton1Click:Connect(function():()window:SetCollapsed(not window.collapsed)
paintChevron()end))table.insert(chromeConnections,bar.InputBegan:Connect(function(input:InputObject):()if input.UserInputType==Enum.UserInputType.MouseButton2 then window:SetCollapsed(not window.
collapsed)paintChevron()end end))table.insert(paintChrome,paintChevron)end header=bar end local body:ScrollingFrame=create('ScrollingFrame',{Parent=root,Name='Body',Active=true,AutomaticCanvasSize=
Enum.AutomaticSize.Y,BackgroundTransparency=1,BorderSizePixel=0,CanvasSize=UDim2.new(),Position=UDim2.fromOffset(0,headerHeight),ScrollBarThickness=3,ScrollingDirection=Enum.ScrollingDirection.Y,Size=
UDim2.new(1,0,1,-headerHeight),ZIndex=2})::ScrollingFrame bind(body,'ScrollBarImageColor3','accentDim')local adopted:any=manager.Adopt(root,{id=id,body=body,header=header,closeButton=closeButton,
headerHeight=headerHeight,dragHandle=bare and root or header,requiresMenuOpen=bare,onClose=definition.onClose})adopted.fullHeight=definition.size.Y for _,connection:RBXScriptConnection in ipairs(
chromeConnections)do table.insert(adopted.connections,connection)trackUiConnection(connection)end window=adopted for _,paint:()->()in ipairs(paintChrome)do paint()end if state and type(state.
registerThemeRefresh)=='function'then state.registerThemeRefresh(function():()if root.Parent==nil then return end for _,paint:()->()in ipairs(paintChrome)do paint()end end,{'arrows','enabled',
'surface'})end if window.collapsed then window:SetCollapsed(true)end manager.byId[id]=window raise(window)return window end function manager.Get(id:string):any return manager.byId[id]end function
manager.SetMenuVisible(visible:boolean):()for _,window:any in ipairs(manager.list)do if not visible and window.pinned then continue end window.root.Visible=visible and window.wasVisible==true if
visible==false then window.wasVisible=window.root.Visible end end end local function intersects(ax:number,ay:number,aw:number,ah:number,bx:number,by:number,bw:number,bh:number):boolean return ax<bx+bw
and bx<ax+aw and ay<by+bh and by<ay+ah end local function obstacleRects(excludeWindow:any):{{number}}local rects:{{number}}={}for _,other:any in ipairs(manager.list)do local root:any=other.root if
other~=excludeWindow and root and root.Visible~=false then local size:Vector2=manager.MeasureWindow(other)table.insert(rects,{root.Position.X.Offset,root.Position.Y.Offset,size.X,size.Y})end end
return rects end local function rectIsFree(x:number,y:number,width:number,height:number,obstacles:{{number}}):boolean local screen:Vector2=logicalViewport()if x<MARGIN or y<MARGIN or x+width>screen.X-
MARGIN or y+height>screen.Y-MARGIN then return false end for _,rect:{number}in ipairs(obstacles)do if intersects(x,y,width,height,rect[1]-PLACEMENT_GAP,rect[2]-PLACEMENT_GAP,rect[3]+PLACEMENT_GAP*2,
rect[4]+PLACEMENT_GAP*2)then return false end end return true end local function anchorRect(preferredAnchor:any):{number}?if type(preferredAnchor)~='table'then return nil end local root:any=
preferredAnchor.root if root then if root.Parent==nil then return nil end local size:Vector2=manager.MeasureWindow(preferredAnchor)return{root.Position.X.Offset,root.Position.Y.Offset,size.X,size.Y}
end local x:number?=tonumber(preferredAnchor.x)local y:number?=tonumber(preferredAnchor.y)if x and y then return{x,y,tonumber(preferredAnchor.width)or 0,tonumber(preferredAnchor.height)or 0}end return
nil end function manager.FindFreeRect(desiredSize:Vector2,preferredAnchor:any?,excludeWindow:any?):Vector2 local width:number=desiredSize.X local height:number=desiredSize.Y local obstacles:{{number}}
=obstacleRects(excludeWindow)local screen:Vector2=logicalViewport()local function fitY(y:number):number return math.max(MARGIN,math.min(y,screen.Y-height-MARGIN))end local anchor:{number}?=anchorRect(
preferredAnchor)if anchor then local candidates:{{number}}={{anchor[1]+anchor[3]+PLACEMENT_GAP,fitY(anchor[2])},{anchor[1]-width-PLACEMENT_GAP,fitY(anchor[2])},{anchor[1],anchor[2]+anchor[4]+
PLACEMENT_GAP}}for _,candidate:{number}in ipairs(candidates)do if rectIsFree(candidate[1],candidate[2],width,height,obstacles)then return Vector2.new(candidate[1],candidate[2])end end end local step:
number=40 local yLimit:number=math.max(64,screen.Y-height-MARGIN)local xLimit:number=math.max(MARGIN,screen.X-width-MARGIN)local y:number=64 while y<=yLimit do local x:number=MARGIN while x<=xLimit do
if rectIsFree(x,y,width,height,obstacles)then return Vector2.new(x,y)end x+=step end y+=step end return clamp(Vector2.new(MARGIN,64),desiredSize)end function manager.PlaceWindow(window:any,options:any
?):()if not window or not window.root then return end local opts:any=options or{}if opts.anchor~=nil then window.placementAnchor=opts.anchor end if window.userMoved and opts.force~=true then manager.
ClampWindow(window)return end local size:Vector2=manager.MeasureWindow(window)local landed:Vector2=manager.FindFreeRect(size,window.placementAnchor,window)window.root.Position=UDim2.fromOffset(landed.
X,landed.Y)manager.ClampWindow(window)manager.SetHome(window,Vector2.new(window.root.Position.X.Offset,window.root.Position.Y.Offset))end function manager.ReflowWindow(window:any):()if not window or
not window.root then return end if window.userMoved then manager.ClampWindow(window)return end local size:Vector2=manager.MeasureWindow(window)local x:number=window.root.Position.X.Offset local y:
number=window.root.Position.Y.Offset if rectIsFree(x,y,size.X,size.Y,obstacleRects(window))then return end local landed:Vector2=manager.FindFreeRect(size,window.placementAnchor,window)window.root.
Position=UDim2.fromOffset(landed.X,landed.Y)manager.ClampWindow(window)manager.SetHome(window,Vector2.new(window.root.Position.X.Offset,window.root.Position.Y.Offset))end local function categoryAnchor
(feature:any):any local row:any=feature and feature.row local rowRoot:any=row and row.Parent and(row.Parent::any).Parent if rowRoot then for _,entry:any in ipairs(manager.list)do if entry.root==
rowRoot then return entry end end end local gui:any=state.clickGui if row and gui and type(gui.categories)=='table'and type(row.GetAttribute)=='function'then local raw:any=row:GetAttribute(
'FeatureCategory')local name:any=raw if type(host.normalizeFeatureCategory)=='function'then name=host.normalizeFeatureCategory(raw)end local record:any=name and gui.categories[name]if record and
record.window then return record.window end end return nil end function manager.OpenFeatureSettings(feature:any):any if type(feature)~='table'or feature.row==nil then return nil end local id:string=
'FeatureSettings_'..tostring(feature.configKey or feature.name):gsub('%W','')local existing:any=manager.byId[id]if existing then existing.root.Visible=true if existing.stalePlacement then existing.
stalePlacement=nil manager.PlaceWindow(existing,{force=true})end feature.expanded=true if feature.syncArrow then feature.syncArrow()end raise(existing)return existing end local bitmapText:any=state.
bitmapText local width:number=200 for _,optionRow:any in ipairs(feature.optionRows or{})do local label:any=optionRow:FindFirstChild('OptionLabel')local text:string=label and tostring(label.Text)or''
local labelWidth:number if bitmapText and type(bitmapText.measure)=='function'then labelWidth=bitmapText.measure({{text=text}},16)else labelWidth=#text*12 end width=math.max(width,8+labelWidth+24+72)
end width=math.min(width,420)local expectedLimit:number=tonumber(state.uiMaxSettingsHeight)or 200 if expectedLimit<=0 then expectedLimit=600 end local expectedHeight:number=math.min(TITLE_HEIGHT+(
feature.optionsHeight or 40),expectedLimit)local anchor:any=categoryAnchor(feature)local slot:Vector2=manager.FindFreeRect(Vector2.new(width,expectedHeight),anchor)local window:any=manager.Create({id=
id,title=tostring(feature.name)..' Settings',size=Vector2.new(width,TITLE_HEIGHT+40),position=slot,closable=true,collapsible=false,onClose=function():()feature.expanded=false if feature.syncArrow then
feature.syncArrow()end end})window.featureSettings=true window.placementAnchor=anchor feature.settingsWindow=window local options:any=feature.options if feature.updateExpandedSize then feature.
updateExpandedSize(false)end if options then options.Visible=true options.Position=UDim2.fromOffset(0,0)options.Size=UDim2.new(1,0,0,feature.optionsHeight or 40)options.Parent=window.body if
bitmapText and type(bitmapText.adoptTree)=='function'then table.insert(window.connections,bitmapText.adoptTree(options))end end local function resizeToOptions():()if options then options.Size=UDim2.
new(1,0,0,feature.optionsHeight or 40)end local wanted:number=TITLE_HEIGHT+(feature.optionsHeight or 40)local limit:number=tonumber(state.uiMaxSettingsHeight)or 200 if limit<=0 then limit=math.huge
end wanted=math.min(wanted,limit)wanted=math.min(wanted,logicalViewport().Y-MARGIN*2)window.fullHeight=wanted if not window.collapsed then window.root.Size=UDim2.fromOffset(width,wanted)end manager.
ReflowWindow(window)end window.resizeToOptions=resizeToOptions local function repackWidth():()if not window.root.Parent then return end local bitmapText:any=state.bitmapText local
widest:number=0 for _,optionRow:any in ipairs(feature.optionRows or{})do local label:any=optionRow:FindFirstChild('OptionLabel')if label then local measured:number=0 if bitmapText and type(bitmapText.
measure)=='function'then measured=tonumber(bitmapText.measure({{text=label.Text}},tonumber(label.TextSize)or 16))or 0 end if measured<=0 then local bounds:any=label.TextBounds measured=bounds and
tonumber(bounds.X)or 0 end if measured>0 then widest=math.max(widest,8+measured+24+72)end end end if widest>0 then local packed:number=math.clamp(widest,160,420)if packed~=width then width=packed
resizeToOptions()end end end window.repackWidth=repackWidth task.defer(repackWidth)feature.expanded=true if feature.syncArrow then feature.syncArrow()end window.root.Visible=true raise(window)return
window end function manager.BringToFront(window:any):()if window then raise(window)end end function manager.DebugRects():any local rects:{any}={}for _,window:any in ipairs(manager.list)do local x:
number,y:number,w:number,h:number=windowRect(window)table.insert(rects,{id=window.id,x=x,y=y,width=w,height=h,visible=window.root.Visible==true,userMoved=window.userMoved==true,collapsed=window.
collapsed==true})end return{scale=layerScale(),viewport=logicalViewport(),windows=rects}end do local bitmapText:any=state.bitmapText if bitmapText and type(bitmapText.onReady)=='table'then table.
insert(bitmapText.onReady,function():()for _,window:any in ipairs(manager.list)do if window.featureSettings and window.repackWidth then window.repackWidth()end end end)end end state.
addMenuVisibilityListener(function(visible:boolean):()for _,window:any in ipairs(manager.list)do if window.featureSettings then if not visible and window.pinned then continue end if visible then
window.root.Visible=window.wasOpenWithMenu==true else window.wasOpenWithMenu=window.root.Visible window.root.Visible=false end end end end)manager.opacity=0.85 manager.bodyTransparency=1-manager.
opacity function manager.SetOpacity(alpha:number):()local wanted:number=tonumber(alpha)or 0.85 if wanted~=wanted then wanted=0.85 end wanted=math.clamp(wanted,0.15,1)manager.opacity=wanted manager.
bodyTransparency=1-wanted for _,window:any in ipairs(manager.list)do local root:any=window.root if root and type(root.IsA)=='function'then if root:IsA('CanvasGroup')then root.GroupTransparency=0 end
local surface:boolean=type(root.GetAttribute)=='function'and root:GetAttribute('WurstPanelSurface')==true if surface then root.BackgroundTransparency=1-wanted end end end end function manager.Reclamp(
):()local placed:{Vector2}={}for _,window:any in ipairs(manager.list)do local x:number,y:number,w:number,h:number=windowRect(window)local home:Vector2?=homePositions[window.id]local base:Vector2=home
or Vector2.new(x,y)local landed:Vector2=clamp(base,Vector2.new(w,h))local attempts:number=0 while attempts<8 do local stacked:boolean=false for _,other:Vector2 in ipairs(placed)do if math.abs(other.X-
landed.X)<6 and math.abs(other.Y-landed.Y)<6 then stacked=true break end end if not stacked then break end attempts+=1 landed=clamp(landed+Vector2.new(26,26),Vector2.new(w,h))end table.insert(placed,
landed)window.root.Position=UDim2.fromOffset(landed.X,landed.Y)end end state.onLayout(function():()manager.Reclamp()end)state.windows=manager activeCleanup=function():()for index:number=#manager.list,
1,-1 do local window:any=manager.list[index]pcall(function():()window.root:Destroy()end)end manager.list={}manager.byId={}state.windows=nil end Module.Initialized=true return manager end function
Module.destroy():()if activeCleanup then pcall(activeCleanup)end activeCleanup=nil Module.Initialized=false end return Module
-- src/Wurst.lua
local RUNTIME_COMPATIBILITY_MARKER:string='Initialization error \u{2014} check executor console'local DEFAULT_FEATURE_CATEGORY:string='Utilities'local OFFICIAL_CATEGORIES:{string}={'Exploits',
'Utilities','World','Render','Blocks','Items'}local CATEGORY_ALIASES:{[string]:string}={}local FEATURE_CATEGORIES:{[string]:string}={}local UI_RADIUS:{[string]:number}={shell=8,panel=5,card=4,control=
3,chip=3}local UI_MOTION:{[string]:number}={fast=0.1,base=0.16,slow=0.22}type LayoutMetrics={mode:string,compact:boolean,touch:boolean,contentPadding:number,cardSpacing:number,cardHeight:number,
cardCorner:number,controlHeight:number,optionRowHeight:number,sectionRowHeight:number,switchWidth:number,switchHeight:number,iconButton:number,keySlotWidth:number,touchTarget:number,titleFontSize:
number,bodyFontSize:number,captionFontSize:number,scrollBar:number}local function computeLayoutMetrics(_viewport:Vector2,touchEnabled:boolean):LayoutMetrics local mode:string='Desktop'if touchEnabled
then mode='Mobile'end local mobile:boolean=mode=='Mobile'return{mode=mode,compact=mobile,touch=touchEnabled,contentPadding=10,cardSpacing=2,cardHeight=22,cardCorner=UI_RADIUS.card,controlHeight=20,
optionRowHeight=22,sectionRowHeight=22,switchWidth=26,switchHeight=14,iconButton=16,keySlotWidth=34,touchTarget=touchEnabled and 32 or 24,titleFontSize=16,bodyFontSize=16,captionFontSize=12,scrollBar=
3}end local RUNTIME_BRANCH:string='arena/0ce50d99-arandommenu'do local override:any=nil local okGenv,genv=pcall(function()return(getgenv::any)()end)if okGenv and type(genv)=='table'and type(genv.
WURST_BRANCH)=='string'and genv.WURST_BRANCH~=''then override=genv.WURST_BRANCH elseif type(_G)=='table'and type((_G::any).WURST_BRANCH)=='string'and(_G::any).WURST_BRANCH~=''then override=(_G::any).
WURST_BRANCH else local fenv:any=(getfenv()::any)if type(fenv)=='table'and type(fenv.WURST_BRANCH)=='string'and fenv.WURST_BRANCH~=''then override=fenv.WURST_BRANCH end end if type(override)=='string'
and override~=''then RUNTIME_BRANCH=(override:gsub('^refs/heads/',''))end end local RUNTIME_RAW_BASE:string=[[https://raw.githubusercontent.com/XzAngel19/ARandomMenu/refs/heads/]]..RUNTIME_BRANCH..'/'
local RUNTIME_SAFETY_SOURCE_URL:string=RUNTIME_RAW_BASE..'src/Wurst.lua'local SOURCE_STAMP:string='v199-20261108-aim-camera-fire'local BUILD_NUMBER:string?=string.match(SOURCE_STAMP,'^v(%d+)')local
BUILD_LABEL:string=if BUILD_NUMBER then'0.'..BUILD_NUMBER else'0.0'local BOOT_STARTED_AT:number=os.clock()local BOOT_FINGERPRINT_SECONDS:number=0 local PENDING_FINGERPRINT:boolean=false local PRODUCT=
{name='Wurst',version=BUILD_LABEL..' Release',logPrefix='Wurst',guiName='Wurst',blurName='WurstBlur',storageFolder='Wurst'}local function createRuntimeDestructor(deps:any):any local state=deps.state
local MenuBlur=deps.MenuBlur local ScreenGui=deps.ScreenGui local allFeatures=deps.allFeatures local featureConnections=deps.featureConnections local TaskManager=deps.TaskManager local
cleanupMM2Runtime=deps.cleanupMM2Runtime local closeConfigSave=deps.closeConfigSave local setMenuCursorUnlocked=deps.setMenuCursorUnlocked local disconnectFeatureConnection=deps.
disconnectFeatureConnection local persistenceGenv=deps.persistenceGenv local logPrint:any=deps.logPrint local reportError:any=deps.warn local runtimeCleaned:boolean=false local runtimeDestructed:
boolean=false local runtimeHandoff:any=nil local function cleanupStep(label:string,step:()->()):()local ok:boolean,failure:any=pcall(step)if not ok then reportError('['..PRODUCT.name..'] destruct/'..
label..': '..tostring(failure))end end local function cleanupRuntime():()if runtimeCleaned then return end runtimeCleaned=true cleanupStep('teleport persistence',function():()if state.teleportPersist
then state.teleportPersist.Close()end end)cleanupStep('blur',function():()if MenuBlur then MenuBlur:Destroy()MenuBlur=nil end end)cleanupStep('cursor',function():()setMenuCursorUnlocked(false)end)
cleanupStep('session',function():()if type(state.destroySessionTracking)=='function'then state.destroySessionTracking()end state.sessionInfo=nil end)for _,feature in ipairs(allFeatures)do if feature.
enabled then cleanupStep('feature '..tostring(feature.name),function():()feature.onToggle(false)end)feature.enabled=false end end cleanupStep('held keybinds',function():()for _,binding:any in pairs(
state.shortcutBindings)do if binding.hold and binding.pressed and binding.onRelease then binding.pressed=false binding.onRelease()end end table.clear(state.shortcutBindings)end)cleanupStep(
'feature connections',function():()for name in pairs(featureConnections)do disconnectFeatureConnection(name)end end)cleanupStep('tooltips',function():()for _,connection:RBXScriptConnection in ipairs(
state.tooltipConnections)do connection:Disconnect()end table.clear(state.tooltipConnections)end)cleanupStep('ui connections',function():()for _,connection:RBXScriptConnection in ipairs(state.
uiConnections)do connection:Disconnect()end table.clear(state.uiConnections)end)cleanupStep('projectile calibration',function():()local services:any=state.frameworkEnvironment and state.
frameworkEnvironment.services local registry:any=services and services.projectileCalibration local calibration:any=registry and registry.get()if calibration and type(calibration.destroy)=='function'
then calibration:destroy()end if registry then registry.set(nil)end end)cleanupStep('task manager',function():()TaskManager:Destroy()end)cleanupStep('global input',function():()for _,key:string in
ipairs({'globalInputChangedConnection','globalInputEndedConnection','globalInputBeganConnection','globalShortcutBeganConnection','globalShortcutEndedConnection'})do local connection:any=state[key]if
connection then connection:Disconnect()state[key]=nil end end end)cleanupStep('colour picker',function():()local picker:any=state.colorPicker if not picker then return end if picker.inputChanged then
picker.inputChanged:Disconnect()end if picker.inputEnded then picker.inputEnded:Disconnect()end end)cleanupStep('MM2 runtime',cleanupMM2Runtime)for _,key:string in ipairs({'cleanupFrameworkRuntime',
'cleanupTRSRuntime','cleanupVDRuntime','cleanupBedFightRuntime','cleanupX2DuelsRuntime'})do local cleanup:any=state[key]if type(cleanup)=='function'then cleanupStep(key,cleanup)end end cleanupStep(
'config',closeConfigSave)local genv:any=persistenceGenv()if genv and genv.WURST_RUNTIME==runtimeHandoff then genv.WURST_RUNTIME=nil end end local function destructRuntime():()if runtimeDestructed then
return end runtimeDestructed=true cleanupStep('session kill switch',function():()if state.teleportPersist then state.teleportPersist.Stop()end end)if runtimeCleaned then return end cleanupStep(
'runtime',cleanupRuntime)cleanupStep('screen gui',function():()if ScreenGui.Parent then ScreenGui:Destroy()end end)logPrint('['..PRODUCT.name..'] Destruct complete.')end local function bindHandoff():(
)runtimeHandoff={Destruct=destructRuntime}local activeGenv:any=persistenceGenv()if activeGenv then activeGenv.WURST_RUNTIME=runtimeHandoff end ScreenGui.AncestryChanged:Connect(function(_:Instance,
parent:Instance?):()if not parent then cleanupRuntime()end end)end return{cleanupRuntime=cleanupRuntime,destructRuntime=destructRuntime,bindHandoff=bindHandoff}end local function bootstrap():()local
executorGlobals:any=getfenv()local DEBUG_LOGS:boolean=(function():boolean local getGenv:any=executorGlobals.getgenv if type(getGenv)=='function'then local ok:boolean,genv:any=pcall(getGenv)if ok and
type(genv)=='table'then return genv.WURST_DEBUG==true end end return false end)()local realPrint:any=executorGlobals.print local realWarn:any=executorGlobals.warn local print=function(...:any):()if
DEBUG_LOGS then realPrint(...)end end local warn=function(...:any):()if DEBUG_LOGS then realWarn(...)end end local function cloneReference(instance:any):any if type(executorGlobals.cloneref)==
'function'then local success:boolean,cloned:any=pcall(executorGlobals.cloneref,instance)if success then return cloned end end return instance end local Players:Players=cloneReference(game:GetService(
'Players'))local UserInputService:UserInputService=cloneReference(game:GetService('UserInputService'))local TweenService:TweenService=cloneReference(game:GetService('TweenService'))local Lighting:
Lighting=cloneReference(game:GetService('Lighting'))local RunService:RunService=cloneReference(game:GetService('RunService'))local HttpService:HttpService=cloneReference(game:GetService('HttpService')
)local CollectionService:CollectionService=cloneReference(game:GetService('CollectionService'))local Stats:Stats=cloneReference(game:GetService('Stats'))local LocalPlayer:Player=(function():Player
local player:any=Players.LocalPlayer while player==nil do task.wait(0.1)player=Players.LocalPlayer end return player::Player end)()local PlayerGui:PlayerGui=LocalPlayer:WaitForChild('PlayerGui')::
PlayerGui local GUI_NAME=PRODUCT.guiName local BLUR_NAME=PRODUCT.blurName local CONFIG_FOLDER=PRODUCT.storageFolder local CONFIG_PROFILE_FOLDER=CONFIG_FOLDER..'/Profiles'local CONFIG_GAME_ID:string=
tostring(game.GameId>0 and game.GameId or game.PlaceId)local CONFIG_FILE_LUA=CONFIG_PROFILE_FOLDER..'/Game_'..CONFIG_GAME_ID..'.lua'local CONFIG_FILE_BAK=CONFIG_FILE_LUA..'.bak'local CONFIG_FILE=
CONFIG_PROFILE_FOLDER..'/Game_'..CONFIG_GAME_ID..'.Config'local REPOSITORY_RAW_BASE=RUNTIME_RAW_BASE local GAME_MODULES:{[number]:string}={[71480482338212]='BedFight',[142823291]='MM2',[14315258385]=
'TRS',[93978595733734]='VD',[92876937625630]='X2Duels',[135856908115931]='X2Duels'}type GameFingerprint={module:string,required:number,markers:{(ReplicatedStorage)->boolean}}local GAME_FINGERPRINTS:{
GameFingerprint}={{module='MM2',required=4,markers={function(replicated:ReplicatedStorage):boolean local remotes:Instance?=replicated:FindFirstChild('Remotes')local gameplay:Instance?=remotes and
remotes:FindFirstChild('Gameplay')if not gameplay then return false end local found:number=0 for _,name:string in ipairs({'RoundStart','RoundEndFade','GetCurrentPlayerData','PlayerDataChanged'})do if
gameplay:FindFirstChild(name)then found+=1 end end return found>=3 end,function(replicated:ReplicatedStorage):boolean local services:Instance?=replicated:FindFirstChild('ClientServices')local
weaponService:Instance?=services and services:FindFirstChild('WeaponService')return weaponService~=nil and weaponService:FindFirstChild('GunFired')~=nil end,function(replicated:ReplicatedStorage):
boolean local mainGui:Instance?=replicated:FindFirstChild('MainGUI')local gameFrame:Instance?=mainGui and mainGui:FindFirstChild('Game')return gameFrame~=nil and gameFrame:FindFirstChild('Timer')~=nil
end,function(_replicated:ReplicatedStorage):boolean local ok:boolean,groups:any=pcall(function():any return game:GetService('PhysicsService'):GetRegisteredCollisionGroups()end)if not ok or type(groups
)~='table'then return false end for _,group:any in ipairs(groups)do if type(group)=='table'and group.name=='Murderer'then return true end end return false end,function(_replicated:ReplicatedStorage):
boolean for _,object:Instance in ipairs(workspace:GetChildren())do if object:IsA('Model')and object.Name~='Lobby'and(object:FindFirstChild('CoinContainer')or object:FindFirstChild('CoinAreas'))then
return true end end return false end}},{module='X2Duels',required=3,markers={function(replicated:ReplicatedStorage):boolean local packages:Instance?=replicated:FindFirstChild('Packages')local
networking:Instance?=packages and packages:FindFirstChild('Networking')local re:Instance?=networking and networking:FindFirstChild('RE')local match:Instance?=re and re:FindFirstChild('Match')if not
match then return false end local found:number=0 for _,name:string in ipairs({'MatchResults','WinNotification','PlayerRespawned','SetDead'})do if match:FindFirstChild(name)then found+=1 end end return
found>=2 end,function(replicated:ReplicatedStorage):boolean local packages:Instance?=replicated:FindFirstChild('Packages')local networking:Instance?=packages and packages:FindFirstChild('Networking')
local re:Instance?=networking and networking:FindFirstChild('RE')local rematch:Instance?=re and re:FindFirstChild('Rematch')return rematch~=nil and rematch:FindFirstChild('RematchVote')~=nil and
rematch:FindFirstChild('RematchTeleport')~=nil end,function(replicated:ReplicatedStorage):boolean local packages:Instance?=replicated:FindFirstChild('Packages')local net:Instance?=packages and
packages:FindFirstChild('Net')local re:Instance?=net and net:FindFirstChild('RE')if not re then return false end local found:number=0 for _,name:string in ipairs({'ThrowEvent','SlashEvent','GunKill',
'KnifeKill'})do if re:FindFirstChild(name)then found+=1 end end return found>=3 end,function(replicated:ReplicatedStorage):boolean local client:Instance?=replicated:FindFirstChild('Client')local
controllers:Instance?=client and client:FindFirstChild('Controllers')return controllers~=nil and controllers:FindFirstChild('MatchController')~=nil and controllers:FindFirstChild(
'GunCrosshairController')~=nil end,function(replicated:ReplicatedStorage):boolean local packages:Instance?=replicated:FindFirstChild('Packages')local networking:Instance?=packages and packages:
FindFirstChild('Networking')local re:Instance?=networking and networking:FindFirstChild('RE')local combat:Instance?=re and re:FindFirstChild('Combat')return combat~=nil and combat:FindFirstChild(
'Beam')~=nil and combat:FindFirstChild('PlaySound')~=nil end,function(replicated:ReplicatedStorage):boolean local client:Instance?=replicated:FindFirstChild('Client')local legacy:Instance?=client and
client:FindFirstChild('LegacyClientScripts')return legacy~=nil and legacy:FindFirstChild('BeamHandle')~=nil and client:FindFirstChild('ClientRuntime')~=nil end}}}local function
detectGameModuleByStructure():(string?,number)local replicated:ReplicatedStorage=game:GetService('ReplicatedStorage')for _,fingerprint:GameFingerprint in ipairs(GAME_FINGERPRINTS)do local score:number
=0 for _,marker:(ReplicatedStorage)->boolean in ipairs(fingerprint.markers)do local ok:boolean,matched:any=pcall(marker,replicated)if ok and matched==true then score+=1 end end if score>=fingerprint.
required then return fingerprint.module,score end end return nil,0 end local ACTIVE_GAME_MODULE:string?=GAME_MODULES[game.PlaceId]local ACTIVE_GAME_SOURCE:string=ACTIVE_GAME_MODULE and'PlaceId'or
'none'PENDING_FINGERPRINT=ACTIVE_GAME_MODULE==nil if not ACTIVE_GAME_MODULE then local fingerprintAt:number=os.clock()local detected:string?,score:number=detectGameModuleByStructure()
BOOT_FINGERPRINT_SECONDS=os.clock()-fingerprintAt if detected then ACTIVE_GAME_MODULE=detected ACTIVE_GAME_SOURCE='fingerprint '..tostring(score)..' markers'PENDING_FINGERPRINT=false end end local
GAME_CHECK={MM2=142823291,BedFight=71480482338212,TRS=14315258385,VD=93978595733734,X2Duels=92876937625630,MM2Active=ACTIVE_GAME_MODULE=='MM2',TRSActive=ACTIVE_GAME_MODULE=='TRS',VDActive=
ACTIVE_GAME_MODULE=='VD',BedFightActive=ACTIVE_GAME_MODULE=='BedFight',X2DuelsActive=ACTIVE_GAME_MODULE=='X2Duels'}function GAME_CHECK.matches(identifier:number):boolean if game.PlaceId==identifier
then return true end for placeId:number,moduleName:string in pairs(GAME_MODULES)do if placeId==identifier then return ACTIVE_GAME_MODULE==moduleName end end return false end GAME_CHECK.source=
ACTIVE_GAME_SOURCE export type ThemeShape={opacity:number,tooltipOpacity:number,radius:number,rowHeight:number,borderThickness:number,maxHeight:number,scale:number}local THEME_PRESETS:{[string]:{[
string]:Color3}}={Wurst={background=Color3.fromRGB(64,64,64),sidebar=Color3.fromRGB(48,48,48),surface=Color3.fromRGB(64,64,64),surfaceRaised=Color3.fromRGB(74,74,74),surfaceHover=Color3.fromRGB(86,86,
86),surfacePressed=Color3.fromRGB(96,96,96),button=Color3.fromRGB(56,56,56),buttonHover=Color3.fromRGB(72,72,72),buttonPressed=Color3.fromRGB(84,84,84),outline=Color3.fromRGB(16,16,16),accent=Color3.
fromRGB(16,16,16),accentDim=Color3.fromRGB(36,36,36),accentFaint=Color3.fromRGB(56,56,56),accentSoft=Color3.fromRGB(40,40,40),accentText=Color3.fromRGB(240,240,240),enabled=Color3.fromRGB(32,175,32),
enabledText=Color3.fromRGB(240,240,240),arrows=Color3.fromRGB(0,204,0),text=Color3.fromRGB(240,240,240),textMuted=Color3.fromRGB(168,168,168),tooltip=Color3.fromRGB(24,24,24)},Monochrome={background=
Color3.fromRGB(8,8,9),sidebar=Color3.fromRGB(13,13,14),surface=Color3.fromRGB(26,26,28),surfaceRaised=Color3.fromRGB(32,32,35),surfaceHover=Color3.fromRGB(40,40,44),surfacePressed=Color3.fromRGB(48,48
,52),button=Color3.fromRGB(32,32,35),buttonHover=Color3.fromRGB(46,46,50),buttonPressed=Color3.fromRGB(56,56,61),outline=Color3.fromRGB(92,92,98),accent=Color3.fromRGB(236,236,238),accentDim=Color3.
fromRGB(196,196,202),accentFaint=Color3.fromRGB(160,160,168),accentSoft=Color3.fromRGB(52,52,56),accentText=Color3.fromRGB(8,8,9),enabled=Color3.fromRGB(236,236,238),enabledText=Color3.fromRGB(8,8,9),
arrows=Color3.fromRGB(0,204,0),text=Color3.fromRGB(245,245,247),textMuted=Color3.fromRGB(158,158,166),tooltip=Color3.fromRGB(16,16,18)}}local Theme:{[string]:Color3}={}for token:string,value:Color3 in
pairs(THEME_PRESETS.Wurst)do Theme[token]=value end Theme.positive=Color3.fromRGB(126,196,138)Theme.negative=Color3.fromRGB(214,106,106)local ThemeEngine={name='Wurst',presets=THEME_PRESETS,tokens=
Theme,shape={opacity=0.86,tooltipOpacity=0.95,radius=0,rowHeight=22,borderThickness=1,maxHeight=340,scale=1}::ThemeShape}local themeBindings:any={}local themeBuckets:any={}local themeWatches:any={}
local themeListeners:{any}={}local function unbindTheme(instance:any)local record=themeBindings[instance]if record then for _,token in pairs(record)do if themeBuckets[token]then themeBuckets[token][
instance]=nil end end end themeBindings[instance]=nil local watch=themeWatches[instance]themeWatches[instance]=nil if watch then watch:Disconnect()end end local function repaintThemeBindings(values:{[
string]:Color3}):()for token,value in pairs(values)do for instance in pairs(themeBuckets[token]or{})do local ok=pcall(function()if instance.Parent==nil then unbindTheme(instance)return end for
property,boundToken in pairs(themeBindings[instance]or{})do if boundToken==token then instance[property]=value end end end)if not ok then unbindTheme(instance)end end end end function ThemeEngine.Bind
(instance:Instance,property:string,token:string):Instance local record=themeBindings[instance]if not record then record={}themeBindings[instance]=record local ok,watch=pcall(function()return instance.
Destroying:Connect(function()unbindTheme(instance)end)end)if ok then themeWatches[instance]=watch end end local old=record[property]record[property]=token if old and old~=token and themeBuckets[old]
then local stillUsed=false for _,bound in pairs(record)do if bound==old then stillUsed=true break end end if not stillUsed then themeBuckets[old][instance]=nil end end themeBuckets[token]=themeBuckets
[token]or{}themeBuckets[token][instance]=true if Theme[token]then(instance::any)[property]=Theme[token]end return instance end function ThemeEngine.OnChange(callback:any,tokens:{string}?):()table.
insert(themeListeners,function(name,changed)if tokens and changed then local relevant=false for _,token in ipairs(tokens)do if changed[token]then relevant=true break end end if not relevant then
return end end callback(name,changed)end)end function ThemeEngine.SetMany(input:{[string]:Color3}):()local values=table.clone(input)if values.accent~=nil or values.background~=nil then local accent=
values.accent or Theme.accent local background=values.background or Theme.background for token,amount in pairs({accentDim=0.72,accentSoft=0.58,accentFaint=0.42})do values[token]=values[token]or Color3
.new(background.R+(accent.R-background.R)*amount,background.G+(accent.G-background.G)*amount,background.B+(accent.B-background.B)*amount)end end local changed={}for token,value in pairs(values)do if
Theme[token]~=value then changed[token]=value Theme[token]=value end end if next(changed)==nil then return end repaintThemeBindings(changed)for _,listener in ipairs(themeListeners)do pcall(listener,
ThemeEngine.name,changed)end end function ThemeEngine.Set(token:string,value:Color3):()ThemeEngine.SetMany({[token]=value})end function ThemeEngine.Apply(name:string):boolean local preset:{[string]:
Color3}?=THEME_PRESETS[name]if not preset then return false end for token:string,value:Color3 in pairs(preset::{[string]:Color3})do Theme[token]=value end ThemeEngine.name=name repaintThemeBindings(
Theme)for _,listener:(string)->()in ipairs(themeListeners)do pcall(listener,name)end return true end local DEFAULT_HEADER_COLOR:Color3=Theme.text local DEFAULT_MENU_COLOR:Color3=Theme.background local
DEFAULT_SURFACE_COLOR:Color3=Theme.surface local DEFAULT_BORDER_COLOR:Color3=Theme.outline local TITLE_FONT:Font=Font.new('rbxasset://fonts/families/BuilderSans.json',Enum.FontWeight.Bold,Enum.
FontStyle.Normal)local THIN_FONT:Font=Font.new('rbxasset://fonts/families/BuilderSans.json',Enum.FontWeight.Regular,Enum.FontStyle.Normal)local CONTROL_FONT:Font=Font.new(
'rbxasset://fonts/families/BuilderSans.json',Enum.FontWeight.SemiBold,Enum.FontStyle.Normal)local VALVE_FONT:Font=CONTROL_FONT local TEXT_BASE_MULTIPLIER:number=1.35 local TEXT_SCALE_MINIMUM:number=
0.8 local TEXT_SCALE_MAXIMUM:number=1.2 local IS_MOBILE_DEVICE:boolean=UserInputService.TouchEnabled and not UserInputService.KeyboardEnabled local DESKTOP_HIDDEN_GUIS:{[string]:boolean}={TouchGui=
true,TouchInteractButtons=true}local function isDesktopMachine():boolean if not UserInputService.KeyboardEnabled then return false end if UserInputService.MouseEnabled then return true end local ok:
boolean,lastInput:any=pcall(function():any return UserInputService:GetLastInputType()end)if not ok then return false end return lastInput==Enum.UserInputType.MouseButton1 or lastInput==Enum.
UserInputType.MouseMovement or lastInput==Enum.UserInputType.MouseWheel or lastInput==Enum.UserInputType.Keyboard end local function suppressTouchControls():()if not UserInputService.TouchEnabled or
not isDesktopMachine()then return end for name in pairs(DESKTOP_HIDDEN_GUIS)do local ok:boolean,gui:any=pcall(function():any return PlayerGui:FindFirstChild(name)end)if ok and gui and gui:IsA(
'ScreenGui')and gui.Enabled then pcall(function():()gui.Enabled=false end)end end end pcall(function():()local function watch(gui:Instance):()pcall(function():()gui:GetPropertyChangedSignal('Enabled')
:Connect(function():()if gui.Enabled then task.defer(suppressTouchControls)end end)end)end suppressTouchControls()PlayerGui.ChildAdded:Connect(function(child:Instance):()if DESKTOP_HIDDEN_GUIS[child.
Name]then watch(child)task.defer(suppressTouchControls)end end)for _,child:Instance in ipairs(PlayerGui:GetChildren())do if DESKTOP_HIDDEN_GUIS[child.Name]then watch(child)end end end)local configData
:any={version=1,states={},values={},ui={}}local configSaveQueued=false local configSaveClosed=false local configWriteAllowed:any=nil local function serializeLuaValue(value:any,depth:number):string
local kind:string=typeof(value)if kind=='string'then return string.format('%q',value)elseif kind=='number'then if value~=value or value==math.huge or value==-math.huge then return'0'end return string.
format('%.14g',value)elseif kind=='boolean'then return value and'true'or'false'elseif kind=='table'and depth<6 then local parts:{string}={}local length:number=#value for index:number=1,length do table
.insert(parts,serializeLuaValue(value[index],depth+1))end for key:any,entry:any in pairs(value)do local keyType:string=typeof(key)local keyText:string?=nil if keyType=='string'then keyText='['..string
.format('%q',key)..']'elseif keyType=='number'then if key%1~=0 or key<1 or key>length then keyText='['..string.format('%.14g',key)..']'end elseif keyType=='boolean'then keyText='['..tostring(key)..']'
end if keyText then table.insert(parts,keyText..' = '..serializeLuaValue(entry,depth+1))end end if#parts==0 then return'{}'end return'{'..table.concat(parts,', ')..'}'end return'nil'end local function
decodeLuaConfig(source:string):any local compiler:any=(getfenv()::any).loadstring if type(compiler)~='function'or type(source)~='string'then return nil end local built:boolean,chunk:any=pcall(compiler
,source,'@WurstConfig')if not built or type(chunk)~='function'then return nil end local ran:boolean,result:any=pcall(chunk)if not ran or type(result)~='table'then return nil end return result end
local function applyLoadedConfig(loaded:any):boolean if type(loaded)~='table'then return false end configData.version=type(loaded.version)=='number'and loaded.version or 1 configData.states=type(
loaded.states)=='table'and loaded.states or{}configData.values=type(loaded.values)=='table'and loaded.values or{}configData.ui=type(loaded.ui)=='table'and loaded.ui or{}return true end local function
loadConfig()if type(isfile)~='function'or type(readfile)~='function'then return end local function readLuaConfig(path:string):any local existsOk:boolean,exists:any=pcall(isfile,path)if not existsOk or
exists~=true then return nil end local readOk:boolean,source:any=pcall(readfile,path)if not readOk or type(source)~='string'then return nil end return decodeLuaConfig(source)end local loaded:any=
readLuaConfig(CONFIG_FILE_LUA)if not applyLoadedConfig(loaded)then loaded=readLuaConfig(CONFIG_FILE_BAK)end if not applyLoadedConfig(loaded)then pcall(function()if isfile(CONFIG_FILE)then local
decoded:any=HttpService:JSONDecode(readfile(CONFIG_FILE))if applyLoadedConfig(decoded)then loaded=decoded end end end)end configData.ui.windowPosition=nil configData.ui.windowSize=nil end
local function saveConfigNow()if configSaveClosed or type(writefile)~='function'or(type(configWriteAllowed)=='function'and not configWriteAllowed())then return end pcall(function()if type(isfolder)==
'function'and type(makefolder)=='function'and not isfolder(CONFIG_FOLDER)then makefolder(CONFIG_FOLDER)end if type(isfolder)=='function'and type(makefolder)=='function'and not isfolder(
CONFIG_PROFILE_FOLDER)then makefolder(CONFIG_PROFILE_FOLDER)end local payload:string='-- Wurst config for game '..CONFIG_GAME_ID..' \183 modules, options, keybinds and UI.\n'..'return {\n'..
'    version = '..serializeLuaValue(configData.version,0)..',\n'..'    states = '..serializeLuaValue(configData.states,0)..',\n'..'    values = '..serializeLuaValue(configData.values,0)..',\n'..
'    ui = '..serializeLuaValue(configData.ui,0)..',\n'..'}\n'if type(isfile)=='function'and type(readfile)=='function'then local existsOk:boolean,exists:any=pcall(isfile,CONFIG_FILE_LUA)if existsOk
and exists==true then local readOk:boolean,previous:any=pcall(readfile,CONFIG_FILE_LUA)if readOk and type(previous)=='string'and#previous>0 then writefile(CONFIG_FILE_BAK,previous)end end end if type(
configWriteAllowed)=='function'and not configWriteAllowed()then return end writefile(CONFIG_FILE_LUA,payload)end)end local configSaveGeneration:number=0 local function queueConfigSave()if
configSaveClosed or configSaveQueued then return end configSaveQueued=true local generation:number=configSaveGeneration+1 configSaveGeneration=generation task.delay(0.35,function()if
configSaveGeneration~=generation then return end configSaveQueued=false saveConfigNow()end)end local function flushConfigSave()configSaveGeneration+=1 configSaveQueued=false saveConfigNow()end
local function closeConfigSave()if configSaveClosed then return end flushConfigSave()configSaveClosed=true end pcall(function():()(game::any):BindToClose(function():()saveConfigNow()end)end)
local function colorFromConfig(value,fallback)if type(value)=='table'then return Color3.fromRGB(math.clamp(tonumber(value[1])or 0,0,255),math.clamp(tonumber(value[2])or 0,0,255),math.clamp(tonumber(
value[3])or 0,0,255))end return fallback end local function colorToConfig(color)return{math.round(color.R*255),math.round(color.G*255),math.round(color.B*255)}end local function sharedRuntimeGenv():
any local getGenv:any=(getfenv()::any).getgenv if type(getGenv)=='function'then local ok:boolean,genv:any=pcall(getGenv)if ok and type(genv)=='table'then return genv end end return nil end local
previousGenv:any=sharedRuntimeGenv()local previousRuntime:any=previousGenv and previousGenv.WURST_RUNTIME if type(previousRuntime)=='table'and type(previousRuntime.Destruct)=='function'then pcall(
previousRuntime.Destruct)end local function findExistingGui()local locations={PlayerGui}local coreGuiOk,coreGui=pcall(function()return game:GetService('CoreGui')end)if coreGuiOk and coreGui then table
.insert(locations,coreGui)end if type(gethui)=='function'then local hiddenGuiOk,hiddenGui=pcall(gethui)if hiddenGuiOk and hiddenGui then table.insert(locations,hiddenGui)end end for _,location in
ipairs(locations)do local existing=location:FindFirstChild(GUI_NAME)while existing do existing:Destroy()existing=location:FindFirstChild(GUI_NAME)end end end findExistingGui()loadConfig()local oldBlur
=Lighting:FindFirstChild(BLUR_NAME)if oldBlur then oldBlur:Destroy()end local MenuBlur:any=nil pcall(function():()MenuBlur=Instance.new('BlurEffect')MenuBlur.Name=BLUR_NAME MenuBlur.Size=14 MenuBlur.
Enabled=false MenuBlur.Parent=Lighting end)local ScreenGui=Instance.new('ScreenGui')ScreenGui.Name=GUI_NAME ScreenGui.ResetOnSpawn=false ScreenGui.IgnoreGuiInset=true ScreenGui.AutoLocalize=false
ScreenGui.DisplayOrder=999999 ScreenGui.ZIndexBehavior=Enum.ZIndexBehavior.Sibling ScreenGui:SetAttribute('BuildId','0001-S17')local function parentScreenGui()local protector:any=(getfenv()::any).syn
if type(protector)=='table'and type(protector.protect_gui)=='function'then pcall(protector.protect_gui,ScreenGui)end if type(gethui)=='function'then local hiddenGuiOk,hiddenGui=pcall(gethui)if
hiddenGuiOk and hiddenGui then local parented=pcall(function()ScreenGui.Parent=hiddenGui end)if parented then return end end end local coreGuiOk,coreGui=pcall(function()return game:GetService(
'CoreGui')end)if coreGuiOk and coreGui then local parented=pcall(function()ScreenGui.Parent=coreGui end)if parented then return end end ScreenGui.Parent=PlayerGui end parentScreenGui()local PopupLayer
:Frame=Instance.new('Frame')PopupLayer.Name='PopupLayer'PopupLayer.Active=false PopupLayer.BackgroundTransparency=1 PopupLayer.BorderSizePixel=0 PopupLayer.Size=UDim2.fromScale(1,1)PopupLayer.ZIndex=
150 PopupLayer.Parent=ScreenGui if configData.ui.unboundKeyDefaultsVersion~=1 then for key:string,value:any in pairs(configData.values)do if string.sub(key,1,9)=='Shortcut.'or(type(value)=='string'and
string.match(string.lower(key),'%.%w*key$'))then configData.values[key]='Unknown'end end configData.ui.unboundKeyDefaultsVersion=1 end if configData.ui.menuKeyRestoreVersion~=1 then if not configData.
ui.toggleKey or configData.ui.toggleKey=='Unknown'then configData.ui.toggleKey='RightShift'end configData.ui.menuKeyRestoreVersion=1 end local savedToggleKey=Enum.KeyCode.RightShift if type(configData
.ui.toggleKey)=='string'then pcall(function()savedToggleKey=Enum.KeyCode[configData.ui.toggleKey]or Enum.KeyCode.RightShift end)end local state:any={debug=DEBUG_LOGS,visible=false,isMobile=
IS_MOBILE_DEVICE,guiStyle='Default',mobileMenuButtonVisible=true,toggleKey=savedToggleKey,waitingForKey=false,keyCaptureCallback=nil,headerColor=DEFAULT_HEADER_COLOR,menuColor=DEFAULT_MENU_COLOR,
hookFunction=(getfenv()::any).hookfunction,hookMetamethod=(getfenv()::any).hookmetamethod,getNamecallMethod=(getfenv()::any).getnamecallmethod,checkCaller=(getfenv()::any).checkcaller,newCClosure=(
getfenv()::any).newcclosure,fireTouchInterest=(getfenv()::any).firetouchinterest,requireModule=require,uiConnections={}::{RBXScriptConnection},styledButtons={}::{TextButton},styledTextBoxes={}::{
TextBox},gameModulePaths={}::{[string]:{string}}}state.activeGameName=ACTIVE_GAME_MODULE state.textScale=configData.ui.textScaleCalibrationVersion==2 and math.clamp(tonumber(configData.ui.textScale)or
1,TEXT_SCALE_MINIMUM,TEXT_SCALE_MAXIMUM)or 1 state.animationsEnabled=configData.ui.animationsEnabled~=false state.mobileButtonSize=math.clamp(tonumber(configData.ui.mobileButtonSize)or 52,28,120)
configData.ui.guiStyle=nil if configData.ui.wurstSchema~=3 then if configData.ui.wurstSchema~=2 then configData.values['ClickGUI.Maxheight']=nil configData.values['ClickGUI.Maxsettingsheight']=nil
configData.values['ClickGUI.Opacity']=nil configData.values['ClickGUI.Tooltipopacity']=nil configData.values['WurstLogo.Background']=nil configData.values['WurstLogo.Text']=nil configData.values[
'WurstLogo.Visibility']=nil end configData.ui.wurstSchema=3 configData.ui.windows={}queueConfigSave()end configData.ui.textScaleCalibrationVersion=2 state.menuStyle='Wurst'do local savedScale:any=
configData.values['UI.MenuScale']if type(savedScale)=='number'and savedScale==savedScale then ThemeEngine.shape.scale=math.clamp(savedScale,0.7,1.3)end end if configData.values['UI.CodeFaceMigration']
~=1 then configData.values['UI.CodeFaceMigration']=1 if configData.values.interfaceFont~=nil and configData.values.interfaceFont~='Code'then configData.values.interfaceFont='Code'end if configData.
values['UI.FontPack']~=nil and configData.values['UI.FontPack']~='Code'then configData.values['UI.FontPack']='Code'end queueConfigSave()end state.fontPack='Code'local persistenceEnvironment:any=
getfenv()local persistenceGenv=sharedRuntimeGenv local function teleportWarn(message:string):()local logger:any=type(realWarn)=='function'and realWarn or warn logger('['..PRODUCT.logPrefix..'] '..
message)end local function resolveTeleportQueue():any for _,name:string in ipairs({'queue_on_teleport','queueonteleport'})do local queue:any=persistenceEnvironment[name]if type(queue)=='function'then
return queue end end for _,providerName:string in ipairs({'syn','fluxus'})do local provider:any=persistenceEnvironment[providerName]if type(provider)=='table'and type(provider.queue_on_teleport)==
'function'then return provider.queue_on_teleport end end return nil end local teleportTokenPath:string=CONFIG_PROFILE_FOLDER..'/Teleport_'..CONFIG_GAME_ID..'.txt'local function newTeleportOwner():
string local ok:boolean,guid:any=pcall(function():string return HttpService:GenerateGUID(false)end)if ok and type(guid)=='string'and#guid>0 then return guid end return tostring(os.time())..':'..
tostring(os.clock())..':'..tostring(math.random(1,1073741824))..':'..tostring({})end local teleportOwner:string=newTeleportOwner()local teleportSerial:number=0 local teleportToken:string=''local
teleportGuarded:boolean=false local function writeTeleportToken(value:string):boolean if type(readfile)~='function'or type(writefile)~='function'then return false end local ok:boolean=pcall(function()
:()if type(isfolder)=='function'and type(makefolder)=='function'then if not isfolder(CONFIG_FOLDER)then makefolder(CONFIG_FOLDER)end if not isfolder(CONFIG_PROFILE_FOLDER)then makefolder(
CONFIG_PROFILE_FOLDER)end end writefile(teleportTokenPath,value)end)if not ok then return false end local readOk:boolean,stored:any=pcall(readfile,teleportTokenPath)return readOk and stored==value end
local function claimTeleportToken():()teleportSerial+=1 teleportToken=teleportOwner..':'..tostring(teleportSerial)teleportGuarded=writeTeleportToken(teleportToken)end local function
revokeTeleportToken():()if not teleportGuarded then return end if type(readfile)~='function'then teleportWarn('could not read the teleport token; revocation is unverified')return end local ok:boolean,
stored:any=pcall(readfile,teleportTokenPath)if not ok then teleportWarn('could not read the teleport token; revocation is unverified')return end if stored~=teleportToken then return end if not
writeTeleportToken('stopped:'..teleportToken)then local checked:boolean,current:any=pcall(readfile,teleportTokenPath)local removed:boolean=false if checked and current==teleportToken and type(delfile)
=='function'then local deleted:boolean,result:any=pcall(delfile,teleportTokenPath)removed=deleted and result~=false end local stillOk:boolean,remaining:any=pcall(readfile,teleportTokenPath)if(stillOk
and remaining==teleportToken)or(not stillOk and not removed)then teleportWarn('could not revoke the teleport token')end end end local function buildTeleportSnippet(token:string,guarded:boolean):string
local armedGameId:number=game.GameId local guardLines:{string}=guarded and{'    if type(readfile) ~= "function" then return false end','    local ok, value = pcall(readfile, '..string.format('%q',
teleportTokenPath)..')','    return ok and value == '..string.format('%q',token)}or{'    return true'}return table.concat({[[local genv = (type(getgenv) == "function" and getgenv()) or {}]],
'local function authorized()',table.concat(guardLines,'\n'),'end','local function allowed()','    return genv.WURST_STOPPED ~= true and game.GameId == '..tostring(armedGameId)..' and authorized()',
'end',[[if genv.WURST_STOPPED == true or not authorized() then return end]],'if not game:IsLoaded() then game.Loaded:Wait() end','if not allowed() then return end','task.wait(1)',
'if not allowed() then return end','local ok, err = pcall(function()','    local source = game:HttpGet('..string.format('%q',RUNTIME_RAW_BASE..'src/Wurst.lua?t=')..' .. tostring(os.time()))',
'    if not allowed() then return end','    genv.WURST_BRANCH = '..string.format('%q',RUNTIME_BRANCH),'    loadstring(source)()','end)',
[[if not ok then warn("[Wurst] teleport reinject failed: " .. tostring(err)) end]]},'\n')end local queueTeleport:any=resolveTeleportQueue()local contract:any={supported=type(queueTeleport)=='function'
,enabled=false,closed=false,queued=false}state.teleportPersist=contract local teleportConnection:any=nil local appliedTeleportSetting:boolean?=nil local initialGenv:any=persistenceGenv()if initialGenv
then initialGenv.WURST_TP_OWNER=teleportOwner end claimTeleportToken()local function ownsTeleportSession():boolean local genv:any=persistenceGenv()return not genv or genv.WURST_TP_OWNER==teleportOwner
end local function ownsRuntimeData():boolean local genv:any=persistenceGenv()if genv and genv.WURST_TP_OWNER~=nil and genv.WURST_TP_OWNER~=teleportOwner then return false end if teleportGuarded and
type(readfile)=='function'then local ok:boolean,value:any=pcall(readfile,teleportTokenPath)if ok then return value==teleportToken or value=='stopped:'..teleportToken end end return true end
configWriteAllowed=ownsRuntimeData state.ownsRuntimeData=ownsRuntimeData local function disconnectTeleport():()if teleportConnection then pcall(function():()teleportConnection:Disconnect()end)
teleportConnection=nil end end local function markTeleportStopped():()local genv:any=persistenceGenv()if genv and genv.WURST_TP_OWNER==teleportOwner then genv.WURST_STOPPED=true genv.WURST_TP_OWNER=
nil end end local function warnUnguardedTeleport():()if not teleportGuarded and not contract._unguardedSaid then contract._unguardedSaid=true teleportWarn('teleport token storage unavailable; '..
[[a queue already submitted during teleport may not be revocable]])end end local function onTeleport(status:any):()if contract.closed or not contract.enabled or not ownsTeleportSession()then return
end if status==Enum.TeleportState.Failed then if contract.queued then contract.queued=false revokeTeleportToken()claimTeleportToken()end return end if status~=Enum.TeleportState.Started and status~=
Enum.TeleportState.WaitingForServer and status~=Enum.TeleportState.InProgress then return end if contract.queued then return end warnUnguardedTeleport()local ok:boolean,result:any=pcall(queueTeleport,
buildTeleportSnippet(teleportToken,teleportGuarded))if ok and result~=false then contract.queued=true else teleportWarn('teleport queue failed: '..tostring(result))end end contract.SetEnabled=function
(enabled:boolean):()if contract.closed or not ownsTeleportSession()then return end local wanted:boolean=enabled==true contract.enabled=wanted if appliedTeleportSetting==wanted then return end if
appliedTeleportSetting~=nil then revokeTeleportToken()claimTeleportToken()end appliedTeleportSetting=wanted contract.queued=false disconnectTeleport()local genv:any=persistenceGenv()if genv and genv.
WURST_TP_OWNER==teleportOwner then genv.WURST_STOPPED=not wanted end if not wanted then return end if not contract.supported then if not contract._unsupportedSaid then contract._unsupportedSaid=true
teleportWarn('queue_on_teleport is not available; '..'the menu cannot follow teleports here')end return end warnUnguardedTeleport()local connected:boolean,result:any=pcall(function():any return
LocalPlayer.OnTeleport:Connect(onTeleport)end)if connected then teleportConnection=result else contract.supported=false teleportWarn('could not observe teleport: '..tostring(result))end end contract.
Sync=function():()contract.SetEnabled(contract.enabled)end contract.Close=function():()if contract.closed then return end local preserve:boolean=contract.queued and contract.enabled and
ownsTeleportSession()contract.closed=true disconnectTeleport()if not preserve then revokeTeleportToken()markTeleportStopped()end contract.enabled=false end contract.Stop=function():()contract.closed=
true contract.enabled=false disconnectTeleport()revokeTeleportToken()markTeleportStopped()end contract.SetEnabled(configData.states['ClickGUI.Keepafterteleport']~=false)state.menuStyleListeners={}::{(
string)->()}state.SetMenuStyle=function(style:string,openImmediately:boolean?):()if style~='Wurst'and style~='Navigator'then return end state.menuStyle=style configData.values['UI.MenuStyle']=style
queueConfigSave()local navigator:any=state.navigator local navigatorOpen:boolean=navigator~=nil and navigator.root~=nil and navigator.root.Visible==true if style=='Wurst'then if navigatorOpen and
type(navigator.Close)=='function'then navigator.Close()end if openImmediately and not state.visible then state.setMenuVisible(true)end else if state.visible then state.setMenuVisible(false)end if
openImmediately and navigator and type(navigator.Open)=='function'then navigator.Open()end end for _,listener:(string)->()in ipairs(state.menuStyleListeners)do pcall(listener,style)end end state.
openMenuSurface=function():()if state.menuStyle=='Navigator'then local navigator:any=state.navigator if navigator and navigator.root and navigator.root.Visible then navigator.Close()return end if
state.visible then state.setMenuVisible(false)end if navigator and type(navigator.Open)=='function'then navigator.Open()return end end local navigator:any=state.navigator if navigator and navigator.
root and navigator.root.Visible and type(navigator.Close)=='function'then navigator.Close()end state.setMenuVisible(not state.visible)end local function validLayoutViewport(viewport:any):boolean
return viewport~=nil and type(viewport.X)=='number'and type(viewport.Y)=='number'and viewport.X>1 and viewport.Y>1 and viewport.X<math.huge and viewport.Y<math.huge end local bootViewport:any=
workspace.CurrentCamera and workspace.CurrentCamera.ViewportSize state.layoutViewport=if validLayoutViewport(bootViewport)then bootViewport else Vector2.new(1280,720)state.layout=computeLayoutMetrics(
state.layoutViewport,UserInputService.TouchEnabled)state.layoutListeners={}::{(LayoutMetrics)->()}state.onLayout=function(listener:(LayoutMetrics)->()):()table.insert(state.layoutListeners,listener)
end state.refreshLayout=function(force:boolean?):boolean local camera:Camera?=workspace.CurrentCamera local viewport:any=camera and camera.ViewportSize if not validLayoutViewport(viewport)then if
force then state.layoutNeedsRefresh=true end return false end local nextLayout:LayoutMetrics=computeLayoutMetrics(viewport,UserInputService.TouchEnabled)local previous:LayoutMetrics=state.layout local
previousViewport:Vector2=state.layoutViewport local forceRefresh:boolean=force==true or state.layoutNeedsRefresh==true state.layoutNeedsRefresh=false if not forceRefresh and previous.mode==nextLayout.
mode and previous.cardHeight==nextLayout.cardHeight and previousViewport.X==viewport.X and previousViewport.Y==viewport.Y then return true end state.layoutViewport=viewport state.layout=nextLayout for
_,listener:(LayoutMetrics)->()in ipairs(state.layoutListeners)do local ok:boolean,failure:any=pcall(listener,nextLayout)if not ok then warn('['..PRODUCT.name..'] layout listener: '..tostring(failure))
end end return true end local function trackUiConnection(connection:RBXScriptConnection):RBXScriptConnection table.insert(state.uiConnections,connection)return connection end local menuCursorState:any
={owned=false,applying=false,behavior=nil,iconEnabled=nil,fights=0,fightStart=0,yielded=false}local CURSOR_FIGHT_LIMIT:number=8 local CURSOR_FIGHT_WINDOW:number=1 local function setMenuCursorUnlocked(
visible:boolean):()if menuCursorState.applying then return end if visible then if menuCursorState.yielded then return end if not menuCursorState.owned then menuCursorState.behavior=UserInputService.
MouseBehavior menuCursorState.iconEnabled=UserInputService.MouseIconEnabled menuCursorState.owned=true end menuCursorState.applying=true pcall(function():()UserInputService.MouseBehavior=Enum.
MouseBehavior.Default UserInputService.MouseIconEnabled=true end)menuCursorState.applying=false return end if not menuCursorState.owned then menuCursorState.yielded=false return end menuCursorState.
applying=true pcall(function():()if UserInputService.MouseBehavior==Enum.MouseBehavior.Default then local saved:any=menuCursorState.behavior if saved==Enum.MouseBehavior.LockCenter or saved==nil then
saved=Enum.MouseBehavior.Default end UserInputService.MouseBehavior=saved end if UserInputService.MouseIconEnabled==true and menuCursorState.iconEnabled~=nil then UserInputService.MouseIconEnabled=
menuCursorState.iconEnabled end end)menuCursorState.applying=false menuCursorState.behavior=nil menuCursorState.iconEnabled=nil menuCursorState.owned=false menuCursorState.fights=0 menuCursorState.
yielded=false end local function enforceMenuCursor():()if menuCursorState.applying then return end if state.visible~=true or not menuCursorState.owned then return end local rightClicking:boolean=false
pcall(function():()rightClicking=UserInputService:IsMouseButtonPressed(Enum.UserInputType.MouseButton2)end)if rightClicking then return end if UserInputService.MouseBehavior==Enum.MouseBehavior.
Default and UserInputService.MouseIconEnabled==true then return end local now:number=os.clock()if now-menuCursorState.fightStart>CURSOR_FIGHT_WINDOW then menuCursorState.fightStart=now menuCursorState
```

```
.fights=0 end menuCursorState.fights+=1 if menuCursorState.fights>CURSOR_FIGHT_LIMIT then menuCursorState.yielded=true menuCursorState.fights=0 return end setMenuCursorUnlocked(true)end pcall(function
():()trackUiConnection(UserInputService:GetPropertyChangedSignal('MouseBehavior'):Connect(enforceMenuCursor))trackUiConnection(UserInputService:GetPropertyChangedSignal('MouseIconEnabled'):Connect(
enforceMenuCursor))trackUiConnection(UserInputService.InputEnded:Connect(function(input:InputObject):()if input.UserInputType==Enum.UserInputType.MouseButton2 and state.visible then task.defer(
function():()if state.visible and not UserInputService:IsMouseButtonPressed(Enum.UserInputType.MouseButton2)then setMenuCursorUnlocked(true)end end)end end))if LocalPlayer then pcall(function():()
trackUiConnection(LocalPlayer.CharacterAdded:Connect(function():()if state.visible~=true then menuCursorState.owned=false menuCursorState.yielded=false end end))end)end state.setChoiceBlur=
function(active:boolean):()state.choiceBlur=active if MenuBlur then MenuBlur.Enabled=state.visible==true or active end end local menuVisibilityListeners:{(boolean)->()}={}state.
addMenuVisibilityListener=function(listener:(boolean)->()):()table.insert(menuVisibilityListeners,listener)end local function setMenuVisibleCore(visible:boolean):()state.visible=visible
setMenuCursorUnlocked(visible)if MenuBlur then pcall(function():()MenuBlur.Enabled=visible==true or state.choiceBlur==true end)end for _,listener:(boolean)->()in ipairs(menuVisibilityListeners)do
pcall(listener,visible)end end state.setMenuVisible=setMenuVisibleCore type AssetDefinition={url:string,fileName:string,fallback:string,version:number?,assetType:string?}do local generatedEnvironment:
any=nil if type(executorGlobals.getgenv)=='function'then local environmentOk:boolean,resolvedEnvironment:any=pcall(executorGlobals.getgenv)if environmentOk and type(resolvedEnvironment)=='table'then
generatedEnvironment=resolvedEnvironment end end local function getEnvironmentValue(name:string):any if generatedEnvironment and generatedEnvironment[name]~=nil then return generatedEnvironment[name]
end return executorGlobals[name]end local AssetManager:any={cacheRoot=CONFIG_FOLDER,cacheFolder=CONFIG_FOLDER..'/Assets',resolved={}::{[string]:string},resolving={}::{[string]:boolean},manifest={
valveFont={url=REPOSITORY_RAW_BASE..'src/guis/Wurst/Assets/Typography/ValveBD.otf',fileName='valve-bd-v1.otf',fallback='',version=1,assetType='font/otf'},wurstLogo={url=REPOSITORY_RAW_BASE..
'src/guis/Wurst/Assets/Wurst/wurst_128.png',fileName='wurst-logo-v1.png',fallback='',version=1,assetType='image/png'},bitmapFontAtlas={url=REPOSITORY_RAW_BASE..
'src/guis/Wurst/Assets/Font/monocraft-16.png',fileName='monocraft-16-v1.png',fallback='',version=1,assetType='image/png'},bitmapFontAtlas8={url=REPOSITORY_RAW_BASE..
'src/guis/Wurst/Assets/Font/monocraft-8.png',fileName='monocraft-8-v1.png',fallback='',version=1,assetType='image/png'},bitmapFontAtlas24={url=REPOSITORY_RAW_BASE..
'src/guis/Wurst/Assets/Font/monocraft-24.png',fileName='monocraft-24-v1.png',fallback='',version=1,assetType='image/png'},bitmapFontAtlas32={url=REPOSITORY_RAW_BASE..
'src/guis/Wurst/Assets/Font/monocraft-32.png',fileName='monocraft-32-v1.png',fallback='',version=1,assetType='image/png'}}::{[string]:AssetDefinition}}local function selectHttpRequest():(any,string)
local synEnvironment:any=getEnvironmentValue('syn')local candidates:{{any}}={{getEnvironmentValue('request'),'request'},{getEnvironmentValue('http_request'),'http_request'},{getEnvironmentValue(
'httprequest'),'httprequest'},{synEnvironment and synEnvironment.request,'syn.request'}}for _,candidate:{any}in ipairs(candidates)do if type(candidate[1])=='function'then return candidate[1],tostring(
candidate[2])end end return nil,'game:HttpGet'end local function validateAssetBytes(bytes:any,definition:AssetDefinition):(boolean,string?)if type(bytes)~='string'then return false,
'non-binary response: '..type(bytes)end if#bytes<64 then return false,'response too short: '..tostring(#bytes)..' bytes'end local prefix:string=string.lower(string.sub(bytes,1,128))if string.find(
prefix,'<!doctype html',1,true)or string.find(prefix,'<html',1,true)or string.find(prefix,'404: not found',1,true)then return false,'the response contains HTML/404'end if definition.assetType==
'font/otf'then local fontSignature:string=string.sub(bytes,1,4)if fontSignature~='OTTO'and fontSignature~=string.char(0,1,0,0)then return false,'invalid OpenType/TrueType signature'end return true,nil
end local isPng:boolean=string.sub(bytes,1,8)=='\137PNG\r\n\26\n'local isJpeg:boolean=string.byte(bytes,1)==0xff and string.byte(bytes,2)==0xd8 and string.byte(bytes,3)==0xff if not isPng and not
isJpeg then return false,'invalid PNG signature'end return true,nil end function AssetManager.ensureCacheFolder():(boolean,string?)local isFolder:any=getEnvironmentValue('isfolder')local makeFolder:
any=getEnvironmentValue('makefolder')local function ensureOne(path:string):(boolean,string?)if type(isFolder)=='function'then local checked:boolean,exists:any=pcall(isFolder,path)if checked and exists
then return true,nil end end if type(makeFolder)~='function'then return false,'makefolder is unavailable'end local created:boolean,createError:any=pcall(makeFolder,path)if not created then return
false,tostring(createError)end return true,nil end local rootOk:boolean,rootError:string?=ensureOne(AssetManager.cacheRoot)if not rootOk then return false,'carpeta principal: '..tostring(rootError)end
local cacheOk:boolean,cacheError:string?=ensureOne(AssetManager.cacheFolder)if not cacheOk then return false,'carpeta Assets: '..tostring(cacheError)end return true,nil end function AssetManager.
sweepStaleCache():number local listFiles:any=getEnvironmentValue('listfiles')local deleteFile:any=getEnvironmentValue('delfile')if type(listFiles)~='function'or type(deleteFile)~='function'then return
0 end local expected:{[string]:boolean}={}for _,definition:any in pairs(AssetManager.manifest)do expected[tostring(definition.fileName)]=true end local listed:boolean,entries:any=pcall(listFiles,
AssetManager.cacheFolder)if not listed or type(entries)~='table'then return 0 end local removed:number=0 for _,path:any in ipairs(entries)do local entry:string=tostring(path)local fileName:string=
entry:gsub('\\','/'):match('([^/]+)$')or entry if not expected[fileName]then if pcall(deleteFile,entry)then removed+=1 end end end return removed end function AssetManager.readVerifiedCache(path:
string,definition:AssetDefinition):(string?,string?)local isFile:any=getEnvironmentValue('isfile')local readFile:any=getEnvironmentValue('readfile')if type(isFile)~='function'or type(readFile)~=
'function'then return nil,'isfile/readfile are unavailable'end local checked:boolean,exists:any=pcall(isFile,path)if not checked or not exists then return nil,'file not found'end local readOk:boolean,
bytes:any=pcall(readFile,path)if not readOk then return nil,'readfile: '..tostring(bytes)end local valid:boolean,validationError:string?=validateAssetBytes(bytes,definition)if not valid then return
nil,validationError end return bytes,nil end function AssetManager.download(definition:AssetDefinition):(string?,string?)local requestFunction:any,_requestName:string=selectHttpRequest()local
lastError:string='no response'for attempt:number=1,3 do local requestOk:boolean,response:any if requestFunction then requestOk,response=pcall(requestFunction,{Url=definition.url,Method='GET',Headers={
Accept=definition.assetType or'*/*'}})else requestOk,response=pcall(function():any return{StatusCode=200,Body=(game::any):HttpGet(definition.url,true)}end)end if requestOk then local statusCode:number
=0 local bytes:any=nil if type(response)=='string'then statusCode=200 bytes=response elseif type(response)=='table'then statusCode=tonumber(response.StatusCode or response.Status or response.
status_code)or 0 bytes=response.Body or response.body or response.ResponseBody end local valid:boolean,validationError:string?=validateAssetBytes(bytes,definition)if statusCode>=200 and statusCode<300
and valid then return bytes,nil end response='HTTP '..tostring(statusCode)..': '..tostring(validationError)end lastError='attempt '..tostring(attempt)..': '..tostring(response)if attempt<3 then task.
wait(0.35*attempt)end end return nil,'download failed after three attempts; '..lastError end function AssetManager.resolve(logicalName:string):(string?,string?)local alreadyResolved:string?=
AssetManager.resolved[logicalName]if alreadyResolved then return alreadyResolved,nil end local definition:AssetDefinition?=AssetManager.manifest[logicalName]if not definition then return nil,
'asset desconocido: '..logicalName end local assetFunction:any=getEnvironmentValue('getcustomasset')or getEnvironmentValue('getsynasset')local writeFile:any=getEnvironmentValue('writefile')if type(
assetFunction)~='function'then return nil,'getcustomasset/getsynasset is unavailable'end if type(writeFile)~='function'then return nil,'writefile is unavailable'end local folderOk:boolean,folderError:
string?=AssetManager.ensureCacheFolder()if not folderOk then return nil,folderError end local path:string=AssetManager.cacheFolder..'/'..definition.fileName local waitTime:number=0 while AssetManager.
resolving[logicalName]and waitTime<12 do task.wait(0.05)waitTime+=0.05 end alreadyResolved=AssetManager.resolved[logicalName]if alreadyResolved then return alreadyResolved,nil end if AssetManager.
resolving[logicalName]then return nil,'timed out waiting for '..logicalName end AssetManager.resolving[logicalName]=true local cachedBytes:string?,cacheError:string?=AssetManager.readVerifiedCache(
path,definition)if not cachedBytes then print('['..PRODUCT.logPrefix..':Assets] cache cannot be reused: '..tostring(cacheError))local downloaded:string?,downloadError:string?=AssetManager.download(
definition)if not downloaded then AssetManager.resolving[logicalName]=false return nil,downloadError end local written:boolean,writeError:any=pcall(writeFile,path,downloaded)if not written then
AssetManager.resolving[logicalName]=false return nil,'writefile: '..tostring(writeError)end local verified:string?,verifyError:string?=AssetManager.readVerifiedCache(path,definition)if not verified
then AssetManager.resolving[logicalName]=false return nil,'post-write verification: '..tostring(verifyError)end end local assetOk:boolean,assetId:any=pcall(assetFunction,path)if not assetOk or type(
assetId)~='string'or assetId==''then AssetManager.resolving[logicalName]=false return nil,'getcustomasset: '..tostring(assetId)end AssetManager.resolved[logicalName]=assetId AssetManager.resolving[
logicalName]=false return assetId,nil end function AssetManager.resolveFont(logicalName:string,fallbackFont:Font,weight:Enum.FontWeight,style:Enum.FontStyle):Font local assetId:string?,loadError:
string?=AssetManager.resolve(logicalName)if not assetId then warn('['..PRODUCT.logPrefix..':Assets] fallback font for '..logicalName..': '..tostring(loadError))return fallbackFont end local fontOk:
boolean,fontFace:any=pcall(Font.new,assetId,weight,style)if not fontOk or typeof(fontFace)~='Font'then warn('['..PRODUCT.logPrefix..':Assets] Font.new fallback para '..logicalName..': '..tostring(
fontFace))return fallbackFont end print('['..PRODUCT.logPrefix..':Assets] font loaded: '..logicalName)return fontFace end VALVE_FONT=AssetManager.resolveFont('valveFont',CONTROL_FONT,Enum.FontWeight.
Bold,Enum.FontStyle.Normal)state.customFontAssets={['Valve']='valveFont'}state.listInterfaceFonts=function():{string}local names:{string}={'Menu default'}for _,item:EnumItem in ipairs(Enum.Font:
GetEnumItems())do if item~=Enum.Font.Unknown then table.insert(names,item.Name)end end for name:string in pairs(state.customFontAssets)do table.insert(names,name)end table.sort(names,function(first:
string,second:string):boolean if first=='Menu default'then return true end if second=='Menu default'then return false end return first<second end)return names end state.resolveInterfaceFont=function(
name:string):Font?if name=='Menu default'then return Font.new('rbxasset://fonts/families/BuilderSans.json',Enum.FontWeight.Regular,Enum.FontStyle.Normal)end local customAsset:string?=state.
customFontAssets[name]if customAsset then return AssetManager.resolveFont(customAsset,CONTROL_FONT,Enum.FontWeight.Regular,Enum.FontStyle.Normal)end local enumOk:boolean,enumItem:any=pcall(function():
any return(Enum.Font::any)[name]end)if not enumOk or typeof(enumItem)~='EnumItem'then return nil end local fontOk:boolean,face:any=pcall(Font.fromEnum,enumItem)if not fontOk then return nil end return
face::Font end state.applyInterfaceFont=function(name:string,persist:boolean?):()local base:Font?=state.resolveInterfaceFont(name)if not base then return end local resolvedBase:Font=base::Font local
family:string=resolvedBase.Family local style:Enum.FontStyle=resolvedBase.Style local hasWeights:boolean=string.find(family,'fonts/families/',1,true)~=nil local titleWeight:Enum.FontWeight=hasWeights
and Enum.FontWeight.Bold or resolvedBase.Weight local controlWeight:Enum.FontWeight=hasWeights and Enum.FontWeight.SemiBold or resolvedBase.Weight local bodyWeight:Enum.FontWeight=hasWeights and Enum.
FontWeight.Regular or resolvedBase.Weight TITLE_FONT=Font.new(family,titleWeight,style)CONTROL_FONT=Font.new(family,controlWeight,style)THIN_FONT=Font.new(family,bodyWeight,style)VALVE_FONT=
CONTROL_FONT for _,descendant:Instance in ipairs(ScreenGui:GetDescendants())do if descendant:IsA('TextLabel')or descendant:IsA('TextButton')or descendant:IsA('TextBox')then local current:Font?=(
descendant::any).FontFace if current then local weight:Enum.FontWeight=hasWeights and(current::Font).Weight or resolvedBase.Weight local ok:boolean=pcall(function():()(descendant::any).FontFace=Font.
new(family,weight,(current::Font).Style)end)if not ok then(descendant::any).FontFace=resolvedBase end else(descendant::any).FontFace=resolvedBase end end end if persist~=false then configData.values.
interfaceFont=name queueConfigSave()end end state.resolveAsset=function(logicalName:string):(string?,string?)return AssetManager.resolve(logicalName)end state.sweepStaleAssets=function():number return
AssetManager.sweepStaleCache()end state.applyAssetImage=function(logicalName:string,image:ImageLabel,onApplied:(()->())?):()task.spawn(function():()local assetId:string?,failure:string?=AssetManager.
resolve(logicalName)if not assetId then warn('['..PRODUCT.logPrefix..':Assets] '..logicalName..': '..tostring(failure))return end if not image.Parent then return end image.Image=assetId::string if
onApplied then onApplied()end end)end end local registeredText={}local SectionManager={entries={}}function SectionManager.register(sectionName,builder)SectionManager.entries[sectionName]={builder=
builder,loaded=false,loading=false}end function SectionManager.initialize(sectionName:string):(boolean,string?)local entry=SectionManager.entries[sectionName]if not entry or entry.loaded or entry.
loading then return entry~=nil and entry.loaded==true,nil end entry.loading=true local success,errorMessage=xpcall(entry.builder,function(message)local text=tostring(message)if debug and type(debug.
traceback)=='function'then return debug.traceback(text,2)end return text end)entry.loading=false if success then entry.loaded=true return true,nil end warn('['..PRODUCT.name..'] '..sectionName..
' failed to initialize: '..tostring(errorMessage))return false,tostring(errorMessage)end function SectionManager.initializeAllAsync():()local sectionNames:{string}={}for sectionName:string in pairs(
SectionManager.entries)do table.insert(sectionNames,sectionName)end table.sort(sectionNames)local total:number=#sectionNames local completed:number=0 local failures:number=0 local function
finishLoading():()print('['..PRODUCT.logPrefix..':Bootstrap] Load complete \183 failures='..tostring(failures))local function finish():()state.menuReady=true if type(state.refreshLauncher)=='function'
then pcall(state.refreshLauncher)end end if type(task)=='table'and type(task.defer)=='function'then task.defer(finish)else finish()end end if total==0 then finishLoading()return end for _,sectionName:
string in ipairs(sectionNames)do task.spawn(function():()print('['..PRODUCT.logPrefix..':Bootstrap] Initializing section '..sectionName)local success:boolean,errorMessage:string?=SectionManager.
initialize(sectionName)if not success then failures+=1 local environment=state.frameworkEnvironment if environment and type(environment.notify)=='function'then environment.notify('Could not load '..
sectionName..' \u{2014} check the console')end warn('['..PRODUCT.logPrefix..':Bootstrap] Section '..sectionName..' failed: '..tostring(errorMessage))end completed+=1 if completed>=total then
finishLoading()end end)end end local function create(className,properties)local object=Instance.new(className)for property,value in pairs(properties or{})do if property~='Parent'then object[property]=
value end end if object:IsA('TextLabel')or object:IsA('TextButton')or object:IsA('TextBox')then object.AutoLocalize=false end if properties and properties.Parent then object.Parent=properties.Parent
end return object end local function cornerRadius(_radius:number):UDim return UDim.new(0,0)end local function addRoundedStyle(object:GuiObject,radius:number,strokeTransparency:number?):(UICorner,
UIStroke)local corner:UICorner=Instance.new('UICorner')corner.Name='StyleCorner'corner.CornerRadius=cornerRadius(radius)corner.Parent=object local stroke:UIStroke=Instance.new('UIStroke')stroke.Name=
'StyleStroke'stroke.ApplyStrokeMode=Enum.ApplyStrokeMode.Border stroke.Color=Theme.outline stroke.Thickness=1 stroke.Transparency=strokeTransparency or 0.45 stroke.Parent=object return corner,stroke
end state.bitmapText={size=16,lineHeight=18,advance=12,ascent=14,source='monocraft-fallback',atlas={ready=false,image=nil::string?,size=16,first=32,last=126,columns=16,cellWidth=11,cellHeight=17,
advance=11,ascent=15,descent=2,generation=0,fallbackGlyph=63},onReady={}::{(()->())},stats={draws=0,skipped=0,glyphsCreated=0,glyphsReused=0,lastDrawMs=0},_live=setmetatable({},{__mode='k'})::any,
_adopted=setmetatable({},{__mode='k'})::any,_pendingAdopt=setmetatable({},{__mode='k'})::any,width=function(text:any):number return state.bitmapText.measure({{text=tostring(text)}},16)end,make=
function(properties:any):TextLabel local resolved:any={}for key:string,value:any in pairs(properties or{})do resolved[key]=value end resolved.FontFace=CONTROL_FONT resolved.TextSize=resolved.TextSize
or 16 if resolved.TextTruncate==nil then resolved.TextTruncate=Enum.TextTruncate.AtEnd end return(create('TextLabel',resolved)::any)::TextLabel end}local fontEnvironment:any=getfenv()state.bitmapText.
capabilities={canHttp=type((game::any).HttpGet)=='function',canRead=type(fontEnvironment.readfile)=='function',canWrite=type(fontEnvironment.writefile)=='function',canCustomAsset=type(fontEnvironment.
getcustomasset)=='function',canSynAsset=type(fontEnvironment.getsynasset)=='function',mobile=state.isMobile,desktop=state.isMobile~=true}state.bitmapText.family={[8]={ready=false,image=nil::
string?,size=8,columns=16,cellWidth=5,cellHeight=9,advance=5,ascent=8,descent=1,asset='bitmapFontAtlas8'},[16]=state.bitmapText.atlas,[24]={ready=false,image=nil::string?,size=24,columns=16,cellWidth=
16,cellHeight=25,advance=16,ascent=22,descent=3,asset='bitmapFontAtlas24'},[32]={ready=false,image=nil::string?,size=32,columns=16,cellWidth=21,cellHeight=33,advance=21,ascent=29,descent=4,asset=
'bitmapFontAtlas32'}}state.bitmapText.layerScale=function():number local instance:any=PopupLayer:FindFirstChild('ClickGuiScale')local value:number=instance and tonumber(instance.Scale)or 1 if value~=
value or value<=0 then value=1 end return value end state.bitmapText.pickAtlas=function(physicalSize:number):any local contract:any=state.bitmapText if contract.rasterFamilyEnabled==false then return
contract.atlas end local smallestCovering:any=nil local largest:any=nil for _,member:any in pairs(contract.family)do if member.ready and member.image then if member.size>=physicalSize and(
smallestCovering==nil or member.size<smallestCovering.size)then smallestCovering=member end if largest==nil or member.size>largest.size then largest=member end end end return smallestCovering or
largest or contract.atlas end state.bitmapText.baselineFor=function(textSize:number?):number local contract:any=state.bitmapText return contract.atlas.ascent*((tonumber(textSize)or 16)/contract.atlas.
size)end state.bitmapText.eachCode=function(text:string,visit:(number)->()):()if utf8.len(text)~=nil then for _,code:number in utf8.codes(text)do visit(code)end else for position:number=1,#text do
visit(string.byte(text,position)or 63)end end end local function contractAdvanceFor(atlas:any,code:number,scale:number):number local advances:any=atlas.advances local specific:any=advances and(
advances[code]or advances[tostring(code)])return(tonumber(specific)or atlas.advance)*scale end state.bitmapText.measure=function(segments:{any},textSize:number?):number local contract:any=state.
bitmapText local atlas:any=contract.atlas local scale:number=(tonumber(textSize)or 16)/contract.size local widest:number=0 local x:number=0 for _,segment:any in ipairs(segments)do contract.eachCode(
tostring(segment.text or''),function(code:number):()if code==10 then x=0 else if atlas.ready then x+=contractAdvanceFor(atlas,code,scale)else x+=contract.advance*scale end widest=math.max(widest,x)end
end)end return widest end state.bitmapText.wrap=function(text:any,wrapWidth:number,textSize:number?):(string,number)local contract:any=state.bitmapText local lines:{string}={}local current:string=''
for word:string in string.gmatch(tostring(text or''),'%S+')do local candidate:string=current==''and word or(current..' '..word)if contract.measure({{text=candidate}},textSize)<=wrapWidth or current==
''then current=candidate else table.insert(lines,current)current=word end end if current~=''then table.insert(lines,current)end if#lines==0 then lines={''}end return table.concat(lines,'\n'),#lines
end state.bitmapText.draw=function(parent:Instance,segments:{any},options:any):Frame local startedAt:number=os.clock()local contract:any=state.bitmapText local resolvedOptions:any=options or{}local
name:string=tostring(resolvedOptions.Name or'BitmapText')local textSize:number=tonumber(resolvedOptions.TextSize)or 16 local zIndex:number=tonumber(resolvedOptions.ZIndex)or 20 local align:string=
tostring(resolvedOptions.align or'Left')local shadow:boolean=resolvedOptions.shadow==true local maxWidth:number?=tonumber(resolvedOptions.maxWidth)local wantsEllipsis:boolean=resolvedOptions.ellipsis
==true local transparency:number=tonumber(resolvedOptions.transparency)or 0 local atlas:any=contract.atlas local deviceScale:number=contract.layerScale()local raster:any=contract.pickAtlas(textSize*
deviceScale)local parts:{string}={tostring(textSize),align,tostring(shadow),tostring(maxWidth),tostring(atlas.ready),tostring(transparency),tostring(wantsEllipsis),string.format('%.4f',deviceScale),
tostring(raster and raster.size),tostring(atlas.generation)}for _,segment:any in ipairs(segments)do local color:Color3=segment.color or Theme.text table.insert(parts,tostring(segment.text or''))table.
insert(parts,string.format('%02X%02X%02X',math.floor(color.R*255+0.5),math.floor(color.G*255+0.5),math.floor(color.B*255+0.5)))end local signature:string=table.concat(parts,'\1')local holder:any=(
parent::any):FindFirstChild(name)local record:any=holder and contract._live[holder]if holder and not record then holder:Destroy()holder=nil end if not holder then holder=create('Frame',{Parent=parent,
Name=name,BackgroundTransparency=1,BorderSizePixel=0})record={glyphs={}::{ImageLabel},fallback=nil::TextLabel?}contract._live[holder]=record end record.request={parent=parent,segments=segments,options
=resolvedOptions}holder.ZIndex=zIndex if resolvedOptions.AnchorPoint or resolvedOptions.Position then holder.AnchorPoint=resolvedOptions.AnchorPoint or Vector2.new(0,0.5)holder.Position=
resolvedOptions.Position or UDim2.new(0,0,0.5,0)elseif align=='Right'then holder.AnchorPoint=Vector2.new(1,0.5)holder.Position=UDim2.new(1,0,0.5,0)elseif align=='Center'then holder.AnchorPoint=Vector2
.new(0.5,0.5)holder.Position=UDim2.new(0.5,0,0.5,0)else holder.AnchorPoint=Vector2.new(0,0.5)holder.Position=UDim2.new(0,0,0.5,0)end if record.signature==signature then contract.stats.skipped+=1
return holder::Frame end record.signature=signature record.paintMap={}local used:number=0 local function nextGlyph():ImageLabel used+=1 local glyph:ImageLabel?=record.glyphs[used]if glyph then
contract.stats.glyphsReused+=1 else glyph=(create('ImageLabel',{Parent=holder,Name='G',BackgroundTransparency=1,BorderSizePixel=0})::any)::ImageLabel pcall(function():()(glyph::any).ResampleMode=Enum.
ResamplerMode.Pixelated end)record.glyphs[used]=glyph contract.stats.glyphsCreated+=1 end(glyph::ImageLabel).Visible=true return glyph::ImageLabel end local width:number=0 local height:number=0 if
atlas.ready and atlas.image then if next(contract._pendingAdopt)~=nil then contract.flushPendingAdopt()end if record.fallback then(record.fallback::TextLabel).Visible=false end local scale:number=
textSize/atlas.size local function snapPx(value:number):number return math.round(value*deviceScale)/deviceScale end local rasterScale:number=textSize/raster.size local pixelStep:number=1/deviceScale
local glyphWidth:number=math.max(pixelStep,snapPx(raster.cellWidth*rasterScale))local glyphHeight:number=math.max(pixelStep,snapPx(raster.cellHeight*rasterScale))local lineAdvance:number=contract.
lineHeight*scale local shadowOffset:number=math.max(pixelStep,snapPx(2*scale))local x:number=0 local y:number=0 local lineCount:number=1 local truncated:boolean=false local contentWidth:number?=
maxWidth if maxWidth~=nil and wantsEllipsis and contract.measure(segments,textSize)>maxWidth then contentWidth=math.max(0,maxWidth-2*contractAdvanceFor(atlas,46,scale))end local lastColor:Color3=Theme
.text local currentSegment=1 local function blit(code:number,paint:Color3,dx:number,dy:number,z:number):()local index:number=code-atlas.first local glyph:ImageLabel=nextGlyph()record.paintMap[used]={
segment=currentSegment,factor=z==zIndex+1 and 0.25 or 1}glyph.Image=raster.image glyph.ImageColor3=paint glyph.ImageTransparency=transparency glyph.ImageRectOffset=Vector2.new((index%raster.columns)*
raster.cellWidth,(index//raster.columns)*raster.cellHeight)glyph.ImageRectSize=Vector2.new(raster.cellWidth,raster.cellHeight)glyph.Position=UDim2.fromOffset(snapPx(dx),snapPx(dy))glyph.Size=UDim2.
fromOffset(glyphWidth,glyphHeight)glyph.ZIndex=z end for segmentIndex,segment:any in ipairs(segments)do currentSegment=segmentIndex local color:Color3=segment.color or Theme.text local shade:Color3=
Color3.new(color.R*0.25,color.G*0.25,color.B*0.25)lastColor=color contract.eachCode(tostring(segment.text or''),function(code:number):()if truncated then return end if code==10 then x=0 y+=lineAdvance
lineCount+=1 return end local advance:number=contractAdvanceFor(atlas,code,scale)if contentWidth~=nil and x+advance>contentWidth then truncated=true return end if code~=32 then local mapped:number=
code if code<atlas.first or code>atlas.last then mapped=atlas.fallbackGlyph end if shadow then blit(mapped,shade,x+shadowOffset,y+shadowOffset,zIndex+1)end blit(mapped,color,x,y,zIndex+2)end x+=
advance width=math.max(width,x)end)end if truncated and wantsEllipsis then local dot:number=46 local dotAdvance:number=contractAdvanceFor(atlas,dot,scale)local budget:number=(maxWidth::number)local
dotX:number=math.min(x,budget-dotAdvance*2)if dotX>=0 then if shadow then blit(dot,Color3.new(lastColor.R*0.25,lastColor.G*0.25,lastColor.B*0.25),dotX+shadowOffset,y+shadowOffset,zIndex+1)blit(dot,
Color3.new(lastColor.R*0.25,lastColor.G*0.25,lastColor.B*0.25),dotX+dotAdvance+shadowOffset,y+shadowOffset,zIndex+1)end blit(dot,lastColor,dotX,y,zIndex+2)blit(dot,lastColor,dotX+dotAdvance,y,zIndex+2
)width=math.max(width,dotX+dotAdvance*2)end end height=glyphHeight+(lineCount-1)*lineAdvance else local composed:{string}={}local estimated:number=0 for _,segment:any in ipairs(segments)do local text:
string=tostring(segment.text or''):gsub('&','&amp;'):gsub('<','&lt;'):gsub('>','&gt;')local color:Color3=segment.color or Theme.text table.insert(composed,string.format(
'<font color="#%02X%02X%02X">%s</font>',math.floor(color.R*255+0.5),math.floor(color.G*255+0.5),math.floor(color.B*255+0.5),text))estimated+=#tostring(segment.text or'')end local fallback:TextLabel?=
record.fallback if not fallback then fallback=contract.make({Parent=holder,Name='Fallback',BackgroundTransparency=1,RichText=true,Size=UDim2.new(1,0,1,0),TextColor3=Theme.text})record.fallback=
fallback contract._adopted[fallback]=true end local label:TextLabel=fallback::TextLabel label.Visible=true label.Text=table.concat(composed)label.TextSize=textSize label.TextTransparency=transparency
label.TextXAlignment=align=='Right'and Enum.TextXAlignment.Right or align=='Center'and Enum.TextXAlignment.Center or Enum.TextXAlignment.Left label.ZIndex=zIndex+1 width=estimated*math.floor(textSize*
0.75)height=math.floor(textSize*1.125)end for index:number=used+1,#record.glyphs do(record.glyphs[index]::ImageLabel).Visible=false end if maxWidth~=nil then width=math.min(width,maxWidth)end holder.
ClipsDescendants=maxWidth~=nil holder.Size=UDim2.fromOffset(width,height)holder:SetAttribute('MeasuredWidth',width)contract.stats.draws+=1 contract.stats.lastDrawMs=(os.clock()-startedAt)*1000 return
holder::Frame end state.bitmapText.recolor=function(holder:any,colors:{Color3}):boolean local contract=state.bitmapText local record=holder and contract._live[holder]if not record or not record.
request then return false end local segments=record.request.segments local dirty=false for index,segment in ipairs(segments)do if colors[index]and segment.color~=colors[index]then segment.color=colors
[index]dirty=true end end if not dirty then return true end record.signature=nil if record.fallback and record.fallback.Visible then local composed={}for _,segment in ipairs(segments)do local text=
tostring(segment.text or''):gsub('&','&amp;'):gsub('<','&lt;'):gsub('>','&gt;')local color=segment.color or Theme.text table.insert(composed,string.format('<font color="#%02X%02X%02X">%s</font>',math.
floor(color.R*255+0.5),math.floor(color.G*255+0.5),math.floor(color.B*255+0.5),text))end record.fallback.Text=table.concat(composed)else for index,binding in ipairs(record.paintMap or{})do local glyph
=record.glyphs[index]local color=segments[binding.segment].color or Theme.text local factor=binding.factor local paint=factor==1 and color or Color3.new(color.R*factor,color.G*factor,color.B*factor)if
glyph and glyph.Parent and glyph.ImageColor3~=paint then glyph.ImageColor3=paint end end end contract.stats.recolors=(contract.stats.recolors or 0)+1 return true end state.bitmapText.release=function(
holder:Instance?):()if holder then state.bitmapText._live[holder]=nil holder:Destroy()end end state.bitmapText.adopt=function(label:any,options:any?):()local contract:any=state.bitmapText if label==
nil or type(label.IsA)~='function'or label:IsA('TextBox')or not(label:IsA('TextLabel')or label:IsA('TextButton'))then return end if label:GetAttribute('NoBitmap')==true then return end if contract.
_adopted[label]then return end if label.Parent~=nil and contract._live[label.Parent]~=nil then return end if not(contract.atlas.ready and contract.atlas.image)then contract._pendingAdopt[label]=
options or true return end contract._pendingAdopt[label]=nil local resolvedOptions:any=options or{}contract._adopted[label]=true label.TextTransparency=1 label.ClipsDescendants=true local function
paint():()if label.Parent==nil then return end local alignment:any=label.TextXAlignment contract.draw(label,{{text=label.Text,color=label.TextColor3}},{Name='Glyphs',TextSize=label.TextSize,ZIndex=(
tonumber(label.ZIndex)or 1)+1,align=alignment==Enum.TextXAlignment.Right and'Right'or alignment==Enum.TextXAlignment.Center and'Center'or'Left',shadow=resolvedOptions.shadow==true,maxWidth=math.max(0,
label.AbsoluteSize.X),ellipsis=label.TextTruncate==Enum.TextTruncate.AtEnd})end paint()pcall(function():()label:GetPropertyChangedSignal('Text'):Connect(paint)label:GetPropertyChangedSignal(
'TextColor3'):Connect(paint)label:GetPropertyChangedSignal('TextSize'):Connect(paint)label:GetPropertyChangedSignal('AbsoluteSize'):Connect(paint)label:GetPropertyChangedSignal('TextXAlignment'):
Connect(paint)end)end state.bitmapText.flushPendingAdopt=function():()local contract:any=state.bitmapText if not(contract.atlas.ready and contract.atlas.image)then return end local parked:{Instance}={
}for label:any in pairs(contract._pendingAdopt)do table.insert(parked,label)end for _,label:any in ipairs(parked)do local options:any=contract._pendingAdopt[label]contract._pendingAdopt[label]=nil if
label.Parent~=nil then contract.adopt(label,type(options)=='table'and options or nil)end end end state.bitmapText.adoptTree=function(root:Instance,options:any?):any local contract:any=state.bitmapText
pcall(function():()for _,descendant:Instance in ipairs(root:GetDescendants())do contract.adopt(descendant,options)end end)local connected:boolean,connection:any=pcall(function():any return root.
DescendantAdded:Connect(function(descendant:Instance):()contract.adopt(descendant,options)end)end)if connected and connection then return connection end return{Connected=false,Disconnect=function():()
end}end state.bitmapText.loadExactAtlas=function(image:string,metrics:any):()local atlas:any=state.bitmapText.atlas if type(image)~='string'or type(metrics)~='table'then return end for _,key:string in
ipairs({'size','first','last','columns','cellWidth','cellHeight','advance'})do if tonumber(metrics[key])then atlas[key]=tonumber(metrics[key])end end if type(metrics.advances)=='table'then atlas.
advances=metrics.advances else atlas.advances=nil end atlas.image=image atlas.ready=true atlas.generation+=1 state.bitmapText.rasterFamilyEnabled=false state.bitmapText.source='minecraft-exact'for
holder:any,record:any in pairs(state.bitmapText._live)do if holder.Parent~=nil and record.request~=nil then pcall(function():()state.bitmapText.draw(record.request.parent,record.request.segments,
record.request.options)end)end end for _,listener:()->()in ipairs(state.bitmapText.onReady)do pcall(listener)end end state.bitmapText._fallback={size=16,first=32,last=126,columns=16,cellWidth=11,
cellHeight=17,advance=11,ascent=15,descent=2}state.bitmapText.rasterFamilyEnabled=true state.bitmapText.restoreFallbackAtlas=function():()local contract:any=state.bitmapText local atlas:any=contract.
atlas for key:string,value:number in pairs(contract._fallback)do atlas[key]=value end atlas.advances=nil atlas.image=contract._fallbackImage atlas.ready=contract._fallbackImage~=nil atlas.generation+=
1 contract.rasterFamilyEnabled=true contract.source=atlas.ready and'monocraft-fallback'or'unavailable'end state.bitmapText.packs={['Minecraft 1.18.1']={folder='MinecraftFont',version='1.18.1',source=
'minecraft-exact'},['Minecraft 26.2']={folder='MinecraftFont-26.2',version='26.2',source='minecraft-26.2-local'}}state.bitmapText.probePack=function(selection:string):boolean local contract:any=state.
bitmapText local pack:any=contract.packs[selection]if not pack then return false end local loaded:boolean=false pcall(function():()if type(isfile)~='function'or type(readfile)~='function'then return
end local manifestPath:string=pack.folder..'/manifest.json'if not isfile(manifestPath)then return end local manifest:any=HttpService:JSONDecode(readfile(manifestPath))if type(manifest)~='table'or
manifest.format~='wurst-minecraft-font-v1'or manifest.version~=pack.version or type(manifest.runtime)~='table'then warn('['..PRODUCT.logPrefix..':Assets] '..pack.folder..
'/ is present but not a Minecraft '..pack.version..' pack; ignoring')return end local imagePath:string=pack.folder..'/'..tostring(manifest.runtime.image or'runtime-ascii.png')if not isfile(imagePath)
then return end local registerAsset:any=fontEnvironment.getcustomasset or fontEnvironment.getsynasset if type(registerAsset)~='function'then return end local assetOk:boolean,assetId:any=pcall(
registerAsset,imagePath)if assetOk and type(assetId)=='string'and assetId~=''then contract.loadExactAtlas(assetId,manifest.runtime)contract.source=pack.source loaded=true end end)return loaded end
state.bitmapText.applyFontPack=function(selection:string):()local contract:any=state.bitmapText state.fontPack=selection contract.restoreFallbackAtlas()if selection=='Code'then contract.atlas.ready=
false contract.atlas.generation+=1 contract.source='code-vector'end if selection~='Monocraft'and selection~='Code'and contract.packs[selection]then if not contract.probePack(selection)and contract.
_packWarned~=selection then contract._packWarned=selection warn('['..PRODUCT.logPrefix..':Assets] font pack "'..selection..'" is not installed; rendering the Monocraft fallback (OFL)')end end contract
.flushPendingAdopt()for holder:any,record:any in pairs(contract._live)do if holder.Parent~=nil and record.request~=nil then pcall(function():()contract.draw(record.request.parent,record.request.
segments,record.request.options)end)end end for _,listener:()->()in ipairs(contract.onReady)do pcall(listener)end end task.spawn(function():()local contract:any=state.bitmapText if state.fontPack==
'Code'then contract.source='code-vector'return end if type(state.resolveAsset)~='function'then state.bitmapText.source='unavailable'return end local packLoaded:boolean=false if state.fontPack~=
'Monocraft'then packLoaded=contract.probePack(state.fontPack)if packLoaded then print('['..PRODUCT.logPrefix..':Assets] font pack loaded: '..tostring(state.fontPack))elseif contract._packWarned~=state
.fontPack then contract._packWarned=state.fontPack warn('['..PRODUCT.logPrefix..':Assets] font pack "'..tostring(state.fontPack)..'" is not installed; rendering the Monocraft fallback (OFL)')end end
local assetId:string?,atlasError:string?=state.resolveAsset('bitmapFontAtlas')if assetId then contract._fallbackImage=assetId if not packLoaded then contract.atlas.image=assetId contract.atlas.ready=
true contract.atlas.generation+=1 contract.source='monocraft-fallback'end for _,member:any in pairs(contract.family)do if member~=contract.atlas and member.asset then pcall(function():()local memberId
:string?=(state.resolveAsset(member.asset))if memberId then member.image=memberId member.ready=true end end)end end elseif not packLoaded then contract.source='unavailable'warn('['..PRODUCT.logPrefix
..':Assets] no bitmap atlas ('..tostring(atlasError)..'); text renders on the vector stand-in')return end contract.flushPendingAdopt()for holder:any,record:any in pairs(contract._live)do if holder.
Parent~=nil and record.request~=nil then pcall(function():()contract.draw(record.request.parent,record.request.segments,record.request.options)end)end end for _,listener:()->()in ipairs(contract.
onReady)do pcall(listener)end end)state.onLayout(function():()for holder:any,record:any in pairs(state.bitmapText._live)do if holder.Parent~=nil and record.request~=nil then pcall(function():()state.
bitmapText.draw(record.request.parent,record.request.segments,record.request.options)end)end end end)state.registeredTextIndex=setmetatable({},{__mode='k'})::any local function registerText(object:
TextLabel|TextButton|TextBox,normalSize:number):TextLabel|TextButton|TextBox local entry:any={object=object,normalSize=normalSize}table.insert(registeredText,entry)state.registeredTextIndex[object]=
entry object.TextSize=math.round(normalSize*TEXT_BASE_MULTIPLIER*state.textScale)return object end local function fitTextSize(object:any,requested:number):number local height:number=object.
AbsoluteSize.Y if height<=0 then local vertical:UDim=object.Size.Y height=vertical.Scale==0 and vertical.Offset or 0 end if height<=0 then return requested end return math.max(7,math.min(requested,
math.floor((height-2)/1.2)))end state.fitTextSize=fitTextSize state.setTextMetric=function(object:any,normalSize:number):()local entry:any=state.registeredTextIndex[object]if entry then entry.
normalSize=normalSize end object.TextSize=fitTextSize(object,math.round(normalSize*TEXT_BASE_MULTIPLIER*state.textScale))end local function makeTextLabel(parent:Instance,text:string,normalSize:number?
):TextLabel return registerText(create('TextLabel',{Parent=parent,BackgroundTransparency=1,BorderSizePixel=0,ClipsDescendants=true,FontFace=THIN_FONT,Text=text,TextColor3=Theme.text,TextTruncate=Enum.
TextTruncate.AtEnd,TextXAlignment=Enum.TextXAlignment.Left,TextYAlignment=Enum.TextYAlignment.Center}),normalSize or 14)::TextLabel end local function makeButton(parent:Instance,text:string,normalSize
:number?):TextButton local isDefaultStyle:boolean=state.guiStyle=='Default'local button=registerText(create('TextButton',{Parent=parent,AutoButtonColor=false,ClipsDescendants=true,TextTruncate=Enum.
TextTruncate.AtEnd,BackgroundColor3=isDefaultStyle and Theme.button or DEFAULT_SURFACE_COLOR,BorderSizePixel=0,FontFace=CONTROL_FONT,Text=text,TextColor3=isDefaultStyle and Theme.text or Theme.text,
TextXAlignment=Enum.TextXAlignment.Center,TextYAlignment=Enum.TextYAlignment.Center}),normalSize or 14)addRoundedStyle(button,9,0.48)button:SetAttribute('IdleColor',isDefaultStyle and Theme.button or
DEFAULT_SURFACE_COLOR)button:SetAttribute('HoverColor',isDefaultStyle and Theme.buttonHover or Theme.surfaceHover)button:SetAttribute('PressedColor',isDefaultStyle and Theme.buttonPressed or Theme.
surfacePressed)table.insert(state.styledButtons,button)trackUiConnection(button.MouseEnter:Connect(function()if button:GetAttribute('NoHover')or button:GetAttribute('Disabled')then return end local
configuredHover:any=button:GetAttribute('HoverColor')local hoverColor:Color3=typeof(configuredHover)=='Color3'and configuredHover or Theme.surfaceHover TweenService:Create(button,TweenInfo.new(
UI_MOTION.base,Enum.EasingStyle.Quad,Enum.EasingDirection.Out),{BackgroundColor3=hoverColor}):Play()end))trackUiConnection(button.MouseLeave:Connect(function()if button:GetAttribute('NoHover')then
return end local selectedColor:any=button:GetAttribute('SelectedColor')local configuredIdle:any=button:GetAttribute('IdleColor')local targetColor:Color3=button:GetAttribute('Selected')and(typeof(
selectedColor)=='Color3'and selectedColor or Theme.accentSoft)or(typeof(configuredIdle)=='Color3'and configuredIdle or DEFAULT_SURFACE_COLOR)TweenService:Create(button,TweenInfo.new(UI_MOTION.base,
Enum.EasingStyle.Quad,Enum.EasingDirection.Out),{BackgroundColor3=targetColor}):Play()end))trackUiConnection(button.MouseButton1Down:Connect(function():()if button:GetAttribute('Disabled')or button:
GetAttribute('NoPressEffect')then return end local configuredPressed:any=button:GetAttribute('PressedColor')local pressedColor:Color3=typeof(configuredPressed)=='Color3'and configuredPressed or Theme.
surfacePressed TweenService:Create(button,TweenInfo.new(0.08,Enum.EasingStyle.Quad),{BackgroundColor3=pressedColor}):Play()end))trackUiConnection(button.MouseButton1Up:Connect(function():()if button:
GetAttribute('Disabled')or button:GetAttribute('NoPressEffect')then return end local selectedColor:any=button:GetAttribute('SelectedColor')local configuredHover:any=button:GetAttribute('HoverColor')
local targetColor:Color3=button:GetAttribute('Selected')and(typeof(selectedColor)=='Color3'and selectedColor or Theme.accentSoft)or(typeof(configuredHover)=='Color3'and configuredHover or Theme.
surfaceHover)TweenService:Create(button,TweenInfo.new(UI_MOTION.fast,Enum.EasingStyle.Quad),{BackgroundColor3=targetColor}):Play()end))return button::TextButton end local function setToggleSwitch(
button:TextButton,enabled:boolean,animate:boolean?):()button:SetAttribute('Checked',enabled)local frame:Frame?=button:FindFirstChild('SwitchTrack')::Frame?local core:Frame?=frame and frame:
FindFirstChild('Knob')::Frame?if not frame or not core then return end local stroke:UIStroke?=frame:FindFirstChild('MarkStroke')::UIStroke?local targetSize:UDim2=enabled and UDim2.fromScale(0.62,0.62)
or UDim2.fromScale(0,0)local targetTransparency:number=enabled and 0.05 or 1 if animate and state.animationsEnabled then TweenService:Create(core,TweenInfo.new(UI_MOTION.base,Enum.EasingStyle.Quart),{
Size=targetSize,BackgroundTransparency=targetTransparency}):Play()else core.Size=targetSize core.BackgroundTransparency=targetTransparency end if stroke then stroke.Transparency=enabled and 0.15 or
0.45 end end local function createToggleSwitch(parent:Instance,enabled:boolean,width:number?):TextButton local markSize:number=math.clamp(width or 15,12,18)local button:TextButton=create('TextButton',
{TextTruncate=Enum.TextTruncate.AtEnd,ClipsDescendants=true,Parent=parent,Active=true,AutoButtonColor=false,BackgroundTransparency=1,BorderSizePixel=0,Size=UDim2.fromOffset(math.max(markSize+14,28),30
),Text=''})::TextButton local frame:Frame=create('Frame',{Parent=button,Name='SwitchTrack',AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.text,BackgroundTransparency=1,BorderSizePixel=0,
Position=UDim2.fromScale(0.5,0.5),Size=UDim2.fromOffset(markSize,markSize),ZIndex=button.ZIndex+1})::Frame create('UIStroke',{Parent=frame,Name='MarkStroke',Color=Theme.outline,Transparency=0.45,
Thickness=1})local _core:Frame=create('Frame',{Parent=frame,Name='Knob',AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.text,BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.
fromScale(0.5,0.5),Size=UDim2.fromScale(0,0),ZIndex=frame.ZIndex+1})::Frame setToggleSwitch(button,enabled,false)return button end local function applyKeySlotStyle(button:TextButton):()button.ZIndex=
math.max(2,button.ZIndex)button.BackgroundColor3=Theme.surface button.BackgroundTransparency=0.54 button.TextColor3=Theme.textMuted button:SetAttribute('ThemeRole','KeySlot')button:SetAttribute(
'IdleColor',Theme.surface)button:SetAttribute('HoverColor',Theme.surfaceHover)button:SetAttribute('PressedColor',Theme.buttonPressed)local buttonStroke:UIStroke?=button:FindFirstChildOfClass(
'UIStroke')if buttonStroke then buttonStroke.Color=Theme.outline buttonStroke.Transparency=0.78 end end state.applyLinesGlyph=function(button:TextButton,dimmed:boolean,active:boolean?):()button.Text=
''local existing:Instance?=button:FindFirstChild('LinesGlyph')local glyph:Frame if existing and existing:IsA('Frame')then glyph=existing::Frame else glyph=create('Frame',{Parent=button,Name=
'LinesGlyph',AnchorPoint=Vector2.new(0.5,0.5),BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.fromScale(0.5,0.5),Size=UDim2.fromOffset(12,8),ZIndex=button.ZIndex+1})::Frame for index:number=
1,3 do local _bar:Frame=create('Frame',{Parent=glyph,Name='Line'..tostring(index),AnchorPoint=Vector2.new(0.5,0.5),BackgroundColor3=Theme.text,BorderSizePixel=0,Position=UDim2.new(0.5,0,0,(index-1)*
3.5+0.5),Size=UDim2.fromOffset(index==2 and 9 or 12,1),ZIndex=button.ZIndex+2})::Frame end end local barTransparency:number=dimmed and 0.74 or(active and 0.02 or 0.3)for _,bar:Instance in ipairs(glyph
:GetChildren())do if bar:IsA('Frame')then bar.BackgroundTransparency=barTransparency end end end state.restyleButtons=function(isDefault:boolean):()for _,button:TextButton in ipairs(state.
styledButtons)do if button.Parent then local role:any=button:GetAttribute('ThemeRole')if role~='Navigation'and role~='Icon'then local idleColor:Color3=isDefault and Theme.button or
DEFAULT_SURFACE_COLOR button:SetAttribute('IdleColor',idleColor)button:SetAttribute('HoverColor',isDefault and Theme.buttonHover or Theme.surfaceHover)button:SetAttribute('PressedColor',isDefault and
Theme.buttonPressed or Theme.surfacePressed)if button:GetAttribute('Selected')~=true then button.BackgroundColor3=idleColor end button.TextColor3=isDefault and Theme.text or Theme.text local stroke:
UIStroke?=button:FindFirstChild('StyleStroke')::UIStroke?if stroke then stroke.Enabled=true stroke.Color=isDefault and Theme.outline or Theme.outline stroke.Transparency=isDefault and 0.32 or 0.48 end
if role=='KeySlot'then button.BackgroundColor3=Theme.surface button.BackgroundTransparency=0.54 button.TextColor3=Theme.textMuted local keyStroke:UIStroke?=button:FindFirstChild('StyleStroke')::
UIStroke?if keyStroke then keyStroke.Color=Theme.outline keyStroke.Transparency=0.78 end end end end end local KEY_DISPLAY_NAMES:{[string]:string}={RightShift='RShift',LeftShift='LShift',
RightControl='RCtrl',LeftControl='LCtrl',RightAlt='RAlt',LeftAlt='LAlt',Backspace='Back',CapsLock='Caps',Return='Enter',Escape='Esc',PageUp='PgUp',PageDown='PgDn',Insert='Ins',Delete='Del',RightMeta=
'RMeta',LeftMeta='LMeta'}state.keyDisplayName=function(name:any):string local resolved:string=tostring(name or'')return KEY_DISPLAY_NAMES[resolved]or resolved end local function setKeySlotCapture(
button:TextButton,active:boolean):()local stroke:UIStroke?=button:FindFirstChild('StyleStroke')::UIStroke?TweenService:Create(button,TweenInfo.new(UI_MOTION.fast,Enum.EasingStyle.Quad,Enum.
EasingDirection.Out),{BackgroundColor3=active and Theme.buttonPressed or Theme.surface,TextColor3=active and Theme.text or Theme.textMuted}):Play()if stroke then TweenService:Create(stroke,TweenInfo.
new(UI_MOTION.fast,Enum.EasingStyle.Quad,Enum.EasingDirection.Out),{Transparency=active and 0.38 or 0.78}):Play()end end local function beginKeyCapture(button:TextButton,callback:(Enum.KeyCode)->()):(
)if state.waitingForKey then state.waitingForKey=false if state.keybindButton then state.keybindButton.Text=state.keyDisplayName(state.toggleKey.Name)setKeySlotCapture(state.keybindButton,false)end
end if state.keyCaptureCancel then state.keyCaptureCancel()end button.Text='...'setKeySlotCapture(button,true)local cancelled:boolean=false state.keyCaptureCancel=function():()if cancelled then return
end cancelled=true state.keyCaptureCallback=nil state.keyCaptureCancel=nil setKeySlotCapture(button,false)if button.Parent and button:GetAttribute('CaptureIdleText')then button.Text=tostring(button:
GetAttribute('CaptureIdleText'))end end state.keyCaptureCallback=function(keyCode:Enum.KeyCode):()if cancelled then return end cancelled=true state.keyCaptureCancel=nil setKeySlotCapture(button,false)
callback(keyCode)end end local function makeTextBox(parent:Instance,defaultText:any):TextBox local isDefaultStyle:boolean=state.guiStyle=='Default'local textBox:TextBox=(registerText(create('TextBox',
{Parent=parent,BackgroundColor3=isDefaultStyle and Theme.buttonPressed or Theme.surface,BorderSizePixel=0,ClearTextOnFocus=false,ClipsDescendants=true,TextTruncate=Enum.TextTruncate.AtEnd,FontFace=
THIN_FONT,PlaceholderColor3=isDefaultStyle and Theme.textMuted or Theme.textMuted,Text=tostring(defaultText),TextColor3=isDefaultStyle and Theme.text or Theme.text,TextXAlignment=Enum.TextXAlignment.
Center}),13)::any)::TextBox addRoundedStyle(textBox,9,0.5)table.insert(state.styledTextBoxes,textBox)trackUiConnection(textBox.Focused:Connect(function():()TweenService:Create(textBox,TweenInfo.new(
UI_MOTION.base,Enum.EasingStyle.Quad),{BackgroundColor3=state.guiStyle=='Default'and Theme.buttonHover or Theme.surfaceRaised}):Play()end))trackUiConnection(textBox.FocusLost:Connect(function():()
TweenService:Create(textBox,TweenInfo.new(UI_MOTION.base,Enum.EasingStyle.Quad),{BackgroundColor3=state.guiStyle=='Default'and Theme.buttonPressed or Theme.surface}):Play()end))return textBox end
state.restyleTextBoxes=function(isDefault:boolean):()for _,textBox:TextBox in ipairs(state.styledTextBoxes)do if textBox.Parent then textBox.BackgroundColor3=isDefault and Theme.buttonPressed or Theme
.surface textBox.TextColor3=isDefault and Theme.text or Theme.text textBox.PlaceholderColor3=isDefault and Theme.textMuted or Theme.textMuted local stroke:UIStroke?=textBox:FindFirstChild(
'StyleStroke')::UIStroke?if stroke then stroke.Color=isDefault and Theme.outline or Theme.outline stroke.Transparency=isDefault and 0.3 or 0.5 end end end end local ToastContainer:Frame=(create(
'Frame',{Parent=ScreenGui,Name='ToastContainer',AnchorPoint=Vector2.new(1,1),BackgroundTransparency=1,BorderSizePixel=0,Position=UDim2.new(1,-12,1,-12),Size=UDim2.fromOffset(state.layout.compact and
220 or 300,300),ZIndex=300})::any)::Frame state.onLayout(function(metrics:LayoutMetrics):()ToastContainer.Size=UDim2.fromOffset(metrics.compact and 220 or 300,300)end)create('UIListLayout',{Parent=
ToastContainer,FillDirection=Enum.FillDirection.Vertical,HorizontalAlignment=Enum.HorizontalAlignment.Right,Padding=UDim.new(0,8),SortOrder=Enum.SortOrder.LayoutOrder,VerticalAlignment=Enum.
VerticalAlignment.Bottom})local toastSequence:number=0 local lastToastText:string=''local lastToastAt:number=-math.huge local TOAST_STACK_LIMIT:number=5 local activeToasts:{any}={}local function
toastAccent(kind:string?):Color3 if kind=='danger'then return Theme.negative end if kind=='good'then return Theme.enabled end return Theme.accent end local function toastLifetime(text:string):number
return math.clamp(2.6+#text*0.022,2.6,7)end local function retireToast(toast:any):()for index:number=#activeToasts,1,-1 do if activeToasts[index]==toast then table.remove(activeToasts,index)end end
end local function notify(message:any,kind:string?):()local text:string=tostring(message)local now=os.clock()if text==lastToastText and now-lastToastAt<3 then return end lastToastText,lastToastAt=text
,now print('['..PRODUCT.name..'] '..text)toastSequence+=1 local toast:CanvasGroup=Instance.new('CanvasGroup')toast.Name='Toast'toast.AutomaticSize=Enum.AutomaticSize.Y toast.BackgroundColor3=state.
guiStyle=='Default'and Theme.surface or Theme.surfaceRaised toast.BackgroundTransparency=state.guiStyle=='Default'and 0.12 or 0.04 toast.BorderSizePixel=0 toast.GroupTransparency=1 toast.LayoutOrder=
toastSequence toast.Size=UDim2.new(1,0,0,state.layout.compact and 46 or 50)toast.ZIndex=301 toast.Parent=ToastContainer create('UICorner',{Parent=toast,CornerRadius=cornerRadius(UI_RADIUS.panel)})
create('UIStroke',{Parent=toast,Color=state.guiStyle=='Default'and Theme.outline or Theme.outline,Transparency=state.guiStyle=='Default'and 0.42 or 0.22,Thickness=1})local _accent:Frame=(create(
'Frame',{Parent=toast,BackgroundColor3=toastAccent(kind),BorderSizePixel=0,Size=UDim2.new(0,4,1,0),ZIndex=302})::any)::Frame local label:TextLabel=(makeTextLabel(toast,text,14)::any)::TextLabel label.
AutomaticSize=Enum.AutomaticSize.Y label.Position=UDim2.fromOffset(16,10)label.Size=UDim2.new(1,-28,0,36)label.TextColor3=state.guiStyle=='Default'and Theme.text or Theme.text label.TextWrapped=true
label.ZIndex=303 TweenService:Create(toast,TweenInfo.new(0.22,Enum.EasingStyle.Quart,Enum.EasingDirection.Out),{GroupTransparency=0}):Play()table.insert(activeToasts,toast)while#activeToasts>
TOAST_STACK_LIMIT do local oldest:any=table.remove(activeToasts,1)if oldest and oldest~=toast and oldest.Parent then pcall(function():()oldest:Destroy()end)end end task.delay(toastLifetime(text),
function():()retireToast(toast)if not toast.Parent then return end local tween:Tween=TweenService:Create(toast,TweenInfo.new(0.2,Enum.EasingStyle.Quad,Enum.EasingDirection.In),{GroupTransparency=1})
tween:Play()tween.Completed:Once(function():()toast:Destroy()end)end)end local function confirmDialog(options:any):Frame?if type(options)~='table'then return nil end local title:string=tostring(
options.title or'Question')local message:string=tostring(options.message or'')local yesText:string=tostring(options.yes or'Yes')local noText:string=tostring(options.no or'No')local onYes:any=options.
onYes local onNo:any=options.onNo local rememberLabel:any=options.rememberLabel local remember:boolean=options.rememberDefault==true local previous:Instance?=ScreenGui:FindFirstChild('ConfirmDialog')
while previous do previous:Destroy()previous=ScreenGui:FindFirstChild('ConfirmDialog')end local backdrop:TextButton=(create('TextButton',{Parent=ScreenGui,Name='ConfirmDialog',Text='',AutoButtonColor=
false,Active=true,BackgroundColor3=Color3.new(0,0,0),BackgroundTransparency=0.4,BorderSizePixel=0,Size=UDim2.fromScale(1,1),ZIndex=400})::any)::TextButton local panel:Frame=(create('Frame',{Parent=
backdrop,Name='Panel',Active=true,AnchorPoint=Vector2.new(0.5,0.5),Position=UDim2.fromScale(0.5,0.5),Size=UDim2.new(1,-32,0,0),AutomaticSize=Enum.AutomaticSize.Y,BackgroundColor3=Theme.surface,
BorderSizePixel=0,ZIndex=401})::any)::Frame create('UISizeConstraint',{Parent=panel,MaxSize=Vector2.new(340,10000)})create('UICorner',{Parent=panel,CornerRadius=cornerRadius(UI_RADIUS.panel)})create(
'UIStroke',{Parent=panel,Color=Theme.outline,Thickness=1})create('UIListLayout',{Parent=panel,FillDirection=Enum.FillDirection.Vertical,SortOrder=Enum.SortOrder.LayoutOrder,Padding=UDim.new(0,6)})
create('UIPadding',{Parent=panel,PaddingTop=UDim.new(0,14),PaddingBottom=UDim.new(0,14),PaddingLeft=UDim.new(0,16),PaddingRight=UDim.new(0,16)})local titleLabel:TextLabel=(create('TextLabel',{Parent=
panel,Name='Title',LayoutOrder=1,BackgroundTransparency=1,FontFace=CONTROL_FONT,Text=title,TextColor3=Theme.text,TextSize=15,TextWrapped=true,TextXAlignment=Enum.TextXAlignment.Left,Size=UDim2.new(1,0
,0,20),ZIndex=402})::any)::TextLabel local messageLabel:TextLabel=(create('TextLabel',{Parent=panel,Name='Message',LayoutOrder=2,AutomaticSize=Enum.AutomaticSize.Y,BackgroundTransparency=1,FontFace=
CONTROL_FONT,Text=message,TextColor3=Theme.textMuted,TextSize=13,TextWrapped=true,TextXAlignment=Enum.TextXAlignment.Left,TextYAlignment=Enum.TextYAlignment.Top,Size=UDim2.new(1,0,0,0),ZIndex=402})::
any)::TextLabel if type(rememberLabel)=='string'and rememberLabel~=''then local row:TextButton=(create('TextButton',{Parent=panel,Name='RememberChoice',LayoutOrder=3,Size=UDim2.new(1,0,0,26),
BackgroundTransparency=1,AutoButtonColor=false,Text='',ZIndex=402})::any)::TextButton local box:Frame=(create('Frame',{Parent=row,Name='Box',AnchorPoint=Vector2.new(0,0.5),Position=UDim2.new(0,0,0.5,0
),Size=UDim2.fromOffset(16,16),BackgroundColor3=Theme.button,BorderSizePixel=0,ZIndex=403})::any)::Frame create('UICorner',{Parent=box,CornerRadius=cornerRadius(UI_RADIUS.control)})create('UIStroke',{
Parent=box,Color=Theme.outline,Thickness=1})local mark:TextLabel=(create('TextLabel',{Parent=box,Name='Check',BackgroundTransparency=1,Size=UDim2.fromScale(1,1),FontFace=CONTROL_FONT,Text='\u{2713}',
TextColor3=Theme.accentText,TextSize=14,Visible=remember,ZIndex=404})::any)::TextLabel create('TextLabel',{Parent=row,Name='Label',BackgroundTransparency=1,Position=UDim2.fromOffset(24,0),Size=UDim2.
new(1,-24,1,0),FontFace=CONTROL_FONT,Text=rememberLabel,TextColor3=Theme.text,TextSize=13,TextXAlignment=Enum.TextXAlignment.Left,ZIndex=403})local function refreshRemember():()mark.Visible=remember
box.BackgroundColor3=remember and Theme.accent or Theme.button end refreshRemember()row.MouseButton1Click:Connect(function():()remember=not remember refreshRemember()end)end local actions:Frame=(
create('Frame',{Parent=panel,Name='Actions',LayoutOrder=4,BackgroundTransparency=1,Size=UDim2.new(1,0,0,30),ZIndex=402})::any)::Frame create('UIListLayout',{Parent=actions,FillDirection=Enum.
FillDirection.Horizontal,HorizontalAlignment=Enum.HorizontalAlignment.Right,VerticalAlignment=Enum.VerticalAlignment.Center,SortOrder=Enum.SortOrder.LayoutOrder,Padding=UDim.new(0,8)})local answered:
boolean=false local function close(answer:boolean?):()if answered then return end answered=true if backdrop.Parent then backdrop:Destroy()end local callback:any=answer==true and onYes or onNo if type(
callback)=='function'then local ok:boolean,failure:any=pcall(callback,remember)if not ok then warn('['..PRODUCT.name..'] confirm dialog: '..tostring(failure))end end end local function answerButton(
text:string,order:number,primary:boolean):TextButton local button:TextButton=(create('TextButton',{Parent=actions,Name=text,LayoutOrder=order,AutoButtonColor=false,BackgroundColor3=primary and Theme.
accent or Theme.button,BorderSizePixel=0,Size=UDim2.fromOffset(primary and 92 or 78,28),Text=text,FontFace=CONTROL_FONT,TextColor3=primary and Theme.accentText or Theme.text,TextSize=13,ZIndex=403})::
any)::TextButton create('UICorner',{Parent=button,CornerRadius=cornerRadius(UI_RADIUS.control)})button.MouseButton1Click:Connect(function():()close(primary)end)return button end answerButton(noText,1,
false)answerButton(yesText,2,true)return panel end state.confirmDialog=confirmDialog local CardBin:Frame=(create('Frame',{Parent=ScreenGui,Name='CardBin',BackgroundTransparency=1,BorderSizePixel=0,
Size=UDim2.fromScale(1,1),Visible=false})::any)::Frame state.universalScroll=CardBin state.trsScroll=CardBin state.vdScroll=CardBin state.mvsdScroll=CardBin state.bedFightScroll=CardBin state.
spoofScroll=CardBin print('['..PRODUCT.logPrefix..':Bootstrap] Base GUI visible; building universal features')local universalFeatures={}local movementFeatures={}state.movementFeatures=movementFeatures
local mm2Features={}state.trsFeatures={}state.spoofFeatures={}state.vdFeatures={}state.bedFightFeatures={}state.bedWarsFeatures={}state.mvsdFeatures={}local allFeatures:{any}={}local
featureConnections={}state.invokeModuleSearch=function(query:string):boolean local search:any=state.moduleSearch if type(search)~='table'or type(search.Execute)~='function'then return false end local
result:any=search.Execute(query)return type(result)=='table'and result.status=='success'end state.onLayout(function():()for _,feature:any in ipairs(allFeatures)do if type(feature.applyLayout)==
'function'then local ok:boolean,failure:any=pcall(feature.applyLayout)if not ok then warn('['..PRODUCT.name..'] card layout: '..tostring(failure))end end end end)state.tooltipConnections={}::{
RBXScriptConnection}local MobileMenuButton=create('TextButton',{TextTruncate=Enum.TextTruncate.AtEnd,ClipsDescendants=true,Parent=ScreenGui,Name='MobileMenuButton',AutoButtonColor=false,
BackgroundColor3=Theme.surface,BackgroundTransparency=0.12,BorderSizePixel=0,FontFace=CONTROL_FONT,Position=UDim2.new(1,-18,0,18),AnchorPoint=Vector2.new(1,0),Size=UDim2.fromOffset(52,52),Text=
'\u{2261}',TextColor3=Theme.text,TextSize=20,Visible=false,ZIndex=190})::TextButton create('UICorner',{Parent=MobileMenuButton,CornerRadius=UDim.new(1,0)})create('UIStroke',{Parent=MobileMenuButton,
Color=Theme.outline,Transparency=0.55,Thickness=1})do local dragInput:InputObject?=nil local dragStart:Vector2=Vector2.zero local startPosition:UDim2=MobileMenuButton.Position local travelled:number=0
local storedX:number?=tonumber(configData.ui.launcherX)local storedY:number?=tonumber(configData.ui.launcherY)if storedX and storedY then MobileMenuButton.Position=UDim2.fromOffset(storedX,storedY)
MobileMenuButton.AnchorPoint=Vector2.new(0,0)end local launcherScale:UIScale=create('UIScale',{Parent=MobileMenuButton,Name='LauncherScale',Scale=1})::UIScale local function pressLauncher(pressed:
boolean):()if not state.animationsEnabled then launcherScale.Scale=1 return end TweenService:Create(launcherScale,TweenInfo.new(pressed and UI_MOTION.fast or 0.34,pressed and Enum.EasingStyle.Quart or
Enum.EasingStyle.Back,Enum.EasingDirection.Out),{Scale=pressed and 0.9 or 1}):Play()end trackUiConnection(MobileMenuButton.InputBegan:Connect(function(input:InputObject):()if input.UserInputType~=Enum
.UserInputType.MouseButton1 and input.UserInputType~=Enum.UserInputType.Touch then return end dragInput=input dragStart=Vector2.new(input.Position.X,input.Position.Y)startPosition=MobileMenuButton.
Position travelled=0 pressLauncher(true)end))trackUiConnection(UserInputService.InputChanged:Connect(function(input:InputObject):()if not dragInput then return end if input.UserInputType~=Enum.
UserInputType.MouseMovement and input.UserInputType~=Enum.UserInputType.Touch then return end local pointer:Vector2=Vector2.new(input.Position.X,input.Position.Y)local delta:Vector2=pointer-dragStart
travelled=math.max(travelled,delta.Magnitude)if travelled<6 then return end MobileMenuButton.AnchorPoint=Vector2.new(0,0)local viewport:Vector2=ScreenGui.AbsoluteSize local size:Vector2=
MobileMenuButton.AbsoluteSize local baseX:number=startPosition.X.Scale*viewport.X+startPosition.X.Offset local baseY:number=startPosition.Y.Scale*viewport.Y+startPosition.Y.Offset MobileMenuButton.
Position=UDim2.fromOffset(math.clamp(math.round(baseX+delta.X),0,math.max(viewport.X-size.X,0)),math.clamp(math.round(baseY+delta.Y),0,math.max(viewport.Y-size.Y,0)))end))trackUiConnection(
UserInputService.InputEnded:Connect(function(input:InputObject):()if input~=dragInput then return end dragInput=nil pressLauncher(false)if travelled>=6 then local viewport:Vector2=ScreenGui.
AbsoluteSize local size:Vector2=MobileMenuButton.AbsoluteSize local currentX:number=MobileMenuButton.Position.X.Offset local currentY:number=MobileMenuButton.Position.Y.Offset local margin:number=14
local nearestX:number=currentX+size.X*0.5<viewport.X*0.5 and margin or math.max(margin,viewport.X-size.X-margin)local restingY:number=math.clamp(currentY,margin,math.max(margin,viewport.Y-size.Y-
margin))local resting:UDim2=UDim2.fromOffset(math.round(nearestX),math.round(restingY))if state.animationsEnabled then TweenService:Create(MobileMenuButton,TweenInfo.new(0.42,Enum.EasingStyle.Back,
Enum.EasingDirection.Out),{Position=resting}):Play()else MobileMenuButton.Position=resting end configData.ui.launcherX=resting.X.Offset configData.ui.launcherY=resting.Y.Offset queueConfigSave()end
end))trackUiConnection(MobileMenuButton.MouseButton1Click:Connect(function():()if travelled>=6 then return end state.openMenuSurface()end))end local function refreshLauncherVisibility():()
MobileMenuButton.Visible=state.menuReady==true and(state.isMobile or state.toggleKey==Enum.KeyCode.Unknown)and state.pill==nil and not state.visible end state.refreshLauncher=refreshLauncherVisibility
refreshLauncherVisibility()state.addMenuVisibilityListener(function():()refreshLauncherVisibility()end)type TaskHandle={Connected:boolean,Disconnect:(self:TaskHandle)->(),label:string}type
ScheduledTask={callback:(...any)->(),handle:TaskHandle,failures:number,label:string,costMs:number,calls:number,warned:boolean}local TASK_WARN_MS:number=3 local PERFORMANCE_EVERY:number=0.5 local
TaskManager:any={tasks={}::{ScheduledTask},renderTasks={}::{ScheduledTask},taskCount=0,renderCount=0,nextId=0,connection=nil::RBXScriptConnection?,renderConnection=nil::RBXScriptConnection?,tasksMs=0,
renderMs=0}state.performance={fps=60,frameMs=1000/60,tasksMs=0,renderMs=0,taskCount=0,renderCount=0,slowestTask=nil::(string|nil),slowestMs=0,spikeMs=0,spikeCount=0,spikeAgo=-1,spikeAt=nil::(number|
nil),spikeWarnedAt=nil::(number|nil),worstSpikeMs=0,peakTask=nil::(string|nil),peakTaskMs=0,peakTaskPhase=nil::(string|nil),peakTaskAt=nil::(number|nil),observedSince=os.clock()}local function
```

```
describeTask(callback:any):string for _,kind:string in ipairs({'n','s'})do local ok:boolean,info:any=pcall(debug.info,callback,kind)if ok and type(info)=='string'and info~=''then local name:string=
tostring(info)if not string.find(name,'/',1,true)and not string.find(name,'@',1,true)and string.find(name,'.rs:',1,true)==nil then if#name>64 then name=string.sub(name,1,64)end return name end end end
return'task'end local function removeFromBucket(bucket:{ScheduledTask},index:number):()local last:number=#bucket if index~=last then bucket[index]=bucket[last]bucket[last].handle.index=index end
bucket[last]=nil end function TaskManager:Connect(callback:(...any)->(),phase:string?,label:string?):TaskHandle self.nextId+=1 local bucket:{ScheduledTask}=phase=='render'and self.renderTasks or self.
tasks local handle:TaskHandle handle={Connected=true,label=label or describeTask(callback),index=0,Disconnect=function(selfHandle:any):()if not selfHandle.Connected then return end selfHandle.
Connected=false local current:{ScheduledTask}=selfHandle.bucket local index:number=selfHandle.index if current and index>=1 and index<=#current and current[index]~=nil and current[index].handle==
selfHandle then removeFromBucket(current,index)end end}::any handle.bucket=bucket handle.index=#bucket+1 bucket[handle.index]={callback=callback,handle=handle,failures=0,label=handle.label,costMs=0,
calls=0,warned=false}if phase=='render'then self.renderCount=#self.renderTasks else self.taskCount=#self.tasks end return handle end function TaskManager:Stats():any local slowest:any=nil local list:{
any}={}for _,bucket:{ScheduledTask}in ipairs({self.tasks,self.renderTasks})do for _,scheduled:ScheduledTask in ipairs(bucket)do table.insert(list,{label=scheduled.label,ms=scheduled.costMs,calls=
scheduled.calls})if not slowest or scheduled.costMs>slowest.ms then slowest={label=scheduled.label,ms=scheduled.costMs}end end end table.sort(list,function(left:any,right:any):boolean return left.ms>
right.ms end)return{taskCount=#self.tasks,renderCount=#self.renderTasks,tasksMs=self.tasksMs,renderMs=self.renderMs,slowest=slowest,top=list}end function TaskManager:Destroy():()for _,bucket:{
ScheduledTask}in ipairs({self.tasks,self.renderTasks})do for _,scheduled:ScheduledTask in ipairs(bucket)do scheduled.handle.Connected=false end table.clear(bucket)end self.taskCount=0 self.renderCount
=0 if self.connection then self.connection:Disconnect()self.connection=nil end if self.renderConnection then self.renderConnection:Disconnect()self.renderConnection=nil end end local function
runBucket(bucket:{ScheduledTask},deltaTime:number,phase:string):number local total:number=0 local index:number=1 while index<=#bucket do local scheduled:ScheduledTask=bucket[index]local startedAt:
number=os.clock()local success:boolean,message:any=pcall(scheduled.callback,deltaTime)local finishedAt:number=os.clock()local cost:number=math.max(0,(finishedAt-startedAt)*1000)local performance:any=
state.performance if cost>(performance.peakTaskMs or 0)then performance.peakTaskMs=cost performance.peakTask=scheduled.label performance.peakTaskPhase=phase performance.peakTaskAt=finishedAt end total
+=cost scheduled.calls+=1 if cost>scheduled.costMs then scheduled.costMs=cost else scheduled.costMs+=(cost-scheduled.costMs)*0.05 end if not success then scheduled.failures+=1 warn('['..PRODUCT.name..
'] scheduled task '..scheduled.label..' failed: '..tostring(message))if scheduled.failures>=3 then scheduled.handle.Connected=false end else scheduled.failures=0 end if not scheduled.warned and
scheduled.costMs>=TASK_WARN_MS then scheduled.warned=true warn('['..PRODUCT.name..':Perf] '..scheduled.label..string.format(' recent callback score %.2f ms',scheduled.costMs)..
[[ (decaying peak; elapsed may include waits). Compare with the module off.]])end if not scheduled.handle.Connected then if bucket[index]==scheduled then removeFromBucket(bucket,index)end else index+=
1 end end return total end local function publishPerformance(deltaTime:number):()local performance:any=state.performance local frameMs:number=math.max(deltaTime,1/1000)*1000 performance.frameMs+=(
frameMs-performance.frameMs)*0.1 performance.fps=1000/math.max(performance.frameMs,1/1000)local stats:any=TaskManager:Stats()performance.taskCount=stats.taskCount performance.renderCount=stats.
renderCount performance.tasksMs=stats.tasksMs performance.renderMs=stats.renderMs performance.slowestTask=stats.slowest and stats.slowest.label or nil performance.slowestMs=stats.slowest and stats.
slowest.ms or 0 performance.spikeAgo=if performance.spikeAt then math.max(0,os.clock()-performance.spikeAt)else-1 performance.updatedAt=os.clock()end local performanceElapsed:number=0 local SPIKE_MS:
number=45 local SPIKE_WARN_EVERY:number=5 TaskManager.connection=RunService.Heartbeat:Connect(function(deltaTime:number):()local total:number=runBucket(TaskManager.tasks,deltaTime,'heartbeat')
TaskManager.tasksMs=total performanceElapsed+=deltaTime if performanceElapsed>=PERFORMANCE_EVERY then performanceElapsed=0 publishPerformance(deltaTime)end local frameMs:number=math.max(0,deltaTime)*
1000 local performance:any=state.performance if frameMs>=SPIKE_MS then performance.spikeCount+=1 performance.spikeMs=frameMs performance.spikeAt=os.clock()if frameMs>performance.worstSpikeMs then
performance.worstSpikeMs=frameMs end local now:number=os.clock()if now-(performance.spikeWarnedAt or-math.huge)>=SPIKE_WARN_EVERY then performance.spikeWarnedAt=now warn(string.format(
'[%s:Perf] frame interval %.0f ms (cause not determined)',PRODUCT.name,frameMs))end end end)TaskManager.renderConnection=RunService.RenderStepped:Connect(function(deltaTime:number):()local total:
number=runBucket(TaskManager.renderTasks,deltaTime,'render')TaskManager.renderMs=total end)state.taskManager=TaskManager local function disconnectFeatureConnection(name)local connection=
featureConnections[name]if connection then connection:Disconnect()featureConnections[name]=nil end end local function getCharacterParts()local character=LocalPlayer.Character if not character then
return nil,nil,nil end return character,character:FindFirstChildOfClass('Humanoid'),character:FindFirstChild('HumanoidRootPart')end do local cachedControls:any=nil local controlsFetchedAt:number=-math
.huge local function getControls():any if cachedControls~=nil or os.clock()-controlsFetchedAt<2 then return cachedControls end controlsFetchedAt=os.clock()local ok:boolean,controls:any=pcall(function(
):any local scripts:Instance?=LocalPlayer:FindFirstChildOfClass('PlayerScripts')local playerModule:Instance?=scripts and scripts:FindFirstChild('PlayerModule')if not playerModule then return nil end
return(state.requireModule(playerModule)::any):GetControls()end)if ok and type(controls)=='table'then cachedControls=controls end return cachedControls end state.getMoveInput=function():(number,number
)local controls:any=getControls()if controls then local ok:boolean,move:any=pcall(controls.GetMoveVector,controls)if ok and typeof(move)=='Vector3'and move.Magnitude>0.001 then return-move.Z,move.X
end end local _,humanoid:Humanoid?=getCharacterParts()if humanoid and humanoid.MoveDirection.Magnitude>0.001 then local camera:Camera?=workspace.CurrentCamera if camera then local forward:Vector3=
camera.CFrame.LookVector local right:Vector3=camera.CFrame.RightVector local flatForward:Vector3=Vector3.new(forward.X,0,forward.Z)local flatRight:Vector3=Vector3.new(right.X,0,right.Z)if flatForward.
Magnitude>0.001 and flatRight.Magnitude>0.001 then local direction:Vector3=humanoid.MoveDirection return direction:Dot(flatForward.Unit),direction:Dot(flatRight.Unit)end end end local forwardScalar:
number=0 local rightScalar:number=0 if UserInputService.KeyboardEnabled then if UserInputService:IsKeyDown(Enum.KeyCode.W)then forwardScalar+=1 end if UserInputService:IsKeyDown(Enum.KeyCode.S)then
forwardScalar-=1 end if UserInputService:IsKeyDown(Enum.KeyCode.D)then rightScalar+=1 end if UserInputService:IsKeyDown(Enum.KeyCode.A)then rightScalar-=1 end end return forwardScalar,rightScalar end
local lastJumpRequestAt:number=-math.huge trackUiConnection(UserInputService.JumpRequest:Connect(function():()lastJumpRequestAt=os.clock()end))state.onJumpRequest=function(callback:()->()):
RBXScriptConnection return UserInputService.JumpRequest:Connect(function():()if state.keyCaptureCallback or state.waitingForKey or UserInputService:GetFocusedTextBox()then return end callback()end)end
state.isJumpHeld=function():boolean if UserInputService:GetFocusedTextBox()then return false end if UserInputService.KeyboardEnabled and UserInputService:IsKeyDown(Enum.KeyCode.Space)then return true
end if UserInputService.GamepadEnabled and UserInputService:IsKeyDown(Enum.KeyCode.ButtonA)then return true end local _,humanoid:Humanoid?=getCharacterParts()if humanoid and humanoid.Jump then return
true end return os.clock()-lastJumpRequestAt<0.2 end state.getAimRay=function():Ray?local camera:Camera?=workspace.CurrentCamera if not camera then return nil end local pointer:Vector2 if
UserInputService.MouseEnabled then pointer=UserInputService:GetMouseLocation()else pointer=camera.ViewportSize/2 end return camera:ViewportPointToRay(pointer.X,pointer.Y)end end local function
getFeatureCardParent(_container:Instance):Instance return CardBin end local function applyFeaturePresentation(feature:any,isDefault:boolean):()if feature.kind=='hold'then feature.enabled=feature.
active==true end if state.applyCardSkin then state.applyCardSkin(feature,false)end if feature.options then feature.options.BackgroundTransparency=1 end if feature.shortcutButton then feature.
shortcutButton.BackgroundColor3=isDefault and Theme.surface or Theme.surfaceRaised local inertSlot:boolean=feature.kind=='group'feature.shortcutButton.BackgroundTransparency=inertSlot and 0.86 or(
isDefault and 0.54 or 1)feature.shortcutButton.TextTransparency=inertSlot and 1 or 0 feature.shortcutButton.TextColor3=isDefault and Theme.text or Theme.text end end state.restyleAllFeatures=function(
isDefault:boolean):()for _,feature:any in ipairs(allFeatures)do applyFeaturePresentation(feature,isDefault)end end state.applyCardSkin=function(feature:any,hovered:boolean?):()local row:Frame?=feature
.row if not row then return end local resolvedRow:Frame=row::Frame local isHovered:boolean=hovered==true local isOn:boolean=feature.enabled==true local surface:number local titleTransparency:number=0
local titleColor:Color3=Theme.text local fill:Color3=Theme.surface if feature.isCategory then surface=isHovered and 0.3 or 0.5 titleColor=Theme.text titleTransparency=0 elseif feature.isAction then
surface=isHovered and 0.3 or 0.5 elseif feature.isHoldAction then fill=isOn and Theme.enabled or Theme.surface surface=isOn and 0 or(isHovered and 0.3 or 0.5)titleColor=isOn and Theme.enabledText or
Theme.text else fill=isOn and Theme.enabled or Theme.surface surface=isOn and 0 or(isHovered and 0.3 or 0.5)titleColor=isOn and Theme.enabledText or Theme.text end resolvedRow.BackgroundColor3=fill
resolvedRow.BackgroundTransparency=surface if feature.drawTitle then if feature.lastTitleColor~=titleColor or feature.lastTitleTransparency~=titleTransparency then feature.lastTitleColor=titleColor
feature.lastTitleTransparency=titleTransparency feature.drawTitle(titleColor,titleTransparency)end elseif feature.title then local titleLabel:TextLabel=feature.title::TextLabel titleLabel.TextColor3=
titleColor titleLabel.TextTransparency=titleTransparency end end ThemeEngine.OnChange(function(_name,changed):()for _,feature:any in ipairs(allFeatures)do if feature.row then pcall(state.applyCardSkin
,feature,false)end end if type(state.optionThemeRefresh)=='function'then pcall(state.optionThemeRefresh,changed)end end)local function refreshFeatureToggle(feature:any):()if feature.isHoldAction then
feature.enabled=feature.active==true end state.applyCardSkin(feature,false)if type(feature.syncBindSquare)=='function'then feature.syncBindSquare()end if state.floatingUi then state.floatingUi.
refreshFeature(feature)end if state.navigator and state.navigator.refreshFeature then state.navigator.refreshFeature(feature)end end local function runActionButton(button:TextButton,idleText:string,
callback:()->(),successMessage:string,errorPrefix:string):()if button:GetAttribute('Running')==true then return end button:SetAttribute('Running',true)button:SetAttribute('Disabled',true)button.Active
=false button.Text='Running\u{2026}'button.BackgroundColor3=state.guiStyle=='Default'and Theme.buttonHover or Theme.accentSoft task.spawn(function():()local running:boolean=true task.spawn(function():
()local frames:{string}={'Running\183','Running\183\183','Running\183\183\183'}local frameIndex:number=1 while running and button.Parent do button.Text=frames[frameIndex]frameIndex=frameIndex%#frames+
1 task.wait(0.22)end end)local success:boolean,errorMessage:any=pcall(callback)running=false if button.Parent then button.Text=success and'Done'or'Failed'button.BackgroundColor3=success and Theme.
accentSoft or Theme.negative end if success then notify(successMessage)else warn(errorPrefix..': '..tostring(errorMessage))notify('Error in '..errorPrefix)end task.wait(0.45)if button.Parent then
button.Text=idleText button.Active=true button:SetAttribute('Disabled',false)button:SetAttribute('Running',false)button.BackgroundColor3=idleText=='RUN'and Theme.accentSoft or Theme.surface end end)
end local function compactShortcutLabel(label:string):string local trimmed:string=string.upper((string.gsub(string.gsub(label,'%s+',' '),'^%s*(.-)%s*$','%1')))if trimmed==''then return'ACTION'end if#
trimmed<=18 then return trimmed end local cut:string=string.sub(trimmed,1,18)local lastSpace:number?=string.find(string.reverse(cut),' ')if lastSpace and lastSpace<8 then cut=string.sub(cut,1,18-
lastSpace)end return cut end state.shortcutBindings={}::{[string]:any}local function shortcutSlotText(binding:any):string if binding.key==Enum.KeyCode.Unknown then return''end return binding.key.Name
end local function setShortcutPressed(binding:any,pressed:boolean):()if binding.pressed==pressed then return end binding.pressed=pressed if pressed then if binding.onPress then task.spawn(binding.
onPress)end elseif binding.onRelease then task.spawn(binding.onRelease)end end local function bindActionShortcut(button:TextButton?,shortcutId:string,label:string,callback:()->(),options:any?):()local
configuration:any=options or{}local optionKey:string='Shortcut.'..shortcutId:gsub('%W','')local binding:any={id=shortcutId,button=button,label=label,hold=configuration.hold==true,key=Enum.KeyCode.
Unknown,pressed=false,onPress=configuration.onPress or callback,onRelease=configuration.onRelease,feedback=configuration.feedback}local savedName:any=configData.values[optionKey]if type(savedName)==
'string'and savedName~=''then local resolved:boolean,savedKey:any=pcall(function():any return(Enum.KeyCode::any)[savedName]end)if resolved and savedKey then binding.key=savedKey end end state.
shortcutBindings[shortcutId]=binding local function refreshButtonText():()if not button then return end(button::TextButton).Text=shortcutSlotText(binding);(button::TextButton):SetAttribute(
'CaptureIdleText',(button::TextButton).Text)end binding.refresh=refreshButtonText refreshButtonText()binding.assign=function(keyCode:Enum.KeyCode):boolean local oldKey:Enum.KeyCode=binding.key local
cleared:boolean=keyCode==oldKey binding.key=cleared and Enum.KeyCode.Unknown or keyCode setShortcutPressed(binding,false)configData.values[optionKey]=binding.key.Name queueConfigSave()for _,other:any
in pairs(state.shortcutBindings)do if other~=binding or(oldKey~=Enum.KeyCode.Unknown and other.key==oldKey)or(binding.key~=Enum.KeyCode.Unknown and other.key==binding.key)then if type(other.refresh)==
'function'then other.refresh()end end end return not cleared end if state.isMobile then if button and state.bindMobileActionPlacement then state.bindMobileActionPlacement(button,'Shortcut'..shortcutId
:gsub('%W',''),compactShortcutLabel(label),function():()if not binding.hold and binding.onPress then binding.onPress()end end,{isActive=function():boolean return binding.hold==true end,onPress=
function():()if binding.hold and binding.onPress then binding.onPress()end end,onRelease=function():()if binding.hold and binding.onRelease then binding.onRelease()end end})end return end if button
then trackUiConnection((button::TextButton).MouseButton1Click:Connect(function():()beginKeyCapture(button,function(keyCode:Enum.KeyCode):()binding.assign(keyCode)end)end))end end state.
bindFeatureActivationKey=function(feature:any,defaultKey:Enum.KeyCode?,onPress:()->(),onRelease:(()->())?):()local binding:any=state.shortcutBindings[feature.configKey]if not binding then return end
binding.onPress=onPress binding.onRelease=onRelease binding.hold=onRelease~=nil if binding.refresh then binding.refresh()end end do local function shortcutInputAllowed(_gameProcessed:boolean):boolean
return not state.keyCaptureCallback and not state.waitingForKey and UserInputService:GetFocusedTextBox()==nil end state.globalShortcutBeganConnection=UserInputService.InputBegan:Connect(function(input
:InputObject,gameProcessed:boolean):()if input.UserInputType~=Enum.UserInputType.Keyboard or not shortcutInputAllowed(gameProcessed)then return end if input.KeyCode==Enum.KeyCode.Unknown then return
end for _,binding:any in pairs(state.shortcutBindings)do if binding.key==input.KeyCode then if binding.hold then setShortcutPressed(binding,true)elseif binding.onPress then task.spawn(function():()
binding.onPress()if type(binding.feedback)=='function'then local ok:boolean,text:any=pcall(binding.feedback)if ok and type(text)=='string'and text~=''then notify(text)end end end)end end end end)state
.globalShortcutEndedConnection=UserInputService.InputEnded:Connect(function(input:InputObject):()if input.UserInputType~=Enum.UserInputType.Keyboard or input.KeyCode==Enum.KeyCode.Unknown then return
end for _,binding:any in pairs(state.shortcutBindings)do if binding.hold and binding.key==input.KeyCode then setShortcutPressed(binding,false)end end end)trackUiConnection(UserInputService.
WindowFocusReleased:Connect(function():()for _,binding:any in pairs(state.shortcutBindings)do if binding.hold then setShortcutPressed(binding,false)end end end))end state.itemEspProviders={}state.
itemEspListeners={}state.gameBridge={roleProvider=nil::any,teamProvider=nil::any,extras={}::{any},listeners={}::{(string,any)->()}}state.onGameBridge=function(listener:(string,any)->()):()->()table.
insert(state.gameBridge.listeners,listener)if state.gameBridge.roleProvider then pcall(listener,'roles',state.gameBridge.roleProvider)end if state.gameBridge.teamProvider then pcall(listener,'team',
state.gameBridge.teamProvider)end for _,extra:any in ipairs(state.gameBridge.extras)do pcall(listener,'extra',extra)end return function():()local index:number?=table.find(state.gameBridge.listeners,
listener)if index then table.remove(state.gameBridge.listeners,index)end end end state.registerTeamProvider=function(provider:any):()if type(provider)~='table'or(type(provider.IsFriendly)~='function'
and type(provider.Get)~='function'and type(provider.GetRelation)~='function')then return end state.gameBridge.teamProvider=provider state.teamProvider=provider for _,listener:(string,any)->()in
ipairs(state.gameBridge.listeners)do pcall(listener,'team',provider)end end state.registerRoleProvider=function(provider:any):()if type(provider)~='table'or type(provider.Get)~='function'then return
end state.gameBridge.roleProvider=provider if type(state.clearRoleMemo)=='function'then state.clearRoleMemo()end for _,listener:(string,any)->()in ipairs(state.gameBridge.listeners)do pcall(listener,
'roles',provider)end end state.unregisterRoleProvider=function(provider:any):()if state.gameBridge.roleProvider~=provider then return end state.gameBridge.roleProvider=nil if state.clearRoleMemo then
state.clearRoleMemo()end for _,listener in ipairs(state.gameBridge.listeners)do pcall(listener,'roles',nil)end end state.registerEspExtra=function(extra:any):()if type(extra)~='table'or type(extra.
Name)~='string'then return end table.insert(state.gameBridge.extras,extra)for _,listener:(string,any)->()in ipairs(state.gameBridge.listeners)do pcall(listener,'extra',extra)end end local roleMemo:{[
Player]:{role:string?,at:number}}=setmetatable({},{__mode='k'})::any local roleColorMemo:{[Player]:{colour:Color3?,at:number}}=setmetatable({},{__mode='k'})::any local ROLE_MEMO_SECONDS:number=1/30
local TEAM_SIDE_BY_ROLE:{[string]:string}={['murderer']='hostile',['killer']='hostile',['traitor']='hostile',['zombie']='hostile',['infected']='hostile',['sheriff']='guard',['hero']='guard',[
'detective']='guard',['innocent']='guard',['survivor']='guard',['crewmate']='guard',['dead']='none'}state.teamCheck={}local TEAM_ATTRIBUTE_NAMES:{string}={'Team','TeamId','TeamID','Faction',
'FactionId','FactionID'}local function normalizeTeamToken(value:any):string?if typeof(value)=='Instance'then value=value.Name end if type(value)~='string'and type(value)~='number'then return nil end
local token:string=string.lower(string.gsub(tostring(value),'^%s*(.-)%s*$','%1'))if token==''or token=='none'or token=='unknown'or token=='outside'then return nil end return token end local function
teamAttribute(player:Player):string?for _,name:string in ipairs(TEAM_ATTRIBUTE_NAMES)do local ok:boolean,value:any=pcall(player.GetAttribute,player,name)local token:string?=ok and normalizeTeamToken(
value)or nil if token then return token end end local ok:boolean,value:any=pcall(function():any local child:Instance?=player:FindFirstChild('Team')return child and(child::any).Value or nil end)return
ok and normalizeTeamToken(value)or nil end local function providerToken(provider:any,player:Player):string?local ok:boolean,value:any=pcall(provider.Get,player)return ok and normalizeTeamToken(value)
or nil end local function isFfaToken(token:string?):boolean return token=='ffa'or token=='freeforall'end function state.teamCheck.analyze(player:Player?):{[string]:any}if not player or player==
LocalPlayer then return{sameTeam=false,method='self',hostile=false}end local provider:any=state.gameBridge.teamProvider if provider and type(provider.GetRelation)=='function'then local ok:boolean,
relation:any=pcall(provider.GetRelation,player)if ok and relation=='Teammate'then return{sameTeam=true,method='provider',hostile=false}end if ok and relation=='Enemy'then return{sameTeam=false,method=
'provider',hostile=true}end return{sameTeam=nil,method='provider-unknown',hostile=nil}end if provider and type(provider.IsFriendly)=='function'then local ok:boolean,friendly:any=pcall(provider.
IsFriendly,player)if ok and type(friendly)=='boolean'then return{sameTeam=friendly,method='provider',hostile=not friendly}end end if provider and type(provider.Get)=='function'then local mine:string?=
providerToken(provider,LocalPlayer)local theirs:string?=providerToken(provider,player::Player)if mine and theirs then if isFfaToken(mine)and isFfaToken(theirs)then return{sameTeam=false,method=
'provider-ffa',hostile=true}end if not isFfaToken(mine)and not isFfaToken(theirs)then return{sameTeam=mine==theirs,method='provider',hostile=mine~=theirs}end end return{sameTeam=nil,method=
'provider-unknown',hostile=nil}end if provider then return{sameTeam=nil,method='provider-unknown',hostile=nil}end local roleProvider:any=state.gameBridge.roleProvider if roleProvider and type(state.
playerRole)=='function'then local myRole:any=state.playerRole(LocalPlayer)local theirRole:any=state.playerRole(player::Player)if type(myRole)=='string'and type(theirRole)=='string'then local mySide:
string?=TEAM_SIDE_BY_ROLE[string.lower(myRole)]local theirSide:string?=TEAM_SIDE_BY_ROLE[string.lower(theirRole)]if mySide and theirSide and mySide~='none'and theirSide~='none'then return{sameTeam=
mySide==theirSide,method='roles',hostile=mySide~=theirSide}end if not mySide and not theirSide and myRole==theirRole and myRole~='Unknown'and myRole~='None'then return{sameTeam=true,method='roles',
hostile=false}end end if roleProvider.ReplacesTeams==true then return{sameTeam=nil,method='roles-unknown',hostile=nil}end end local myAttribute:string?=teamAttribute(LocalPlayer)local theirAttribute:
string?=teamAttribute(player::Player)if myAttribute or theirAttribute then if not myAttribute or not theirAttribute then return{sameTeam=nil,method='attributes-unknown',hostile=nil}end if isFfaToken(
myAttribute)and isFfaToken(theirAttribute)then return{sameTeam=false,method='attributes-ffa',hostile=true}end if isFfaToken(myAttribute)or isFfaToken(theirAttribute)then return{sameTeam=nil,method=
'attributes-unknown',hostile=nil}end return{sameTeam=myAttribute==theirAttribute,method='attributes',hostile=myAttribute~=theirAttribute}end if LocalPlayer.Neutral==true or(player::Player).Neutral==
true then return{sameTeam=false,method='neutral',hostile=true}end local myTeam:Team?=LocalPlayer.Team local theirTeam:Team?=(player::Player).Team if myTeam~=nil or theirTeam~=nil then if myTeam==nil
or theirTeam==nil then return{sameTeam=false,method='teams',hostile=true}end return{sameTeam=(myTeam::Team)==(theirTeam::Team),method='teams',hostile=(myTeam::Team)~=(theirTeam::Team)}end local
teamsService:any=game:FindService('Teams')if teamsService and#teamsService:GetTeams()>0 then local sameColour:boolean=LocalPlayer.TeamColor==(player::Player).TeamColor return{sameTeam=sameColour,
method='teamcolor',hostile=not sameColour}end return{sameTeam=nil,method='unknown',hostile=nil}end function state.teamCheck.isSameTeam(player:Player?):boolean local verdict:any=state.teamCheck.
analyze(player)return verdict.sameTeam==true end function state.teamCheck.isEnemy(player:Player?):boolean local verdict:any=state.teamCheck.analyze(player)if verdict.hostile~=nil then return verdict.
hostile end if verdict.sameTeam==false then return true end return false end state.playerRole=function(player:Player):string?local provider:any=state.gameBridge.roleProvider if not provider then
return nil end local now:number=os.clock()local cached:any=roleMemo[player]if cached and now-cached.at<ROLE_MEMO_SECONDS then return cached.role end local resolved:string?=nil local ok:boolean,role:
any=pcall(provider.Get,player)if ok and type(role)=='string'and role~=''then resolved=role end roleMemo[player]={role=resolved,at=now}return resolved end state.playerRoleColor=function(player:Player):
Color3?local provider:any=state.gameBridge.roleProvider if not provider then return nil end local now:number=os.clock()local cached:any=roleColorMemo[player]if cached and now-cached.at<
ROLE_MEMO_SECONDS then return cached.colour end local role:string?=state.playerRole(player)local resolved:Color3?=nil if role then if type(provider.GetColor)=='function'then local ok:boolean,colour:
any=pcall(provider.GetColor,role)if ok and typeof(colour)=='Color3'then resolved=colour end end if not resolved then local colour:any=provider.Colors and provider.Colors[role]if typeof(colour)==
'Color3'then resolved=colour end end end roleColorMemo[player]={colour=resolved,at=now}return resolved end state.playerPerk=function(player:Player):string?local provider:any=state.gameBridge.
roleProvider if not provider or type(provider.GetPerk)~='function'then return nil end local ok:boolean,perk:any=pcall(provider.GetPerk,player)if ok and type(perk)=='string'and perk~=''then return perk
end return nil end local function clearRoleMemo():()table.clear(roleMemo::any)table.clear(roleColorMemo::any)end state.clearRoleMemo=clearRoleMemo state.invalidateGameRoles=function():()clearRoleMemo(
)for _,listener:(string,any)->()in ipairs(state.gameBridge.listeners)do pcall(listener,'roleUpdate',state.gameBridge.roleProvider)end end state.emitGameBridgeEvent=function(eventType:string,payload:
any):()for _,listener:(string,any)->()in ipairs(state.gameBridge.listeners)do pcall(listener,eventType,payload)end end state.featureTooltip=create('TextLabel',{TextTruncate=Enum.TextTruncate.AtEnd,
ClipsDescendants=true,Parent=PopupLayer,Name='FeatureTooltip',Active=false,AutomaticSize=Enum.AutomaticSize.None,BackgroundColor3=Theme.surface,BackgroundTransparency=0.04,BorderColor3=Color3.fromRGB(
58,58,58),BorderSizePixel=1,FontFace=THIN_FONT,Size=UDim2.fromOffset(320,30),Text='',TextColor3=Color3.fromRGB(205,205,205),TextSize=11,TextWrapped=true,Visible=false,ZIndex=250})::TextLabel create(
'UIPadding',{Parent=state.featureTooltip,PaddingBottom=UDim.new(0,5),PaddingLeft=UDim.new(0,8),PaddingRight=UDim.new(0,8),PaddingTop=UDim.new(0,5)})state.tooltipToken=0 local function
showFeatureTooltip(row:Frame,text:string,pointer:Vector2?):()state.tooltipToken=state.tooltipToken+1 state.featureTooltip.Text=text state.featureTooltip.TextTransparency=1 local bitmap:any=state.
bitmapText local innerLimit:number=488 local wrapped:string,lineTotal:number=bitmap.wrap(text,innerLimit,16)local measured:number=bitmap.measure({{text=wrapped}},16)local width:number=math.clamp(
measured+24,210,520)local height:number=math.max(30,lineTotal*bitmap.lineHeight+12)state.featureTooltip.Size=UDim2.fromOffset(width,height)bitmap.draw(state.featureTooltip,{{text=wrapped,color=Color3.
fromRGB(205,205,205)}},{Name='Glyphs',TextSize=16,ZIndex=251,AnchorPoint=Vector2.new(0,0.5),Position=UDim2.new(0,0,0.5,0)})local manager:any=state.windows local viewport:Vector2 if manager and type(
manager.LogicalViewport)=='function'then viewport=manager.LogicalViewport()else local camera:Camera?=workspace.CurrentCamera viewport=camera and camera.ViewportSize or Vector2.new(1280,720)end local
anchor:Vector2=pointer or Vector2.new(row.AbsolutePosition.X+16,row.AbsolutePosition.Y+row.AbsoluteSize.Y*0.5)if manager and type(manager.ScreenToLogical)=='function'then anchor=manager.
ScreenToLogical(anchor)end local x:number=math.clamp(anchor.X+12,8,math.max(8,viewport.X-width-8))local y:number=math.clamp(anchor.Y-height-8,8,math.max(8,viewport.Y-height-8))state.featureTooltip.
Position=UDim2.fromOffset(x,y)state.featureTooltip.Visible=true end local function hideFeatureTooltip():()state.tooltipToken=state.tooltipToken+1 state.featureTooltip.Visible=false end local
COMPACT_FEATURE_CARDS:boolean=true local function addFeatureTooltip(feature:any,text:string):()if COMPACT_FEATURE_CARDS then return end feature.tooltipText=text if feature.tooltipBound then return end
feature.tooltipBound=true local row:Frame=feature.row::Frame local heldInput:InputObject?=nil local heldStart:Vector2?=nil table.insert(state.tooltipConnections,row.MouseLeave:Connect(function():()
heldInput=nil heldStart=nil hideFeatureTooltip()end))table.insert(state.tooltipConnections,row.InputBegan:Connect(function(input:InputObject):()if input.UserInputType~=Enum.UserInputType.Touch and
input.UserInputType~=Enum.UserInputType.MouseButton1 then return end heldInput=input heldStart=Vector2.new(input.Position.X,input.Position.Y)state.tooltipToken=state.tooltipToken+1 local pressToken:
number=state.tooltipToken task.delay(0.42,function():()if heldInput==input and state.tooltipToken==pressToken then local pointer:Vector2=input.UserInputType==Enum.UserInputType.Touch and Vector2.new(
input.Position.X,input.Position.Y)or UserInputService:GetMouseLocation()showFeatureTooltip(row,feature.tooltipText,pointer)end end)end))table.insert(state.tooltipConnections,row.InputChanged:Connect(
function(input:InputObject):()if heldInput==input and heldStart then local current:Vector2=Vector2.new(input.Position.X,input.Position.Y)if(current-heldStart).Magnitude>12 then heldInput=nil heldStart
=nil hideFeatureTooltip()end end end))table.insert(state.tooltipConnections,row.InputEnded:Connect(function(input:InputObject):()if heldInput==input then heldInput=nil heldStart=nil
hideFeatureTooltip()end end))end type FlyRuntime={yLevel:number?,burstAt:number,platform:BasePart?,walkSpeed:number?}local findPlayerByText:(string)->Player?=function(text:string):Player?local query:
string=string.lower(text)if query==''then return nil end local players:{Player}=Players:GetPlayers()for pass:number=1,3 do for _,player:Player in ipairs(players)do local name:string=string.lower(
player.Name)local displayName:string=string.lower(player.DisplayName)local matched:boolean=pass==1 and(name==query or displayName==query)or pass==2 and(name:sub(1,#query)==query or displayName:sub(1,#
query)==query)or(name:find(query,1,true)~=nil or displayName:find(query,1,true)~=nil)if matched then return player end end end return nil end local performFling:(Player)->()=function(_target:Player):(
)end state.isProtectedTarget=function(_targetPlayer:Player):boolean return false end local MM2Scroll:Frame=CardBin local cleanupMM2Runtime=function()end type GameModule={Name:string,PlaceId:number,
Events:{[string]:any},Initialized:boolean,init:(runtime:any)->any,destroy:()->()}type ModuleDiagnostics={uiBindings:number,taskBindings:number}state.logBootstrap=function(message:string):()print('['..
PRODUCT.logPrefix..':Bootstrap] '..message)end print('['..PRODUCT.logPrefix..'] build loader-v3 \183 '..SOURCE_STAMP)state.logBootstrapError=function(message:string):()warn('['..PRODUCT.logPrefix..
':Bootstrap] '..message)end local widgetBuilders:any=nil local function createGameModuleEnvironment(diagnostics:ModuleDiagnostics):any local baseEnvironment:any=getfenv()local function recordBinding(
callback:(...any)->any):(...any)->any return function(...:any):any diagnostics.uiBindings+=1 return callback(...)end end local environment:any={game=game,workspace=workspace,print=print,warn=warn,
Players=Players,UserInputService=UserInputService,TweenService=TweenService,Lighting=Lighting,StarterGui=game:GetService('StarterGui'),RunService=RunService,HttpService=HttpService,CollectionService=
CollectionService,Stats=Stats,LocalPlayer=LocalPlayer,PlayerGui=PlayerGui,ScreenGui=ScreenGui,PRODUCT=PRODUCT,Theme=Theme,ThemeEngine=ThemeEngine,GAME_CHECK=GAME_CHECK,CONTROL_FONT=CONTROL_FONT,
TITLE_FONT=TITLE_FONT,THIN_FONT=THIN_FONT,DEFAULT_HEADER_COLOR=DEFAULT_HEADER_COLOR,DEFAULT_SURFACE_COLOR=DEFAULT_SURFACE_COLOR,DEFAULT_BORDER_COLOR=DEFAULT_BORDER_COLOR,state=state,services=nil,
configData=configData,queueConfigSave=queueConfigSave,flushConfigSave=flushConfigSave,create=create,BUILD_LABEL=BUILD_LABEL,suppressTouchControls=suppressTouchControls,makeTextLabel=makeTextLabel,
makeButton=makeButton,makeTextBox=makeTextBox,notify=notify,isMenuOwned=function(instance:Instance):boolean return instance:IsDescendantOf(ScreenGui)end,TaskManager=TaskManager,featureConnections=
featureConnections,disconnectFeatureConnection=disconnectFeatureConnection,getCharacterParts=getCharacterParts,addFeatureTooltip=addFeatureTooltip,cloneReference=cloneReference,PopupLayer=PopupLayer,
UI_RADIUS=UI_RADIUS,UI_MOTION=UI_MOTION,cornerRadius=cornerRadius,setKeySlotCapture=setKeySlotCapture,SettingsPage=CardBin,getFeatureCardParent=getFeatureCardParent,REPOSITORY_RAW_BASE=
REPOSITORY_RAW_BASE,RUNTIME_SAFETY_SOURCE_URL=RUNTIME_SAFETY_SOURCE_URL,RUNTIME_COMPATIBILITY_MARKER=RUNTIME_COMPATIBILITY_MARKER,SOURCE_STAMP=SOURCE_STAMP,TEXT_BASE_MULTIPLIER=TEXT_BASE_MULTIPLIER,
TEXT_SCALE_MINIMUM=TEXT_SCALE_MINIMUM,TEXT_SCALE_MAXIMUM=TEXT_SCALE_MAXIMUM,registeredText=registeredText,initialViewport=initialViewport,FEATURE_CATEGORIES=FEATURE_CATEGORIES,categories={order=
OFFICIAL_CATEGORIES,aliases=CATEGORY_ALIASES,default=DEFAULT_FEATURE_CATEGORY},DEFAULT_FEATURE_CATEGORY=DEFAULT_FEATURE_CATEGORY,OFFICIAL_CATEGORIES=OFFICIAL_CATEGORIES,normalizeFeatureCategory=
normalizeFeatureCategory,universalFeatures=universalFeatures,movementFeatures=movementFeatures,applyFeaturePresentation=applyFeaturePresentation,refreshFeatureToggle=refreshFeatureToggle,VALVE_FONT=
VALVE_FONT,trackUiConnection=trackUiConnection,createToggleSwitch=createToggleSwitch,setToggleSwitch=setToggleSwitch,applyKeySlotStyle=applyKeySlotStyle,beginKeyCapture=beginKeyCapture,
bindActionShortcut=bindActionShortcut,runActionButton=runActionButton,colorFromConfig=colorFromConfig,colorToConfig=colorToConfig,registerRoleProvider=state.registerRoleProvider,registerTeamProvider=
state.registerTeamProvider,registerEspExtra=state.registerEspExtra,onGameBridge=state.onGameBridge,playerRole=state.playerRole,teamCheck=state.teamCheck,findPlayerByText=findPlayerByText,performFling=
performFling,MM2Scroll=MM2Scroll,mm2Features=mm2Features,vdFeatures=state.vdFeatures,bedFightFeatures=state.bedFightFeatures,bedWarsFeatures=state.bedWarsFeatures,mvsdFeatures=state.mvsdFeatures,
allFeatures=allFeatures,cleanupMM2Runtime=cleanupMM2Runtime}for builderName:string,builder:any in pairs(widgetBuilders or{})do environment[builderName]=recordBinding(builder)end return setmetatable(
environment,{__index=baseEnvironment})end local function loadGameModule(placeId:number):(GameModule?,any,ModuleDiagnostics,string?)local executorEnvironment:any=getfenv()local diagnostics:
ModuleDiagnostics={uiBindings=0,taskBindings=0}if type(executorEnvironment.loadstring)~='function'or type(executorEnvironment.setfenv)~='function'then return nil,nil,diagnostics,
'The executor does not support loadstring/setfenv'end local moduleName:string=GAME_MODULES[placeId]or ACTIVE_GAME_MODULE or tostring(placeId)local paths:{string}=state.gameModulePaths[moduleName]or{
'src/games/'..moduleName..'/base.lua'}local moduleEnvironment:any=createGameModuleEnvironment(diagnostics)moduleEnvironment.services=state.moduleServices local loadedModules:{GameModule}={}local
skipped:{string}={}for _,path:string in paths do local source:any=nil local failure:any=nil if type(state.fetchRepositorySource)=='function'then source,failure=state.fetchRepositorySource(path)else
local ok:boolean ok,source=pcall(function():string return(game::any):HttpGet(REPOSITORY_RAW_BASE..path..'?v='..SOURCE_STAMP,true)end)if not ok then failure=source source=nil end end if type(source)~=
'string'or#source<64 then local readFile:any=executorEnvironment.readfile local localPath:string='Wurst/'..path local ok:boolean=false if type(readFile)=='function'then ok,source=pcall(readFile,
localPath)end if not ok or type(source)~='string'then table.insert(skipped,path..' -> '..tostring(failure))continue end end local chunk:any,compileError:any=executorEnvironment.loadstring(source,'@'..
path)if type(chunk)~='function'then table.insert(skipped,path..' -> '..tostring(compileError))continue end executorEnvironment.setfenv(chunk,moduleEnvironment)local ok:boolean,loaded:any=pcall(chunk)
if not ok or type(loaded)~='table'or type(loaded.init)~='function'or type(loaded.destroy)~='function'then table.insert(skipped,path..' -> '..tostring(loaded))continue end table.insert(loadedModules,
loaded)end if#loadedModules==0 then return nil,nil,diagnostics,'No '..moduleName..' file loaded ('..table.concat(skipped,' | ')..')'end for _,reason:string in ipairs(skipped)do state.
logBootstrapError('Skipped '..moduleName..' file '..reason)end local composite:any={Name=moduleName,Initialized=false}function composite.init(runtime:any):any local initializedCount:number=0 local
failures:{string}={}for _,loaded:GameModule in loadedModules do local ok:boolean,failure:any=pcall(loaded.init,runtime)if ok then initializedCount+=1 else pcall(loaded.destroy)table.insert(failures,
tostring(loaded.Name)..': '..tostring(failure))state.logBootstrapError(moduleName..' module '..tostring(loaded.Name)..' failed: '..tostring(failure))end end if initializedCount==0 then error(table.
concat(failures,' | '))end composite.Initialized=true return composite end function composite.destroy():()for index:number=#loadedModules,1,-1 do pcall(loadedModules[index].destroy)end composite.
Initialized=false end return composite::GameModule,moduleEnvironment,diagnostics,nil end local function registerPlaceModule(tabName:string,placeId:number):()if game.PlaceId~=placeId and
ACTIVE_GAME_MODULE~=GAME_MODULES[placeId]then return end state.logBootstrap('Detected PlaceId: '..tostring(game.PlaceId)..' \183 module '..tostring(ACTIVE_GAME_MODULE)..' ('..ACTIVE_GAME_SOURCE..')')
SectionManager.register(tabName,function():()local loadedModule:GameModule?,_moduleEnvironment:any,diagnostics:ModuleDiagnostics,loadError:string?=loadGameModule(GAME_MODULES[game.PlaceId]and game.
PlaceId or placeId)if not loadedModule then state.logBootstrapError('Could not load '..tabName..': '..tostring(loadError))error(loadError or'Could not load the module')end local taskCountBefore:number
=TaskManager.nextId local featureCountBefore:number=#allFeatures local connectionsBefore:{[string]:boolean}={}for connectionName:string in pairs(featureConnections)do connectionsBefore[connectionName]
=true end local runtime:any={Menu={Root=CardBin,ScreenGui=ScreenGui,Pages={},Components=state.components,Notify=notify},State=state,Services=state.frameworkEnvironment.services,TaskManager=TaskManager
,Diagnostics=diagnostics}local initialized:boolean,initError:any=pcall(loadedModule.init,runtime)if not initialized then loadedModule.Initialized=true local rolledBack:boolean,rollbackError:any=pcall(
loadedModule.destroy)if not rolledBack then state.logBootstrapError(tabName..' rollback failed: '..tostring(rollbackError))end for connectionName:string in pairs(featureConnections)do if not
connectionsBefore[connectionName]then disconnectFeatureConnection(connectionName)end end for taskId:number,scheduled:ScheduledTask in pairs(TaskManager.tasks)do if taskId>taskCountBefore then
scheduled.handle:Disconnect()end end for index:number=#allFeatures,featureCountBefore+1,-1 do local feature:any=allFeatures[index]if feature.enabled then pcall(feature.onToggle,false)end if feature.
registry then feature.registry[feature.name]=nil end if feature.row and feature.row.Parent then feature.row:Destroy()end if feature.settingsWindow then pcall(function():()feature.settingsWindow:
Destroy()end)feature.settingsWindow=nil end table.remove(allFeatures,index)if state.moduleSearch and type(state.moduleSearch.Unregister)=='function'then state.moduleSearch.Unregister(feature)end if
feature.configKey and state.shortcutBindings then state.shortcutBindings[feature.configKey]=nil end if state.floatingUi and state.floatingUi.unregisterFeature then state.floatingUi.unregisterFeature(
feature)end end state.logBootstrapError(tabName..' init failed: '..tostring(initError))error(initError)end diagnostics.taskBindings=TaskManager.nextId-taskCountBefore state.activeGameModule=
loadedModule state.logBootstrap(tabName..' active \183 UI callbacks='..tostring(diagnostics.uiBindings)..' \183 Task callbacks='..tostring(diagnostics.taskBindings))local function destroyModule():()
local destroyed:boolean,destroyError:any=pcall(loadedModule.destroy)if not destroyed then state.logBootstrapError(tabName..' destroy failed: '..tostring(destroyError))end end if tabName=='MM2'then
cleanupMM2Runtime=destroyModule elseif tabName=='TRS'then state.cleanupTRSRuntime=destroyModule elseif tabName=='VD'then state.cleanupVDRuntime=destroyModule elseif tabName=='BedFight'then state.
cleanupBedFightRuntime=destroyModule elseif tabName=='X2Duels'then state.cleanupX2DuelsRuntime=destroyModule end end)end state.logBootstrap('Initializing PlaceId='..tostring(game.PlaceId)..
' \183 module='..tostring(ACTIVE_GAME_MODULE or'Universal'))registerPlaceModule('MM2',GAME_CHECK.MM2)registerPlaceModule('TRS',GAME_CHECK.TRS)registerPlaceModule('VD',GAME_CHECK.VD)
registerPlaceModule('BedFight',GAME_CHECK.BedFight)registerPlaceModule('X2Duels',GAME_CHECK.X2Duels)if PENDING_FINGERPRINT then task.spawn(function():()local attempts:number=0 while attempts<6 do
attempts+=1 task.wait(0.5)local detected:string?,score:number=detectGameModuleByStructure()if detected then ACTIVE_GAME_MODULE=detected ACTIVE_GAME_SOURCE='fingerprint '..tostring(score)..' markers'
state.activeGameName=detected GAME_CHECK.MM2Active=detected=='MM2'GAME_CHECK.TRSActive=detected=='TRS'GAME_CHECK.VDActive=detected=='VD'GAME_CHECK.BedFightActive=detected=='BedFight'GAME_CHECK.
X2DuelsActive=detected=='X2Duels'GAME_CHECK.source=ACTIVE_GAME_SOURCE PENDING_FINGERPRINT=false registerPlaceModule(detected,GAME_CHECK[detected]or 0)SectionManager.initialize(detected)break end end
end)end state.boot={stamp=SOURCE_STAMP,bundle=nil::{[string]:string}?,sources={}::{[string]:string},timing={}::{[string]:number},started=BOOT_STARTED_AT}state.boot.timing.gui=os.clock()-
BOOT_STARTED_AT state.markBoot=function(phase:string,startedAt:number):()state.boot.timing[phase]=os.clock()-startedAt end state.summarizeBoot=function():()local timing:any=state.boot.timing local
total:number=os.clock()-(state.boot.started or os.clock())print(string.format('['..PRODUCT.logPrefix..
':Timing] fingerprint %.2fs \183 gui %.2fs \183 bundle %.2fs \183 manifest %.2fs \183 widgets %.2fs \183 framework %.2fs \183 libraries %.2fs \183 modules %.2fs (%d/%d) \183 game module %.2fs \183 total %.2fs'
,BOOT_FINGERPRINT_SECONDS,tonumber(timing.gui)or 0,tonumber(timing.bundle)or 0,tonumber(timing.manifest)or 0,tonumber(timing.widgets)or 0,tonumber(timing.framework)or 0,tonumber(timing.libraries)or 0,
tonumber(timing.modules)or 0,tonumber(timing.moduleCount)or 0,tonumber(timing.moduleTotal)or 0,tonumber(timing.game)or 0,total))end state.sourceLooksValid=function(source:any,minimum:number):boolean
if type(source)~='string'or#source<minimum then return false end local head:string=string.lower(string.sub(source,1,512))return string.find(head,'404: not found',1,true)==nil and string.find(head,
'<!doctype html',1,true)==nil and string.find(head,'<html',1,true)==nil end state.describeHttpFailure=function(body:any):string local text:string=''if type(body)=='string'then text=string.sub(body,1,
512)else text=tostring(body)end local lowered:string=string.lower(text)local cause:string if string.find(lowered,'404: not found',1,true)then cause='HTTP 404'elseif string.find(lowered,
'<!doctype html',1,true)or string.find(lowered,'<html',1,true)then cause='HTML answer, not Lua'elseif#text<32 then cause='empty answer ('..tostring(#text)..' bytes)'else cause=string.sub(string.gsub(
text,'%s+',' '),1,160)end return cause..' \183 branch '..RUNTIME_BRANCH end state.httpGetPath=function(path:string):(string?,string?)local url:string=REPOSITORY_RAW_BASE..path..'?v='..SOURCE_STAMP
local ok:boolean,body:any=pcall(function():string return(game::any):HttpGet(url,true)end)if ok and state.sourceLooksValid(body,32)then return body::string,nil end if not ok then return nil,
'HttpGet threw: '..tostring(body)..' \183 branch '..RUNTIME_BRANCH end return nil,state.describeHttpFailure(body)end state.fetchRepositorySource=function(path:string):(string?,string?)local bundled:
any=state.boot.bundle and state.boot.bundle[path]if type(bundled)=='string'then return bundled,nil end local remembered:string?=state.boot.sources[path]if remembered then return remembered,nil end
local downloaded:string?,downloadError:string?=state.httpGetPath(path)if downloaded then state.boot.sources[path]=downloaded return downloaded,nil end return nil,downloadError or('Download failed: '..
path)end state.purgeLegacySourceCache=function():number local environment:any=getfenv()local listFiles:any=environment.listfiles local deleteFile:any=environment.delfile local isFolder:any=environment
.isfolder if type(listFiles)~='function'or type(deleteFile)~='function'then return 0 end local removed:number=0 local function purge(folder:string):()local listed:boolean,entries:any=pcall(listFiles,
folder)if not listed or type(entries)~='table'then return end for _,raw:any in ipairs(entries)do local entry:string=tostring(raw)local directory:boolean=false if type(isFolder)=='function'then local
checked:boolean,verdict:any=pcall(isFolder,entry)directory=checked and verdict==true end if directory then purge(entry)else if pcall(deleteFile,entry)then removed+=1 end end end end purge('Wurst/src')
pcall(deleteFile,'Wurst/.stamp')return removed end state.compileRepositorySource=function(path:string,source:string):(any,string?)local environment:any=getfenv()if type(environment.loadstring)~=
'function'then return nil,'The executor does not support loadstring'end local chunk:any,compileError:any=environment.loadstring(source,'@'..path)if type(chunk)~='function'then return nil,
'Compilation failed ('..path..'): '..tostring(compileError)end if type(environment.setfenv)=='function'and state.frameworkEnvironment then environment.setfenv(chunk,state.frameworkEnvironment)end
local ran:boolean,result:any=pcall(chunk)if not ran then return nil,'Execution failed ('..path..'): '..tostring(result)end return result,nil end state.fetchRepositoryChunk=function(path:string):(any,
string?)local source:string?,sourceError:string?=state.fetchRepositorySource(path)if not source then return nil,sourceError end return state.compileRepositorySource(path,source::string)end state.
downloadSourcesParallel=function(paths:{string}):()local pending:number=#paths if pending==0 then return end local index:number=0 local workers:number=math.min(8,pending)local timedOut:boolean=false
for _=1,workers do task.spawn(function():()while not timedOut do index+=1 local current:number=index if current>#paths then return end local path:string=paths[current]if not state.boot.sources[path]
then local ok:boolean,source:any=pcall(state.fetchRepositorySource,path)if ok and type(source)=='string'then state.boot.sources[path]=source end end pending-=1 end end)end local deadline:number=os.
clock()+12 while pending>0 and os.clock()<deadline do task.wait(0.03)end if pending>0 then timedOut=true for _,path:string in ipairs(paths)do if not state.boot.sources[path]then local source:string?=
state.fetchRepositorySource(path)if source then state.boot.sources[path]=source end end end end end do(function():()local diagnostics:ModuleDiagnostics={uiBindings=0,taskBindings=0}state.
frameworkEnvironment=createGameModuleEnvironment(diagnostics)type OwnershipService={set:(string,any?)->(),apply:(Instance?)->()}type ModuleServices={movementInput:any,aim:any,menu:any,mobileActions:
any,protectedTargets:any,fovOwnership:OwnershipService,platformStandOwnership:OwnershipService,activity:any,spoofAvatar:any,projectileCalibration:any,gameBridge:any,shortcuts:any,registries:any}
local function makePropertyOwnership(propertyName:string):OwnershipService local owners:{[string]:any}={}local baselines:any=setmetatable({},{__mode='k'})local sequence:number=0 local service:any={}
function service.set(ownerId:string,value:any?):()if value==nil then owners[ownerId]=nil if next(owners)==nil then for instance:any,baseline:any in pairs(baselines)do if instance.Parent then instance[
propertyName]=baseline end end baselines=setmetatable({},{__mode='k'})end return end local owner:any=owners[ownerId]if not owner then sequence+=1 owner={sequence=sequence,value=value}owners[ownerId]=
owner else owner.value=value end end function service.apply(instance:Instance?):()if not instance then return end local target:any=instance if baselines[target]==nil then baselines[target]=target[
propertyName]end local selected:any=nil for _,owner:any in pairs(owners)do if not selected or owner.sequence>selected.sequence then selected=owner end end if selected then target[propertyName]=
selected.value end end return service::OwnershipService end local protectedTargetProvider:(Player?)->boolean=function(player:Player?):boolean return player==nil or player==LocalPlayer end local
activityFlags:{[string]:boolean}={}local spoofEmotes:{any}={}local spoofDescription:HumanoidDescription?=nil local spoofRig:Model?=nil local projectileCalibrationOwner:any=nil local
SCREEN_CAPTURE_NAMES:{string}={'screenshot','captureScreenshot','capture_screenshot','getScreenshot'}local function findScreenCapture():any local environment:any=getfenv()for _,name:string in ipairs(
SCREEN_CAPTURE_NAMES)do local candidate:any=environment[name]if type(candidate)=='function'then return candidate end end return nil end local screenCaptureService:any={isAvailable=function():boolean
return findScreenCapture()~=nil end,capture=function(path:string):(boolean,string?)local environment:any=getfenv()local capture:any=findScreenCapture()if type(capture)~='function'then return false,
'executor has no screen capture API'end local folder:string?=string.match(path,'^(.*)/[^/]+$')local isFolder:any=environment.isfolder local makeFolder:any=environment.makefolder if folder and type(
isFolder)=='function'and type(makeFolder)=='function'then local parent:string?=string.match(folder::string,'^(.*)/[^/]+$')if parent and not isFolder(parent)then pcall(makeFolder,parent)end if not
isFolder(folder::string)then pcall(makeFolder,folder)end end local ok:boolean,result:any=pcall(capture,path)if not ok then ok,result=pcall(capture)end if not ok then return false,tostring(result)end
if result==false then return false,'screen capture API declined the request'end if type(result)=='string'and result~=''and type(environment.writefile)=='function'then pcall(environment.writefile,path,
result)end return true,nil end,writeText=function(path:string,contents:string):boolean local environment:any=getfenv()if type(environment.writefile)~='function'then return false end return pcall(
environment.writefile,path,contents)==true end}local moduleServices:ModuleServices={movementInput={getVector=state.getMoveInput,isJumpHeld=state.isJumpHeld,onJumpRequest=state.onJumpRequest},aim={
getRay=state.getAimRay},screenCapture=screenCaptureService,menu={isVisible=function():boolean return state.visible==true end,isCapturingInput=function():boolean return state.keyCaptureCallback~=nil or
state.waitingForKey==true end,setVisible=function(visible:boolean):()state.setMenuVisible(visible)end},mobileActions={bindPlacement=function(...:any):any local bindPlacement:any=state.
bindMobileActionPlacement if type(bindPlacement)=='function'then return bindPlacement(...)end return nil end},protectedTargets={isProtected=function(player:Player?):boolean return
protectedTargetProvider(player)end,setProvider=function(provider:((Player?)->boolean)?):()protectedTargetProvider=provider or function(player:Player?):boolean return player==nil or player==LocalPlayer
end end},fovOwnership=makePropertyOwnership('FieldOfView'),platformStandOwnership=makePropertyOwnership('PlatformStand'),activity={set=function(name:string,active:boolean):()activityFlags[name]=active
end,isActive=function(name:string):boolean return activityFlags[name]==true end},spoofAvatar={getEmotes=function():{any}return spoofEmotes end,setEmotes=function(emotes:{any}):()spoofEmotes=emotes end
,getDescription=function():HumanoidDescription?return spoofDescription end,setDescription=function(value:HumanoidDescription?):()spoofDescription=value end,getRig=function():Model?return spoofRig end,
setRig=function(value:Model?):()spoofRig=value end},projectileCalibration={set=function(owner:any):()projectileCalibrationOwner=owner end,get=function():any return projectileCalibrationOwner end},
itemEsp={register=function(provider:any):(()->())?if type(provider)~='table'or type(provider.name)~='string'or#provider.name==0 or type(provider.setEnabled)~='function'then return nil end for _,
existing:any in ipairs(state.itemEspProviders)do if existing.name==provider.name then return nil end end table.insert(state.itemEspProviders,provider)for _,listener:()->()in ipairs(state.
itemEspListeners)do pcall(listener)end return function():()local index:number?=table.find(state.itemEspProviders,provider)if index then table.remove(state.itemEspProviders,index::number)for _,listener
:()->()in ipairs(state.itemEspListeners)do pcall(listener)end end end end,providers=function():{any}return state.itemEspProviders end,onChanged=function(listener:()->()):(()->())?table.insert(state.
itemEspListeners,listener)return function():()local index:number?=table.find(state.itemEspListeners,listener)if index then table.remove(state.itemEspListeners,index::number)end end end},gameBridge={
onEvent=state.onGameBridge,teamCheck=state.teamCheck,playerRole=state.playerRole,playerRoleColor=state.playerRoleColor,playerPerk=state.playerPerk,registerRoles=state.registerRoleProvider,
unregisterRoles=state.unregisterRoleProvider,silentShot=function():any return state.lastSilentShotInfo end},shortcuts={bindActivation=state.bindFeatureActivationKey},registries={wurstOptions=function(
):any return state.wurstOptions end,moduleSearch=function():any return state.moduleSearch end}}state.frameworkEnvironment.services=moduleServices state.moduleServices=moduleServices local bundleAt:
number=os.clock()local packed:any,packedError:string?=state.fetchRepositoryChunk('src/libraries/bundle.lua')if type(packed)=='table'and packed.stamp==SOURCE_STAMP and type(packed.files)=='table'then
state.boot.bundle=packed.files state.logBootstrap('Bundle accepted \183 stamp='..SOURCE_STAMP)task.spawn(function():()local purgedSources:number=state.purgeLegacySourceCache()local staleAssets:number=
state.sweepStaleAssets()if purgedSources+staleAssets>0 then print('['..PRODUCT.logPrefix..':Cache] purged '..tostring(purgedSources)..' legacy source file(s), '..tostring(staleAssets)..
' stale asset(s)')end end)else if packedError then state.logBootstrapError('Bundle: '..packedError)elseif type(packed)=='table'then state.logBootstrapError('Bundle stamp mismatch: got '..tostring(
packed.stamp)..' expected '..SOURCE_STAMP)end state.boot.bundle=nil end state.markBoot('bundle',bundleAt)local manifestAt:number=os.clock()local manifest:any,manifestError:string?=state.
fetchRepositoryChunk('src/libraries/Manifest.lua')if type(manifest)~='table'or type(manifest.modules)~='table'then if manifestError then state.logBootstrapError('Manifest: '..manifestError)end state.
logBootstrapError([[Manifest: src/libraries/Manifest.lua did not load; the menu cannot boot]])if type(state.notify)=='function'then pcall(state.notify,'Wurst: the module catalog could not be loaded')
end return end local catalogCategories:any=manifest.categories if type(catalogCategories)=='table'then if type(catalogCategories.order)=='table'and#catalogCategories.order>0 then OFFICIAL_CATEGORIES=
catalogCategories.order end if type(catalogCategories.aliases)=='table'then CATEGORY_ALIASES=catalogCategories.aliases end end if type(manifest.cards)=='table'then FEATURE_CATEGORIES=manifest.cards
end state.categories={order=OFFICIAL_CATEGORIES,aliases=CATEGORY_ALIASES,default=DEFAULT_FEATURE_CATEGORY}state.markBoot('manifest',manifestAt)state.gameModulePaths=manifest.gameModules or{}for _,
entry:any in ipairs(manifest.gamePlaces or{})do if type(entry)=='table'and type(entry.placeId)=='number'and type(entry.module)=='string'then GAME_MODULES[entry.placeId]=entry.module end end local
loadList:{string}={}table.insert(loadList,manifest.widgets or'src/guis/Wurst/Code/Widgets.lua')for _,path:string in ipairs(manifest.core or{})do table.insert(loadList,path)end for _,path:string in
ipairs(manifest.libraries or{})do table.insert(loadList,path)end for _,entry:any in ipairs(manifest.modules or{})do table.insert(loadList,type(entry)=='table'and entry.path or tostring(entry))end if
not state.boot.bundle then state.downloadSourcesParallel(loadList)end local loaded:{any}={}local function loadPiece(path:string):any?local piece:any,failure:string?=state.fetchRepositoryChunk(path)if
type(piece)~='table'or type(piece.init)~='function'then state.logBootstrapError('Invalid module: '..path..' \183 '..tostring(failure))return nil end return piece end local widgetsAt:number=os.clock()
local widgetPath:string=manifest.widgets or'src/guis/Wurst/Code/Widgets.lua'local widgetPiece:any?=loadPiece(widgetPath)if widgetPiece then local widgetOk:boolean,builders:any=pcall(widgetPiece.init,
state.frameworkEnvironment)if widgetOk and type(builders)=='table'then table.insert(loaded,widgetPiece)widgetBuilders=builders for builderName:string,builder:any in pairs(builders)do state.
frameworkEnvironment[builderName]=function(...:any):any diagnostics.uiBindings+=1 return builder(...)end end else state.logBootstrapError('Widgets init failed: '..tostring(builders))end end state.
markBoot('widgets',widgetsAt)if not widgetBuilders then state.logBootstrapError([[Widget library unavailable; modules cannot build their settings]])notify('Could not load the widget library')state.
summarizeBoot()return end local frameworkAt:number=os.clock()local frameworkPiece:any?=loadPiece(manifest.core[1]or'src/libraries/Framework.lua')if not frameworkPiece then state.logBootstrapError(
'Framework unavailable; external modules skipped')state.summarizeBoot()return end local frameworkOk:boolean,framework:any=pcall(frameworkPiece.init,state.frameworkEnvironment)if not frameworkOk or
type(framework)~='table'then state.logBootstrapError('Framework init failed: '..tostring(framework))state.summarizeBoot()return end table.insert(loaded,frameworkPiece)state.framework=framework state.
markBoot('framework',frameworkAt)local librariesAt:number=os.clock()local libraries:any={}for _,path:string in ipairs(manifest.libraries or{})do local piece:any?=loadPiece(path)if not piece then
continue end local startedOk:boolean,library:any=pcall(piece.init,state.frameworkEnvironment)if not startedOk or type(library)~='table'then state.logBootstrapError('Library failed ('..path..'): '..
tostring(library))continue end table.insert(loaded,piece)local key:string=string.lower(string.match(path,'([^/]+)%.lua$')or path)libraries[key]=library end local cardLibrary:any=libraries.cards if
cardLibrary and type(cardLibrary.createUniversalFeature)=='function'then widgetBuilders.createUniversalFeature=cardLibrary.createUniversalFeature state.frameworkEnvironment.createUniversalFeature=
function(...:any):any diagnostics.uiBindings+=1 return cardLibrary.createUniversalFeature(...)end else state.logBootstrapError('Card factory unavailable; modules cannot create cards')notify(
'Could not load the card factory')return end local entityLibrary:any=libraries.entity state.weaponLibrary=libraries.weapons state.renderLibrary=libraries.render state.markBoot('libraries',librariesAt)
local context:any={framework=framework,entity=entityLibrary,weapons=libraries.weapons,render=libraries.render,libraries=libraries,host=state.frameworkEnvironment,services=moduleServices}local
modulesAt:number=os.clock()local moduleCount:number=0 for _,entry:any in ipairs(manifest.modules)do local path:string=type(entry)=='table'and entry.path or tostring(entry)local piece:any?=loadPiece(
path)if piece then local started:boolean,failure:any=pcall(piece.init,context)if started then table.insert(loaded,piece)moduleCount+=1 state.logBootstrap('Module loaded: '..path)else state.
logBootstrapError('Module failed ('..path..'): '..tostring(failure))end end end state.boot.timing.moduleCount=moduleCount state.boot.timing.moduleTotal=#manifest.modules state.markBoot('modules',
modulesAt)if moduleCount<#manifest.modules then local missing:number=#manifest.modules-moduleCount state.logBootstrapError(tostring(missing)..' de '..tostring(#manifest.modules)..
' modules did not start; look for "Module failed" above')notify(tostring(missing)..' modules did not load \u{2014} check the console')end state.cleanupFrameworkRuntime=function():()for index:number=#
loaded,1,-1 do local piece:any=loaded[index]if type(piece.destroy)=='function'then pcall(piece.destroy)end end table.clear(loaded)state.framework=nil state.renderLibrary=nil end state.summarizeBoot()
end)()end do local savedFont:string=tostring(configData.values.interfaceFont or'')if savedFont==''or savedFont=='Menu default'or savedFont=='Minecraft (Monocraft)'or savedFont=='Candy Fruits'then
savedFont='Code'end task.spawn(function():()state.applyInterfaceFont(savedFont,false)end)end do local function initializeWindowInput():()local function reflowForViewport(force:boolean?):()task.defer(
function():()if not ScreenGui.Parent then return end if not state.refreshLayout(force)then return end if state.floatingUi and state.floatingUi.reclamp then state.floatingUi.reclamp()end end)end local
viewportConnection:RBXScriptConnection?=nil local function bindViewport():()if viewportConnection then viewportConnection:Disconnect()viewportConnection=nil end local camera:Camera?=workspace.
CurrentCamera if not camera then return end viewportConnection=trackUiConnection(camera:GetPropertyChangedSignal('ViewportSize'):Connect(function():()reflowForViewport(false)end))end bindViewport()
trackUiConnection(workspace:GetPropertyChangedSignal('CurrentCamera'):Connect(function():()bindViewport()reflowForViewport(true)end))state.globalInputBeganConnection=UserInputService.InputBegan:
Connect(function(input)if state.keyCaptureCallback then if input.UserInputType==Enum.UserInputType.Keyboard and input.KeyCode~=Enum.KeyCode.Unknown then if input.KeyCode==Enum.KeyCode.Escape then if
state.keyCaptureCancel then state.keyCaptureCancel()end return end local callback=state.keyCaptureCallback callback(input.KeyCode)task.defer(function():()if state.keyCaptureCallback==callback then
state.keyCaptureCallback=nil end end)end return end if state.waitingForKey then if input.UserInputType==Enum.UserInputType.Keyboard and input.KeyCode~=Enum.KeyCode.Unknown then if input.KeyCode==Enum.
KeyCode.Escape then state.waitingForKey=false if state.keybindButton then state.keybindButton.Text=state.keyDisplayName(state.toggleKey.Name)setKeySlotCapture(state.keybindButton,false)end return end
state.toggleKey=input.KeyCode if state.keybindButton then state.keybindButton.Text=state.keyDisplayName(input.KeyCode.Name)setKeySlotCapture(state.keybindButton,false)end configData.ui.toggleKey=input
.KeyCode.Name queueConfigSave()notify('Menu key: '..input.KeyCode.Name)task.defer(function():()state.waitingForKey=false end)end return end if UserInputService:GetFocusedTextBox()then return end if
input.UserInputType==Enum.UserInputType.Keyboard and input.KeyCode==state.toggleKey then state.openMenuSurface()end end)end initializeWindowInput()task.defer(function():()state.refreshLayout(true)end)
end local runtime:any=createRuntimeDestructor({state=state,MenuBlur=MenuBlur,ScreenGui=ScreenGui,allFeatures=allFeatures,featureConnections=featureConnections,TaskManager=TaskManager,cleanupMM2Runtime
=cleanupMM2Runtime,closeConfigSave=closeConfigSave,setMenuCursorUnlocked=setMenuCursorUnlocked,disconnectFeatureConnection=disconnectFeatureConnection,persistenceGenv=persistenceGenv,logPrint=print,
warn=warn})state.destruct=runtime.destructRuntime state.cleanupRuntime=runtime.cleanupRuntime runtime.bindHandoff()SectionManager.initializeAllAsync()task.defer(saveConfigNow)print(PRODUCT.name..' v'
..PRODUCT.version..' | Toggle key: '..state.toggleKey.Name..' | build '..SOURCE_STAMP..' | branch '..RUNTIME_BRANCH)end local initialized:boolean,bootstrapError:any=xpcall(bootstrap,function(message:
any):string local text:string=tostring(message)if debug and type(debug.traceback)=='function'then return debug.traceback('['..PRODUCT.logPrefix..':Bootstrap] '..text,2)end return'['..PRODUCT.logPrefix
..':Bootstrap] '..text end)if not initialized then local reportFatal:any=(getfenv()::any).warn if type(reportFatal)=='function'then reportFatal('['..PRODUCT.logPrefix..
':Bootstrap] Initialization stopped:\n'..tostring(bootstrapError))end pcall(function():()local players:any=game:GetService('Players')local playerGui:any=players.LocalPlayer:WaitForChild('PlayerGui')
local gui:any=Instance.new('ScreenGui')gui.Name='WurstBootError'gui.ZIndexBehavior=Enum.ZIndexBehavior.Sibling gui.DisplayOrder=9999 gui.Parent=playerGui local box:any=Instance.new('Frame')box.
BackgroundColor3=Color3.fromRGB(26,24,30)box.BorderSizePixel=0 box.Size=UDim2.fromOffset(440,150)box.Position=UDim2.new(0.5,-220,0.06,0)box.Parent=gui Instance.new('UICorner',box).CornerRadius=UDim.
new(0,8)local label:any=Instance.new('TextLabel')label.BackgroundTransparency=1 label.Size=UDim2.new(1,-28,1,-16)label.Position=UDim2.fromOffset(14,8)label.Font=Enum.Font.Gotham label.TextSize=13
label.TextXAlignment=Enum.TextXAlignment.Left label.TextYAlignment=Enum.TextYAlignment.Top label.TextWrapped=true label.TextColor3=Color3.fromRGB(255,122,122)label.Text='[Wurst] no arranco:\n'..
tostring(bootstrapError):sub(1,400)label.Parent=box end)end
```